

# **$q$ -Series and Inverse $q$ -Analog**

## **A Proof-Oriented Synthesis**

Human-readable research synthesis for  
Vladimir Reshetnikov's ProveIt repository

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# Abstract

The finite and infinite  $q$ -Pochhammer symbols and the Gaussian, or  $q$ -binomial, coefficients form a small set of notational primitives from which a remarkable amount of mathematics can be built. They simultaneously encode weighted subsets, partitions, lattice paths, finite-field geometry,  $q$ -difference operators, basic hypergeometric summations, theta products, cyclotomic arithmetic, geometric interpolation, and the Fabius–Rvachev functions.

This volume gives one proof-oriented account of both the forward theory and its branch-aware inverse problems. The forward parts develop the algebraic, combinatorial, analytic, arithmetic, and asymptotic structure of  $q$ -series; the inverse part develops regular and ramified reversion, Bell-polynomial and Lagrange–Good coefficient engines, Lambert- $W$  asymptotic transfer, finite- and infinite-product branch atlases,  $q$ -special-function inversion, certified numerical enclosures, and parameter recovery from Fabius–Rvachev observables. Repeated source results are stated once with the strongest proved hypotheses. Every retained theorem, lemma, proposition, and corollary has a human-readable proof; genuinely open claims occur only in explicitly labelled conjecture environments. The appendices distinguish exact Lean counterparts, partial formal infrastructure, human-proved frontier results, source records, and conjectures.

# Preface and source map

The aim is not merely to list identities, but to expose the mechanisms that repeatedly generate them. Three such mechanisms dominate:

1. finite products split into residue classes, blocks, or initial and tail factors;
2. coefficient extraction turns product factorizations into convolution identities;
3. functional equations determine analytic  $q$ -series from one initial value.

Once these mechanisms are visible, many formulas that initially appear unrelated become instances of the same principle.

The supplied archive was used throughout. In particular, the weighted subset framework of Konvalina motivates [Chapter 4](#); Kupershmidt’s treatment of the  $q$ -Newton binomial informs [Chapters 12](#) and [20](#); Straub’s seminar on  $q$ -binomial congruences is reflected in [Chapter 19](#); the formalized  $q$ -series project of Lau, Lee, and Ono motivates the algebraic-versus-analytic distinctions in [Chapter 24](#) and supplies a modern Bailey-pair route to the Rogers–Ramanujan identities; and the central  $q$ -binomial paper of Chern, Dilcher, and Jiu is developed, with a complete proof, in [Chapter 21](#). The archive’s reference chapters and encyclopedia snapshots were checked against the standard sources listed in the bibliography. The current *ProveIt* Fabius-function documentation [43], including its main exposition and non-formalized research-frontier drafts, supplies the geometric-node interpolation rows, exact residual moments, half-base identities, formal filters, and dyadic Thue–Morse connections expanded independently in [Chapters 25](#) to [27](#). Online research also led to the recent automated identity-translation framework of Zhong and Zhao [45] and the Isabelle formalization of combinatorial  $q$ -analogues by Eberl [41]. Additional material was drawn from the classical books and articles of Andrews, Askey, Bailey, Gasper, Rahman, Rota, Goldman, Reiner, Stanton, White, and others, as well as the NIST Digital Library of Mathematical Functions.

## Two regimes for $q$

Throughout the finite algebraic chapters,  $q$  may be an indeterminate and identities are understood in a polynomial or rational-function ring. In analytic chapters, unless stated otherwise,

$$q \in \mathbb{C}, \quad 0 < |q| < 1.$$

For the Jackson integral and positivity statements we usually assume  $0 < q < 1$ . A capital  $Q$  is used for the cardinality of a finite field, so  $Q$  is a prime power and normally  $Q > 1$ .

## Conventions

Empty products equal 1, empty sums equal 0, and

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = 0 \quad \text{when } k < 0 \text{ or } k > n$$

for nonnegative integral  $n$ . Multiple parameters are compressed as

$$(a_1, \dots, a_r; q)_n = \prod_{j=1}^r (a_j; q)_n.$$

The notation  $(a; q)_\infty$  is analytic when  $|q| < 1$ ; when formal power series are intended, that will be said explicitly.

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## **Part I**

# **The finite algebra**

# Chapter 1

## The algebra of $q$ -shifted factorials

### 1.1 Definitions and first examples

**Definition 1.1.** For  $n \in \mathbb{N}_0$  the  $q$ -shifted factorial, or  $q$ -Pochhammer symbol, is

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad (a; q)_0 = 1.$$

For parameters  $a_1, \dots, a_r$  we write

$$(a_1, \dots, a_r; q)_n = \prod_{i=1}^r (a_i; q)_n.$$

When  $0 < |q| < 1$ , the infinite symbol is

$$(a; q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j).$$

The word “shifted” refers to the geometric progression  $a, aq, aq^2, \dots$ . The first few values are

$$(a; q)_1 = 1 - a, \quad (a; q)_2 = (1 - a)(1 - aq), \quad (a; q)_3 = (1 - a)(1 - aq)(1 - aq^2).$$

Important specializations include

$$(q; q)_n = \prod_{j=1}^n (1 - q^j), \quad (-q; q)_n = \prod_{j=1}^n (1 + q^j), \quad (a; q^r)_n = \prod_{j=0}^{n-1} (1 - aq^{rj}).$$

### 1.2 Concatenation, dissection, and reversal

**Proposition 1.2** (Concatenation law). For  $m, n \in \mathbb{N}_0$ ,

$$(a; q)_{m+n} = (a; q)_m (aq^m; q)_n.$$

Consequently, whenever the denominator is nonzero,

$$\frac{(a; q)_{m+n}}{(a; q)_m} = (aq^m; q)_n.$$

*Proof.* Split the defining product at  $j = m$ :

$$\prod_{j=0}^{m+n-1} (1 - aq^j) = \left( \prod_{j=0}^{m-1} (1 - aq^j) \right) \left( \prod_{j=m}^{m+n-1} (1 - aq^j) \right).$$

In the second product put  $j = m + r$ , where  $0 \leq r < n$ . □

**Proposition 1.3** (Dissection into residue classes). *For integers  $r, n \geq 1$ ,*

$$(a; q)_{rn} = \prod_{s=0}^{r-1} (aq^s; q^r)_n.$$

*If  $|q| < 1$ , then also*

$$(a; q)_{\infty} = \prod_{s=0}^{r-1} (aq^s; q^r)_{\infty}.$$

*Proof.* Every integer  $j$  with  $0 \leq j < rn$  has a unique representation  $j = s + rt$  with  $0 \leq s < r$  and  $0 \leq t < n$ . Reordering the finite product gives the first identity. For the infinite identity, the factors are absolutely summably close to 1 because  $\sum_{j \geq 0} |aq^j| < \infty$ , so the product may be regrouped into the  $r$  residue classes. □

**Corollary 1.4** (Dissection with a remainder). *Let  $r \geq 1$  and  $n \geq 0$  be integers, let  $0 \leq u < r$ , and set  $N = rn + u$ . Then*

$$(a; q)_N = \prod_{s=0}^{u-1} (aq^s; q^r)_{n+1} \prod_{s=u}^{r-1} (aq^s; q^r)_n.$$

*For  $u = 0$  this is [Proposition 1.3](#).*

*Proof.* Division with remainder by  $r$  is a bijection between the nonnegative integers  $j$  and the pairs  $(s, \ell)$  with  $0 \leq s < r$  and  $\ell \geq 0$ , given by  $j = s + r\ell$ . Fix a residue  $s$  and count the  $\ell$  for which  $j = s + r\ell$  lies in the exponent set  $\{0, \dots, N - 1\}$  of the defining product. The condition  $s + r\ell < rn + u$  reads

$$r\ell < rn + (u - s).$$

If  $s < u$ , then  $0 < u - s \leq u < r$ , so

$$rn < rn + (u - s) \leq rn + r = r(n + 1).$$

As  $r\ell$  runs through the multiples of  $r$ , the inequality  $r\ell < rn + (u - s)$  therefore holds exactly for  $\ell = 0, 1, \dots, n$ , which is  $n + 1$  values.

If  $s \geq u$ , then  $u - s \leq 0$ , so the inequality implies  $r\ell < rn$ , that is  $\ell \leq n - 1$ ; conversely every such  $\ell$  satisfies  $s + r\ell \leq s + r(n - 1) = rn + (s - r) < rn \leq N$ . Hence  $\ell$  ranges over  $0, \dots, n - 1$ , which is  $n$  values.

The defining product is finite and its factors commute, so it may be reordered according to the residue of  $j$  modulo  $r$ :

$$(a; q)_N = \prod_{s=0}^{u-1} \prod_{\ell=0}^n (1 - aq^s(q^r)^\ell) \prod_{s=u}^{r-1} \prod_{\ell=0}^{n-1} (1 - aq^s(q^r)^\ell).$$

Each inner product is by definition a  $q^r$ -shifted factorial with argument  $aq^s$ , namely  $(aq^s; q^r)_{n+1}$  in the first family and  $(aq^s; q^r)_n$  in the second. This is the asserted identity. When  $u = 0$  the first family is empty and the second has all  $r$  residues, recovering [Proposition 1.3](#). □

**Remark 1.5.** Two consistency checks are worth recording. The total number of linear factors on the right is

$$u(n+1) + (r-u)n = rn + u = N,$$

as it must be, since  $(a; q)_N$  is a product of  $N$  linear factors in  $a$ . At the other extreme  $n = 0$  the corollary reads  $(a; q)_u = \prod_{s=0}^{u-1} (aq^s; q^r)_1 = \prod_{s=0}^{u-1} (1 - aq^s)$ , again the definition. Combining the corollary with [Proposition 1.2](#) in the form  $(a; q)_{rn+u} = (a; q)_{rn} (aq^{rn}; q)_u$  and dividing by [Proposition 1.3](#) recovers the identity  $\prod_{s=0}^{u-1} (1 - aq^{s+rn}) = (aq^{rn}; q)_u$ , so the two decompositions agree.

**Proposition 1.6** (Base reversal). *For  $n \geq 0$  and nonzero  $a, q$  in a field,*

$$(a; q)_n = (-a)^n q^{\binom{n}{2}} (a^{-1}; q^{-1})_n.$$

*Equivalently,*

$$(a; q^{-1})_n = (-a)^n q^{-\binom{n}{2}} (a^{-1}; q)_n.$$

*Proof.* Factor  $-aq^j$  from each term:

$$1 - aq^j = -aq^j (1 - a^{-1}q^{-j}).$$

Multiplying for  $0 \leq j < n$  gives the factor  $(-a)^n q^{0+1+\dots+(n-1)} = (-a)^n q^{\binom{n}{2}}$ . The remaining product is  $(a^{-1}; q^{-1})_n$ . The assumptions  $a \neq 0$  and  $q \neq 0$  are exactly what makes the two displayed inverses legitimate; equivalently one may formulate the same proof for unit parameters over a commutative ring.  $\square$

### 1.3 Negative indices

The concatenation law suggests the only extension to negative integers that preserves

$$(a; q)_{m+n} = (a; q)_m (aq^m; q)_n.$$

**Definition 1.7.** For  $n \geq 1$ , define

$$(a; q)_{-n} = \frac{1}{(aq^{-n}; q)_n},$$

provided no factor in the denominator vanishes.

**Proposition 1.8** (Closed form for a negative index). *For  $n \geq 1$  and  $a \neq 0$ ,*

$$(a; q)_{-n} = \frac{(-1)^n a^{-n} q^{\binom{n+1}{2}}}{(q/a; q)_n}.$$

*Moreover, for all integers  $m, n$  for which both sides are defined,*

$$(a; q)_{m+n} = (a; q)_m (aq^m; q)_n.$$

*Proof.* By definition,

$$(aq^{-n}; q)_n = \prod_{j=0}^{n-1} (1 - aq^{-n+j}).$$

Factor  $-aq^{-n+j}$  from each factor. Their product is

$$(-a)^n q^{-n^2 + \binom{n}{2}} = (-a)^n q^{-\binom{n+1}{2}},$$

and the remaining product is

$$\prod_{j=0}^{n-1} (1 - a^{-1}q^{n-j}) = (q/a; q)_n.$$

Taking the reciprocal proves the formula. The extended concatenation identity follows by cancelling the overlapping finite factors; the cases in which one of  $m, n, m+n$  is negative reduce directly to the definition.  $\square$

**Corollary 1.9** (A terminating factor). *Let  $q$  be a nonzero element of a field. If  $0 \leq k \leq N$  and  $(q; q)_{N-k} \neq 0$ , then*

$$(q^{-N}; q)_k = (-1)^k q^{-Nk + \binom{k}{2}} \frac{(q; q)_N}{(q; q)_{N-k}}.$$

*Independently of that denominator condition,  $(q^{-N}; q)_k = 0$  for  $k > N$ .*

*Proof.* The denominator-cleared identity

$$q^{Nk} (q^{-N}; q)_k = (-1)^k q^{\binom{k}{2}} (q^{N-k+1}; q)_k$$

holds for  $k \leq N$  once  $q \neq 0$ . Concatenation gives

$$(q; q)_N = (q; q)_{N-k} (q^{N-k+1}; q)_k.$$

Division by the nonzero product  $q^{Nk} (q; q)_{N-k}$  gives the displayed quotient. For  $k > N$ , the defining product contains its index- $N$  factor  $1 - q^{-N}q^N = 0$ ; here only  $q \neq 0$  is needed.  $\square$

## 1.4 Partial fractions and logarithmic derivatives

**Theorem 1.10** (Finite partial-fraction decomposition). *Assume  $q^i \neq q^j$  for  $0 \leq i < j \leq n$ . Then*

$$\frac{1}{(z; q)_{n+1}} = \sum_{j=0}^n \frac{(-1)^j q^{\binom{j+1}{2}}}{(q; q)_j (q; q)_{n-j}} \frac{1}{1 - zq^j}.$$

*Proof.* If  $q = 0$ , pairwise distinctness of  $q^0, \dots, q^n$  forces  $n \leq 1$ , and the formula is immediate in the two possible cases. Assume henceforth that  $q \neq 0$ . The poles of the left-hand side are the distinct points  $z = q^{-j}$ ,  $0 \leq j \leq n$ , and the coefficient of  $(1 - zq^j)^{-1}$  in its partial-fraction expansion is

$$A_j = \frac{1 - zq^j}{(z; q)_{n+1}} \Big|_{z=q^{-j}} = \frac{1}{\prod_{r \neq j} (1 - q^{r-j})}.$$

For  $r < j$ , put  $s = j - r$  and obtain

$$\prod_{r=0}^{j-1} (1 - q^{r-j}) = \prod_{s=1}^j (1 - q^{-s}) = (-1)^j q^{-\binom{j+1}{2}} (q; q)_j.$$

The factors with  $r > j$  multiply to  $(q; q)_{n-j}$ . Thus  $A_j$  is the displayed coefficient. After subtracting the proposed right-hand side, all finite poles are removable. The difference is therefore a polynomial; it is  $O(z^{-1})$  at infinity, so it is identically zero.  $\square$

**Proposition 1.11** (Logarithmic derivatives). *Away from its zeros,*

$$\frac{\partial}{\partial a} \log(a; q)_n = - \sum_{j=0}^{n-1} \frac{q^j}{1 - aq^j}.$$



If  $a = a(x)$  is differentiable, then

$$\frac{d}{dx} \log(a(x); q)_n = -a'(x) \sum_{j=0}^{n-1} \frac{q^j}{1 - a(x)q^j}.$$

*Proof.* Differentiate the logarithm of the finite product term by term.  $\square$

## 1.5 Infinite products and Lambert series

**Theorem 1.12** (Convergence and zero set). *Let  $0 < |q| < 1$ . The product  $(a; q)_\infty$  converges locally uniformly as a function of  $a \in \mathbb{C}$  and therefore defines an entire function. Its zeros are precisely*

$$a = q^{-j}, \quad j = 0, 1, 2, \dots,$$

*and all are simple.*

*Proof.* On a compact set  $K \subset \mathbb{C}$  let  $M = \max_{a \in K} |a|$ . Then

$$\sum_{j=0}^{\infty} \sup_{a \in K} |aq^j| \leq \frac{M}{1 - |q|} < \infty.$$

The standard convergence criterion for infinite products shows that  $\prod_j (1 - aq^j)$  converges uniformly on  $K$ ; equivalently, one may apply uniform convergence to the tails after choosing an analytic branch of the logarithm near 1. Uniform limits of holomorphic functions are holomorphic, so the limit is entire.

If  $a = q^{-j}$ , one factor vanishes. Conversely, if no factor vanishes, then the tail product is nonzero because the sum of the absolute deviations from 1 converges, while the finite initial product is nonzero. At  $a = q^{-j}$  exactly one factor vanishes and its derivative is  $-q^j \neq 0$ , whereas all other factors are nonzero; the zero is simple.  $\square$

### 1.5.1 Complex orders

Finite and negative integral indices extend naturally to a complex order once an infinite product is available. Fix  $0 < |q| < 1$  and a branch of the logarithm at  $q$ , denoted by  $\log q$ , and write  $q^\alpha = \exp(\alpha \log q)$ .

**Definition 1.13.** For  $\alpha \in \mathbb{C}$ , define

$$(a; q)_\alpha := \frac{(a; q)_\infty}{(aq^\alpha; q)_\infty}, \tag{1.1}$$

whenever the denominator is nonzero.

**Proposition 1.14** (Complex concatenation). *Whenever the displayed quantities are defined,*

$$(a; q)_{\alpha+\beta} = (a; q)_\alpha (aq^\alpha; q)_\beta. \tag{1.2}$$

*In particular,*

$$(a; q)_{\alpha+1} = (1 - aq^\alpha)(a; q)_\alpha. \tag{1.3}$$

*For integral  $\alpha$ , (1.1) agrees with the finite and negative-index definitions already given.*

*Proof.* By definition,

$$(a; q)_\alpha (aq^\alpha; q)_\beta = \frac{(a; q)_\infty}{(aq^\alpha; q)_\infty} \frac{(aq^\alpha; q)_\infty}{(aq^{\alpha+\beta}; q)_\infty} = \frac{(a; q)_\infty}{(aq^{\alpha+\beta}; q)_\infty}.$$

This is (1.2). Taking  $\beta = 1$  gives (1.3). If  $\alpha = n \geq 0$ , the tail factorization  $(a; q)_\infty = (a; q)_n (aq^n; q)_\infty$  recovers the finite product. If  $\alpha = -n < 0$ , the same relation, applied after shifting by  $-n$ , gives  $(a; q)_{-n} = 1/(aq^{-n}; q)_n$ .  $\square$

**Proposition 1.15** (Derivative with respect to the order). *Fix  $a \in \mathbb{C}$  with  $(a; q)_\infty \neq 0$ , and let  $U \subset \mathbb{C}$  be open with*

$$aq^{\alpha+j} \neq 1 \quad (\alpha \in U, j \geq 0).$$

*Then  $\alpha \mapsto (a; q)_\alpha$  is holomorphic and nonvanishing on  $U$ , and its logarithmic derivative is*

$$\frac{\partial_\alpha (a; q)_\alpha}{(a; q)_\alpha} = (\log q) \sum_{j=0}^{\infty} \frac{aq^{\alpha+j}}{1 - aq^{\alpha+j}}. \quad (1.4)$$

*Equivalently, the left-hand side is  $\partial_\alpha \log(a; q)_\alpha$  for every local holomorphic branch of the logarithm. The series converges locally uniformly on  $U$ .*

*Proof.* Put  $G(\alpha) = (aq^\alpha; q)_\infty$ . The infinite-product theorem and the displayed nonvanishing condition show that  $G$  is holomorphic and nonzero on  $U$ . If  $K \Subset U$ , then  $\sup_{\alpha \in K} |aq^\alpha| < \infty$ . Choose  $J$  so that  $|aq^{\alpha+j}| \leq \frac{1}{2}$  for  $\alpha \in K$  and  $j \geq J$ . The corresponding summands are bounded by  $2 \sup_K |aq^\alpha| |q|^j$ ; the finitely many earlier denominators have a positive minimum on  $K$ . Thus the series converges uniformly on  $K$ , and termwise logarithmic differentiation of the normally convergent product gives

$$\frac{G'(\alpha)}{G(\alpha)} = -(\log q) \sum_{j=0}^{\infty} \frac{aq^{\alpha+j}}{1 - aq^{\alpha+j}}.$$

Since  $(a; q)_\alpha = (a; q)_\infty / G(\alpha)$  and the numerator is nonzero and independent of  $\alpha$ , the claimed formula follows.  $\square$

**Theorem 1.16** (Lambert-series expansion). *If  $|q| < 1$  and  $|a| < 1$ , then*

$$\log(a; q)_\infty = - \sum_{m=1}^{\infty} \frac{a^m}{m(1 - q^m)}.$$

*Consequently,*

$$\frac{\partial}{\partial a} \log(a; q)_\infty = - \sum_{m=1}^{\infty} \frac{a^{m-1}}{1 - q^m} = - \sum_{j=0}^{\infty} \frac{q^j}{1 - aq^j}.$$

*All series converge locally uniformly in  $|a| < 1$ .*

*Proof.* For  $|a| < 1$ , use  $\log(1 - u) = - \sum_{m \geq 1} u^m / m$  and absolute convergence:

$$\sum_{j \geq 0} \sum_{m \geq 1} \frac{|a|^m |q|^{jm}}{m} = \sum_{m \geq 1} \frac{|a|^m}{m(1 - |q|^m)} < \infty.$$

Thus the order of summation may be exchanged:

$$\sum_{j \geq 0} \log(1 - aq^j) = - \sum_{m \geq 1} \frac{a^m}{m} \sum_{j \geq 0} q^{jm}.$$

The geometric sum gives the first formula. Local uniform convergence permits termwise differentiation. Expanding  $(1 - aq^j)^{-1}$  geometrically gives the second form of the derivative.  $\square$

**Corollary 1.17** (Divisor sums). *As a formal power series,*

$$-q \frac{d}{dq} \log(q; q)_\infty = \sum_{n=1}^{\infty} \sigma_1(n) q^n,$$

where  $\sigma_1(n) = \sum_{d|n} d$ .

*Proof.* Logarithmic differentiation gives

$$-q \frac{d}{dq} \log(q; q)_\infty = \sum_{j \geq 1} \frac{j q^j}{1 - q^j} = \sum_{j \geq 1} \sum_{r \geq 1} j q^{jr}.$$

The coefficient of  $q^n$  is the sum of all  $j$  dividing  $n$ . □

## 1.6 The dilogarithmic limit

The singular regime  $q \rightarrow 1^-$  turns an infinite product into an exponential of order  $(1 - q)^{-1}$ . The dilogarithm is the natural leading term.

**Theorem 1.18** (Asymptotic expansion near  $q = 1$ ). *Fix  $0 < \rho < 1$  and an integer  $N \geq 1$ . Uniformly for  $|a| \leq \rho$ , as  $t \rightarrow 0^+$ ,*

$$\begin{aligned} \log(a; e^{-t})_\infty = & -\frac{\text{Li}_2(a)}{t} + \frac{1}{2} \log(1 - a) \\ & - \sum_{r=1}^{N-1} \frac{B_{2r}}{(2r)!} t^{2r-1} \text{Li}_{2-2r}(a) + O_{\rho,N}(t^{2N-1}), \end{aligned}$$

where  $B_{2r}$  are Bernoulli numbers and  $\text{Li}_s(a) = \sum_{m \geq 1} a^m / m^s$  for  $|a| < 1$ .

*Proof.* By [Theorem 1.16](#),

$$\log(a; e^{-t})_\infty = - \sum_{m \geq 1} \frac{a^m}{m(1 - e^{-mt})}.$$

The Bernoulli expansion at the origin is

$$\frac{1}{1 - e^{-x}} = \frac{1}{x} + \frac{1}{2} + \sum_{r=1}^{N-1} \frac{B_{2r}}{(2r)!} x^{2r-1} + R_N(x),$$

with  $R_N(x) = O_N(x^{2N-1})$  for  $0 \leq x \leq 1$ . Substitution of the displayed polynomial terms gives respectively

$$-\frac{1}{t} \sum_{m \geq 1} \frac{a^m}{m^2}, \quad -\frac{1}{2} \sum_{m \geq 1} \frac{a^m}{m} = \frac{1}{2} \log(1 - a),$$

and the polylogarithmic terms in the statement.

It remains to bound the total remainder. Split the  $m$ -sum at  $m \leq t^{-1}$ . In this range,

$$\sum_{m \leq t^{-1}} \frac{|a|^m}{m} |R_N(mt)| \leq C_N t^{2N-1} \sum_{m \geq 1} \rho^m m^{2N-2} = O_{\rho,N}(t^{2N-1}).$$

For  $m > t^{-1}$ , subtracting finitely many asymptotic terms leaves a quantity bounded by a polynomial in  $mt$  plus  $(1 - e^{-mt})^{-1}$ ; the factor  $\rho^m$  makes the tail exponentially small in  $1/t$ , hence smaller than every power of  $t$ . This proves the uniform remainder estimate. □

## 1.7 Classical shifted factorials as a limit

**Proposition 1.19** (Finite classical limit). *Let  $U$  be a sufficiently small simply connected neighborhood of 1 not containing 0, and choose the holomorphic logarithm on  $U$  with  $\log 1 = 0$ . For  $q \in U$  put  $q^x = \exp(x \log q)$ . Then, for fixed  $x \in \mathbb{C}$  and  $n \in \mathbb{N}_0$ ,*

$$\lim_{q \rightarrow 1} \frac{(q^x; q)_n}{(1 - q)^n} = x(x + 1) \cdots (x + n - 1) = (x)_n.$$

*Proof.* Set  $h = \log q$ , so  $h \rightarrow 0$  and  $q = \exp h$ . For every fixed  $u \in \mathbb{C}$ ,

$$\frac{1 - q^u}{1 - q} = \frac{1 - \exp(uh)}{1 - \exp h} \rightarrow u$$

by the Taylor expansion of the exponential, also when  $u = 0$ . Applying this to  $u = x + j$  in

$$\frac{(q^x; q)_n}{(1 - q)^n} = \prod_{j=0}^{n-1} \frac{1 - q^{x+j}}{1 - q}$$

proves the result. □

## Chapter 2

# $q$ -integers and Gaussian coefficients

### 2.1 Definitions

**Definition 2.1.** For  $n \in \mathbb{N}_0$  define the  $q$ -integer and  $q$ -factorial by

$$[n]_q = \frac{1 - q^n}{1 - q} = 1 + q + \cdots + q^{n-1}, \quad [n]_q! = \prod_{j=1}^n [j]_q.$$

For  $0 \leq k \leq n$ , the *Gaussian coefficient* is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}} = \frac{[n]_q!}{[k]_q! [n-k]_q!}.$$

Outside  $0 \leq k \leq n$  it is defined to be 0.

The notation is initially a rational expression, but the recurrences below show that it is a polynomial in  $q$  with nonnegative integral coefficients.

**Proposition 2.2** (Product forms). For  $0 \leq k \leq n$ ,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{n-k+1}; q)_k}{(q; q)_k} = \prod_{j=1}^k \frac{1 - q^{n-k+j}}{1 - q^j}.$$

*Proof.* Cancel  $(q; q)_{n-k}$  from  $(q; q)_n$  using [Proposition 1.2](#). □

### 2.2 The two Pascal recurrences

**Theorem 2.3** ( $q$ -Pascal recurrences). For  $1 \leq k \leq n-1$ ,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q, \tag{2.1}$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + \begin{bmatrix} n-1 \\ k \end{bmatrix}_q. \tag{2.2}$$

Both identities remain valid at the boundary under the zero convention.

*Proof.* Put the two terms on the right of (2.1) over the common denominator  $(q; q)_k (q; q)_{n-k}$ . Their numerator is

$$(q; q)_{n-1} ((1 - q^k) + q^k (1 - q^{n-k})) = (q; q)_{n-1} (1 - q^n) = (q; q)_n.$$

The proof of (2.2) is the same, using  $q^{n-k} (1 - q^k) + (1 - q^{n-k}) = 1 - q^n$ . □

**Corollary 2.4** (Polynomiality and positivity). *For all  $0 \leq k \leq n$ , the Gaussian coefficient  $\begin{bmatrix} n \\ k \end{bmatrix}_q$  belongs to  $\mathbb{N}_0[q]$ .*

*Proof.* Induct on  $n$ . The boundary values are 1. Either recurrence in [Theorem 2.3](#) expresses an interior coefficient as a sum of two polynomials with nonnegative integral coefficients.  $\square$

## 2.3 Symmetry, reciprocity, and degree

**Theorem 2.5** (Elementary structure). *For  $0 \leq k \leq n$ ,*

- (i)  $\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q$ ;
- (ii)  $\begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}} = q^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_q$ ;
- (iii)  $\begin{bmatrix} n \\ k \end{bmatrix}_q$  has degree  $k(n-k)$ , constant term 1, and leading coefficient 1;
- (iv) its coefficient sequence is palindromic.

*Proof.* Work first in the rational-function field  $\mathbb{Q}(q)$ . Symmetry is immediate from the factorial quotient. For reciprocity, use

$$(q^{-1}; q^{-1})_m = (-1)^m q^{-\binom{m+1}{2}} (q; q)_m$$

for  $m = n, k, n-k$ . The signs cancel and the exponent is

$$-\binom{n+1}{2} + \binom{k+1}{2} + \binom{n-k+1}{2} = -k(n-k).$$

Both sides of the resulting cleared identity are polynomials, so the identity holds identically, including at roots of unity where the factorial quotient itself may display a removable  $0/0$ .

The polynomiality already proved in [Corollary 2.4](#) makes the degree calculation legitimate. Since  $(q; q)_m$  has degree  $\binom{m+1}{2}$ , constant term 1, and leading coefficient  $(-1)^m$ , the quotient in  $\mathbb{Q}(q)$  has degree

$$\binom{n+1}{2} - \binom{k+1}{2} - \binom{n-k+1}{2} = k(n-k),$$

with constant and leading coefficients both equal to 1. Finally, reciprocity says

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}.$$

Writing  $\begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{j=0}^{k(n-k)} c_j q^j$  and comparing powers gives  $c_j = c_{k(n-k)-j}$ , which is precisely palindromicity.  $\square$

**Remark 2.6** (Exact reciprocal core in Lean). Items (i) and (ii) have exact machine-checked counterparts. Symmetry is `Fabius.gaussianBinomial_symm`. Reciprocity is `Fabius.gaussianBinomial_reciprocity_units` in `QBinomialReciprocity.lean`: for every unit  $u$  in every commutative semiring and all  $n, k \in \mathbb{N}_0$ , including the zero extension above the diagonal,

$$u^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{u^{-1}} = \begin{bmatrix} n \\ k \end{bmatrix}_u.$$

The wrapper `Fabius.gaussianBinomial_reciprocity` specializes this to a nonzero element of an arbitrary semifield. Unlike the quotient proof above, the formal proof is a denominator-free induction on the two Pascal recurrences, so it remains valid with zero divisors. Items (iii) and (iv) are not yet represented as polynomial-degree and coefficient theorems; consequently [Theorem 2.5](#) remains **partial**.

**Corollary 2.7** (Odd reciprocal degree at  $q = -1$ ). *Let  $R$  be a commutative ring. If  $k(n - k)$  is odd, then*

$$\begin{bmatrix} n \\ k \end{bmatrix}_{-1} = 0 \quad \text{in } R.$$

*In particular, for every  $a, b \in \mathbb{N}_0$ ,*

$$\begin{bmatrix} 2a \\ 2b + 1 \end{bmatrix}_{-1} = 0,$$

*where a column above the row is interpreted by the zero extension.*

*Proof.* First evaluate the universal integer Gaussian polynomial at  $-1$ . Reciprocity changes its value to its negative when  $k(n - k)$  is odd. Hence twice that integer is zero, and torsion-freeness of  $\mathbb{Z}$  makes the integer itself zero. The recursive Gaussian coefficient commutes with every semiring homomorphism, so the canonical map  $\mathbb{Z} \rightarrow R$  transports this zero to every commutative ring, including characteristic two. For the displayed specialization, if  $b < a$ , then both  $2b + 1$  and  $2a - (2b + 1) = 2(a - b - 1) + 1$  are odd, so their product is odd. If  $b \geq a$ , then  $2b + 1 > 2a$  and the zero extension applies directly.  $\square$

**Remark 2.8** (Formal version). The two statements are respectively `Fabius.gaussianBinomial_neg_one_eq_zero_of_odd_degree` and `Fabius.gaussianBinomial_neg_one_even_odd_eq_zero` in `QBinomialReciprocity.lean`. Both are total in the natural indices and hold over every commutative ring.

**Corollary 2.9** (Classical limit). *For fixed  $n, k$ ,*

$$\lim_{q \rightarrow 1} \begin{bmatrix} n \\ k \end{bmatrix}_q = \binom{n}{k}.$$

*Proof.* Use the  $q$ -factorial form and  $[j]_q \rightarrow j$ .  $\square$

## 2.4 Multinomial coefficients and flags

**Definition 2.10.** If  $n_1, \dots, n_r \geq 0$  and  $n_1 + \dots + n_r = n$ , define

$$\begin{bmatrix} n \\ n_1, \dots, n_r \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_{n_1} \cdots (q; q)_{n_r}}.$$

**Proposition 2.11** (Factorization into binomials). *Let  $N_j = n_j + n_{j+1} + \dots + n_r$ . Then*

$$\begin{bmatrix} n \\ n_1, \dots, n_r \end{bmatrix}_q = \prod_{j=1}^{r-1} \begin{bmatrix} N_j \\ n_j \end{bmatrix}_q.$$

*In particular, every  $q$ -multinomial coefficient is a polynomial with nonnegative integral coefficients.*

*Proof.* Write out the factorial ratios. Consecutive numerator and denominator factors telescope:

$$\prod_{j=1}^{r-1} \frac{(q; q)_{N_j}}{(q; q)_{n_j} (q; q)_{N_{j+1}}} = \frac{(q; q)_n}{\prod_{j=1}^r (q; q)_{n_j}}.$$

Positivity follows from [Corollary 2.4](#).  $\square$

## 2.5 Adjacent ratios and efficient recurrences

**Proposition 2.12.** *Let  $q$  lie in a field. For  $0 \leq k < n$ , if  $\begin{bmatrix} n \\ k \end{bmatrix}_q \neq 0$  and  $1 - q^{k+1} \neq 0$ , then*

$$\frac{\begin{bmatrix} n \\ k+1 \end{bmatrix}_q}{\begin{bmatrix} n \\ k \end{bmatrix}_q} = \frac{1 - q^{n-k}}{1 - q^{k+1}}.$$

*For the same range, if  $\begin{bmatrix} n-1 \\ k \end{bmatrix}_q \neq 0$  and  $1 - q^{n-k} \neq 0$ , then*

$$\frac{\begin{bmatrix} n \\ k \end{bmatrix}_q}{\begin{bmatrix} n-1 \\ k \end{bmatrix}_q} = \frac{1 - q^n}{1 - q^{n-k}}.$$

*Proof.* The denominator-free cross identities

$$(1 - q^{k+1}) \begin{bmatrix} n \\ k+1 \end{bmatrix}_q = (1 - q^{n-k}) \begin{bmatrix} n \\ k \end{bmatrix}_q$$

and

$$(1 - q^{n-k}) \begin{bmatrix} n \\ k \end{bmatrix}_q = (1 - q^n) \begin{bmatrix} n-1 \\ k \end{bmatrix}_q$$

hold over every commutative ring and, by zero extension, are total in all natural  $n, k$ : on and above the row boundary both sides vanish. In the strict range  $k < n$ , division over a field by the two explicitly assumed nonzero denominators in each line gives the stated ratios.  $\square$

These ratios are useful numerically, but at roots of unity their denominators may vanish and the quotients need not be defined. The cross-multiplied identities in the proof and the polynomial recurrences of [Theorem 2.3](#) remain valid there; unlike the quotients, the two cross identities also include the row boundary and every above-row index.

## 2.6 Coefficient moments

Write

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{j=0}^D c_j q^j, \quad D = k(n-k),$$

and choose an exponent  $X$  with probability  $\Pr(X = j) = c_j / \begin{pmatrix} n \\ k \end{pmatrix}$ . The combinatorial meaning of  $X$  will be established in [Chapter 5](#); its moments can already be found algebraically.

**Theorem 2.13** (Mean and variance). *For the preceding distribution,*

$$\mathbb{E}X = \frac{k(n-k)}{2}, \quad \text{Var}(X) = \frac{k(n-k)(n+1)}{12}.$$

*Equivalently,*

$$\left. \frac{d}{dq} \begin{bmatrix} n \\ k \end{bmatrix}_q \right|_{q=1} = \frac{k(n-k)}{2} \begin{pmatrix} n \\ k \end{pmatrix}.$$

*Proof.* The palindromic identity  $c_j = c_{D-j}$  immediately gives  $\mathbb{E}X = D/2$  and the derivative formula.

For the variance, use the moment-generating function

$$M(t) = \frac{\begin{bmatrix} n \\ k \end{bmatrix}_e e^t}{\begin{pmatrix} n \\ k \end{pmatrix}} = \prod_{i=1}^k \frac{e^{(n-k+i)t} - 1}{(n-k+i)t} \frac{it}{e^{it} - 1}.$$



The local expansion

$$\log \frac{e^u - 1}{u} = \frac{u}{2} + \frac{u^2}{24} + O(u^3)$$

gives

$$\log M(t) = \frac{t}{2} \sum_{i=1}^k (n - k) + \frac{t^2}{24} \sum_{i=1}^k ((n - k + i)^2 - i^2) + O(t^3).$$

The first sum is  $k(n - k)$ . The second equals

$$k(n - k)^2 + 2(n - k) \frac{k(k + 1)}{2} = k(n - k)(n + 1).$$

Since the first and second derivatives of  $\log M$  at 0 are the mean and variance, respectively, the result follows.  $\square$

## Chapter 3

# Finite $q$ -binomial theorems and inversion

### 3.1 Euler's finite product expansions

**Theorem 3.1** (Finite  $q$ -binomial theorem). *For  $n \geq 0$ ,*

$$(z; q)_n = \sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k, \quad (3.1)$$

$$(-z; q)_n = \sum_{k=0}^n q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k. \quad (3.2)$$

*Proof.* It is enough to prove the first identity; replace  $z$  by  $-z$  for the second. The case  $n = 0$  is clear. Suppose the formula holds for  $n$ . Then

$$(z; q)_{n+1} = (1 - zq^n)(z; q)_n.$$

The coefficient of  $z^k$  on the right is

$$(-1)^k q^{\binom{k}{2}} \left( \begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n-k+1} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q \right).$$

By the second  $q$ -Pascal recurrence, the parenthesis is  $\begin{bmatrix} n+1 \\ k \end{bmatrix}_q$ . This completes the induction.  $\square$

**Corollary 3.2** (Alternating sums). *For  $n > 0$ ,*

$$\sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q = 0.$$

*More generally,*

$$\sum_{k=0}^n (-1)^k q^{\binom{k}{2} + mk} \begin{bmatrix} n \\ k \end{bmatrix}_q = (q^m; q)_n.$$

*Proof.* Set  $z = 1$  and  $z = q^m$ , respectively, in (3.1).  $\square$

**Remark 3.3** (Formal version). Over every commutative ring, including  $q = 0$ , roots of unity, zero divisors, and positive characteristic, `Fabius.sum_gaussianBinomial_alternating_mul_pow` proves the second display for all  $m, n \in \mathbb{N}_0$ , and `Fabius.sum_gaussianBinomial_alternating` proves the first with its exact row-zero boundary 1. Both declarations are in `QBinomialInversion.lean`; neither uses a quotient or cancellation hypothesis.

**Corollary 3.4** (Weighted subset sum). For  $0 \leq k \leq n$ ,

$$\sum_{0 \leq i_1 < \dots < i_k \leq n-1} q^{i_1 + \dots + i_k} = q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

Equivalently,

$$\sum_{1 \leq j_1 < \dots < j_k \leq n} q^{j_1 + \dots + j_k} = q^{\binom{k+1}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

*Proof.* Expand  $\prod_{i=0}^{n-1} (1 + zq^i)$  by choosing either 1 or  $zq^i$  from each factor, and compare the coefficient of  $z^k$  with (3.2).  $\square$

**Remark 3.5** (Formal version). The zero-based and one-based formulas are respectively `Fabius.sum_pow_sum_powersetCard_eq_gaussianBinomial` and `Fabius.sum_pow_sum_powersetCard_Icc_eq_gaussianBinomial` in `BitPositionQBinoimial.lean`. They are total in  $n, k \in \mathbb{N}_0$  over every commutative ring; the first uses `range n`, and the second literally uses the interval `Icc 1 n`. The latter is obtained by translating each chosen index upward by one, which contributes the extra factor  $q^k$ .

## 3.2 Reciprocal finite products

**Theorem 3.6** (Reciprocal finite  $q$ -binomial theorem). For  $n \geq 0$  and  $|z| < 1$ ,

$$\frac{1}{(z; q)_{n+1}} = \sum_{k=0}^{\infty} \begin{bmatrix} n+k \\ k \end{bmatrix}_q z^k.$$

The identity also holds formally in  $\mathbb{Z}[q][[z]]$ .

*Proof.* Let  $H_n(z) = \sum_{k \geq 0} \begin{bmatrix} n+k \\ k \end{bmatrix}_q z^k$ . The second  $q$ -Pascal recurrence gives

$$\begin{bmatrix} n+k \\ k \end{bmatrix}_q - q^n \begin{bmatrix} n+k-1 \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q.$$

Therefore

$$(1 - zq^n)H_n(z) = H_{n-1}(z),$$

where  $H_0(z) = \sum_{k \geq 0} z^k = (1 - z)^{-1}$ . Iterating yields

$$H_n(z) = \frac{1}{(1-z)(1-zq) \cdots (1-zq^n)}.$$

All operations are valid formally; analytically they are justified for  $|z| < 1$ .  $\square$

**Remark 3.7** (Formal version and analytic boundary). For every commutative ring  $R$ , every  $q \in R$ , and every  $n \in \mathbb{N}_0$ , `Fabius.prod_one_sub_qPow_X_mul_gaussianBinomialGeneratingSeries` in `SymmetricFunctionGenerating.lean` proves

$$\left( \prod_{j=0}^n (1 - q^j X) \right) \text{PowerSeries.mk} \left( k \mapsto \begin{bmatrix} n+k \\ k \end{bmatrix}_q \right) = 1 \quad \text{in } R[[X]].$$

This is denominator-free and total at  $q = 0$ , roots of unity, zero divisors, positive characteristic, and the zero ring. It proves the formal power-series half of [Theorem 3.6](#); no current declaration here evaluates that series at a complex  $z$  or establishes convergence for  $|z| < 1$ . The theorem therefore remains only partially formalized as stated.

### 3.3 $q$ -Vandermonde convolution

**Theorem 3.8** ( $q$ -Vandermonde convolution). For  $m, n, r \in \mathbb{N}_0$ ,

$$\begin{bmatrix} m+n \\ r \end{bmatrix}_q = \sum_{j \in \mathbb{Z}} q^{(m-j)(r-j)} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ r-j \end{bmatrix}_q, \quad (3.3)$$

$$= \sum_{j \in \mathbb{Z}} q^{j(n-r+j)} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ r-j \end{bmatrix}_q. \quad (3.4)$$

Only finitely many terms are nonzero.

*Proof.* Factor

$$(-z; q)_{m+n} = (-z; q)_m (-zq^m; q)_n$$

and use (3.2). The coefficient of  $z^r$  on the left is  $q^{\binom{r}{2}} \begin{bmatrix} m+n \\ r \end{bmatrix}_q$ . The term with  $j$  chosen from the first factor and  $r-j$  from the second has exponent

$$\binom{j}{2} + m(r-j) + \binom{r-j}{2}.$$

Subtracting  $\binom{r}{2}$  gives  $(m-j)(r-j)$ , proving (3.3). Splitting the product in the reverse order gives (3.4).  $\square$

**Corollary 3.9** (Central form). For  $m, n \geq 0$ ,

$$\begin{bmatrix} m+n \\ m \end{bmatrix}_q = \sum_{j=0}^{\min(m,n)} q^{j^2} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ j \end{bmatrix}_q.$$

*Proof.* Set  $r = m$  in (3.3) and put  $\ell = m - j$ . Then the exponent becomes  $\ell^2$ , while symmetry gives  $\begin{bmatrix} m \\ m-\ell \end{bmatrix}_q = \begin{bmatrix} m \\ \ell \end{bmatrix}_q$  and the second factor is  $\begin{bmatrix} n \\ \ell \end{bmatrix}_q$ .  $\square$

**Corollary 3.10** (Shifted central form). For  $N \in \mathbb{N}_0$  and  $k \in \mathbb{Z}$ ,

$$\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q = \sum_{\ell \in \mathbb{Z}} q^{\ell^2 + \ell k} \begin{bmatrix} N \\ \ell \end{bmatrix}_q \begin{bmatrix} N \\ \ell+k \end{bmatrix}_q, \quad (3.5)$$

where a Gaussian coefficient whose lower index falls outside  $\{0, 1, \dots, \text{upper index}\}$  is read as 0, so that only finitely many terms are nonzero. The case  $k = 0$  is [Corollary 3.9](#) with  $m = n = N$ .

*Proof.* If  $|k| > N$  both sides vanish: the left side because  $N+k \notin \{0, \dots, 2N\}$ , and the right side because a nonzero term requires both  $0 \leq \ell \leq N$  and  $0 \leq \ell+k \leq N$ , which forces  $|k| \leq N$ .

Assume therefore  $-N \leq k \leq N$  and set  $r = N+k \in \{0, \dots, 2N\}$ . [Theorem 3.8](#), in the form (3.3) with  $m = n = N$ , gives

$$\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q = \sum_{j \in \mathbb{Z}} q^{(N-j)(N+k-j)} \begin{bmatrix} N \\ j \end{bmatrix}_q \begin{bmatrix} N \\ N+k-j \end{bmatrix}_q.$$

Substitute  $\ell = N - j$ . Then  $N+k-j = \ell+k$ , so the exponent becomes  $\ell(\ell+k) = \ell^2 + \ell k$ ; the symmetry  $\begin{bmatrix} N \\ N-\ell \end{bmatrix}_q = \begin{bmatrix} N \\ \ell \end{bmatrix}_q$  of [Theorem 2.5](#) (i) turns the first Gaussian coefficient into  $\begin{bmatrix} N \\ \ell \end{bmatrix}_q$ , and the second is  $\begin{bmatrix} N \\ \ell+k \end{bmatrix}_q$ . This is (3.5).  $\square$

**Remark 3.11** (Formal version). `FiniteQBinomialCore.lean` defines the zero-extended integer coefficient and its four totality laws.

*Definition*

`Fabius.gaussianBinomialInt`

*Cast, boundary, and reflection laws*

`Fabius.gaussianBinomialInt_ofNat`

`Fabius.gaussianBinomialInt_eq_zero_of_neg`

`Fabius.gaussianBinomialInt_eq_zero_of_lt`

`Fabius.gaussianBinomialInt_symm`

`QBinomialVandermonde.lean` proves the signed formula in two total forms.

*Finite range*  $0 \leq \ell \leq N$

`Fabius.gaussianBinomial_two_mul_int_shifted_central`

*Literal finite sum over all  $\ell \in \mathbb{Z}$*

`Fabius.gaussianBinomial_two_mul_int_shifted_central_fsum`

The Lean power is written  $q^{\ell^{\natural}(\ell+k)^{\natural}}$ , with  $t^{\natural} = \max(t, 0)$ ; every possibly nonzero term has  $\ell, \ell + k \geq 0$ , so its exponent is exactly  $\ell(\ell + k)$ . Thus the formal theorem is total in  $k$ , including  $|k| > N$ , over every commutative semiring and at  $q = 0$ .

### 3.4 $q$ -binomial inversion

**Theorem 3.12** ( $q$ -binomial inversion). *For sequences  $(a_n)$  and  $(b_n)$  in a commutative ring containing  $q$ , the relations*

$$b_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q a_k$$

*are equivalent to*

$$a_n = \sum_{k=0}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q b_k.$$

*Proof.* Substitute the first relation into the proposed inverse. The coefficient of  $a_j$  becomes

$$\sum_{k=j}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q.$$

The factorial definition gives

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q = \begin{bmatrix} n \\ j \end{bmatrix}_q \begin{bmatrix} n-j \\ k-j \end{bmatrix}_q.$$

Put  $r = n - k$ . The coefficient is

$$\begin{bmatrix} n \\ j \end{bmatrix}_q \sum_{r=0}^{n-j} (-1)^r q^{\binom{r}{2}} \begin{bmatrix} n-j \\ r \end{bmatrix}_q.$$

By [Corollary 3.2](#), this equals 0 when  $n > j$  and 1 when  $n = j$ . Thus the composition is the identity. The same argument in the opposite order proves equivalence.  $\square$

**Corollary 3.13** (Matrix orthogonality). *For  $0 \leq j \leq n$ ,*

$$\sum_{k=j}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q = \delta_{n,j}.$$

*Proof.* This is the kernel computation in the preceding proof.  $\square$

*Remark 3.14* (Formal version and scaled strengthening). The scalar algebra is formalized in `QBinomialInversion.lean`. Its public surface consists exactly of the following definitions and theorems.

*Definitions*

`Fabius.gaussianBinomialKernel`  
`Fabius.gaussianBinomialInverseKernel`  
`Fabius.scaledGaussianBinomialKernel`  
`Fabius.scaledGaussianBinomialInverseKernel`

*Theorems*

`Fabius.gaussianBinomial_mul`  
`Fabius.sum_gaussianBinomial_alternating_mul_pow`  
`Fabius.sum_gaussianBinomial_alternating`  
`Fabius.gaussianBinomialKernel_left_orthogonality`  
`Fabius.gaussianBinomialKernel_right_orthogonality`  
`Fabius.scaledGaussianBinomialKernel_left_orthogonality`  
`Fabius.scaledGaussianBinomialKernel_right_orthogonality`

The chain identity already holds over every commutative semiring. The alternating and orthogonality results hold over every commutative ring. Both multiplication orders are stated on the total interval  $I_{cc} \ j \ n$ , including the empty case  $n < j$ . The scaled forward and inverse entries are

$$s^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_q, \quad (-s)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q,$$

for an arbitrary independent  $s$ ; neither  $s$  nor  $q$  is assumed nonzero or invertible.

The common finite-sum argument is isolated in `FiniteTriangularTransform.lean`, whose exhaustive public surface is the definition `Fabius.lowerTriangularTransform` and the theorem `Fabius.lowerTriangularTransform_comp`. For  $[Semiring \ R] \ [AddCommMonoid \ M] \ [Module \ R \ M]$ , the theorem converts a total ordered kernel convolution on  $I_{cc} \ j \ n$  into equality of the whole composite sequence function with its input. It assumes neither commutativity of  $R$  nor additive inverses in  $M$ . The Gaussian kernels are zero-extended above the row, so the empty case  $n < j$  fits this generic total interface without a separate range hypothesis.

The module-valued sequence theorem is formalized in `QBinomialTransform.lean`, whose exact public surface is:

*Definitions*

`Fabius.scaledGaussianBinomialTransform`  
`Fabius.scaledGaussianBinomialInverseTransform`  
`Fabius.gaussianBinomialTransform`  
`Fabius.gaussianBinomialInverseTransform`

*Theorems*

`Fabius.scaledGaussianBinomialInverseTransform_transform`  
`Fabius.scaledGaussianBinomialTransform_inverseTransform`  
`Fabius.scaledGaussianBinomial_inversion`  
`Fabius.gaussianBinomial_inversion`

The two forward-transform definitions require only  $[Semiring \ R] \ [AddCommMonoid \ M] \ [Module \ R \ M]$ ; the two signed inverse definitions require  $[Ring \ R]$  with the same target assumptions. All four theorems require exactly  $[CommRing \ R] \ [AddCommMonoid \ M] \ [Module \ R \ M]$ . The first two give both compositions as equalities of whole sequence functions and the last two give the scaled and classical iff statements. Taking  $M = R$  yields [Theorem 3.12](#); the formal theorem is stronger in allowing arbitrary module-valued sequences and requires no topology or convergence.

`QDifferenceAnnihilation.lean` formalizes the scaled characteristic polynomial and the  $q$ -difference functional; its exact public surface consists of four theorems.

`Fabius.sum_scaledGaussianBinomialInverseKernel_mul_pow`  
`Fabius.sum_gaussianBinomialInverseKernel_mul_geometric_pow`  
`Fabius.qDifference_sum_eval2_eq_map_coeff_mul`  
`Fabius.qDifference_sum_eval2_eq_zero_of_degree_lt`

Writing

$$K_{q,s}^{-1}(n, k) = (-s)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q,$$

the first theorem is the homogeneous identity

$$\sum_{k=0}^n K_{q,s}^{-1}(n, k) z^k = \prod_{j=0}^{n-1} (z - sq^j). \quad (3.6)$$

It holds over every commutative ring for arbitrary  $q, s, z$ . With  $s = 1$  and  $z = q^d$ , the second theorem gives the complete moment table

$$\sum_{k=0}^n K_{q,1}^{-1}(n, k) (q^k)^d = \prod_{j=0}^{n-1} (q^d - q^j), \quad (3.7)$$

so the row annihilates every monomial of degree  $d < n$ .

The last two theorems make the polynomial statement strictly stronger than the scalar lemma usually quoted in Bailey inversion. If  $A$  is a semiring,  $R$  a commutative ring,  $\varphi: A \rightarrow R$  a semiring homomorphism, and  $P \in A[X]$  has degree at most  $n$ , then

$$\sum_{k=0}^n K_{q,1}^{-1}(n, k) P^\varphi(q^k) = \varphi([X^n]P) \prod_{j=0}^{n-1} (q^n - q^j), \quad (3.8)$$

where  $P^\varphi$  denotes coefficientwise scalar extension. If  $\deg P < n$ , the left side is zero. These degree statements include  $n = 0$  and the zero polynomial. Nodes may collide and the surviving factor in (3.8) may vanish: no division, distinctness, nonzero or invertible base, domain, characteristic, topology, or convergence hypothesis is present.

The global geometric-numerator bridge is owned by `GeometricQBinomialLagrange.lean`:

```
Fabius.geometricQBinomialWeightNumerator_eq_scaledGaussianBinomialInverseKernel
```

It identifies the denominator-free numerator with  $K_{q,q}^{-1}(n, k)$  for every pair of natural indices, including indices above the diagonal; equivalently, its generating polynomial is the  $s = q$  specialization of (3.6). This global identity does not remove the field, finite-node injectivity, or in-range assumptions from separately normalized Lagrange quotients.

Finally, `QBinomialInversionSpecializations.lean` has exactly two public definitions and four public theorems.

*Definitions*

```
Fabius.qGaussianResidualCoeff
```

```
Fabius.qGaussianReconstructionCoeff
```

*Theorems*

```
Fabius.qGaussianResidualCoeff_eq
```

```
Fabius.qGaussianReconstructionCoeff_eq
```

```
Fabius.qGaussianReconstructionCoeff_residualCoeff_delta
```

```
Fabius.qGaussianResidualCoeff_reconstructionCoeff_delta
```

At base  $q^2$  and scale  $-q$ , the coefficients are

$$(-q)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_{q^2}, \quad q^{(n-k)^2} \begin{bmatrix} n \\ k \end{bmatrix}_{q^2},$$

The two definitions and these two pointwise closed forms require only an arbitrary, possibly noncommutative ring. Exactly the two convolution-delta theorems require a commutative ring, and prove that both total finite orders are the Kronecker delta.

### 3.5 A $q$ -Cauchy product identity

**Theorem 3.15** ( $q$ -Cauchy identity). *For  $u, v$  in a commutative ring,*

$$(uv; q)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (u; q)_k v^k (v; q)_{n-k}.$$

*Proof.* Induct on  $n$ . The assertion is clear at  $n = 0$ . Let the right side be  $C_n(u, v)$ . Use

$$\begin{bmatrix} n+1 \\ k \end{bmatrix}_q = q^{n+1-k} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q + \begin{bmatrix} n \\ k \end{bmatrix}_q$$

and split  $C_{n+1}$  accordingly. In the first part shift  $k \mapsto k+1$ , use  $(u; q)_{k+1} = (u; q)_k(1 - uq^k)$ , and combine with the second part after using  $(v; q)_{n+1-k} = (v; q)_{n-k}(1 - vq^{n-k})$ . The bracket that remains is

$$(1 - vq^{n-k}) + vq^{n-k}(1 - uq^k) = 1 - uvq^n,$$

independent of  $k$ . Therefore

$$C_{n+1}(u, v) = (1 - uvq^n)C_n(u, v) = (uv; q)_{n+1}.$$

□

*Remark 3.16.* Taking  $u = 0$  recovers the tautology  $(0; q)_k = 1$  inside a weighted expansion; taking special limits of  $u$  or  $v$  produces several finite convolution formulas used later in the proof of the  $q$ -Pfaff–Saalschütz summation.

*Remark 3.17* (Formal version and strengthenings). `QBinomialCauchy.lean` provides the theorem and its compatibility spelling.

```
Fabius.finite_qCauchy_identity
Fabius.finiteQPochhammerIn_mul_eq_sum_gaussianBinomial
```

It is stated for arbitrary  $q, u, v$  in every commutative ring and every  $n \in \mathbb{N}_0$ , so it includes  $n = 0$ ,  $q = 0$ , roots of unity, positive characteristic, and zero divisors. The proof is entirely polynomial and assumes neither cancellation nor nonvanishing. The reflected strengthening `Fabius.finite_qCauchy_identity_reflected` states

$$(uv; q)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (v; q)_k v^{n-k} (u; q)_{n-k}.$$

The same module defines the denominator-free  $q$ -Bernstein term

$$B_{n,k}^{(q)}(x) = \begin{bmatrix} n \\ k \end{bmatrix}_q x^k (x; q)_{n-k}$$

as `Fabius.qBernsteinBasis` and proves the total partition of unity

$$\sum_{k=0}^n B_{n,k}^{(q)}(x) = 1$$

as `Fabius.sum_qBernsteinBasis`.

Together with `Fabius.finite_qCauchy_second_identity`, described at [Lemma 9.7](#), these are exhaustively the one public definition and five public theorems of `QBinomialCauchy.lean`.



## Chapter 4

# Weighted binomial coefficients and symmetric functions

The ordinary and Gaussian binomial coefficients are instances of a broader construction in which each available object carries a weight. This point of view, developed systematically by Konvalina, places subset selection, multiset selection, Stirling numbers, and Gaussian coefficients in one framework.

### 4.1 Elementary and complete weighted coefficients

**Definition 4.1.** Let  $\mathbf{w} = (w_1, \dots, w_n)$  be elements of a commutative ring. Define

$$C_k^n(\mathbf{w}) = \sum_{1 \leq i_1 < \dots < i_k \leq n} w_{i_1} \cdots w_{i_k},$$

$$S_k^n(\mathbf{w}) = \sum_{1 \leq i_1 \leq \dots \leq i_k \leq n} w_{i_1} \cdots w_{i_k}.$$

We set  $C_0^n = S_0^n = 1$ . Thus  $C_k^n$  is the elementary symmetric function  $e_k(w_1, \dots, w_n)$  and  $S_k^n$  is the complete homogeneous symmetric function  $h_k(w_1, \dots, w_n)$ .

**Theorem 4.2** (Weighted Pascal recurrences). For  $n, k \geq 1$ ,

$$C_k^n(\mathbf{w}) = C_k^{n-1}(\mathbf{w}) + w_n C_{k-1}^{n-1}(\mathbf{w}), \quad (4.1)$$

$$S_k^n(\mathbf{w}) = S_k^{n-1}(\mathbf{w}) + w_n S_{k-1}^{n-1}(\mathbf{w}). \quad (4.2)$$

*Proof.* A  $k$ -element subset of  $\{1, \dots, n\}$  either omits  $n$ , contributing  $C_k^{n-1}$ , or contains  $n$ , in which case removal of  $n$  leaves a  $(k-1)$ -element subset and the weight acquires a factor  $w_n$ . This proves (4.1).

For multisets, separate those with no occurrence of  $n$  from those with at least one occurrence. Removing one occurrence of  $n$  from the latter leaves an arbitrary multiset of size  $k-1$  drawn from all  $n$  types. This proves (4.2).  $\square$

**Theorem 4.3** (Generating products). One has

$$\prod_{i=1}^n (1 + w_i z) = \sum_{k=0}^n C_k^n(\mathbf{w}) z^k, \quad (4.3)$$

$$\prod_{i=1}^n \frac{1}{1 - w_i z} = \sum_{k=0}^{\infty} S_k^n(\mathbf{w}) z^k. \quad (4.4)$$

The second identity is formal in  $z$ , or analytic when  $|w_i z| < 1$  for all  $i$ .

*Proof.* In the first product, choosing  $w_i z$  from exactly the factors indexed by a subset  $I$  produces  $z^{|I|} \prod_{i \in I} w_i$ . Summing over subsets gives (4.3). For the second, expand each factor geometrically:

$$\frac{1}{1 - w_i z} = \sum_{r_i \geq 0} w_i^{r_i} z^{r_i}.$$

The coefficient of  $z^k$  sums over weak compositions  $r_1 + \cdots + r_n = k$ , which are equivalent to multisets of size  $k$ .  $\square$

**Corollary 4.4** (Elementary–complete orthogonality). *For  $m \geq 0$ ,*

$$\sum_{j=0}^m (-1)^j C_j^m(\mathbf{w}) S_{m-j}^n(\mathbf{w}) = \delta_{m,0}.$$

*Proof.* Multiply  $\prod_i (1 - w_i z)$  by its reciprocal and compare the coefficient of  $z^m$ , using Theorem 4.3 with  $z$  replaced by  $-z$  in the elementary generating function.  $\square$

**Remark 4.5** (Formal version and exact boundary). For a finite family  $a : \iota \rightarrow R$ , the evaluator and its bridge to Mathlib’s universal `MvPolynomial.esymm` are

```
elementarySymmetricEval
elementarySymmetricEval_eq_eval_esymm.
```

The adjoining-variable recurrences are

```
elementarySymmetricEval_option_succ
completeHomogeneousEval_option_succ.
```

They prove, respectively,

$$e_{r+1}(x, a) = x e_r(a) + e_{r+1}(a), \quad h_{r+1}(x, a) = x h_r(x, a) + h_{r+1}(a).$$

Thus both weighted Pascal recurrences of Theorem 4.2 are exact, including their zero-degree boundaries, over every commutative semiring.

The exhaustive public surface of `SymmetricFunctionOrthogonality.lean` is its one definition above and the following six theorems, all in namespace `Fabius`:

```
elementarySymmetricEval_eq_eval_esymm
elementarySymmetricEval_zero
elementarySymmetricEval_comp_equiv
elementarySymmetricEval_option_succ
elementarySymmetricEval_fin_succ
sum_elementarySymmetricEval_mul_completeHomogeneousEval.
```

The first five structural theorems hold over a commutative semiring; the last proves exactly

$$\sum_{j=0}^m (-1)^j e_j(a) h_{m-j}(a) = \begin{cases} 1, & m = 0, \\ 0, & m > 0, \end{cases}$$

over every commutative ring, for every finite index type and every  $m \in \mathbb{N}_0$ . It therefore includes the empty family and degree zero and uses no division, domain, characteristic, or nonvanishing hypothesis.

The exhaustive public surface of `SymmetricFunctionGenerating.lean` consists of the two definitions

```
elementarySymmetricGeneratingSeries
completeHomogeneousGeneratingSeries
```

and the six theorems

```
coeff_elementarySymmetricGeneratingSeries
coeff_completeHomogeneousGeneratingSeries
elementarySymmetricGeneratingSeries_eq_prod
elementarySymmetricGeneratingSeries_neg_mul_completeHomogeneousGeneratingSeries
completeHomogeneousGeneratingSeries_eq_invOfUnit_elementarySymmetricGeneratingSeries_neg
prod_one_sub_qPow_X_mul_gaussianBinomialGeneratingSeries.
```

The definitions, coefficient formulas, and finite product  $E_a(X) = \prod_i (1 + a_i X)$  hold over every commutative semiring. Over every commutative ring the remaining theorems prove

$$E_{-a}(X)H_a(X) = 1, \quad H_a(X) = \text{invOfUnit}(E_{-a}(X), 1),$$

as well as the geometric-alphabet reciprocal identity recorded after [Theorem 3.6](#). Hence the polynomial elementary identity and the formal-power-series version of the complete identity in [Theorem 4.3](#) are exact. The theorem as labelled remains **partial** because its analytic evaluation and convergence assertion under  $|w_i z| < 1$  has not been formalized.

The complementary finite-index interface in `CompleteHomogeneousGenerating.lean` consists of the definition `completeHomogeneousGeneratingSeriesOn` and the six theorems

```
coeff_completeHomogeneousGeneratingSeriesOn
completeHomogeneousGeneratingSeriesOn_empty
completeHomogeneousGeneratingSeriesOn_insert
one_sub_mul_completeHomogeneousGeneratingSeriesOn_insert
prod_one_sub_mul_completeHomogeneousGeneratingSeriesOn
completeHomogeneousGeneratingSeriesOn_eq_invOfUnit_prod.
```

The coefficient, empty-family, and adjoining-variable recurrence results hold over every commutative semiring. Over every commutative ring, the remaining results clear one or all finite linear denominators and identify the series with `Mathlib`'s canonical `PowerSeries.invOfUnit`; zero divisors are allowed. Thus this module and `SymmetricFunctionGenerating.lean` give complementary formal proofs of the complete-product half of [Theorem 4.3](#), while the latter also supplies the elementary product and elementary-complete reciprocity.

There is also a finite-family analytic companion. The sole public theorem `completeHomogeneousEvalOn_isBigO_pow` in `CompleteHomogeneousAsymptotics.lean` transfers coordinatewise  $\mathcal{O}g$  bounds through every fixed degree  $r$ : the corresponding complete homogeneous evaluation is  $\mathcal{O}g^r$ . It includes degree zero and requires no nonvanishing assumption on  $g$ . This asymptotic bound does not evaluate either formal series at  $z$ , establish its convergence, or close the analytic clause of the labelled theorem.

## 4.2 Classical, Stirling, and Gaussian specializations

**Theorem 4.6** (A specialization table). *The following identities hold.*

(i) If  $w_i = 1$ , then

$$C_k^n = \binom{n}{k}, \quad S_k^n = \binom{n+k-1}{k}.$$

(ii) If  $w_i = q^{i-1}$ , then

$$C_k^n = q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q, \quad S_k^n = \begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q.$$

(iii) For the unsigned Stirling numbers of the first kind  $c(n, k)$  and Stirling numbers of the second kind  $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ ,

$$c(n+1, k+1) = C_{n-k}^n(1, 2, \dots, n),$$

$$\left\{ \begin{matrix} n+k \\ n \end{matrix} \right\} = S_k^n(1, 2, \dots, n).$$

*Proof.* For  $w_i = 1$ ,  $C_k^n$  counts  $k$ -subsets. The multiset count is the stars-and-bars number  $\binom{n+k-1}{k}$ .

For  $w_i = q^{i-1}$ , the elementary identity is [Corollary 3.4](#). For the complete identity, [Theorem 4.3](#) gives

$$\prod_{i=0}^{n-1} \frac{1}{1 - zq^i} = \frac{1}{(z; q)_n}.$$

The coefficient of  $z^k$  is  $\left[ \begin{matrix} n+k-1 \\ k \end{matrix} \right]_q$  by [Theorem 3.6](#).

Finally,

$$x(x+1) \cdots (x+n) = \sum_{j=0}^{n+1} c(n+1, j) x^j.$$

The coefficient of  $x^{k+1}$  is  $e_{n-k}(1, 2, \dots, n)$ . The standard ordinary generating function

$$\prod_{i=1}^n \frac{1}{1 - iz} = \sum_{k \geq 0} \left\{ \begin{matrix} n+k \\ n \end{matrix} \right\} z^k$$

follows from the recurrence for Stirling numbers of the second kind, or by induction on  $n$ ; comparison with (4.4) proves the last formula.  $\square$

### 4.3 Principal specialization of Schur polynomials

The two Gaussian entries of [Theorem 4.6\(ii\)](#) are the one-column and the one-row case of a single product formula. Stating it requires no symmetric-function theory beyond determinants, because a Schur polynomial can be *defined* by its bialternant.

**Definition 4.7.** Let  $m \geq 1$  and let  $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_m \geq 0)$  be a partition with at most  $m$  parts, padded by zeros. Put

$$\mu_j = \lambda_j + m - j \quad (1 \leq j \leq m), \quad \delta = (m-1, m-2, \dots, 0),$$

so that  $\mu_1 > \mu_2 > \cdots > \mu_m \geq 0$ , and  $\delta$  is the value of  $\mu$  at  $\lambda = 0$ . For indeterminates  $x_1, \dots, x_m$  set

$$a_\mu(x_1, \dots, x_m) = \det(x_i^{\mu_j})_{i,j=1}^m,$$

and define the *Schur polynomial*

$$s_\lambda(x_1, \dots, x_m) = \frac{a_\mu(x_1, \dots, x_m)}{a_\delta(x_1, \dots, x_m)}.$$

Finally put  $n(\lambda) = \sum_{j=1}^m (j-1)\lambda_j$ .

**Lemma 4.8** (Vandermonde alternant). *For indeterminates  $y_1, \dots, y_m$ ,*

$$\det(y_i^{j-1})_{i,j=1}^m = \prod_{1 \leq i < j \leq m} (y_j - y_i).$$

*Consequently  $a_\delta(x_1, \dots, x_m) = \prod_{1 \leq i < j \leq m} (x_i - x_j)$ , and  $a_\delta$  divides  $a_\mu$  in  $\mathbb{Z}[x_1, \dots, x_m]$ , so  $s_\lambda$  is a polynomial with integer coefficients.*

*Proof.* Induct on  $m$ ; for  $m = 1$  both sides equal 1. Regard  $D = \det(y_i^{j-1})_{i,j=1}^m$  as a polynomial in  $y_m$  of degree at most  $m - 1$ . If  $y_m = y_i$  for some  $i < m$ , two rows coincide and  $D = 0$ ; hence  $\prod_{i < m} (y_m - y_i)$  divides  $D$ , and by the degree bound  $D = c \prod_{i < m} (y_m - y_i)$  with  $c$  free of  $y_m$ . Expanding along the last row, the coefficient of  $y_m^{m-1}$  in  $D$  is the cofactor at position  $(m, m)$ , namely  $\det(y_i^{j-1})_{i,j=1}^{m-1}$ , which by induction equals  $\prod_{1 \leq i < j \leq m-1} (y_j - y_i)$ . Multiplying proves the formula.

Reversing the order of the columns of  $(x_i^{m-j})_{i,j}$  turns the exponents  $m - j$  into  $j - 1$  and multiplies the determinant by the sign  $(-1)^{\binom{m}{2}}$  of the reversal permutation, so

$$a_\delta = (-1)^{\binom{m}{2}} \prod_{1 \leq i < j \leq m} (x_j - x_i) = \prod_{1 \leq i < j \leq m} (x_i - x_j).$$

For the divisibility, interchanging  $x_i$  and  $x_j$  interchanges two rows of  $a_\mu$ , so  $a_\mu$  changes sign; setting  $x_i = x_j$  therefore gives  $a_\mu = 0$ , and  $x_i - x_j$  divides  $a_\mu$  in  $\mathbb{Z}[x_1, \dots, x_m]$ . The polynomials  $x_i - x_j$  with  $i < j$  are pairwise non-associate primes in that unique factorization domain, so their product, which is  $a_\delta$ , divides  $a_\mu$ .  $\square$

**Theorem 4.9** (Principal specialization of a Schur polynomial). *Let  $\lambda$  have at most  $m$  parts. Then, as an identity in  $\mathbb{Q}(q)$ ,*

$$s_\lambda(1, q, q^2, \dots, q^{m-1}) = q^{n(\lambda)} \prod_{1 \leq i < j \leq m} \frac{1 - q^{\lambda_i - \lambda_j + j - i}}{1 - q^{j - i}}.$$

The left side lies in  $\mathbb{Z}[q]$ , so by [Corollary 24.11](#) the identity persists at every complex  $q$  once the right-hand quotient has been reduced; as displayed it is to be read at those  $q$  with  $q^d \neq 1$  for  $1 \leq d \leq m - 1$ .

*Proof.* Specialize  $x_i = q^{i-1}$ , so that  $x_i^{\mu_j} = q^{(i-1)\mu_j}$ . Writing  $y_j = q^{\mu_j}$ , the matrix  $(x_i^{\mu_j})_{i,j}$  has entries  $y_j^{i-1}$ , hence is the transpose of  $(y_i^{j-1})_{i,j}$ . A determinant is invariant under transposition, so [Lemma 4.8](#) gives

$$a_\mu(1, q, \dots, q^{m-1}) = \prod_{1 \leq i < j \leq m} (q^{\mu_j} - q^{\mu_i}).$$

Because  $\lambda$  is weakly decreasing we have  $\mu_1 > \dots > \mu_m \geq 0$ , so  $\mu_i - \mu_j > 0$  whenever  $i < j$ , and we extract from each difference the *smaller* power:

$$q^{\mu_j} - q^{\mu_i} = q^{\mu_j} (1 - q^{\mu_i - \mu_j}).$$

Hence

$$a_\mu(1, q, \dots, q^{m-1}) = q^{A(\mu)} \prod_{1 \leq i < j \leq m} (1 - q^{\mu_i - \mu_j}), \quad A(\mu) = \sum_{1 \leq i < j \leq m} \mu_j = \sum_{j=1}^m (j-1)\mu_j,$$

the last equality because  $\mu_j$  is contributed once for each  $i < j$ . Applying this to  $\mu = \delta$ , where  $\delta_i - \delta_j = j - i$ , gives

$$a_\delta(1, q, \dots, q^{m-1}) = q^{A(\delta)} \prod_{1 \leq i < j \leq m} (1 - q^{j-i}),$$

a nonzero element of  $\mathbb{Q}(q)$ . Dividing,

$$s_\lambda(1, q, \dots, q^{m-1}) = q^{A(\mu) - A(\delta)} \prod_{1 \leq i < j \leq m} \frac{1 - q^{\mu_i - \mu_j}}{1 - q^{j-i}}.$$

Finally  $\mu_i - \mu_j = \lambda_i - \lambda_j + j - i$  and

$$A(\mu) - A(\delta) = \sum_{j=1}^m (j-1)(\mu_j - \delta_j) = \sum_{j=1}^m (j-1)\lambda_j = n(\lambda).$$

The left side is a polynomial with integer coefficients by [Lemma 4.8](#); the transfer to arbitrary complex  $q$  is [Lemma 24.10](#) together with [Corollary 24.11](#).  $\square$

**Corollary 4.10** (The one-column and one-row shadows). *Let  $m = n \geq 1$ .*

(i) *For  $0 \leq k \leq n$  and  $\lambda = (1^k) = (1, \dots, 1, 0, \dots, 0)$ ,*

$$s_{(1^k)}(1, q, \dots, q^{n-1}) = q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

(ii) *For  $k \geq 0$  and  $\lambda = (k)$ ,*

$$s_{(k)}(1, q, \dots, q^{n-1}) = \begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q.$$

Since  $s_{(1^k)} = e_k$  and  $s_{(k)} = h_k$  [22], these are exactly the two Gaussian entries of [Theorem 4.6\(ii\)](#), that is,  $C_k^n(\mathbf{w})$  and  $S_k^n(\mathbf{w})$  at  $w_i = q^{i-1}$ .

*Proof.* (i) Here  $\lambda_i = 1$  for  $i \leq k$  and  $\lambda_i = 0$  for  $i > k$ , so  $n(\lambda) = \sum_{i \leq k} (i-1) = \binom{k}{2}$ , and  $\lambda_i - \lambda_j$  vanishes except when  $i \leq k < j$ , where it equals 1. The product of [Theorem 4.9](#) therefore collapses to

$$\prod_{i=1}^k \prod_{j=k+1}^n \frac{1 - q^{j-i+1}}{1 - q^{j-i}} = \prod_{i=1}^k \frac{1 - q^{n-i+1}}{1 - q^{k-i+1}},$$

the inner product telescoping. The numerators are

$$(1 - q^n)(1 - q^{n-1}) \cdots (1 - q^{n-k+1}) = \frac{(q; q)_n}{(q; q)_{n-k}},$$

$$(1 - q^k) \cdots (1 - q) = (q; q)_k,$$

so the product is  $\begin{bmatrix} n \\ k \end{bmatrix}_q$ .

(ii) Here  $\lambda_1 = k$  and  $\lambda_j = 0$  for  $j \geq 2$ , so  $n(\lambda) = 0$  and only the pairs  $(1, j)$  contribute:

$$\prod_{j=2}^n \frac{1 - q^{k+j-1}}{1 - q^{j-1}} = \frac{(q; q)_{n+k-1} / (q; q)_k}{(q; q)_{n-1}} = \begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q. \quad \square$$

*Remark 4.11.* Both hypotheses of [Theorem 4.9](#) are needed as stated. If  $\lambda$  had more than  $m$  parts,  $\mu$  would not be strictly decreasing and  $a_\mu$  would vanish identically. And the displayed quotient is meaningful only where  $q^{j-i} \neq 1$  for all  $1 \leq i < j \leq m$ ; at the excluded roots of unity, and at  $q = 1$ , one must cancel first, exactly as prescribed by [Corollary 24.11](#). After cancellation the limit  $q \rightarrow 1$  is

$$s_\lambda(1, 1, \dots, 1) = \prod_{1 \leq i < j \leq m} \frac{\lambda_i - \lambda_j + j - i}{j - i},$$

by  $\lim_{q \rightarrow 1} (1 - q^a) / (1 - q^b) = a/b$ ; this is Weyl's dimension formula for  $\mathrm{GL}_m$ , and it stands to [Theorem 4.9](#) exactly as [Corollary 2.9](#) stands to the Gaussian coefficient.

## 4.4 Weighted inversion

**Theorem 4.12** (Symmetric-function inversion). *Let  $(a_m)$  and  $(b_m)$  be sequences. The transformations*

$$b_m = \sum_{j=0}^m S_{m-j}^n(\mathbf{w}) a_j$$

and

$$a_m = \sum_{j=0}^m (-1)^{m-j} C_{m-j}^n(\mathbf{w}) b_j$$

are inverse to each other.

*Proof.* The generating functions  $A(z) = \sum a_m z^m$  and  $B(z) = \sum b_m z^m$  satisfy

$$B(z) = A(z) \prod_{i=1}^n (1 - w_i z)^{-1}.$$

Multiplication by  $\prod_i (1 - w_i z)$  gives the proposed inverse coefficient formula. Equivalently, use the orthogonality in [Corollary 4.4](#).  $\square$

**Remark 4.13** (Formal version and module-valued strengthening). The exhaustive public surface of `SymmetricFunctionTransform.lean` is the four definitions

```
Fabius.completeHomogeneousKernel
Fabius.signedElementaryKernel
Fabius.completeHomogeneousTransform
Fabius.signedElementaryTransform
```

and the five theorems

```
Fabius.completeHomogeneousKernel_left_orthogonality
Fabius.completeHomogeneousKernel_right_orthogonality
Fabius.signedElementaryTransform_completeHomogeneousTransform
Fabius.completeHomogeneousTransform_signedElementaryTransform
Fabius.weightedSymmetricFunction_inversion.
```

For a finite family  $w : \iota \rightarrow R$ , the complete-homogeneous kernel and transform need only `[CommSemiring R]`; the signed-elementary kernel and transform require `[CommRing R]`. Both kernel orthogonalitys, both inverse compositions, and the inversion iff use exactly `[CommRing R] [AddCommMonoid M] [Module R M]` with `[Fintype  $\iota$ ]`. The compositions are equalities of whole sequence functions, so `Fabius.weightedSymmetricFunction_inversion` proves [Theorem 4.12](#) exactly and strengthens it to arbitrary module-valued sequences.

Both kernels are extended by zero above the diagonal, including the empty outer interval. Their composition proofs reuse `Fabius.lowerTriangularTransform_comp`; together with the definition `Fabius.lowerTriangularTransform`, these are exhaustively the one theorem and one definition of `FiniteTriangularTransform.lean`. The generic theorem itself only assumes a semiring acting on an additive commutative monoid and a total ordered finite-convolution identity. No division, topology, domain, characteristic, or nonvanishing condition occurs.

## 4.5 The subset–subspace analogy

The choice  $w_i = q^{i-1}$  is more than a convenient specialization. It replaces the Boolean lattice of subsets by the lattice of subspaces over a finite field. Ordinary binomial coefficients count subsets by cardinality; Gaussian coefficients count subspaces by dimension. The two-term weighted recurrence mirrors the decomposition of a subspace according to its relation with a fixed hyperplane. This analogy is made exact in [Chapter 6](#).

## Chapter 5

# Partitions, words, paths, and inversion statistics

### 5.1 Binary words

Let  $\mathcal{W}_{n,k}$  be the set of binary words of length  $n$  containing  $k$  ones. Define

$$\text{inv}(w) = \# \{(i, j) : i < j, w_i = 1, w_j = 0\}.$$

**Theorem 5.1** (Inversion generating function). *For  $0 \leq k \leq n$ ,*

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_q = \sum_{w \in \mathcal{W}_{n,k}} q^{\text{inv}(w)}.$$

*Proof.* Let  $F_{n,k}(q)$  denote the right side. If the last letter is 1, deletion of it gives a word in  $\mathcal{W}_{n-1,k-1}$  and creates no new inversion. If the last letter is 0, deletion gives a word in  $\mathcal{W}_{n-1,k}$  and the final zero forms an inversion with each of the  $k$  ones, contributing a factor  $q^k$ . Hence

$$F_{n,k} = F_{n-1,k-1} + q^k F_{n-1,k}.$$

The boundary values are 1, so  $F_{n,k}$  and  $\left[ \begin{matrix} n \\ k \end{matrix} \right]_q$  obey the same recurrence and initial conditions.  $\square$

**Corollary 5.2** (Inversion-law interpretation). *For  $0 \leq k \leq n$ , the random variable in [Theorem 2.13](#) is the inversion number of a uniformly random word in  $\mathcal{W}_{n,k}$ .*

*Proof.* The coefficient of  $q^j$  in [Theorem 5.1](#) counts words with  $j$  inversions, while  $|\mathcal{W}_{n,k}| = \binom{n}{k}$ .  $\square$

### 5.2 Partitions inside a rectangle

A partition  $\lambda = (\lambda_1, \dots, \lambda_k)$  is said to fit in a  $k \times m$  rectangle if

$$m \geq \lambda_1 \geq \dots \geq \lambda_k \geq 0.$$

Its size is  $|\lambda| = \lambda_1 + \dots + \lambda_k$ .

**Theorem 5.3** (Rectangular partition generating function). *For  $m, k \geq 0$ ,*

$$\left[ \begin{matrix} m+k \\ k \end{matrix} \right]_q = \sum_{\lambda \subseteq k \times m} q^{|\lambda|}.$$



*Proof.* Take a word  $w \in \mathcal{W}_{m+k,k}$  and let the positions of its ones be  $1 \leq i_1 < \cdots < i_k \leq m+k$ . Define

$$\lambda_j = m + j - i_j \quad (1 \leq j \leq k).$$

Then  $0 \leq \lambda_j \leq m$  and

$$\lambda_j - \lambda_{j+1} = i_{j+1} - i_j - 1 \geq 0.$$

Thus  $\lambda$  fits in the rectangle. Conversely,  $i_j = m + j - \lambda_j$  recovers a strictly increasing sequence of positions, so this is a bijection. The number  $\lambda_j$  is exactly the number of zeros to the right of the  $j$ th one, hence  $|\lambda| = \text{inv}(w)$ . Apply [Theorem 5.1](#).  $\square$

**Corollary 5.4** (Conjugation and complement). *Conjugating Ferrers diagrams proves  $\left[ \begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q = \left[ \begin{smallmatrix} m+k \\ m \end{smallmatrix} \right]_q$ . Taking the complement in the rectangle proves*

$$\left[ \begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_{q^{-1}} = q^{-mk} \left[ \begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q.$$

*Proof.* Conjugation exchanges a  $k \times m$  rectangle with an  $m \times k$  rectangle without changing size. Complementation sends size  $r$  to  $mk - r$ , so the generating polynomial is reciprocal of degree  $mk$ .  $\square$

### 5.3 Lattice paths and area

Encode a binary word by a path from  $(0, 0)$  to  $(m, k)$ , reading 0 as an east step and 1 as a north step. With a suitable choice of side, the number of unit squares between the path and the boundary is the inversion number.

**Corollary 5.5** (Area generating function). *The polynomial  $\left[ \begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q$  is the generating function of monotone lattice paths from  $(0, 0)$  to  $(m, k)$  by the area under the path:*

$$\left[ \begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q = \sum_P q^{\text{area}(P)}.$$

*Proof.* Under the word–path encoding, every north step contributes the number of later east steps, which is precisely the number of squares in its row between the path and the chosen boundary. Summing over north steps gives  $\text{inv}(w)$ .  $\square$

### 5.4 Durfee squares and the central convolution

The largest square in the northwest corner of a partition diagram is its Durfee square.

*Remark 5.6* (Durfee-square proof of the central convolution). The identity in [Corollary 3.9](#) has a direct partition proof. A partition in an  $m \times n$  rectangle with Durfee square of side  $j$  consists of the  $j \times j$  square, a partition to its right fitting in a  $j \times (n - j)$  rectangle, and a partition below it fitting in an  $(m - j) \times j$  rectangle. By [Theorem 5.3](#), their generating functions are

$$q^{j^2} \left[ \begin{smallmatrix} n \\ j \end{smallmatrix} \right]_q \left[ \begin{smallmatrix} m \\ j \end{smallmatrix} \right]_q.$$

The Durfee size is unique, so summing over  $j$  counts every partition exactly once. Thus the algebraic proof above and the Durfee decomposition prove one and the same canonical result, rather than two separately labelled assertions.

## 5.5 Words with several letters

Let  $\mathcal{W}(n_1, \dots, n_r)$  be the words containing  $n_i$  copies of the letter  $i$ , ordered by  $1 < \dots < r$ . Let  $\text{inv}(w)$  count pairs  $i < j$  of positions with  $w_i > w_j$ .

**Theorem 5.7** (Multiset inversion enumerator). *If  $n = n_1 + \dots + n_r$ , then*

$$\left[ \begin{matrix} n \\ n_1, \dots, n_r \end{matrix} \right]_q = \sum_{w \in \mathcal{W}(n_1, \dots, n_r)} q^{\text{inv}(w)}.$$

*Proof.* Choose the positions of the largest letter  $r$ . If their binary indicator has inversion number  $s$ , then  $s$  counts precisely the inversions in which the larger letter is  $r$ . The remaining positions contain a word on  $1, \dots, r-1$ , and its inversions are independent. Therefore the generating function factors as

$$\left[ \begin{matrix} n \\ n_r \end{matrix} \right]_q \left[ \begin{matrix} n - n_r \\ n_1, \dots, n_{r-1} \end{matrix} \right]_q.$$

Iterating and using [Proposition 2.11](#) proves the formula.  $\square$

## 5.6 Rogers–Szegő polynomials

**Definition 5.8.** The Rogers–Szegő polynomial is

$$H_n(z; q) = \sum_{k=0}^n \left[ \begin{matrix} n \\ k \end{matrix} \right]_q z^k.$$

**Proposition 5.9** (Basic recurrence). *One has*

$$H_{n+1}(z; q) = zH_n(z; q) + H_n(qz; q), \quad H_0(z; q) = 1.$$

Moreover, if  $|q| < 1$ , then

$$\sum_{n=0}^{\infty} \frac{H_n(z; q)}{(q; q)_n} t^n = \frac{1}{(t; q)_{\infty} (zt; q)_{\infty}}$$

for  $|t| < 1$  and  $|zt| < 1$ .

*Proof.* Insert the first  $q$ -Pascal recurrence into  $H_{n+1}$ :

$$\sum_k \left( \left[ \begin{matrix} n \\ k-1 \end{matrix} \right]_q + q^k \left[ \begin{matrix} n \\ k \end{matrix} \right]_q \right) z^k = zH_n(z; q) + H_n(qz; q).$$

For the generating function, use

$$\frac{\left[ \begin{matrix} n \\ k \end{matrix} \right]_q}{(q; q)_n} = \frac{1}{(q; q)_k (q; q)_{n-k}}.$$

Writing  $n = k + j$  and separating the double sum gives

$$\left( \sum_{j \geq 0} \frac{t^j}{(q; q)_j} \right) \left( \sum_{k \geq 0} \frac{(zt)^k}{(q; q)_k} \right).$$

Each factor equals the reciprocal infinite product by Euler's identity, proved later in [Corollary 8.2](#); alternatively, the equality follows already by taking  $n \rightarrow \infty$  in [Theorem 3.6](#). This gives the stated product.  $\square$

**Proposition 5.10** (Three-term recurrence). *For  $n \geq 1$ ,*

$$H_{n+1}(z; q) = (1 + z)H_n(z; q) - z(1 - q^n)H_{n-1}(z; q).$$

*Proof.* Both sides are polynomials in  $z$ , so it suffices to compare the coefficient of  $z^k$  for every  $k \in \mathbb{Z}$ , using the convention  $\begin{bmatrix} n \\ k \end{bmatrix}_q = 0$  for  $k < 0$  or  $k > n$ . On the left that coefficient is  $\begin{bmatrix} n+1 \\ k \end{bmatrix}_q$ ; on the right it is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q + \begin{bmatrix} n \\ k-1 \end{bmatrix}_q - (1 - q^n) \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q.$$

We claim that for every  $k \in \mathbb{Z}$  and every  $n \geq 1$ ,

$$(1 - q^n) \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q = (1 - q^{n+1-k}) \begin{bmatrix} n \\ k-1 \end{bmatrix}_q.$$

Granting the claim, the right-hand coefficient equals

$$\begin{bmatrix} n \\ k \end{bmatrix}_q + (1 - (1 - q^{n+1-k})) \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n+1-k} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q,$$

which is the second  $q$ -Pascal recurrence (2.2) with  $n$  replaced by  $n+1$ ; by Theorem 2.3 that recurrence is valid for all  $k$  under the zero convention, and the right-hand coefficient is therefore  $\begin{bmatrix} n+1 \\ k \end{bmatrix}_q$ , as required.

To prove the claim, let first  $1 \leq k \leq n$ , so that  $k-1 \geq 0$  and  $n-k \geq 0$  and both Gaussian coefficients are given by factorial quotients. From  $(q; q)_n = (1 - q^n)(q; q)_{n-1}$  and  $(q; q)_{n-k+1} = (1 - q^{n-k+1})(q; q)_{n-k}$ ,

$$\frac{\begin{bmatrix} n \\ k-1 \end{bmatrix}_q}{\begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q} = \frac{(q; q)_n (q; q)_{n-k}}{(q; q)_{n-1} (q; q)_{n-k+1}} = \frac{1 - q^n}{1 - q^{n-k+1}},$$

and cross-multiplication gives the claim. For  $k \leq 0$  both sides vanish, since then  $k-1 < 0$  and both Gaussian coefficients are zero. For  $k = n+1$  the left side vanishes because  $\begin{bmatrix} n-1 \\ n \end{bmatrix}_q = 0$  and the right side vanishes because  $1 - q^{n+1-k} = 1 - q^0 = 0$ . For  $k > n+1$  every term is zero. This exhausts all  $k$  and completes the proof.  $\square$

**Corollary 5.11** (Dilation formula). *For  $n \geq 1$ ,*

$$H_n(qz; q) = H_n(z; q) - z(1 - q^n)H_{n-1}(z; q).$$

*Proof.* By Proposition 5.9,  $H_n(qz; q) = H_{n+1}(z; q) - zH_n(z; q)$ . Substituting Proposition 5.10 for  $H_{n+1}(z; q)$  and cancelling  $zH_n(z; q)$  gives the formula.  $\square$

*Remark 5.12.* The recurrence of Proposition 5.9 moves the argument, from  $z$  to  $qz$ ; the one above does not. It is therefore a three-term recurrence in the index  $n$  alone, of the shape familiar from orthogonal polynomial sequences, and it evaluates  $H_0, H_1, \dots, H_N$  at a fixed numerical  $z$  in  $\mathcal{O}(N)$  ring operations, without ever forming the Gaussian coefficients. At  $q = 0$  every  $\begin{bmatrix} n \\ k \end{bmatrix}_q$  equals 1 by Proposition 22.11, the recurrence becomes  $H_{n+1} = (1 + z)H_n - zH_{n-1}$ , and its solution is the geometric sum  $1 + z + \dots + z^n$ , as it must be.

## Chapter 6

# Finite fields, Grassmannians, and rank enumeration

### 6.1 Counting subspaces

**Theorem 6.1** (Gaussian coefficients count subspaces). *Let  $Q$  be a prime power. The number of  $k$ -dimensional linear subspaces of  $\mathbb{F}_Q^n$  is*

$$\begin{bmatrix} n \\ k \end{bmatrix}_Q.$$

*Proof.* Count ordered linearly independent  $k$ -tuples in  $\mathbb{F}_Q^n$ . The first vector may be any nonzero vector, the second any vector outside the span of the first, and so on, giving

$$(Q^n - 1)(Q^n - Q) \cdots (Q^n - Q^{k-1}).$$

Each  $k$ -dimensional subspace has

$$(Q^k - 1)(Q^k - Q) \cdots (Q^k - Q^{k-1})$$

ordered bases. Dividing and cancelling  $Q^i$  in corresponding factors yields

$$\prod_{i=0}^{k-1} \frac{Q^{n-i} - 1}{Q^{k-i} - 1} = \prod_{j=1}^k \frac{1 - Q^{n-k+j}}{1 - Q^j} = \begin{bmatrix} n \\ k \end{bmatrix}_Q.$$

□

**Theorem 6.2** (Geometric Pascal recurrence). *Let  $H$  be a fixed hyperplane in  $\mathbb{F}_Q^n$ . Then*

$$\begin{bmatrix} n \\ k \end{bmatrix}_Q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_Q + Q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_Q.$$

*Proof.* A  $k$ -subspace  $U$  is either contained in  $H$  or not. The contained subspaces contribute  $\begin{bmatrix} n-1 \\ k \end{bmatrix}_Q$ . If  $U \not\subseteq H$ , then  $W = U \cap H$  has dimension  $k-1$ . Fixing  $W$ , the quotient  $\mathbb{F}_Q^n/W$  has dimension  $n-k+1$ , and  $H/W$  is a hyperplane. The subspace  $U/W$  is a line not contained in  $H/W$ . The total number of lines in  $\mathbb{F}_Q^{n-k+1}$  minus those in its hyperplane is

$$\frac{Q^{n-k+1} - 1}{Q - 1} - \frac{Q^{n-k} - 1}{Q - 1} = Q^{n-k}.$$

There are  $\begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_Q$  choices for  $W$ , proving the formula. □

## 6.2 Flags and $q$ -multinomial coefficients

**Theorem 6.3** (Partial flags). *Let  $d_i = n_1 + \cdots + n_i$ . The number of flags*

$$0 = V_0 \subset V_1 \subset \cdots \subset V_r = \mathbb{F}_Q^n, \quad \dim(V_i/V_{i-1}) = n_i,$$

is

$$\begin{bmatrix} n \\ n_1, \dots, n_r \end{bmatrix}_Q.$$

*Proof.* Choose  $V_1$  in  $\begin{bmatrix} n \\ n_1 \end{bmatrix}_Q$  ways. In the quotient  $\mathbb{F}_Q^n/V_1$ , choose  $V_2/V_1$  in  $\begin{bmatrix} n-n_1 \\ n_2 \end{bmatrix}_Q$  ways, and continue. The product is the factorization in [Proposition 2.11](#).  $\square$

## 6.3 Matrices of fixed rank

**Theorem 6.4** (Rank enumeration). *The number of  $m \times n$  matrices of rank  $r$  over  $\mathbb{F}_Q$  is*

$$N_{m,n,r}(Q) = \begin{bmatrix} m \\ r \end{bmatrix}_Q \begin{bmatrix} n \\ r \end{bmatrix}_Q |\mathrm{GL}_r(\mathbb{F}_Q)|,$$

where

$$|\mathrm{GL}_r(\mathbb{F}_Q)| = \prod_{i=0}^{r-1} (Q^r - Q^i).$$

Equivalently,

$$N_{m,n,r}(Q) = \frac{\prod_{i=0}^{r-1} (Q^m - Q^i)(Q^n - Q^i)}{\prod_{i=0}^{r-1} (Q^r - Q^i)}.$$

*Proof.* Every rank- $r$  matrix  $A$  can be factored as  $A = UV$ , where  $U$  is an  $m \times r$  matrix of full column rank and  $V$  is an  $r \times n$  matrix of full row rank. There are  $\prod_{i=0}^{r-1} (Q^m - Q^i)$  choices for  $U$  and  $\prod_{i=0}^{r-1} (Q^n - Q^i)$  choices for  $V$ . Two pairs  $(U, V)$  and  $(U', V')$  give the same  $A$  exactly when

$$U' = UG, \quad V' = G^{-1}V$$

for a unique  $G \in \mathrm{GL}_r(\mathbb{F}_Q)$ . Thus every matrix is counted  $|\mathrm{GL}_r|$  times. Substituting the subspace-count formulas gives the first expression.  $\square$

## 6.4 Schubert cells and a topological meaning

Let  $\mathrm{Gr}(k, n)$  be the complex Grassmannian of  $k$ -planes in  $\mathbb{C}^n$ . Its Schubert cell decomposition is indexed by partitions inside a  $k \times (n - k)$  rectangle.

**Theorem 6.5** (Poincaré polynomial of the Grassmannian). *The integral cohomology of  $\mathrm{Gr}(k, n)$  is concentrated in even degrees, and its Poincaré polynomial is*

$$P_{\mathrm{Gr}(k,n)}(t) = \sum_{j \geq 0} \mathrm{rank} H^{2j}(\mathrm{Gr}(k, n); \mathbb{Z}) t^{2j} = \begin{bmatrix} n \\ k \end{bmatrix}_{t^2}.$$

*Proof.* Fix the standard flag in  $\mathbb{C}^n$ . Every  $k$ -plane has a unique row-reduced echelon matrix. Its pivot positions determine a binary word with  $k$  ones, hence a partition  $\lambda$  inside the  $k \times (n - k)$  rectangle. Once the pivots are fixed, the entries allowed to vary freely occupy exactly the boxes of  $\lambda$ ; the corresponding stratum is therefore an affine complex space  $\mathbb{C}^{|\lambda|}$ . Ordering the strata by inclusion of diagrams yields a CW decomposition with one real cell of dimension  $2|\lambda|$  for each  $\lambda$ . Since all cells have even dimension, every cellular boundary map is zero. Thus the rank of  $H^{2j}$  is the number of such partitions of size  $j$ . Apply [Theorem 5.3](#) with  $q = t^2$ .  $\square$

*Remark 6.6* (The “same polynomial” principle). For a prime power  $Q$ , evaluating  $\begin{bmatrix} n \\ k \end{bmatrix}_q$  at  $q = Q$  counts rational points of a Grassmannian over  $\mathbb{F}_Q$ . Evaluating it at  $q = t^2$  records the even Betti numbers of the complex Grassmannian. This is an elementary instance of the broad principle that polynomial point counts over finite fields often shadow topological invariants over  $\mathbb{C}$ .

## 6.5 Growth over finite fields

**Proposition 6.7** (Dimension-dominant bounds). *For  $Q > 1$  and  $0 \leq k \leq n$ ,*

$$Q^{k(n-k)} \leq \begin{bmatrix} n \\ k \end{bmatrix}_Q \leq \frac{Q^{k(n-k)}}{(Q^{-1}; Q^{-1})_\infty}.$$

*Proof.* Rewrite the product as

$$\begin{bmatrix} n \\ k \end{bmatrix}_Q = Q^{k(n-k)} \prod_{j=1}^k \frac{1 - Q^{-(n-k+j)}}{1 - Q^{-j}}.$$

Each ratio is at least 1 because  $n - k + j \geq j$ , proving the lower bound. For the upper bound, discard the numerator factors and note

$$\prod_{j=1}^k (1 - Q^{-j}) \geq \prod_{j=1}^{\infty} (1 - Q^{-j}) = (Q^{-1}; Q^{-1})_\infty.$$

□

## Chapter 7

# Symmetry and unimodality

Palindromicity follows immediately from reciprocity. Unimodality is subtler: the coefficients rise toward the middle and then fall. A concise conceptual proof comes from the representation theory of  $\mathfrak{sl}_2$ .

### 7.1 A representation-theoretic lemma

Let  $\mathfrak{sl}_2(\mathbb{C})$  have generators  $E, F, H$  with

$$[H, E] = 2E, \quad [H, F] = -2F, \quad [E, F] = H.$$

**Lemma 7.1** (Structure of finite-dimensional  $\mathfrak{sl}_2$ -modules). *Every finite-dimensional complex  $\mathfrak{sl}_2$ -module is a direct sum of irreducibles  $L_d$  indexed by  $d \in \mathbb{N}_0$ . The module  $L_d$  has dimension  $d + 1$  and  $H$ -weights*

$$d, d - 2, \dots, -d,$$

*each with multiplicity one.*

*Proof.* We include the standard argument. Restrict a representation to the compact real form  $\mathfrak{su}_2$ . A finite-dimensional Lie-algebra representation integrates to the simply connected compact group  $SU(2)$  by exponentiating the representing matrices; the Lie relations ensure consistency of the local exponentials, and simple connectedness removes monodromy. Starting from any Hermitian inner product, average it over  $SU(2)$  with normalized Haar measure. The result is invariant. Hence the orthogonal complement of every invariant subspace is invariant, proving complete reducibility.

It remains to classify an irreducible module  $V$ . In the invariant Hermitian structure, a suitable scalar multiple of  $H$  is self-adjoint, so  $H$  is diagonalizable. Choose an eigenvector  $v$  whose eigenvalue  $d$  has maximal real part. Since  $E$  raises an  $H$ -weight by 2, maximality gives  $Ev = 0$ . The commutation relations imply by induction

$$HF^j v = (d - 2j)F^j v, \quad EF^j v = j(d - j + 1)F^{j-1} v.$$

Because  $V$  is finite-dimensional, let  $m$  be maximal with  $F^m v \neq 0$ . Applying  $E$  to  $F^{m+1} v = 0$  gives

$$(m + 1)(d - m)F^m v = 0,$$

so  $d = m \in \mathbb{N}_0$ . The vectors  $v, Fv, \dots, F^d v$  have distinct weights and are linearly independent. Their span is invariant under  $E, F, H$  and is nonzero; irreducibility makes it all of  $V$ . The displayed weights and multiplicities follow.  $\square$

## 7.2 Exterior powers and Gaussian coefficients

Let  $L_{n-1}$  have a weight basis  $v_0, \dots, v_{n-1}$  with  $Hv_i = (n-1-2i)v_i$ .

**Proposition 7.2** (Exterior-power character). *For  $0 \leq k \leq n$ ,*

$$\text{ch} \left( \bigwedge^k L_{n-1} \right) (x) = x^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{x^{-2}}.$$

*Proof.* A basis of the exterior power consists of  $v_{i_1} \wedge \dots \wedge v_{i_k}$  with  $0 \leq i_1 < \dots < i_k \leq n-1$ . Its weight is

$$k(n-1) - 2(i_1 + \dots + i_k).$$

Therefore

$$\text{ch} = x^{k(n-1)} \sum_{i_1 < \dots < i_k} (x^{-2})^{i_1 + \dots + i_k}.$$

Use [Corollary 3.4](#) with  $q = x^{-2}$ . The prefactor becomes

$$x^{k(n-1)} x^{-2 \binom{k}{2}} = x^{k(n-k)}.$$

□

**Theorem 7.3** (Unimodality of Gaussian coefficients). *If*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = c_0 + c_1 q + \dots + c_D q^D, \quad D = k(n-k),$$

*then*

$$c_0 \leq c_1 \leq \dots \leq c_{\lfloor D/2 \rfloor}$$

*and  $c_j = c_{D-j}$ . Thus the coefficient sequence is symmetric and unimodal.*

*Proof.* Symmetry was proved in [Theorem 2.5](#). By [Lemma 7.1](#), decompose

$$\bigwedge^k L_{n-1} \cong \bigoplus_{d \geq 0} m_d L_d.$$

All weights in this exterior power have the same parity as  $D$ . The character of one summand  $L_d$ , placed inside the ambient sequence of weights

$$D, D-2, \dots, -D,$$

contributes a consecutive block of ones centered at weight 0: it is zero until weight  $d$  is reached, then one at every step down to  $-d$ , then zero again. As one moves from weight  $D$  toward 0, such a block can turn on but cannot turn off. A nonnegative sum of these centered blocks therefore has weakly increasing multiplicities toward the middle.

By [Proposition 7.2](#), the multiplicity of weight  $D-2j$  is exactly  $c_j$ . Hence  $c_0 \leq c_1 \leq \dots \leq c_{\lfloor D/2 \rfloor}$ . □

*Remark 7.4.* The proof explains more than the inequality: the successive differences  $c_j - c_{j-1}$  record multiplicities of irreducible summands with prescribed highest weight. Stronger strict-unimodality results require additional arguments and have a small family of exceptional cases; ordinary unimodality is exactly what the  $\mathfrak{sl}_2$  decomposition supplies for free.



## **Part II**

# **Infinite series and basic hypergeometric summation**

## Chapter 8

# The infinite $q$ -binomial theorem

### 8.1 A functional-equation proof

**Theorem 8.1** (Infinite  $q$ -binomial theorem). *Let  $|q| < 1$  and  $|z| < 1$ . Then*

$$\sum_{n=0}^{\infty} \frac{(a; q)_n}{(q; q)_n} z^n = \frac{(az; q)_{\infty}}{(z; q)_{\infty}}.$$

*The convergence is locally uniform in  $(a, z)$  on compact subsets of  $\mathbb{C} \times \{|z| < 1\}$ .*

*Proof.* Set

$$F(z) = \sum_{n \geq 0} c_n z^n, \quad c_n = \frac{(a; q)_n}{(q; q)_n}.$$

Since  $c_n/c_{n-1} = (1 - aq^{n-1})/(1 - q^n) \rightarrow 1$ , the radius of convergence is 1 unless the series terminates; in every case it converges for  $|z| < 1$ . The coefficient recurrence is equivalent to

$$(1 - q^n)c_n = (1 - aq^{n-1})c_{n-1}.$$

Comparing coefficients shows that

$$(1 - z)F(z) = (1 - az)F(qz), \quad F(0) = 1.$$

Now put  $G(z) = (az; q)_{\infty}/(z; q)_{\infty}$ . Using  $(z; q)_{\infty} = (1 - z)(qz; q)_{\infty}$  gives the same functional equation and  $G(0) = 1$ .

To prove uniqueness, iterate the functional equation:

$$F(z) = \prod_{j=0}^{N-1} \frac{1 - azq^j}{1 - zq^j} F(q^N z).$$

As  $N \rightarrow \infty$ ,  $q^N z \rightarrow 0$ , so  $F(q^N z) \rightarrow 1$ . The finite product tends to  $G(z)$ . Hence  $F = G$ .  $\square$

**Corollary 8.2** (Euler's two infinite identities). *For  $|q| < 1$ ,*

$$(z; q)_{\infty} = \sum_{n=0}^{\infty} \frac{(-1)^n q^{\binom{n}{2}}}{(q; q)_n} z^n, \quad z \in \mathbb{C}, \quad (8.1)$$

$$\frac{1}{(z; q)_{\infty}} = \sum_{n=0}^{\infty} \frac{z^n}{(q; q)_n}, \quad |z| < 1. \quad (8.2)$$

*Proof.* Set  $a = 0$  in [Theorem 8.1](#) to obtain (8.2). For (8.1), start with the finite identity

$$(z; q)_N = \sum_{n=0}^N (-1)^n q^{\binom{n}{2}} \begin{bmatrix} N \\ n \end{bmatrix}_q z^n.$$

For fixed  $n$ ,  $\begin{bmatrix} N \\ n \end{bmatrix}_q \rightarrow 1/(q; q)_n$  as  $N \rightarrow \infty$ . The limiting series converges for every  $z$  because its successive-term ratio has magnitude asymptotic to  $|z| |q|^n$ . To justify passage to the limit uniformly on  $|z| \leq R$ , note that

$$\left| \begin{bmatrix} N \\ n \end{bmatrix}_q \right| \leq \frac{\prod_{j \geq 1} (1 + |q|^j)}{(\inf_{r \geq 0} |(q; q)_r|)^2},$$

where the denominator is positive because  $(q; q)_r \rightarrow (q; q)_\infty \neq 0$ . Thus the summands, extended by zero for  $n > N$ , are dominated by a constant multiple of  $|q|^{\binom{n}{2}} R^n$ . Dominated convergence, together with local uniform convergence of the products, now permits passage to the limit.  $\square$

**Corollary 8.3** (A ratio expansion). *If  $|b_j z| < 1$  for  $1 \leq j \leq s$ , then*

$$\frac{(a_1 z; q)_\infty \cdots (a_r z; q)_\infty}{(b_1 z; q)_\infty \cdots (b_s z; q)_\infty} = \sum_{N \geq 0} c_N z^N,$$

where

$$c_N = \sum_{u_1 + \cdots + u_r + v_1 + \cdots + v_s = N} \prod_{i=1}^r \frac{(-1)^{u_i} q^{\binom{u_i}{2}} a_i^{u_i}}{(q; q)_{u_i}} \prod_{j=1}^s \frac{b_j^{v_j}}{(q; q)_{v_j}}.$$

When  $s = 0$  there is no restriction on  $z$ .

*Proof.* Each numerator factor has the entire Euler expansion, while the reciprocal expansion of  $(b_j z; q)_\infty^{-1}$  converges absolutely for  $|b_j z| < 1$ . Hence all finitely many series converge absolutely in the stated common domain, so repeated Cauchy multiplication is legitimate. Collecting the terms of total degree  $N$  gives the displayed finite convolution.  $\square$

## 8.2 Formal and analytic interpretations

**Proposition 8.4** (Formal version). *Let  $A$  be a commutative ring complete and separated for the  $I$ -adic topology, let  $q \in I$ , and let  $a \in A$ . Then the coefficientwise  $I$ -adic products are defined in  $A[[z]]$ ,  $(z; q)_\infty$  is a unit there, and*

$$\frac{(az; q)_\infty}{(z; q)_\infty} = \sum_{n \geq 0} \frac{(a; q)_n}{(q; q)_n} z^n$$

in  $A[[z]]$ . Here every  $(q; q)_n$  is a unit. In particular, with  $a$  and  $q$  indeterminates, the identity holds in  $\mathbb{Z}[a][[q]][[z]]$ .

*Proof.* For  $j \geq 1$  the  $j$ th product factor differs from 1 by a coefficient in  $I^j$ . Thus the finite products form coefficientwise  $I$ -adic Cauchy sequences and have limits in  $A[[z]]$ . Their constant coefficient in  $z$  is 1, so  $(z; q)_\infty$  is invertible. Also  $q^n \in I$ , hence

$$(1 - q^n)^{-1} = \sum_{r \geq 0} q^{nr}$$

converges  $I$ -adically, and therefore  $(q; q)_n$  is a unit.

Write the quotient as  $F(z) = \sum_{n \geq 0} c_n z^n$ . Splitting off the initial factors gives

$$(1 - z)F(z) = (1 - az)F(qz), \quad c_0 = 1,$$

and hence

$$(1 - q^n)c_n = (1 - aq^{n-1})c_{n-1}, \quad n \geq 1.$$

Because  $1 - q^n$  is a unit, this recurrence has a unique solution, namely  $c_n = (a; q)_n / (q; q)_n$ .  $\square$

### 8.3 Limits in the number of factors

The stronger effective statement [Theorem 22.6](#) below gives both the fixed-column limit and its shifted form, together with a geometric error term. This is the stable finite-to-infinite passage used here.

*Remark 8.5.* The two Euler identities are therefore the stable limits of the finite product and reciprocal-product formulas. This stability is one reason finite  $q$ -binomial identities so often serve as polynomial approximations to infinite  $q$ -series identities.

## Chapter 9

# Basic hypergeometric series

### 9.1 Definition and convergence

**Definition 9.1.** The basic hypergeometric series is

$${}_r\phi_s \left[ \begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; q, z \right] = \sum_{n=0}^{\infty} \frac{(a_1, \dots, a_r; q)_n}{(q, b_1, \dots, b_s; q)_n} \left( (-1)^n q^{\binom{n}{2}} \right)^{1+s-r} z^n,$$

provided no denominator factor vanishes. If one numerator parameter is  $q^{-N}$ , the series terminates at  $n = N$ .

**Proposition 9.2** (Elementary convergence classification). *Assume the series does not terminate and the denominator parameters avoid poles.*

- (i) *If  $r \leq s$ , the series converges for every  $z$ .*
- (ii) *If  $r = s + 1$ , its radius of convergence is 1.*
- (iii) *If  $r > s + 1$ , the terms generally fail to tend to zero for  $z \neq 0$ ; such series are normally interpreted only when terminating or formally.*

*Proof.* Let  $T_n$  be the  $n$ th summand. Since  $(1 - aq^n) \rightarrow 1$  for each fixed parameter,

$$\frac{T_{n+1}}{T_n} = z \frac{\prod_{i=1}^r (1 - a_i q^n)}{(1 - q^{n+1}) \prod_{j=1}^s (1 - b_j q^n)} (-1)^{1+s-r} q^{n(1+s-r)}.$$

The ratio tends to 0 when  $r \leq s$ , to  $z$  when  $r = s + 1$ , and has unbounded magnitude when  $r > s + 1$  and  $z \neq 0$ . Apply the ratio test.  $\square$

The infinite  $q$ -binomial theorem is the evaluation

$${}_1\phi_0 \left[ \begin{matrix} a \\ - \end{matrix}; q, z \right] = \frac{(az; q)_{\infty}}{(z; q)_{\infty}}.$$

When  $s = r - 1$  the factor  $((-1)^n q^{\binom{n}{2}})^{1+s-r}$  in the definition is identically 1, so that

$${}_r\phi_{r-1} \left[ \begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_{r-1} \end{matrix}; q, z \right] = \sum_{n=0}^{\infty} \frac{(a_1; q)_n \cdots (a_r; q)_n}{(q; q)_n (b_1; q)_n \cdots (b_{r-1}; q)_n} z^n.$$

Under the scaling  $a_i = q^{\alpha_i}$ ,  $b_j = q^{\beta_j}$  this family degenerates as  $q \rightarrow 1^-$  to the ordinary hypergeometric series

$${}_rF_{r-1} \left( \begin{matrix} \alpha_1, \dots, \alpha_r \\ \beta_1, \dots, \beta_{r-1} \end{matrix}; z \right) := \sum_{n=0}^{\infty} \frac{(\alpha_1)_n \cdots (\alpha_r)_n}{n! (\beta_1)_n \cdots (\beta_{r-1})_n} z^n, \quad (\alpha)_n = \alpha(\alpha + 1) \cdots (\alpha + n - 1),$$

and – this is the point of the next theorem – the degeneration is valid for the full nonterminating series, not only for the terminating specializations tabulated in [Chapter C](#).

**Theorem 9.3** (Classical limit of a nonterminating  ${}_r\phi_{r-1}$ ). *Let  $r \geq 1$  and let  $\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_{r-1} \in \mathbb{C}$  satisfy*

$$\beta_j \notin \{0, -1, -2, \dots\} \quad (1 \leq j \leq r-1).$$

*For  $0 < q < 1$  let  $q^c = e^{c \log q}$  with the real logarithm. Then there is  $q_0 \in (0, 1)$  such that for  $q_0 < q < 1$  no denominator factor of*

$$\mathcal{H}_q(z) := {}_r\phi_{r-1} \left[ \begin{matrix} q^{\alpha_1}, \dots, q^{\alpha_r} \\ q^{\beta_1}, \dots, q^{\beta_{r-1}} \end{matrix}; q, z \right]$$

*vanishes, and for every  $\rho \in (0, 1)$*

$$\lim_{q \rightarrow 1^-} \mathcal{H}_q(z) = {}_rF_{r-1} \left( \begin{matrix} \alpha_1, \dots, \alpha_r \\ \beta_1, \dots, \beta_{r-1} \end{matrix}; z \right)$$

*uniformly for  $|z| \leq \rho$ . Both sides are analytic in  $|z| < 1$ . If no  $\alpha_i$  lies in  $\{0, -1, -2, \dots\}$ , both series have radius of convergence exactly 1: for the basic hypergeometric function  $\mathcal{H}_q$  this is [Proposition 9.2\(ii\)](#), and for  ${}_rF_{r-1}$  the ratio of consecutive terms is*

$$\frac{z \prod_{i=1}^r (\alpha_i + n)}{(n+1) \prod_{j=1}^{r-1} (\beta_j + n)} \longrightarrow z \quad (n \rightarrow \infty).$$

*If some  $\alpha_i$  is a nonpositive integer, both series terminate at the same degree, and the asserted limit follows term by term; see [Remark 9.4](#).*

*Proof.* Throughout put  $\beta_r := 1$ , so that  $(q; q)_n = (q^{\beta_r}; q)_n$  and the  $n$ th summand

$$T_n(q) := \frac{\prod_{i=1}^r (q^{\alpha_i}; q)_n}{\prod_{i=1}^r (q^{\beta_i}; q)_n} z^n$$

has equally many Pochhammer factors above and below.

*Step 1: no denominator vanishes near  $q = 1$ .* A factor  $1 - q^{\beta+m}$  with  $m \geq 0$  vanishes exactly when  $(\beta + m) \log q \in 2\pi i\mathbb{Z}$ . Writing  $\beta + m = \sigma + i\tau$  and using that  $\log q$  is real and nonzero, this forces  $\sigma = 0$  together with  $\tau \log q \in 2\pi\mathbb{Z}$ . If  $\tau = \Im\beta = 0$ , the two conditions say  $\beta = -m$ , which the hypothesis excludes for  $\beta = \beta_1, \dots, \beta_{r-1}$  and which is false for  $\beta_r = 1$ . If  $\Im\beta \neq 0$ , then  $\tau \log q$  is a nonzero multiple of  $2\pi$ , whence  $|\log q| \geq 2\pi/|\Im\beta|$ . Taking

$$q_0 = \max \{ \exp(-2\pi/|\Im\beta_j|) : \Im\beta_j \neq 0 \}$$

(and  $q_0$  arbitrary in  $(0, 1)$  if all  $\beta_j$  are real) rules out every vanishing for  $q_0 < q < 1$ .

*Step 2: termwise convergence.* Fix  $n$  and divide each of the  $2r$  Pochhammer symbols by  $(1 - q)^n$ ; the powers of  $(1 - q)$  cancel, so

$$T_n(q) = \frac{\prod_{i=1}^r (q^{\alpha_i}; q)_n (1 - q)^{-n}}{\prod_{i=1}^r (q^{\beta_i}; q)_n (1 - q)^{-n}} z^n.$$

By [Proposition 1.19](#) the  $i$ th numerator factor tends to  $(\alpha_i)_n$  and the  $i$ th denominator factor to  $(\beta_i)_n$ , with  $(\beta_r)_n = (1)_n = n!$ . The hypothesis on the  $\beta_j$  makes every  $(\beta_i)_n$  nonzero, so

$$\lim_{q \rightarrow 1^-} T_n(q) = t_n := \frac{(\alpha_1)_n \cdots (\alpha_r)_n}{n! (\beta_1)_n \cdots (\beta_{r-1})_n} z^n.$$

*Step 3: two elementary estimates.* Put  $t = -\log q > 0$ , so that  $q^w = e^{-wt}$ , and set  $h(u) = 1 - e^{-ut}$  for  $u \geq 0$ .

(a) If  $w = \sigma + i\tau$  with  $\sigma > 0$ , then

$$h(\sigma) \leq |1 - e^{-wt}| \leq h(\sigma) + |\tau| t.$$

Indeed, with  $\varrho = e^{-\sigma t} \in (0, 1)$  and  $\phi = -\tau t$ ,  $|1 - \varrho e^{i\phi}|^2 = (1 - \varrho)^2 + 2\varrho(1 - \cos \phi)$ . Discarding the nonnegative second term gives the lower bound; using  $1 - \cos \phi \leq \phi^2/2$  and  $\varrho \leq 1$  gives  $|1 - \varrho e^{i\phi}|^2 \leq (1 - \varrho)^2 + (\tau t)^2 \leq ((1 - \varrho) + |\tau| t)^2$ .

(b)  $h$  is increasing and concave on  $[0, \infty)$  with  $h(0) = 0$ , so  $u \mapsto h(u)/u$  is nonincreasing and therefore

$$\frac{h(v)}{h(u)} \leq \max\left(1, \frac{v}{u}\right) \quad (u, v > 0).$$

Moreover  $e^{ut} \geq 1 + ut$  gives  $h(u) \geq ut/(1 + ut)$ , that is,  $h(u)^{-1} \leq (ut)^{-1} + 1$ .

*Step 4: a majorant uniform in  $q$  and in  $|z| \leq \rho$ .* Set  $K = 1 + \max_i |\alpha_i| + \max_i |\beta_i|$  and fix an integer  $m_0 \geq 2K$ . For  $m \geq m_0$  the real numbers  $\sigma_i = \Re \alpha_i + m$  and  $\gamma_i = \Re \beta_i + m$  are at least  $m/2 > 0$ . The ratio of consecutive summands is

$$\frac{T_{m+1}(q)}{T_m(q)} = z \prod_{i=1}^r \frac{1 - q^{\alpha_i+m}}{1 - q^{\beta_i+m}},$$

and for  $m \geq m_0$  and every  $t > 0$ , Step 3 gives

$$\left| \frac{1 - q^{\alpha_i+m}}{1 - q^{\beta_i+m}} \right| \leq \frac{h(\sigma_i) + |\Im \alpha_i| t}{h(\gamma_i)} \leq \max\left(1, \frac{\sigma_i}{\gamma_i}\right) + |\Im \alpha_i| t \left(\frac{1}{\gamma_i t} + 1\right) \leq 1 + \frac{6K}{m} + Kt,$$

because  $\max(1, \sigma_i/\gamma_i) \leq 1 + |\Re \alpha_i - \Re \beta_i|/\gamma_i \leq 1 + 4K/m$  and  $|\Im \alpha_i|/\gamma_i \leq 2K/m$ . Put  $C = 6rK$ , fix  $\rho \in (0, 1)$ , choose  $t_1 > 0$  with  $e^{rKt_1} \leq \rho^{-1/2}$ , and set  $\theta = \rho^{1/2} < 1$ . For  $|z| \leq \rho$ ,  $0 < t \leq t_1$  and  $m \geq m_0$ ,

$$\left| \frac{T_{m+1}(q)}{T_m(q)} \right| \leq \rho \left(1 + \frac{6K}{m} + Kt_1\right)^r \leq \rho e^{rKt_1} e^{C/m} \leq \theta e^{C/m}.$$

Multiplying these bounds and using  $\sum_{m=m_0}^{n-1} m^{-1} \leq 1 + \log(n/m_0)$ ,

$$|T_n(q)| \leq |T_{m_0}(q)| \theta^{n-m_0} e^C \left(\frac{n}{m_0}\right)^C \quad (n > m_0).$$

By Step 2 the finitely many functions  $T_0, \dots, T_{m_0}$  converge as  $q \rightarrow 1^-$ , hence are bounded, say by  $B$ , on some interval  $(q_1, 1)$ ; the bound is uniform in  $|z| \leq \rho$  because  $T_n(q)$  is  $z^n$  times a function of  $q$  alone. Therefore, with

$$M_n := B \quad (n \leq m_0), \quad M_n := B e^C m_0^{-C} n^C \theta^{n-m_0} \quad (n > m_0),$$

one has  $|T_n(q)| \leq M_n$  for all  $n \geq 0$ , all  $|z| \leq \rho$  and all  $q$  with  $\max(q_0, q_1, e^{-t_1}) < q < 1$ , while  $\sum_{n \geq 0} M_n < \infty$  because  $\theta < 1$  dominates the polynomial factor  $n^C$ .

*Step 5: interchange.* The series  $\sum_n T_n(q)$  is dominated by the summable sequence  $(M_n)$  uniformly in  $q$ , so it converges uniformly in  $q$  on that interval; each term has the limit  $t_n$  by Step 2. Dominated convergence with respect to counting measure (Tannery's theorem) therefore gives

$$\lim_{q \rightarrow 1^-} \sum_{n \geq 0} T_n(q) = \sum_{n \geq 0} t_n,$$

which is the asserted limit. For the uniformity, note first that letting  $q \rightarrow 1^-$  in  $|T_n(q)| \leq M_n$  gives  $|t_n| \leq M_n$  as well. Let  $\varepsilon > 0$  and choose  $N > m_0$  with  $\sum_{n > N} M_n < \varepsilon/3$ . Writing  $T_n(q) = z^n g_n(q)$  and  $t_n = z^n g_n(1^-)$ , where  $g_n$  does not involve  $z$ , one has for every  $|z| \leq \rho$

$$\left| \sum_{n \geq 0} T_n(q) - \sum_{n \geq 0} t_n \right| \leq \sum_{n \leq N} |g_n(q) - g_n(1^-)| + 2 \sum_{n > N} M_n,$$

because  $|z|^n \leq 1$ . The first sum is a finite sum of quantities independent of  $z$ , each tending to 0 by Step 2, hence is  $< \varepsilon/3$  once  $q$  is close enough to 1. The convergence is therefore uniform on  $|z| \leq \rho$ .  $\square$

*Remark 9.4* (The terminating case). If some  $\alpha_i$  is a nonpositive integer  $-N$ , the numerator factor  $(q^{-N}; q)_n$  vanishes for  $n > N$ , the series is a finite sum, and neither the restriction  $|z| < 1$  nor Steps 3–5 are needed: applying [Proposition 1.19](#) to each of the  $N + 1$  terms suffices. That elementary mechanism is what underlies the itemized list in [Chapter C](#); [Theorem 9.3](#) is its nonterminating counterpart, and the analytic work in Steps 3–5 is exactly the price of dropping the terminating hypothesis.

## 9.2 Heine's transformation

**Theorem 9.5** (Heine transformation). *Initially for  $|z| < 1$  and  $|b| < 1$ ,*

$${}_2\phi_1 \left[ \begin{matrix} a, b \\ c \end{matrix}; q, z \right] = \frac{(b, az; q)_\infty}{(c, z; q)_\infty} {}_2\phi_1 \left[ \begin{matrix} c/b, z \\ az \end{matrix}; q, b \right].$$

*The identity extends wherever both sides are defined by analytic continuation.*

*Proof.* Use the tail-factor identity to write

$$\frac{(b; q)_n}{(c; q)_n} = \frac{(b; q)_\infty (cq^n; q)_\infty}{(c; q)_\infty (bq^n; q)_\infty}.$$

By the infinite  $q$ -binomial theorem,

$$\frac{(cq^n; q)_\infty}{(bq^n; q)_\infty} = \sum_{k \geq 0} \frac{(c/b; q)_k}{(q; q)_k} b^k q^{nk}.$$

Insert this into the  ${}_2\phi_1$  series and interchange the absolutely convergent sums:

$$\frac{(b; q)_\infty}{(c; q)_\infty} \sum_{k \geq 0} \frac{(c/b; q)_k}{(q; q)_k} b^k \sum_{n \geq 0} \frac{(a; q)_n}{(q; q)_n} (zq^k)^n.$$

The inner sum is

$$\frac{(azq^k; q)_\infty}{(zq^k; q)_\infty} = \frac{(az; q)_\infty}{(z; q)_\infty} \frac{(z; q)_k}{(az; q)_k}.$$

Collecting factors yields the claimed transformation. Both sides are meromorphic functions of the parameters, so equality extends from the initial open domain by the identity theorem.  $\square$

## 9.3 The $q$ -Gauss sum

**Theorem 9.6** ( $q$ -Gauss summation). *If  $|c/(ab)| < 1$ , then*

$${}_2\phi_1 \left[ \begin{matrix} a, b \\ c \end{matrix}; q, \frac{c}{ab} \right] = \frac{(c/a, c/b; q)_\infty}{(c, c/(ab); q)_\infty}.$$

*Proof.* Set  $z = c/(ab)$  in [Theorem 9.5](#). Then  $az = c/b$ , so one numerator parameter of the transformed series cancels its denominator parameter:

$${}_2\phi_1 \left[ \begin{matrix} c/b, c/(ab) \\ c/b \end{matrix}; q, b \right] = {}_1\phi_0 \left[ \begin{matrix} c/(ab) \\ - \end{matrix}; q, b \right] = \frac{(c/a; q)_\infty}{(b; q)_\infty}.$$

Multiplication by the prefactor in Heine's transformation gives the result. The initially stronger restrictions are removed by analytic continuation within the convergence domain.  $\square$



## 9.4 A second finite Cauchy identity

**Lemma 9.7** (Cauchy convolution II). *For  $a, b, c$  in a commutative ring,*

$$(ac; q)_n (bc; q)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (a; q)_k (b; q)_k c^k (c; q)_{n-k} (abcq^k; q)_{n-k}.$$

*Proof.* Apply the  $q$ -Cauchy identity [Theorem 3.15](#) with  $u = bq^k$ ,  $v = ac$ , and length  $n - k$ :

$$(abcq^k; q)_{n-k} = \sum_{r=0}^{n-k} \begin{bmatrix} n-k \\ r \end{bmatrix}_q (bq^k; q)_r (ac)^r (ac; q)_{n-k-r}.$$

Insert this in the right side of the proposed identity. Use

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} n-k \\ r \end{bmatrix}_q = \begin{bmatrix} n \\ k+r \end{bmatrix}_q \begin{bmatrix} k+r \\ k \end{bmatrix}_q, \quad (b; q)_k (bq^k; q)_r = (b; q)_{k+r}.$$

Put  $m = k + r$  and use

$$(c; q)_{n-k} = (c; q)_{n-m} (cq^{n-m}; q)_{m-k}.$$

The double sum becomes

$$\sum_{m=0}^n \begin{bmatrix} n \\ m \end{bmatrix}_q (b; q)_m c^m (c; q)_{n-m} (ac; q)_{n-m} \sum_{k=0}^m \begin{bmatrix} m \\ k \end{bmatrix}_q (a; q)_k a^{m-k} (cq^{n-m}; q)_{m-k}.$$

In the inner sum replace  $j = m - k$  and apply [Theorem 3.15](#) with  $u = cq^{n-m}$  and  $v = a$ ; it equals  $(acq^{n-m}; q)_m$ . Hence

$$(ac; q)_{n-m} (acq^{n-m}; q)_m = (ac; q)_n$$

can be factored out. The remaining sum is

$$(ac; q)_n \sum_{m=0}^n \begin{bmatrix} n \\ m \end{bmatrix}_q (b; q)_m c^m (c; q)_{n-m} = (ac; q)_n (bc; q)_n$$

by one final application of the  $q$ -Cauchy identity. □

*Remark 9.8* (Formal version). The displayed identity is `Fabius.finite_qCauchy_second_identity` in `QBinomialCauchy.lean`. It holds for arbitrary  $q, a, b, c$  in every commutative ring and every  $n \in \mathbb{N}_0$ , including  $n = 0$ ,  $q = 0$ , roots of unity, positive characteristic, and rings with zero divisors. The formal proof expands and reindexes a finite triangle and applies the primary and reflected finite  $q$ -Cauchy identities; it uses no quotient, cancellation, or nonvanishing assumption. This exact finite convolution does not by itself formalize the quotient-bearing  $q$ -Pfaff–Saalschütz theorem that follows.

## 9.5 The $q$ -Pfaff–Saalschütz sum

**Theorem 9.9** ( $q$ -Pfaff–Saalschütz summation). *For  $n \in \mathbb{N}_0$  and parameters for which all displayed expressions are defined,*

$${}_3\phi_2 \left[ \begin{matrix} a, b, q^{-n} \\ c, abq^{1-n}/c \end{matrix}; q, q \right] = \frac{(c/a, c/b; q)_n}{(c, c/(ab); q)_n}.$$

*Proof.* In Lemma 9.7, replace its parameters by

$$a \mapsto a, \quad b \mapsto b, \quad c \mapsto \frac{c}{ab}.$$

This gives

$$\frac{(c/a, c/b; q)_n}{(c, c/(ab); q)_n} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (a, b; q)_k \left( \frac{c}{ab} \right)^k \frac{(c/(ab); q)_{n-k} (cq^k; q)_{n-k}}{(c, c/(ab); q)_n}.$$

We show that the  $k$ th summand is the  $k$ th term of the stated  ${}_3\phi_2$ . The tail identities give

$$\frac{(cq^k; q)_{n-k}}{(c; q)_n} = \frac{1}{(c; q)_k}.$$

Also, reversal of the two finite products gives

$$(q^{-n}; q)_k = (-1)^k q^{-nk + \binom{k}{2}} (q^{n-k+1}; q)_k,$$

$$(abq^{1-n}/c; q)_k = \left( -\frac{ab}{c} \right)^k q^{k(1-n) + \binom{k}{2}} (cq^{n-k}/(ab); q)_k.$$

Therefore

$$\begin{aligned} & \frac{(q^{-n}; q)_k q^k}{(c; q)_k (abq^{1-n}/c; q)_k} \frac{(c/(ab); q)_n}{(c/(ab); q)_{n-k}} \\ &= \left( \frac{c}{ab} \right)^k \frac{(q^{n-k+1}; q)_k}{(q; q)_k} \frac{(c/(ab); q)_n}{(c/(ab); q)_{n-k} (cq^{n-k}/(ab); q)_k} \\ &= \begin{bmatrix} n \\ k \end{bmatrix}_q \left( \frac{c}{ab} \right)^k. \end{aligned}$$

Substitution yields exactly

$$\sum_{k=0}^n \frac{(a, b, q^{-n}; q)_k}{(q, c, abq^{1-n}/c; q)_k} q^k.$$

□

## 9.6 The two $q$ -Chu–Vandermonde sums

**Corollary 9.10** ( $q$ -Chu–Vandermonde). *For  $n \in \mathbb{N}_0$ , wherever the displayed rational expressions are defined,*

$${}_2\phi_1 \left[ \begin{matrix} q^{-n}, a \\ c \end{matrix}; q, \frac{cq^n}{a} \right] = \frac{(c/a; q)_n}{(c; q)_n}, \quad (9.1)$$

$${}_2\phi_1 \left[ \begin{matrix} q^{-n}, a \\ c \end{matrix}; q, q \right] = a^n \frac{(c/a; q)_n}{(c; q)_n}. \quad (9.2)$$

*Proof.* For (9.1), put  $b = q^{-n}$  in the  $q$ -Gauss sum. The infinite products telescope:

$$\frac{(cq^n, c/a; q)_\infty}{(c, cq^n/a; q)_\infty} = \frac{(c/a; q)_n}{(c; q)_n}.$$

On the nonempty seed domain  $|cq^n/a| < 1$ , substitution of  $b = q^{-n}$  is legitimate directly in Theorem 9.6; its convergence hypothesis becomes precisely that inequality, and  $(q^{-n}; q)_k = 0$  for

$k > n$ . Both sides of (9.1) are rational in  $(a, c)$ , and they agree on this open set wherever the displayed denominators are nonzero. After taking their difference and clearing denominators, the resulting polynomial vanishes on a nonempty open subset of  $\mathbb{C}^2$ , hence vanishes identically. Thus the identity holds wherever both sides are defined. Termination alone would not give this: it makes the two sides defined, not equal.

For (9.2), let  $b \rightarrow 0$  in Theorem 9.9, with its other numerator parameter equal to  $a$ . Then  $(b; q)_k \rightarrow 1$  and  $(abq^{1-n}/c; q)_k \rightarrow 1$ , so the left side becomes the displayed  ${}_2\phi_1$ . On the right,

$$\frac{(c/b; q)_n}{(c/(ab); q)_n} \rightarrow a^n$$

by comparing the leading factors of the two degree- $n$  polynomials in  $1/b$ .  $\square$

**Lemma 9.11** (Reversal of a terminating  ${}_2\phi_1$ ). *Let  $n \in \mathbb{N}_0$ , let  $q \neq 0$ , and let  $a, c, z$  be nonzero with  $(c; q)_n \neq 0$  and  $(q^{1-n}/a; q)_n \neq 0$ . Then*

$${}_2\phi_1 \left[ \begin{matrix} q^{-n}, a \\ c \end{matrix}; q, z \right] = (-1)^n q^{-\binom{n+1}{2}} z^n \frac{(a; q)_n}{(c; q)_n} {}_2\phi_1 \left[ \begin{matrix} q^{-n}, q^{1-n}/c \\ q^{1-n}/a \end{matrix}; q, \frac{cq^{n+1}}{az} \right]. \quad (9.3)$$

Both series terminate at  $n$ , so (9.3) is an identity between rational functions of  $a, c, z$  over  $\mathbb{Q}(q)$ ; no convergence hypothesis enters. The substitution  $(a, c) \mapsto (q^{1-n}/c, q^{1-n}/a)$  is an involution, and applying (9.3) twice returns the original series.

*Proof.* Since  $r = 2$  and  $s = 1$ , the factor  $((-1)^j q^{\binom{j}{2}})^{1+s-r}$  in the definition of  ${}_2\phi_1$  equals 1, so the left side is  $\sum_{j=0}^n T_j$  with

$$T_j = \frac{(q^{-n}; q)_j (a; q)_j}{(q; q)_j (c; q)_j} z^j.$$

We reverse the order of summation, writing  $j = n - i$  with  $0 \leq i \leq n$ .

First record the reversal rule for a single finite shifted factorial. Let  $x \neq 0$ . The concatenation law Proposition 1.2 gives  $(x; q)_n = (x; q)_{n-i} (xq^{n-i}; q)_i$ , and base reversal Proposition 1.6, applied to the second factor, gives

$$(xq^{n-i}; q)_i = (-x)^i q^{(n-i)i + \binom{i}{2}} (q^{1-n}/x; q)_i.$$

Because  $(n-i)i + \binom{i}{2} = ni - \binom{i+1}{2}$ , these combine into

$$(x; q)_{n-i} = (x; q)_n \frac{(-1)^i x^{-i} q^{\binom{i+1}{2} - ni}}{(q^{1-n}/x; q)_i}. \quad (9.4)$$

Now apply (9.4) to each of the four shifted factorials occurring in  $T_{n-i}$ . The four bases  $q^{-n}, a, q, c$  reflect to  $q, q^{1-n}/a, q^{-n}, q^{1-n}/c$  respectively, the first and third of which are again members of the list (the reflection of  $q^{-n}$  is  $q$  and that of  $q$  is  $q^{-n}$ , while  $q^{1-n}/a$  and  $q^{1-n}/c$  are new). Explicitly,

$$\begin{aligned} (q^{-n}; q)_{n-i} &= (q^{-n}; q)_n \frac{(-1)^i q^{\binom{i+1}{2}}}{(q; q)_i}, & (a; q)_{n-i} &= (a; q)_n \frac{(-1)^i a^{-i} q^{\binom{i+1}{2} - ni}}{(q^{1-n}/a; q)_i}, \\ (q; q)_{n-i} &= (q; q)_n \frac{(-1)^i q^{\binom{i+1}{2} - ni - i}}{(q^{-n}; q)_i}, & (c; q)_{n-i} &= (c; q)_n \frac{(-1)^i c^{-i} q^{\binom{i+1}{2} - ni}}{(q^{1-n}/c; q)_i}. \end{aligned}$$

Form the quotient  $[(q^{-n}; q)_{n-i} (a; q)_{n-i}] / [(q; q)_{n-i} (c; q)_{n-i}]$ . The four factors  $(-1)^i$  cancel in pairs. The four powers of  $q$  contribute

$$\binom{i+1}{2} + \left( \binom{i+1}{2} - ni \right) - \left( \binom{i+1}{2} - ni - i \right) - \left( \binom{i+1}{2} - ni \right) = ni + i,$$

the scalars contribute  $a^{-i}c^i$ , and the reflected shifted factorials contribute

$$\frac{(q^{-n}; q)_i (q^{1-n}/c; q)_i}{(q; q)_i (q^{1-n}/a; q)_i}.$$

Multiplying by  $z^{n-i} = z^n z^{-i}$  and collecting the  $i$ -dependent scalars as  $(cq^{n+1}/(az))^i$ , we obtain

$$T_{n-i} = \frac{(q^{-n}; q)_n (a; q)_n}{(q; q)_n (c; q)_n} z^n \frac{(q^{-n}; q)_i (q^{1-n}/c; q)_i}{(q; q)_i (q^{1-n}/a; q)_i} \left( \frac{cq^{n+1}}{az} \right)^i.$$

By [Corollary 1.9](#) with  $N = k = n$ ,

$$(q^{-n}; q)_n = (-1)^n q^{-n^2 + \binom{n}{2}} (q; q)_n = (-1)^n q^{-\binom{n+1}{2}} (q; q)_n,$$

so the constant in front is  $(-1)^n q^{-\binom{n+1}{2}} z^n (a; q)_n / (c; q)_n$ . Summing over  $0 \leq i \leq n$  gives (9.3). The final assertion follows because the reflected parameters are  $a' = q^{1-n}/c$  and  $c' = q^{1-n}/a$ , so that  $c'q^{n+1}/(a'z) = [(q^{1-n}/a)q^{n+1}] / [(q^{1-n}/c) \cdot cq^{n+1}/(az)] = z$ , and the reflection is therefore an involution on the parameter pair.  $\square$

**Proposition 9.12** (The second  $q$ -Chu–Vandermonde sum by reversal). *Identity (9.2) is a formal consequence of (9.1): it follows from [Lemma 9.11](#) alone, without [Theorem 9.9](#), without [Lemma 9.7](#), and without any limiting process.*

*Proof.* Fix  $n \in \mathbb{N}_0$  and nonzero parameters  $A, C$  with  $(A; q)_n \neq 0$  and  $(C; q)_n \neq 0$ . Put

$$a = \frac{q^{1-n}}{C}, \quad c = \frac{q^{1-n}}{A}, \quad z = \frac{cq^n}{a} = \frac{Cq^n}{A}.$$

These are nonzero, and by [Proposition 1.6](#) the hypotheses of [Lemma 9.11](#) hold:  $(c; q)_n = (q^{1-n}/A; q)_n = (-A)^{-n} q^{-\binom{n}{2}} (A; q)_n \neq 0$  and  $(q^{1-n}/a; q)_n = (C; q)_n \neq 0$ . Since  $az = cq^n$ , the reflected argument in (9.3) is  $cq^{n+1}/(az) = q$ , and the reflected parameters are  $q^{1-n}/c = A$  and  $q^{1-n}/a = C$ . Hence (9.3) reads

$${}_2\phi_1 \left[ \begin{matrix} q^{-n}, a \\ c \end{matrix}; q, \frac{cq^n}{a} \right] = (-1)^n q^{-\binom{n+1}{2}} \left( \frac{cq^n}{a} \right)^n \frac{(a; q)_n}{(c; q)_n} {}_2\phi_1 \left[ \begin{matrix} q^{-n}, A \\ C \end{matrix}; q, q \right].$$

The left side is  $(c/a; q)_n / (c; q)_n$  by (9.1), and  $c/a = C/A$ . Solving for the remaining series and using  $a/(cq^n) = A/(Cq^n)$ ,

$${}_2\phi_1 \left[ \begin{matrix} q^{-n}, A \\ C \end{matrix}; q, q \right] = (-1)^n q^{\binom{n+1}{2}} \left( \frac{A}{Cq^n} \right)^n \frac{(C/A; q)_n}{(a; q)_n}.$$

By [Proposition 1.6](#) once more,  $(a; q)_n = (q^{1-n}/C; q)_n = (-C)^{-n} q^{-\binom{n}{2}} (C; q)_n$ . Substituting and cancelling  $C^n$  and the two signs,

$${}_2\phi_1 \left[ \begin{matrix} q^{-n}, A \\ C \end{matrix}; q, q \right] = q^{\binom{n+1}{2} + \binom{n}{2} - n^2} A^n \frac{(C/A; q)_n}{(C; q)_n} = A^n \frac{(C/A; q)_n}{(C; q)_n},$$

because  $\binom{n+1}{2} + \binom{n}{2} = \frac{n^2+n}{2} + \frac{n^2-n}{2} = n^2$ . This is (9.2) at the pair  $(A, C)$ , which was an arbitrary admissible pair. The auxiliary condition  $(A; q)_n \neq 0$  excludes only a proper algebraic subset of the parameter space, and both sides of (9.2) are rational functions of  $(A, C)$  there; agreeing off a proper algebraic subset, they agree wherever both are defined. This is the same identity-theorem step used for (9.1) in the proof of [Corollary 9.10](#). Every step used only finite products, so, after multiplication by  $A^{-n}$  is undone and denominators are cleared, (9.2) holds over any commutative ring in which  $q$  and  $A$  are invertible.  $\square$

## 9.7 A hierarchy of mechanisms

The proofs in this chapter form a hierarchy:

$$\text{finite } q\text{-Pascal} \implies q\text{-Cauchy} \implies q\text{-Pfaff-Saalschütz},$$

while

$$\begin{aligned} & \text{infinite } q\text{-binomial} \implies \text{Heine} \implies q\text{-Gauss} \\ & \implies \text{first } q\text{-Chu-Vandermonde} \implies \text{second } q\text{-Chu-Vandermonde}, \end{aligned}$$

the last arrow being [Proposition 9.12](#). The finite chain is valid in a commutative ring after denominators are cleared. The infinite chain requires convergence or a suitable complete formal topology, but only as far as its third arrow: the first  $q$ -Chu-Vandermonde sum is already a rational, hence purely algebraic, identity, as the proof of [Corollary 9.10](#) records, and by [Lemma 9.11](#) the fourth arrow is finite-product algebra. Thus (9.2) does not depend on [Theorem 9.9](#), and the two derivations of it given here are genuinely independent. This distinction becomes important in computer formalization; see [Chapter 24](#).

## Chapter 10

# Euler telescoping and very-well-poised summations

The summations in this chapter are finite. Their natural first home is therefore a rational-function field, where no convergence hypothesis is needed and every nonzero auxiliary rational function may legitimately be cancelled. Only after the rational identity has been proved do we specialize the parameters to complex numbers. This order matters: normalizing a sum by its conjectured product can introduce auxiliary denominators whose zeros are not poles of the original identity.

The main result is Jackson's terminating balanced very-well-poised  ${}_8\phi_7$  summation. Its proof has two elementary layers. A six-variable rational transformation supplies the algebraic certificate, and Euler's product-difference lemma turns that certificate into a finite telescoping sum. The terminal  ${}_6\phi_5$  identity is then an honest finite limit.

### 10.1 Euler's product-difference telescope

**Lemma 10.1** (Euler's telescoping lemma). *Let  $K$  be a field and let  $N \in \mathbb{N}_0$ . Choose  $u_0, \dots, u_N \in K$  and  $v_0, \dots, v_N \in K$ . Put  $w_k = u_k - v_k$  and assume  $w_0 v_1 \cdots v_N \neq 0$ . Then*

$$\sum_{k=0}^N \frac{w_k}{w_0} \frac{u_0 u_1 \cdots u_{k-1}}{v_1 v_2 \cdots v_k} = \frac{1}{w_0} \left( \frac{u_0 u_1 \cdots u_N}{v_1 v_2 \cdots v_N} - v_0 \right), \quad (10.1)$$

where empty products are 1.

*Proof.* For  $k \geq 1$  the  $k$ th summand is

$$\frac{1}{w_0} \left( \frac{u_0 u_1 \cdots u_k}{v_1 v_2 \cdots v_k} - \frac{u_0 u_1 \cdots u_{k-1}}{v_1 v_2 \cdots v_{k-1}} \right),$$

because  $w_k = u_k - v_k$ . The  $k = 0$  summand is  $(u_0 - v_0)/w_0$ . Adding the  $N + 1$  differences cancels every intermediate product and leaves exactly the right-hand side of (10.1).  $\square$

The application below has  $u_N = v_0 = 0$ , so the right-hand side vanishes. The advantage of Lemma 10.1 is that the telescoping antiderivative is encoded by the two product sequences; it need not be guessed term by term.

### 10.2 Jackson's rational certificate

**Lemma 10.2** (Jackson's six-variable rational certificate). *In the rational-function field*

$$\mathbb{Q}(A, B, C, D, E, F)$$

one has

$$\begin{aligned}
& 1 - \frac{(1-B)(1-C)(1-D)(1-E)(1-F)(1-A^3/(BCDEF))}{(1-A/B)(1-A/C)(1-A/D)(1-A/E)(1-A/F)(1-BCDEF/A^2)} \\
&= \frac{(1-A)(1-D)(1-A^2/(BCDE))(1-A^2/(BCDF))(1-A^2/(BDEF))(1-A^2/(CDEF))}{(1-A/B)(1-A/C)(1-A/E)(1-A/F)(1-A^2/(BCDEF))(1-A^3/(BCD^2EF))} \\
&\times \left[ 1 - \frac{(1-A/(BD))(1-A/(CD))(1-A/(DE))(1-A/(DF))(1-A^2/(BCDEF))(1-A^3/(BCDEF))}{(1-1/D)(1-A/D)(1-A^2/(BCDE))(1-A^2/(BCDF))(1-A^2/(BDEF))(1-A^2/(CDEF))} \right]. \quad (10.2)
\end{aligned}$$

Consequently the identity holds after every specialization to a field for which all rational expressions displayed in (10.2) are defined.

*Proof.* For a finite list set

$$\Pi(x_1, \dots, x_r) := \prod_{i=1}^r (1 - x_i),$$

and abbreviate the left-hand side of (10.2) by

$$J(A; B, C, D, E, F) := 1 - \frac{\Pi(B, C, D, E, F, A^3/(BCDEF))}{\Pi(A/B, A/C, A/D, A/E, A/F, BCDEF/A^2)}.$$

Define also

$$R(A; B, C, D, E, F) := \frac{\Pi(A, A/(EF), A^2/(BCDE), A^2/(BCDF))}{\Pi(A/E, A/F, A^2/(BCD), A^2/(BCDEF))}$$

and the rational substitution

$$\sigma(A; B, C, D, E, F) := \left( \frac{A^2}{BCD}; E, F, \frac{A}{BC}, \frac{A}{CD}, \frac{A}{BD} \right). \quad (10.3)$$

We first prove the seed transformation

$$J(A; B, C, D, E, F) = R(A; B, C, D, E, F) J(\sigma(A; B, C, D, E, F)). \quad (10.4)$$

Let  $D_0, N_0$  be respectively the denominator and numerator products in the definition of  $J$ :

$$\begin{aligned}
D_0 &= \Pi(A/B, A/C, A/D, A/E, A/F, BCDEF/A^2), \\
N_0 &= \Pi(B, C, D, E, F, A^3/(BCDEF)).
\end{aligned}$$

After the substitution (10.3), the analogous products, before common factors are cancelled, may be written

$$\begin{aligned}
D_1 &= \Pi(A/B, A/C, A/D, A^2/(BCDE), A^2/(BCDF), EF/A), \\
N_1 &= \Pi(A/(BC), A/(BD), A/(CD), E, F, A^3/(BCDEF)).
\end{aligned}$$

Put  $\Delta_i = D_i - N_i$ . We claim

$$\Delta_0 = \lambda \Delta_1, \quad \lambda := \frac{B^2 C^2 D^2 (A-1)}{A(A^2 - BCD)}. \quad (10.5)$$

This claim has a short coefficient proof. Put  $L = A^3/(BCDE)$  and regard  $\Delta_0, \Delta_1$  as Laurent polynomials in  $F$ . Directly pairing the constant and reciprocal factors shows that each has the form

$$\Delta_i = r_i F + s_i + \frac{L r_i}{F}.$$

Introduce the two four-term polynomials

$$\begin{aligned} U &:= A^2 - A(B + C + D) + A + BCD, \\ V &:= A^2 + ABCD - A(BC + BD + CD) + BCD. \end{aligned}$$

Expanding only in the affine variable  $E$  gives

$$\begin{aligned} r_0 &= \frac{A-1}{A^2} (-AU + EV), \\ r_1 &= \frac{A^2 - BCD}{AB^2C^2D^2} (-AU + EV). \end{aligned}$$

Hence  $r_0 = \lambda r_1$ .

It remains to compare the constant coefficients. At  $F = 1$  both  $N_0$  and  $N_1$  vanish, and cancelling their common factors gives

$$\begin{aligned} \frac{\Delta_0(1)}{\Delta_1(1)} &= \frac{(1 - A/E)(1 - A)(1 - BCDE/A^2)}{(1 - A^2/(BCDE))(1 - A^2/(BCD))(1 - E/A)} \\ &= \frac{B^2C^2D^2(A-1)}{A(A^2 - BCD)} = \lambda. \end{aligned}$$

The coefficients of  $F$  and  $F^{-1}$  already have ratio  $\lambda$ , so the equality at  $F = 1$  forces  $s_0 = \lambda s_1$ . This proves (10.5). A second direct cancellation gives

$$R(A; B, C, D, E, F) \frac{D_0}{D_1} = \lambda.$$

Since  $J = \Delta_0/D_0$  and  $J \circ \sigma = \Delta_1/D_1$ , the seed transformation (10.4) follows.

Apply the seed transformation twice. Its second iterate is

$$\sigma^2(A; B, C, D, E, F) = \left( \frac{A^3}{BCD^2EF}, \frac{A}{CD}, \frac{A}{BD}, \frac{A^2}{BCDEF}, \frac{A}{DF}, \frac{A}{DE} \right). \quad (10.6)$$

Substitution and cancellation in the two prefactors gives

$$\begin{aligned} &R(A; B, C, D, E, F) R(\sigma(A; B, C, D, E, F)) \\ &= \frac{\Pi(A, D, A^2/(BCDE), A^2/(BCDF), A^2/(BDEF), A^2/(CDEF))}{\Pi(A/B, A/C, A/E, A/F, A^2/(BCDEF), A^3/(BCD^2EF))}. \end{aligned}$$

Likewise (10.6) gives

$$\begin{aligned} J(\sigma^2) &= 1 - \frac{(1 - A/(BD))(1 - A/(CD))(1 - A/(DE))(1 - A/(DF))}{(1 - 1/D)(1 - A/D)(1 - A^2/(BCDE))(1 - A^2/(BCDF))} \\ &\quad \times \frac{(1 - A^2/(BCDEF))(1 - A^3/(BCDEF))}{(1 - A^2/(BDEF))(1 - A^2/(CDEF))}. \end{aligned}$$

Thus  $J = R(R \circ \sigma)(J \circ \sigma^2)$  is precisely (10.2). Every step took place in the rational-function field, so the final specialization assertion follows by applying a field homomorphism wherever the displayed denominators remain nonzero.  $\square$



### 10.3 Jackson's terminating ${}_8\phi_7$ sum

**Theorem 10.3** (Jackson's terminating very-well-poised  ${}_8\phi_7$  sum). *For  $N \in \mathbb{N}_0$ , the following is an identity in  $\mathbb{Q}(q, a, b, c, d)$ :*

$$\begin{aligned} \sum_{k=0}^N \frac{1 - aq^{2k}}{1 - a} \frac{(a, b, c, d, a^2q^{N+1}/(bcd), q^{-N}; q)_k}{(q, aq/b, aq/c, aq/d, bcdq^{-N}/a, aq^{N+1}; q)_k} q^k \\ = \frac{(aq, aq/(bc), aq/(bd), aq/(cd); q)_N}{(aq/b, aq/c, aq/d, aq/(bcd); q)_N}. \end{aligned} \quad (10.7)$$

Consequently (10.7) holds for  $q, a, b, c, d \in \mathbb{C}^\times$  whenever

$$(1 - a)(q, aq/b, aq/c, aq/d, bcdq^{-N}/a, aq^{N+1}, aq/(bcd); q)_N \neq 0.$$

Equivalently, it holds at every nonzero complex specialization at which the terms in the displayed identity are defined.

*Proof.* Work first in the field  $K = \mathbb{Q}(q, a, b, c, d)$ . Let  $F(n, k)$  be the  $k$ th summand divided by the product on the right-hand side of (10.7); explicitly,

$$\begin{aligned} F(n, k) &= \frac{(aq/b, aq/c, aq/d, aq/(bcd); q)_n}{(aq, aq/(bc), aq/(bd), aq/(cd); q)_n} \frac{1 - aq^{2k}}{1 - a} q^k \\ &\quad \times \frac{(a, b, c, d, a^2q^{n+1}/(bcd), q^{-n}; q)_k}{(q, aq/b, aq/c, aq/d, bcdq^{-n}/a, aq^{n+1}; q)_k}. \end{aligned}$$

Set  $F(n, k) = 0$  outside  $0 \leq k \leq n$ . Define

$$\begin{aligned} u_k &= (1 - aq^k)(1 - bq^k)(1 - cq^k)(1 - dq^k) \left(1 - \frac{a^2q^{n+k+1}}{bcd}\right) (1 - q^{-n-1+k}), \\ v_k &= \left(1 - \frac{aq^k}{b}\right) \left(1 - \frac{aq^k}{c}\right) \left(1 - \frac{aq^k}{d}\right) \left(1 - \frac{bcdq^{-n+k-1}}{a}\right) \\ &\quad \times (1 - aq^{n+k+1})(1 - q^k), \quad w_k := u_k - v_k. \end{aligned}$$

Then  $u_{n+1} = 0$ ,  $v_0 = 0$ , and

$$w_0 = (1 - a)(1 - b)(1 - c)(1 - d) \left(1 - \frac{a^2q^{n+1}}{bcd}\right) (1 - q^{-n-1}). \quad (10.8)$$

Apply Lemma 10.2 with

$$A = aq^{2k}, \quad B = \frac{aq^k}{b}, \quad C = \frac{aq^k}{c}, \quad D = aq^{n+k+1}, \quad E = \frac{aq^k}{d}, \quad F = q^k. \quad (10.9)$$

The fraction subtracted from 1 on the certificate's left-hand side becomes  $v_k/u_k$ . Cancelling the monomials on its right-hand side yields

$$w_k = bcd \frac{1 - aq^{2k}}{1 - a^2q^{2n+2}/(bcd)} \frac{q^k}{aq^{n+1}} \mathcal{B}_{n,k}, \quad (10.10)$$

where

$$\begin{aligned} \mathcal{B}_{n,k} &= (1 - q^{-n-1})(1 - aq^{n+1}/b)(1 - aq^{n+1}/c)(1 - aq^{n+1}/d) \\ &\quad \times (1 - bcdq^{-n+k-1}/a)(1 - a^2q^{n+k+1}/(bcd)) \frac{aq^{n+1}}{bcd} \\ &\quad + (1 - q^{-n+k-1})(1 - aq^{n+k+1})(1 - aq^{n+1}/(bc))(1 - aq^{n+1}/(bd)) \\ &\quad \times (1 - aq^{n+1}/(cd))(1 - a^2q^{n+1}/(bcd)). \end{aligned} \quad (10.11)$$

The leading factor  $bcd$  in (10.10) is essential.

We next verify the finite-difference certificate without an unprinted multivariate expansion. Put

$$\begin{aligned} \mathcal{C}_{n,k} := & -\frac{1-aq^{2k}}{1-a}q^k \frac{(a,b,c,d;q)_k}{(q,aq/b,aq/c,aq/d;q)_k} \frac{(aq/b,aq/c,aq/d,aq/(bcd);q)_n}{(aq,aq/(bc),aq/(bd),aq/(cd);q)_{n+1}} \\ & \times \frac{(q^{-n-1},a^2q^{n+1}/(bcd);q)_k}{(1-q^{-n-1})(1-a^2q^{n+1}/(bcd))(bcdq^{-n}/a,aq^{n+2};q)_k}. \end{aligned}$$

We claim that

$$F(n+1,k) - F(n,k) = \mathcal{C}_{n,k} \mathcal{B}_{n,k} \quad (0 \leq k \leq n+1). \quad (10.12)$$

For  $0 \leq k \leq n$ , repeated use of

$$\frac{(xq;q)_k}{(x;q)_k} = \frac{1-xq^k}{1-x}$$

gives

$$\begin{aligned} \frac{F(n+1,k)}{F(n,k)} - 1 = & -\frac{\mathcal{B}_{n,k}}{(1-q^{-n+k-1})(1-aq^{n+1}/(bc))(1-aq^{n+1}/(bd))} \\ & \times \frac{1}{(1-aq^{n+1}/(cd))(1-a^2q^{n+1}/(bcd))(1-aq^{n+k+1})}. \end{aligned}$$

Multiplying by the displayed formula for  $F(n,k)$  and cancelling one initial and one terminal factor in each shifted factorial gives (10.12). For  $k = n+1$  one has  $(q^{-n};q)_{n+1} = 0$ , so  $F(n,n+1) = 0$ ; the second summand of  $\mathcal{B}_{n,n+1}$  also vanishes, and direct cancellation in the first summand gives the same formula.

On the other hand,

$$\frac{u_0 u_1 \cdots u_{k-1}}{v_1 v_2 \cdots v_k} = \frac{(a,b,c,d,a^2q^{n+1}/(bcd),q^{-n-1};q)_k}{(aq/b,aq/c,aq/d,bcdq^{-n}/a,aq^{n+2},q;q)_k}. \quad (10.13)$$

Define

$$\begin{aligned} D_n = & \frac{(aq/b,aq/c,aq/d,aq/(bcd);q)_n}{(aq,aq/(bc),aq/(bd),aq/(cd);q)_{n+1}} \\ & \times \left[ -\frac{aq^{n+1}}{bcd}(1-b)(1-c)(1-d) \left( 1 - \frac{a^2q^{2n+2}}{bcd} \right) \right]. \end{aligned}$$

The factor  $bcd$  in the denominator of  $D_n$  is essential. Substituting (10.8), (10.10), and (10.13), and cancelling the common factors, gives

$$D_n \frac{w_k}{w_0} \frac{u_0 u_1 \cdots u_{k-1}}{v_1 v_2 \cdots v_k} = \mathcal{C}_{n,k} \mathcal{B}_{n,k}.$$

Together with (10.12), this proves the exact finite-difference certificate

$$F(n+1,k) - F(n,k) = D_n \frac{w_k}{w_0} \frac{u_0 u_1 \cdots u_{k-1}}{v_1 v_2 \cdots v_k}. \quad (10.14)$$

Use Lemma 10.1 with upper index  $n+1$ . Since  $u_{n+1} = v_0 = 0$ , its right-hand side is zero. Summing (10.14) over  $0 \leq k \leq n+1$  therefore gives

$$\sum_k F(n+1,k) = \sum_k F(n,k).$$

At  $n=0$  only  $k=0$  occurs and  $F(0,0) = 1$ . Hence the normalized sum is 1 for every  $n$ , which is exactly (10.7) in  $K$ .

Finally, clear only the denominators occurring in the original identity. Equality in  $K$  then specializes at every complex tuple satisfying the hypotheses of the theorem, even when an auxiliary normalization factor used inside the proof vanishes.  $\square$

*Remark 10.4* (The removable value  $a = 1$ ). The statement above excludes  $a = 1$  because the very-well-poised factor is displayed as a quotient. The singularity is removable. For  $k \geq 1$ ,

$$\frac{1 - aq^{2k}}{1 - a}(a; q)_k = (1 - aq^{2k})(aq; q)_{k-1},$$

whose value at  $a = 1$  is  $(1 - q^{2k})(q; q)_{k-1}$ ; for  $k = 0$  the combined factor is 1. Thus (10.7) extends to  $a = 1$  whenever the remaining displayed denominators are nonzero.

## 10.4 The terminating ${}_6\phi_5$ limit

**Corollary 10.5** (Terminating very-well-poised  ${}_6\phi_5$ ). *For  $N \in \mathbb{N}_0$ , the identity*

$$\begin{aligned} \sum_{k=0}^N \frac{1 - aq^{2k}}{1 - a} \frac{(a, b, c, q^{-N}; q)_k}{(q, aq/b, aq/c, aq^{N+1}; q)_k} \left( \frac{aq^{N+1}}{bc} \right)^k \\ = \frac{(aq, aq/(bc); q)_N}{(aq/b, aq/c; q)_N} \end{aligned} \quad (10.15)$$

*holds in  $\mathbb{Q}(q, a, b, c)$ . Consequently it holds for  $q, a, b, c \in \mathbb{C}^\times$  whenever*

$$(1 - a)(q, aq/b, aq/c, aq^{N+1}; q)_N \neq 0.$$

*It also has the removable  $a = 1$  extension described in Remark 10.4.*

*Proof.* In (10.7), regard  $x = d^{-1}$  as an indeterminate over  $\mathbb{Q}(q, a, b, c)$  and take the value at  $x = 0$  after cancelling the common leading powers of  $d$ . For each fixed  $k$ ,

$$\begin{aligned} (d; q)_k &= (-d)^k q^{\binom{k}{2}} (1 + O(d^{-1})), \\ (bcdq^{-N}/a; q)_k &= \left( -\frac{bcdq^{-N}}{a} \right)^k q^{\binom{k}{2}} (1 + O(d^{-1})). \end{aligned}$$

Therefore

$$q^k \frac{(d; q)_k}{(bcdq^{-N}/a; q)_k} \longrightarrow \left( \frac{aq^{N+1}}{bc} \right)^k.$$

The factors with arguments  $a^2q^{N+1}/(bcd)$  and  $aq/d$  tend to 1. On the product side the four factors involving  $d$  likewise tend to 1, leaving the quotient in (10.15). The sum has only  $N + 1$  terms, so taking this coefficientwise rational limit is legitimate and proves the identity in  $\mathbb{Q}(q, a, b, c)$ . The specialization and removable-value statements follow exactly as in Theorem 10.3 and remark 10.4.  $\square$

## Chapter 11

# Bilateral series and Ramanujan's ${}_1\psi_1$ sum

### 11.1 Definition

For integral  $n$ , negative  $q$ -shifted factorials are understood as in [Proposition 1.8](#).

**Definition 11.1** (Bilateral basic hypergeometric series). Define

$${}_1\psi_1 \left[ \begin{matrix} a \\ b \end{matrix}; q, z \right] = \sum_{n \in \mathbb{Z}} \frac{(a; q)_n}{(b; q)_n} z^n.$$

The positive tail behaves like a geometric series in  $z$ , while the negative tail behaves like a geometric series in  $b/(az)$ .

**Proposition 11.2** (Convergence annulus). *If denominator factors do not vanish, the  ${}_1\psi_1$  series converges absolutely in*

$$|b/a| < |z| < 1.$$

*Proof.* For  $n \rightarrow +\infty$ , the ratio of consecutive terms tends to  $z$ . For the negative tail put  $n = -m$  and use the negative-index formula. The ratio of successive terms in  $m$  tends to  $b/(az)$ . The ratio test gives the annulus.  $\square$

**Theorem 11.3** (Ramanujan's  ${}_1\psi_1$  summation). *For  $|b/a| < |z| < 1$ ,*

$${}_1\psi_1 \left[ \begin{matrix} a \\ b \end{matrix}; q, z \right] = \frac{(q, b/a, az, q/(az); q)_\infty}{(b, q/a, z, b/(az); q)_\infty}.$$

*Proof.* We first prove the formula for  $b = q^{m+1}$ , where  $m \in \mathbb{N}_0$ . In that case  $1/(b; q)_n = 0$  for  $n < -m$  in the limiting sense, so shift  $n = k - m$ :

$$\begin{aligned} \sum_{n \in \mathbb{Z}} \frac{(a; q)_n}{(q^{m+1}; q)_n} z^n &= z^{-m} \frac{(a; q)_{-m}}{(q^{m+1}; q)_{-m}} \sum_{k=0}^{\infty} \frac{(aq^{-m}; q)_k}{(q; q)_k} z^k \\ &= z^{-m} \frac{(q; q)_m}{(aq^{-m}; q)_m} \frac{(aq^{-m}z; q)_\infty}{(z; q)_\infty}. \end{aligned}$$

The second equality is the infinite  $q$ -binomial theorem. Reverse the finite products:

$$\begin{aligned} (aq^{-m}; q)_m &= (-a)^m q^{-m(m+1)/2} (q/a; q)_m, \\ (aq^{-m}z; q)_m &= (-az)^m q^{-m(m+1)/2} (q/(az); q)_m. \end{aligned}$$

Since  $(aq^{-m}z; q)_\infty = (aq^{-m}z; q)_m (az; q)_\infty$ , we obtain

$$\frac{(q; q)_m}{(q/a; q)_m} \frac{(q/(az); q)_m (az; q)_\infty}{(z; q)_\infty}.$$

On the other hand, substituting  $b = q^{m+1}$  in the proposed product and cancelling tails gives exactly the same expression:

$$\frac{(q; q)_\infty}{(q^{m+1}; q)_\infty} \frac{(q^{m+1}/a; q)_\infty}{(q/a; q)_\infty} \frac{(q/(az); q)_\infty}{(q^{m+1}/(az); q)_\infty} \frac{(az; q)_\infty}{(z; q)_\infty}.$$

For fixed  $a, z$  with  $|z| < 1$ , both sides are analytic functions of  $b$  in a neighborhood of 0 contained in  $|b| < |az|$ . Absolute convergence is locally uniform there by [Proposition 11.2](#). For all sufficiently large  $m$ , the points  $b = q^{m+1}$  lie in that neighborhood, and they accumulate at 0. The identity theorem therefore proves equality near 0. Both sides are meromorphic functions of  $b$  throughout the disk  $|b| < |az|$ : the series gives the meromorphic continuation termwise on compacta avoiding denominator zeros, and the product quotient is visibly meromorphic. Uniqueness of meromorphic continuation proves the identity everywhere in that disk where both sides are defined, which is precisely the asserted annulus for fixed  $a$  and  $z$ .  $\square$

*Remark 11.4.* Ramanujan's sum contains the infinite  $q$ -binomial theorem and several theta-product identities as limiting or specialized cases. It is a model example of a bilateral identity: the second inequality in the convergence annulus is controlled by the positive tail, while the first is controlled by the negative tail.

## Chapter 12

# $q$ -difference calculus and the quantum binomial

### 12.1 The $q$ -derivative

Assume  $q \neq 1$ . For  $x \neq 0$  define

$$D_q f(x) = \frac{f(x) - f(qx)}{(1 - q)x}.$$

At  $x = 0$  one uses the continuous extension when it exists.

**Proposition 12.1** (Elementary rules). *For  $n \geq 1$ ,*

$$D_q x^n = [n]_q x^{n-1}.$$

*The product rules are*

$$\begin{aligned} D_q(fg)(x) &= g(x)D_q f(x) + f(qx)D_q g(x), \\ &= f(x)D_q g(x) + g(qx)D_q f(x). \end{aligned}$$

*Moreover,*

$$D_q(f(q^r x)) = q^r (D_q f)(q^r x)$$

*for integral  $r$ .*

*Proof.* The monomial identity follows from  $(x^n - q^n x^n)/((1 - q)x) = [n]_q x^{n-1}$ . For the first product rule, insert and subtract  $f(qx)g(x)$  in  $f(x)g(x) - f(qx)g(qx)$ . For the second, insert and subtract  $f(x)g(qx)$ . The scaling rule follows directly from the definition.  $\square$

**Theorem 12.2** ( $q$ -Leibniz rule). *For  $n \in \mathbb{N}_0$ ,*

$$D_q^n(fg)(x) = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^k f)(q^{n-k}x)(D_q^{n-k} g)(x).$$

*Proof.* Induct on  $n$ . The assertion is the ordinary product at  $n = 0$ . Apply  $D_q$  to the  $k$ th term at level  $n$ . By the product and scaling rules, it contributes

$$q^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^{k+1} f)(q^{n-k}x)(D_q^{n-k} g)(x)$$

to index  $k + 1$ , and

$$\begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^k f)(q^{n-k+1}x)(D_q^{n-k+1} g)(x)$$

to index  $k$ . The total coefficient of index  $k$  is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n-k+1} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q$$

by the second  $q$ -Pascal recurrence.  $\square$

## 12.2 $q$ -Taylor and Newton interpolation

Define the  $q$ -falling power based at  $a$  by

$$(x-a)_q^m = \prod_{j=0}^{m-1} (x - q^j a), \quad (x-a)_q^0 = 1.$$

**Lemma 12.3.** For  $m \geq 1$ ,

$$D_q(x-a)_q^m = [m]_q (x-a)_q^{m-1}.$$

Consequently,

$$D_q^j(x-a)_q^m = \frac{[m]_q!}{[m-j]_q!} (x-a)_q^{m-j}$$

for  $0 \leq j \leq m$ .

*Proof.* Let  $P_m(x) = (x-a)_q^m$ . Then

$$P_m(qx) = q^m (x - q^{-1}a) \prod_{j=0}^{m-2} (x - q^j a).$$

Subtracting from  $P_m(x) = (x - q^{m-1}a) \prod_{j=0}^{m-2} (x - q^j a)$  gives

$$P_m(x) - P_m(qx) = (1 - q^m)x P_{m-1}(x).$$

Divide by  $(1-q)x$ . Iteration proves the second formula.  $\square$

**Theorem 12.4** ( $q$ -Taylor formula). Assume  $[j]_q \neq 0$  for  $1 \leq j \leq N$ . If  $f$  is a polynomial of degree at most  $N$ , then

$$f(x) = \sum_{k=0}^N \frac{(D_q^k f)(a)}{[k]_q!} (x-a)_q^k.$$

*Proof.* The polynomials  $(x-a)_q^k$ ,  $0 \leq k \leq N$ , are monic of distinct degrees and hence form a basis. Write  $f = \sum c_k (x-a)_q^k$ . Apply  $D_q^j$  and set  $x = a$ . By Lemma 12.3, every term with  $k > j$  still contains the factor  $x-a$ , terms with  $k < j$  vanish after too many derivatives, and the  $k = j$  term equals  $c_j [j]_q!$ . Thus  $c_j = (D_q^j f)(a) / [j]_q!$ .  $\square$

**Remark 12.5.** When the nodes  $a, aq, aq^2, \dots$  are distinct, this is a Newton interpolation formula on that geometric grid. In every case covered by the theorem, its basis zeros follow a multiplicative rather than an additive progression.

### 12.3 The quantum plane

**Theorem 12.6** (Noncommutative  $q$ -binomial theorem). *Let  $X, Y$  belong to an associative algebra and satisfy*

$$YX = qXY.$$

*Then*

$$(X + Y)^n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q X^{n-k} Y^k.$$

*Proof.* Induct on  $n$  and multiply on the right by  $X + Y$ . Since  $Y^k X = q^k X Y^k$ , the coefficient of  $X^{n+1-k} Y^k$  becomes

$$q^k \begin{bmatrix} n \\ k \end{bmatrix}_q + \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q$$

by the first  $q$ -Pascal recurrence. □

**Corollary 12.7** (Noncommutative  $q$ -multinomial theorem). *Let  $r \geq 1$  and let  $x_1, \dots, x_r$  be elements of an associative algebra over a commutative ring in which  $q$  is a central element, and suppose that*

$$x_j x_i = q x_i x_j \quad (1 \leq i < j \leq r).$$

*Then, for every  $n \geq 0$ ,*

$$(x_1 + \dots + x_r)^n = \sum_{\substack{n_1, \dots, n_r \geq 0 \\ n_1 + \dots + n_r = n}} \begin{bmatrix} n \\ n_1, \dots, n_r \end{bmatrix}_q x_1^{n_1} x_2^{n_2} \dots x_r^{n_r},$$

*where on the right the factors are written in increasing order of index. Equivalently, the  $q$ -multinomial coefficients are exactly the structure constants that expand powers of  $x_1 + \dots + x_r$  in the ordered monomial basis of the quantum affine  $r$ -space.*

*Proof.* Induct on  $r$ . For  $r = 1$  there is nothing to prove, and  $r = 2$  is [Theorem 12.6](#).

Let  $r \geq 2$  and set  $X = x_1$  and  $Y = x_2 + \dots + x_r$ . Because  $x_j x_1 = q x_1 x_j$  for every  $j \geq 2$  and  $q$  is central,

$$YX = \sum_{j=2}^r x_j x_1 = q \sum_{j=2}^r x_1 x_j = qXY,$$

so [Theorem 12.6](#) applies to the pair  $(X, Y)$  and gives

$$(x_1 + \dots + x_r)^n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q x_1^{n-k} (x_2 + \dots + x_r)^k.$$

Substituting  $n_1 = n - k$  and using the symmetry  $\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q$  of [Theorem 2.5](#), this reads

$$(x_1 + \dots + x_r)^n = \sum_{n_1=0}^n \begin{bmatrix} n \\ n_1 \end{bmatrix}_q x_1^{n_1} (x_2 + \dots + x_r)^{n-n_1}.$$

The elements  $x_2, \dots, x_r$  satisfy the same commutation relations among themselves, so the inductive hypothesis applies to them and gives, with  $m = n - n_1$ ,

$$(x_2 + \dots + x_r)^m = \sum_{\substack{n_2, \dots, n_r \geq 0 \\ n_2 + \dots + n_r = m}} \begin{bmatrix} m \\ n_2, \dots, n_r \end{bmatrix}_q x_2^{n_2} \dots x_r^{n_r}.$$



Since the coefficients are central, substitution produces the ordered monomials. For the multi-index  $(n_1, \dots, n_r)$ , the coefficient is

$$\begin{bmatrix} n \\ n_1 \end{bmatrix}_q \begin{bmatrix} n - n_1 \\ n_2, \dots, n_r \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_{n_1} (q; q)_{n-n_1}} \cdot \frac{(q; q)_{n-n_1}}{(q; q)_{n_2} \cdots (q; q)_{n_r}} = \begin{bmatrix} n \\ n_1, \dots, n_r \end{bmatrix}_q$$

by the definition of the  $q$ -multinomial coefficient preceding [Proposition 2.11](#); the factor  $(q; q)_{n-n_1}$  telescopes away, exactly as in the proof of that proposition. No factor was transposed at any stage after the expansion, so the monomials are produced in increasing order of index, as asserted.  $\square$

*Remark 12.8* (The same coefficient read combinatorially). Expanding  $(x_1 + \cdots + x_r)^n$  without using any relation produces the  $r^n$  words  $x_{w_1} x_{w_2} \cdots x_{w_n}$  with  $w \in \{1, \dots, r\}^n$ . Sorting such a word into increasing order uses one adjacent transposition for each inversion of  $w$ , and each transposition contributes exactly one factor  $q$ ; hence

$$x_{w_1} x_{w_2} \cdots x_{w_n} = q^{\text{inv}(w)} x_1^{n_1} \cdots x_r^{n_r},$$

where  $n_i$  is the multiplicity of the letter  $i$  in  $w$ . Comparing coefficients with [Corollary 12.7](#) recovers the multiset inversion enumerator of [Theorem 5.7](#), and conversely that theorem proves [Corollary 12.7](#) again. The quantum affine space is therefore a purely algebraic encoding of the inversion statistic, just as [Theorem 12.6](#) encodes [Theorem 5.1](#).

**Corollary 12.9** (Factorization of  $q$ -exponentials). *Assume  $|q| < 1$ . Define*

$$e_q(x) = \sum_{n=0}^{\infty} \frac{x^n}{[n]_q!} = \frac{1}{((1-q)x; q)_{\infty}}, \quad |(1-q)x| < 1,$$

and

$$E_q(x) = \sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}} x^n}{[n]_q!} = (-(1-q)x; q)_{\infty}, \quad x \in \mathbb{C}.$$

Then  $e_q(x)E_q(-x) = 1$  wherever the first series is defined. If  $YX = qXY$ , then, as an identity of formal series in total degree,

$$e_q(X + Y) = e_q(X)e_q(Y).$$

*Proof.* The product formulas and the stated convergence domains follow from Euler's identities after replacing  $z$  by  $(1-q)x$ . Their product is 1. For the noncommutative factorization, apply [Theorem 12.6](#) degree by degree and use

$$\frac{1}{[n]_q!} \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{[n-k]_q! [k]_q!}.$$

Reindex with  $r = n - k$  to obtain the Cauchy product of  $e_q(X)$  and  $e_q(Y)$ , with all  $X$  factors preceding all  $Y$  factors. Since every total degree receives only finitely many contributions, the formal rearrangement is valid.  $\square$

**Proposition 12.10** (Eigenfunction properties). *Under the hypotheses and within the convergence domains of [Corollary 12.9](#), for scalar  $\lambda$ ,*

$$D_q e_q(\lambda x) = \lambda e_q(\lambda x), \quad D_q E_q(\lambda x) = \lambda E_q(q\lambda x).$$

*Proof.* Differentiate the series term by term and use  $D_q x^n = [n]_q x^{n-1}$ .  $\square$

## 12.4 Jackson's $q$ -integral

Assume  $0 < q < 1$ . Define

$$\int_0^a f(t) d_q t = (1-q)a \sum_{n=0}^{\infty} q^n f(aq^n),$$

whenever the series converges.

**Theorem 12.11** (Fundamental theorem of  $q$ -calculus). *Let  $\Omega \subset \mathbb{C}$  satisfy  $q\Omega \subset \Omega$ , and suppose*

$$F(x) = (1-q)x \sum_{n \geq 0} q^n f(xq^n)$$

*is defined for  $x \in \Omega$ , for example because the series converges locally uniformly. Then*

$$D_q F(x) = f(x) \quad (x \in \Omega \setminus \{0\}).$$

*If  $0 \in \Omega$  and  $f$  is continuous at 0, set  $F(0) = 0$ ; then  $D_q F$  has the continuous extension  $D_q F(0) = f(0)$ .*

*Conversely, if  $a \neq 0$ ,  $G$  is defined on  $\{0, a, aq, aq^2, \dots\}$ , and  $G(aq^n) \rightarrow G(0)$ , then the Jackson sum of  $D_q G$  converges and*

$$\int_0^a D_q G(t) d_q t = G(a) - G(0).$$

*Proof.* For  $x \neq 0$ , shifting the convergent series gives

$$F(qx) = (1-q)x \sum_{n \geq 1} q^n f(xq^n).$$

Thus  $F(x) - F(qx) = (1-q)xf(x)$  and  $D_q F(x) = f(x)$ . If  $f$  is continuous at 0, this identity itself gives the asserted continuous extension.

For the converse, the  $N$ th partial Jackson sum is exactly

$$\sum_{n=0}^N (G(aq^n) - G(aq^{n+1})) = G(a) - G(aq^{N+1}).$$

Letting  $N \rightarrow \infty$  proves both convergence and the formula.  $\square$

**Corollary 12.12** (Jackson integration by parts). *Let  $a \neq 0$ . Suppose the two Jackson series below converge and*

$$f(aq^N)g(aq^N) \longrightarrow f(0)g(0).$$

*Then*

$$\int_0^a f(t) D_q g(t) d_q t = f(a)g(a) - f(0)g(0) - \int_0^a g(qt) D_q f(t) d_q t.$$

*In particular, the endpoint condition holds when  $f$  and  $g$  are continuous at 0.*

*Proof.* For every  $N$ , the product rule gives the finite identity

$$\begin{aligned} & (1-q)a \sum_{n=0}^N q^n [f(aq^n) D_q g(aq^n) + g(aq^{n+1}) D_q f(aq^n)] \\ &= \sum_{n=0}^N (f(aq^n)g(aq^n) - f(aq^{n+1})g(aq^{n+1})) \\ &= f(a)g(a) - f(aq^{N+1})g(aq^{N+1}). \end{aligned}$$

Let  $N \rightarrow \infty$  and use the assumed convergence and endpoint limit.  $\square$

**Example 12.13** (Monomials). For  $\Re(s) > -1$ ,

$$\int_0^a t^s d_q t = (1 - q)a^{s+1} \sum_{n \geq 0} q^{n(s+1)} = \frac{a^{s+1}}{[s+1]_q},$$

where  $[u]_q = (1 - q^u)/(1 - q)$  for complex  $u$ .

## Chapter 13

# The $q$ -gamma and $q$ -beta functions

Throughout this chapter  $0 < q < 1$  unless stated otherwise.

### 13.1 Definition and functional equation

**Definition 13.1.** For  $x > 0$ , define

$$\Gamma_q(x) = (1 - q)^{1-x} \frac{(q; q)_\infty}{(q^x; q)_\infty}.$$

With  $q^z = \exp(z \log q)$ , the same product quotient defines a meromorphic function of  $z \in \mathbb{C}$ ; its possible poles occur where  $q^{z+j} = 1$  for some  $j \in \mathbb{N}_0$ . Whenever complex arguments of  $\Gamma_q$  occur below, this meromorphic continuation is intended.

**Theorem 13.2** (Basic properties). For  $x > 0$ ,

$$\Gamma_q(1) = 1, \quad \Gamma_q(x+1) = [x]_q \Gamma_q(x).$$

Consequently, for  $n \in \mathbb{N}_0$ ,

$$\Gamma_q(n) = [n-1]_q!.$$

The recurrence extends as an identity of meromorphic functions of a complex argument.

*Proof.* The value at 1 is immediate. The tail identity  $(q^x; q)_\infty = (1 - q^x)(q^{x+1}; q)_\infty$  gives

$$\frac{\Gamma_q(x+1)}{\Gamma_q(x)} = (1 - q)^{-1}(1 - q^x) = [x]_q.$$

Iteration gives the integral values. The same tail identity is valid for complex  $x$ , and hence proves the meromorphic recurrence wherever both sides are finite; equality elsewhere follows in the sense of meromorphic continuation.  $\square$

**Proposition 13.3** (Logarithmic derivatives and convexity). For  $x > 0$ ,

$$\begin{aligned} \frac{d}{dx} \log \Gamma_q(x) &= -\log(1 - q) + (\log q) \sum_{n=0}^{\infty} \frac{q^{n+x}}{1 - q^{n+x}}, \\ \frac{d^2}{dx^2} \log \Gamma_q(x) &= (\log q)^2 \sum_{n=0}^{\infty} \frac{q^{n+x}}{(1 - q^{n+x})^2} > 0. \end{aligned}$$

Thus  $\Gamma_q$  is strictly log-convex.

*Proof.* Differentiate the locally uniformly convergent logarithm of the defining infinite product. A second differentiation gives the positive series.  $\square$

## 13.2 A $q$ -Bohr–Mollerup theorem

**Theorem 13.4** ( $q$ -Bohr–Mollerup characterization). *The function  $\Gamma_q$  is the unique function  $f : (0, \infty) \rightarrow (0, \infty)$  satisfying*

$$f(1) = 1, \quad f(x+1) = [x]_q f(x),$$

*and such that  $\log f$  is convex.*

*Proof.* Existence follows from [Theorem 13.2](#) and [proposition 13.3](#). For uniqueness let  $f$  have the stated properties and fix  $0 < x \leq 1$ . Convexity on the interval from  $n$  to  $n+1$  gives

$$f(n+x) \leq f(n)^{1-x} f(n+1)^x.$$

A comparison of the adjacent secant slopes on  $[n+x, n+1]$  and  $[n+1, n+2]$  gives

$$f(n+x) \geq \frac{f(n+1)}{[n+1]_q^{1-x}}.$$

Using the recurrence,

$$f(n+x) = f(x) \prod_{j=0}^{n-1} [x+j]_q, \quad f(n) = [n-1]_q!, \quad f(n+1) = [n]_q!,$$

we obtain

$$\frac{[n]_q!}{[n+1]_q^{1-x} \prod_{j=0}^{n-1} [x+j]_q} \leq f(x) \leq \frac{[n-1]_q! [n]_q^x}{\prod_{j=0}^{n-1} [x+j]_q}.$$

The ratio of the upper bound to the lower bound is

$$\left( \frac{[n+1]_q}{[n]_q} \right)^{1-x} \longrightarrow 1.$$

Hence the bounds determine  $f(x)$  uniquely for  $0 < x \leq 1$ , and the recurrence determines it on all of  $(0, \infty)$ . Since  $\Gamma_q$  satisfies the same conditions,  $f = \Gamma_q$ .  $\square$

## 13.3 Classical limit

**Theorem 13.5** (Limit to Euler's gamma function). *For every  $x > 0$ ,*

$$\lim_{q \rightarrow 1^-} \Gamma_q(x) = \Gamma(x).$$

*The convergence is locally uniform on  $(0, \infty)$ .*

*Proof.* It suffices by the functional equations to consider  $0 < x \leq 1$ . The bounds in the proof of [Theorem 13.4](#) hold for  $\Gamma_q(x)$ . Fix  $n$  and let  $q \rightarrow 1^-$ . Since  $[u]_q \rightarrow u$ , they become

$$\begin{aligned} \frac{n!}{(n+1)^{1-x} x(x+1) \cdots (x+n-1)} &\leq \liminf_{q \rightarrow 1^-} \Gamma_q(x) \\ &\leq \limsup_{q \rightarrow 1^-} \Gamma_q(x) \leq \frac{(n-1)! n^x}{x(x+1) \cdots (x+n-1)}. \end{aligned}$$

Both outer expressions tend to  $\Gamma(x)$  as  $n \rightarrow \infty$ . To see this directly, use Euler's beta integral:

$$B(x, n) = \int_0^1 t^{x-1} (1-t)^{n-1} dt = \frac{(n-1)!}{x(x+1) \cdots (x+n-1)}.$$

After  $t = u/n$ , dominated convergence gives  $n^x B(x, n) \rightarrow \int_0^\infty u^{x-1} e^{-u} du = \Gamma(x)$ . The lower expression differs from the upper by a factor tending to 1. This proves pointwise convergence. We record the uniformity argument. On a compact interval  $x \in [\delta, 1]$ , the upper outer bound can be written as

$$n^x B(x, n) = \int_0^n u^{x-1} \left(1 - \frac{u}{n}\right)_+^{n-1} du.$$

For  $n \geq 2$ , its integrand is dominated uniformly in  $x$  by  $u^{\delta-1}$  on  $(0, 1]$  and by  $e^{-u/2}$  on  $[1, \infty)$ . More explicitly, after extending the kernel by 0 for  $u > n$ ,

$$\sup_{\delta \leq x \leq 1} \left| u^{x-1} \left(1 - \frac{u}{n}\right)_+^{n-1} - u^{x-1} e^{-u} \right| \rightarrow 0$$

for every  $u > 0$ , while this supremum is bounded by an integrable multiple of  $u^{\delta-1} \mathbf{1}_{(0,1]}(u) + e^{-u/2} \mathbf{1}_{[1,\infty)}(u)$ . The dominated convergence theorem therefore gives

$$\sup_{\delta \leq x \leq 1} |n^x B(x, n) - \Gamma(x)| \rightarrow 0.$$

The lower outer bound differs by the factor  $(n/(n+1))^{1-x}$ , which also tends uniformly to 1. For fixed  $n$ , the original  $q$ -bounds converge uniformly as  $q \rightarrow 1^-$  because they contain only finitely many continuous factors. A uniform squeeze proves local uniform convergence on  $(0, 1]$ , and the recurrence extends it to arbitrary compact subsets of  $(0, \infty)$ .  $\square$

### 13.4 The Jackson $q$ -beta integral

**Definition 13.6.** For  $x, y > 0$ , define

$$B_q(x, y) = \int_0^1 t^{x-1} \frac{(qt; q)_\infty}{(q^y t; q)_\infty} d_q t.$$

**Theorem 13.7** ( $q$ -beta evaluation). For  $x, y > 0$ ,

$$B_q(x, y) = \frac{\Gamma_q(x) \Gamma_q(y)}{\Gamma_q(x+y)}.$$

*Proof.* Expand the Jackson integral:

$$\begin{aligned} B_q(x, y) &= (1-q) \sum_{n \geq 0} q^{nx} \frac{(q^{n+1}; q)_\infty}{(q^{n+y}; q)_\infty} \\ &= (1-q) \frac{(q; q)_\infty}{(q^y; q)_\infty} \sum_{n \geq 0} \frac{(q^y; q)_n}{(q; q)_n} q^{nx}. \end{aligned}$$

The infinite  $q$ -binomial theorem evaluates the sum as  $(q^{x+y}; q)_\infty / (q^x; q)_\infty$ . Thus

$$B_q(x, y) = (1-q) \frac{(q; q)_\infty (q^{x+y}; q)_\infty}{(q^x; q)_\infty (q^y; q)_\infty},$$

which is exactly the gamma quotient after substituting the definition of  $\Gamma_q$ .  $\square$

**Corollary 13.8** (Recurrences). For  $x, y > 0$ ,

$$B_q(x+1, y) = \frac{[x]_q}{[x+y]_q} B_q(x, y), \quad B_q(x, y+1) = \frac{[y]_q}{[x+y]_q} B_q(x, y).$$

*Proof.* Apply the functional equation for  $\Gamma_q$  to the quotient in [Theorem 13.7](#).  $\square$

### 13.5 Multiplication and reflection-type formulas

**Theorem 13.9** ( $q$ -multiplication formula). *For an integer  $m \geq 1$  and  $x > 0$ ,*

$$\Gamma_q(mx) \prod_{j=1}^{m-1} \Gamma_{q^m}\left(\frac{j}{m}\right) = [m]_q^{mx-1} \prod_{j=0}^{m-1} \Gamma_{q^m}\left(x + \frac{j}{m}\right).$$

*Proof.* Multiply the definitions on the right. By dissection,

$$\prod_{j=0}^{m-1} (q^{mx+j}; q^m)_\infty = (q^{mx}; q)_\infty.$$

Hence

$$\prod_{j=0}^{m-1} \Gamma_{q^m}\left(x + \frac{j}{m}\right) = (1 - q^m)^{(m+1)/2 - mx} \frac{(q^m; q^m)_\infty^m}{(q^{mx}; q)_\infty}.$$

Similarly,

$$\prod_{j=1}^{m-1} \Gamma_{q^m}\left(\frac{j}{m}\right) = (1 - q^m)^{(m-1)/2} \frac{(q^m; q^m)_\infty^m}{(q; q)_\infty}.$$

Taking the ratio and using  $[m]_q = (1 - q^m)/(1 - q)$  produces the formula.  $\square$

**Proposition 13.10** (Reflection-type product). *For  $0 < x < 1$ ,*

$$\Gamma_q(x) \Gamma_q(1 - x) = (1 - q) \frac{(q; q)_\infty^2}{(q^x, q^{1-x}; q)_\infty}.$$

*Proof.* Multiply the two defining products; the powers of  $1 - q$  combine to one.  $\square$

*Remark 13.11.* Unlike Euler's reflection formula, the right side is naturally a theta quotient rather than an elementary trigonometric function. Jacobi's triple product in the next part makes that theta structure explicit.

## **Part III**

# **Theta products and partition identities**



## Chapter 14

# Jacobi's triple product

### 14.1 A bilateral theta series

For  $z \neq 0$  and  $|q| < 1$ , define

$$\Theta(z; q) = \sum_{n \in \mathbb{Z}} q^{\binom{n}{2}} z^n.$$

The series converges locally uniformly on  $\mathbb{C}^*$  because its terms decay quadratically in both directions.

**Theorem 14.1** (Jacobi triple product). *For  $z \neq 0$  and  $|q| < 1$ ,*

$$\sum_{n \in \mathbb{Z}} q^{\binom{n}{2}} z^n = (-z, -q/z, q; q)_\infty, \quad (14.1)$$

$$\sum_{n \in \mathbb{Z}} (-1)^n q^{\binom{n}{2}} z^n = (z, q/z, q; q)_\infty. \quad (14.2)$$

*Proof.* It suffices to prove (14.1); replace  $z$  by  $-z$  for (14.2). Put

$$F(z) = (-z; q)_\infty (-q/z; q)_\infty (q; q)_\infty.$$

The products converge locally uniformly on  $\mathbb{C}^*$ , so  $F$  has a Laurent expansion  $F(z) = \sum_{n \in \mathbb{Z}} c_n z^n$ . The shift identities give

$$F(qz) = \frac{1 + z^{-1}}{1 + z} F(z) = z^{-1} F(z).$$

Comparing Laurent coefficients yields

$$c_{n+1} = q^n c_n, \quad c_n = q^{\binom{n}{2}} c_0$$

for every integer  $n$ .

It remains to find  $c_0$ . Euler's product expansions give

$$(-z; q)_\infty = \sum_{r \geq 0} \frac{q^{\binom{r}{2}} z^r}{(q; q)_r}, \quad (-q/z; q)_\infty = \sum_{s \geq 0} \frac{q^{\binom{s+1}{2}} z^{-s}}{(q; q)_s}.$$

Hence

$$c_0 = (q; q)_\infty \sum_{r \geq 0} \frac{q^{r^2}}{(q; q)_r^2}.$$

The sum is the generating function obtained by decomposing an arbitrary partition according to the side  $r$  of its Durfee square: the square contributes  $q^{r^2}$ , while the diagrams to its right and below are independent partitions with at most  $r$  parts, each having generating function  $1/(q; q)_r$ . Therefore

$$\sum_{r \geq 0} \frac{q^{r^2}}{(q; q)_r^2} = \frac{1}{(q; q)_\infty},$$

so  $c_0 = 1$ . Thus  $F(z) = \sum_{n \in \mathbb{Z}} q^{\binom{n}{2}} z^n$ .  $\square$

**Corollary 14.2** (Quasi-periodicity and zeros). *The theta function satisfies*

$$\Theta(qz; q) = z^{-1} \Theta(z; q).$$

*Its zeros are the simple points  $z = -q^m$ ,  $m \in \mathbb{Z}$ .*

*Proof.* The functional equation follows either from reindexing the bilateral series or from the proof above. The product in (14.1) vanishes exactly when  $1 + zq^j = 0$  or  $1 + q^{j+1}/z = 0$ , which together give  $z = -q^m$  for  $m \in \mathbb{Z}$ . Exactly one factor vanishes at each such point.  $\square$

## 14.2 A finite triple product

Jacobi's identity has an exact polynomial refinement. The truncated product  $(z; q)_N (q/z; q)_N$  is again a bilateral theta-like sum, with the factor  $1/(q; q)_\infty$  replaced by a central Gaussian coefficient; letting  $N \rightarrow \infty$  then reproves Theorem 14.1 from finite algebra plus a single application of dominated convergence.

**Theorem 14.3** (Finite triple product). *For  $N \in \mathbb{N}_0$  and  $z \neq 0$ ,*

$$(z; q)_N (q/z; q)_N = \sum_{k=-N}^N (-1)^k q^{\binom{k}{2}} \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q z^k. \quad (14.3)$$

Here  $\binom{k}{2}$  denotes  $k(k-1)/2$  for every integer  $k$ , negative  $k$  included; equivalently  $\binom{-m}{2} = \binom{m+1}{2}$  for  $m \geq 0$ . Both sides lie in  $\mathbb{Z}[q][z, z^{-1}]$  and (14.3) is an identity there; in particular it carries no convergence hypothesis and holds with  $q$  an indeterminate.

*Proof.* By Theorem 3.1, (3.1),

$$(z; q)_N = \sum_{i=0}^N (-1)^i q^{\binom{i}{2}} \begin{bmatrix} N \\ i \end{bmatrix}_q z^i,$$

and, replacing  $z$  by  $q/z$  in the same identity and using  $\binom{j}{2} + j = \binom{j+1}{2}$ ,

$$(q/z; q)_N = \sum_{j=0}^N (-1)^j q^{\binom{j+1}{2}} \begin{bmatrix} N \\ j \end{bmatrix}_q z^{-j}.$$

Multiply the two expansions and collect the pairs contributing to  $z^k$ , namely those with  $i = j + k$ . Since  $(-1)^{j+k}(-1)^j = (-1)^k$ , and since  $\begin{bmatrix} N \\ j+k \end{bmatrix}_q = 0$  unless  $0 \leq j+k \leq N$ , the inner sum may be extended over all  $j \in \mathbb{Z}$ :

$$(z; q)_N (q/z; q)_N = \sum_{k \in \mathbb{Z}} (-1)^k z^k \sum_{j \in \mathbb{Z}} q^{\binom{j+k}{2} + \binom{j+1}{2}} \begin{bmatrix} N \\ j+k \end{bmatrix}_q \begin{bmatrix} N \\ j \end{bmatrix}_q.$$

The two binomial exponents combine cleanly. As an identity between polynomials in  $j$  and  $k$ ,

$$\binom{j+k}{2} + \binom{j+1}{2} - \binom{k}{2} = \frac{1}{2} [(j+k)(j+k-1) + j(j+1) - k(k-1)] = j^2 + jk,$$

and being a polynomial identity it is valid for negative  $k$  as well, with the convention above. On a term with  $\begin{bmatrix} N \\ j \end{bmatrix}_q \begin{bmatrix} N \\ j+k \end{bmatrix}_q \neq 0$  one has  $0 \leq j \leq N$  and  $0 \leq j+k \leq N$ , so  $j^2 + jk = j(j+k) \geq 0$  and

no negative power of  $q$  is created; terms with  $j(j+k) < 0$  carry a vanishing Gaussian coefficient. Hence the inner sum equals

$$q^{\binom{k}{2}} \sum_{j \in \mathbb{Z}} q^{j^2+jk} \begin{bmatrix} N \\ j \end{bmatrix}_q \begin{bmatrix} N \\ j+k \end{bmatrix}_q = q^{\binom{k}{2}} \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q$$

by [Corollary 3.10](#). Therefore

$$(z; q)_N (q/z; q)_N = \sum_{k \in \mathbb{Z}} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q z^k,$$

and the summand vanishes for  $|k| > N$ , which is (14.3). Both sides are Laurent polynomials in  $z$  with coefficients in  $\mathbb{Z}[q]$  because each Gaussian coefficient is such a polynomial by [Theorem 2.5](#).  $\square$

**Example 14.4.** At  $N = 1$  the coefficients are  $\begin{bmatrix} 2 \\ 0 \end{bmatrix}_q = \begin{bmatrix} 2 \\ 2 \end{bmatrix}_q = 1$  and  $\begin{bmatrix} 2 \\ 1 \end{bmatrix}_q = 1 + q$ , so (14.3) reads

$$(1 - z)(1 - q/z) = -qz^{-1} + (1 + q) - z,$$

which is correct. The coefficient  $q^{\binom{-1}{2}} = q$  of  $z^{-1}$  shows the convention for negative  $k$  at work. At  $N = 2$  one gets

$(z; q)_2 (q/z; q)_2 = q^3 z^{-2} - q(1 + q + q^2 + q^3) z^{-1} + (1 + q + 2q^2 + q^3 + q^4) - (1 + q + q^2 + q^3) z + qz^2$ ,  
the coefficients being  $\begin{bmatrix} 4 \\ 0 \end{bmatrix}_q = \begin{bmatrix} 4 \\ 4 \end{bmatrix}_q = 1$ ,  $\begin{bmatrix} 4 \\ 1 \end{bmatrix}_q = \begin{bmatrix} 4 \\ 3 \end{bmatrix}_q = 1 + q + q^2 + q^3$  and  $\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q = 1 + q + 2q^2 + q^3 + q^4$ ,  
with the outer  $q$ -powers  $q^{\binom{-2}{2}} = q^3$ ,  $q^{\binom{-1}{2}} = q$  and  $q^{\binom{2}{2}} = q$ .

**Corollary 14.5** (Finite-to-infinite triple-product convergence). *Let  $|q| < 1$  and  $z \neq 0$ . As  $N \rightarrow \infty$ , the two sides of (14.3) converge uniformly on every closed annulus  $r \leq |z| \leq R$  with  $0 < r \leq R$ . Their common limit is  $(z; q)_\infty (q/z; q)_\infty$ ; multiplying the limiting series by  $(q; q)_\infty$  recovers Jacobi's triple product (14.2).*

*Proof.* Extend the summand of (14.3) by zero, that is, read  $\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q = 0$  for  $|k| > N$ , so that the right side is a sum over all  $k \in \mathbb{Z}$ .

A bound uniform in both  $N$  and  $k$ . By [Proposition 2.2](#),

$$\begin{bmatrix} 2N \\ N+k \end{bmatrix}_q = \frac{(q; q)_{2N}}{(q; q)_{N+k} (q; q)_{N-k}} \quad (|k| \leq N).$$

Put

$$\mu = \inf_{m \geq 0} |(q; q)_m|, \quad M = \prod_{j \geq 1} (1 + |q|^j) < \infty.$$

No factor of  $(q; q)_m$  vanishes, because  $|q| < 1$  forces  $q^j \neq 1$  for  $j \geq 1$ , and  $(q; q)_m \rightarrow (q; q)_\infty$ , which is nonzero by [Theorem 1.12](#). A convergent sequence of nonzero numbers with nonzero limit is bounded away from 0, so  $\mu > 0$ ; and  $|(q; q)_m| \leq M$  for every  $m$ . Hence

$$\left| \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q \right| \leq \frac{M}{\mu^2} \quad \text{for all } N \geq 0 \text{ and all } k \in \mathbb{Z}. \quad (14.4)$$

*A summable dominating function.* For every integer  $k$  one has  $\binom{k}{2} = \frac{1}{2}(k^2 - k) \geq \frac{1}{2}(k^2 - |k|)$ , with equality for  $k \geq 0$ , so  $|q|^{\binom{k}{2}} \leq |q|^{(k^2 - |k|)/2}$ . On the annulus  $r \leq |z| \leq R$  we have  $|z^k| \leq R^{|k|} + r^{-|k|}$ . Therefore, for all  $N$  and all  $z$  in the annulus,

$$\left| (-1)^k q^{\binom{k}{2}} \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q z^k \right| \leq g(k) := \frac{M}{\mu^2} |q|^{(k^2 - |k|)/2} (R^{|k|} + r^{-|k|}),$$

and  $\sum_{k \in \mathbb{Z}} g(k) < \infty$ , since  $|q|^{k^2/2}$  decays faster than any geometric sequence grows.

*The termwise limit.* For each fixed  $k$ , using  $(q; q)_m \rightarrow (q; q)_\infty \neq 0$  again,

$$\left[ \begin{matrix} 2N \\ N+k \end{matrix} \right]_q = \frac{(q; q)_{2N}}{(q; q)_{N+k} (q; q)_{N-k}} \xrightarrow{N \rightarrow \infty} \frac{(q; q)_\infty}{(q; q)_\infty^2} = \frac{1}{(q; q)_\infty}.$$

This is [Theorem 22.7](#) when  $k = 0$ ; the argument is unchanged for fixed  $k \neq 0$ , since  $N \pm k \rightarrow \infty$  with  $N$ . Dominated convergence for series, applied on the counting measure of  $\mathbb{Z}$  with dominating function  $g$ , therefore gives

$$\lim_{N \rightarrow \infty} \sum_{k \in \mathbb{Z}} (-1)^k q^{\binom{k}{2}} \left[ \begin{matrix} 2N \\ N+k \end{matrix} \right]_q z^k = \frac{1}{(q; q)_\infty} \sum_{k \in \mathbb{Z}} (-1)^k q^{\binom{k}{2}} z^k,$$

The convergence is moreover uniform on the annulus: given  $\varepsilon > 0$ , choose  $K$  with  $\sum_{|k| > K} g(k) < \varepsilon/3$ , which bounds the two tails simultaneously for every  $N$  and every  $z$  in the annulus because  $g$  depends on neither; the remaining  $2K + 1$  terms converge because  $\left[ \begin{matrix} 2N \\ N+k \end{matrix} \right]_q \rightarrow 1/(q; q)_\infty$  is a limit of numbers, and  $|z^k| \leq R^K + r^{-K}$  there. On the left of (14.3),  $(z; q)_N (q/z; q)_N \rightarrow (z; q)_\infty (q/z; q)_\infty$  locally uniformly on  $\mathbb{C}^*$  by [Theorem 1.12](#). Multiplying by  $(q; q)_\infty$  we obtain

$$(q, z, q/z; q)_\infty = \sum_{k \in \mathbb{Z}} (-1)^k q^{\binom{k}{2}} z^k,$$

which is (14.2). □

*Remark 14.6* (Two mechanisms for one identity). [Theorem 14.1](#) was proved above from a functional equation together with the Durfee-square evaluation of the constant Laurent coefficient. [Corollary 14.5](#) instead proves the stronger local-uniform convergence statement from the polynomial identity (14.3), whose only inputs are [Theorem 3.1](#) and [Theorem 3.8](#); Jacobi's identity is then a consequence, not a second canonical assertion. This route separates algebra from analysis completely: (14.3) holds in  $\mathbb{Z}[q][z, z^{-1}]$ , and the entire analytic content is the single limit  $\left[ \begin{matrix} 2N \\ N+k \end{matrix} \right]_q \rightarrow 1/(q; q)_\infty$ , which is the same finite-to-infinite stabilization recorded coefficientwise in [Proposition 16.6](#) and columnwise in [Theorem 22.6](#). That separation is exactly the architecture recommended in [Chapter 24](#). Note also that (14.3) is genuinely finer than (14.2): its coefficients are the central Gaussian coefficients  $\left[ \begin{matrix} 2N \\ N+k \end{matrix} \right]_q$ , which by [Theorem 2.5](#) are palindromic polynomials of degree  $(N+k)(N-k)$ , whereas in the limit every one of them collapses to the single  $k$ -independent factor  $1/(q; q)_\infty$ ; what survives of the polynomial is not its constant term but its stabilised coefficient sequence, namely the partition numbers of [Proposition 16.6](#).

### 14.3 Euler's pentagonal number theorem

**Theorem 14.7** (Pentagonal number theorem). *For  $|q| < 1$ ,*

$$(q; q)_\infty = \sum_{n \in \mathbb{Z}} (-1)^n q^{n(3n-1)/2} = 1 + \sum_{n=1}^{\infty} (-1)^n \left( q^{n(3n-1)/2} + q^{n(3n+1)/2} \right).$$

*Proof.* In (14.2), replace the base  $q$  by  $q^3$  and set  $z = q$ . The product side is

$$(q; q^3)_\infty (q^2; q^3)_\infty (q^3; q^3)_\infty = (q; q)_\infty$$

by dissection. The series exponent is

$$3 \binom{n}{2} + n = \frac{n(3n-1)}{2}.$$

Pair the terms  $n$  and  $-n$  to obtain the second form. □

**Corollary 14.8** (Euler's partition recurrence). *Let  $p(0) = 1$  and  $p(n) = 0$  for  $n < 0$ . Then for  $n \geq 1$ ,*

$$p(n) = \sum_{j=1}^{\infty} (-1)^{j-1} \left( p\left(n - \frac{j(3j-1)}{2}\right) + p\left(n - \frac{j(3j+1)}{2}\right) \right).$$

*Proof.* The generating function for partitions is  $\sum_{n \geq 0} p(n)q^n = 1/(q; q)_{\infty}$ . Multiply this by the pentagonal expansion and compare the coefficient of  $q^n$ .  $\square$

## 14.4 Jacobi's cubic identity

**Theorem 14.9** (Jacobi identity). *For  $|q| < 1$ ,*

$$(q; q)_{\infty}^3 = \sum_{n=0}^{\infty} (-1)^n (2n+1) q^{n(n+1)/2}.$$

*Proof.* Differentiate (14.2) with respect to  $z$  and set  $z = 1$ . Since

$$(z; q)_{\infty} = (1-z)(qz; q)_{\infty},$$

the derivative of the product side at  $z = 1$  is  $-(q; q)_{\infty}^3$ . On the series side it is

$$\sum_{m \in \mathbb{Z}} m (-1)^m q^{\binom{m}{2}}.$$

Pair  $m = -n$  with  $m = n+1$  for  $n \geq 0$ . Their sum is

$$-(2n+1)(-1)^n q^{n(n+1)/2}.$$

Cancelling the common minus sign proves the identity.  $\square$

## 14.5 A theta form of the $q$ -gamma reflection product

**Corollary 14.10.** *For  $0 < x < 1$ ,*

$$\Gamma_q(x) \Gamma_q(1-x) = (1-q) \frac{(q; q)_{\infty}^3}{\sum_{n \in \mathbb{Z}} (-1)^n q^{\binom{n}{2} + xn}}.$$

*Proof.* Set  $z = q^x$  in (14.2):

$$\sum_{n \in \mathbb{Z}} (-1)^n q^{\binom{n}{2} + xn} = (q^x, q^{1-x}, q; q)_{\infty}.$$

Substitute this in Proposition 13.10.  $\square$

## Chapter 15

# Theta modularity, lattice dissection, and sums of squares

This chapter collects the theta-function consequences that are not needed for the elementary development of the Jacobi triple product in [Chapter 14](#), but which reveal its modular and arithmetic force. The arguments are deliberately explicit: Poisson summation supplies modular inversion, lattice dissection supplies product identities, and unique factorization or logarithmic differentiation converts theta coefficients into divisor sums. Throughout the analytic sections,  $0 < |q| < 1$ .

### 15.1 Watson's quintuple product

**Theorem 15.1** (Watson's quintuple product). *For  $z \neq 0$ ,*

$$\begin{aligned} (q, -z, -q/z; q)_\infty (qz^2, q/z^2; q^2)_\infty \\ = \sum_{n \in \mathbb{Z}} (-1)^n q^{n(3n-1)/2} z^{3n} (1 + zq^n). \end{aligned} \quad (15.1)$$

*Proof.* By [Theorem 14.1](#),

$$\begin{aligned} (-z, -q/z, q; q)_\infty &= \sum_{r \in \mathbb{Z}} q^{r(r-1)/2} z^r, \\ (qz^2, q/z^2, q^2; q^2)_\infty &= \sum_{s \in \mathbb{Z}} (-1)^s q^{s^2} z^{2s}. \end{aligned}$$

Consequently, the left side of (15.1), multiplied by  $(q^2; q^2)_\infty$ , is the product of these two bilateral series. Normal convergence on compact annuli permits coefficient extraction. The coefficient of  $z^m$  is

$$q^{m(m-1)/2} \sum_{s \in \mathbb{Z}} (-1)^s q^{3s^2 + (1-2m)s}. \quad (15.2)$$

Apply the minus-sign form of the triple product with base  $Q = q^6$ . Quasi-periodicity gives, when  $m = 3n$ ,

$$\sum_{s \in \mathbb{Z}} (-1)^s q^{3s^2 + (1-6n)s} = (-1)^n q^{-3n^2 + n} (q^2; q^2)_\infty.$$

Thus (15.2) equals  $(-1)^n q^{n(3n-1)/2} (q^2; q^2)_\infty$ . When  $m = 3n + 1$ , the same calculation gives  $(-1)^n q^{n(3n+1)/2} (q^2; q^2)_\infty$ . When  $m = 3n + 2$ , the remaining Jacobi factor is evaluated at  $Q^{-n}$  and vanishes, because the product contains the factor  $1 - Q^0$  after an integral quasi-periodic shift.

After division by the nonzero product  $(q^2; q^2)_\infty$ , only the powers  $z^{3n}$  and  $z^{3n+1}$  remain. Their coefficients are respectively

$$(-1)^n q^{n(3n-1)/2}, \quad (-1)^n q^{n(3n+1)/2},$$

which combine to the right side of (15.1).  $\square$

*Remark 15.2* (Coefficient dissection and root-system products). The proof illustrates a mechanism that persists in higher-rank product identities. Multiplying two Jacobi factors first produces a lattice sum; fixing the exponent of  $z$  restricts that lattice to an affine slice; and the remaining one-dimensional theta factor then uses its zeros and quasi-periodicity to distinguish residue classes. In Macdonald-type root-system products, higher-dimensional lattice cosets and Weyl symmetries play analogous roles. This is a structural comparison, not an invocation or proof of a root-system identity.

## 15.2 Jacobi theta functions and the heat equation

Let  $\tau$  lie in the upper half-plane and put

$$q = e^{\pi i \tau}, \quad \zeta = e^{2\pi i z}.$$

Whenever a rational power of  $q$  occurs in the theta or eta formulas, it is defined by

$$q^\alpha := e^{\pi i \alpha \tau};$$

it is not a power obtained from an unspecified logarithm of the complex number  $q$ . Whenever  $\zeta^{1/2}$  occurs below, it means the single-valued expression  $e^{\pi i z}$ , not an unspecified square root.

**Definition 15.3** (Jacobi theta functions).

$$\vartheta_3(z | \tau) = \sum_{n \in \mathbb{Z}} e^{\pi i n^2 \tau + 2\pi i n z} = \sum_{n \in \mathbb{Z}} q^{n^2} \zeta^n, \quad (15.3)$$

$$\vartheta_4(z | \tau) = \sum_{n \in \mathbb{Z}} (-1)^n e^{\pi i n^2 \tau + 2\pi i n z}, \quad (15.4)$$

$$\vartheta_2(z | \tau) = \sum_{n \in \mathbb{Z}} e^{\pi i (n+1/2)^2 \tau + 2\pi i (n+1/2) z}. \quad (15.5)$$

These series and all their termwise derivatives converge normally on compact subsets of  $\mathbb{C} \times \{\text{Im } \tau > 0\}$ .

**Proposition 15.4** (Theta product forms). *One has*

$$\vartheta_3(z | \tau) = (q^2, -q\zeta, -q/\zeta; q^2)_\infty, \quad (15.6)$$

$$\vartheta_4(z | \tau) = (q^2, q\zeta, q/\zeta; q^2)_\infty, \quad (15.7)$$

$$\vartheta_2(z | \tau) = q^{1/4} \zeta^{1/2} (q^2, -q^2\zeta, -\zeta^{-1}; q^2)_\infty. \quad (15.8)$$

*In particular,*

$$\vartheta_2(0 | \tau) = 2q^{1/4} (q^2; q^2)_\infty (-q^2; q^2)_\infty^2, \quad (15.9)$$

$$\vartheta_3(0 | \tau) = (q^2; q^2)_\infty (-q; q^2)_\infty^2, \quad (15.10)$$

$$\vartheta_4(0 | \tau) = (q^2; q^2)_\infty (q; q^2)_\infty^2. \quad (15.11)$$

*Proof.* Apply [Theorem 14.1](#) with base  $q^2$ . The substitution  $z = -q\zeta$  there works because

$$(-1)^n (q^2)^{\binom{n}{2}} (-q\zeta)^n = q^{n^2} \zeta^n;$$

the substitution  $z = q\zeta$  gives the second formula. Finally,

$$\vartheta_2(z | \tau) = q^{1/4} \zeta^{1/2} \sum_{n \in \mathbb{Z}} q^{n(n+1)} \zeta^n,$$

and the triple product with argument  $-q^2\zeta$  gives (15.8). Set  $z = 0$  and use  $(-1; q^2)_\infty = 2(-q^2; q^2)_\infty$  to obtain the three null values.  $\square$

**Proposition 15.5** (Heat equation). *Each  $\vartheta_j$ , for  $j = 2, 3, 4$ , satisfies*

$$\frac{\partial \vartheta_j}{\partial \tau} = \frac{1}{4\pi i} \frac{\partial^2 \vartheta_j}{\partial z^2}. \quad (15.12)$$

*Proof.* For a summand with shift  $\alpha \in \{0, 1/2\}$ , differentiation in  $\tau$  multiplies by  $\pi i(n + \alpha)^2$ , whereas two differentiations in  $z$  multiply by  $-4\pi^2(n + \alpha)^2$ . Since  $-4\pi^2/(4\pi i) = \pi i$ , the identities hold term by term; normal convergence justifies both differentiations.  $\square$

### 15.3 Poisson summation and modular inversion

**Definition 15.6** (Fourier transform convention). For an integrable function  $f : \mathbb{R} \rightarrow \mathbb{C}$  and  $\xi \in \mathbb{R}$ , write

$$\widehat{f}_{2\pi}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx.$$

**Theorem 15.7** (Poisson summation). *For every Schwartz function  $f : \mathbb{R} \rightarrow \mathbb{C}$ , one has*

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{k \in \mathbb{Z}} \widehat{f}_{2\pi}(k). \quad (15.13)$$

*Proof.* The periodization  $P(x) = \sum_{n \in \mathbb{Z}} f(x + n)$  and all its derivatives converge uniformly. Thus  $P$  is smooth and 1-periodic. Its  $k$ th Fourier coefficient is

$$\begin{aligned} \int_0^1 P(x) e^{-2\pi i k x} dx &= \sum_{n \in \mathbb{Z}} \int_0^1 f(x + n) e^{-2\pi i k x} dx \\ &= \int_{\mathbb{R}} f(u) e^{-2\pi i k u} du = \widehat{f}_{2\pi}(k), \end{aligned}$$

because  $e^{2\pi i k n} = 1$ . Repeated integration by parts makes these Fourier coefficients rapidly decreasing. The Fourier inversion theorem for smooth periodic functions therefore gives an absolutely and uniformly convergent Fourier series for  $P$ . Evaluating it at  $x = 0$  proves (15.13).  $\square$

**Lemma 15.8** (Fourier transform of a complex Gaussian). *If  $\text{Im } \tau > 0$  and  $u \in \mathbb{C}$ , then, with the principal square-root branch,*

$$\int_{\mathbb{R}} e^{\pi i \tau x^2 + 2\pi i u x} dx = (-i\tau)^{-1/2} e^{-\pi i u^2 / \tau}. \quad (15.14)$$

*Proof.* First let  $\tau = it$  with  $t > 0$ , and for real  $u$  put

$$I_t(u) = \int_{\mathbb{R}} e^{-\pi t x^2 + 2\pi i u x} dx.$$

Differentiation under the integral is justified by Gaussian domination. Since

$$\frac{d}{dx} e^{-\pi t x^2} = -2\pi t x e^{-\pi t x^2},$$

integration by parts, with vanishing boundary term, gives

$$\begin{aligned} I'_t(u) &= 2\pi i \int_{\mathbb{R}} x e^{-\pi t x^2 + 2\pi i u x} dx \\ &= -\frac{i}{t} \int_{\mathbb{R}} \frac{d}{dx} (e^{-\pi t x^2}) e^{2\pi i u x} dx \\ &= -\frac{2\pi u}{t} I_t(u). \end{aligned}$$



Because the ordinary Gaussian integral gives  $I_t(0) = t^{-1/2}$ , this differential equation yields

$$I_t(u) = t^{-1/2} e^{-\pi u^2/t} \quad (u \in \mathbb{R}).$$

For  $u$  in a compact subset of  $\mathbb{C}$ , the integrand and all its  $u$ -derivatives are dominated by a constant times

$$e^{-\pi t x^2 + C|x|},$$

so the integral is entire in  $u$ . The identity theorem therefore extends the formula to every complex  $u$ .

Finally, if  $K$  is a compact subset of the upper half-plane and  $u$  is fixed, then for some  $\delta > 0$  and all  $\tau \in K$ ,

$$\left| e^{\pi i \tau x^2 + 2\pi i u x} \right| \leq e^{-\pi \delta x^2 + 2\pi |\operatorname{Im} u| |x|}.$$

The same bound with an additional polynomial factor handles every  $\tau$ -derivative. Thus the integral is holomorphic in  $\tau$ . The identity theorem extends the formula from the positive imaginary axis to the whole upper half-plane. Since  $-i\tau$  lies in the right half-plane, its principal square root is holomorphic there and is characterized by

$$(-i \cdot i t)^{1/2} = t^{1/2}.$$

□

**Theorem 15.9** (Jacobi imaginary transformation). *For  $\operatorname{Im} \tau > 0$  and  $z \in \mathbb{C}$ ,*

$$\vartheta_3\left(\frac{z}{\tau} \middle| -\frac{1}{\tau}\right) = (-i\tau)^{1/2} e^{\pi i z^2/\tau} \vartheta_3(z \mid \tau). \quad (15.15)$$

At  $z = 0$ ,

$$\vartheta_2(0 \mid -1/\tau) = (-i\tau)^{1/2} \vartheta_4(0 \mid \tau), \quad (15.16)$$

$$\vartheta_3(0 \mid -1/\tau) = (-i\tau)^{1/2} \vartheta_3(0 \mid \tau), \quad (15.17)$$

$$\vartheta_4(0 \mid -1/\tau) = (-i\tau)^{1/2} \vartheta_2(0 \mid \tau). \quad (15.18)$$

*Proof.* Apply [Theorem 15.7](#) to  $f(x) = e^{\pi i \tau x^2 + 2\pi i z x}$ . For fixed  $z$ , the Gaussian decay supplied by  $\operatorname{Im} \tau > 0$  shows that  $f$  is a Schwartz function. By [Lemma 15.8](#),

$$\widehat{f}_{2\pi}(k) = (-i\tau)^{-1/2} e^{-\pi i (z-k)^2/\tau}.$$

After factoring out  $e^{-\pi i z^2/\tau}$ , Poisson summation is exactly (15.15). For the shifted Gaussian  $f(x) = e^{\pi i \tau (x+1/2)^2}$ , translation by  $1/2$  multiplies the  $k$ th Fourier transform by  $(-1)^k$ . Poisson summation then gives

$$\vartheta_2(0 \mid \tau) = (-i\tau)^{-1/2} \vartheta_4(0 \mid -1/\tau).$$

This is (15.18); replacing  $\tau$  by  $-1/\tau$  and using the chosen branch gives (15.16). □

**Definition 15.10** (Dedekind eta function). For  $\operatorname{Im} \tau > 0$ ,

$$\eta(\tau) = e^{\pi i \tau/12} \prod_{n=1}^{\infty} (1 - e^{2\pi i n \tau}) = q^{1/12} (q^2; q^2)_{\infty}. \quad (15.19)$$

**Theorem 15.11** (Theta-eta product and eta modularity). *One has*

$$\vartheta_2(0 \mid \tau) \vartheta_3(0 \mid \tau) \vartheta_4(0 \mid \tau) = 2\eta(\tau)^3, \quad (15.20)$$

$$\eta(\tau + 1) = e^{\pi i/12} \eta(\tau), \quad (15.21)$$

$$\eta(-1/\tau) = (-i\tau)^{1/2} \eta(\tau). \quad (15.22)$$

*Proof.* Multiply the three null-value formulas in [Proposition 15.4](#). The elementary product relations

$$(-q; q^2)_\infty (q; q^2)_\infty = (q^2; q^4)_\infty, \quad (-q^2; q^2)_\infty = \frac{(q^4; q^4)_\infty}{(q^2; q^2)_\infty},$$

and  $(q^2; q^2)_\infty = (q^2; q^4)_\infty (q^4; q^4)_\infty$  reduce the product to  $2q^{1/4}(q^2; q^2)_\infty^3 = 2\eta(\tau)^3$ .

The translation formula follows immediately from the definition. Applying the three modular formulas in [Theorem 15.9](#) to the theta-eta product yields

$$\eta(-1/\tau)^3 = (-i\tau)^{3/2}\eta(\tau)^3.$$

Moreover,  $\eta$  has no zeros in the upper half-plane. Indeed,

$$\sum_{n \geq 1} |e^{2\pi i n \tau}| < \infty,$$

and none of the factors  $1 - e^{2\pi i n \tau}$  vanishes; hence the convergent infinite product in (15.19) is nonzero. Since  $-i\tau$  lies in the right half-plane, its principal square root is also holomorphic and nonzero there. The quotient

$$h(\tau) = \frac{\eta(-1/\tau)}{(-i\tau)^{1/2}\eta(\tau)}$$

is holomorphic and takes values among the cube roots of unity, hence is constant. At  $\tau = i$  it equals 1, proving (15.22) with no unresolved root of unity.  $\square$

## 15.4 Ramanujan's theta function and lattice dissection

**Definition 15.12** (Ramanujan's general theta function). For  $a, b \in \mathbb{C}$  with  $|ab| < 1$ , set

$$f(a, b) = \sum_{n \in \mathbb{Z}} a^{n(n+1)/2} b^{n(n-1)/2}. \quad (15.23)$$

Here a zeroth power is 1, including at a zero base. Pairing the terms indexed by  $n$  and  $-n$  gives the unambiguous form

$$f(a, b) = 1 + \sum_{n=1}^{\infty} (a^n + b^n)(ab)^{n(n-1)/2}.$$

This series converges normally on compact subsets of  $\{(a, b) : |ab| < 1\}$ : on such a compact set its  $n$ th paired term is bounded by  $2M^n \rho^{n(n-1)/2}$  for suitable  $M < \infty$  and  $\rho < 1$ .

**Proposition 15.13** (Ramanujan's product). For  $a, b \in \mathbb{C}$  with  $|ab| < 1$ ,

$$f(a, b) = (-a, -b, ab; ab)_\infty. \quad (15.24)$$

At base zero, the infinite symbol is interpreted by its natural extension  $(c; 0)_\infty := 1 - c$ .

*Proof.* If  $ab \neq 0$ , then both  $a$  and  $b$  are nonzero. In [Theorem 14.1](#), take the base to be  $ab$  and the argument to be  $-a$ . The  $n$ th summand becomes

$$(-1)^n (ab)^{\binom{n}{2}} (-a)^n = a^{n(n+1)/2} b^{n(n-1)/2},$$

and the product is  $(-a, -b, ab; ab)_\infty$ .

If  $ab = 0$ , the paired form of the defining series gives  $f(a, b) = 1 + a + b$ . With the stated zero-base convention, the product side is

$$(1 + a)(1 + b) = 1 + a + b + ab = 1 + a + b.$$

Thus the identity also holds in the degenerate cases.  $\square$

For  $|Q| < 1$  and  $x \in \mathbb{C}^\times$  write

$$\mathcal{T}(x; Q) = \sum_{n \in \mathbb{Z}} Q^{n^2} x^n.$$

The quadratic exponent makes this series absolutely convergent for every such  $Q$  and  $x$ .

**Theorem 15.14** (Schröter lattice dissection). *Let  $a, b$  be positive integers, let  $0 < |q| < 1$ , and let  $x, y \in \mathbb{C}^\times$ . Then*

$$\begin{aligned} \mathcal{T}(x; q^a) \mathcal{T}(y; q^b) &= \sum_{k=0}^{a+b-1} y^k q^{bk^2} \mathcal{T}(xyq^{2bk}; q^{a+b}) \\ &\quad \times \mathcal{T}(y^a x^{-b} q^{2abk}; q^{ab(a+b)}). \end{aligned} \quad (15.25)$$

*Proof.* Expand the left side over  $(n, m) \in \mathbb{Z}^2$  and partition the lattice by the unique  $k \in \{0, \dots, a+b-1\}$  for which  $m - n = k + (a+b)s$  with  $s \in \mathbb{Z}$ . In that residue class the change of variables

$$n = r - bs, \quad m = k + r + as$$

is a bijection with  $(r, s) \in \mathbb{Z}^2$ . Direct substitution gives

$$\begin{aligned} x^n y^m &= y^k (xy)^r (y^a x^{-b})^s, \\ an^2 + bm^2 &= (a+b)r^2 + 2bkr + bk^2 + ab(a+b)s^2 + 2abks. \end{aligned}$$

The  $r$ - and  $s$ -sums are therefore the two displayed theta series, and the remaining factor is  $y^k q^{bk^2}$ . Absolute convergence justifies the partition and rearrangement.  $\square$

**Remark 15.15** (Sublattice cosets and diagonalization). For fixed  $k$ , the linear part of the change of variables is

$$\begin{pmatrix} n \\ m - k \end{pmatrix} = \begin{pmatrix} 1 & -b \\ 1 & a \end{pmatrix} \begin{pmatrix} r \\ s \end{pmatrix},$$

whose determinant is  $a+b$ . Its image is the index- $(a+b)$  sublattice on which the second coordinate minus the first is divisible by  $a+b$ . Translating this image by  $(0, k)$  gives the coset  $m - n \equiv k \pmod{a+b}$ , and the values  $k = 0, \dots, a+b-1$  give all cosets. Equivalently, this is the rank-two Smith-normal-form description of the congruence map  $(n, m) \mapsto m - n \pmod{a+b}$ . At the same time, the mixed  $rs$  term in  $an^2 + bm^2$  cancels under this basis change. The remaining linear terms, which depend on the coset representative  $k$ , are absorbed into the arguments of the two theta series. Thus Schröter dissection is simultaneously a sublattice decomposition and a diagonalization of the quadratic form; this remark records the mechanism and asserts no additional identity.

## 15.5 Theta powers and sums of squares

Let  $r_d(n)$  be the number of ordered  $d$ -tuples  $(x_1, \dots, x_d) \in \mathbb{Z}^d$  with  $x_1^2 + \dots + x_d^2 = n$ . For  $d = 0$  we use the unique empty tuple, so  $r_0(0) = 1$  and  $r_0(n) = 0$  for  $n \geq 1$ .

**Proposition 15.16** (Universal sums-of-squares generating function). *For every nonnegative integer  $d$  and  $|q| < 1$ ,*

$$\left( \sum_{m \in \mathbb{Z}} q^{m^2} \right)^d = \sum_{n=0}^{\infty} r_d(n) q^n. \quad (15.26)$$

*The same identity holds coefficientwise in  $\mathbb{Z}[[q]]$ .*

*Proof.* In the  $d$ -fold Cauchy product, choosing  $q^{x_j^2}$  from the  $j$ th factor produces  $q^{x_1^2 + \dots + x_d^2}$ , and every ordered representation occurs exactly once. Analytically, absolute convergence permits the Cauchy product. Formally, a fixed coefficient receives only finitely many contributions because  $x_j^2 \leq n$  whenever the tuple contributes to  $q^n$ .  $\square$

**Warning 15.17** (Generating functions are not coefficient evaluations). The proposition identifies  $r_d(n)$  as a theta-power coefficient, but by itself gives no divisor-sum formula for that coefficient. In particular, the classical evaluations of  $r_6(n)$  and  $r_8(n)$  require additional input: one proves modularity of the relevant theta powers, determines a basis (or a dimension bound) for the appropriate modular-form space, decomposes into Eisenstein and possible cusp components, and only then compares Fourier coefficients. Such evaluations must not be inferred from the Cauchy-product argument alone. The separate two- and four-square proofs below supply their own additional arithmetic and analytic input.

Define the real Dirichlet character modulo 4 by

$$\chi_4(d) = \begin{cases} 0, & 2 \mid d, \\ 1, & d \equiv 1 \pmod{4}, \\ -1, & d \equiv 3 \pmod{4}. \end{cases}$$

**Lemma 15.18** (Gaussian unique factorization). *The ring  $\mathbb{Z}[i]$  is Euclidean for  $N(x + iy) = x^2 + y^2$ , hence is a unique-factorization domain. For every rational prime  $p$ :*

- (i)  $2 = -i(1 + i)^2$ ;
- (ii) if  $p \equiv 3 \pmod{4}$ , then  $p$  is irreducible in  $\mathbb{Z}[i]$ ;
- (iii) if  $p \equiv 1 \pmod{4}$ , then  $p = \pi\bar{\pi}$  for a Gaussian prime  $\pi$ , unique up to a unit and conjugation.

*Proof.* For nonzero  $\beta$ , round the real and imaginary parts of  $\alpha/\beta$  to obtain  $\gamma \in \mathbb{Z}[i]$ . Then

$$|\alpha/\beta - \gamma|^2 \leq \frac{1}{2},$$

and therefore

$$N(\alpha - \gamma\beta) = N(\beta) |\alpha/\beta - \gamma|^2 \leq \frac{1}{2}N(\beta) < N(\beta).$$

Thus Euclidean division holds. Descent in the norm gives factorization into irreducibles, and Bézout's identity gives Euclid's lemma, hence uniqueness up to order and the units  $\{\pm 1, \pm i\}$ .

The factorization of 2 is immediate. If  $p \equiv 3 \pmod{4}$  factored into nonunits, the norms of the factors would force  $p = x^2 + y^2$ , impossible modulo 4. Thus  $p$  is irreducible.

Now let  $p \equiv 1 \pmod{4}$  and put

$$t = \left(\frac{p-1}{2}\right)!.$$

Pairing  $j$  with  $p-j$  in Wilson's theorem gives

$$-1 \equiv (p-1)! \equiv (-1)^{(p-1)/2} \left(\left(\frac{p-1}{2}\right)!\right)^2 \equiv t^2 \pmod{p},$$

because  $(p-1)/2$  is even. Hence  $p \mid (t+i)(t-i)$  in the Gaussian integers. If  $p$  were irreducible, it would be prime and would divide one factor, which is impossible because its imaginary coordinate is  $\pm 1$ . Therefore  $p = \pi\rho$  with nonunits. Norms force  $N(\pi) = N(\rho) = p$ . An element of prime norm is irreducible, so  $\pi$  and  $\rho$  are Gaussian primes. Moreover,

$$\pi\bar{\pi} = N(\pi) = p = \pi\rho,$$

and cancellation gives  $\rho = \bar{\pi}$ . If  $p = \sigma\bar{\sigma}$  is another such factorization, uniqueness of factorization applied to  $\pi\bar{\pi} = \sigma\bar{\sigma}$  shows that  $\sigma$  is associated either to  $\pi$  or to  $\bar{\pi}$ . This proves uniqueness up to a unit and conjugation.  $\square$

**Theorem 15.19** (Jacobi's two-square theorem). *For every  $n \geq 1$ ,*

$$r_2(n) = 4 \sum_{d|n} \chi_4(d). \quad (15.27)$$

*Equivalently, if*

$$n = 2^a \prod_{j=1}^u p_j^{\alpha_j} \prod_{\ell=1}^v r_\ell^{\beta_\ell}, \quad p_j \equiv 1 \pmod{4}, \quad r_\ell \equiv 3 \pmod{4},$$

*then  $r_2(n) = 0$  if some  $\beta_\ell$  is odd, and otherwise  $r_2(n) = 4 \prod_{j=1}^u (\alpha_j + 1)$ .*

*Proof.* A representation  $n = x^2 + y^2$  is a Gaussian integer of norm  $n$ . By [Lemma 15.18](#), the prime above 2 has forced exponent up to a unit. A prime  $r_\ell \equiv 3 \pmod{4}$  is self-conjugate and contributes an even exponent to a norm, so  $\beta_\ell$  must be even and then its exponent in  $x + iy$  is forced. For  $p_j = \pi_j \bar{\pi}_j$ , the exponent of  $\pi_j$  may be any  $e_j \in \{0, \dots, \alpha_j\}$  and the exponent of  $\bar{\pi}_j$  is then  $\alpha_j - e_j$ . Finally there are four units. Unique factorization shows that these choices are exhaustive and nonduplicative, proving the product formula.

The divisor sum is multiplicative. Its local factor is 1 at 2,  $\alpha_j + 1$  at  $p_j \equiv 1 \pmod{4}$ , and  $1 - 1 + \dots + (-1)^{\beta_\ell}$  at  $r_\ell \equiv 3 \pmod{4}$ . These are exactly the factors in the product formula, proving [\(15.27\)](#).  $\square$

**Corollary 15.20** (Two-square Lambert series). *For  $|q| < 1$ ,*

$$\left( \sum_{m \in \mathbb{Z}} q^{m^2} \right)^2 = 1 + 4 \sum_{d=1}^{\infty} \chi_4(d) \frac{q^d}{1 - q^d} = 1 + 4 \sum_{j=0}^{\infty} (-1)^j \frac{q^{2j+1}}{1 - q^{2j+1}}. \quad (15.28)$$

*Proof.* By [Proposition 15.16](#) and [Theorem 15.19](#), the coefficient of  $q^n$  in the first Lambert series is  $4 \sum_{d|n} \chi_4(d) = r_2(n)$ . Since  $\chi_4$  vanishes on even integers and equals  $(-1)^j$  at  $2j + 1$ , the second form follows.  $\square$

**Theorem 15.21** (Jacobi's four-square theorem). *For every  $n \geq 1$ ,*

$$r_4(n) = 8 \sum_{\substack{d|n \\ 4 \nmid d}} d. \quad (15.29)$$

*Proof.* Fix  $x \in \mathbb{C}$  with  $0 < |x| < 1$ . Apply the minus-sign form of [Theorem 14.1](#) with base  $x$  and argument  $a^2 x$ , and multiply by  $a$ . This gives

$$(a - a^{-1}) \prod_{\nu \geq 1} (1 - a^2 x^\nu)(1 - a^{-2} x^\nu)(1 - x^\nu) = \sum_{m \in \mathbb{Z}} (-1)^m a^{2m+1} x^{m(m+1)/2}. \quad (15.30)$$

For this fixed  $x$ , both sides and their  $a$ -derivatives converge normally on compact annuli in the  $a$ -plane. As functions of  $x$ , all theta series, products, and logarithmic-derivative series used below converge locally uniformly for  $|x| < 1$ , so the subsequent differentiations, rearrangements, and parity dissections are legitimate. Differentiate at  $a = 1$  and divide by 2:

$$(x; x)_\infty^3 = \frac{1}{2} \sum_{m \in \mathbb{Z}} (-1)^m (2m + 1) x^{m(m+1)/2}. \quad (15.31)$$

After squaring [\(15.31\)](#), the summand indexed by  $(m, n)$  is

$$\frac{1}{4} (-1)^{m+n} (2m + 1)(2n + 1) x^{(m^2 + n^2 + m + n)/2}.$$

If  $m + n$  is even, put

$$m = r + s, \quad n = r - s.$$

Then

$$(2m + 1)(2n + 1) = (2r + 1)^2 - (2s)^2, \quad \frac{m^2 + n^2 + m + n}{2} = r^2 + s^2 + r.$$

If  $m + n$  is odd, put

$$m = r + s, \quad n = s - r - 1.$$

Now  $(-1)^{m+n} = -1$ , while

$$(2m + 1)(2n + 1) = (2s)^2 - (2r + 1)^2.$$

Thus the parity sign reverses this coefficient, and the exponent is again  $r^2 + s^2 + r$ . Both changes of variables are bijections from their respective parity classes to  $\mathbb{Z}^2$ , so the two classes contribute equally. Consequently,

$$(x; x)_\infty^6 = \frac{1}{2} \sum_{r, s \in \mathbb{Z}} ((2r + 1)^2 - (2s)^2) x^{r^2 + s^2 + r}. \quad (15.32)$$

Let  $D = x d/dx$  and set

$$A(x) = \sum_{s \in \mathbb{Z}} x^{s^2}, \quad B(x) = \sum_{r \in \mathbb{Z}} x^{r^2 + r}.$$

Then (15.32) is

$$(x; x)_\infty^6 = \frac{1}{2} [A(x)(1 + 4D)B(x) - 4B(x)DA(x)]. \quad (15.33)$$

The triple product gives

$$A(x) = \prod_{\nu \geq 1} (1 + x^{2\nu-1})^2 (1 - x^{2\nu}), \quad (15.34)$$

$$B(x) = 2 \prod_{\nu \geq 1} (1 + x^{2\nu})^2 (1 - x^{2\nu}). \quad (15.35)$$

Using

$$D \log(1 + x^j) = \frac{jx^j}{1 + x^j}, \quad D \log(1 - x^j) = -\frac{jx^j}{1 - x^j},$$

the two copies of  $D \log \prod_{\nu \geq 1} (1 - x^{2\nu})$  cancel, and

$$D \log B - D \log A = 2 \sum_{\nu \geq 1} \left( \frac{2\nu x^{2\nu}}{1 + x^{2\nu}} - \frac{(2\nu - 1)x^{2\nu-1}}{1 + x^{2\nu-1}} \right).$$

Since  $A(x)B(x)/2$  is the product on the first line below, (15.33) becomes

$$\begin{aligned} (x; x)_\infty^6 &= \prod_{\nu \geq 1} (1 + x^{2\nu-1})^2 (1 + x^{2\nu})^2 (1 - x^{2\nu})^2 \\ &\quad \times \left[ 1 - 8 \sum_{\nu \geq 1} \left( \frac{(2\nu - 1)x^{2\nu-1}}{1 + x^{2\nu-1}} - \frac{2\nu x^{2\nu}}{1 + x^{2\nu}} \right) \right]. \end{aligned} \quad (15.36)$$

We first verify the product identity

$$\prod_{\nu \geq 1} (1 + x^\nu)^4 (1 - x^\nu)^2 = \prod_{\nu \geq 1} (1 + x^{2\nu-1})^2 (1 + x^{2\nu})^2 (1 - x^{2\nu})^2.$$

Indeed, the quotient of the left product by the right product is

$$\left[ \prod_{\nu \geq 1} (1 - x^{4\nu-2}) \prod_{\nu \geq 1} (1 + x^{2\nu}) \right]^2 = 1,$$

because

$$\prod_{\nu \geq 1} (1 + x^{2\nu}) = \frac{(x^4; x^4)_\infty}{(x^2; x^2)_\infty} = \frac{1}{(x^2; x^4)_\infty} = \prod_{\nu \geq 1} (1 - x^{4\nu-2})^{-1}.$$

This identity therefore turns (15.36) into

$$\prod_{\nu \geq 1} \left( \frac{1 - x^\nu}{1 + x^\nu} \right)^4 = 1 - 8 \sum_{\nu \geq 1} \left( \frac{(2\nu - 1)x^{2\nu-1}}{1 + x^{2\nu-1}} - \frac{2\nu x^{2\nu}}{1 + x^{2\nu}} \right). \quad (15.37)$$

The minus-sign triple product with base  $x^2$  and argument  $x$  gives

$$\sum_{m \in \mathbb{Z}} (-1)^m x^{m^2} = (x, x, x^2; x^2)_\infty.$$

Furthermore,

$$(x, x, x^2; x^2)_\infty = (x; x^2)_\infty^2 (x^2; x^2)_\infty = \frac{(x; x)_\infty}{(-x; x)_\infty} = \prod_{\nu \geq 1} \frac{1 - x^\nu}{1 + x^\nu},$$

where the middle equality follows from

$$(x; x)_\infty = (x; x^2)_\infty (x^2; x^2)_\infty, \quad (x; x)_\infty (-x; x)_\infty = (x^2; x^2)_\infty.$$

Replace  $x$  by  $-x$  in (15.37). Since  $m^2 \equiv m \pmod{2}$ , the preceding triple-product specialization turns the left side into the fourth power of  $\sum_{m \in \mathbb{Z}} x^{m^2}$ . The right side becomes

$$1 + 8 \left[ \sum_{\substack{j \geq 1 \\ j \text{ odd}}} \frac{jx^j}{1 - x^j} + \sum_{\nu \geq 1} \frac{2\nu x^{2\nu}}{1 + x^{2\nu}} \right].$$

For the even part,

$$\frac{2\nu x^{2\nu}}{1 + x^{2\nu}} = \frac{2\nu x^{2\nu}}{1 - x^{2\nu}} - \frac{4\nu x^{4\nu}}{1 - x^{4\nu}},$$

and hence

$$\sum_{\nu \geq 1} \frac{2\nu x^{2\nu}}{1 + x^{2\nu}} = \sum_{\substack{j \geq 1 \\ 2 \nmid j}} \frac{jx^j}{1 - x^j} - \sum_{\substack{j \geq 1 \\ 4 \nmid j}} \frac{jx^j}{1 - x^j}.$$

Adding the odd-index sum leaves exactly the indices not divisible by 4. Thus

$$\left( \sum_{m \in \mathbb{Z}} x^{m^2} \right)^4 = 1 + 8 \sum_{\substack{j \geq 1 \\ 4 \nmid j}} \frac{jx^j}{1 - x^j}. \quad (15.38)$$

By Proposition 15.16, the coefficient of  $x^n$  on the left is  $r_4(n)$ ; on the right it is  $8 \sum_{d|n, 4 \nmid d} d$ . This proves (15.29).  $\square$

*Remark 15.22.* The two-square proof is arithmetic and the four-square proof analytic, but both terminate in Lambert series. This is the common mechanism: a theta product exposes geometric denominators, and coefficient extraction turns them into divisor sums.

# Chapter 16

## Partitions and product–sum identities

### 16.1 Unrestricted, distinct, and odd parts

**Definition 16.1.** A *partition* of  $n$  is a weakly decreasing finite sequence of positive integers with sum  $n$ . Let  $p(n)$  count unrestricted partitions,  $d(n)$  partitions into distinct parts, and  $o(n)$  partitions into odd parts; set  $p(0) = d(0) = o(0) = 1$  and all three counts equal to 0 for  $n < 0$ .

**Theorem 16.2** (Basic partition products). *With the notation of Definition 16.1,*

$$\begin{aligned}\sum_{n \geq 0} p(n)q^n &= \frac{1}{(q; q)_\infty}, \\ \sum_{n \geq 0} d(n)q^n &= (-q; q)_\infty, \\ \sum_{n \geq 0} o(n)q^n &= \frac{1}{(q; q^2)_\infty}.\end{aligned}$$

Moreover,  $d(n) = o(n)$  for every  $n$ .

*Proof.* For unrestricted partitions, the part  $j$  may occur  $0, 1, 2, \dots$  times and contributes the geometric factor  $(1 - q^j)^{-1}$ . For distinct partitions, the part  $j$  occurs zero or one time and contributes  $1 + q^j$ . For odd parts, retain only the odd geometric factors. Finally,

$$(-q; q)_\infty = \prod_{j \geq 1} \frac{1 - q^{2j}}{1 - q^j} = \frac{(q^2; q^2)_\infty}{(q; q)_\infty} = \frac{1}{(q; q^2)_\infty},$$

so the two generating functions agree coefficientwise. □

*Remark 16.3* (A bijection). The equality  $d(n) = o(n)$  also has a direct proof. In a partition into distinct parts, write each part uniquely as  $2^r u$  with  $u$  odd. Replace that part by  $2^r$  copies of  $u$ . Distinct powers of two ensure that the process is reversible by binary expansion of the multiplicity of every odd part.

### 16.2 Bounded partitions

**Proposition 16.4.** *The following generating functions are equal:*

$$\frac{1}{(q; q)_r} = \sum_{n \geq 0} p_{\leq r}(n)q^n,$$

where  $p_{\leq r}(n)$  counts partitions with largest part at most  $r$ , or equivalently partitions with at most  $r$  parts.



*Proof.* The product allows parts  $1, \dots, r$  with arbitrary multiplicity. Conjugation of Ferrers diagrams exchanges the condition “largest part at most  $r$ ” with “at most  $r$  parts.”  $\square$

**Corollary 16.5** (Durfee decomposition of all partitions).

$$\frac{1}{(q; q)_{\infty}} = \sum_{r=0}^{\infty} \frac{q^{r^2}}{(q; q)_r^2}.$$

*Proof.* Decompose every partition by the side  $r$  of its Durfee square. The diagrams to the right and below are independent partitions with at most  $r$  parts. This is the identity used in the proof of Jacobi’s triple product.  $\square$

### 16.3 Finite approximations and stabilization

**Proposition 16.6.** *For fixed  $N$ , the coefficient of  $q^N$  in  $\left[ \begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q$  equals  $p(N)$  whenever  $m, k \geq N$ .*

*Proof.* The Gaussian coefficient counts partitions inside a  $k \times m$  rectangle. Any partition of  $N$  has at most  $N$  parts and largest part at most  $N$ , so for  $m, k \geq N$  the rectangle restriction is vacuous.  $\square$

*Remark 16.7.* Thus  $\left[ \begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q$  converges coefficientwise to  $1/(q; q)_{\infty}$  as both sides of the rectangle tend to infinity. This finite-to-infinite passage is a recurring bridge between Gaussian polynomials and partition identities.

## Chapter 17

# Bailey pairs and the Rogers–Ramanujan identities

### 17.1 Bailey pairs

For brevity, write  $(a)_n = (a; q)_n$  in this chapter.

**Definition 17.1** (Bailey pair). Sequences  $(\alpha_n)_{n \geq 0}$  and  $(\beta_n)_{n \geq 0}$  form a *Bailey pair relative to  $a$*  if

$$\beta_n = \sum_{r=0}^n \frac{\alpha_r}{(q)_{n-r}(aq)_{n+r}} \quad (n \geq 0). \quad (17.1)$$

### 17.2 Bailey inversion

**Lemma 17.2** ( $q$ -difference annihilation). *If  $P(x)$  is a polynomial of degree less than  $N$ , then*

$$\sum_{s=0}^N (-1)^{N-s} q^{\binom{N-s}{2}} \begin{bmatrix} N \\ s \end{bmatrix}_q P(q^s) = 0.$$

*Proof.* By linearity it suffices to take  $P(x) = x^m$  with  $0 \leq m < N$ . Put  $t = N - s$  and use the finite  $q$ -binomial theorem:

$$\begin{aligned} \sum_{s=0}^N (-1)^{N-s} q^{\binom{N-s}{2}} \begin{bmatrix} N \\ s \end{bmatrix}_q q^{ms} &= q^{mN} \sum_{t=0}^N (-1)^t q^{\binom{t}{2}} \begin{bmatrix} N \\ t \end{bmatrix}_q q^{-mt} \\ &= q^{mN} (q^{-m}; q)_N = 0, \end{aligned}$$

for the product contains the factor  $1 - q^0$ . □

**Remark 17.3** (Lean counterpart and boundary). This lemma is `exact`, by `Fabius.qDifference_sum_eval2_eq_zero_of_degree_lt` in `QDifferenceAnnihilation.lean`. The formal result is stronger than the displayed field-style proof: coefficients may lie in any semiring and be mapped into an arbitrary commutative ring before evaluation,  $q$  need not be nonzero or invertible, the geometric nodes may collide, and no use of  $q^{-m}$  or division is made. Its degree comparison includes  $N = 0$  and the zero polynomial. The same module proves the scaled characteristic identity (3.6), every monomial moment (3.7), and the exact scalar-extension top-coefficient formula (3.8).

**Theorem 17.4** (Bailey inversion). *Assume  $a \neq 1$  initially. The Bailey-pair relation is equivalent to*

$$\alpha_n = \frac{1 - aq^{2n}}{1 - a} \sum_{j=0}^n \frac{(a)_{n+j}}{(q)_{n-j}} (-1)^{n-j} q^{\binom{n-j}{2}} \beta_j.$$

*At  $a = 1$ , the right side is interpreted by its removable limit.*

*Proof.* Substitute the definition of  $\beta_j$  into the proposed inverse. The coefficient of  $\alpha_r$  is

$$\frac{1 - aq^{2n}}{1 - a} \sum_{j=r}^n \frac{(a)_{n+j}}{(q)_{n-j}(q)_{j-r}(aq)_{j+r}} (-1)^{n-j} q^{\binom{n-j}{2}}.$$

If  $n = r$ , only  $j = n$  occurs, and

$$\frac{1 - aq^{2n}}{1 - a} \frac{(a)_{2n}}{(aq)_{2n}} = 1.$$

Suppose  $N = n - r > 0$  and put  $j = r + s$ . Then

$$\frac{(a)_{n+r+s}}{(aq)_{2r+s}} = (1 - a)(aq^{2r+s+1}; q)_{N-1}.$$

The coefficient becomes

$$\frac{1 - aq^{2n}}{(q)_N} \sum_{s=0}^N (-1)^{N-s} q^{\binom{N-s}{2}} \begin{bmatrix} N \\ s \end{bmatrix}_q P(q^s),$$

where  $P(x) = (aq^{2r+1}x; q)_{N-1}$  has degree  $N - 1$ . It vanishes by [Lemma 17.2](#). Thus the two triangular matrices are inverse. Since every expression is rational in  $a$  and the singularity at  $a = 1$  is removable, the limiting statement follows.  $\square$

**Corollary 17.5** (Two unit Bailey pairs). *Let  $\beta_n = \delta_{n,0}$ .*

(i) *Relative to  $a = 1$ ,*

$$\alpha_0 = 1, \quad \alpha_n = (-1)^n q^{\binom{n}{2}} (1 + q^n) \quad (n > 0).$$

(ii) *Relative to  $a = q$ ,*

$$\alpha_n = (-1)^n q^{\binom{n}{2}} [2n + 1]_q \quad (n \geq 0).$$

*Proof.* With  $\beta_j = \delta_{j,0}$ , Bailey inversion gives

$$\alpha_n = \frac{1 - aq^{2n}}{1 - a} \frac{(a)_n}{(q)_n} (-1)^n q^{\binom{n}{2}}.$$

For  $a \rightarrow 1$  and  $n > 0$ ,  $(a)_n/(1 - a) \rightarrow (q)_{n-1}$ , so the prefactor reduces to  $(1 - q^{2n})/(1 - q^n) = 1 + q^n$ . For  $a = q$ , the ratio  $(a)_n/(q)_n$  is 1 and the remaining factor is  $[2n + 1]_q$ .  $\square$

### 17.3 The limiting Bailey lemma

**Lemma 17.6** (Auxiliary finite identity). *For  $N \geq 0$ ,*

$$\sum_{k=0}^N z^k q^{k(k-1)} \begin{bmatrix} N \\ k \end{bmatrix}_q (zq^k; q)_{N-k} = 1.$$

*Proof.* Call the left side  $S_N(z)$ . Using the second  $q$ -Pascal recurrence and splitting the last factor of  $(zq^k; q)_{N+1-k}$  gives

$$S_{N+1}(z) = (1 - zq^N)S_N(z) + zq^N S_N(qz).$$

The identity is immediate for  $N = 0$ . If  $S_N$  is identically 1, the recurrence gives  $S_{N+1} = 1 - zq^N + zq^N = 1$ . Induction completes the proof.  $\square$

**Theorem 17.7** (Limiting Bailey lemma, finite form). *If  $(\alpha_n, \beta_n)$  is a Bailey pair relative to  $a$ , define*

$$\alpha'_n = a^n q^{n^2} \alpha_n, \quad \beta'_n = \sum_{j=0}^n \frac{a^j q^{j^2}}{(q)_{n-j}} \beta_j.$$

*Then  $(\alpha'_n, \beta'_n)$  is also a Bailey pair relative to  $a$ .*

*Proof.* Substitute the Bailey relation for  $\beta_j$  and interchange the finite sums. The coefficient of  $\alpha_r$  in  $\beta'_n$  is

$$a^r q^{r^2} \sum_{s=0}^N \frac{a^s q^{s^2+2rs}}{(q)_{N-s} (q)_s (aq)_{2r+s}}, \quad N = n - r.$$

Let  $A = aq^{2r+1}$ . Since

$$a^s q^{s^2+2rs} = A^s q^{s(s-1)}, \quad (aq)_{2r+s} = (aq)_{2r} (A)_s,$$

the desired Bailey coefficient  $a^r q^{r^2} / ((q)_N (aq)_{n+r})$  is equivalent, after clearing factors, to

$$\sum_{s=0}^N \begin{bmatrix} N \\ s \end{bmatrix}_q A^s q^{s(s-1)} (Aq^s)_{N-s} = 1.$$

This is [Lemma 17.6](#). □

**Corollary 17.8** (Limiting Bailey lemma, infinite form). *If the series converge absolutely, then every Bailey pair relative to  $a$  satisfies*

$$\sum_{n=0}^{\infty} a^n q^{n^2} \beta_n = \frac{1}{(aq; q)_{\infty}} \sum_{n=0}^{\infty} a^n q^{n^2} \alpha_n.$$

*Proof.* The transformed pair in [Theorem 17.7](#) satisfies

$$\beta'_N = \sum_{r=0}^N \frac{\alpha'_r}{(q)_{N-r} (aq)_{N+r}}.$$

Let  $N \rightarrow \infty$ . On the left,

$$\beta'_N \rightarrow \frac{1}{(q)_{\infty}} \sum_{j \geq 0} a^j q^{j^2} \beta_j.$$

On the right,

$$\beta'_N \rightarrow \frac{1}{(q)_{\infty} (aq)_{\infty}} \sum_{r \geq 0} a^r q^{r^2} \alpha_r.$$

Absolute convergence permits dominated convergence. Cancel  $(q)_{\infty}^{-1}$ . □

## 17.4 The Rogers–Ramanujan identities

**Theorem 17.9** (Rogers–Ramanujan). *For  $|q| < 1$ ,*

$$\sum_{n=0}^{\infty} \frac{q^{n^2}}{(q; q)_n} = \frac{1}{(q; q^5)_{\infty} (q^4; q^5)_{\infty}}, \quad (17.2)$$

$$\sum_{n=0}^{\infty} \frac{q^{n(n+1)}}{(q; q)_n} = \frac{1}{(q^2; q^5)_{\infty} (q^3; q^5)_{\infty}}. \quad (17.3)$$

*Proof.* For the first identity, begin with the unit pair relative to  $a = 1$  in [Corollary 17.5](#). Apply the finite limiting lemma once. The new pair is

$$\alpha'_n = q^{n^2} \alpha_n, \quad \beta'_n = \frac{1}{(q)_n}.$$

Applying the infinite limiting lemma to this new pair gives

$$\begin{aligned} \sum_{n \geq 0} \frac{q^{n^2}}{(q)_n} &= \frac{1}{(q)_\infty} \left( 1 + \sum_{r \geq 1} (-1)^r q^{2r^2 + \binom{r}{2}} (1 + q^r) \right) \\ &= \frac{1}{(q)_\infty} \sum_{r \in \mathbb{Z}} (-1)^r q^{(5r^2 - r)/2}. \end{aligned}$$

By Jacobi's triple product with base  $q^5$  and  $z = q^2$ ,

$$\sum_{r \in \mathbb{Z}} (-1)^r q^{(5r^2 - r)/2} = (q^2, q^3, q^5; q^5)_\infty.$$

Dissection gives

$$(q; q)_\infty = (q, q^2, q^3, q^4, q^5; q^5)_\infty,$$

which proves (17.2).

For the second identity use the unit pair relative to  $a = q$ . After one finite limiting step,

$$\alpha'_n = (-1)^n q^{n^2 + n + \binom{n}{2}} [2n + 1]_q, \quad \beta'_n = \frac{1}{(q)_n}.$$

The infinite limiting lemma gives

$$\begin{aligned} \sum_{n \geq 0} \frac{q^{n^2 + n}}{(q)_n} &= \frac{1}{(q^2; q)_\infty} \sum_{r \geq 0} q^{r^2 + r} \alpha'_r \\ &= \frac{1}{(q; q)_\infty} \sum_{r \geq 0} (-1)^r q^{(5r^2 + 3r)/2} (1 - q^{2r+1}) \\ &= \frac{1}{(q; q)_\infty} \sum_{r \in \mathbb{Z}} (-1)^r q^{(5r^2 + 3r)/2}. \end{aligned}$$

In the last step, the subtracted term indexed by  $r \geq 0$  is the term with bilateral index  $-r - 1$ . Jacobi's triple product with base  $q^5$  and  $z = q^4$  gives

$$\sum_{r \in \mathbb{Z}} (-1)^r q^{(5r^2 + 3r)/2} = (q, q^4, q^5; q^5)_\infty.$$

Divide by the fivefold dissection of  $(q; q)_\infty$  to obtain (17.3). □

*Remark 17.10* (Partition interpretation). The product side of (17.2) counts partitions into parts congruent to 1 or 4 modulo 5; the product side of (17.3) counts parts congruent to 2 or 3 modulo 5. The sum sides count partitions with successive parts differing by at least 2, with an additional smallest-part condition in the second identity. Establishing the latter combinatorial interpretations requires a separate bijective or finite-polynomial argument; the proof above instead reveals the hypergeometric engine.

## Chapter 18

# Bailey transforms and the Andrews–Gordon identities

The limiting Bailey lemma and the two Rogers–Ramanujan series–product identities already appear, with proofs, in [Chapter 17](#). This chapter does not duplicate those results. It adds the genuinely stronger nonlimiting Bailey transform, the parameter-lowering operation which moves between Bailey chains, the combinatorial and continued-fraction forms of Rogers–Ramanujan theory, and every member of the Andrews–Gordon family.

Throughout this chapter,

$$q \in \mathbb{C}, \quad 0 < |q| < 1,$$

and  $(a_1, \dots, a_s; q)_n$  abbreviates  $\prod_{\nu=1}^s (a_\nu; q)_n$ . Thus  $(q; q)_n \neq 0$ . As in [Chapter 17](#), a Bailey pair relative to  $a$  means the single canonical relation in [Definition 17.1](#), equivalently (17.1). We recall the terminology here but do not introduce a second definition.

Whenever a quotient is displayed below, its denominator is assumed nonzero. The individual statements spell out the only parameter restrictions beyond  $0 < |q| < 1$ .

### 18.1 The full Bailey lemma

**Theorem 18.1** (Full Bailey lemma). *Let  $a, \rho_1, \rho_2 \in \mathbb{C}$ , with  $\rho_1 \rho_2 \neq 0$ , and suppose that*

$$(aq; q)_m, \quad (aq/\rho_1; q)_m, \quad (aq/\rho_2; q)_m$$

*are nonzero for every  $m \geq 1$ . Let  $(\alpha, \beta)$  be a Bailey pair relative to  $a$ , and put*

$$A = \frac{aq}{\rho_1 \rho_2}.$$

*Define*

$$\alpha_n^* = \frac{(\rho_1, \rho_2; q)_n A^n}{(aq/\rho_1, aq/\rho_2; q)_n} \alpha_n, \tag{18.1}$$

$$\beta_n^* = \sum_{j=0}^n \frac{(\rho_1, \rho_2; q)_j (A; q)_{n-j} A^j}{(q; q)_{n-j} (aq/\rho_1, aq/\rho_2; q)_n} \beta_j. \tag{18.2}$$

*Then  $(\alpha^*, \beta^*)$  is a Bailey pair relative to  $a$ .*

*Proof.* Insert (17.1) into (18.2) and interchange the two finite sums. For fixed  $r$ , set

$$N = n - r, \quad j = r + k, \quad c = aq^{2r+1}.$$

The coefficient of  $\alpha_r$  is

$$\frac{(\rho_1, \rho_2; q)_r A^r}{(aq/\rho_1, aq/\rho_2; q)_n (aq; q)_{2r}} \sum_{k=0}^N \frac{(\rho_1 q^r, \rho_2 q^r; q)_k (A; q)_{N-k} A^k}{(q; q)_k (q; q)_{N-k} (c; q)_k}. \quad (18.3)$$

The finite reversal identity

$$\frac{(A; q)_{N-k}}{(q; q)_{N-k}} = \frac{(A; q)_N}{(q; q)_N} \frac{(q^{-N}; q)_k}{(q^{1-N}/A; q)_k} \left(\frac{q}{A}\right)^k \quad (18.4)$$

follows by reversing the last  $k$  factors in the two products. Consequently the sum in (18.3) equals

$$\frac{(A; q)_N}{(q; q)_N} {}_3\phi_2 \left[ \begin{matrix} \rho_1 q^r, \rho_2 q^r, q^{-N} \\ c, q^{1-N}/A \end{matrix}; q, q \right].$$

This series is balanced: if  $x = \rho_1 q^r$  and  $y = \rho_2 q^r$ , then

$$\frac{xyq^{1-N}}{c} = \frac{\rho_1 \rho_2 q^{-N}}{a} = \frac{q^{1-N}}{A}.$$

The already proved  $q$ -Pfaff–Saalschütz summation [Theorem 9.9](#) therefore evaluates the last display as

$$\frac{(aq^{r+1}/\rho_1, aq^{r+1}/\rho_2; q)_N}{(q; q)_N (c; q)_N}. \quad (18.5)$$

Now use the product splittings

$$(aq/\rho_\nu; q)_n = (aq/\rho_\nu; q)_r (aq^{r+1}/\rho_\nu; q)_N \quad (\nu = 1, 2)$$

and

$$(aq; q)_{2r} (aq^{2r+1}; q)_N = (aq; q)_{n+r}.$$

After cancellation, the coefficient (18.3) becomes

$$\frac{1}{(q; q)_{n-r} (aq; q)_{n+r}} \frac{(\rho_1, \rho_2; q)_r A^r}{(aq/\rho_1, aq/\rho_2; q)_r}.$$

The second factor is exactly the multiplier which defines  $\alpha_r^*$  from  $\alpha_r$  in (18.1). Thus, after multiplication by  $\alpha_r$ , the coefficient contribution is the Bailey kernel times  $\alpha_r^*$ . Summing over  $r$  gives

$$\beta_n^* = \sum_{r=0}^n \frac{\alpha_r^*}{(q; q)_{n-r} (aq; q)_{n+r}},$$

which is the required Bailey-pair relation.

The reversal and Pfaff–Saalschütz calculation was written on the Zariski-open set where all temporary denominators are nonzero. On that set it proves an identity of rational functions in  $a, \rho_1, \rho_2$ . Multiplying by a common denominator turns it into a polynomial identity, so it holds identically. Therefore the coefficient formula extends to every parameter choice satisfying the permanent nonvanishing assumptions in the theorem, including choices at which one of the temporary factors vanishes.  $\square$

*Remark 18.2* (The limiting theorem is not restated). Letting  $\rho_1, \rho_2$  tend to infinity in [Theorem 18.1](#), with  $n$  fixed, gives

$$\frac{(\rho_1, \rho_2; q)_n}{(aq/\rho_1, aq/\rho_2; q)_n} \left(\frac{aq}{\rho_1 \rho_2}\right)^n \longrightarrow a^n q^{n^2}.$$

The beta summands have the corresponding finite limits. The resulting pair transformation is precisely [Theorem 17.7](#); its absolutely convergent infinite consequence is already [Corollary 17.8](#). Thus the present theorem strictly supplies the two free parameters which the backbone's limiting form omits.

## 18.2 Lowering the relative parameter

**Lemma 18.3** (Parameter-lowering shift). *Let  $(\alpha, \beta)$  be a Bailey pair relative to  $a$ , and assume  $(a; q)_m \neq 0$  for every  $m \geq 1$ . Define*

$$\tilde{\alpha}_0 = \alpha_0, \quad (18.6)$$

$$\tilde{\alpha}_n = (1 - a) \left( \frac{\alpha_n}{1 - aq^{2n}} - \frac{aq^{2n-2}\alpha_{n-1}}{1 - aq^{2n-2}} \right) \quad (n \geq 1), \quad (18.7)$$

$$\tilde{\beta}_n = \beta_n. \quad (18.8)$$

*Then  $(\tilde{\alpha}, \tilde{\beta})$  is a Bailey pair relative to  $a/q$ .*

*Proof.* A Bailey pair relative to  $a/q$  has kernel

$$\frac{1}{(q; q)_{n-j}(a; q)_{n+j}}.$$

Substitute (18.6) and (18.7) into the sum against this kernel, and reindex the part containing  $\alpha_{j-1}$ . For fixed  $0 \leq r \leq n$ , the coefficient of  $\alpha_r$  is

$$\begin{aligned} & \frac{1-a}{1-aq^{2r}} \frac{1}{(q; q)_{n-r}(a; q)_{n+r}} \\ & - \frac{(1-a)aq^{2r}}{1-aq^{2r}} \frac{1}{(q; q)_{n-r-1}(a; q)_{n+r+1}}, \end{aligned}$$

where the second term is absent if  $r = n$ . Put both terms over

$$(q; q)_{n-r}(a; q)_{n+r+1}(1 - aq^{2r}).$$

Their numerator, apart from the common factor  $1 - a$ , is

$$(1 - aq^{n+r}) - aq^{2r}(1 - q^{n-r}) = 1 - aq^{2r}.$$

This formula also covers  $r = n$ , because then  $1 - q^{n-r} = 0$ . After cancellation, the coefficient is

$$\frac{1-a}{(q; q)_{n-r}(a; q)_{n+r+1}} = \frac{1}{(q; q)_{n-r}(aq; q)_{n+r}}.$$

It is exactly the original Bailey kernel. Therefore the transformed sum is  $\beta_n = \tilde{\beta}_n$ , proving the claim.  $\square$

*Remark 18.4.* Lemma 18.3 is more general than the specialization needed below. With  $a = q$ , it changes a pair relative to  $q$  into a pair relative to 1 without changing its beta sequence. Moving that single shift to different depths of a Bailey chain is what creates the different linear tails in the Andrews–Gordon sums.

## 18.3 Iterating the limiting chain

Let  $S_a$  denote exactly the finite pair transformation proved in Theorem 17.7. We use a superscript  $(t)$  for the result of applying  $S_a$   $t$  times.



**Lemma 18.5** (Iterated Bailey chain). *For every Bailey pair  $(\alpha, \beta)$  relative to  $a$  and every  $t \in \mathbb{N}_0$ , put  $(\alpha^{(0)}, \beta^{(0)}) = (\alpha, \beta)$ . For  $t \geq 1$ ,*

$$\alpha_n^{(t)} = a^{tn} q^{tn^2} \alpha_n, \quad (18.9)$$

$$\beta_n^{(t)} = \sum_{n \geq r_1 \geq \dots \geq r_t \geq 0} \frac{a^{r_1 + \dots + r_t} q^{r_1^2 + \dots + r_t^2} \beta_{r_t}}{(q; q)_{n-r_1} \prod_{\ell=1}^{t-1} (q; q)_{r_\ell - r_{\ell+1}}}. \quad (18.10)$$

*In particular, if  $\beta_n = \delta_{n,0}$ , then  $\beta_n^{(1)} = 1/(q; q)_n$ , while for  $t \geq 2$ ,*

$$\beta_n^{(t)} = \sum_{n \geq r_1 \geq \dots \geq r_{t-1} \geq 0} \frac{a^{r_1 + \dots + r_{t-1}} q^{r_1^2 + \dots + r_{t-1}^2}}{(q; q)_{n-r_1} \prod_{\ell=1}^{t-2} (q; q)_{r_\ell - r_{\ell+1}} (q; q)_{r_{t-1}}}, \quad (18.11)$$

*Proof.* The case  $t = 0$  is tautological. One application of [Theorem 17.7](#) multiplies the alpha component by  $a^n q^{n^2}$ , which proves (18.9) by induction.

For the beta component, one application of the same transformation gives (18.10) at depth 1, where the product is empty. Suppose the formula holds at some depth  $t \geq 1$ . The next application of  $S_a$  gives

$$\beta_n^{(t+1)} = \sum_{j=0}^n \frac{a^j q^{j^2}}{(q; q)_{n-j}} \beta_j^{(t)}.$$

Insert the induction hypothesis with  $j$  as its outer bound and rename  $j = r_1$ . This adds precisely the new outer inequality  $n \geq r_1$ , the factor  $a^{r_1} q^{r_1^2}$ , and the denominator  $(q; q)_{n-r_1}$ , yielding (18.10) at depth  $t + 1$ . Finally, if  $\beta_{r_t} = \delta_{r_t,0}$ , only  $r_t = 0$  survives; deleting that forced index gives (18.11).  $\square$

## 18.4 Rogers–Ramanujan partitions and the continued fraction

Write

$$G(q) = \sum_{n \geq 0} \frac{q^{n^2}}{(q; q)_n}, \quad H(q) = \sum_{n \geq 0} \frac{q^{n^2+n}}{(q; q)_n}.$$

Their product evaluations are exactly [Theorem 17.9](#), so we use rather than restate that theorem.

**Corollary 18.6** (Rogers–Ramanujan partition interpretations). *For every  $N \geq 0$ :*

- (i) *the number of partitions of  $N$  whose successive parts differ by at least 2 equals the number of partitions of  $N$  into parts congruent to 1 or 4 (mod 5);*
- (ii) *the number of partitions of  $N$  whose successive parts differ by at least 2 and whose smallest part is at least 2 equals the number of partitions of  $N$  into parts congruent to 2 or 3 (mod 5).*

*Proof.* Fix the number  $n$  of parts. A partition with successive differences at least 2 contains the staircase

$$(2n - 1, 2n - 3, \dots, 3, 1),$$

whose size is  $n^2$ . Subtracting it componentwise leaves an arbitrary weakly decreasing  $n$ -tuple of nonnegative integers, equivalently a partition with at most  $n$  parts. By [Proposition 16.4](#), its generating function is  $1/(q; q)_n$ . Thus the generating function for the partitions in part (i) is  $G(q)$ .

If the smallest part must be at least 2, subtract instead

$$(2n, 2n - 2, \dots, 4, 2),$$

whose size is  $n^2 + n$ . The remainder is again an arbitrary partition with at most  $n$  parts, so the generating function is  $H(q)$ .

By Theorem 17.9,  $G(q)$  is the reciprocal product over positive integers congruent to 1 or 4 (mod 5), and  $H(q)$  is the reciprocal product over those congruent to 2 or 3 (mod 5). Each reciprocal factor permits an arbitrary multiplicity of its part. All these products and series are holomorphic at  $q = 0$  throughout the open unit disk. The analytic identities there therefore imply, by uniqueness of Taylor expansions, equality coefficient by coefficient. This proves both claims.  $\square$

For each  $m \in \mathbb{N}_0$ , introduce the single family

$$F_m(q) = \sum_{n \geq 0} \frac{q^{n^2 + mn}}{(q; q)_n}. \quad (18.12)$$

Thus  $F_0 = G$  and  $F_1 = H$ .

**Lemma 18.7** (Rogers recurrence). *For every  $m \in \mathbb{N}_0$ ,*

$$F_m(q) = F_{m+1}(q) + q^{m+1} F_{m+2}(q). \quad (18.13)$$

*Proof.* The  $n = 0$  terms cancel when  $F_m - F_{m+1}$  is formed. For  $n \geq 1$ , use  $(1 - q^n)/(q; q)_n = 1/(q; q)_{n-1}$ :

$$\begin{aligned} F_m - F_{m+1} &= \sum_{n \geq 1} \frac{q^{n^2 + mn}(1 - q^n)}{(q; q)_n} \\ &= \sum_{n \geq 1} \frac{q^{n^2 + mn}}{(q; q)_{n-1}} \\ &= q^{m+1} \sum_{r \geq 0} \frac{q^{r^2 + (m+2)r}}{(q; q)_r} = q^{m+1} F_{m+2}, \end{aligned}$$

where  $r = n - 1$ . Absolute convergence justifies the subtraction.  $\square$

**Theorem 18.8** (Rogers–Ramanujan continued fraction). *For  $|q| < 1$ , the single-valued ratio satisfies*

$$\frac{F_1(q)}{F_0(q)} = \frac{1}{1 + \frac{q}{1 + \frac{q^2}{1 + \frac{q^3}{1 + \ddots}}}}, \quad (18.14)$$

*and the continued fraction converges locally uniformly in the unit disk. On any simply connected subdomain of  $0 < |q| < 1$ , choose a branch of  $q^{1/5}$  and define*

$$R(q) := q^{1/5} \frac{F_1(q)}{F_0(q)} = q^{1/5} \frac{(q, q^4; q^5)_\infty}{(q^2, q^3; q^5)_\infty}. \quad (18.15)$$

*Proof.* For  $N \in \mathbb{N}_0$ , consider the finite continuant

$$C_N = 1 + \frac{q}{1 + \frac{q^2}{\ddots + \frac{q^N}{1}}} = \frac{P_N}{Q_N}.$$

Its numerator and denominator satisfy

$$\begin{aligned} P_{-1} &= 1, & P_0 &= 1, & P_N &= P_{N-1} + q^N P_{N-2}, \\ Q_{-1} &= 0, & Q_0 &= 1, & Q_N &= Q_{N-1} + q^N Q_{N-2}. \end{aligned}$$

We claim

$$P_N = \sum_{j=0}^{\lfloor (N+1)/2 \rfloor} q^{j^2} \begin{bmatrix} N+1-j \\ j \end{bmatrix}_q, \quad (18.16)$$

$$Q_N = \sum_{j=0}^{\lfloor N/2 \rfloor} q^{j^2+j} \begin{bmatrix} N-j \\ j \end{bmatrix}_q. \quad (18.17)$$

The formulas give the displayed initial values. For (18.16), apply the second  $q$ -Pascal recurrence from Theorem 2.3:

$$\begin{bmatrix} N+1-j \\ j \end{bmatrix}_q = \begin{bmatrix} N-j \\ j \end{bmatrix}_q + q^{N+1-2j} \begin{bmatrix} N-j \\ j-1 \end{bmatrix}_q.$$

After multiplication by  $q^{j^2}$ , the first term sums to  $P_{N-1}$ . In the second term put  $h = j - 1$ ; its power is

$$j^2 + N + 1 - 2j = N + h^2,$$

so it sums to  $q^N P_{N-2}$ . This proves (18.16) by induction. The same argument, using

$$\begin{bmatrix} N-j \\ j \end{bmatrix}_q = \begin{bmatrix} N-1-j \\ j \end{bmatrix}_q + q^{N-2j} \begin{bmatrix} N-1-j \\ j-1 \end{bmatrix}_q,$$

proves (18.17); after  $h = j - 1$ , the second exponent is  $N + h^2 + h$ .

For fixed  $j$ ,

$$\begin{bmatrix} N+1-j \\ j \end{bmatrix}_q = \frac{(q^{N+2-2j}; q)_j}{(q; q)_j} \longrightarrow \frac{1}{(q; q)_j},$$

and the analogous limit holds in (18.17). The convergence may be passed through the sums locally uniformly. Indeed, on  $|q| \leq r < 1$ , every Gaussian coefficient occurring above is bounded in modulus by

$$C_r = \frac{(-r; r)_\infty}{(r; r)_\infty},$$

because all powers in its numerator are positive; the remaining factors  $r^{j^2}$  and  $r^{j^2+j}$  are summable. Dominated convergence gives

$$P_N \longrightarrow F_0(q), \quad Q_N \longrightarrow F_1(q)$$

locally uniformly.

The products in Theorem 17.9 show that  $F_0$  and  $F_1$  are nonzero for  $|q| < 1$ . Let  $K_0$  be a compact subset of the unit disk. Continuity and zero-freeness give

$$\eta_{K_0} = \min_{q \in K_0} |F_0(q)| > 0.$$

Uniform convergence  $P_N \rightarrow F_0$  gives  $|P_N - F_0| < \eta_{K_0}/2$  on  $K_0$  for all sufficiently large  $N$ . Those  $P_N$  are consequently zero-free on  $K_0$ , and

$$C_N^{-1} = \frac{Q_N}{P_N} \longrightarrow \frac{F_1(q)}{F_0(q)}.$$

The convergence is uniform on  $K_0$ . We define each finite truncation by the rational continuant  $Q_N/P_N$ ; where the nested expression is defined, the two agree, and the continuant supplies its analytic continuation across removable intermediate singularities. This proves (18.14) and its local uniform convergence. Finally, on any chosen fifth-root branch, taking the ratio of the two already proved product evaluations in Theorem 17.9 and multiplying by  $q^{1/5}$  gives (18.15).  $\square$

*Remark 18.9* (Schur-type finitizations). The polynomials  $P_N$  and  $Q_N$  in (18.16)–(18.17) are Schur-type finitizations of  $F_0$  and  $F_1$ . Thus the same recurrence has three compatible realizations: a finite rational continuant, an exact finite sum of Gaussian polynomials, and, after  $N \rightarrow \infty$ , a Rogers–Ramanujan series. The finite level is useful in its own right: it retains coefficient data, exhibits positivity, and permits exact induction before any analytic limit is taken. This is a reusable proof pattern. When a quotient of  $q$ -series is governed by a short recurrence, one should first look for its continuants and then for Gaussian-polynomial formulas for those continuants; convergence and the infinite identity may then emerge as the final step of finite algebra.

## 18.5 The full Andrews–Gordon family

**Theorem 18.10** (Andrews–Gordon identities). *Let  $k, i \in \mathbb{N}_0$  satisfy  $k \geq 2$  and  $1 \leq i \leq k$ . For  $n_1, \dots, n_{k-1} \geq 0$ , put*

$$N_j = n_j + n_{j+1} + \dots + n_{k-1} \quad (1 \leq j \leq k-1).$$

Then

$$\sum_{n_1, \dots, n_{k-1} \geq 0} \frac{q^{N_1^2 + \dots + N_{k-1}^2 + N_i + \dots + N_{k-1}}}{(q; q)_{n_1} \dots (q; q)_{n_{k-1}}} = \frac{(q^i, q^{2k+1-i}, q^{2k+1}; q^{2k+1})_\infty}{(q; q)_\infty}, \quad (18.18)$$

where  $N_i + \dots + N_{k-1}$  is the empty sum 0 when  $i = k$ .

*Proof.* Put

$$K = k - 1, \quad M = 2k + 1,$$

and use decreasing variables

$$r_j = N_j \quad (1 \leq j \leq K), \quad r_{K+1} = 0.$$

Then  $n_j = r_j - r_{j+1}$ , so the left side of (18.18) is

$$\sum_{r_1 \geq \dots \geq r_K \geq 0} \frac{q^{r_1^2 + \dots + r_K^2 + r_i + \dots + r_K}}{(q; q)_{r_1 - r_2} \dots (q; q)_{r_{K-1} - r_K} (q; q)_{r_K}}. \quad (18.19)$$

We derive this expression from the unit pair relative to  $q$  in Corollary 17.5.

*The case  $i = 1$ .* Apply  $S_q$  exactly  $K$  times, and then apply the infinite limiting identity Corollary 17.8, still relative to  $q$ . By Lemma 18.5, the beta side has  $K$  decreasing variables after the final outer summation. Every one has weight  $q^{r_j^2 + r_j}$ ; its denominator is exactly that in (18.19). Hence the beta side is the desired sum for  $i = 1$ .

The unit alpha sequence relative to  $q$ , from Corollary 17.5, is

$$\alpha_n^U = (-1)^n q^{\binom{n}{2}} \frac{1 - q^{2n+1}}{1 - q}.$$

The  $K$  chain steps and the final limiting identity together multiply this by  $q^{k(n^2+n)}$ . Since  $(q^2; q)_\infty = (q; q)_\infty / (1 - q)$ , the alpha side is

$$\frac{1}{(q; q)_\infty} \sum_{n \geq 0} (-1)^n q^{\{Mn^2 + (M-2)n\}/2} (1 - q^{2n+1}) = \frac{1}{(q; q)_\infty} \sum_{m \in \mathbb{Z}} (-1)^m q^{\{Mm^2 + (M-2)m\}/2}.$$

The term containing  $-q^{2n+1}$  becomes the negative-index term  $m = -n - 1$ . Apply the Jacobi triple product [Theorem 14.1](#) with base  $q^M$  and parameter  $q^{M-1}$ . The bilateral sum is

$$(q^{M-1}, q, q^M; q^M)_\infty,$$

which is the numerator in [\(18.18\)](#) for  $i = 1$ .

The cases  $2 \leq i \leq k$ . Set

$$t = k - i + 1, \quad u = i - 2.$$

Thus  $t \geq 1$ ,  $u \geq 0$ , and  $t + u = K$ . Starting with the same unit pair relative to  $q$ , perform, in order,

$$\underbrace{S_q \circ \cdots \circ S_q}_{t \text{ times}}, \quad \text{the lowering shift of [Lemma 18.3](#),} \quad \underbrace{S_1 \circ \cdots \circ S_1}_{u \text{ times}},$$

and finally apply [Corollary 17.8](#) relative to 1. The lowering shift leaves beta unchanged. The first  $S_q$ , applied to the delta beta sequence, creates the terminal factor  $1/(q; q)_n$ ; the remaining  $t - 1 = k - i$   $q$ -steps create the innermost variables, with weights  $q^{r_j^2 + r_j}$ . The  $u$  subsequent 1-steps and the final limiting sum create the outer  $u + 1 = i - 1$  variables, with weights  $q^{r_j^2}$ . Thus the linear weights occur exactly for  $r_i, \dots, r_K$ , and the beta side is [\(18.19\)](#).

It remains to compute the alpha side. After the first  $t$  steps, write

$$\gamma_n = q^{t(n^2+n)} \alpha_n^U.$$

The lowering formula gives  $\delta_0 = 1$ , while for  $n \geq 1$ ,

$$\begin{aligned} \delta_n &= (1 - q) \left( \frac{\gamma_n}{1 - q^{2n+1}} - \frac{q^{2n-1} \gamma_{n-1}}{1 - q^{2n-1}} \right) \\ &= (-1)^n q^{\{(2t+1)n^2 - (2t-1)n\}/2} \left( 1 + q^{(2t-1)n} \right). \end{aligned} \quad (18.20)$$

For completeness, the two powers before factoring in the last line are

$$t(n^2 + n) + \binom{n}{2} \quad \text{and} \quad 2n - 1 + t(n^2 - n) + \binom{n-1}{2};$$

their difference is  $(2t - 1)n$ , which proves [\(18.20\)](#).

The  $u$  transformations  $S_1$  and the final limiting identity multiply  $\delta_n$  by  $q^{(u+1)n^2} = q^{(i-1)n^2}$ . Since

$$2(t + i) - 1 = M, \quad 2t - 1 = M - 2i,$$

the alpha side becomes

$$\begin{aligned} \frac{1}{(q; q)_\infty} \left( 1 + \sum_{n \geq 1} (-1)^n q^{\{Mn^2 - (M-2i)n\}/2} \left( 1 + q^{(M-2i)n} \right) \right) \\ = \frac{1}{(q; q)_\infty} \sum_{m \in \mathbb{Z}} (-1)^m q^{\{Mm^2 - (M-2i)m\}/2}. \end{aligned}$$

Here the two positive-index terms pair  $m = n$  with  $m = -n$ . The Jacobi triple product with base  $q^M$  and parameter  $q^i$  evaluates the bilateral sum as

$$(q^i, q^{M-i}, q^M; q^M)_\infty.$$

This proves [\(18.18\)](#) in every case.

All finite Bailey transformations above are finite rearrangements. For  $|q| \leq R < 1$ ,

$$|(q; q)_m| \geq \prod_{\nu=1}^m (1 - R^\nu) \geq (R; R)_\infty,$$

uniformly in  $m$ . Since  $K = k - 1$  is fixed, every multiple-series summand is therefore bounded by a constant depending only on  $R$  and  $K$ , times  $R^{r_1^2 + \dots + r_K^2}$ . Dropping the monotonicity conditions gives the summable majorant

$$C_{R,K} \left( \sum_{n \geq 0} R^{n^2} \right)^K.$$

The alpha and bilateral sums are bounded similarly by  $C'_R R^{cn^2 - Cn}$  with  $c > 0$ . Hence all final series converge absolutely and locally uniformly, and every limiting passage used in the proof satisfies the convergence hypothesis of [Corollary 17.8](#).  $\square$

**Corollary 18.11** (The modulus-seven specialization). *For  $r, s \geq 0$ , put  $N_1 = r + s$  and  $N_2 = s$ . Then*

$$\sum_{r,s \geq 0} \frac{q^{N_1^2 + N_2^2 + N_1 + N_2}}{(q; q)_r (q; q)_s} = \frac{1}{(q^2, q^3, q^4, q^5, q^7)_\infty}, \quad (18.21)$$

$$\sum_{r,s \geq 0} \frac{q^{N_1^2 + N_2^2 + N_2}}{(q; q)_r (q; q)_s} = \frac{1}{(q, q^3, q^4, q^6, q^7)_\infty}, \quad (18.22)$$

$$\sum_{r,s \geq 0} \frac{q^{N_1^2 + N_2^2}}{(q; q)_r (q; q)_s} = \frac{1}{(q, q^2, q^5, q^6, q^7)_\infty}. \quad (18.23)$$

*Equivalently, the coefficient of  $q^n$  in the  $i$ -th sum ( $i = 1, 2, 3$ ) counts partitions of  $n$  into parts not congruent to 0 or  $\pm i \pmod{7}$ .*

*Proof.* Take  $k = 3$  and respectively  $i = 1, 2, 3$  in [Theorem 18.10](#). The three numerator products are

$$(q, q^6, q^7; q^7)_\infty, \quad (q^2, q^5, q^7; q^7)_\infty, \quad (q^3, q^4, q^7; q^7)_\infty.$$

Dissect

$$(q; q)_\infty = (q, q^2, q^3, q^4, q^5, q^6, q^7; q^7)_\infty$$

and cancel the three displayed residue classes. This gives (18.21)–(18.23). Each remaining reciprocal factor  $1/(1 - q^m)$  records an arbitrary multiplicity of the part  $m$ , and all these products and series are holomorphic at  $q = 0$  throughout the open unit disk. Since the analytic identities hold there, uniqueness of Taylor expansions proves the coefficient interpretation.  $\square$

**Remark 18.12** (Odd-modulus specialization map). The modulus in [Theorem 18.10](#) is  $2k + 1$ . Consequently  $k = 4$  gives four identities of odd modulus 9, and  $k = 13$  gives thirteen identities of odd modulus 27. These are direct specializations of the theorem just proved and require no new product argument. Modulus 14 does not arise by “substituting 14 for  $2k + 1$ ”: no integral  $k$  satisfies that equation. Its natural setting is the parity-sensitive even-modulus theory of Bressoud, whose extra parity condition is absent from the Andrews–Gordon sums above. Constructing an equally elementary Bailey-lattice route to that theory is the [even-modulus lowering open problem](#) recorded in the next chapter.

**Remark 18.13** (Logical dependencies). No external identity is being used as an unproved black box in this chapter. The full Bailey lemma reduces to the proved  $q$ -Pfaff–Saalschütz summation [Theorem 9.9](#); the continued fraction reduces to the proved  $q$ -Pascal recurrence and [Theorem 17.9](#); and the Andrews–Gordon products reduce to the proved Jacobi triple product [Theorem 14.1](#). The limiting Bailey results invoked above are precisely the finite and infinite theorems already proved in [Chapter 17](#).

## **Part IV**

# **Cyclotomic arithmetic and extended indices**

## Chapter 19

# Cyclotomic factorization and congruences

The arithmetic of Gaussian coefficients is governed by cyclotomic polynomials. This is the precise sense in which a primitive root of unity plays the role of a prime. The point of this chapter is that the essential congruences do not require mysterious cancellation: they follow from product factorization, coefficient extraction, and a careful accounting of blocks.

### 19.1 Cyclotomic valuations and carries

Let  $\Phi_d(q)$  denote the  $d$ th cyclotomic polynomial, normalized by

$$q^n - 1 = \prod_{d|n} \Phi_d(q).$$

For a nonzero polynomial  $F \in \mathbb{Z}[q]$ , write  $\nu_d(F)$  for the exponent of  $\Phi_d(q)$  in its factorization.

**Theorem 19.1** (Cyclotomic factorization of a shifted factorial). *For every  $n \geq 0$ ,*

$$(q; q)_n = (-1)^n \prod_{d=1}^n \Phi_d(q)^{\lfloor n/d \rfloor}.$$

Consequently,

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_q = \prod_{d=1}^n \Phi_d(q)^{e_d(n,k)}, \quad e_d(n,k) = \lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n-k)/d \rfloor. \quad (16.1)$$

Moreover  $e_d(n, k)$  is always either 0 or 1.

*Proof.* Since  $1 - q^j = -(q^j - 1)$  and  $q^j - 1 = \prod_{d|j} \Phi_d(q)$ ,

$$(q; q)_n = \prod_{j=1}^n (1 - q^j) = (-1)^n \prod_{j=1}^n \prod_{d|j} \Phi_d(q).$$

For fixed  $d$ , the multiples of  $d$  among  $1, \dots, n$  are  $d, 2d, \dots, \lfloor n/d \rfloor d$ , so the exponent of  $\Phi_d$  is  $\lfloor n/d \rfloor$ . Dividing the factorizations of  $(q; q)_n$ ,  $(q; q)_k$ , and  $(q; q)_{n-k}$  gives (16.1); the signs cancel.

Write  $k = ad + r$  and  $n - k = bd + s$  with  $0 \leq r, s < d$ . Then

$$e_d(n, k) = \lfloor (r + s)/d \rfloor,$$

which is 0 or 1. □



**Corollary 19.2** (Square-free cyclotomic structure). *Every Gaussian polynomial is a product of distinct cyclotomic polynomials. More precisely,*

$$\Phi_d(q) \mid \begin{bmatrix} n \\ k \end{bmatrix}_q \iff (k \bmod d) > (n \bmod d). \quad (16.2)$$

Equivalently,  $\Phi_d$  occurs exactly when adding  $k$  and  $n - k$  produces a carry in the units place in base  $d$ .

*Proof.* The first assertion follows because every exponent in (16.1) belongs to  $\{0, 1\}$ . If  $n = cd + t$  and  $k = ad + r$  with  $0 \leq r, t < d$ , then the residue of  $n - k$  is  $t - r$  when  $r \leq t$  and  $t - r + d$  when  $r > t$ . Thus the two residues add to  $t$  in the first case and to  $t + d$  in the second. The latter is precisely the carry case, and by the proof of Theorem 19.1 it is equivalent to  $e_d(n, k) = 1$ .  $\square$

**Example 19.3.** For  $n = 10$  and  $k = 4$ , the values of  $d$  for which  $4 \bmod d > 10 \bmod d$  are  $d = 5, 7, 8, 9$ . Hence

$$\begin{bmatrix} 10 \\ 4 \end{bmatrix}_q = \Phi_5(q)\Phi_7(q)\Phi_8(q)\Phi_9(q).$$

The degrees add to  $4 + 6 + 4 + 6 = 20 = 4(10 - 4)$ , as required by reciprocity.

**Corollary 19.4** (MacMahon's  $q$ -Catalan polynomial). *For  $n \geq 0$  put*

$$C_n(q) := \frac{1}{[n+1]_q} \begin{bmatrix} 2n \\ n \end{bmatrix}_q.$$

*Then  $C_n(q) \in \mathbb{Z}[q]$ ; it has degree  $n(n-1)$ , and  $C_n(1) = \frac{1}{n+1} \binom{2n}{n}$ .*

*Proof.* Since  $q^N - 1 = \prod_{d \mid N} \Phi_d(q)$  and  $\Phi_1(q) = q - 1$ ,

$$[n+1]_q = \frac{q^{n+1} - 1}{q - 1} = \prod_{\substack{d \mid n+1 \\ d > 1}} \Phi_d(q).$$

Hence  $\nu_d([n+1]_q) = 1$  exactly for those  $d > 1$  dividing  $n+1$ , and  $\nu_d([n+1]_q) = 0$  for every other  $d$ , the case  $d = 1$  included: indeed  $[n+1]_q$  takes the value  $n+1 \neq 0$  at  $q = 1$ , so  $q - 1$  does not divide it.

Let  $d > 1$  divide  $n+1$ . Then  $n \equiv -1 \pmod{d}$ , so  $n \bmod d = d - 1$ , while  $2n \equiv -2 \pmod{d}$  together with  $0 \leq d - 2 < d$  gives  $2n \bmod d = d - 2$ . Therefore

$$n \bmod d = d - 1 > d - 2 = 2n \bmod d,$$

and the carry criterion (16.2) of Corollary 19.2, applied with upper index  $2n$  and lower index  $n$ , gives  $\Phi_d(q) \mid \begin{bmatrix} 2n \\ n \end{bmatrix}_q$ , that is,  $\nu_d(\begin{bmatrix} 2n \\ n \end{bmatrix}_q) = 1$ . Combining the two paragraphs,

$$\nu_d\left(\begin{bmatrix} 2n \\ n \end{bmatrix}_q\right) \geq \nu_d([n+1]_q) \quad \text{for every } d \geq 1,$$

so  $[n+1]_q$  divides  $\begin{bmatrix} 2n \\ n \end{bmatrix}_q$  in  $\mathbb{Q}[q]$  and  $C_n \in \mathbb{Q}[q]$ . Moreover  $[n+1]_q = 1 + q + \cdots + q^n$  is monic and  $\begin{bmatrix} 2n \\ n \end{bmatrix}_q \in \mathbb{Z}[q]$  by Corollary 2.4, so division with remainder in  $\mathbb{Z}[q]$  produces a quotient in  $\mathbb{Z}[q]$ ; by uniqueness of the quotient in  $\mathbb{Q}[q]$  that quotient is  $C_n$ , whence  $C_n \in \mathbb{Z}[q]$ .

Finally,  $\deg \begin{bmatrix} 2n \\ n \end{bmatrix}_q = n \cdot n = n^2$  by Theorem 2.5(iii) and  $\deg [n+1]_q = n$ , so  $\deg C_n = n^2 - n$ . Evaluating the polynomial identity  $[n+1]_q C_n(q) = \begin{bmatrix} 2n \\ n \end{bmatrix}_q$  at  $q = 1$  and using Corollary 2.9 gives  $(n+1)C_n(1) = \binom{2n}{n}$ .  $\square$

*Remark 19.5.* The argument above establishes integrality and says nothing about signs. The coefficients of  $C_n(q)$  are in fact nonnegative —  $C_n$  enumerates the Dyck paths of semilength  $n$  by a suitable major-index statistic, for which see [33] — but that is a combinatorial theorem and not a consequence of cyclotomic divisibility. Compare Corollary 5.5, where the corresponding statistic for unrestricted monotone lattice paths is the area and the enumeration is the unconstrained  $\begin{bmatrix} 2n \\ n \end{bmatrix}_q$ .

## 19.2 The $q$ -Lucas congruence

The following theorem is often called the  $q$ -Lucas theorem. It is naturally proved by grouping a finite product into complete root-of-unity blocks.

**Lemma 19.6** (A complete root-of-unity block). *If  $\zeta$  is a primitive  $d$ th root of unity, then*

$$\prod_{j=0}^{d-1} (1 + x\zeta^j) = 1 - (-x)^d. \quad (16.3)$$

*Proof.* The roots in  $x$  of the left side are  $-\zeta^{-j}$ , which are exactly the roots of  $1 - (-x)^d$ . Both polynomials have constant term 1, so they are equal.  $\square$

**Theorem 19.7** ( $q$ -Lucas). *Let  $a, r \geq 0$ , let  $d \geq 1$ , and let  $0 \leq b, s < d$ . Then*

$$\begin{bmatrix} ad+b \\ rd+s \end{bmatrix}_q \equiv \binom{a}{r} \begin{bmatrix} b \\ s \end{bmatrix}_q \pmod{\Phi_d(q)}. \quad (16.4)$$

Here the right side is understood as 0 when  $r > a$  or  $s > b$ .

*Proof.* It suffices to prove equality after substituting an arbitrary primitive  $d$ th root  $\zeta$  for  $q$ , because  $\Phi_d$  is the minimal polynomial of  $\zeta$  over  $\mathbb{Q}$ .

The finite  $q$ -binomial theorem gives

$$\prod_{j=0}^{ad+b-1} (1 + x\zeta^j) = \sum_{h=0}^{ad+b} \zeta^{\binom{h}{2}} \begin{bmatrix} ad+b \\ h \end{bmatrix}_\zeta x^h. \quad (16.5)$$

Group the product into  $a$  complete blocks and one incomplete block. By [Lemma 19.6](#),

$$\prod_{j=0}^{ad+b-1} (1 + x\zeta^j) = (1 - (-x)^d)^a \prod_{j=0}^{b-1} (1 + x\zeta^j).$$

The coefficient of  $x^{rd+s}$  on the right is

$$\binom{a}{r} (-1)^{r(d+1)} \zeta^{\binom{s}{2}} \begin{bmatrix} b \\ s \end{bmatrix}_\zeta. \quad (16.6)$$

Comparing (16.5) and (16.6) leaves only the phase. We claim that

$$(-1)^{r(d+1)} \zeta^{\binom{s}{2} - \binom{rd+s}{2}} = 1. \quad (16.7)$$

Indeed,

$$2 \left( \binom{rd+s}{2} - \binom{s}{2} \right) = rd(rd+2s-1).$$

If  $d$  is odd,  $r(rd+2s-1)$  is even, so the exponent difference is a multiple of  $d$ , while  $r(d+1)$  is even. If  $d$  is even, then  $\zeta^{d/2} = -1$  and  $rd+2s-1$  is odd; hence the second factor in (16.7) equals  $(-1)^r$ , as does the first. Their product is 1. The coefficient comparison therefore gives (16.4) at  $q = \zeta$ , completing the proof.  $\square$

**Corollary 19.8** (Evaluation at  $q = -1$ ). *For  $0 \leq k \leq n$ ,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \Big|_{q=-1} = \begin{cases} \binom{n/2}{k/2}, & n \text{ even and } k \text{ even,} \\ 0, & n \text{ even and } k \text{ odd,} \\ \binom{(n-1)/2}{\lfloor k/2 \rfloor}, & n \text{ odd.} \end{cases}$$

*Proof.* Apply [Theorem 19.7](#) with  $d = 2$ . Write  $n = 2a + b$  and  $k = 2r + s$ , where  $b, s \in \{0, 1\}$ . Since  $\begin{bmatrix} b \\ s \end{bmatrix}_{-1}$  is 1 when  $s \leq b$  and 0 otherwise, the three cases follow.  $\square$

**Remark 19.9** (Exact Lean crosswalk). The three binomial-valued parity cases are respectively `Fabius.gaussianBinomial_neg_one_even_even`, `Fabius.gaussianBinomial_neg_one_odd_even`, and `Fabius.gaussianBinomial_neg_one_odd_odd` in `GaussianBinomialAtNegOne.lean`. Together with `Fabius.gaussianBinomial_neg_one_even_odd_eq_zero` from `QBinomialReciprocity.lean`, they prove exactly the displayed value formula over every commutative ring. All four declarations are total in  $a, b \in \mathbb{N}_0$ : when a column lies above its row, the recursive Gaussian coefficient and the ordinary binomial coefficient both use their zero extensions. This exact crosswalk is independent of the proof through  $q$ -Lucas; [Theorem 19.7](#) itself remains unformalized.

### 19.3 Cyclic sieving for subsets

A triple  $(X, C, F(q))$ , where a finite cyclic group  $C = \langle c \rangle$  acts on a finite set  $X$  and  $F(q) \in \mathbb{Z}[q]$ , exhibits the *cyclic sieving phenomenon* when

$$\# \text{Fix}(c^j) = F(\omega^j)$$

for every  $j$ , where  $\omega$  is a primitive  $|C|$ th root of unity.

**Theorem 19.10** (Cyclic sieving for  $k$ -subsets). *Let  $X$  be the set of  $k$ -element subsets of  $\mathbb{Z}/n\mathbb{Z}$ , and let  $C_n$  act by translation. Then*

$$\left( X, C_n, \begin{bmatrix} n \\ k \end{bmatrix}_q \right)$$

*exhibits the cyclic sieving phenomenon.*

*Proof.* Fix  $j$ , and let  $d$  be the order of translation by  $j$ . Its orbits on  $\mathbb{Z}/n\mathbb{Z}$  all have size  $d$ , and there are  $n/d$  such orbits. A subset is fixed precisely when it is a union of these orbits. Thus

$$\# \text{Fix}(c^j) = \begin{cases} \begin{pmatrix} n/d \\ k/d \end{pmatrix}, & d \mid k, \\ 0, & d \nmid k. \end{cases} \quad (16.8)$$

The number  $\omega^j$  is a primitive  $d$ th root. Write  $n = (n/d)d$  and  $k = rd + s$  with  $0 \leq s < d$ . By  $q$ -Lucas,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \Big|_{q=\omega^j} = \begin{pmatrix} n/d \\ r \end{pmatrix} \begin{bmatrix} 0 \\ s \end{bmatrix}_{\omega^j}.$$

This is 0 if  $s > 0$ , and it is  $\begin{pmatrix} n/d \\ k/d \end{pmatrix}$  if  $s = 0$ , exactly as in (16.8).  $\square$

### 19.4 A $q$ -Babbage congruence modulo a cyclotomic square

The modulus  $\Phi_n(q)^2$  records not only the value at a primitive  $n$ th root, but also first-order contact there. The next congruence is a particularly transparent example. Its block proof is also a useful template for stronger  $q$ -congruences; see Andrews [3] and Straub [34].

**Theorem 19.11** ( $q$ -Babbage congruence). *For integers  $a \geq b \geq 0$  and  $n \geq 1$ ,*

$$\begin{bmatrix} an \\ bn \end{bmatrix}_q \equiv \begin{bmatrix} a \\ b \end{bmatrix}_{q^{n^2}} \pmod{\Phi_n(q)^2}. \quad (16.9)$$

*Proof.* Split the finite product into  $a$  consecutive blocks of length  $n$ :

$$\prod_{j=0}^{an-1} (1 + xq^j) = \prod_{i=0}^{a-1} \prod_{j=0}^{n-1} (1 + xq^{in+j}).$$

Apply the finite  $q$ -binomial theorem to each block. Comparing the coefficient of  $x^{bn}$  gives

$$q^{\binom{bn}{2}} \begin{bmatrix} an \\ bn \end{bmatrix}_q = \sum_{\substack{0 \leq k_0, \dots, k_{a-1} \leq n \\ k_0 + \dots + k_{a-1} = bn}} q^{\sum_i \binom{k_i}{2} + n \sum_i i k_i} \prod_{i=0}^{a-1} \begin{bmatrix} n \\ k_i \end{bmatrix}_q. \quad (16.10)$$

If  $0 < k_i < n$ , then  $\Phi_n(q)$  divides  $\begin{bmatrix} n \\ k_i \end{bmatrix}_q$  by [Corollary 19.2](#). A tuple contributing to (16.10) cannot have exactly one such intermediate entry: all other entries would be 0 or  $n$ , making the sum of the  $k_i$  nonzero modulo  $n$ . Therefore every tuple containing an intermediate entry contributes a multiple of  $\Phi_n(q)^2$ .

Modulo  $\Phi_n(q)^2$ , only tuples with each  $k_i \in \{0, n\}$  survive. Such a tuple is specified by a  $b$ -element subset  $S \subseteq \{0, 1, \dots, a-1\}$ . After division by  $q^{\binom{bn}{2}}$ , its exponent is

$$n^2 \sum_{i \in S} i + b \binom{n}{2} - \binom{bn}{2} = n^2 \left( \sum_{i \in S} i - \binom{b}{2} \right).$$

Hence

$$\begin{bmatrix} an \\ bn \end{bmatrix}_q \equiv \sum_{\substack{S \subseteq \{0, \dots, a-1\} \\ |S|=b}} (q^{n^2})^{\sum_{i \in S} i - \binom{b}{2}} \pmod{\Phi_n(q)^2}.$$

The subset form of the finite  $q$ -binomial theorem identifies the sum as  $\begin{bmatrix} a \\ b \end{bmatrix}_{q^{n^2}}$ . □

*Remark 19.12* (Why the square appears). The decisive observation in the proof is combinatorial: because the target degree is a multiple of the block length, nontrivial block choices occur in pairs. Each nontrivial block supplies one factor  $\Phi_n(q)$ , so two of them supply the square. Congruences modulo  $\Phi_n^3$  require genuinely finer first- and second-order information; the  $q$ -Ljunggren congruences of Straub and subsequent refinements belong to that deeper layer.

**Corollary 19.13** (Primitive-root value and derivative). *Let  $\zeta$  be a primitive  $n$ th root of unity. Then*

$$\begin{bmatrix} an \\ bn \end{bmatrix}_\zeta = \binom{a}{b},$$

*and the derivatives at  $q = \zeta$  of the two sides of (16.9) are equal.*

*Proof.* The first assertion follows either from  $q$ -Lucas or from (16.9), because  $\zeta^{n^2} = 1$ . If a polynomial is divisible by  $\Phi_n^2$ , it has a zero of order at least two at every primitive  $n$ th root, so both its value and first derivative vanish there. □

## Chapter 20

# Negative upper indices and geometric Newton interpolation

Ordinary binomial coefficients acquire a useful extension to negative upper arguments. Gaussian coefficients do as well, but the powers of  $q$  are essential: omitting them destroys the Pascal recurrences. The same algebra leads naturally to interpolation on a geometric grid.

### 20.1 Generalized Gaussian coefficients

**Definition 20.1.** For an integer  $N$  and  $k \geq 0$ , define

$$\begin{bmatrix} N \\ k \end{bmatrix}_q := \frac{(q^{N-k+1}; q)_k}{(q; q)_k}. \quad (17.1)$$

For  $N \geq 0$  this agrees with the usual Gaussian coefficient, including the value 0 when  $k > N$ .

**Proposition 20.2** (Negative-index reflection). For  $m \geq 1$  and  $k \geq 0$ ,

$$\begin{bmatrix} -m \\ k \end{bmatrix}_q = (-1)^k q^{-mk - \binom{k}{2}} \begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q. \quad (17.2)$$

*Proof.* From the definition,

$$\begin{aligned} (q^{-m-k+1}; q)_k &= \prod_{j=0}^{k-1} (1 - q^{-(m+k-1-j)}) \\ &= (-1)^k q^{-\sum_{j=0}^{k-1} (m+k-1-j)} \prod_{j=0}^{k-1} (1 - q^{m+k-1-j}) \\ &= (-1)^k q^{-mk - \binom{k}{2}} (q^m; q)_k. \end{aligned}$$

After division by  $(q; q)_k$ , the final ratio is  $\begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q$ . □

**Theorem 20.3** (Pascal recurrences for all integral upper indices). For every integer  $N$  and every  $k \geq 1$ ,

$$\begin{bmatrix} N \\ k \end{bmatrix}_q = \begin{bmatrix} N-1 \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} N-1 \\ k \end{bmatrix}_q, \quad (17.3)$$

$$= q^{N-k} \begin{bmatrix} N-1 \\ k-1 \end{bmatrix}_q + \begin{bmatrix} N-1 \\ k \end{bmatrix}_q. \quad (17.4)$$

Both formulas are identities in  $\mathbb{Q}(q)$ .

*Proof.* Put the two terms on the right of (17.3) over the common denominator  $(q; q)_k$ . Since

$$\left[ \begin{matrix} N-1 \\ k-1 \end{matrix} \right]_q = \frac{(q^{N-k+1}; q)_{k-1}(1-q^k)}{(q; q)_k}$$

the numerator of the right side is

$$\begin{aligned} & (q^{N-k+1}; q)_{k-1} \left[ (1-q^k) + q^k(1-q^{N-k}) \right] \\ &= (q^{N-k+1}; q)_{k-1}(1-q^N) = (q^{N-k+1}; q)_k, \end{aligned}$$

which proves (17.3). For (17.4), the corresponding bracket is

$$q^{N-k}(1-q^k) + (1-q^{N-k}) = 1 - q^N.$$

The same common-denominator calculation applies.  $\square$

**Corollary 20.4** (Reciprocal product series). *For  $m \geq 1$  and  $|z| < 1$ ,*

$$\frac{1}{(z; q)_m} = \sum_{k=0}^{\infty} \left[ \begin{matrix} m+k-1 \\ k \end{matrix} \right]_q z^k = \sum_{k=0}^{\infty} (-1)^k q^{mk+\binom{k}{2}} \left[ \begin{matrix} -m \\ k \end{matrix} \right]_q z^k. \quad (17.5)$$

*The first equality also holds in the formal power-series ring  $\mathbb{Z}[q][[z]]$ .*

*Proof.* The reciprocal finite  $q$ -binomial theorem, already proved in Theorem 3.6, gives the first expansion. Substitute (17.2) to obtain the second. Formal validity follows because each coefficient of  $z^k$  is a polynomial and multiplication by the finite product  $(z; q)_m$  can be checked coefficientwise.  $\square$

## 20.2 An arbitrary upper parameter

When a branch of  $q^\alpha$  has been fixed, it is often convenient to define

$$\left[ \begin{matrix} \alpha \\ k \end{matrix} \right]_q = \frac{(q^{\alpha-k+1}; q)_k}{(q; q)_k}. \quad (17.6)$$

The proofs of the Pascal recurrences are purely algebraic in  $q^\alpha$  and therefore remain valid. A related, and usually more analytic, expansion is the following.

**Theorem 20.5** (Generalized upper-parameter series). *For  $|q| < 1$ ,  $|z| < 1$ , and any parameter  $\alpha$  for which the factors are defined,*

$$\frac{(q^{\alpha+1}z; q)_\infty}{(z; q)_\infty} = \sum_{k=0}^{\infty} \left[ \begin{matrix} \alpha+k \\ k \end{matrix} \right]_q z^k. \quad (17.7)$$

*Proof.* Apply the infinite  $q$ -binomial theorem with  $a = q^{\alpha+1}$ :

$$\frac{(q^{\alpha+1}z; q)_\infty}{(z; q)_\infty} = \sum_{k \geq 0} \frac{(q^{\alpha+1}; q)_k}{(q; q)_k} z^k.$$

The coefficient is exactly  $\left[ \begin{matrix} \alpha+k \\ k \end{matrix} \right]_q$  by (17.6).  $\square$

### 20.3 Two complex arguments and generalized binomial series

The extension in [Definition 1.13](#) also permits both arguments of a Gaussian coefficient to be complex. Throughout this section take  $0 < q < 1$  and set  $q^z = \exp(z \log q)$ , using the real logarithm of  $q$ .

**Definition 20.6.** For  $\alpha, \beta \in \mathbb{C}$ , define

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q := \frac{(q^{\beta+1}, q^{\alpha-\beta+1}; q)_\infty}{(q, q^{\alpha+1}; q)_\infty}, \quad (17.8)$$

whenever the denominator is nonzero.

**Theorem 20.7** (Gamma representation and Pascal laws). *Wherever the expressions are defined,*

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q = \frac{\Gamma_q(\alpha + 1)}{\Gamma_q(\beta + 1)\Gamma_q(\alpha - \beta + 1)}, \quad (17.9)$$

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q = \begin{bmatrix} \alpha \\ \alpha - \beta \end{bmatrix}_q, \quad (17.10)$$

$$\begin{bmatrix} \alpha + 1 \\ \beta \end{bmatrix}_q = q^\beta \begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q + \begin{bmatrix} \alpha \\ \beta - 1 \end{bmatrix}_q, \quad (17.11)$$

$$= \begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q + q^{\alpha+1-\beta} \begin{bmatrix} \alpha \\ \beta - 1 \end{bmatrix}_q. \quad (17.12)$$

If  $\beta = k \in \mathbb{N}_0$ , this definition agrees with [\(17.6\)](#).

*Proof.* Substitution of the product definition of  $\Gamma_q$  proves [\(17.9\)](#); all powers of  $1 - q$  cancel. Symmetry is immediate either from that quotient or directly from [\(17.8\)](#). For  $\beta = k \in \mathbb{N}_0$ , tail cancellation gives

$$\frac{(q^{k+1}; q)_\infty}{(q; q)_\infty} = \frac{1}{(q; q)_k}, \quad \frac{(q^{\alpha-k+1}; q)_\infty}{(q^{\alpha+1}; q)_\infty} = (q^{\alpha-k+1}; q)_k,$$

which is exactly [\(17.6\)](#).

For the first Pascal law, divide every term by  $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q$ . The gamma functional equation gives

$$\frac{\begin{bmatrix} \alpha+1 \\ \beta \end{bmatrix}_q}{\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q} = \frac{[\alpha + 1]_q}{[\alpha - \beta + 1]_q}, \quad \frac{\begin{bmatrix} \alpha \\ \beta-1 \end{bmatrix}_q}{\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q} = \frac{[\beta]_q}{[\alpha - \beta + 1]_q}.$$

The identity  $[\alpha + 1]_q = q^\beta [\alpha - \beta + 1]_q + [\beta]_q$  proves [\(17.11\)](#). The alternative identity  $[\alpha + 1]_q = [\alpha - \beta + 1]_q + q^{\alpha+1-\beta} [\beta]_q$  proves [\(17.12\)](#).  $\square$

**Corollary 20.8** (Classical generalized binomial). *If  $\alpha > \beta > -1$  are real, then*

$$\lim_{q \rightarrow 1^-} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q = \frac{\Gamma(\alpha + 1)}{\Gamma(\beta + 1)\Gamma(\alpha - \beta + 1)}.$$

*Proof.* Apply the locally uniform  $q$ -gamma limit in [Theorem 13.5](#) to [\(17.9\)](#).  $\square$

**Theorem 20.9** (Generalized finite and reciprocal  $q$ -binomial theorems). *Let  $0 < q < 1$ . If  $|zq^\alpha| < 1$ , then*

$$(z; q)_\alpha = \sum_{k=0}^{\infty} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} \alpha \\ k \end{bmatrix}_q z^k. \quad (17.13)$$

If  $|z| < 1$ , then

$$\frac{1}{(z; q)_\alpha} = \sum_{k=0}^{\infty} \begin{bmatrix} \alpha + k - 1 \\ k \end{bmatrix}_q z^k. \quad (17.14)$$

For a nonnegative integral  $\alpha$ , the first series terminates and becomes Euler's finite theorem; in that case the restriction  $|zq^\alpha| < 1$  can be removed because both sides are polynomials. For a positive integral  $\alpha$ , the second becomes the reciprocal finite theorem.

*Proof.* Apply the infinite  $q$ -binomial theorem with  $a = q^{-\alpha}$  and argument  $zq^\alpha$ :

$$\frac{(z; q)_\infty}{(zq^\alpha; q)_\infty} = \sum_{k \geq 0} \frac{(q^{-\alpha}; q)_k}{(q; q)_k} (zq^\alpha)^k.$$

The left side is  $(z; q)_\alpha$ . Reversing the finite numerator product gives

$$\frac{(q^{-\alpha}; q)_k}{(q; q)_k} q^{\alpha k} = (-1)^k q^{\binom{k}{2}} \begin{bmatrix} \alpha \\ k \end{bmatrix}_q,$$

which proves (17.13).

For the reciprocal formula, apply the infinite theorem with  $a = q^\alpha$  and argument  $z$ :

$$\frac{(q^\alpha z; q)_\infty}{(z; q)_\infty} = \sum_{k \geq 0} \frac{(q^\alpha; q)_k}{(q; q)_k} z^k.$$

The left side is  $(z; q)_\alpha^{-1}$  and the coefficient is  $\begin{bmatrix} \alpha + k - 1 \\ k \end{bmatrix}_q$ . The integral specializations follow from the consistency results already proved.  $\square$

## 20.4 Newton interpolation on the grid $1, q, q^2, \dots$

The usual Newton basis is adapted to an arbitrary sequence of distinct nodes. At geometric nodes its basis polynomials are  $q$ -Pochhammer symbols.

**Theorem 20.10** (Geometric Newton interpolation). *Assume  $q^i \neq q^j$  whenever  $0 \leq i < j \leq n$ . For arbitrary values  $y_0, \dots, y_n$  there is a unique polynomial  $P$  of degree at most  $n$  satisfying*

$$P(q^j) = y_j \quad (0 \leq j \leq n).$$

*It has the Newton expansion*

$$P(x) = \sum_{k=0}^n c_k \prod_{r=0}^{k-1} (x - q^r) = \sum_{k=0}^n c_k (-1)^k q^{\binom{k}{2}} (x; q^{-1})_k, \quad (17.15)$$

where

$$c_k = \sum_{j=0}^k \frac{y_j}{\prod_{\substack{0 \leq r \leq k \\ r \neq j}} (q^j - q^r)} \quad (17.16)$$

$$= \sum_{j=0}^k \frac{(-1)^j q^{-jk + \binom{j+1}{2}}}{(q; q)_j (q; q)_{k-j}} y_j. \quad (17.17)$$



*Proof.* For distinct nodes  $x_0, \dots, x_n$ , Lagrange interpolation gives existence and uniqueness. The coefficient of the Newton basis polynomial  $\prod_{r=0}^{k-1} (x - x_r)$  is the divided difference

$$y[x_0, \dots, x_k] = \sum_{j=0}^k \frac{y_j}{\prod_{r \neq j} (x_j - x_r)};$$

this follows by comparing the coefficient of  $x^k$  in the Lagrange interpolant through the first  $k+1$  nodes. Taking  $x_j = q^j$  proves (17.16).

It remains to simplify the denominator. Splitting the factors below and above  $j$  gives

$$\begin{aligned} \prod_{\substack{0 \leq r \leq k \\ r \neq j}} (q^j - q^r) &= \left[ \prod_{r=0}^{j-1} -q^r (1 - q^{j-r}) \right] \left[ \prod_{r=j+1}^k q^j (1 - q^{r-j}) \right] \\ &= (-1)^j q^{\binom{j}{2} + j(k-j)} (q; q)_j (q; q)_{k-j} \\ &= (-1)^j q^{jk - \binom{j+1}{2}} (q; q)_j (q; q)_{k-j}. \end{aligned}$$

Taking reciprocals yields (17.17). Finally,

$$\prod_{r=0}^{k-1} (x - q^r) = (-1)^k q^{\binom{k}{2}} \prod_{r=0}^{k-1} (1 - xq^{-r}) = (-1)^k q^{\binom{k}{2}} (x; q^{-1})_k,$$

which proves the second form of (17.15).  $\square$

**Corollary 20.11** (Triangular reconstruction). *The coefficients in (17.15) can be recovered successively by*

$$c_0 = y_0, \quad c_k = \frac{y_k - \sum_{r=0}^{k-1} c_r \prod_{j=0}^{r-1} (q^k - q^j)}{\prod_{j=0}^{k-1} (q^k - q^j)}. \quad (17.18)$$

*Proof.* Evaluate (17.15) at  $x = q^k$ . Every term with index greater than  $k$  vanishes, while the coefficient multiplying  $c_k$  is nonzero by the distinct-node hypothesis. Solving for  $c_k$  gives the formula.  $\square$

*Remark 20.12.* Kupersmidt's  $q$ -Newton viewpoint emphasizes that the classical binomial theorem, Newton interpolation, and  $q$ -difference calculus are not separate constructions: each is controlled by a basis whose successive factors form a geometric progression. The finite formula above is algebraic and needs no convergence hypothesis; infinite Newton series require additional growth conditions on the data.

## **Part V**

# **Central sums, limits, and effective methods**

## Chapter 21

# Central $q$ -binomial sums and Lambert series

Central Gaussian coefficients frequently become tractable only after a seemingly extraneous factor is inserted. The elementary identity in the next lemma shows why the factor  $(-q; q)_{2k}$  is natural: it converts a central coefficient at base  $q^2$  into a single quotient of shifted factorials. This chapter develops both a regular  $q$ -beta evaluation and a deleted-singularity identity proved recently by Chern, Dilcher, and Jiu [7].

### 21.1 The central-factor reduction

**Lemma 21.1.** *For every  $k \geq 0$ ,*

$$\left[ \begin{matrix} 2k \\ k \end{matrix} \right]_{q^2} \frac{1}{(-q; q)_{2k}} = \frac{(q; q^2)_k}{(q^2; q^2)_k}. \quad (18.1)$$

*Proof.* The elementary product identity

$$(q; q)_{2k}(-q; q)_{2k} = (q^2; q^2)_{2k}$$

gives

$$\begin{aligned} \left[ \begin{matrix} 2k \\ k \end{matrix} \right]_{q^2} \frac{1}{(-q; q)_{2k}} &= \frac{(q^2; q^2)_{2k}}{(q^2; q^2)_k^2} \frac{(q; q)_{2k}}{(q^2; q^2)_{2k}} \\ &= \frac{(q; q^2)_k (q^2; q^2)_k}{(q^2; q^2)_k^2}, \end{aligned}$$

which is the claimed quotient. □

For a complex parameter  $z$  and  $0 < q < 1$ , we use the analytic extension

$$[z]_q = \frac{1 - q^z}{1 - q},$$

with the real logarithm in  $q^z = \exp(z \log q)$ .

## 21.2 A $q$ -beta evaluation

**Theorem 21.2** (A regular central  $q$ -binomial sum). *Let  $0 < q < 1$ , and let  $\alpha \in \mathbb{C}$  be such that none of the denominators below vanishes. Then*

$$\begin{aligned} \sum_{k=0}^{\infty} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [2k+1+\alpha]_q} \\ = \frac{\Gamma_{q^2}(3/2) \Gamma_{q^2}((\alpha+1)/2)}{\Gamma_{q^2}((\alpha+2)/2)}. \end{aligned} \quad (18.2)$$

In particular, for real  $\alpha$  it is enough to exclude  $\alpha = -1, -3, -5, \dots$

*Proof.* By Lemma 21.1 and the telescoping quotient

$$\frac{1}{[2k+1+\alpha]_q} = \frac{1}{[\alpha+1]_q} \frac{(q^{\alpha+1}; q^2)_k}{(q^{\alpha+3}; q^2)_k},$$

the left side is

$$\frac{1}{[\alpha+1]_q} {}_2\phi_1 \left[ \begin{matrix} q, q^{\alpha+1} \\ q^{\alpha+3} \end{matrix}; q^2, q \right].$$

The  $q$ -Gauss sum from Theorem 9.6, with base  $q^2$ , gives

$$\frac{1}{[\alpha+1]_q} \frac{(q^{\alpha+2}, q^2; q^2)_{\infty}}{(q^{\alpha+3}, q; q^2)_{\infty}}. \quad (18.3)$$

Now

$$(q^{\alpha+1}; q^2)_{\infty} = (1 - q^{\alpha+1})(q^{\alpha+3}; q^2)_{\infty}, \quad (q; q^2)_{\infty} = (1 - q)(q^3; q^2)_{\infty}.$$

Using these two relations to simplify (18.3), we obtain

$$\frac{(q^2; q^2)_{\infty} (q^{\alpha+2}; q^2)_{\infty}}{(q^3; q^2)_{\infty} (q^{\alpha+1}; q^2)_{\infty}}. \quad (18.4)$$

Finally insert

$$\Gamma_{q^2}(z) = (1 - q^2)^{1-z} \frac{(q^2; q^2)_{\infty}}{(q^{2z}; q^2)_{\infty}}.$$

In the quotient on the right of (18.2), all powers of  $1 - q^2$  cancel, and the remaining products are exactly (18.4).  $\square$

**Corollary 21.3** (Classical limit of the regular sum). *For  $\alpha \notin \{-1, -3, -5, \dots\}$ ,*

$$\sum_{k=0}^{\infty} \binom{2k}{k} \frac{1}{4^k (2k+1+\alpha)} = \frac{\Gamma(3/2) \Gamma((\alpha+1)/2)}{\Gamma((\alpha+2)/2)}. \quad (18.5)$$

*Proof.* The series is

$$\frac{1}{\alpha+1} {}_2F_1 \left( \begin{matrix} 1/2, (\alpha+1)/2 \\ (\alpha+3)/2 \end{matrix}; 1 \right).$$

Gauss's classical summation gives

$$\frac{1}{\alpha+1} \frac{\Gamma((\alpha+3)/2) \Gamma(1/2)}{\Gamma((\alpha+2)/2)}.$$

Since  $\Gamma((\alpha+3)/2) = ((\alpha+1)/2) \Gamma((\alpha+1)/2)$  and  $\Gamma(1/2)/2 = \Gamma(3/2)$ , this is (18.5). The same identity is also the  $q \rightarrow 1^-$  limit of Theorem 21.2 by the  $q$ -gamma limit in Theorem 13.5.  $\square$

### 21.3 Deleting a singular term

The regular formula becomes singular when  $\alpha = -2m - 1$ . Removing the offending  $k = m$  term does not merely produce a finite part of the same gamma quotient; it yields an alternating Lambert-series tail.

**Theorem 21.4** (Deleted-singularity central identity). *Let  $m \geq 0$  and  $0 < q < 1$ . Then*

$$\begin{aligned} \sum_{\substack{k \geq 0 \\ k \neq m}} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k - m]_{q^2}} \\ = \begin{bmatrix} 2m \\ m \end{bmatrix}_{q^2} \frac{(1 + q)q^m}{(-q; q)_{2m}} \sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1} q^r}{[r]_q}. \end{aligned} \quad (18.6)$$

*Proof.* By Lemma 21.1, the left side is

$$(1 - q^2) \sum_{\substack{k \geq 0 \\ k \neq m}} \frac{(q; q^2)_k q^k}{(q^2; q^2)_k (1 - q^{2k-2m})}. \quad (18.7)$$

Introduce an auxiliary parameter  $a$  and define

$$\begin{aligned} F(a, q) := \frac{1}{1 - aq^{-2m}} \left( {}_2\phi_1 \left[ \begin{matrix} q, aq^{-2m} \\ aq^{-2m+2}; q^2, q \end{matrix} \right] \right. \\ \left. - \frac{(q, aq^{-2m}; q^2)_m q^m}{(q^2, aq^{-2m+2}; q^2)_m} \right). \end{aligned} \quad (18.8)$$

The elementary cancellation

$$\frac{(aq^{-2m}; q^2)_k}{(aq^{-2m+2}; q^2)_k} = \frac{1 - aq^{-2m}}{1 - aq^{2k-2m}}$$

shows that

$$F(a, q) = \sum_{\substack{k \geq 0 \\ k \neq m}} \frac{(q; q^2)_k q^k}{(q^2; q^2)_k (1 - aq^{2k-2m})}. \quad (18.9)$$

Thus (18.7) equals  $(1 - q^2) \lim_{a \rightarrow 1} F(a, q)$ .

Apply  $q$ -Gauss with base  $q^2$  to the  ${}_2\phi_1$  in (18.8). We obtain

$$\begin{aligned} F(a, q) &= \frac{1}{1 - aq^{-2m}} \left( \frac{(q^2, aq^{-2m+1}; q^2)_{\infty}}{(q, aq^{-2m+2}; q^2)_{\infty}} - \frac{(q, aq^{-2m}; q^2)_m q^m}{(q^2, aq^{-2m+2}; q^2)_m} \right) \\ &= \frac{H(a) - C_m}{1 - a}, \end{aligned} \quad (18.10)$$

where

$$\begin{aligned} H(a) &= \frac{(q^2, aq; q^2)_{\infty} (aq^{-2m+1}; q^2)_m}{(q, aq^2; q^2)_{\infty} (aq^{-2m}; q^2)_m}, \\ C_m &= \frac{(q; q^2)_m q^m}{(q^2; q^2)_m}. \end{aligned}$$

The reversal identity for finite shifted factorials gives

$$\frac{(q^{-2m+1}; q^2)_m}{(q^{-2m}; q^2)_m} = q^m \frac{(q; q^2)_m}{(q^2; q^2)_m},$$

so  $H(1) = C_m$ . Hence L'Hospital's rule yields

$$\lim_{a \rightarrow 1} F(a, q) = -H'(1) = -H(1) \left. \frac{\partial}{\partial a} \log H(a) \right|_{a=1}. \quad (18.11)$$

Differentiating the logarithms of the product factors gives

$$\lim_{a \rightarrow 1} F(a, q) = C_m \left( \sum_{r=1}^{\infty} \frac{(-1)^{r-1} q^r}{1 - q^r} + \sum_{r=1}^{2m} \frac{(-1)^{r-1} q^{-r}}{1 - q^{-r}} \right). \quad (18.12)$$

For  $r \geq 1$ ,

$$\frac{q^{-r}}{1 - q^{-r}} = -\frac{1}{1 - q^r}.$$

Consequently, the two summands in (18.12) combine, for  $1 \leq r \leq 2m$ , to  $(-1)^r$ . Their sum is 0 because  $2m$  is even. Therefore

$$\lim_{a \rightarrow 1} F(a, q) = C_m \sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1} q^r}{1 - q^r}. \quad (18.13)$$

Multiply by  $1 - q^2 = (1 + q)(1 - q)$ , use  $(1 - q)/(1 - q^r) = 1/[r]_q$ , and finally use (18.1) at  $k = m$  to rewrite

$$C_m = q^m \frac{(q; q^2)_m}{(q^2; q^2)_m} = q^m \left[ \begin{matrix} 2m \\ m \end{matrix} \right]_{q^2} \frac{1}{(-q; q)_{2m}}.$$

The result is exactly (18.6).  $\square$

*Remark 21.5* (The mechanism). The deleted term is encoded by the auxiliary parameter  $a$ . The  $q$ -Gauss product has a removable zero-over-zero singularity at  $a = 1$ , and its logarithmic derivative turns products into Lambert series. This parameter-differentiation mechanism recurs throughout the theory of harmonic-number and Lambert-series refinements of hypergeometric summations.

## 21.4 Separate finite and infinite components

The deleted sum in Theorem 21.4 naturally splits at the missing index. Each component has a simpler one-sided Lambert-series evaluation.

**Theorem 21.6** (Infinite and finite components). *Let  $m \geq 0$  and  $0 < q < 1$ , and put*

$$P_m(q) = \left[ \begin{matrix} 2m \\ m \end{matrix} \right]_{q^2} \frac{(1 + q)q^m}{(-q; q)_{2m}}.$$

*Then*

$$\begin{aligned} & \sum_{k=m+1}^{\infty} \left[ \begin{matrix} 2k \\ k \end{matrix} \right]_{q^2} \frac{q^k}{(-q; q)_{2k} [k - m]_{q^2}} \\ &= P_m(q) \left( \sum_{r=1}^{\infty} \frac{q^{2r-1}}{[2r-1]_q} - \sum_{r=1}^{\infty} \frac{q^{2r+2m}}{[2r+2m]_q} \right), \end{aligned} \quad (18.14)$$

*and*

$$\sum_{k=0}^{m-1} \left[ \begin{matrix} 2k \\ k \end{matrix} \right]_{q^2} \frac{q^k}{(-q; q)_{2k} [k - m]_{q^2}} = -P_m(q) \sum_{r=1}^m \frac{q^{2r-1}}{[2r-1]_q}. \quad (18.15)$$

*For  $m = 0$ , the finite sum and the sum on its right are both empty.*

*Proof.* By Lemma 21.1, shifting  $k \mapsto k + m$  in the infinite component gives

$$\begin{aligned} & \sum_{k=m+1}^{\infty} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k-m]_{q^2}} \\ &= (1-q^2) q^m \frac{(q; q^2)_m}{(q^2; q^2)_m} \sum_{k=1}^{\infty} \frac{(q^{2m+1}; q^2)_k q^k}{(q^{2m+2}; q^2)_k (1-q^{2k})}. \end{aligned}$$

Introduce

$$U(a, q) = \frac{1}{1-a} \left( {}_2\phi_1 \left[ \begin{matrix} a, q^{2m+1} \\ q^{2m+2} \end{matrix}; q^2, a^{-1}q \right] - 1 \right).$$

For  $a$  near 1, absolute convergence permits termwise passage to the limit. Since

$$\frac{(a; q^2)_k}{1-a} = (aq^2; q^2)_{k-1} \quad (k \geq 1),$$

we obtain

$$\lim_{a \rightarrow 1} U(a, q) = \sum_{k=1}^{\infty} \frac{(q^{2m+1}; q^2)_k q^k}{(q^{2m+2}; q^2)_k (1-q^{2k})}.$$

The  $q$ -Gauss summation, at base  $q^2$ , evaluates the auxiliary series as

$$U(a, q) = \frac{1}{1-a} \left( \frac{(q, a^{-1}q^{2m+2}; q^2)_{\infty}}{(a^{-1}q, q^{2m+2}; q^2)_{\infty}} - 1 \right).$$

The product quotient tends to 1 as  $a \rightarrow 1$ . L'Hospital's rule and termwise logarithmic differentiation, justified locally uniformly, give

$$\lim_{a \rightarrow 1} U(a, q) = \sum_{r=1}^{\infty} \frac{q^{2r-1}}{1-q^{2r-1}} - \sum_{r=1}^{\infty} \frac{q^{2r+2m}}{1-q^{2r+2m}}.$$

Multiplication by the prefactor above, followed by  $(1-q)/(1-q^j) = 1/[j]_q$  and Lemma 21.1 at  $k = m$ , proves (18.14).

To obtain the finite component, subtract (18.14) from the full deleted sum (18.6). Indeed,

$$\begin{aligned} \sum_{j=2m+1}^{\infty} \frac{(-1)^{j-1} q^j}{[j]_q} &= \sum_{r=m+1}^{\infty} \frac{q^{2r-1}}{[2r-1]_q} - \sum_{r=m+1}^{\infty} \frac{q^{2r}}{[2r]_q} \\ &= \sum_{r=1}^{\infty} \frac{q^{2r-1}}{[2r-1]_q} - \sum_{r=1}^{\infty} \frac{q^{2r+2m}}{[2r+2m]_q} - \sum_{r=1}^m \frac{q^{2r-1}}{[2r-1]_q}. \end{aligned}$$

The first two sums constitute the infinite component, leaving exactly (18.15). □

## 21.5 The classical finite part

Define the alternating harmonic number

$$\overline{H}_N = \sum_{r=1}^N \frac{(-1)^{r-1}}{r}.$$

**Theorem 21.7** (Classical deleted-singularity identity). *For every  $m \geq 0$ ,*

$$\sum_{\substack{k \geq 0 \\ k \neq m}} \binom{2k}{k} \frac{1}{4^k (k-m)} = \binom{2m}{m} \frac{\log 2 - \overline{H}_{2m}}{2^{2m-1}}. \quad (18.16)$$

*Proof.* Put  $c_k = (1/2)_k/k! = \binom{2k}{k}/4^k$ . For  $a$  away from the nonpositive integers, the series

$$T(a) = \sum_{k=0}^{\infty} \frac{c_k}{k+a}$$

converges locally uniformly and defines a meromorphic function. For  $\Re a > 0$ , termwise integration of  $(1-x)^{-1/2} = \sum_{k \geq 0} c_k x^k$  gives

$$T(a) = \int_0^1 x^{a-1} (1-x)^{-1/2} dx = B(a, 1/2) = \frac{\Gamma(a)\Gamma(1/2)}{\Gamma(a+1/2)}. \quad (18.17)$$

Analytic continuation makes (18.17) valid wherever both sides are defined. The desired deleted sum is the finite part at  $a = -m$ :

$$\lim_{a \rightarrow -m} \left( T(a) - \frac{c_m}{a+m} \right). \quad (18.18)$$

Write  $a = -m + \varepsilon$ . The standard expansions of the gamma function at a negative integer and at a regular point are

$$\begin{aligned} \Gamma(-m + \varepsilon) &= \frac{(-1)^m}{m! \varepsilon} (1 + (H_m - \gamma)\varepsilon + O(\varepsilon^2)), \\ \Gamma(1/2 - m + \varepsilon) &= \Gamma(1/2 - m) (1 + \psi(1/2 - m)\varepsilon + O(\varepsilon^2)). \end{aligned}$$

Moreover,

$$\frac{(-1)^m \Gamma(1/2)}{m! \Gamma(1/2 - m)} = c_m.$$

Thus the finite part in (18.18) is

$$c_m (H_m - \gamma - \psi(1/2 - m)). \quad (18.19)$$

Using  $\psi(1/2) = -\gamma - 2 \log 2$  and the recurrence  $\psi(z+1) = \psi(z) + 1/z$ , we find

$$\psi(1/2 - m) = -\gamma - 2 \log 2 + 2 \sum_{r=1}^m \frac{1}{2r-1}.$$

Since

$$\overline{H}_{2m} = \sum_{r=1}^m \frac{1}{2r-1} - \frac{1}{2} H_m,$$

the bracket in (18.19) is  $2(\log 2 - \overline{H}_{2m})$ . Hence the finite part equals

$$2 \frac{\binom{2m}{m}}{4^m} (\log 2 - \overline{H}_{2m}),$$

which is (18.16). □

**Proposition 21.8** (Limit of the Lambert tail). *For fixed  $m \geq 0$ ,*

$$\lim_{q \rightarrow 1^-} \sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1} q^r}{[r]_q} = \log 2 - \overline{H}_{2m}. \quad (18.20)$$



*Proof.* For  $0 < q < 1$ , set  $a_r(q) = q^r/[r]_q$ . These numbers are positive and decrease with  $r$ , since

$$\frac{a_{r+1}(q)}{a_r(q)} = q \frac{[r]_q}{[r+1]_q} < 1.$$

Moreover,

$$a_r(q) = \frac{q^r}{1 + q + \cdots + q^{r-1}} \leq \frac{1}{r},$$

because every summand in the denominator is at least  $q^{r-1}$ . The alternating-series remainder after  $N$  terms therefore has absolute value at most  $a_{N+1}(q) \leq 1/(N+1)$ , uniformly in  $q$ . For each fixed  $r$ ,  $a_r(q) \rightarrow 1/r$  as  $q \rightarrow 1^-$ . First pass to the limit in a finite partial sum and then let  $N \rightarrow \infty$ . The result is the tail of the alternating harmonic series,

$$\sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1}}{r} = \log 2 - \overline{H}_{2m}.$$

□

## Chapter 22

# Asymptotic regimes and Bernoulli expansions

A  $q$ -analogue should explain both the deformation near  $q = 1$  and behavior far from it. Gaussian coefficients have especially clean asymptotic descriptions because their logarithms are finite sums of a single elementary function.

### 22.1 A complete expansion as $q \rightarrow 1$

Let the Bernoulli numbers be defined by

$$\frac{x}{e^x - 1} = \sum_{r=0}^{\infty} B_r \frac{x^r}{r!}.$$

**Lemma 22.1** (The basic logarithmic expansion). As  $x \rightarrow 0$ ,

$$\log \frac{1 - e^{-x}}{x} = -\frac{x}{2} + \sum_{r=1}^{\infty} \frac{B_{2r}}{2r(2r)!} x^{2r}. \quad (19.1)$$

The series converges for  $|x| < 2\pi$ .

*Proof.* Differentiate the left side:

$$\frac{d}{dx} \log \frac{1 - e^{-x}}{x} = \frac{1}{e^x - 1} - \frac{1}{x}.$$

By the Bernoulli generating function,

$$\frac{1}{e^x - 1} - \frac{1}{x} = -\frac{1}{2} + \sum_{r=1}^{\infty} B_{2r} \frac{x^{2r-1}}{(2r)!}.$$

Integrate term by term and use the value 0 at  $x = 0$ . The nearest nonzero singularities are at  $x = \pm 2\pi i$ , giving radius  $2\pi$ .  $\square$

**Theorem 22.2** (Bernoulli expansion of a Gaussian coefficient). Let  $0 \leq k \leq n$ , put  $D = k(n - k)$ , and set  $q = e^{-t}$ . As  $t \rightarrow 0$ ,

$$\log \frac{[n]_q!}{[k]_q! [n-k]_q!} = -\frac{D}{2}t + \sum_{r=1}^{\infty} \frac{B_{2r} t^{2r}}{2r(2r)!} \sum_{j=1}^k ((n - k + j)^{2r} - j^{2r}). \quad (19.2)$$

For fixed  $n, k$ , the displayed series converges whenever  $|t| < 2\pi/n$ .

*Proof.* From the product formula,

$$\begin{aligned} \left[ \begin{matrix} n \\ k \end{matrix} \right]_{e^{-t}} &= \prod_{j=1}^k \frac{1 - e^{-(n-k+j)t}}{1 - e^{-jt}} \\ &= \binom{n}{k} \prod_{j=1}^k \frac{(1 - e^{-(n-k+j)t})/((n-k+j)t)}{(1 - e^{-jt})/(jt)}. \end{aligned}$$

Take logarithms and apply [Lemma 22.1](#) to every factor. The linear term is

$$-\frac{t}{2} \sum_{j=1}^k (n-k) = -\frac{1}{2} k(n-k)t = -\frac{D}{2}t,$$

and the even terms give the sum in (19.2). Every argument of the basic series has modulus below  $2\pi$  when  $|t| < 2\pi/n$ , so the finite sum converges there.  $\square$

*Remark 22.3* (Power-sum form of the coefficients). Let  $S_m(N) = \sum_{j=1}^N j^m$  denote the  $m$ th power sum. For every integer  $m \geq 1$ , reindexing the first half of an inner sum by  $i = n - k + j$  gives

$$\sum_{j=1}^k ((n-k+j)^m - j^m) = \sum_{i=n-k+1}^n i^m - \sum_{j=1}^k j^m = S_m(n) - S_m(k) - S_m(n-k).$$

Applied with  $m = 2r$  to the inner sum of (19.2), and with  $m = 1$  to the sum  $\sum_{j=1}^k ((n-k+j) - j) = \sum_{j=1}^k (n-k)$  that produced the linear term in the proof of [Theorem 22.2](#), this turns the expansion into

$$\log \frac{\left[ \begin{matrix} n \\ k \end{matrix} \right]_{e^{-t}}}{\binom{n}{k}} = -\frac{t}{2} (S_1(n) - S_1(k) - S_1(n-k)) + \sum_{r=1}^{\infty} \frac{B_{2r} t^{2r}}{2r(2r)!} (S_{2r}(n) - S_{2r}(k) - S_{2r}(n-k)),$$

still for  $|t| < 2\pi/n$ , the factor  $-\frac{1}{2}$  being the coefficient of  $x$  in (19.1). The case  $m = 1$  of the display evaluates the linear bracket outright:  $S_1(n) - S_1(k) - S_1(n-k) = \sum_{j=1}^k (n-k) = k(n-k) = D$ .

Three features become visible in this form. First, each coefficient is a difference of Faulhaber polynomials, hence itself a polynomial in  $n$  and  $k$  with rational coefficients, of total degree exactly  $2r+1$ : since  $S_{2r}(N) = N^{2r+1}/(2r+1) + O(N^{2r})$ , the homogeneous part of top degree is  $(n^{2r+1} - k^{2r+1} - (n-k)^{2r+1})/(2r+1)$ , whose coefficient of  $n^{2r}k$  is  $-\frac{1}{2r+1} \binom{2r+1}{1} (-1) = 1$ , so no cancellation lowers the degree. Second, the expression is manifestly invariant under  $k \mapsto n-k$ , as it must be by the symmetry recorded in [Theorem 2.5](#). Third, at  $r = 1$  the identity  $S_2(N) = N(N+1)(2N+1)/6$  gives

$$\begin{aligned} S_2(n) - S_2(k) - S_2(n-k) &= \frac{n(n+1)(2n+1) - k(k+1)(2k+1)}{6} \\ &\quad - \frac{(n-k)(n-k+1)(2n-2k+1)}{6} \\ &= k(n-k)(n+1) = D(n+1), \end{aligned}$$

the value used in [Corollary 22.4](#) below.

**Corollary 22.4** (First terms). *With  $D = k(n-k)$ ,*

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_{e^{-t}} = \binom{n}{k} \exp \left( -\frac{D}{2}t + \frac{D(n+1)}{24}t^2 + O(t^4) \right), \quad (19.3)$$

$$= \binom{n}{k} \left( 1 - \frac{D}{2}t + \left( \frac{D^2}{8} + \frac{D(n+1)}{24} \right) t^2 + O(t^3) \right). \quad (19.4)$$

*Proof.* For  $r = 1$  in (19.2),

$$\begin{aligned} \sum_{j=1}^k ((n-k+j)^2 - j^2) &= k(n-k)^2 + (n-k)k(k+1) \\ &= k(n-k)(n+1) = D(n+1). \end{aligned}$$

Since  $B_2/(2 \cdot 2!) = 1/24$ , the logarithmic formula follows. Expanding the exponential gives the second formula.  $\square$

*Remark 22.5* (Reciprocity and centering). The factor  $e^{-Dt/2}$  is forced by reciprocity:

$$\begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}} = e^{-Dt} \begin{bmatrix} n \\ k \end{bmatrix}_{e^t}.$$

Therefore the centered quantity  $e^{Dt/2} \begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}}$  is an even analytic function of  $t$ . This explains, before any calculation, why all odd powers beyond the linear term vanish from the logarithm.

## 22.2 Fixed $q$ and growing parameters

**Theorem 22.6** (Fixed-column limit). *If  $|q| < 1$  and  $k \in \mathbb{N}_0$  is fixed, then*

$$\lim_{n \rightarrow \infty} \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q; q)_k}, \quad (19.5)$$

$$\lim_{n \rightarrow \infty} \begin{bmatrix} n+k \\ k \end{bmatrix}_q = \frac{1}{(q; q)_k}. \quad (22.1)$$

*If moreover  $0 < |q| < 1$ , then, as  $n \rightarrow \infty$ ,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q; q)_k} \left( 1 + O(|q|^{n-k+1}) \right), \quad (19.6)$$

$$\begin{bmatrix} n+k \\ k \end{bmatrix}_q = \frac{1}{(q; q)_k} \left( 1 + O(|q|^{n+1}) \right). \quad (22.2)$$

*Proof.* If  $q = 0$ , then  $(q; q)_k = 1$  and both Gaussian coefficients are 1 once  $n \geq k$ , so both limits are immediate. Assume henceforth that  $0 < |q| < 1$ . The product formula gives

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{n-k+1}; q)_k}{(q; q)_k}.$$

The numerator tends to 1. Since the number of factors is fixed,

$$\prod_{j=0}^{k-1} (1 - q^{n-k+1+j}) = 1 + O(|q|^{n-k+1}),$$

which proves (19.5) and (19.6). Replacing  $n$  by  $n+k$  gives (22.1) and (22.2).  $\square$

**Theorem 22.7** (Fixed- $q$  limit with both indices growing). *Let  $0 < |q| < 1$ . For all integers  $0 \leq k \leq n$ ,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{k+1}; q)_\infty (q^{n-k+1}; q)_\infty}{(q; q)_\infty (q^{n+1}; q)_\infty},$$

and consequently

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q; q)_\infty} \left( 1 + O(|q|^{\min(k, n-k)+1}) \right),$$

the implied constant depending only on  $q$ . In particular,

$$\lim_{\min(k, n-k) \rightarrow \infty} \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q; q)_\infty}, \quad (19.7)$$

and for every fixed  $r \in \mathbb{Z}$ ,

$$\lim_{m \rightarrow \infty} \begin{bmatrix} 2m \\ m+r \end{bmatrix}_q = \frac{1}{(q; q)_\infty},$$

the coefficient being read for  $m \geq |r|$ , so that  $m+r$  and  $m-r$  are admissible indices.

*Proof.* By [Theorem 1.12](#) applied at  $a = q$ , the product  $(q; q)_\infty$  converges, and it is nonzero: the zeros of  $a \mapsto (a; q)_\infty$  are the points  $q^{-j}$  with  $j \geq 0$ , and  $q = q^{-j}$  would give  $q^{j+1} = 1$ , hence  $|q| = 1$ . The same theorem at  $a = q^{N+1}$  gives  $(q^{N+1}; q)_\infty \neq 0$  for every  $N \geq 0$ . The tail factorization of [Proposition 1.2](#),

$$(q; q)_\infty = (q; q)_N (q^{N+1}; q)_\infty,$$

therefore yields  $(q; q)_N = (q; q)_\infty / (q^{N+1}; q)_\infty$ . Substituting this at  $N = n$ ,  $N = k$ , and  $N = n - k$  in the factorial form  $\begin{bmatrix} n \\ k \end{bmatrix}_q = (q; q)_n / ((q; q)_k (q; q)_{n-k})$  gives the first display.

For the error term, let  $M \geq 0$ . By the elementary product estimate,

$$|(q^{M+1}; q)_\infty - 1| \leq \prod_{j \geq M+1} (1 + |q|^j) - 1 \leq \exp\left(\frac{|q|^{M+1}}{1 - |q|}\right) - 1 \leq C_q |q|^{M+1}, \quad C_q = \frac{e^{1/(1-|q|)}}{1 - |q|},$$

since  $|q|^{M+1} / (1 - |q|) \leq 1 / (1 - |q|)$  and  $e^x - 1 \leq xe^{x_0}$  for  $0 \leq x \leq x_0$ . Moreover

$$|(q^{M+1}; q)_\infty| \geq \prod_{j \geq 1} (1 - |q|^j) = (|q|; |q|)_\infty > 0$$

uniformly in  $M$ . As  $\min(k, n - k) \leq n$ , applying both bounds at  $M = k$ ,  $M = n - k$ , and  $M = n$  to the first display gives the second.

The two limits follow at once: in the first,  $\min(k, n - k) \rightarrow \infty$  by hypothesis; in the second,  $n = 2m$  and  $k = m + r$  give  $\min(k, n - k) = m - |r| \rightarrow \infty$ , which is also what makes both indices admissible once  $m \geq |r|$ .  $\square$

*Remark 22.8.* The off-centre column in the regime  $q > 1$  needs no new argument. By [\(19.9\)](#) at upper index  $2m$  and lower index  $m + r$ , whose exponent is  $k(n - k) = (m + r)(m - r) = m^2 - r^2$ ,

$$\begin{bmatrix} 2m \\ m+r \end{bmatrix}_q = q^{m^2-r^2} \begin{bmatrix} 2m \\ m+r \end{bmatrix}_{q^{-1}},$$

and [Theorem 22.7](#) applied with  $q^{-1}$  in place of  $q$ , which is legitimate because  $0 < q^{-1} < 1$ , gives  $\begin{bmatrix} 2m \\ m+r \end{bmatrix}_{q^{-1}} \rightarrow (q^{-1}; q^{-1})_\infty^{-1}$ . Hence

$$\begin{bmatrix} 2m \\ m+r \end{bmatrix}_q \sim \frac{q^{m^2-r^2}}{(q^{-1}; q^{-1})_\infty},$$

which is [\(19.11\)](#) at  $r = 0$ .

**Proposition 22.9** (Uniform elementary bound). *For  $0 < q < 1$  and  $0 \leq k \leq n$ ,*

$$1 \leq \begin{bmatrix} n \\ k \end{bmatrix}_q \leq \frac{1}{(q; q)_\infty}. \quad (19.8)$$

*Proof.* The lower bound follows from the partition interpretation: the constant term is 1 and every coefficient is nonnegative. The same interpretation says that  $\left[ \begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q$  is the generating function of partitions fitting in a  $k$  by  $(n - k)$  rectangle. Dropping the rectangle restriction embeds this set into all partitions, whose generating function is  $(q; q)_\infty^{-1}$ . Evaluating at a positive  $q < 1$  gives the upper bound.  $\square$

### 22.3 The regime $q > 1$

**Corollary 22.10** (Transfer through reciprocity). *Let  $q > 1$  and  $D = k(n - k)$ . Then*

$$\left[ \begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q = q^D \left[ \begin{smallmatrix} n \\ k \end{smallmatrix} \right]_{q^{-1}}. \quad (19.9)$$

Consequently, for fixed  $k$ ,

$$\left[ \begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q = \frac{q^{k(n-k)}}{(q^{-1}; q^{-1})_k} \left( 1 + O(q^{-n+k-1}) \right), \quad (19.10)$$

and

$$\left[ \begin{smallmatrix} 2m \\ m \end{smallmatrix} \right]_q \sim \frac{q^{m^2}}{(q^{-1}; q^{-1})_\infty}. \quad (19.11)$$

*Proof.* The reciprocity identity was proved in [Theorem 2.5](#). Apply [Theorems 22.6](#) and [22.7](#) with  $q^{-1}$  in place of  $q$ .  $\square$

### 22.4 Extreme specializations

**Proposition 22.11.** *For  $0 \leq k \leq n$ ,*

$$\left[ \begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q \Big|_{q=0} = 1, \quad \lim_{q \rightarrow 1} \left[ \begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q = \binom{n}{k},$$

*and the leading term as  $q \rightarrow \infty$  is  $q^{k(n-k)}$  with coefficient 1.*

*Proof.* The partition generating polynomial has constant term 1, proving the first assertion. The second follows by replacing every ratio  $(1 - q^a)/(1 - q^b)$  in the product formula by its limit  $a/b$ . Palindromicity and degree  $k(n - k)$  show that the leading coefficient equals the constant coefficient, namely 1.  $\square$

## Chapter 23

# Exact limits, reciprocity, and research frontiers

This chapter isolates three principles that are easy to use informally and equally easy to overstate: modular acceleration near  $q = 1$ , passage from finite identities to formal or analytic limits, and stabilization at the two ends of a reciprocal polynomial. Each proved statement below records the topology in which its limit is taken. The final section then lists open programs as problems rather than silently promoting them to conjectures.

### 23.1 An exact modular acceleration

The eta transformation in [Theorem 15.11](#) gives more than the leading singular term of  $(q; q)_\infty$ . It replaces a slowly convergent product with one whose nome is exponentially small.

**Theorem 23.1** (Exact modular form of the Euler product). *Let  $t > 0$ , and put*

$$q = e^{-t}, \quad Q = e^{-4\pi^2/t}.$$

*Then*

$$(q; q)_\infty = \sqrt{\frac{2\pi}{t}} \exp\left(-\frac{\pi^2}{6t} + \frac{t}{24}\right) (Q; Q)_\infty. \quad (23.1)$$

*Consequently, as  $t \rightarrow 0^+$ ,*

$$(q; q)_\infty = \sqrt{\frac{2\pi}{t}} \exp\left(-\frac{\pi^2}{6t} + \frac{t}{24}\right) \left(1 + O\left(e^{-4\pi^2/t}\right)\right), \quad (23.2)$$

$$\log(q; q)_\infty = -\frac{\pi^2}{6t} + \frac{1}{2} \log \frac{2\pi}{t} + \frac{t}{24} + O\left(e^{-4\pi^2/t}\right). \quad (23.3)$$

*The error terms are real and uniform for  $0 < t \leq t_0$ , for every fixed  $t_0 > 0$  small enough.*

*Proof.* Set  $\tau = it/(2\pi)$ . The definition [\(15.19\)](#) gives

$$\eta\left(\frac{it}{2\pi}\right) = e^{-t/24} (e^{-t}; e^{-t})_\infty. \quad (23.4)$$

Since  $-1/\tau = 2\pi i/t$ , the inversion formula [\(15.22\)](#), with its positive square root on the positive imaginary axis, reads

$$\eta\left(\frac{it}{2\pi}\right) = \sqrt{\frac{2\pi}{t}} \eta\left(\frac{2\pi i}{t}\right).$$

Applying the definition once more at  $2\pi i/t$  yields

$$\eta\left(\frac{2\pi i}{t}\right) = e^{-\pi^2/(6t)}(e^{-4\pi^2/t}; e^{-4\pi^2/t})_\infty.$$

Combining the last two identities with (23.4) proves the exact formula (23.1).

It remains to justify the stated exponential remainder rather than merely observe that  $Q \rightarrow 0$ . For  $0 < Q \leq 1/2$ , the elementary inequality  $-\log(1-u) \leq u/(1-u)$ ,  $0 \leq u < 1$ , gives

$$0 \leq -\log(Q; Q)_\infty = \sum_{m \geq 1} -\log(1 - Q^m) \leq \sum_{m \geq 1} \frac{Q^m}{1 - Q^m} \leq \frac{Q}{(1 - Q)^2} \leq 4Q.$$

Thus  $\log(Q; Q)_\infty = O(Q)$ . If  $L = -\log(Q; Q)_\infty$ , then  $0 \leq L \leq 4Q$  and

$$|(Q; Q)_\infty - 1| = 1 - e^{-L} \leq L \leq 4Q.$$

Substitution in the exact formula gives the first asymptotic, and taking the real logarithm gives the second. The displayed inequalities also provide the asserted uniformity.  $\square$

## 23.2 Coefficientwise and $q$ -adic limits

For a series  $G(q) = \sum_{d \geq 0} g_d q^d$ , write  $\text{ord}_q G = \min\{d : g_d \neq 0\}$ , with  $\text{ord}_q 0 = \infty$ . The congruence  $G \equiv H \pmod{q^{N+1}}$  means that their coefficients agree through degree  $N$ . These conventions make the finiteness behind the usual “truncate and pass to the limit” argument explicit.

**Proposition 23.2** (Coefficientwise passage to an analytic limit). *Let  $R$  be a commutative ring and let  $P_N \in R[[q]]$ . Suppose that for every  $d \geq 0$  there are  $c_d \in R$  and  $N_d$  such that*

$$[q^d]P_N = c_d \quad (N \geq N_d).$$

*Then there is a unique series  $F = \sum_{d \geq 0} c_d q^d \in R[[q]]$ , and  $P_N \rightarrow F$  coefficientwise.*

*If  $R = \mathbb{C}$  and the polynomials  $P_N$ , viewed as holomorphic functions, converge locally uniformly on the unit disk to a function  $P$ , then  $P$  is holomorphic and*

$$c_d = \frac{1}{2\pi i} \int_{|z|=r} \frac{P(z)}{z^{d+1}} dz = \frac{P^{(d)}(0)}{d!} \quad (0 < r < 1). \quad (23.5)$$

*In particular, the formal coefficientwise limit is precisely the Taylor series of the analytic limit.*

*Proof.* The first assertion is coefficientwise: the eventual value  $c_d$  determines the coefficient of degree  $d$ , so the displayed series exists and is the only possible coefficientwise limit.

For the analytic assertion, local uniform convergence of holomorphic functions makes  $P$  holomorphic. Fix  $0 < r < 1$ . Cauchy’s coefficient formula and uniform convergence on the circle  $|z| = r$  give

$$[q^d]P_N = \frac{1}{2\pi i} \int_{|z|=r} \frac{P_N(z)}{z^{d+1}} dz \longrightarrow \frac{1}{2\pi i} \int_{|z|=r} \frac{P(z)}{z^{d+1}} dz.$$

The left side is eventually equal to  $c_d$ . A second application of Cauchy’s formula identifies the last integral with  $P^{(d)}(0)/d!$ , proving (23.5).  $\square$

**Proposition 23.3** (Exact truncation principle). *Let  $(e_j)_{j \geq 1}$  be an arbitrary sequence of integers, and interpret negative powers by their binomial expansions in  $\mathbb{Z}[[q]]$ . The finite products*

$$F_M(q) = \prod_{j=1}^M (1 - q^j)^{e_j}$$



have a well-defined coefficientwise limit  $F \in \mathbb{Z}[[q]]$ , and for every  $N \geq 0$ ,

$$F(q) \equiv \prod_{j=1}^N (1 - q^j)^{e_j} \pmod{q^{N+1}}. \quad (23.6)$$

More generally, let  $R$  be a commutative ring and let  $(S_m)_{m \geq 0} \subset R[[q]]$  satisfy  $\text{ord}_q S_m \rightarrow \infty$ , in the precise sense that for each  $N$  only finitely many  $m$  have  $\text{ord}_q S_m \leq N$ . Then the  $q$ -adic sum  $S = \sum_{m \geq 0} S_m$  is well defined and

$$S(q) \equiv \sum_{\text{ord}_q S_m \leq N} S_m(q) \pmod{q^{N+1}}. \quad (23.7)$$

*Proof.* For every  $j \geq 1$  and  $e \in \mathbb{Z}$ ,

$$(1 - q^j)^e \equiv 1 \pmod{q^j}.$$

For  $e \geq 0$  this follows from the finite binomial theorem. For  $e = -r < 0$ , it follows from

$$(1 - q^j)^{-r} = \sum_{k \geq 0} \binom{k+r-1}{r-1} q^{jk} \quad (r \geq 1).$$

Consequently, if  $M \geq N$ , every factor with  $j > N$  is congruent to 1 modulo  $q^{N+1}$ , and hence

$$F_M(q) \equiv F_N(q) \pmod{q^{N+1}}.$$

Thus every coefficient of  $F_M$  stabilizes; apply [Proposition 23.2](#) with  $R = \mathbb{Z}$ . Passing to the stabilized coefficients gives (23.6).

For the series assertion, fix a degree  $d$ . Only finitely many  $S_m$  have order at most  $d$ , so the coefficient  $\sum_m [q^d] S_m$  is a finite sum in  $R$ . These finite sums define a unique element  $S \in R[[q]]$ . Terms with order greater than  $N$  vanish modulo  $q^{N+1}$ , and the remaining index set is finite by hypothesis. This proves (23.7).  $\square$

### 23.3 Borwein dissections and two-ended stabilization

For  $n \geq 1$ , define

$$P_n(q) = \frac{(q; q)_{3n}}{(q^3; q^3)_n} = \prod_{j=1}^n (1 - q^{3j-2})(1 - q^{3j-1}). \quad (23.8)$$

Grouping coefficients by their exponent modulo 3 gives unique polynomials  $A_n, B_n, C_n \in \mathbb{Z}[x]$  such that

$$P_n(q) = A_n(q^3) - qB_n(q^3) - q^2C_n(q^3). \quad (23.9)$$

The minus signs are part of the normalization; none of the arguments below assumes a sign for the coefficients of the three polynomials.

**Proposition 23.4** (Reciprocity of the Borwein dissection). *For every  $n \geq 1$ ,*

$$A_n(x) = x^{n^2} A_n(x^{-1}), \quad (23.10)$$

$$B_n(x) = x^{n^2-1} C_n(x^{-1}), \quad (23.11)$$

$$C_n(x) = x^{n^2-1} B_n(x^{-1}). \quad (23.12)$$

In particular,

$$B_n(1) = C_n(1), \quad A_n(1) = 2B_n(1).$$

*Proof.* The degree of  $P_n$  is

$$\sum_{j=1}^n ((3j-2) + (3j-1)) = 3n^2.$$

For positive integers  $a, b$ ,  $(1 - q^{-a})(1 - q^{-b}) = q^{-a-b}(1 - q^a)(1 - q^b)$ . There are two factors in every  $j$ -block, so their signs cancel, and

$$P_n(q) = q^{3n^2} P_n(q^{-1}).$$

Insert (23.9) on both sides and put  $x = q^3$ . The residue-zero terms give  $A_n(x) = x^{n^2} A_n(x^{-1})$ . The residue-one terms on the left match the reflected residue-two terms on the right, giving  $B_n(x) = x^{n^2-1} C_n(x^{-1})$ ; the remaining residue class gives the third identity. This comparison is legitimate in the Laurent-polynomial ring  $\mathbb{Z}[q, q^{-1}]$ , where the three residue classes modulo 3 form a direct sum.

Putting  $x = 1$  in the last two identities gives  $B_n(1) = C_n(1)$ . Since  $P_n(1) = 0$ , equation (23.9) at  $q = 1$  gives  $A_n(1) = B_n(1) + C_n(1) = 2B_n(1)$ .  $\square$

**Proposition 23.5** (Exact stabilization at both reciprocal ends). *Write*

$$a_{n,r} = [x^r]A_n(x), \quad b_{n,r} = [x^r]B_n(x), \quad c_{n,r} = [x^r]C_n(x).$$

*For every fixed  $d \geq 0$  and every  $n \geq d+1$ , the lower-end coefficients have already stabilized:*

$$a_{n,d} = a_{d+1,d}, \quad b_{n,d} = b_{d+1,d}, \quad c_{n,d} = c_{d+1,d}.$$

*The coefficients at the same fixed distance from the reciprocal ends satisfy the exact formulas*

$$a_{n,n^2-d} = a_{d+1,d}, \tag{23.13}$$

$$b_{n,n^2-1-d} = c_{d+1,d}, \tag{23.14}$$

$$c_{n,n^2-1-d} = b_{d+1,d}. \tag{23.15}$$

*Thus every fixed-distance coefficient at either end of each polynomial is eventually constant. No coefficientwise nonnegativity assertion is needed.*

*Proof.* Fix  $N \leq n$ . Factoring the product at  $N$  gives

$$P_n(q) = P_N(q) \prod_{j=N+1}^n (1 - q^{3j-2})(1 - q^{3j-1}).$$

For  $j > N$ , both nonconstant exponents in the  $j$ th new block are at least  $3N+1$ . Hence the entire tail product is congruent to 1 modulo  $q^{3N+1}$ , and

$$P_n(q) \equiv P_N(q) \pmod{q^{3N+1}}. \tag{23.16}$$

Take  $N = d+1$ . The exponents  $3d$ ,  $3d+1$ , and  $3d+2$  are all less than  $3N+1 = 3d+4$ , so their coefficients in  $P_n$  and  $P_{d+1}$  agree. By (23.9), those three coefficients are respectively  $a_{n,d}$ ,  $-b_{n,d}$ , and  $-c_{n,d}$ . This proves the three lower-end stabilization identities.

The reciprocity identities in Proposition 23.4 give, for every  $n$ ,

$$a_{n,n^2-d} = a_{n,d}, \quad b_{n,n^2-1-d} = c_{n,d}, \quad c_{n,n^2-1-d} = b_{n,d}.$$

Substituting the already proved lower-end values yields (23.15) and completes the proof.  $\square$

**Historical status of the sign question.** The donor manuscript stated coefficientwise nonnegativity of  $A_n, B_n, C_n$  as a Borwein conjecture. That assertion is not retained here as a conjecture: the relevant Borwein positivity conjectures are historically resolved in the literature [35, 36]. Their external proofs are not reproduced and are not used in this chapter. The unconditional conclusion needed here is the sign-free stabilization theorem above, proved directly from finite-product  $q$ -adic stabilization and reciprocity.

**Historical status of the bilateral-lattice prompt.** The donor manuscript also asked for bilateral Bailey lattices combining a parameter-lowering step with Ramanujan’s  ${}_1\psi_1$  summation. That broad request is not retained as an open problem. Dousse, Jouhet, and Konan [9] established bilateral Bailey lattices and used them to derive Andrews–Gordon- and Bressoud-type identities. A genuine new frontier would therefore have to specify a transformation, residue-class family, or analytic regime not covered by that theory; the unspecialized donor prompt makes no such distinction.

## 23.4 Archival product records and finite quality control

One donor guide normalized fifteen further Rogers–Ramanujan–Slater-type formulas from its supplied archive. They were deliberately not presented as theorems there, and they are not promoted to theorems here. In the displays below, the decorated relation  $\stackrel{\text{record}}{=}$  means “the source records this identity”; it does not assert that this volume has proved the infinite identity. None of these records is used as a premise. Slater’s classical list and the modern survey of McLaughlin, Sills, and Zimmer provide the appropriate literature map [31, 23].

### 23.4.1 Bailey’s modulus-nine trio

$$\sum_{n \geq 0} \frac{q^{3n^2}(q; q)_{3n}}{(q^3; q^3)_n (q^3; q^3)_{2n}} \stackrel{\text{record}}{=} \frac{(q^4, q^5, q^9; q^9)_\infty}{(q^3; q^3)_\infty}, \quad (\text{SR-B9.1})$$

$$\sum_{n \geq 0} \frac{q^{3n^2+3n}(q; q)_{3n}(1 - q^{3n+2})}{(q^3; q^3)_n (q^3; q^3)_{2n+1}} \stackrel{\text{record}}{=} \frac{(q^2, q^7, q^9; q^9)_\infty}{(q^3; q^3)_\infty}, \quad (\text{SR-B9.2})$$

$$\sum_{n \geq 0} \frac{q^{3n^2+3n}(q; q)_{3n+1}}{(q^3; q^3)_n (q^3; q^3)_{2n+1}} \stackrel{\text{record}}{=} \frac{(q, q^8, q^9; q^9)_\infty}{(q^3; q^3)_\infty}. \quad (\text{SR-B9.3})$$

The surviving product-side residue pairs are respectively  $\{4, 5\}$ ,  $\{2, 7\}$ , and  $\{1, 8\}$  modulo (9).

### 23.4.2 Dyson’s modulus-twenty-seven quartet

$$1 + \sum_{n \geq 1} \frac{q^{n^2}(q^3; q^3)_{n-1}}{(q; q)_n (q; q)_{2n-1}} \stackrel{\text{record}}{=} \frac{(q^{12}, q^{15}, q^{27}; q^{27})_\infty}{(q; q)_\infty}, \quad (\text{SR-D27.1})$$

$$\sum_{n \geq 0} \frac{q^{n^2+n}(q^3; q^3)_n}{(q; q)_n (q; q)_{2n+1}} \stackrel{\text{record}}{=} \frac{(q^9; q^9)_\infty}{(q; q)_\infty}, \quad (\text{SR-D27.2})$$

$$\sum_{n \geq 0} \frac{q^{n^2+2n}(q^3; q^3)_n}{(q; q)_n (q; q)_{2n+2}} \stackrel{\text{record}}{=} \frac{(q^6, q^{21}, q^{27}; q^{27})_\infty}{(q; q)_\infty}, \quad (\text{SR-D27.3})$$

$$\sum_{n \geq 0} \frac{q^{n^2+3n}(q^3; q^3)_n}{(q; q)_n (q; q)_{2n+2}} \stackrel{\text{record}}{=} \frac{(q^3, q^{24}, q^{27}; q^{27})_\infty}{(q; q)_\infty}. \quad (\text{SR-D27.4})$$

The second product is the exceptional member: it collapses to the simpler modulus-nine factor  $(q^9; q^9)_\infty$ .

### 23.4.3 Rogers’s modulus-fourteen trio

$$\sum_{n \geq 0} \frac{q^{n^2}}{(q; q)_n (q; q^2)_n} \stackrel{\text{record}}{=} \frac{(q^6, q^8, q^{14}; q^{14})_\infty}{(q; q)_\infty}, \quad (\text{SR-R14.1})$$

$$\sum_{n \geq 0} \frac{q^{n^2+n}}{(q; q)_n (q; q^2)_{n+1}} \stackrel{\text{record}}{=} \frac{(q^4, q^{10}, q^{14}; q^{14})_\infty}{(q; q)_\infty}, \quad (\text{SR-R14.2})$$

$$\sum_{n \geq 0} \frac{q^{n^2+2n}}{(q; q)_n (q; q^2)_{n+1}} \stackrel{\text{record}}{=} \frac{(q^2, q^{12}, q^{14}; q^{14})_\infty}{(q; q)_\infty}. \quad (\text{SR-R14.3})$$

The mixed bases  $q$  and  $q^2$  on the sum side expose the parity-sensitive feature that separates even-modulus theory from the odd Andrews–Gordon family.

### 23.4.4 Rogers–Selberg, Jackson–Slater, and Fine records

$$\sum_{n \geq 0} \frac{q^{2n^2}}{(q^2; q^2)_n (-q; q)_{2n}} \stackrel{\text{record}}{=} \frac{(q^3, q^4, q^7; q^7)_\infty}{(q^2; q^2)_\infty}, \quad (\text{SR-RS.1})$$

$$\sum_{n \geq 0} \frac{q^{2n^2+2n}}{(q^2; q^2)_n (-q; q)_{2n}} \stackrel{\text{record}}{=} \frac{(q^2, q^5, q^7; q^7)_\infty}{(q^2; q^2)_\infty}, \quad (\text{SR-RS.2})$$

$$\sum_{n \geq 0} \frac{q^{2n^2+2n}}{(q^2; q^2)_n (-q; q)_{2n+1}} \stackrel{\text{record}}{=} \frac{(q, q^6, q^7; q^7)_\infty}{(q^2; q^2)_\infty}, \quad (\text{SR-RS.3})$$

$$\sum_{n \geq 0} \frac{q^{2n^2}}{(q; q)_{2n}} \stackrel{\text{record}}{=} \frac{(q, q^7, q^8; q^8)_\infty (q^6, q^{10}; q^{16})_\infty}{(q; q)_\infty}. \quad (\text{SR-JS})$$

The Jackson–Slater product is a useful mixed-modulus record: its right side uses moduli (8) and (16) simultaneously. Fine’s isolated eta-quotient record is

$$\frac{(q^2; q^2)_\infty (q^3; q^3)_\infty (q^8; q^8)_\infty (q^{12}; q^{12})_\infty}{(q; q)_\infty (q^{24}; q^{24})_\infty} \stackrel{\text{record}}{=} 1 + \sum_{k \geq 1} \frac{q^k (1 + q^{4k}) (1 + q^{6k})}{1 + q^{12k}}. \quad (\text{SR-F})$$

Its sum side is Lambert-like, while the product side becomes an eta quotient after  $q = e^{2\pi i \tau}$ .

### 23.4.5 What the finite audit proves—and what it does not

The retained source-only checker `audit/verify_archival_identity_records.py` works in the exact quotient ring  $\mathbb{Z}[[q]]/(q^{101})$ . It truncates every product factor above degree 100, expands reciprocal factors by their exact geometric or binomial series, and includes only sum-side terms whose  $q$ -adic order is at most 100. It compares both sides coefficient by coefficient:

| Family              | Records | Coefficients compared per record |
|---------------------|---------|----------------------------------|
| Bailey modulus (9)  | (3)     | (101)                            |
| Dyson modulus (27)  | (4)     | (101)                            |
| Rogers modulus (14) | (3)     | (101)                            |
| Rogers–Selberg      | (3)     | (101)                            |
| Jackson–Slater      | (1)     | (101)                            |
| Fine                | (1)     | (101)                            |
| Total               | (15)    | (1515) coefficient equalities    |

This exact finite computation is strong transcription-quality control, but it is not an infinite proof: two formal series can agree through any prescribed finite order and then differ. A record may be promoted to a theorem only after a finite degree bound, a convergent analytic argument, or another complete proof supplies the missing infinite step.

## 23.5 Precisely delimited open programs

The following items are requests for constructions or classifications, not assertions presented as true. They are therefore recorded as prose open problems rather than as theorem-like conjectures.

**Open problem (finite Andrews–Gordon certificates).** For every  $k \geq 2$  and  $1 \leq i \leq k$ , construct a finite polynomial identity whose coefficientwise limit is the corresponding Andrews–Gordon identity, whose sum-side terms are manifestly coefficientwise nonnegative, and whose verification reduces to Gaussian recurrences together with a bounded collection of explicit telescoping-certificate types independent of the truncation order. Seek a canonical truncation for which the product side has a direct lattice-path or cylindric-partition interpretation.

**Open problem (even-modulus lowering).** Construct a parity-sensitive parameter-lowering identity for Bailey pairs that produces the even-modulus Bressoud family from finite algebra, with modulus 14 as the first nontrivial test. The parity restriction should be visible in the finite kernel itself, and the proof should separate the terminating identity, coefficientwise limit, and analytic specialization.

**Open problem (total positivity of the Bailey kernel).** For  $0 < a, q < 1$ , consider the lower-triangular kernel

$$M_{n,r}(a, q) = \frac{1}{(q; q)_{n-r}(aq; q)_{n+r}} \quad (0 \leq r \leq n).$$

Determine whether every minor of  $M$  is nonnegative (equivalently, whether this holds after any positive diagonal normalization). If total positivity fails, determine the first obstructing minor and the largest order of total positivity that holds uniformly in  $a, q$ . Either outcome should explain which variation-diminishing or coefficient-sign consequences survive for Bailey transforms.

**Open problem (root-of-unity asymptotics).** Let  $\zeta$  be a primitive root of unity and approach it through  $q = \zeta e^{-t}$ ,  $t \rightarrow 0^+$ , with all logarithmic branches stated explicitly. Develop uniform asymptotic expansions for  $q$ -Pochhammer products, Bailey sums, and their locally defined inverse branches on compact parameter sets avoiding moving poles and zeros. Separate the dilogarithmic exponential, algebraic prefactor, and exponentially small sectors, and determine how the resulting Stokes data governs the singularities of the inverse  $q$ -analogs.

## Chapter 24

# Computation, certification, and formal proof architecture

The identities in this book are well suited to exact computation because their natural objects are polynomials, rational functions, and formal power series. They are also well suited to proof assistants, provided algebraic identities are separated cleanly from analytic convergence. Lau, Lee, and Ono’s 2026 formalization project [21] demonstrates this architecture for  $q$ -Pochhammer symbols, Bailey pairs, Jacobi’s triple product, and the Rogers–Ramanujan identities.

### 24.1 Dynamic programming for Gaussian polynomials

A polynomial is represented here by its coefficient vector. Multiplication by  $q^h$  is therefore a shift by  $h$  places.

**Algorithm 24.1** (Pascal dynamic program). Given  $n \geq 0$  and  $0 \leq k \leq n$ :

1. Replace  $k$  by  $\min(k, n - k)$ , using symmetry.
2. Initialize  $G_0(q) = 1$  and  $G_s(q) = 0$  for  $1 \leq s \leq k$ .
3. For  $r = 1, 2, \dots, n$ , update in descending order

$$G_s(q) \leftarrow G_s(q) + q^{r-s} G_{s-1}(q) \quad (s = \min(r, k), \dots, 1).$$

4. Return  $G_k(q)$ .

**Theorem 24.2** (Correctness and complexity). *Algorithm 24.1 returns  $\begin{bmatrix} n \\ k \end{bmatrix}_q$ . If  $D = k(n - k)$  after the symmetry reduction, its coefficient-level arithmetic cost is  $O(nkD)$  and its storage cost is  $O(kD)$ .*

*Proof.* Let  $G_s^{(r)}$  denote the value after row  $r$ . We prove by induction on  $r$  that

$$G_s^{(r)} = \begin{bmatrix} r \\ s \end{bmatrix}_q \quad (0 \leq s \leq \min(r, k)).$$

The assertion is true at  $r = 0$ . Descending updates ensure that both  $G_s$  and  $G_{s-1}$  on the right are still values from row  $r - 1$ . Hence the update is

$$G_s^{(r)} = \begin{bmatrix} r-1 \\ s \end{bmatrix}_q + q^{r-s} \begin{bmatrix} r-1 \\ s-1 \end{bmatrix}_q = \begin{bmatrix} r \\ s \end{bmatrix}_q$$

by the second  $q$ -Pascal recurrence. At  $r = n$  this gives the desired output.

After replacing  $k$  by  $\min(k, n - k)$ , we have  $k \leq n/2$ . Every intermediate polynomial  $\begin{bmatrix} r \\ s \end{bmatrix}_q$  with  $s \leq k$  has degree  $s(r - s) \leq k(n - k) = D$ . There are at most  $nk$  shift-and-add updates, each touching at most  $D + 1$  coefficients. Storing  $k + 1$  such vectors uses  $O(kD)$  coefficients.  $\square$

**Remark 24.3** (Numerical specialization). When only a numerical value is required, one should not first expand the polynomial. For positive  $q$  near 1, the stable formula is

$$\log \left[ \begin{matrix} n \\ k \end{matrix} \right]_q = \sum_{j=1}^k [\log(-\text{expm1}((n-k+j)\log q)) - \log(-\text{expm1}(j\log q))],$$

where  $\text{expm1}$  avoids cancellation in  $e^x - 1$ . For exact symbolic work, the Pascal algorithm or the cyclotomic factorization is preferable.

## 24.2 A factorization algorithm

**Algorithm 24.4** (Cyclotomic factor list). For each  $1 \leq d \leq n$ , compute

$$e_d = \lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n-k)/d \rfloor.$$

Return the list of  $d$  for which  $e_d = 1$ .

**Theorem 24.5** (Correctness of the factor list). *The output of Algorithm 24.4 is exactly the set of cyclotomic factors of  $\left[ \begin{matrix} n \\ k \end{matrix} \right]_q$ , each with multiplicity one.*

*Proof.* This is Theorem 19.1: the exponent of  $\Phi_d(q)$  is  $e_d$ , and the floor calculation in that theorem proves  $e_d \in \{0, 1\}$ .  $\square$

**Remark 24.6.** The factor-list algorithm takes only  $O(n)$  integer floor operations. It is therefore far cheaper than expanding  $\left[ \begin{matrix} n \\ k \end{matrix} \right]_q$  when the purpose is root-of-unity evaluation, divisibility testing, or congruence work. Polynomial expansion may be postponed until the selected cyclotomic factors are actually multiplied.

## 24.3 Telescoping certificates for finite sums

A term  $T(n, k)$  is called *q-hypergeometric* if the ratios

$$\frac{T(n, k+1)}{T(n, k)}, \quad \frac{T(n+1, k)}{T(n, k)}$$

are rational functions of  $q^n$  and  $q^k$  (and of any fixed parameters). Shifted-factorial terms have this property because adjacent factors cancel.

**Theorem 24.7** (Finite telescoping certificate). *Let*

$$S_n = \sum_{k=L(n)}^{U(n)} T(n, k).$$

*Suppose there are functions  $A_0(n), \dots, A_r(n)$  and a certificate  $G(n, k)$  such that*

$$\sum_{j=0}^r A_j(n) T(n+j, k) = G(n, k+1) - G(n, k) \tag{20.1}$$

*for all relevant  $k$ , after extending  $T$  by zero outside its natural finite range. If  $G(n, k)$  vanishes at both ends of the resulting finite summation interval, then*

$$\sum_{j=0}^r A_j(n) S_{n+j} = 0. \tag{20.2}$$

*Proof.* Sum (20.1) over an interval containing the support of every  $T(n+j, k)$ . The left side becomes  $\sum_j A_j(n)S_{n+j}$ . The right side telescopes to the terminal value of  $G$  minus its initial value, which is 0 by hypothesis.  $\square$

**Corollary 24.8** (Identity certification). *If two finite sums satisfy the same recurrence of order  $r$  and agree for  $r$  consecutive initial values at every nonsingular parameter choice, then they are equal wherever the recurrence determines subsequent values.*

*Proof.* Subtract the two sequences. The difference satisfies the homogeneous recurrence and has  $r$  consecutive zero initial values. Induction through the recurrence gives zero thereafter.  $\square$

**Remark 24.9** (Creative telescoping). The  $q$ -Zeilberger method [25] searches algorithmically for a rational multiple  $G(n, k) = R(q^n, q^k)T(n, k)$  satisfying (20.1). Once found, the certificate is independently checkable by rational-function arithmetic. The search may be sophisticated, but verification is elementary: divide by  $T(n, k)$  and clear denominators.

## 24.4 Denominator clearing and specialization

**Lemma 24.10** (Polynomial identity principle). *Let  $R(q), S(q) \in \mathbb{Q}(q)$ . If  $R(q) = S(q)$  at infinitely many complex values of  $q$  where both sides are defined, then  $R = S$  in  $\mathbb{Q}(q)$ .*

*Proof.* Write  $R - S = A/B$  with  $A, B \in \mathbb{Q}[q]$  and  $B \neq 0$ . The hypothesis says that  $A$  has infinitely many zeros. A nonzero polynomial has only finitely many zeros, so  $A = 0$ .  $\square$

**Corollary 24.11** (Safe generic specialization). *A rational  $q$ -identity proved under generic nonvanishing assumptions remains valid after any specialization at which the denominator-cleared identity is defined. Apparent excluded values may therefore be recovered by cancellation or by taking limits.*

*Proof.* Clear all denominators to obtain a polynomial identity. Polynomial evaluation is valid at every specialization. If the original rational expressions have removable singularities, divide out common factors or take their uniquely determined limits.  $\square$

This principle is particularly important at roots of unity. One should first rewrite a Gaussian coefficient as its polynomial, not substitute a root into the factorial quotient and accept an indeterminate  $0/0$ .

## 24.5 Integer-exponent translation and automated identity discovery

A large class of Gaussian identities can be generated by starting from a terminating basic hypergeometric summation and replacing every shifted factorial whose parameter is an integral power of  $q$  by factorial blocks. This old method is now amenable to systematic symbolic search; the identity finder of Zhong and Zhao [45] organizes the necessary side conditions and backtracking choices. The proof certificate, however, remains elementary substitution.

**Lemma 24.12** (Two translation rules). *For nonnegative integers  $a, n, m, k$  with  $0 \leq k \leq m$ ,*

$$\frac{(q^{a+1}; q)_n}{(q; q)_n} = \left[ \begin{matrix} a+n \\ n \end{matrix} \right]_q, \quad (24.1)$$

$$\frac{(q^{-m}; q)_k}{(q; q)_k} = (-1)^k q^{-mk + \binom{k}{2}} \left[ \begin{matrix} m \\ k \end{matrix} \right]_q. \quad (24.2)$$

More generally,

$$(q^{a+1}; q)_n = \frac{(q; q)_{a+n}}{(q; q)_a}. \quad (24.3)$$



*Proof.* The factors in (24.3) are precisely  $1 - q^{a+1}, \dots, 1 - q^{a+n}$ . Division by  $(q; q)_n$  gives (24.1). For the negative rule, factor  $-q^{-m+j}$  from the  $j$ th factor:

$$\begin{aligned} (q^{-m}; q)_k &= \prod_{j=0}^{k-1} (1 - q^{-m+j}) \\ &= (-1)^k q^{-mk + \binom{k}{2}} \prod_{j=0}^{k-1} (1 - q^{m-j}) \\ &= (-1)^k q^{-mk + \binom{k}{2}} \frac{(q; q)_m}{(q; q)_{m-k}}. \end{aligned}$$

Divide by  $(q; q)_k$  to obtain (24.2).  $\square$

**Theorem 24.13** (Certified integral-parameter translation). *Consider a terminating identity in which every summand and the closed form are products and quotients of monomials in  $q$  and shifted factorials of the forms*

$$(q^{a+1}; q)_n, \quad (q^{-m}; q)_k, \quad (q; q)_r,$$

*with integer-valued parameters satisfying side conditions that keep all displayed finite indices nonnegative and all original denominators nonzero. Replacing the factors by (24.1)–(24.3), cancelling common factorial blocks, and collecting monomial powers of  $q$  produces an algebraically equivalent identity in Gaussian coefficients and residual  $q$ -factorials. If all residual factorial multiplicities cancel, the output is a pure  $q$ -binomial identity.*

*Proof.* Each replacement is an equality by Lemma 24.12. Products of equal rational functions are equal, cancellation of a common nonzero factor preserves equality under the stated generic assumptions, and collecting monomials is ordinary exponent arithmetic. Hence the transformed summand agrees term by term with the original summand, and the transformed closed form agrees with the original closed form. Summing the finitely many terms preserves equality. Finally, clearing denominators turns the result into a polynomial identity, so removable exceptional specializations are recovered by Corollary 24.11.  $\square$

**Remark 24.14** (Constraint generation). The nontrivial search problem is not verification but representation: the same factorial block can participate in several Gaussian coefficients, and the integer parameters must satisfy inequalities such as  $0 \leq k \leq m$ . An automated finder records the multiplicity of each  $(q; q)_r$ , searches for compatible triples

$$\frac{(q; q)_u}{(q; q)_v (q; q)_{u-v}} = \begin{bmatrix} u \\ v \end{bmatrix}_q,$$

and carries the corresponding linear inequalities as side conditions. Once a candidate decomposition is printed, Theorem 24.13 reduces its proof to exact algebra.

**Corollary 24.15** (Mechanically checkable certificates). *A computer-generated integral-parameter translation can be certified without trusting the search program: expand every Gaussian coefficient into factorial quotients, apply base reversal to every negative parameter, clear denominators, and compare the resulting Laurent monomials and finite products.*

*Proof.* The prescribed steps reverse the equalities in Lemma 24.12. After denominator clearing, equality is decidable by polynomial arithmetic. Therefore the search code may be treated as a conjecture generator while the emitted certificate is checked independently.  $\square$

*Remark 24.16* (Bailey inversion as a discovery engine). The same certificate-first philosophy makes Bailey-pair discovery deterministic. Begin with a plausible beta sequence, apply the triangular Bailey inversion of [Theorem 17.4](#) to compute its unique candidate alpha sequence, factor that expression, and test whether the result has a short theta-like form. If the final bilateral exponent is  $(Mn^2 + Bn)/2$ , the coefficient of  $n^2$  predicts product modulus  $M$ , while the linear term predicts the two nontrivial residue classes. This is a search heuristic, not a proof: the recovered pair must still be verified by the finite inversion identity, and every limiting transform must satisfy its convergence hypotheses. For finite positivity questions one should retain the Gaussian-polynomial form as long as possible, because reciprocity, unimodality, and coefficientwise signs can be obscured when an infinite limit is taken too early.

*Historical note 24.17.* Recent formal libraries reinforce the same separation of concerns. Eberl’s Isabelle development [41] formalizes Gaussian polynomials, finite and infinite  $q$ -Pochhammer symbols, Euler expansions, the  $q$ -binomial theorem, and  $q$ -Vandermonde. The Lean developments cited earlier and the Fabius-function repository [43] similarly distinguish denominator-free polynomial recurrences from analytic infinite products. This architecture is exactly what makes automated discovery compatible with proof-level reliability.

## 24.6 Formal series versus analytic series

**Theorem 24.18** (Formal-to-analytic transfer). *Suppose*

$$\sum_{n \geq 0} a_n z^n = \sum_{n \geq 0} b_n z^n$$

*coefficientwise in  $\mathbb{C}[[z]]$ . If both series converge for  $|z| < R$ , then they define the same analytic function on that disk. Conversely, if two power series converge in a disk and define the same function on a subset with an accumulation point in the disk, then their coefficients are equal.*

*Proof.* The forward implication is immediate because the two partial sums agree term by term. For the converse, the identity theorem makes the analytic functions equal throughout the disk; differentiating  $n$  times at 0 gives  $n!a_n = n!b_n$ .  $\square$

**Principle 24.19** (A scalable proof architecture). *For a mixed algebraic-analytic  $q$ -identity, the following order minimizes hidden assumptions:*

1. *define finite shifted factorials and prove their algebraic laws;*
2. *prove terminating identities after denominator clearing;*
3. *define formal infinite products by coefficient stabilization when possible;*
4. *separately establish analytic convergence for  $|q| < 1$ ;*
5. *transfer the formal coefficient identity to analytic functions;*
6. *only then take limits, differentiate parameters, or specialize at roots of unity.*

*Proof.* Each step is justified by results already proved: finite product algebra in [Chapter 1](#); polynomial identity and specialization by [Lemma 24.10](#) and [corollary 24.11](#); formal-to-analytic transfer by [Theorem 24.18](#); and the convergence theorems in [Chapters 8, 9](#) and [11](#). The ordering prevents an analytic operation from being used before its convergence hypothesis has been supplied.  $\square$

## 24.7 Necessary conditions for a proposed identity

The results of this book supply, besides proofs, a set of cheap falsification tests. They are worth isolating, because each is a *necessary* condition that can be checked without expanding anything.

**Principle 24.20** (Falsification tests). *Let  $L = R$  be a proposed identity whose two sides are built from Gaussian coefficients, shifted factorials, and explicit powers of  $q$ . Each of the following is a necessary condition for  $L = R$ .*

1. Degree and order. *If  $L$  and  $R$  are polynomials in  $q$ , then  $\deg L = \deg R$  and  $\text{ord}_{q=0} L = \text{ord}_{q=0} R$ . For a monomial  $F = q^\alpha \prod_i \begin{bmatrix} n_i \\ k_i \end{bmatrix}_q$  these are  $\alpha + \sum_i k_i(n_i - k_i)$  and  $\alpha$ , by [Theorem 2.5\(iii\)](#).*
2. Reversal. *Substituting  $q^{-1}$  for  $q$  in a true identity yields a true identity, and by [Theorem 2.5\(ii\),\(iv\)](#) that substitution acts on a monomial  $F$  as above by an explicit power:  $F(q^{-1}) = q^{-c_F} F(q)$  with  $c_F = \deg F + \text{ord } F$ . If  $L$  and  $R$  are such monomials with  $c_L \neq c_R$ , then  $L = R$  forces  $L = R = 0$ .*
3. The value  $q = 0$ . *After denominators have been cleared, the substitution  $q = 0$  must give a true numerical identity; here  $\begin{bmatrix} n \\ k \end{bmatrix}_q \big|_{q=0} = 1$  by [Proposition 22.11](#).*
4. The classical shadow. *After division by the appropriate power of  $1 - q$ , the limit  $q \rightarrow 1$  must be a true identity between ordinary binomial coefficients and factorials, since  $\begin{bmatrix} n \\ k \end{bmatrix}_q \rightarrow \binom{n}{k}$  by [Corollary 2.9](#); the standard scalings are collected in [Chapter C](#).*
5. Small indices. *Once the degree bound of item (i) is known, each side is determined by finitely many coefficients, so agreement at sufficiently many small indices is a genuine test; and if both sides satisfy one and the same certified recurrence, [Corollary 24.8](#) promotes finitely many agreements to a proof.*
6. Summand ratio. *If a side is presented as a basic hypergeometric series, the quotient of consecutive summands must be the rational function of  $q^k$  that the claimed  ${}_r\phi_s$  normalization predicts; this is the defining property of a  $q$ -hypergeometric term used in [Theorem 24.7](#). A mismatch localizes the error in the parameters rather than in the evaluation.*

*Proof.* Equal polynomials have equal degrees and equal orders at  $q = 0$ , and for a monomial  $F = q^\alpha \prod_i \begin{bmatrix} n_i \\ k_i \end{bmatrix}_q$  each factor has constant term 1 and degree  $k_i(n_i - k_i)$  by [Theorem 2.5\(iii\)](#); this proves item (i). For item (ii), [Theorem 2.5\(ii\)](#) gives  $\begin{bmatrix} n_i \\ k_i \end{bmatrix}_{q^{-1}} = q^{-k_i(n_i - k_i)} \begin{bmatrix} n_i \\ k_i \end{bmatrix}_q$ , so

$$F(q^{-1}) = q^{-\alpha} \prod_i q^{-k_i(n_i - k_i)} \begin{bmatrix} n_i \\ k_i \end{bmatrix}_q = q^{-c_F} F(q), \quad c_F = 2\alpha + \sum_i k_i(n_i - k_i) = \deg F + \text{ord } F,$$

using  $F(q) = q^\alpha \prod_i \begin{bmatrix} n_i \\ k_i \end{bmatrix}_q$ . If  $L = R$ , then applying the substitution gives  $q^{-c_L} L = q^{-c_R} R = q^{-c_R} L$ , so  $(q^{-c_L} - q^{-c_R})L = 0$  in  $\mathbb{Q}(q)$ ; when  $c_L \neq c_R$  the first factor is nonzero and hence  $L = 0$ . Items (iii) and (iv) are the specializations recorded in [Proposition 22.11](#) and [Corollary 2.9](#), legitimate after denominator clearing by [Lemma 24.10](#) and [Corollary 24.11](#). Item (v) is [Corollary 24.8](#). Item (vi) is the definition of a  $q$ -hypergeometric term given before [Theorem 24.7](#).  $\square$

**Remark 24.21.** The six conditions are necessary and not sufficient: passing all of them does not prove a candidate, and only the certified recurrence of item (v), or a proof organized as in [Principle 24.19](#), does. Their value is that every failure is cheap and localizing. A misplaced  $q^{\binom{k}{2}}$  or  $q^{k(n-k)}$  is caught by (i) and (ii) before anything is expanded, a wrong sign or normalization by (iii), and a wrong classical analogue by (iv). The algorithmic search for the certificate of item (v) is the subject of [25].

*Historical note 24.22.* Formal verification forces distinctions that handwritten mathematics often suppresses: a finite product indexed by a natural number is not the same object as a bilateral product; a polynomial Gaussian coefficient is not definitionally the same as a rational factorial quotient; and a formal power-series identity does not automatically contain a proof of analytic convergence. The Lean development of Lau, Lee, and Ono [21] turns these distinctions into reusable interfaces rather than obstacles.

## **Part VI**

# **Geometric filters, dyadic structure, and the Fabius bridge**

## Chapter 25

# Geometric interpolation and exact $q$ -filters

The geometric grid

$$1, q, q^2, \dots, q^n$$

is the natural interpolation grid for  $q$ -calculus. The Newton basis on this grid was developed in [Chapter 20](#). The complementary Lagrange viewpoint is equally useful: its weights are signed Gaussian coefficients, and every moment beyond the interpolation range is again a Gaussian coefficient. This gives exact coefficient filters, closed error formulas for geometric Richardson extrapolation, and the algebraic mechanism behind several constructions in the Fabius-function literature.

### 25.1 A reversed Gaussian row

For  $0 \leq k \leq n$  define

$$N_{n,k}(q) := (-1)^{n-k} q^{\binom{n-k+1}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q, \quad (25.1)$$

and put  $N_{n,k}(q) = 0$  outside this range. Although the definition is written with a quotient notation,  $N_{n,k}(q)$  is a polynomial in  $q$  with integral coefficients up to the displayed monomial and sign.

**Theorem 25.1** (Reversed finite  $q$ -binomial theorem). *For every  $n \geq 0$ ,*

$$\sum_{k=0}^n N_{n,k}(q) z^k = \prod_{j=1}^n (z - q^j). \quad (25.2)$$

Consequently, for every integer  $d$  with  $0 \leq d \leq n$ ,

$$\sum_{k=0}^n N_{n,k}(q) q^{kd} = \begin{cases} (q; q)_n, & d = 0, \\ 0, & 1 \leq d \leq n. \end{cases} \quad (25.3)$$

*Proof.* Write the product on the right as

$$\prod_{j=1}^n (z - q^j) = z^n \prod_{j=1}^n \left(1 - \frac{q^j}{z}\right) = z^n (q/z; q)_n.$$

The finite  $q$ -binomial theorem, [Theorem 3.1](#), gives

$$z^n (q/z; q)_n = \sum_{r=0}^n (-1)^r q^{\binom{r}{2}+r} \begin{bmatrix} n \\ r \end{bmatrix}_q z^{n-r}.$$

Put  $k = n - r$ . Since  $\begin{bmatrix} n \\ r \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q$  by symmetry and  $\binom{r}{2} + r = \binom{r+1}{2} = \binom{n-k+1}{2}$ , this is exactly (25.2).

Now substitute  $z = q^d$ . At  $d = 0$  the product is  $\prod_{j=1}^n (1 - q^j) = (q; q)_n$ . If  $1 \leq d \leq n$ , the factor with  $j = d$  vanishes. This proves (25.3).  $\square$

**Remark 25.2** (Denominator-free meaning). The identity (25.2) belongs to  $\mathbb{Z}[q, z]$  and therefore remains valid after mapping  $q$  into any commutative ring. No division by  $1 - q^j$  is needed. This is the robust way to use Gaussian coefficients when  $q$  is a root of unity or when the coefficient ring has zero divisors.

## 25.2 Lagrange weights at zero

Assume henceforth that  $q$  lies in a field  $K$  and that

$$q^i \neq q^j \quad (0 \leq i < j \leq n).$$

Equivalently, the nodes  $1, q, \dots, q^n$  are distinct. Define

$$\lambda_{n,k}^{(q)} := \frac{N_{n,k}(q)}{(q; q)_n} = \frac{(-1)^{n-k} q^{\binom{n-k+1}{2}}}{(q; q)_n} \begin{bmatrix} n \\ k \end{bmatrix}_q. \quad (25.4)$$

**Theorem 25.3** (Evaluation at zero on a geometric grid). *For every polynomial  $p \in K[X]$  of degree at most  $n$ ,*

$$p(0) = \sum_{k=0}^n \lambda_{n,k}^{(q)} p(q^k). \quad (25.5)$$

Thus  $\lambda_{n,k}^{(q)}$  are precisely the Lagrange interpolation weights for evaluation at 0 on the nodes  $1, q, \dots, q^n$ .

*Proof.* It is enough to verify (25.5) on the basis  $1, X, \dots, X^n$ . For  $p(X) = 1$ , (25.3) divided by  $(q; q)_n$  gives 1. For  $p(X) = X^d$  with  $1 \leq d \leq n$ , the same identity gives  $0 = p(0)$ . Linearity proves the formula for every polynomial of degree at most  $n$ .

The Lagrange interpolation functional at 0 is the unique linear combination of the point evaluations  $p \mapsto p(q^k)$  that agrees with  $p \mapsto p(0)$  on all polynomials of degree at most  $n$ : uniqueness follows from invertibility of the Vandermonde matrix of the distinct nodes. Hence the displayed coefficients are the Lagrange weights.  $\square$

**Corollary 25.4** (Explicit product form). *For  $0 \leq k \leq n$ ,*

$$\lambda_{n,k}^{(q)} = \prod_{\substack{0 \leq j \leq n \\ j \neq k}} \frac{-q^j}{q^k - q^j}. \quad (25.6)$$

*Proof.* The right side is the usual Lagrange basis polynomial associated with the node  $q^k$ , evaluated at 0. By uniqueness in Theorem 25.3, it equals (25.4). One can also simplify the product directly by splitting the factors with  $j < k$  and  $j > k$ .  $\square$

**Example 25.5.** For  $n = 2$  the weights are

$$\lambda_{2,0}^{(q)} = \frac{q^3}{(1-q)(1-q^2)}, \quad \lambda_{2,1}^{(q)} = -\frac{q(1+q)}{(1-q)(1-q^2)}, \quad \lambda_{2,2}^{(q)} = \frac{1}{(1-q)(1-q^2)}.$$

They satisfy

$$\sum_{k=0}^2 \lambda_{2,k}^{(q)} = 1, \quad \sum_{k=0}^2 \lambda_{2,k}^{(q)} q^k = 0, \quad \sum_{k=0}^2 \lambda_{2,k}^{(q)} q^{2k} = 0.$$

### 25.3 All residual moments

The vanishing moments in (25.3) are only the beginning. Every later moment has a closed form. The key fact is a universal remainder formula involving complete homogeneous symmetric functions.

**Theorem 25.6** (Universal Lagrange residual formula). *Let  $v_1, \dots, v_N$  be distinct elements of a field, let*

$$\ell_i(x) = \prod_{\substack{1 \leq j \leq N \\ j \neq i}} \frac{x - v_j}{v_i - v_j},$$

and let  $h_r$  denote the complete homogeneous symmetric polynomial, with  $h_0 = 1$ . For every  $r \geq 0$ ,

$$\sum_{i=1}^N \ell_i(x) v_i^{N+r} = x^{N+r} - \prod_{i=1}^N (x - v_i) h_r(x, v_1, \dots, v_N). \quad (25.7)$$

For exponents  $0 \leq d < N$ , the left side with  $N + r$  replaced by  $d$  is simply  $x^d$ .

*Proof.* The assertion for  $d < N$  is ordinary Lagrange interpolation. For  $r \geq 0$ , put

$$P(T) = \prod_{i=1}^N (T - v_i), \quad H_r(T) = T^{N+r} - P(T) h_r(T, v_1, \dots, v_N).$$

We first show that  $\deg H_r < N$ . Write  $e_j = e_j(v_1, \dots, v_N)$  and  $h_j = h_j(v_1, \dots, v_N)$ . Then

$$P(T) = \sum_{j=0}^N (-1)^j e_j T^{N-j}, \quad h_r(T, v_1, \dots, v_N) = \sum_{s=0}^r T^s h_{r-s}.$$

For  $0 \leq \ell \leq r$ , the coefficient of  $T^{N+r-\ell}$  in their product is

$$\sum_{j=0}^{\ell} (-1)^j e_j h_{\ell-j}.$$

By elementary-complete orthogonality, Corollary 4.4, this is 1 when  $\ell = 0$  and 0 for  $1 \leq \ell \leq r$ . Hence the terms of degrees  $N, N+1, \dots, N+r$  in  $P(T)h_r(T, v_1, \dots, v_N)$  agree with those of  $T^{N+r}$ , and therefore  $\deg H_r < N$ .

At every node  $v_i$  we have  $P(v_i) = 0$ , so  $H_r(v_i) = v_i^{N+r}$ . Since  $H_r$  has degree less than  $N$ , Lagrange interpolation gives

$$H_r(x) = \sum_{i=1}^N \ell_i(x) v_i^{N+r}.$$

Substituting the definition of  $H_r$  proves (25.7).  $\square$

**Theorem 25.7** (Complete geometric moment formula). *Under the distinct-node hypothesis, for every integer  $d \geq 0$ ,*

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} q^{kd} = \begin{cases} 1, & d = 0, \\ 0, & 1 \leq d \leq n, \\ (-1)^n q^{\binom{n+1}{2}} \begin{bmatrix} d-1 \\ n \end{bmatrix}_q, & d \geq n+1. \end{cases} \quad (25.8)$$

Equivalently, with the convention that a Gaussian coefficient is zero outside its natural range,

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} q^{kd} = (-1)^n q^{\binom{n+1}{2}} \begin{bmatrix} d-1 \\ n \end{bmatrix}_q \quad (d > 0). \quad (25.9)$$



*Proof.* The cases  $0 \leq d \leq n$  were proved in [Theorem 25.3](#). Let  $d = n + 1 + r$  with  $r \geq 0$ . Apply [Theorem 25.6](#) with target  $x = 0$  and nodes  $v_k = q^k$  for  $0 \leq k \leq n$ . There are  $N = n + 1$  nodes, and the weights  $\ell_k(0)$  are  $\lambda_{n,k}^{(q)}$ . Since

$$\prod_{k=0}^n (0 - q^k) = (-1)^{n+1} q^{\binom{n+1}{2}},$$

we obtain

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} q^{k(n+1+r)} = (-1)^n q^{\binom{n+1}{2}} h_r(1, q, \dots, q^n).$$

The geometric specialization in [Theorem 4.6](#) gives

$$h_r(1, q, \dots, q^n) = \begin{bmatrix} n+r \\ n \end{bmatrix}_q.$$

Because  $n + r = d - 1$ , this is the desired formula.  $\square$

**Corollary 25.8** (Scaled geometric nodes). *Let  $c \neq 0$  and use the same weights  $\lambda_{n,k}^{(q)}$  on the nodes  $cq^k$ . Then for  $d > 0$ ,*

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} (cq^k)^d = c^d (-1)^n q^{\binom{n+1}{2}} \begin{bmatrix} d-1 \\ n \end{bmatrix}_q.$$

*In particular, these weights evaluate every polynomial of degree at most  $n$  at 0 independently of the nonzero scale  $c$ .*

*Proof.* Factor  $c^d$  from the moment sum and apply [Theorem 25.7](#). Exactness on polynomials follows by linearity.  $\square$

## 25.4 Formal coefficient filters

For a formal power series  $f(X) = \sum_{m \geq 0} a_m X^m$ , define

$$(\mathcal{G}_{n,q} f)(X) := \sum_{k=0}^n \lambda_{n,k}^{(q)} f(q^k X). \quad (25.10)$$

The sum is finite, so this is well defined over any coefficient field in which the weights exist.

**Theorem 25.9** (Diagonal action and exact tail). *For every formal power series  $f(X) = \sum_{m \geq 0} a_m X^m$ ,*

$$[X^m] \mathcal{G}_{n,q} f = \left( \sum_{k=0}^n \lambda_{n,k}^{(q)} q^{km} \right) a_m, \quad (25.11)$$

$$\mathcal{G}_{n,q} f(X) = a_0 + (-1)^n q^{\binom{n+1}{2}} \sum_{r \geq 0} \begin{bmatrix} n+r \\ n \end{bmatrix}_q a_{n+1+r} X^{n+1+r}. \quad (25.12)$$

*Thus the filter deletes the coefficients of degrees  $1, \dots, n$  and multiplies every later coefficient by an explicit Gaussian polynomial.*

*Proof.* Substitution gives

$$\mathcal{G}_{n,q} f(X) = \sum_{k=0}^n \lambda_{n,k}^{(q)} \sum_{m \geq 0} a_m q^{km} X^m.$$

The outer sum is finite, so coefficient extraction may be interchanged with it, proving (25.11). Apply [Theorem 25.7](#): the multiplier is 1 at  $m = 0$ , is 0 for  $1 \leq m \leq n$ , and at  $m = n + 1 + r$  is  $(-1)^n q^{\binom{n+1}{2}} \begin{bmatrix} n+r \\ n \end{bmatrix}_q$ . This gives (25.12).  $\square$

**Corollary 25.10** (Geometric Richardson extrapolation). *Let  $f$  be analytic at 0. The values*

$$f(z), f(qz), \dots, f(q^n z)$$

*produce the extrapolation formula*

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} f(q^k z) = f(0) + O(z^{n+1}) \quad (z \rightarrow 0),$$

*and the complete asymptotic error is the convergent specialization of (25.12).*

*Proof.* Apply [Theorem 25.9](#) to the Taylor series of  $f$ . The first nonconstant surviving term has degree  $n + 1$ .  $\square$

**Theorem 25.11** (A quantitative analytic bound). *Assume  $0 < q < 1$  and  $f(z) = \sum_{m \geq 0} a_m z^m$  with  $|a_m| \leq M R^{-m}$ . If  $|z| < R$ , then*

$$\left| \sum_{k=0}^n \lambda_{n,k}^{(q)} f(q^k z) - f(0) \right| \leq \frac{M q^{\binom{n+1}{2}} (|z|/R)^{n+1}}{(q; q)_n (1 - |z|/R)}. \quad (25.13)$$

*Proof.* For  $r \geq 0$ ,

$$\begin{bmatrix} n+r \\ n \end{bmatrix}_q = \frac{(q^{r+1}; q)_n}{(q; q)_n} \leq \frac{1}{(q; q)_n},$$

because every factor in  $(q^{r+1}; q)_n$  lies in  $(0, 1)$ . Take absolute values in (25.12), use the coefficient bound, and sum the resulting geometric series.  $\square$

*Remark 25.12.* The exact multiplier in (25.12), rather than merely the order estimate  $O(z^{n+1})$ , is the principal advantage of a geometric grid. The complete tail is controlled by one Gaussian row. This phenomenon is used repeatedly in geometric extrapolation, in formal power-series filters, and in the dyadic identities of the next chapter.

## Chapter 26

# Half-base arithmetic and Prouhet–Thue–Morse cancellation

The specialization  $q = 1/2$  turns factors  $1 - q^j$  into normalized Mersenne numbers. It is especially important in exact formulas for the Fabius and Rvachev functions. At this base, the finite  $q$ -binomial theorem produces a compact  $n + 1$ -point Prouhet functional, while the classical Thue–Morse product produces a  $2^n$ -point functional. Both annihilate all polynomials of degree below  $n$ , but their weights and node geometries are very different.

### 26.1 Mersenne normalization

Define

$$\mathfrak{m}_0 := 1, \quad \mathfrak{m}_n := \prod_{j=1}^n (2^j - 1) \quad (n \geq 1). \quad (26.1)$$

**Theorem 26.1** (Half-base factorial and Gaussian formulas). *For  $n \geq 0$  and  $0 \leq k \leq n$ ,*

$$(1/2; 1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{m}_n, \quad (26.2)$$

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2} = 2^{-k(n-k)} \frac{\mathfrak{m}_n}{\mathfrak{m}_k \mathfrak{m}_{n-k}}. \quad (26.3)$$

*Proof.* Each factor satisfies

$$1 - 2^{-j} = 2^{-j} (2^j - 1).$$

Multiplication for  $1 \leq j \leq n$  gives (26.2), because  $1 + 2 + \cdots + n = \binom{n+1}{2}$ . Substitute this expression into

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2} = \frac{(1/2; 1/2)_n}{(1/2; 1/2)_k (1/2; 1/2)_{n-k}}.$$

The power of 2 in the quotient is

$$-\binom{n+1}{2} + \binom{k+1}{2} + \binom{n-k+1}{2} = -k(n-k),$$

which proves (26.3). □

**Theorem 26.2** (Dyadic evaluation of the finite product). *For integers  $m, n \geq 0$ ,*

$$(2^m; 1/2)_n = \begin{cases} 0, & m < n, \\ (-1)^n \frac{\mathfrak{m}_m}{\mathfrak{m}_{m-n}}, & m \geq n. \end{cases} \quad (26.4)$$

Equivalently,

$$\sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2} 2^{mk} = \begin{cases} 0, & m < n, \\ (-1)^n \frac{\mathbf{m}_m}{\mathbf{m}_{m-n}}, & m \geq n. \end{cases} \quad (26.5)$$

*Proof.* By definition,

$$(2^m; 1/2)_n = \prod_{j=0}^{n-1} (1 - 2^{m-j}).$$

If  $m < n$ , the factor with  $j = m$  is zero. If  $m \geq n$ , every factor is negative and

$$\prod_{j=0}^{n-1} (1 - 2^{m-j}) = (-1)^n \prod_{r=m-n+1}^m (2^r - 1) = (-1)^n \frac{\mathbf{m}_m}{\mathbf{m}_{m-n}}.$$

The finite  $q$ -binomial theorem at  $q = 1/2$  and  $z = 2^m$  turns the left side into the sum in (26.5).  $\square$

**Corollary 26.3** (Root locus). *The polynomial*

$$\sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2} z^k$$

*has the simple roots  $1, 2, 4, \dots, 2^{n-1}$  and no others.*

*Proof.* It is  $(z; 1/2)_n = \prod_{j=0}^{n-1} (1 - z/2^j)$ , a degree- $n$  polynomial with exactly the displayed roots.  $\square$

## 26.2 A compressed geometric Prouhet functional

Define nodes and weights

$$\xi_k := -2^k, \quad w_{n,k} := (-1)^k 2^{-\binom{k}{2}} \left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2} \quad (0 \leq k \leq n). \quad (26.6)$$

**Theorem 26.4** (Geometric Prouhet identity). *For every polynomial  $p$  of degree at most  $n$ ,*

$$\sum_{k=0}^n w_{n,k} p(\xi_k) = \mathbf{m}_n [X^n] p(X). \quad (26.7)$$

*In particular, the functional on the left annihilates every polynomial of degree less than  $n$ .*

*Proof.* It suffices to take  $p(X) = X^d$  for  $0 \leq d \leq n$ . With  $q = 1/2$ ,

$$\begin{aligned} \sum_{k=0}^n w_{n,k} \xi_k^d &= (-1)^d \sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \left[ \begin{matrix} n \\ k \end{matrix} \right]_q q^{-kd} \\ &= (-1)^d (q^{-d}; q)_n \end{aligned}$$

by the finite  $q$ -binomial theorem. If  $d < n$ , the factor with index  $j = d$  in  $(q^{-d}; q)_n$  is zero. At  $d = n$ , base reversal gives

$$(q^{-n}; q)_n = (-1)^n q^{-\binom{n+1}{2}} (q; q)_n.$$

Therefore

$$(-1)^n (q^{-n}; q)_n = 2^{\binom{n+1}{2}} (1/2; 1/2)_n = \mathbf{m}_n$$

by (26.2). Linearity now proves (26.7).  $\square$

**Corollary 26.5** (Affine form). *For constants  $a, b$  and every polynomial  $p$  of degree at most  $n$ ,*

$$\sum_{k=0}^n w_{n,k} p(a + b\xi_k) = b^n \mathbf{m}_n[X^n] p(X).$$

*Proof.* Apply [Theorem 26.4](#) to the polynomial  $X \mapsto p(a + bX)$ . Its leading coefficient is  $b^n[X^n]p(X)$ .  $\square$

*Remark 26.6.* The identity uses only  $n + 1$  geometrically spaced nodes. The price for this compression is that the weights are nonuniform rational numbers involving the half-base Gaussian row. The Thue–Morse construction below uses  $2^n$  equally weighted signs on an arithmetic grid instead.

## 26.3 The Thue–Morse product

For a nonnegative integer  $r$ , let  $\text{wt}_2(r)$  be the sum of the binary digits of  $r$  and put

$$\tau_r := (-1)^{\text{wt}_2(r)}.$$

This is the signed Thue–Morse sequence.

**Theorem 26.7** (Finite Thue–Morse product). *For every  $n \geq 0$ ,*

$$\sum_{r=0}^{2^n-1} \tau_r z^r = \prod_{j=0}^{n-1} (1 - z^{2^j}). \quad (26.8)$$

*Proof.* Every  $r$  with  $0 \leq r < 2^n$  has a unique binary expansion  $r = \sum_{j=0}^{n-1} \varepsilon_j 2^j$  with  $\varepsilon_j \in \{0, 1\}$ . Expanding the product on the right, choose either 1 or  $-z^{2^j}$  from the  $j$ th factor. The choice vector  $(\varepsilon_0, \dots, \varepsilon_{n-1})$  contributes

$$(-1)^{\varepsilon_0 + \dots + \varepsilon_{n-1}} z^{\varepsilon_0 + 2\varepsilon_1 + \dots + 2^{n-1}\varepsilon_{n-1}} = \tau_r z^r.$$

Summing over all binary choices gives the identity.  $\square$

**Theorem 26.8** (Prouhet cancellation and first surviving moment). *For  $n \geq 1$ ,*

$$\sum_{r=0}^{2^n-1} \tau_r r^d = \begin{cases} 0, & 0 \leq d < n, \\ (-1)^n 2^{\binom{n}{2}} n!, & d = n. \end{cases} \quad (26.9)$$

*More generally, for every polynomial  $p$  of degree at most  $n$ ,*

$$\sum_{r=0}^{2^n-1} \tau_r p(a + br) = (-1)^n 2^{\binom{n}{2}} n! b^n [X^n] p(X). \quad (26.10)$$

*Proof.* Let  $P_n(z) = \sum_{r < 2^n} \tau_r z^r$ . By [Theorem 26.7](#),  $P_n$  has a zero of order exactly  $n$  at  $z = 1$ , because every factor  $1 - z^{2^j}$  has a simple zero there. The Euler operator  $D = z \frac{d}{dz}$  satisfies

$$D^d P_n(1) = \sum_{r < 2^n} \tau_r r^d.$$

For  $d < n$ , this is zero because  $D^d$  is a linear combination of ordinary derivatives of orders at most  $d$ , all of which vanish at 1.

For  $d = n$ , the coefficient of the ordinary  $n$ th derivative in  $D^n$  is  $z^n$ , while all lower derivatives still vanish at 1. Hence

$$D^n P_n(1) = P_n^{(n)}(1).$$

The leading coefficient of the local expansion at 1 is the product of the first derivatives of the  $n$  factors:

$$P_n^{(n)}(1) = n! \prod_{j=0}^{n-1} \frac{d}{dz} (1 - z^{2^j}) \Big|_{z=1} = n! \prod_{j=0}^{n-1} (-2^j) = (-1)^n 2^{\binom{n}{2}} n!.$$

This proves (26.9). Apply it termwise to the polynomial  $X \mapsto p(a + bX)$ ; all terms below degree  $n$  vanish, and its leading coefficient is  $b^n [X^n]p(X)$ . This gives (26.10).  $\square$

**Corollary 26.9** (Equal power sums). *Partition  $\{0, 1, \dots, 2^n - 1\}$  according to the parity of  $\text{wt}_2(r)$ . The two classes have equal sums of  $d$ th powers for every  $0 \leq d < n$ .*

*Proof.* Move the negative-sign terms in (26.9) to the other side.  $\square$

## 26.4 Two cancellation mechanisms

**Principle 26.10** (Geometric versus binary Prouhet filters). *There are two complementary exact functionals of order  $n$ :*

$$\begin{aligned} \sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} p(-2^k) &= \mathfrak{m}_n[X^n]p, \\ \sum_{r=0}^{2^n-1} (-1)^{\text{wt}_2(r)} p(r) &= (-1)^n 2^{\binom{n}{2}} n! [X^n]p, \end{aligned}$$

*valid for  $\deg p \leq n$ . The first uses a Gaussian row on a geometric grid; the second uses uniform signs generated by binary digits on an arithmetic grid.*

*Proof.* These are Theorems 26.4 and 26.8.  $\square$

The common product mechanism is now visible. In the geometric case the annihilating polynomial is  $(q^{-d}; q)_n$  evaluated at integer  $d$ ; in the binary case it is the product  $\prod_{j < n} (1 - z^{2^j})$  and its order of vanishing at  $z = 1$ . Both encode a hierarchy of moment cancellations by forcing a product to acquire prescribed zeros.

## Chapter 27

# Geometric uniform sums, $q$ -cumulants, and the Fabius–Rvachev functions

A second bridge from  $q$ -products to the Fabius theory is probabilistic. Replace discrete binary digits by independent uniform coordinates and weight them geometrically. The resulting random series has a smooth compactly supported density, a Fourier transform given by an infinite product of sinc functions, and cumulants containing the factors  $1 - q^{2^m}$ . At  $q = 1/2$  its distribution function is the Fabius function; after the affine recentering and rescaling  $Z = 2X_{1/2} - 1$ , its density is the Rvachev up function [42, 44, 39, 43].

### 27.1 The geometric uniform random series

Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space, let  $0 < q < 1$ , and let  $U_0, U_1, \dots : \Omega \rightarrow [0, 1]$  be independent random variables with the uniform law on  $[0, 1]$ . Define

$$X_q := (1 - q) \sum_{j=0}^{\infty} q^j U_j. \quad (27.1)$$

**Theorem 27.1** (Convergence, support, symmetry, and self-similarity). *The series (27.1) converges absolutely for every outcome and satisfies  $0 \leq X_q \leq 1$ . Its topological support is  $[0, 1]$ , and*

$$\mathbb{E}X_q = \frac{1}{2}, \quad X_q \stackrel{d}{=} 1 - X_q. \quad (27.2)$$

If  $X'_q$  is an independent copy of  $X_q$ , then

$$X_q \stackrel{d}{=} (1 - q)U_0 + qX'_q. \quad (27.3)$$

Consequently its distribution function  $F_q(x) = \mathbb{P}(X_q \leq x)$  satisfies

$$F_q(x) = \int_0^1 F_q\left(\frac{x - (1 - q)u}{q}\right) du, \quad (27.4)$$

where  $F_q(y) = 0$  for  $y \leq 0$  and  $F_q(y) = 1$  for  $y \geq 1$ .

*Proof.* Because  $0 \leq U_j \leq 1$ ,

$$0 \leq (1 - q) \sum_{j=0}^{N-1} q^j U_j \leq (1 - q) \sum_{j=0}^{N-1} q^j \leq 1.$$

The tail is bounded by

$$(1 - q) \sum_{j=N}^{\infty} q^j = q^N,$$

so the series converges absolutely and takes values in  $[0, 1]$ .

Every  $x \in [0, 1]$  belongs to the support: if all of the first  $N$  variables lie sufficiently close to  $x$ , then their weighted partial sum lies close to  $x(1 - q^N)$ , and the remaining tail is at most  $q^N$ ; the corresponding finite cylinder has positive probability. No point outside  $[0, 1]$  can belong to the support.

Linearity of expectation and the geometric sum give

$$\mathbb{E}X_q = (1 - q) \sum_{j \geq 0} q^j \mathbb{E}U_j = \frac{1 - q}{2} \frac{1}{1 - q} = \frac{1}{2}.$$

Since  $1 - U_j$  are again independent uniform variables,

$$1 - X_q = (1 - q) \sum_{j \geq 0} q^j (1 - U_j)$$

has the same distribution as  $X_q$ . Splitting off the first term and reindexing the tail proves (27.3). Conditioning on  $U_0 = u$  then gives

$$\mathbb{P}((1 - q)u + qX'_q \leq x) = F_q\left(\frac{x - (1 - q)u}{q}\right),$$

and integration over  $u \in [0, 1]$  proves (27.4).  $\square$

## 27.2 Fourier product and cumulants

Use the entire function

$$\text{sinc}_{\text{rad}} z := \begin{cases} \frac{\sin z}{z}, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

**Theorem 27.2** (Characteristic function). *For real  $t$ ,*

$$\varphi_q(t) := \mathbb{E} e^{itX_q} = e^{it/2} \prod_{j=0}^{\infty} \text{sinc}_{\text{rad}}\left(\frac{(1 - q)q^j t}{2}\right). \quad (27.5)$$

*The product converges uniformly on compact subsets of the complex  $t$ -plane and therefore defines an entire function.*

*Proof.* For a uniform variable  $U$  on  $[0, 1]$  and any complex  $s$ ,

$$\mathbb{E} e^{sU} = \int_0^1 e^{su} du = e^{s/2} \frac{\sinh(s/2)}{s/2}.$$

At  $s = ita$  this becomes  $e^{ita/2} \text{sinc}_{\text{rad}}(at/2)$ . Independence gives the finite product for the partial sum. The phases combine because  $\sum_{j \geq 0} (1 - q)q^j = 1$ .

For compact  $t$ -sets, the arguments of the sinc factors are  $O(q^j)$ . Since  $\text{sinc}_{\text{rad}} z = 1 + O(z^2)$  near 0, the series of deviations from 1 is bounded by a constant multiple of  $\sum q^{2j}$ , uniformly on the compact set. The standard uniform convergence criterion for infinite products proves locally uniform convergence to an entire function. Passing to the limit in the finite characteristic functions gives (27.5).  $\square$



**Lean boundary for Theorem 27.2.** The real-frequency identity and the standalone analytic product are now machine-checked in complementary declarations. For  $q, z \in \mathbb{C}$  with  $\|q\| < 1$ , writing  $S_q(z) = \prod_{n \geq 0} \text{sinc}_{\text{rad}}(\pi q^n z)$ , the theorems `hasProdLocallyUniformly_geometricSincProduct`, `geometricSincProductFactors_multipliable`, `hasProd_geometricSincProduct`, and `geometricSincProduct_differentiable` prove locally uniform product convergence, genuine pointwise `Multipliable` and `HasProd` witnesses with value `geometricSincProduct`, and entire-ness. For real  $|q| < 1$ ,  $t \in \mathbb{R}$ , and  $z_q(t) = (1 - q)t/(2\pi)$ , including  $q = 0$  and negative contractions, `charFun_geometricUniformDistribution_eq_phase_mul_geometricSincProduct` and `charFun_geometricUniformDistribution_eq_phase_mul_geometricReciprocalGamma` give

$$\begin{aligned}\varphi_q(t) &= e^{it/2} S_q(z_q(t)) \\ &= e^{it/2} G_q(z_q(t)) G_q(-z_q(t)), \\ G_q(z) &= \text{geometricReciprocalGamma}(q, z).\end{aligned}$$

Moreover, `tendstoLocallyUniformly_prefix_sinc_charFun` proves local uniform convergence on the real frequency line of the full phase-bearing finite-prefix sequence used above, and `tendstoUniformlyOn_prefix_sinc_charFun` supplies the explicit uniform convergence statement on every compact  $K \subset \mathbb{R}$ . Under this monograph's exact-counterpart rule, the labelled theorem remains partial: no single declaration packages its complex- $t$  continuation together with complex-frequency locally uniform convergence of the full phase-bearing finite-prefix sequence. The real- $t$  formula, real-frequency local and compact uniform convergence, and the locally uniform entire sinc product are nevertheless exact subclaims, not merely dyadic specializations.

Let  $Y_q = X_q - 1/2$ . We use Bernoulli numbers in the convention

$$\frac{z}{e^z - 1} = \sum_{m=0}^{\infty} B_m \frac{z^m}{m!}. \quad (27.6)$$

The cumulants  $\kappa_m(Y_q)$  are defined by

$$\log \mathbb{E} e^{tY_q} = \sum_{m \geq 1} \kappa_m(Y_q) \frac{t^m}{m!}$$

near  $t = 0$ .

**Theorem 27.3** ( $q$ -cumulants of the geometric uniform sum). *All odd cumulants of  $Y_q$  vanish, and for  $m \geq 1$ ,*

$$\kappa_{2m}(Y_q) = \frac{B_{2m}}{2m} \frac{(1 - q)^{2m}}{1 - q^{2m}}. \quad (27.7)$$

*Equivalently,*

$$\log \mathbb{E} e^{tY_q} = \sum_{m=1}^{\infty} \frac{B_{2m}}{2m(2m)!} \frac{(1 - q)^{2m}}{1 - q^{2m}} t^{2m} \quad (27.8)$$

*for  $t$  in a neighborhood of 0.*

*Proof.* For  $U$  uniform on  $[0, 1]$ ,

$$\mathbb{E} e^{t(U-1/2)} = \frac{\sinh(t/2)}{t/2}.$$

Differentiating the logarithm gives

$$\frac{d}{dt} \log \frac{\sinh(t/2)}{t/2} = \frac{1}{2} \coth(t/2) - \frac{1}{t}.$$

The Bernoulli generating function implies the classical Laurent expansion

$$\frac{1}{2} \coth(t/2) - \frac{1}{t} = \sum_{m=1}^{\infty} B_{2m} \frac{t^{2m-1}}{(2m)!}.$$

Integrating from 0 to  $t$  yields

$$\log \frac{\sinh(t/2)}{t/2} = \sum_{m=1}^{\infty} \frac{B_{2m}}{2m(2m)!} t^{2m}. \quad (27.9)$$

Now

$$Y_q = (1 - q) \sum_{j \geq 0} q^j (U_j - 1/2).$$

Cumulant generating functions add under independence. Substitute  $t(1 - q)q^j$  into (27.9), sum over  $j$ , and use  $\sum_{j \geq 0} q^{2mj} = 1/(1 - q^{2m})$ . This proves (27.8); comparison of coefficients gives (27.7). Only even powers occur, so all odd cumulants vanish.  $\square$

**Definition 27.4** (Complete exponential Bell polynomials). The polynomials  $\mathcal{B}_n(x_1, \dots, x_n)$  are defined by

$$\exp \left( \sum_{r \geq 1} x_r \frac{t^r}{r!} \right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{t^n}{n!}. \quad (27.10)$$

**Corollary 27.5** (All moments). Let  $\kappa_r = \kappa_r(Y_q)$ , with the values in Theorem 27.3. Then

$$\mathbb{E}(Y_q^n) = \mathcal{B}_n(\kappa_1, \dots, \kappa_n). \quad (27.11)$$

The moments can also be generated recursively from  $\mu_0 = 1$  by

$$\mu_n = \sum_{r=1}^n \binom{n-1}{r-1} \kappa_r \mu_{n-r}, \quad \mu_n := \mathbb{E}(Y_q^n). \quad (27.12)$$

Finally,

$$\mathbb{E}(X_q^n) = \sum_{j=0}^n \binom{n}{j} 2^{j-n} \mu_j. \quad (27.13)$$

*Proof.* The moment generating function is the exponential of the cumulant generating function. Comparing it with the defining identity (27.10) proves (27.11). If  $M(t) = \sum \mu_n t^n / n!$  and  $K(t) = \sum \kappa_r t^r / r!$ , then  $M = e^K$  and  $M' = K' M$ . Comparing the coefficient of  $t^{n-1}$  gives (27.12). The last formula is the binomial expansion of  $(Y_q + 1/2)^n$ .  $\square$

**Corollary 27.6** (The half-base cumulants). At  $q = 1/2$ ,

$$\kappa_{2m}(Y_{1/2}) = \frac{B_{2m}}{2m(2^{2m} - 1)}, \quad \kappa_{2m+1}(Y_{1/2}) = 0. \quad (27.14)$$

Hence all moments of  $X_{1/2}$  are rational and are computable to arbitrary order by (27.12).

*Proof.* In (27.7),

$$\frac{(1 - 1/2)^{2m}}{1 - 2^{-2m}} = \frac{1}{2^{2m} - 1}.$$

The Bernoulli numbers are rational, so the recurrence uses only rational operations.  $\square$

### 27.3 Superpolynomial Fourier decay and smoothness

**Theorem 27.7** (Log-Gaussian decay in frequency). *Put*

$$A = \frac{1-q}{2}, \quad b = \log(1/q) > 0.$$

For sufficiently large  $|t|$ , with  $L = \log(A|t|)$ ,

$$|\varphi_q(t)| \leq \exp\left(-\frac{L^2}{2b} + \frac{L}{2}\right). \quad (27.15)$$

In particular, for every  $M \geq 0$ ,

$$|t|^M \varphi_q(t) \in L^1(\mathbb{R}).$$

*Proof.* The phase  $e^{it/2}$  has absolute value 1, and  $|\operatorname{sinc}_{\text{rad}} u| \leq \min(1, 1/|u|)$ . Choose

$$N = \lfloor L/b \rfloor + 1.$$

For  $0 \leq j < N$  we have  $A|t|q^j \geq 1$ . Hence

$$\begin{aligned} |\varphi_q(t)| &\leq \prod_{j=0}^{N-1} \frac{1}{A|t|q^j} \\ &= \exp\left(-NL + \frac{bN(N-1)}{2}\right). \end{aligned}$$

Because  $L \geq b(N-1)$ , the exponent is at most  $-bN(N-1)/2$ . For  $L$  sufficiently large,  $N \geq L/b$  and  $N-1 \geq L/b-1$ , so

$$-\frac{bN(N-1)}{2} \leq -\frac{L^2}{2b} + \frac{L}{2}.$$

This proves (27.15).

After the change of variables  $L = \log(At)$ , the right side is a Gaussian in  $L$  multiplied by  $e^{L/2}$ . It is therefore dominated by every negative power of  $t$ . More explicitly, for each  $M$  the exponent  $-L^2/(2b) + (M+3/2)L$  tends to  $-\infty$  quadratically, which proves integrability of  $|t|^M \varphi_q(t)$ .  $\square$

**Theorem 27.8** (Smooth compactly supported density). *For every  $0 < q < 1$ , the distribution of  $X_q$  has a  $C^\infty$  density  $f_q$  supported on  $[0, 1]$ . For every  $m \geq 0$ ,*

$$f_q^{(m)}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} (-it)^m e^{-itx} \varphi_q(t) dt. \quad (27.16)$$

All derivatives of  $f_q$  vanish at the endpoints 0 and 1.

*Proof.* By Theorem 27.7,  $t^m \varphi_q(t)$  is integrable for every  $m$ . Fourier inversion for a finite measure with integrable characteristic function gives the density

$$f_q(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} \varphi_q(t) dt.$$

For completeness, one may obtain this directly by convolving the distribution with a centered Gaussian of variance  $\varepsilon$ : its density is the inverse transform of  $\varphi_q(t)e^{-\varepsilon t^2/2}$ . Dominated convergence as  $\varepsilon \downarrow 0$  yields the displayed  $f_q$  uniformly. Against every compactly supported continuous test function, the Gaussian convolutions converge both to integration against  $f_q$  and weakly to the original measure; hence that measure has density  $f_q$ .

Because  $t^m \varphi_q(t)$  is integrable, differentiation under the integral is justified by dominated convergence and gives (27.16). Thus  $f_q \in C^\infty(\mathbb{R})$ . The random variable is supported on  $[0, 1]$ , so the density is zero on the two open complementary half-lines. Smoothness then forces every derivative to have value 0 at both boundary points.  $\square$

## 27.4 Finite spline approximants

Put

$$X_{q,N} := (1-q) \sum_{j=0}^{N-1} q^j U_j, \quad a_j := (1-q)q^j. \quad (27.17)$$

For  $x \in \mathbb{R}$  and an integer  $r \geq 0$ , write  $x_+^r = (\max(x, 0))^r$ ; when  $r = 0$ , interpret  $x_+^0$  as the indicator of  $x > 0$  away from the knots.

The following nonidentically scaled uniform-sum formula is a box-spline form of the generalized Irwin–Hall distribution; compare Bradley and Gupta [40].

**Theorem 27.9** (Generalized uniform-sum spline and the weighted cube cut). *Let  $N \geq 1$ , let  $c_0, \dots, c_{N-1} > 0$ , let  $U_0, \dots, U_{N-1}$  be independent and uniform on  $[0, 1]$ , and put  $S_{\mathbf{c}} := \sum_{j=0}^{N-1} c_j U_j$ . Then the density of  $S_{\mathbf{c}}$  is*

$$f_{\mathbf{c}}(x) = \frac{1}{(N-1)! \prod_{j=0}^{N-1} c_j} \sum_{A \subseteq \{0, \dots, N-1\}} (-1)^{|A|} \left( x - \sum_{j \in A} c_j \right)_+^{N-1}, \quad (27.18)$$

a compactly supported piecewise polynomial of degree at most  $N-1$ , and its distribution function is the volume of the corresponding section of the unit cube: for every  $t \in \mathbb{R}$ ,

$$V_{\mathbf{c}}(t) := \text{vol} \left\{ u \in [0, 1]^N : \sum_{j=0}^{N-1} c_j u_j \leq t \right\} = \frac{1}{N! \prod_{j=0}^{N-1} c_j} \sum_{A \subseteq \{0, \dots, N-1\}} (-1)^{|A|} \left( t - \sum_{j \in A} c_j \right)_+^N. \quad (27.19)$$

In particular, at the geometric weights  $c_j = a_j = (1-q)q^j$  of (27.17) the density of  $X_{q,N}$  is

$$f_{q,N}(x) = \frac{1}{(N-1)! \prod_{j=0}^{N-1} a_j} \sum_{A \subseteq \{0, \dots, N-1\}} (-1)^{|A|} \left( x - \sum_{j \in A} a_j \right)_+^{N-1}, \quad (27.20)$$

and its Fourier transform is the corresponding finite product in (27.5).

*Proof.* The density of  $c_j U_j$  is  $c_j^{-1} \mathbf{1}_{[0, c_j]}$ , so  $f_{\mathbf{c}}$  is the convolution of  $N$  box densities and, for  $s > 0$ , its Laplace transform is

$$\prod_{j=0}^{N-1} \frac{1 - e^{-c_j s}}{c_j s},$$

which uses nothing about the numbers  $c_j$  beyond  $c_j > 0$ . On the other hand

$$\int_0^\infty e^{-sx} \frac{(x-c)_+^{N-1}}{(N-1)!} dx = \frac{e^{-cs}}{s^N},$$

so the Laplace transform of the right-hand side of (27.18) is

$$\frac{1}{s^N \prod_j c_j} \sum_A (-1)^{|A|} e^{-s \sum_{j \in A} c_j} = \prod_{j=0}^{N-1} \frac{1 - e^{-c_j s}}{c_j s},$$

because the subset sum factors as  $\prod_j (1 - e^{-c_j s})$ . Uniqueness of the Laplace transform proves (27.18), and piecewise polynomiality is immediate from the truncated-power representation.

Since  $(U_0, \dots, U_{N-1})$  is uniform on the cube,  $V_{\mathbf{c}}(t) = \mathbb{P}(S_{\mathbf{c}} \leq t) = \int_{-\infty}^t f_{\mathbf{c}}(x) dx$ . The sum in (27.18) is finite, so it may be integrated term by term, and

$$\int_{-\infty}^t (x-c)_+^{N-1} dx = \frac{(t-c)_+^N}{N} \quad (N \geq 1)$$

gives (27.19). Specializing  $c_j = a_j$  gives (27.20), and the Fourier product follows either by transforming the box convolutions or by the finite version of the proof of Theorem 27.2.  $\square$

**Remark 27.10** (The dyadic instance). For the integer weights  $c_j = 2^{n-j}$ ,  $1 \leq j \leq n$  – that is,  $2^n$  times the half-base weights  $a_{j-1} = 2^{-j}$  of (27.17) at  $q = 1/2$  – the map  $A \mapsto r = \sum_{j \in A} 2^{n-j}$  is binary expansion, hence a bijection from the subsets of  $\{1, \dots, n\}$  onto  $\{0, 1, \dots, 2^n - 1\}$  with  $|A| = \text{wt}_2(r)$ ; consequently  $(-1)^{|A|} = \tau_r$  and  $\prod_{j=1}^n c_j = 2^{\binom{n}{2}}$ , so (27.19) becomes

$$V_{\mathbf{c}}(t) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{r=0}^{2^n-1} \tau_r (t-r)_+^n. \quad (27.21)$$

This is the cumulative counterpart of [Theorem 27.11](#), which records the same signs in the density. For  $0 \leq t \leq 1$  every term with  $r \geq 1$  vanishes and (27.21) reduces to  $t^n / (n! 2^{\binom{n}{2}})$ : that reduction is the simplex step of [Remark 28.11](#), and the terms it discards are exactly what makes (28.14) strict for  $q > 1/2$ .

**Theorem 27.11** (Dyadic Thue–Morse spline). *At  $q = 1/2$ ,*

$$f_{1/2,N}(x) = \frac{2^{\binom{N+1}{2}}}{(N-1)!} \sum_{r=0}^{2^N-1} (-1)^{\text{wt}_2(r)} \left(x - \frac{r}{2^N}\right)_+^{N-1}. \quad (27.22)$$

*Thus, after translating by the mean  $(1 - 2^{-N})/2$ , the inverse Fourier transform of the  $N$ th partial sinc product is a Thue–Morse signed spline on the dyadic grid.*

*Proof.* Here  $a_j = 2^{-j-1}$ , so

$$\prod_{j=0}^{N-1} a_j = 2^{-\binom{N+1}{2}}.$$

Every subset  $A$  corresponds to a binary word  $(\varepsilon_0, \dots, \varepsilon_{N-1})$  and

$$\sum_{j \in A} 2^{-j-1} = \sum_{j=0}^{N-1} \varepsilon_j 2^{-j-1}.$$

Reversing the order of the bits identifies this number with  $r/2^N$  for a unique  $0 \leq r < 2^N$ , while  $|A|$  and  $\text{wt}_2(r)$  have the same parity. Substitute these facts into (27.20).  $\square$

The constant  $\|f_q\|_\infty$  left implicit in (27.29) can be evaluated exactly. The mechanism is the fixed point (27.3) read as a convolution.

**Proposition 27.12** (Exact density on the central band, and the sharp sup bound). *Let  $0 < q < 1$  and let  $f_q$  be the density of [Theorem 27.8](#). Then:*

1. *for every  $x \in \mathbb{R}$ ,*

$$f_q(x) = \frac{1}{1-q} \left[ F_q\left(\frac{x}{q}\right) - F_q\left(\frac{x-1+q}{q}\right) \right]; \quad (27.23)$$

2.  $\|f_q\|_\infty \leq \frac{1}{1-q}$ ;
3. *if  $0 < q \leq 1/2$ , then*

$$f_q(x) = \frac{1}{1-q} F_q\left(\frac{x}{q}\right) \quad (0 \leq x \leq 1-q), \quad (27.24)$$

*so that  $f_q$  is identically  $1/(1-q)$  on the band  $[q, 1-q]$  and*

$$\|f_q\|_\infty = \frac{1}{1-q}; \quad (27.25)$$

4. if  $1/2 < q < 1$ , the inequality in (ii) is strict.

At  $q = 1/2$ , (27.23) is (27.34) and (27.24) is (27.35).

*Proof.* By (27.3) we may write  $X_q = (1 - q)U_0 + qX'_q$  with  $U_0$  uniform on  $[0, 1]$  independent of the copy  $X'_q$ . Conditionally on  $X'_q = y$  the variable  $X_q$  is uniform on  $[qy, qy + 1 - q]$ , whose density is the constant  $1/(1 - q)$ ; averaging over  $y$  gives

$$f_q(x) = \frac{1}{1 - q} \mathbb{P}(x - (1 - q) \leq qX'_q \leq x) \quad (27.26)$$

for almost every  $x$ , hence for every  $x$ : the left side is continuous by Theorem 27.8 and the right side is continuous because  $X'_q$  has no atoms. For the same reason the probability equals  $F_q(x/q) - F_q((x - 1 + q)/q)$ , which is (27.23), and a probability is at most 1, which is (ii).

Let  $q \leq 1/2$  and  $0 \leq x \leq 1 - q$ . Then  $x - (1 - q) \leq 0$ , so the event in (27.26) reduces to  $qX'_q \leq x$  and its probability is  $F_q(x/q)$ ; this is (27.24). If moreover  $x \geq q$  then  $x/q \geq 1$  and  $F_q(x/q) = 1$ , so  $f_q \equiv 1/(1 - q)$  on  $[q, 1 - q]$ , an interval that is nonempty precisely because  $q \leq 1/2$ ; with (ii) this gives (27.25).

Finally let  $q > 1/2$ . The window  $[x - (1 - q), x]$  has length  $1 - q < q$ , while  $qX'_q$  has support  $[0, q]$  by Theorem 27.1. An interval shorter than  $q$  cannot contain  $[0, q]$ , so its complement in  $[0, q]$  contains a nonempty relatively open set, of positive probability; hence the probability in (27.26) is  $< 1$  for every  $x$ . It is continuous and tends to 0 as  $|x| \rightarrow \infty$ , so its supremum is attained and is  $< 1$ , which is the strict inequality.  $\square$

**Corollary 27.13** (A coupling convergence bound, with its constant). *Almost surely,*

$$0 \leq X_q - X_{q,N} \leq q^N. \quad (27.27)$$

Consequently

$$F_q(x - q^N) \leq F_{q,N}(x) \leq F_q(x + q^N), \quad (27.28)$$

and in particular

$$\sup_x |F_{q,N}(x) - F_q(x)| \leq \|f_q\|_\infty q^N. \quad (27.29)$$

By Proposition 27.12,  $F_q$  is Lipschitz with constant  $1/(1 - q)$ , so

$$\sup_{x \in \mathbb{R}} |F_{q,N}(x) - F_q(x)| \leq \frac{q^N}{1 - q}, \quad (27.30)$$

and for  $0 < q \leq 1/2$  the constant  $1/(1 - q)$  is the exact Lipschitz constant of  $F_q$ . At  $q = 1/2$  this reads  $\|f\|_\infty = 2$  and

$$\sup_{x \in \mathbb{R}} |F_{1/2,N}(x) - F(x)| \leq 2^{1-N}. \quad (27.31)$$

*Proof.* The omitted tail is nonnegative and bounded by  $(1 - q) \sum_{j \geq N} q^j = q^N$ , which is (27.27). The inclusions

$$\{X_q \leq x - q^N\} \subseteq \{X_{q,N} \leq x\} \subseteq \{X_q \leq x + q^N\}$$

give (27.28). Since  $F_q$  has a bounded continuous density,  $|F_q(x + h) - F_q(x)| \leq \|f_q\|_\infty |h|$ ; applying this on both sides gives (27.29), and Proposition 27.12(ii) turns it into (27.30). Exactness of the Lipschitz constant for  $q \leq 1/2$  is (27.25), and  $q = 1/2$  gives (27.31).  $\square$

**Remark 27.14** (The two normalizations at  $q = 1/2$ ). The value  $\|f\|_\infty = 2$  is the Rvachev normalization seen through Proposition 28.1:  $f(x) = 2 \operatorname{up}(2x - 1) = 2F(1 - |2x - 1|)$  on  $[0, 1]$ , so  $\|\operatorname{up}\|_\infty = \operatorname{up}(0) = F(1) = 1$  and the probability normalization  $\int_{\mathbb{R}} \operatorname{up} = 1$  agrees with the classical normalization  $\operatorname{up}(0) = 1$ .

## 27.5 The Fabius and Rvachev specializations

Set  $q = 1/2$ , write  $X = X_{1/2}$ , and let

$$F(x) := \mathbb{P}(X \leq x),$$

extended by  $F(x) = 0$  for  $x \leq 0$  and  $F(x) = 1$  for  $x \geq 1$ .

**Theorem 27.15** (The probabilistic Fabius equation). *The function  $F$  is  $C^\infty$ , satisfies*

$$F(1 - x) = 1 - F(x), \quad (27.32)$$

$$F(x) = \int_{2x-1}^{2x} F(u) du, \quad (27.33)$$

$$F'(x) = 2F(2x) - 2F(2x - 1). \quad (27.34)$$

In particular, for  $0 \leq x \leq 1/2$ ,

$$F'(x) = 2F(2x). \quad (27.35)$$

Together with  $F(0) = 0$ ,  $F(1) = 1$ , and symmetry, this is the standard Fabius function.

*Proof.* Smoothness follows from Theorem 27.8. Since  $X$  has no atoms and  $X \stackrel{d}{=} 1 - X$ ,

$$F(1 - x) = \mathbb{P}(1 - X \leq 1 - x) = \mathbb{P}(X \geq x) = 1 - F(x),$$

which proves (27.32).

At  $q = 1/2$ , the fixed-point equation is  $X \stackrel{d}{=} (U + X')/2$ . Conditioning on  $U = u$  gives

$$F(x) = \int_0^1 F(2x - u) du = \int_{2x-1}^{2x} F(v) dv,$$

proving (27.33). Differentiate using the fundamental theorem of calculus to obtain (27.34). On  $0 \leq x \leq 1/2$  the term  $F(2x - 1)$  is zero, giving (27.35).  $\square$

Let  $f = F'$  and define the centered density

$$\text{up}(y) := \begin{cases} \frac{1}{2} f\left(\frac{y+1}{2}\right), & -1 \leq y \leq 1, \\ 0, & |y| > 1. \end{cases} \quad (27.36)$$

This is the Rvachev up function in the probability normalization  $\int_{\mathbb{R}} \text{up}(y) dy = 1$ .

**Theorem 27.16** (Infinite sinc product and the up equation). *With the Fourier convention  $\widehat{g}_{\text{ang}}(-t) = \int_{\mathbb{R}} e^{ity} g(y) dy$ ,*

$$\widehat{\text{up}}_{\text{ang}}(-t) = \prod_{k=1}^{\infty} \text{sinc}_{\text{rad}}\left(\frac{t}{2^k}\right). \quad (27.37)$$

Moreover,

$$\text{up}'(y) = 2 \text{up}(2y + 1) - 2 \text{up}(2y - 1) \quad (y \in \mathbb{R}). \quad (27.38)$$

*Proof.* The random variable with density  $\text{up}$  is  $Y = 2X - 1$ . Writing  $V_j = 2U_j - 1$ , we have independent uniform variables on  $[-1, 1]$  and

$$Y = \sum_{j=0}^{\infty} 2^{-j-1} V_j.$$

The characteristic function of  $V_j$  is  $\text{sinc}_{\text{rad}} t$ . Independence therefore gives

$$\mathbb{E}e^{itY} = \prod_{j=0}^{\infty} \text{sinc}_{\text{rad}}(t/2^{j+1}),$$

which is (27.37).

From (27.36),  $f(x) = 2 \text{up}(2x - 1)$ . Differentiate (27.34) to obtain

$$f'(x) = 4f(2x) - 4f(2x - 1).$$

Set  $x = (y + 1)/2$ . Because  $\text{up}'(y) = f'((y + 1)/2)/4$ , substitution of  $f(t) = 2 \text{up}(2t - 1)$  gives

$$\text{up}'(y) = 2 \text{up}(2y + 1) - 2 \text{up}(2y - 1).$$

The identity remains valid outside the support because all boundary derivatives vanish.  $\square$

*Remark 27.17* (Where the  $q$ -objects enter). The parameter  $q$  controls four structures at once:

1. the geometric coordinate weights  $(1 - q)q^j$ ;
2. the factors  $1 - q^{2^m}$  in the cumulants (27.7);
3. the geometric interpolation nodes and Gaussian residual moments of Chapter 25;
4. at  $q = 1/2$ , the Mersenne products, dyadic Thue–Morse splines, and the Fabius–Rvachev functional equations.

Thus the appearance of  $q$ -Pochhammer and Gaussian coefficients in exact Fabius formulas is not accidental: both arise from the same geometric scaling law.



## Chapter 28

# Dyadic Gaussian–Thue–Morse structure and the values of $F$

The three preceding chapters built the geometric filters, the half-base arithmetic and the geometric uniform law separately. This chapter runs them together in the one case where everything is exactly computable: the base  $q = 1/2$ , where the geometric uniform sum is the Fabius variable, the Prouhet filter is the Thue–Morse sign, and the resulting closed formula evaluates  $F$  at every inverse power of two.

**Proposition 28.1** (The up function is a Fabius tail). *For  $|y| \leq 1$ ,*

$$\text{up}(y) = F(1 - |y|), \quad (28.1)$$

*and in particular, for  $0 \leq t \leq 1$ ,*

$$\text{up}(t) = 1 - F(t) = \mathbb{P}(X \geq t). \quad (28.2)$$

*Proof.* Let  $-1 \leq y \leq 0$  and put  $x = (y + 1)/2 \in [0, 1/2]$ . By (27.35),  $f(x) = 2F(2x) = 2F(y + 1)$ , so (27.36) gives  $\text{up}(y) = \frac{1}{2}f(x) = F(1 + y) = F(1 - |y|)$ .

Differentiating the symmetry (27.32) gives  $f(1 - x) = f(x)$ , so  $\text{up}(-y) = \frac{1}{2}f((1 - y)/2) = \frac{1}{2}f((1 + y)/2) = \text{up}(y)$ ; thus  $\text{up}$  is even and (28.1) holds on all of  $[-1, 1]$ . Finally, for  $0 \leq t \leq 1$  the symmetry (27.32) turns  $F(1 - t)$  into  $1 - F(t)$ , and  $X$  has a density, so  $1 - F(t) = \mathbb{P}(X > t) = \mathbb{P}(X \geq t)$ .  $\square$

**Corollary 28.2** (Tail form of the moments). *For every  $n \geq 1$ ,*

$$\mathbb{E}X^n = n \int_0^1 t^{n-1} \text{up}(t) \, dt. \quad (28.3)$$

*Proof.* Since  $X \geq 0$ , Tonelli's theorem applied to the nonnegative integrand  $nt^{n-1}\mathbf{1}_{\{t < X\}}$  gives

$$\mathbb{E}X^n = \mathbb{E} \int_0^X nt^{n-1} \, dt = \int_0^1 nt^{n-1} \mathbb{P}(X \geq t) \, dt,$$

because  $0 \leq X \leq 1$ . Now apply (28.2).  $\square$

Write

Write

$$\Lambda(z) := \int_0^1 e^{zu} \, du = \frac{e^z - 1}{z}, \quad \Lambda(0) := 1, \quad (28.4)$$

an entire function with  $\Lambda(z) = \sum_{r \geq 0} z^r / (r+1)!$  and  $\Lambda(z) = 1 + \mathcal{O}(z)$  at the origin, whose zeros are exactly the points  $2\pi i k$  with  $k \in \mathbb{Z} \setminus \{0\}$ ; in particular  $\Lambda$  has no zero in the disc  $|z| < 2\pi$ . For  $q \in \mathbb{C}$  with  $|q| < 1$  set

$$\mathcal{A}_q(z) := \prod_{j=0}^{\infty} \Lambda((1-q)q^j z) = \sum_{n \geq 0} \alpha_n^{(q)} z^n. \quad (28.5)$$

**Theorem 28.3** (Moment product, dilation equation, coefficient recurrence). *Let  $q \in \mathbb{C}$  with  $|q| < 1$ . Then:*

1. *the product in (28.5) converges locally uniformly on  $\mathbb{C}$  and defines an entire function with  $\mathcal{A}_q(0) = 1$ ;*
2.  *$\mathcal{A}_q$  satisfies the dilation equation*

$$\mathcal{A}_q(z) = \Lambda((1-q)z) \mathcal{A}_q(qz), \quad (28.6)$$

*and is the unique formal power series with constant term 1 that does so; its coefficients satisfy  $\alpha_0^{(q)} = 1$  and*

$$(1 - q^n) \alpha_n^{(q)} = \sum_{r=0}^{n-1} \frac{(1-q)^{n-r} q^r}{(n-r+1)!} \alpha_r^{(q)} \quad (n \geq 1); \quad (28.7)$$

3. *consequently each  $\alpha_n^{(q)}$  is a rational function of  $q$  with rational coefficients, its poles are roots of unity, and it is regular on the open unit disc;*
4. *for real  $0 < q < 1$  the function  $\mathcal{A}_q$  is the moment generating function of  $X_q$ :*

$$\mathcal{A}_q(z) = \mathbb{E} e^{zX_q} = \prod_{j=0}^{\infty} \Lambda((1-q)q^j z), \quad \alpha_n^{(q)} = \frac{\mathbb{E} X_q^n}{n!}, \quad (28.8)$$

*and  $\varphi_q(t) = \mathcal{A}_q(it)$  for every real  $t$ .*

*Proof.* (i) On a compact set  $K \subset \mathbb{C}$  the arguments  $(1-q)q^j z$  are  $\mathcal{O}(|q|^j)$  uniformly, and  $\Lambda(w) - 1 = \mathcal{O}(w)$  near the origin, so  $\sum_{j \geq 0} \sup_{z \in K} |\Lambda((1-q)q^j z) - 1|$  is dominated by a constant multiple of  $\sum_{j \geq 0} |q|^j < \infty$ . The standard criterion for infinite products gives locally uniform convergence to an entire function, and every factor equals 1 at  $z = 0$ .

(ii) Replacing  $z$  by  $qz$  shifts the index by one:

$$\Lambda((1-q)z) \mathcal{A}_q(qz) = \Lambda((1-q)z) \prod_{j \geq 0} \Lambda((1-q)q^{j+1}z) = \prod_{j \geq 0} \Lambda((1-q)q^j z) = \mathcal{A}_q(z),$$

which is (28.6). Comparing the coefficient of  $z^n$  in

$$\sum_{n \geq 0} \alpha_n^{(q)} z^n = \left( \sum_{s \geq 0} \frac{(1-q)^s z^s}{(s+1)!} \right) \left( \sum_{r \geq 0} \alpha_r^{(q)} q^r z^r \right)$$

gives  $\alpha_n^{(q)} = \sum_{r=0}^n (1-q)^{n-r} q^r \alpha_r^{(q)} / (n-r+1)!$ ; the term  $r = n$  on the right is  $q^n \alpha_n^{(q)}$ , and moving it to the left proves (28.7). Since  $1 - q^n \neq 0$  for  $n \geq 1$ , that recurrence determines every coefficient from its predecessors, which is the asserted formal uniqueness.

(iii) Induct on (28.7). One has  $\alpha_0^{(q)} = 1$ , and  $\alpha_n^{(q)}$  is obtained from  $\alpha_0^{(q)}, \dots, \alpha_{n-1}^{(q)}$  by rational operations with the single new denominator  $1 - q^n$ . Hence  $\alpha_n^{(q)} \in \mathbb{Q}(q)$  and its poles lie among the zeros of  $\prod_{j=1}^n (1 - q^j)$ , all of which are roots of unity, and none of those lies in  $|q| < 1$ .

(iv) Let  $0 < q < 1$ . By [Theorem 27.1](#) we have  $0 \leq X_q \leq 1$ , so  $\mathbb{E}X_q^n \leq 1$  and  $\sum_n |z|^n \mathbb{E}X_q^n/n! \leq e^{|z|}$ ; hence  $z \mapsto \mathbb{E}e^{zX_q}$  is entire and term-by-term integration is legitimate. For the truncation  $X_{q,N}$  of [\(27.17\)](#), independence gives

$$\mathbb{E}e^{zX_{q,N}} = \prod_{j=0}^{N-1} \int_0^1 e^{z(1-q)q^j u} du = \prod_{j=0}^{N-1} \Lambda((1-q)q^j z).$$

By [\(27.27\)](#),  $0 \leq X_q - X_{q,N} \leq q^N$  pointwise, so  $e^{zX_{q,N}} \rightarrow e^{zX_q}$  uniformly on the sample space while remaining bounded by  $e^{|z|}$ ; dominated convergence identifies the limit with  $\mathcal{A}_q(z)$  and proves [\(28.8\)](#). Substituting  $z = it$  reproduces  $\varphi_q$ , and comparison of Taylor coefficients gives  $\alpha_n^{(q)} = \mathbb{E}X_q^n/n!$ .  $\square$

*Remark 28.4* (Dilation, refinement, Mahler). Equation [\(28.6\)](#) carries three names in three literatures: it is a *dilation* or *refinement* equation in approximation theory and wavelet analysis, and a *Mahler functional equation* in transcendence theory and in the theory of automatic sequences. This book uses them interchangeably. Part (iii) is visible already in the first cases, which [\(28.7\)](#) produces as

$$\alpha_1^{(q)} = \frac{1}{2}, \quad \alpha_2^{(q)} = \frac{q+2}{12(q+1)}, \quad \alpha_3^{(q)} = \frac{1}{24(q+1)};$$

the half-base values are collected in [Example 28.6](#), and the systematic clearing of the denominators is [Definition 28.61](#).

**Corollary 28.5** (The half-base moment generating function). *Write  $X := X_{1/2}$ ,  $\mathcal{A} := \mathcal{A}_{1/2}$  and  $\alpha_n := \alpha_n^{(1/2)} = \mathbb{E}X^n/n!$ . Then  $\mathcal{A}$  is entire,  $\mathcal{A}(it) = \varphi_{1/2}(t)$ ,*

$$\mathcal{A}(z) = \prod_{j=1}^{\infty} \Lambda\left(\frac{z}{2^j}\right), \quad (28.9)$$

$$\mathcal{A}(2z) = \Lambda(z) \mathcal{A}(z), \quad (28.10)$$

and  $\alpha_0 = 1$  with

$$(2^n - 1) \alpha_n = \sum_{r=0}^{n-1} \frac{\alpha_r}{(n-r+1)!} \quad (n \geq 1). \quad (28.11)$$

*Proof.* Put  $q = 1/2$  in [Theorem 28.3](#). The product [\(28.8\)](#) becomes  $\prod_{j \geq 0} \Lambda(2^{-1}2^{-j}z)$ , which is [\(28.9\)](#); replacing  $z$  by  $2z$  in [\(28.6\)](#) gives [\(28.10\)](#). In [\(28.7\)](#) every summand acquires the factor  $(1/2)^{n-r}(1/2)^r = 2^{-n}$  and the left side is  $(1 - 2^{-n})\alpha_n$ ; multiplication by  $2^n$  gives [\(28.11\)](#). Finally  $\Lambda(is) = e^{is/2} \text{sinc}_{\text{rad}}(s/2)$ , so [\(28.9\)](#) at  $z = it$  is [\(27.5\)](#) at  $q = 1/2$ ; this is the sinc product of [Theorem 27.16](#).  $\square$

**Example 28.6** (The first coefficients and the first dyadic values). Recurrence [\(28.11\)](#) gives

$$\alpha_0 = 1, \quad \alpha_1 = \frac{1}{2}, \quad \alpha_2 = \frac{5}{36}, \quad \alpha_3 = \frac{1}{36}, \quad \alpha_4 = \frac{143}{32400}, \quad \alpha_5 = \frac{19}{32400},$$

equivalently  $\mathbb{E}X^2 = 2!\alpha_2 = 5/18$  and  $\mathbb{E}X^4 = 4!\alpha_4 = 143/1350$ . These agree with the cumulant route of [Corollaries 27.5](#) and [27.6](#): for instance  $\kappa_2(Y_{1/2}) = B_2/(2 \cdot 3) = 1/36$ , so  $\mathbb{E}X^2 = \frac{1}{36} + \frac{1}{4} = \frac{5}{18}$ .

By [Theorem 28.9](#) below,  $F(2^{-n}) = 2^{-\binom{n}{2}} \alpha_n$ , so the same six rationals are the first inverse-dyadic values of the Fabius function:

$$F(1) = 1, \quad F\left(\frac{1}{2}\right) = \frac{1}{2}, \quad F\left(\frac{1}{4}\right) = \frac{5}{72}, \quad F\left(\frac{1}{8}\right) = \frac{1}{288},$$

$$F\left(\frac{1}{16}\right) = \frac{143}{2073600}, \quad F\left(\frac{1}{32}\right) = \frac{19}{33177600}.$$

The value  $F(1/2) = 1/2$  is forced by [\(27.32\)](#); the remaining entries are the first concrete instances of the rationality asserted, without values, by [Corollary 27.6](#). The powers  $2^{-\binom{n}{2}}$  are what makes the values fall off so much faster than the coefficients.

**Remark 28.7** (Two routes to the same rationals). Unlike the moment–cumulant recursion (27.12), which is stated for the centred variable and needs Bernoulli input, the recursion (28.11) is a triangular rational recursion in the raw coefficients alone: one rational division per index, no Bernoulli numbers and no Bell polynomials. It is the cheaper and the numerically stable route to the values below.

**Lemma 28.8** (Derivative iterates of the Fabius function). *For every  $m \geq 0$ ,*

$$F^{(m)}(x) = 2^{\binom{m+1}{2}} F(2^m x) \quad (0 \leq x \leq 2^{-m}). \quad (28.12)$$

*In particular  $F^{(m)}(0) = 0$  for every  $m \geq 0$ .*

*Proof.* The case  $m = 0$  is trivial, and (27.35) is the case  $m = 1$ . Assume (28.12) for  $m$  and let  $0 \leq x \leq 2^{-m-1}$ , so that  $0 \leq 2^m x \leq 1/2$ . Differentiating and applying (27.35) at  $2^m x$  gives

$$F^{(m+1)}(x) = 2^{\binom{m+1}{2}} 2^m F'(2^m x) = 2^{\binom{m+1}{2} + m + 1} F(2^{m+1} x),$$

and  $\binom{m+1}{2} + m + 1 = \binom{m+2}{2}$ . Since  $F(0) = 0$ , evaluating (28.12) at  $x = 0$  gives  $F^{(m)}(0) = 0$ .  $\square$

**Theorem 28.9** (Half-moment evaluation). *For every  $n \geq 0$ ,*

$$F(2^{-n}) = 2^{-\binom{n}{2}} \alpha_n = \frac{\mathbb{E} X^n}{n! 2^{\binom{n}{2}}}. \quad (28.13)$$

*Proof.* For  $n = 0$  both sides are  $F(1) = 1$ ; let  $n \geq 1$ . By Lemma 28.8 we have  $F^{(j)}(0) = 0$  for  $0 \leq j < n$ , and  $F \in C^\infty$  by Theorem 27.15, so Taylor's formula with integral remainder at the point  $2^{-n}$  reduces to the remainder alone:

$$F(2^{-n}) = \frac{1}{(n-1)!} \int_0^{2^{-n}} (2^{-n} - t)^{n-1} F^{(n)}(t) dt = \frac{2^{\binom{n+1}{2}}}{(n-1)!} \int_0^{2^{-n}} (2^{-n} - t)^{n-1} F(2^n t) dt,$$

where (28.12) was used on the whole interval of integration. Substituting  $u = 2^n t$  produces the factor  $2^{-n(n-1)}$  from  $(2^{-n} - t)^{n-1}$  and the factor  $2^{-n}$  from  $dt$ , and

$$\binom{n+1}{2} - n(n-1) - n = \frac{n(n+1)}{2} - n^2 = -\binom{n}{2},$$

so that

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{(n-1)!} \int_0^1 (1-u)^{n-1} F(u) du.$$

By Proposition 28.1,  $F(u) = \text{up}(1-u)$  for  $0 \leq u \leq 1$ ; substituting  $t = 1-u$  and applying Corollary 28.2,

$$\frac{1}{(n-1)!} \int_0^1 t^{n-1} \text{up}(t) dt = \frac{\mathbb{E} X^n}{n(n-1)!} = \alpha_n. \quad \square$$

**Proposition 28.10** (Inverse-power values at a general base). *Let  $0 < q < 1$  and  $n \geq 0$ . Then*

$$F_q(q^n) \leq \frac{q^{\binom{n+1}{2}}}{(1-q)^n n!} \mathbb{E}(X_q^n), \quad (28.14)$$

*with equality for every  $n$  if and only if  $q \leq 1/2$ ; for  $q > 1/2$  and  $n \geq 1$  the inequality is strict. Thus, for  $0 < q \leq 1/2$ ,*

$$F_q(q^n) = \frac{q^{\binom{n+1}{2}}}{(1-q)^n n!} \mathbb{E}(X_q^n) = \frac{q^{\binom{n+1}{2}}}{(1-q)^n} \alpha_n^{(q)}, \quad (28.15)$$

*with  $\alpha_n^{(q)}$  as in (28.5), and  $F_q(1-q^n) = 1 - F_q(q^n)$ . At  $q = 1/2$  the prefactor is  $2^{-\binom{n}{2}}$  and (28.15) is (28.13).*

*Proof.* For  $n = 0$  both sides are  $F_q(1) = 1$ ; let  $n \geq 1$ . Split the series (27.1) after  $n$  terms,

$$X_q = \sum_{j=0}^{n-1} a_j U_j + q^n X'_q, \quad a_j := (1-q)q^j, \quad X'_q := (1-q) \sum_{i \geq 0} q^i U_{n+i}, \quad (28.16)$$

with the  $a_j$  of (27.17) and  $X'_q$  a copy of  $X_q$  independent of  $U_0, \dots, U_{n-1}$ ; note  $\prod_{j < n} a_j = (1-q)^n q^{\binom{n}{2}}$ . Conditionally on  $X'_q = y$ , the event  $\{X_q \leq q^n\}$  reads  $\sum_{j < n} a_j U_j \leq t$  with  $t = q^n(1-y) \in [0, q^n]$ , so its conditional probability is the cube section

$$V(t) := \text{vol} \left\{ u \in [0, 1]^n : \sum_{j=0}^{n-1} a_j u_j \leq t \right\}.$$

Dropping the constraints  $u_j \leq 1$  enlarges the region to a simplex, so

$$V(t) \leq \frac{t^n}{n! \prod_{j < n} a_j} = \frac{t^n}{n! (1-q)^n q^{\binom{n}{2}}}, \quad (28.17)$$

with equality precisely when the simplex lies inside the cube, that is when  $t \leq \min_j a_j = (1-q)q^{n-1}$ : if  $t > (1-q)q^{n-1}$ , the point with  $u_{n-1} = t/a_{n-1} > 1$  and all other coordinates 0 lies in the simplex but not in the cube, and so does a neighbourhood of it, of positive volume. Averaging (28.17) over  $y$  with  $t = q^n(1-y)$  and using  $1 - X_q \stackrel{d}{=} X'_q$  from (27.2),

$$F_q(q^n) \leq \frac{q^{n^2} \mathbb{E}[(1 - X_q)^n]}{n! (1-q)^n q^{\binom{n}{2}}} = \frac{q^{n^2 - \binom{n}{2}}}{n! (1-q)^n} \mathbb{E}(X_q^n),$$

and  $n^2 - \binom{n}{2} = \binom{n+1}{2}$ ; this is (28.14).

If  $q \leq 1/2$ , then every admissible threshold satisfies  $t \leq q^n = q \cdot q^{n-1} \leq (1-q)q^{n-1}$ , so (28.17) is an equality almost surely and (28.15) follows. If  $q > 1/2$  and  $n \geq 1$ , then  $(1-q)q^{n-1} < q^n$  and

$$\mathbb{P}(t > (1-q)q^{n-1}) = \mathbb{P}\left(1 - X'_q > \frac{1-q}{q}\right) > 0,$$

because  $(1-q)/q < 1$  and  $1 - X'_q$  has support  $[0, 1]$  by Theorem 27.1; on that event (28.17) is strict, and so therefore is the average. Finally  $X_q$  has no atoms and  $X_q \stackrel{d}{=} 1 - X_q$ , which gives  $F_q(1 - q^n) = 1 - F_q(q^n)$ .  $\square$

*Remark 28.11* (The half-base cube cut). At  $q = 1/2$ , Proposition 28.10 is a second proof of (28.13) which uses neither Proposition 28.1 nor the smoothness of  $F$ . In scaled coordinates it reads as follows. Write  $X = \sum_{j \geq 1} 2^{-j} V_j$  with  $V_1, V_2, \dots$  independent and uniform on  $[0, 1]$ , multiply by  $2^n$  and split the series after  $n$  terms:

$$2^n X = \sum_{j=1}^n c_j V_j + X', \quad c_j := 2^{n-j}, \quad X' := \sum_{i \geq 1} 2^{-i} V_{n+i}, \quad (28.18)$$

where  $X'$  is independent of  $V_1, \dots, V_n$  and distributed as  $X$ . Conditionally on  $X' = x$ , the event  $2^n X \leq 1$  is  $\sum_j c_j V_j \leq t$  with  $t := 1 - x \in [0, 1]$ . Each  $c_j$  is an integer with  $c_j \geq 1$ , so no upper constraint is active, the admissible region is the entire simplex, and its volume is  $t^n / (n! \prod_j c_j) = (1-x)^n / (n! 2^{\binom{n}{2}})$  since  $\prod_{j=1}^n 2^{n-j} = 2^{\binom{n}{2}}$ . Averaging over  $X'$  and using  $X \stackrel{d}{=} 1 - X$  returns (28.13).

**Corollary 28.12** (Bell-polynomial form of the Fabius values). *Put  $\kappa_1 := 1/2$  and  $\kappa_r := \kappa_r(Y_{1/2})$  for  $r \geq 2$ , the half-base cumulants (27.14). Then for every  $n \geq 0$*

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{n!} \mathcal{B}_n(\kappa_1, \dots, \kappa_n) = 2^{-\binom{n}{2}} \sum_{\substack{\nu_1, \dots, \nu_n \geq 0 \\ \nu_1 + 2\nu_2 + \dots + n\nu_n = n}} \prod_{r=1}^n \frac{1}{\nu_r!} \left( \frac{\kappa_r}{r!} \right)^{\nu_r}. \quad (28.19)$$

Only  $\kappa_1$  and the even  $\kappa_r$  contribute, and every value  $F(2^{-n})$  is rational.

*Proof.* Cumulants of order  $r \geq 2$  are translation invariant, so  $X = Y_{1/2} + 1/2$  has  $\kappa_1(X) = \mathbb{E}X = 1/2$  by (27.2) and  $\kappa_r(X) = \kappa_r$  for  $r \geq 2$ . The moment generating function is the exponential of the cumulant generating function, so the defining identity (27.10) gives  $\mathbb{E}(X^n) = \mathcal{B}_n(\kappa_1, \dots, \kappa_n)$ , exactly as in the proof of Corollary 27.5 but for the uncentred variable; now apply (28.13). The second expression is the explicit expansion of  $\mathcal{B}_n$  obtained by multiplying out  $\exp(\sum_r \kappa_r t^r / r!)$ , and the odd cumulants beyond the first vanish by (27.14).  $\square$

*Remark 28.13* (Sharpness of the hypothesis  $q \leq 1/2$ ). For  $q > 1/2$  the closed form (28.15) fails, and can fail grossly: at  $q = 4/5$  and  $n = 1$  its right side is  $q/(2(1-q)) = 2$ , which is not even a probability. At  $q = 3/5$  the right sides for  $n = 1, 2, 3$  exceed the true values by  $1.3 \cdot 10^{-2}$ ,  $5.6 \cdot 10^{-4}$  and  $1.2 \cdot 10^{-5}$ ; in particular  $F_{3/5}(3/5) = 0.73738\dots$  against  $3/4$ . The reason is the one isolated in the proof: for  $q > 1/2$  the smallest coefficient  $(1-q)q^{n-1}$  drops below the threshold  $q^n$  and the cube truncates the simplex, the deficit being exactly the inclusion–exclusion terms of Theorem 27.9 that (28.18) annihilates.

## 28.1 Reciprocal nodes and the leading-coefficient filter

The filter  $\mathcal{G}_{n,q}$  of (25.10) reproduces the *constant* coefficient of a series and annihilates the degrees  $1, \dots, n$ . Its companion reproduces the coefficient of degree  $n$  and annihilates the degrees  $0, \dots, n-1$ . That companion lives on the reciprocal nodes  $-q^{-k}$ , its weights are again a signed Gaussian row, and its residual tail is again controlled by a single Gaussian row. It is this second normalization, not the evaluation-at-zero one, that produces the exact dyadic formulas of Chapter 27.

Throughout this section  $q$  lies in a field  $K$  and  $q^j \neq 1$  for  $1 \leq j \leq n$ ; from Theorem 28.21 onwards  $K$  is also assumed to have characteristic zero. Put

$$x_k := -q^{-k}, \quad W_{n,k}(q) := (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q \quad (0 \leq k \leq n). \quad (28.20)$$

The nodes are distinct, because  $q^i = q^j$  with  $0 \leq i < j \leq n$  would force  $q^{j-i} = 1$ .

**Theorem 28.14** (Exact geometric moment transform). *For every integer  $d \geq 0$ ,*

$$\sum_{k=0}^n W_{n,k}(q) x_k^d = (-1)^d (q^{-d}; q)_n. \quad (28.21)$$

*In particular*

$$\sum_{k=0}^n W_{n,k}(q) x_k^d = 0 \quad (0 \leq d < n), \quad (28.22)$$

*and*

$$\sum_{k=0}^n W_{n,k}(q) x_k^n = q^{-\binom{n+1}{2}} (q; q)_n. \quad (28.23)$$

*Proof.* Since  $x_k^d = (-1)^d q^{-kd}$ , the left side of (28.21) equals

$$(-1)^d \sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q (q^{-d})^k,$$

which is  $(-1)^d (q^{-d}; q)_n$  by (3.1) at  $z = q^{-d}$ . If  $0 \leq d < n$ , the factor with  $j = d$  in

$$(q^{-d}; q)_n = \prod_{j=0}^{n-1} (1 - q^{j-d})$$

is zero, which gives (28.22). For  $d = n$ , extract  $-q^{j-n}$  from each factor:

$$(q^{-n}; q)_n = (-1)^n q^{-\sum_{j=0}^{n-1} (n-j)} \prod_{j=0}^{n-1} (1 - q^{n-j}) = (-1)^n q^{-\binom{n+1}{2}} (q; q)_n,$$

because  $\sum_{j=0}^{n-1} (n-j) = \binom{n+1}{2}$ . Multiplication by the outer sign  $(-1)^n$  proves (28.23).  $\square$

**Corollary 28.15** (Leading-coefficient functional). *If  $p \in K[x]$  has degree at most  $n$ , then*

$$[x^n]p(x) = \frac{q^{\binom{n+1}{2}}}{(q; q)_n} \sum_{k=0}^n W_{n,k}(q) p(-q^{-k}). \quad (28.24)$$

*Proof.* Write  $p(x) = \sum_{d=0}^n b_d x^d$  and apply the functional termwise. Every term with  $d < n$  vanishes by (28.22), and the term  $d = n$  contributes  $b_n$  by (28.23).  $\square$

**Remark 28.16** (Three faces of one weight row). The functional (28.24) is not a new interpolation device. It is the divided difference  $y[x_0, \dots, x_n]$  of Theorem 20.10 after the substitution  $q \mapsto q^{-1}$ ,  $x \mapsto -x$ ; in particular the closed evaluation of  $\prod_{j \neq k} (x_k - x_j)$  needed for a Lagrange derivation, and the uniqueness of a functional returning the leading coefficient, are already contained in that theorem and in Theorem 25.3. The Vandermonde determinant of the nodes differs only by  $x \mapsto -x$  and a column reversal from the alternant  $a_\delta(1, q^{-1}, \dots, q^{-n})$  evaluated inside the proof of Theorem 4.9. At  $q = 1/2$  the data (28.20) become the data (26.6):  $x_k = \xi_k$  and  $W_{n,k}(1/2) = w_{n,k}$ , so Theorem 26.4 is the case  $q = 1/2$  of Corollary 28.15.

**Theorem 28.17** (The reciprocal filter and its exact tail). *For a formal power series  $f(X) = \sum_{m \geq 0} a_m X^m$  over  $K$  put*

$$(\mathcal{D}_{n,q}f)(X) := \frac{q^{\binom{n+1}{2}}}{(q; q)_n} \sum_{k=0}^n W_{n,k}(q) f(x_k X). \quad (28.25)$$

*Then*

$$[X^m] \mathcal{D}_{n,q}f = \frac{q^{\binom{n+1}{2}}}{(q; q)_n} (-1)^m (q^{-m}; q)_n a_m, \quad (28.26)$$

$$(\mathcal{D}_{n,q}f)(X) = \sum_{r \geq 0} (-1)^r q^{-nr} \begin{bmatrix} n+r \\ n \end{bmatrix}_q a_{n+r} X^{n+r}. \quad (28.27)$$

Thus  $\mathcal{D}_{n,q}$  deletes the coefficients of degrees  $0, \dots, n-1$ , reproduces  $a_n$ , and multiplies every later coefficient by an explicit signed Gaussian polynomial. If  $\deg f \leq n$ , then  $(\mathcal{D}_{n,q}f)(1) = a_n$ , in agreement with Corollary 28.15.



*Proof.* The outer sum in (28.25) is finite, so coefficient extraction commutes with it, and (28.26) is Theorem 28.14. The multiplier vanishes for  $m < n$  by (28.22). Let  $m = n + r$  with  $r \geq 0$ . Extracting  $-q^{j-n-r}$  from each factor of  $(q^{-n-r}; q)_n$  gives

$$(q^{-n-r}; q)_n = (-1)^n q^{\binom{n}{2} - n(n+r)} (q^{r+1}; q)_n,$$

since  $\sum_{j=0}^{n-1} (j - n - r) = \binom{n}{2} - n(n+r)$  and  $\prod_{j=0}^{n-1} (1 - q^{n+r-j}) = (q^{r+1}; q)_n$ . Hence the sign becomes  $(-1)^{n+r} (-1)^n = (-1)^r$ , the total exponent of  $q$  becomes

$$\binom{n+1}{2} + \binom{n}{2} - n(n+r) = -nr,$$

and

$$\frac{(q^{r+1}; q)_n}{(q; q)_n} = \frac{(q; q)_{n+r}}{(q; q)_r (q; q)_n} = \left[ \begin{matrix} n+r \\ n \end{matrix} \right]_q,$$

which is (28.27).  $\square$

*Remark 28.18* (The two exact tails). Theorems 25.9 and 28.17 are companions, not restatements. Both multiply the surviving coefficients by one Gaussian row, but they extract different coefficients on different node sets:  $\mathcal{G}_{n,q}$  evaluates at 0 on the nodes  $q^k$  and returns  $a_0$  with tail multiplier  $(-1)^n q^{\binom{n+1}{2}} \left[ \begin{matrix} n+r \\ n \end{matrix} \right]_q$ , whereas  $\mathcal{D}_{n,q}$  extracts the top coefficient on the nodes  $-q^{-k}$  and returns  $a_n$  with tail multiplier  $(-1)^r q^{-nr} \left[ \begin{matrix} n+r \\ n \end{matrix} \right]_q$ . The  $r$ -dependent sign and the negative power of  $q$  have no counterpart in (25.12): reversing a polynomial exchanges the two functionals only in the finite-degree case, and a power series has no top coefficient to reverse. For  $0 < q < 1$  the estimate  $\left[ \begin{matrix} n+r \\ n \end{matrix} \right]_q \leq 1/(q; q)_n$  used in Theorem 25.11 applies verbatim to (28.27): if  $f$  is analytic on  $|z| \leq R$  with  $R > q^{-n}$  and  $M_R = \max_{|z|=R} |f|$ , then

$$|(\mathcal{D}_{n,q} f)(1) - a_n| \leq \frac{M_R}{(q; q)_n R^n} \cdot \frac{q^{-n}/R}{1 - q^{-n}/R}.$$

**Corollary 28.19** (Integrality and the exact 2-adic valuation). For  $0 \leq k \leq n$ ,

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2} = 2^{-k(n-k)} \left[ \begin{matrix} n \\ k \end{matrix} \right]_2. \quad (28.28)$$

The integer  $\left[ \begin{matrix} n \\ k \end{matrix} \right]_2$  is odd; consequently  $2^{k(n-k)} \left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2}$  is an odd integer and

$$\nu_2 \left( \left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2} \right) = -k(n-k). \quad (28.29)$$

*Proof.* Reciprocity, Theorem 2.5(ii), reads  $\left[ \begin{matrix} n \\ k \end{matrix} \right]_{q^{-1}} = q^{-k(n-k)} \left[ \begin{matrix} n \\ k \end{matrix} \right]_q$ ; taking  $q = 2$  gives (28.28). By Corollary 2.4 the Gaussian coefficient lies in  $\mathbb{N}_0[q]$ , and by Theorem 2.5(iii) its constant term is 1; every other monomial  $q^j$  with  $j \geq 1$  is even at  $q = 2$ , so  $\left[ \begin{matrix} n \\ k \end{matrix} \right]_2 \equiv 1 \pmod{2}$ . An odd integer has 2-adic valuation 0, so (28.29) follows from (28.28).  $\square$

*Remark 28.20* (Reading the valuation). Comparing (28.28) with (26.3) identifies the odd integer explicitly:

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_2 = \frac{\mathfrak{m}_n}{\mathfrak{m}_k \mathfrak{m}_{n-k}},$$

and by Theorem 6.1 it counts the  $k$ -dimensional subspaces of  $\mathbb{F}_2^n$ . Thus the half-base Gaussian row is a finite-geometric count divided by an explicit power of two, and the exponent of that power is  $k(n-k)$ , the dimension of the Grassmannian. In the same way (26.2) exhibits  $(1/2; 1/2)_n$  as a power-of-two normalization of a product of Mersenne numbers, with  $\nu_2((1/2; 1/2)_n) = -\binom{n+1}{2}$  because  $\mathfrak{m}_n$  is odd.



**Theorem 28.21** (Gaussian coefficient transfer). *Let  $K$  have characteristic zero, let  $q \in K^\times$  satisfy  $q^j \neq 1$  for  $1 \leq j \leq n$ , and let*

$$U(z) = \sum_{m \geq 0} u_m z^m, \quad u_0 = 1, \quad V(z) = \sum_{m \geq 0} v_m z^m$$

*be power series in  $K[[z]]$ . For each  $0 \leq k \leq n$  there is exactly one series  $\Psi_k(z) = \sum_{m \geq 0} c_{k,m} z^m \in K[[z]]$  satisfying the bridge relation*

$$\Psi_k(z) U(z) = e^{-z} V(-q^{-k}z), \quad (28.30)$$

*namely  $\Psi_k(z) = e^{-z} V(-q^{-k}z)/U(z)$ , the quotient being legitimate in  $K[[z]]$  because  $U$  has constant term 1. For that family, with the weights of (28.20),*

$$\sum_{k=0}^n W_{n,k}(q) c_{k,d} = 0 \quad (0 \leq d < n), \quad (28.31)$$

$$\sum_{k=0}^n W_{n,k}(q) c_{k,n} = q^{-\binom{n+1}{2}}(q; q)_n v_n. \quad (28.32)$$

*Of the two series, only the single coefficient  $v_n$  is ever seen;  $U$  enters through its constant term alone. In the self-referential case  $U(z) = A(-z)$ ,  $V = A$  with  $a_0 = 1$ , in which (28.30) reads  $\Psi_k(z)A(-z) = e^{-z}A(-q^{-k}z)$  and  $v_n = a_n$ , the theorem therefore extracts the  $n$ th coefficient of  $A$  from the family attached to  $A$  itself; that is the form used in Theorem 28.28.*

*Proof.* Write  $S_d := \sum_{k=0}^n W_{n,k}(q) c_{k,d}$ . Comparing the coefficient of  $z^d$  on the two sides of (28.30) gives, for each fixed  $k$ ,

$$\sum_{i=0}^d u_{d-i} c_{k,i} = \sum_{j=0}^d \frac{(-1)^j}{j!} v_{d-j} (-q^{-k})^{d-j}. \quad (28.33)$$

Multiply (28.33) by  $W_{n,k}(q)$  and sum over  $k$ . Since  $x_k = -q^{-k}$ , the right-hand side becomes

$$\sum_{j=0}^d \frac{(-1)^j}{j!} v_{d-j} \sum_{k=0}^n W_{n,k}(q) x_k^{d-j}, \quad (28.34)$$

and the left-hand side becomes  $\sum_{i=0}^d u_{d-i} S_i$ .

Let  $0 \leq d < n$  and induct on  $d$ . Every exponent occurring in (28.34) satisfies  $0 \leq d-j \leq d < n$ , so the entire right-hand side vanishes by (28.22). On the left, the terms with  $i < d$  vanish by the induction hypothesis (for  $d = 0$  there are none), and the term  $i = d$  equals  $u_0 S_d = S_d$ . Hence  $S_d = 0$ , which is (28.31).

Now take  $d = n$ . In (28.34) every term with  $j \geq 1$  carries an exponent  $n-j < n$  and vanishes, while the term  $j = 0$  equals

$$v_n \sum_{k=0}^n W_{n,k}(q) x_k^n = q^{-\binom{n+1}{2}}(q; q)_n v_n$$

by (28.23). On the left, the terms with  $i < n$  vanish by (28.31) and the term  $i = n$  equals  $u_0 S_n = S_n$ . This proves (28.32).  $\square$

**Remark 28.22** (On the normalization  $u_0 = 1$ ). The proof uses  $u_0 = 1$  twice: once for the invertibility of  $U$ , and once at the last step of each induction stage. Only invertibility of  $u_0$  is actually needed: for arbitrary  $u_0 \neq 0$  the same argument gives  $u_0 S_d = 0$  for  $d < n$  and  $u_0 S_n = q^{-\binom{n+1}{2}}(q; q)_n v_n$ . This is not a restriction, because (28.30) is unchanged when the pair  $(U, V)$  is divided by  $u_0$ . In every application below the normalization is automatic,  $U$  being  $\mathcal{A}(-z)$  or  $A(-z)$  with  $A(0) = 1$ .

**Corollary 28.23** (Invariance under a common translation). *Under the hypotheses of Theorem 28.21, let  $\alpha \in K$  and put  $\tilde{\Psi}_k(z) = e^{\alpha z} \Psi_k(z)$ . Then*

$$\sum_{k=0}^n W_{n,k}(q) [z^n] \tilde{\Psi}_k(z) = \sum_{k=0}^n W_{n,k}(q) c_{k,n} = q^{-\binom{n+1}{2}} (q; q)_n v_n. \quad (28.35)$$

*Proof.* Convolution gives  $[z^n] \tilde{\Psi}_k = \sum_{j=0}^n \alpha^j c_{k,n-j} / j!$ . Weighting by  $W_{n,k}(q)$  and summing over  $k$ , the term with index  $j$  carries the factor  $S_{n-j}$ ; every  $j \geq 1$  has  $n-j < n$  and therefore vanishes by (28.31). Only  $j = 0$  survives, and (28.32) evaluates it.  $\square$

**Remark 28.24** (Two nested triangular systems). The proof runs two triangular cancellations in series. First the finite  $q$ -binomial theorem annihilates every node power of degree below  $n$ . Second, the coefficient comparison (28.33) propagates that annihilation successively through  $c_{k,0}, c_{k,1}, \dots, c_{k,n-1}$ . The first quantity surviving both is  $a_n$ . This double cancellation is what makes Theorem 28.28 possible, and Corollary 28.23 is the reason a free parameter may be inserted there.

## 28.2 The normalized Thue–Morse block

The Thue–Morse polynomial of Theorem 26.7 has a zero of order exactly  $k$  at  $z = 1$ . Dividing that zero out produces an entire function whose Taylor coefficients are precisely the Thue–Morse power sums, and which bridges  $\mathcal{A}$  to its own dilate: that bridge is the hypothesis (28.30) of Theorem 28.21 at  $q = 1/2$ , where  $q^{-k} = 2^k$ . Throughout,  $\mathcal{A} := \mathcal{A}_{1/2}$ ,  $\alpha_n := \alpha_n^{(1/2)}$  and  $\tau_r = (-1)^{\text{wt}_2(r)}$ .

For  $k \geq 0$  put

$$\Psi_k(z) := e^{-z} \prod_{j=0}^{k-1} \Lambda(-2^j z) = \sum_{m \geq 0} c_{k,m} z^m, \quad (28.36)$$

and, for a translation parameter  $\theta \in \mathbb{C}$  and an integer  $d \geq 0$ , write

$$\tilde{T}_{k,d}(\theta) := \sum_{r=0}^{2^k-1} \tau_r (r + \theta)^d, \quad T_{k,d}(\theta) := \tilde{T}_{k,d}(\theta - 2^k) = \sum_{r=0}^{2^k-1} \tau_r (r - 2^k + \theta)^d. \quad (28.37)$$

**Lemma 28.25** (Thue–Morse blocks as dilation bridges). *For every  $k \geq 0$  the function  $\Psi_k$  is entire,*

$$\Psi_k(z) = e^{-2^k z} \prod_{j=0}^{k-1} \Lambda(2^j z), \quad (28.38)$$

$$\Psi_k(z) \mathcal{A}(-z) = e^{-z} \mathcal{A}(-2^k z), \quad (28.39)$$

and for every  $\theta \in \mathbb{C}$

$$e^{(1-\theta)z} \Psi_k(z) = \frac{1}{2^{\binom{k}{2}} z^k} \sum_{r=0}^{2^k-1} \tau_r e^{-(r+\theta)z}, \quad (28.40)$$

the singularity of the right-hand side at  $z = 0$  being removable. Consequently

$$[z^m] e^{(1-\theta)z} \Psi_k(z) = \frac{(-1)^{m+k}}{2^{\binom{k}{2}} (m+k)!} \tilde{T}_{k,m+k}(\theta) \quad (m \geq 0). \quad (28.41)$$

*Proof.*  $\Lambda$  is entire, hence so is  $\Psi_k$ . The reflection identity

$$\Lambda(-u) = \frac{e^{-u} - 1}{-u} = e^{-u} \frac{e^u - 1}{u} = e^{-u} \Lambda(u)$$

turns  $\prod_{j < k} \Lambda(-2^j z)$  into  $e^{-(2^k-1)z} \prod_{j < k} \Lambda(2^j z)$ , because  $\sum_{j < k} 2^j = 2^k - 1$ ; multiplying by  $e^{-z}$  gives (28.38).

Iterating (28.10) at the arguments  $-z, -2z, \dots, -2^{k-1}z$  gives

$$\mathcal{A}(-2^k z) = \mathcal{A}(-z) \prod_{j=0}^{k-1} \Lambda(-2^j z),$$

and multiplication by  $e^{-z}$  is (28.39).

Next,  $\Lambda(-2^j z) = (1 - e^{-2^j z})/(2^j z)$  and  $\sum_{j < k} j = \binom{k}{2}$ , so

$$e^{(1-\theta)z} \Psi_k(z) = \frac{e^{-\theta z}}{2^{\binom{k}{2}} z^k} \prod_{j=0}^{k-1} (1 - e^{-2^j z}).$$

Applying (26.8) with its variable set to  $e^{-z}$  turns the product into  $\sum_{r < 2^k} \tau_r e^{-rz}$ , and multiplication by  $e^{-\theta z}$  gives (28.40). The numerator vanishes to order exactly  $k$  at  $z = 0$  by (26.9), so the quotient is entire.

Finally, expand  $e^{-(r+\theta)z} = \sum_{p \geq 0} (-1)^p (r+\theta)^p z^p / p!$ . For  $p < k$  the polynomial  $X \mapsto (X+\theta)^p$  has degree  $p < k$ , so  $\tilde{T}_{k,p}(\theta) = 0$  by (26.10) with  $n = k$ ,  $a = \theta$ ,  $b = 1$ . Hence

$$\sum_{r < 2^k} \tau_r e^{-(r+\theta)z} = \sum_{p \geq k} \frac{(-1)^p z^p}{p!} \tilde{T}_{k,p}(\theta),$$

and dividing by  $2^{\binom{k}{2}} z^k$  and writing  $p = m + k$  gives (28.41). □

**Corollary 28.26** (Centred form and the coefficient dictionary). *For every  $k \geq 0$ ,*

$$\sum_{r=0}^{2^k-1} \tau_r e^{(r-2^k)z} = (-1)^k 2^{\binom{k}{2}} z^k \Psi_k(z), \quad (28.42)$$

*and the Taylor coefficients of  $\Psi_k$  itself are the centred Thue–Morse power sums:*

$$c_{k,m} = (-1)^k 2^{-\binom{k}{2}} \frac{T_{k,m+k}(0)}{(m+k)!} = \frac{(-1)^{m+k}}{2^{\binom{k}{2}} (m+k)!} \tilde{T}_{k,m+k}(1) \quad (m \geq 0). \quad (28.43)$$

*Proof.* Substituting  $z \mapsto e^z$  in (26.8) gives  $\sum_{r < 2^k} \tau_r e^{rz} = \prod_{j < k} (1 - e^{2^j z})$ , and  $1 - e^u = -u\Lambda(u)$ , so

$$\prod_{j=0}^{k-1} (1 - e^{2^j z}) = \prod_{j=0}^{k-1} (-2^j z \Lambda(2^j z)) = (-1)^k 2^{\binom{k}{2}} z^k \prod_{j=0}^{k-1} \Lambda(2^j z).$$

Multiplying by  $e^{-2^k z}$  and applying (28.38) proves (28.42). Its left-hand side is the exponential generating function  $\sum_{d \geq 0} T_{k,d}(0) z^d / d!$ , so comparing the coefficient of  $z^{m+k}$  gives the first equality in (28.43). For the second, note that  $2^k - 1 - r$  is the bitwise complement of  $r$  in  $k$  binary digits, whence  $\text{wt}_2(2^k - 1 - r) = k - \text{wt}_2(r)$  and  $\tau_{2^k-1-r} = (-1)^k \tau_r$ ; replacing  $r$  by  $2^k - 1 - r$  in  $T_{k,d}(0) = \sum_r \tau_r (r - 2^k)^d$  therefore gives  $T_{k,d}(0) = (-1)^{d+k} \tilde{T}_{k,d}(1)$ . Alternatively, the second equality is (28.41) at  $\theta = 1$ , where the factor  $e^{(1-\theta)z}$  is 1. □

**Remark 28.27** (Centred and raw coordinates). The two displays (28.40) and (28.42) are one identity in two coordinate systems. Indeed (28.38) says  $\Psi_k(z) = e^{-(2^k+1)z} \Psi_k(-z)$ , and substituting  $z \mapsto -z$  in (28.40) at  $\theta = -2^k$  returns (28.42). The centred sums  $T_{k,d}$  are the natural coefficients of  $\Psi_k$  itself, while the raw sums  $\tilde{T}_{k,d}(\theta)$  with a free  $\theta$  are what Corollary 28.23 feeds into Theorem 28.28; by (28.37) the dictionary between them is  $T_{k,d}(\theta) = (-1)^{d+k} \tilde{T}_{k,d}(1 - \theta)$ .

**Theorem 28.28** (Dyadic Gaussian–Thue–Morse coefficient formula). *For every  $n \geq 0$  and every  $\theta \in \mathbb{C}$ ,*

$$a_n = \frac{(-1)^n}{\mathfrak{m}_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} (r+\theta)^{n+k}. \quad (28.44)$$

*Every inner power sum depends on  $\theta$ ; the total does not.*

*Proof.* Fix  $\theta$  and set  $\tilde{\Psi}_k(z) = e^{(1-\theta)z} \Psi_k(z)$ . By (28.39) the family  $\Psi_0, \dots, \Psi_n$  satisfies the bridge relation (28.30) with  $q = 1/2$ , since then  $q^{-k} = 2^k$ ; and  $A$  has constant term 1 by Corollary 28.5. Hence Theorem 28.21 applies, and Corollary 28.23 with  $\alpha = 1 - \theta$  gives

$$\sum_{k=0}^n W_{n,k}(1/2) [z^n] \tilde{\Psi}_k(z) = 2^{\binom{n+1}{2}} \left(\frac{1}{2}; \frac{1}{2}\right)_n a_n = \mathfrak{m}_n a_n, \quad (28.45)$$

the last equality by (26.2). Comparing (28.20) with (26.6),

$$W_{n,k}(1/2) = (-1)^k 2^{-\binom{k}{2}} \left[ \begin{matrix} n \\ k \end{matrix} \right]_{1/2} = w_{n,k},$$

while (28.41) at  $m = n$  gives

$$[z^n] \tilde{\Psi}_k(z) = \frac{(-1)^{n+k}}{2^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \tau_r (r+\theta)^{n+k}.$$

In the product of the two, the signs combine to  $(-1)^k (-1)^{n+k} = (-1)^n$ , independent of  $k$ , and the powers of two combine to  $2^{-\binom{k}{2}} 2^{-\binom{k}{2}} = 4^{-\binom{k}{2}}$ . Substituting into (28.45) and dividing by  $\mathfrak{m}_n$  proves (28.44). The left side of (28.45) does not depend on  $\theta$ , by Corollary 28.23; this is the asserted invariance.  $\square$

**Example 28.29** (The first two cases). Take  $\theta = 1$ . For  $n = 1$  one has  $\mathfrak{m}_1 = 1$  and  $\left[ \begin{smallmatrix} 1 \\ 0 \end{smallmatrix} \right]_{1/2} = \left[ \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} \right]_{1/2} = 1$ ; the inner sums are  $1^1 = 1$  at  $k = 0$  and  $1^2 - 2^2 = -3$  at  $k = 1$ , so

$$a_1 = - \left( \frac{1}{1!} + \frac{-3}{2!} \right) = - \left( 1 - \frac{3}{2} \right) = \frac{1}{2}.$$

For  $n = 2$  one has  $\mathfrak{m}_2 = 3$  and the half-base Gaussian row  $1, \frac{3}{2}, 1$ ; the inner sums are  $1, 1^3 - 2^3 = -7$ , and  $1^4 - 2^4 - 3^4 + 4^4 = 160$ , so

$$a_2 = \frac{1}{3} \left( \frac{1}{2!} + \frac{3}{2} \cdot \frac{-7}{3!} + \frac{160}{4! \cdot 4!} \right) = \frac{1}{3} \left( \frac{1}{2} - \frac{7}{4} + \frac{5}{3} \right) = \frac{1}{3} \cdot \frac{5}{12} = \frac{5}{36},$$

in agreement with the value produced by (28.11).

**Theorem 28.30** (Inverse-dyadic values of the Fabius function). *By Theorem 28.9,  $F(2^{-n}) = 2^{-\binom{n}{2}} a_n$  with  $a_n = \mathbb{E}(X^n)/n!$ . Consequently, for every  $n \geq 0$  and every  $\theta \in \mathbb{C}$ ,*

$$F(2^{-n}) = \frac{(-1)^n}{2^{n^2} \left(\frac{1}{2}; \frac{1}{2}\right)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} (r+\theta)^{n+k}. \quad (28.46)$$

*Proof.* Only the closed form has to be proved.

For (28.46), multiply (28.44) by  $2^{-\binom{n}{2}}$  and use (26.2) together with  $\binom{n}{2} + \binom{n+1}{2} = n^2$ :

$$2^{\binom{n}{2}} \mathfrak{m}_n = 2^{\binom{n}{2} + \binom{n+1}{2}} \left(\frac{1}{2}; \frac{1}{2}\right)_n = 2^{n^2} \left(\frac{1}{2}; \frac{1}{2}\right)_n. \quad \square$$

The blocks of (28.46) may also be centred at their own right endpoints. This is the normalization in which the inner sums are the bare Taylor coefficients of the Thue–Morse blocks  $\Psi_k$ , and it costs one application of digit complementation.

The blocks of (28.46) may also be centred at their own right endpoints. This is the normalization in which the inner sums are the bare Taylor coefficients of the Thue–Morse blocks  $\Psi_k$ , it costs one application of digit complementation, and — unlike the raw normalization — it extends from the inverse powers  $2^{-n}$  to every dyadic point of  $[0, 1]$  at resolution  $2^{-n}$ , the numerator entering only through the length of the Thue–Morse block.

For integers  $m, k, d \geq 0$  and  $\theta \in \mathbb{C}$  define the *long* Thue–Morse block

$$T_{m;k,d}(\theta) := \sum_{r=0}^{m2^k-1} \tau_r (r - m2^k + \theta)^d, \quad \tau_r = (-1)^{\text{wt}_2(r)}, \quad (28.47)$$

an empty sum being 0; thus  $T_{0;k,d} = 0$  and  $T_{1;k,d} = T_{k,d}$  is the centred power sum of (28.37).

**Corollary 28.31** (Centred form, at an arbitrary dyadic point). *For  $n, m \geq 0$  and  $\theta \in \mathbb{C}$  put*

$$\mathcal{N}_{m,n}(\theta) := \sum_{k=0}^n \frac{\lfloor \frac{n}{k} \rfloor_{1/2}}{4^{\binom{k}{2}} (n+k)!} T_{m;k,n+k}(\theta), \quad \mathcal{N}_n(\theta) := \mathcal{N}_{1,n}(\theta). \quad (28.48)$$

*Then  $\mathcal{N}_{m,n}(\theta)$  does not depend on  $\theta$ , and for every integer  $m$  with  $0 \leq m \leq 2^n$ ,*

$$F\left(\frac{m}{2^n}\right) = \frac{\mathcal{N}_{m,n}(\theta)}{2^{\binom{n}{2}} \mathbf{m}_n} = \frac{1}{2^{n^2} \left(\frac{1}{2}; \frac{1}{2}\right)_n} \sum_{k=0}^n \frac{\lfloor \frac{n}{k} \rfloor_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{m2^k-1} \tau_r (r - m2^k + \theta)^{n+k}. \quad (28.49)$$

*In particular  $\mathcal{N}_n(\theta) = \mathbf{m}_n \alpha_n$  for every  $\theta$ , where  $\alpha_n = \mathbb{E}X^n/n!$  is the coefficient called  $a_n$  in Theorem 28.28, and the case  $m = 1$  reads*

$$F(2^{-n}) = \frac{\mathcal{N}_n(\theta)}{2^{\binom{n}{2}} \mathbf{m}_n} = \frac{1}{2^{n^2} \left(\frac{1}{2}; \frac{1}{2}\right)_n} \sum_{k=0}^n \frac{\lfloor \frac{n}{k} \rfloor_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \tau_r (r - 2^k + \theta)^{n+k}, \quad (28.50)$$

*with no sign  $(-1)^n$  in front. The hypothesis  $m \leq 2^n$  cannot be dropped: at  $n = 2, m = 5$  the right-hand side of (28.49) equals  $67/72$ , whereas  $F(5/4) = 1$ .*

*Proof.* For  $n = 0$  the admissible numerators are  $m = 0$  and  $m = 1$ , and  $\left(\frac{1}{2}; \frac{1}{2}\right)_0 = 1$ ; the single term  $k = 0$  is  $\sum_{r < m} \tau_r$ , which is 0 for  $m = 0$  and  $\tau_0 = 1$  for  $m = 1$ , in agreement with  $F(0) = 0$  and  $F(1) = 1$ . Assume  $n \geq 1$ .

*Step 1: the value is a moment functional.* Put

$$\mathcal{K}_m(z) := \sum_{h=0}^{m-1} \tau_h e^{(m-h-1)z}, \quad \mathcal{P}_m(z) := \mathcal{A}(z) \mathcal{K}_m(z) = \sum_{d \geq 0} p_{m,d} z^d, \quad (28.51)$$

so that  $\mathcal{P}_1 = \mathcal{A}$  and, by (28.5),

$$p_{m,n} = \frac{1}{n!} \sum_{h=0}^{m-1} \tau_h \mathbb{E}(X + m - h - 1)^n. \quad (28.52)$$

We claim that

$$F\left(\frac{m}{2^n}\right) = 2^{-\binom{n}{2}} p_{m,n} \quad (0 \leq m \leq 2^n). \quad (28.53)$$

Integrating (27.22) term by term, with  $\int_0^x (t-c)_+^{n-1} dt = (x-c)_+^n/n$ , gives the distribution function of the truncation  $X_{1/2,n}$  of (27.17):

$$F_{1/2,n}(x) = \frac{2^{\binom{n+1}{2}}}{n!} \sum_{r=0}^{2^n-1} \tau_r \left(x - \frac{r}{2^n}\right)_+^n.$$

By (28.18),  $X = X_{1/2,n} + 2^{-n}X'$  with  $X'$  an independent copy of  $X$ , so conditioning on  $X'$  and using  $x \mapsto 2^{-n}(m-x)$ ,

$$F\left(\frac{m}{2^n}\right) = \mathbb{E}\left[F_{1/2,n}(2^{-n}(m-X'))\right] = \frac{2^{\binom{n+1}{2}-n^2}}{n!} \sum_{r=0}^{2^n-1} \tau_r \mathbb{E}[(m-r-X')_+^n],$$

the factor  $2^{-n \cdot n}$  coming from the  $n$ th power of  $2^{-n}$ . Since  $0 \leq X' \leq 1$  and  $m \leq 2^n$ , a term with  $r \geq m$  has  $m-r-X' \leq 0$  and drops out, while a term with  $r < m$  has  $m-r-X' \geq 0$  and keeps its positive part. With  $\binom{n+1}{2} - n^2 = -\binom{n}{2}$  and  $1-X' \stackrel{d}{=} X$  from (27.2),

$$F\left(\frac{m}{2^n}\right) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{r=0}^{m-1} \tau_r \mathbb{E}(m-r-X')^n = \frac{2^{-\binom{n}{2}}}{n!} \sum_{r=0}^{m-1} \tau_r \mathbb{E}(X+m-r-1)^n,$$

which is (28.53) by (28.52). At  $m=1$  this is Theorem 28.9 again, since  $p_{1,n} = \alpha_n$ .

*Step 2: the long block is a dilated bridge.* Write  $r = h2^k + s$  with  $0 \leq h < m$  and  $0 \leq s < 2^k$ ; this is a bijection onto  $\{0, \dots, m2^k - 1\}$ , and the binary digits of  $h2^k$  and of  $s$  occupy disjoint positions, so  $\tau_{h2^k+s} = \tau_h \tau_s$ . Hence

$$\begin{aligned} \sum_{d \geq 0} T_{m;k,d}(0) \frac{z^d}{d!} &= e^{-m2^k z} \sum_{r=0}^{m2^k-1} \tau_r e^{rz} = e^{-m2^k z} \left( \sum_{h=0}^{m-1} \tau_h e^{h2^k z} \right) \left( \sum_{s=0}^{2^k-1} \tau_s e^{sz} \right) \\ &= \mathcal{K}_m(-2^k z) \left( e^{-2^k z} \sum_{s=0}^{2^k-1} \tau_s e^{sz} \right) = (-1)^k 2^{\binom{k}{2}} z^k \Psi_{m,k}(z), \end{aligned}$$

where  $\Psi_{m,k}(z) := \mathcal{K}_m(-2^k z) \Psi_k(z)$ . The third equality uses

$$\mathcal{K}_m(-2^k z) = e^{-(m-1)2^k z} \sum_{h < m} \tau_h e^{h2^k z},$$

and the fourth is (28.42). Consequently

$$[z^n] \Psi_{m,k}(z) = \frac{(-1)^k}{2^{\binom{k}{2}} (n+k)!} T_{m;k,n+k}(0), \quad (28.54)$$

and multiplying (28.39) by  $\mathcal{K}_m(-2^k z)$  gives

$$\Psi_{m,k}(z) \mathcal{A}(-z) = e^{-z} \mathcal{K}_m(-2^k z) \mathcal{A}(-2^k z) = e^{-z} \mathcal{P}_m(-2^k z).$$

*Step 3: transfer.* Apply Theorem 28.21 at  $q = 1/2$ , where  $q^{-k} = 2^k$ , with  $U(z) = \mathcal{A}(-z)$ , whose constant term is 1 by Corollary 28.5, with  $V = \mathcal{P}_m$ , and with the family  $\Psi_{m,0}, \dots, \Psi_{m,n}$ , for which Step 2 is the bridge relation (28.30). By (28.32) and (26.2),

$$\sum_{k=0}^n W_{n,k}(1/2) [z^n] \Psi_{m,k} = 2^{\binom{n+1}{2}} \left(\frac{1}{2}; \frac{1}{2}\right)_n p_{m,n} = \mathfrak{m}_n p_{m,n},$$

and inserting (28.54) together with  $W_{n,k}(1/2) = (-1)^k 2^{-\binom{k}{2}} \frac{[n]}{[k]}_{1/2}$  makes the two signs cancel and the two triangular powers of 2 combine into  $4^{-\binom{k}{2}}$ :

$$\mathcal{N}_{m,n}(0) = \mathbf{m}_n p_{m,n}.$$

Dividing by  $2^{\binom{n}{2}} \mathbf{m}_n$  and using (28.53) proves (28.49) at  $\theta = 0$ ; the second expression follows from  $2^{\binom{n}{2}} \mathbf{m}_n = 2^{n^2} (\frac{1}{2}; \frac{1}{2})_n$ , which is (26.2) with  $\binom{n}{2} + \binom{n+1}{2} = n^2$ .

*The free parameter.* Translating the inner argument multiplies the exponential generating function of Step 2 by  $e^{\theta z}$ , so  $[z^n](e^{\theta z} \Psi_{m,k}) = (-1)^k 2^{-\binom{k}{2}} T_{m;k,n+k}(\theta)/(n+k)!$ , while Corollary 28.23 with  $\alpha = \theta$  leaves the weighted sum unchanged. Hence  $\mathcal{N}_{m,n}(\theta) = \mathcal{N}_{m,n}(0)$  for every  $\theta$ .

*Consistency with the raw form.* For  $m = 1$  the two normalizations are exchanged by digit complementation. Substituting  $r \mapsto 2^k - 1 - r$  in the inner sum of  $\mathcal{N}_n(1 - \theta)$  and using  $\text{wt}_2(2^k - 1 - r) = k - \text{wt}_2(r)$ , hence  $\tau_{2^k-1-r} = (-1)^k \tau_r$ , together with  $(2^k - 1 - r) - 2^k + (1 - \theta) = -(r + \theta)$ , gives

$$\sum_{r=0}^{2^k-1} \tau_r (r - 2^k + 1 - \theta)^{n+k} = (-1)^{n+2k} \sum_{r=0}^{2^k-1} \tau_r (r + \theta)^{n+k},$$

and  $(-1)^{n+2k} = (-1)^n$  does not depend on  $k$ , so it factors out of the sum over  $k$ :  $\mathcal{N}_n(1 - \theta)$  is  $(-1)^n$  times the sum in (28.44), which is  $\mathbf{m}_n \alpha_n$ . This recovers the case  $m = 1$  a second time, and it is the  $k$ -independent sign referred to in Remark 28.33.  $\square$

*Remark 28.32* (Resolution and numerator are separated). In (28.49) the half-base Gaussian row  $\frac{[n]}{[k]}_{1/2}/4^{\binom{k}{2}}$  depends only on the resolution  $2^{-n}$ , while the numerator  $m$  enters only through the length  $m2^k$  of the Thue–Morse block. It is instructive to compare Theorem 27.11, which writes the density of the truncation  $X_{1/2,N}$  as a Thue–Morse signed spline: there the Thue–Morse signs appear alone and the identity is exact only for the truncation, whereas here they appear weighted by a Gaussian row and the identity is exact for the limit function  $F$  itself. The formula returns 0 at  $m = 0$  and 1 at  $m = 2^n$ , and since  $F(m/2^n) = F(2m/2^{n+1})$  it is heavily overdetermined. The row  $n = 3$  is

$$0, \quad \frac{1}{288}, \quad \frac{5}{72}, \quad \frac{73}{288}, \quad \frac{1}{2}, \quad \frac{215}{288}, \quad \frac{67}{72}, \quad \frac{287}{288}, \quad 1,$$

symmetric about  $1/2$  as (27.32) requires.

*Remark 28.33* (The shift must be common, or geometric). The freedom in  $\theta$  in (28.44), (28.46) and (28.50) is Corollary 28.23 at  $q = 1/2$ : one exponential  $e^{\alpha z}$  is pulled out of all  $n + 1$  blocks *simultaneously*, after which (28.31) annihilates every convolution term but one. It is not a symmetry of an individual block. To see what is and is not allowed, give each block its own shift and set

$$\mathcal{N}_n(\theta_0, \dots, \theta_n) := \sum_{k=0}^n \frac{[n]}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \tau_r (r - 2^k + \theta_k)^{n+k}. \quad (28.55)$$

Two one-parameter families of shifts leave this constant, and they are exactly the two displays above:

$$\mathcal{N}_n(\theta, \dots, \theta) = \mathbf{m}_n \alpha_n, \quad \mathcal{N}_n(2^0 + \theta, \dots, 2^n + \theta) = (-1)^n \mathbf{m}_n \alpha_n,$$

the first by Corollary 28.31, the second by (28.44) after cancelling  $-2^k$  against  $2^k$ . The second family is genuinely block-dependent, and it survives only because digit complementation converts it into the first, at the cost of the  $k$ -independent sign  $(-1)^n$  computed in the proof of Corollary 28.31.

These two families are not independent of each other, and they are not the only structured ones. Both are members of the two-parameter geometric family  $\theta_k = \eta - 2^k \vartheta$  of Theorem 28.99, at  $\vartheta = 0$



and at  $\vartheta = -1$  respectively; and on the whole of that family the double cancellation of [Remark 28.24](#) survives and evaluates the sum in closed form,

$$\mathcal{N}_n(\eta - 2^0\vartheta, \dots, \eta - 2^n\vartheta) = \mathfrak{m}_n \frac{A_n(\vartheta)}{n!},$$

a polynomial of degree  $n$  in  $\vartheta$ , free of  $\eta$ , whose value at  $\vartheta = 0$  is the invariant  $\mathfrak{m}_n\alpha_n$ . What fails away from  $\vartheta \in \{0, -1\}$  is therefore not the cancellation but the constancy: the transfer theorem still extracts exactly one coefficient, only of  $e^{\vartheta u}\mathcal{A}(u)$  rather than of  $\mathcal{A}(u)$ .

Outside that family, unequal shifts do destroy the extraction. For  $n = 3$  take  $(\theta_0, \theta_1, \theta_2, \theta_3) = (0, 1, 2, 3)$ , which is not of the form  $\eta - 2^k\vartheta$ : matching the first two entries forces  $\vartheta = \eta = -1$ , and that choice produces  $(0, 1, 3, 7)$  instead. One finds

$$\mathcal{N}_3(0, 1, 2, 3) = \frac{1}{32}, \quad \mathcal{N}_3(0, 1, 3, 7) = -\frac{7}{12}, \quad \mathfrak{m}_3\alpha_3 = 21 \cdot \frac{1}{36} = \frac{7}{12},$$

the middle value being  $\mathfrak{m}_3 A_3(-1)/3!$ , as [Theorem 28.99](#) requires, and the first being nothing in particular.

*Remark 28.34* (Anatomy and conditioning of the dyadic formula). The ingredients of (28.46) separate cleanly, and each has a single identifiable origin. One factor  $2^{-\binom{k}{2}}$ , together with the row  $\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$ , is the geometric difference stencil  $w_{n,k}$  on the nodes  $-2^k$  of [Theorem 26.4](#); the second factor  $2^{-\binom{k}{2}}$  appears when the order- $k$  Thue–Morse zero is removed in (28.40). Their product is the  $4^{-\binom{k}{2}}$  of the statement: the duplication is structural, not a misprint. The product  $(\frac{1}{2}; \frac{1}{2})_n$  is the normalized geometric Vandermonde denominator of that stencil, equal by (26.2) to the odd Mersenne product  $\mathfrak{m}_n = \prod_{j=1}^n (2^j - 1)$  divided by  $2^{\binom{n+1}{2}}$ , and the remaining  $2^{-\binom{n}{2}}$  is the Jacobian  $\prod_{j=1}^n 2^{n-j}$  of the cube cut (28.18). The factorials  $(n+k)!$  are contributed by the exponential generating functions, and each inner sum is an integer, being a power sum of Thue–Morse signs whose first  $k$  moments vanish.

Arithmetically, therefore, four independent sources meet in one line: dyadic dilation powers, of which (28.28) isolates the part carried by the Gaussian row,  $\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} = 2^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_2$  with an *odd* integer on the right ([Corollary 28.19](#)); the odd Mersenne product  $\mathfrak{m}_n$ ; ordinary factorials; and integer Thue–Morse power sums. The freedom in  $\theta$  is not an accident among those power sums: by [Corollary 28.23](#) it says precisely that the Gaussian stencil has already annihilated everything below degree  $n$ , so only the top coefficient of the shifted test polynomial is ever seen.

The formula is an exact rational certificate, but it is ill conditioned in floating point, and the loss occurs *inside* the blocks rather than between them. The outer cancellation over  $k$  is mild: at  $\theta = 1$  the largest block of (28.46) exceeds the value of the whole sum by the factors

$$\begin{array}{c} 3.00, 4.20, 4.69, 4.42, 4.09, \\ 3.84, 3.68, 3.59, 3.53, 3.50 \end{array}$$

for  $n = 1, \dots, 10$ , peaking at  $n = 3$  and decreasing thereafter, and other choices of  $\theta$  alter these ratios only by a bounded factor. The digits are lost within a single block, in the alternating Thue–Morse power sum: for  $k = n$  the largest summand exceeds  $\sum_{r < 2^k} (-1)^{\text{wt}_2(r)} (r + \theta)^{n+k}$  by a factor 4.8 at  $n = 4$ ,  $5.2 \cdot 10^3$  at  $n = 8$ ,  $3.9 \cdot 10^6$  at  $n = 10$  and  $2.9 \cdot 10^{10}$  at  $n = 12$ , and this ratio grows without bound. For numerical work the triangular recursion (28.11) is preferable: it delivers  $a_n$ , hence  $F(2^{-n}) = 2^{-\binom{n}{2}} a_n$ , with one rational division per index and no cancellation at all.

**Proposition 28.35** (The block is a Taylor polynomial). *Put*

$$\mathcal{M}_n(\theta) := \frac{\mathcal{N}_n(\theta)}{2^{\binom{n}{2}} (1/2; 1/2)_n} = 2^n \alpha_n = 2^{\binom{n+1}{2}} F(2^{-n}), \quad (28.56)$$



and, for  $m \geq 0$ ,

$$\mathcal{T}_m(\theta, y) := 2^{-\binom{m+1}{2}} \sum_{n=0}^m \frac{(2^{m+1}y)^{m-n}}{(m-n)!} \mathcal{M}_n(\theta). \quad (28.57)$$

Then  $\mathcal{T}_m$  does not depend on  $\theta$ , contains no  $q$ -object, and is exactly the  $m$ th Taylor polynomial of  $F$  at the point  $2^{-m}$ :

$$\mathcal{T}_m(\theta, y) = \sum_{r=0}^m 2^{\binom{r+1}{2} - \binom{m-r}{2}} \alpha_{m-r} \frac{y^r}{r!} = \sum_{r=0}^m F^{(r)}(2^{-m}) \frac{y^r}{r!}. \quad (28.58)$$

*Proof.* Independence of  $\theta$  and the first equality in (28.56) are Corollary 28.23 combined with  $\mathcal{N}_n(0) = \mathfrak{m}_n \alpha_n$  and  $\mathfrak{m}_n = 2^{\binom{n+1}{2}} (1/2; 1/2)_n$ ; the second equality is Theorem 28.9. Substituting  $\mathcal{M}_n = 2^n \alpha_n$  into (28.57) and writing  $r = m - n$  gives the exponent

$$-\binom{m+1}{2} + (m+1)r + (m-r) = mr - \binom{m}{2} = \binom{r+1}{2} - \binom{m-r}{2},$$

which is the first equality in (28.58). For the second, Theorem 28.9 gives  $2^{-\binom{m-r}{2}} \alpha_{m-r} = F(2^{r-m})$ , while (28.12) at  $x = 2^{-m}$ , legitimate because  $r \leq m$ , gives  $F^{(r)}(2^{-m}) = 2^{\binom{r+1}{2}} F(2^{r-m})$ .  $\square$

*Remark 28.36.* Proposition 28.35 explains Corollary 28.23 concretely: a common translation perturbs each block by terms of lower degree, and those are exactly the terms annihilated by the order- $n$  geometric divided difference of Theorem 26.4.

### 28.3 Composition and determinant forms

Expanding the product (28.8) instead of inverting it produces two further closed forms. The combinatorial input is an ordered geometric sum.

**Lemma 28.37** (Geometric simplex sum). *Let  $0 < |q| < 1$ , let  $r_1, \dots, r_\ell$  be positive integers, and put  $t_h := r_h + r_{h+1} + \dots + r_\ell$ . Then*

$$\sum_{1 \leq j_1 < j_2 < \dots < j_\ell} q^{j_1 r_1 + \dots + j_\ell r_\ell} = \prod_{h=1}^{\ell} \frac{q^{t_h}}{1 - q^{t_h}}. \quad (28.59)$$

*Proof.* Every strictly increasing  $\ell$ -tuple of positive integers has a unique gap representation  $j_i = i + g_1 + \dots + g_i$  with  $g_h \geq 0$ . Then

$$\sum_{i=1}^{\ell} j_i r_i = \sum_{i=1}^{\ell} i r_i + \sum_{h=1}^{\ell} g_h t_h,$$

because  $g_h$  occurs in  $j_i$  exactly for  $i \geq h$ . Summing the independent geometric series in  $g_1, \dots, g_\ell$  gives

$$q^{\sum_i i r_i} \prod_{h=1}^{\ell} \frac{1}{1 - q^{t_h}},$$

and  $\sum_i i r_i = \sum_h t_h$ , which converts the prefactor into  $\prod_h q^{t_h}$ .  $\square$

An ordered composition  $\rho = (r_1, \dots, r_\ell) \models n$  is a sequence of positive integers with  $r_1 + \dots + r_\ell = n$ ; write  $\ell(\rho) = \ell$  and  $s_j := r_1 + \dots + r_j$ .

**Theorem 28.38** (Composition representation). *For  $0 < q < 1$  and  $n \geq 1$ ,*

$$\alpha_n^{(q)} = \left(\frac{1-q}{q}\right)^n \sum_{\rho \models n} \prod_{j=1}^{\ell(\rho)} \frac{q^{s_j}}{(1-q^{s_j})(r_j+1)!}. \quad (28.60)$$

*In particular, at  $q = 1/2$ ,*

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{\rho \models n} \prod_{j=1}^{\ell(\rho)} \frac{1}{(2^{s_j}-1)(r_j+1)!}. \quad (28.61)$$

*Proof.* Expand each factor of (28.8) as  $\Lambda((1-q)q^j z) = \sum_{r \geq 0} (1-q)^r q^{jr} z^r / (r+1)!$  and collect the coefficient of  $z^n$ . A nonzero contribution chooses an index set  $0 \leq j_1 < \dots < j_\ell$  of factors and positive exponents  $r_1, \dots, r_\ell$  with  $\sum r_i = n$ , so that

$$\alpha_n^{(q)} = (1-q)^n \sum_{\ell \geq 1} \sum_{\substack{r_1 + \dots + r_\ell = n \\ r_i \geq 1}} \frac{1}{\prod_i (r_i + 1)!} \sum_{0 \leq j_1 < \dots < j_\ell} q^{\sum_i j_i r_i}.$$

Shifting  $j_i \mapsto j_i + 1$  multiplies the inner sum by  $q^{-n}$  and puts it in the form (28.59), so the inner sum equals  $q^{-n} \prod_h q^{t_h} / (1 - q^{t_h})$ . Reversing the composition  $(r_1, \dots, r_\ell) \mapsto (r_\ell, \dots, r_1)$  is a bijection of the compositions of  $n$  that turns the tail sums  $t_h$  into the prefix sums  $s_j$ . This proves (28.60). At  $q = 1/2$  the prefactor is 1 and  $q^{s_j} / (1 - q^{s_j}) = 1 / (2^{s_j} - 1)$ ; multiply by  $2^{-\binom{n}{2}}$  and apply Theorem 28.9.  $\square$

**Theorem 28.39** (Lower-Hessenberg determinant). *Put  $\beta_{n,k} := \frac{1}{(2^n - 1)(n - k + 1)!}$  for  $0 \leq k \leq n - 1$ , and let  $\mathcal{H}_n$  be the  $n \times n$  lower-Hessenberg matrix whose  $(i, k)$  entry is  $\beta_{i,k}$  for  $0 \leq k \leq i - 1$ , is  $-1$  for  $k = i$ , and is 0 for  $k > i$  (rows indexed by  $1 \leq i \leq n$ , columns by  $0 \leq k \leq n - 1$ ):*

$$\mathcal{H}_n = \begin{pmatrix} \beta_{1,0} & -1 & 0 & \cdots & 0 \\ \beta_{2,0} & \beta_{2,1} & -1 & \ddots & \vdots \\ \beta_{3,0} & \beta_{3,1} & \beta_{3,2} & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & -1 \\ \beta_{n,0} & \beta_{n,1} & \beta_{n,2} & \cdots & \beta_{n,n-1} \end{pmatrix}. \quad (28.62)$$

*Then, for  $n \geq 1$ ,*

$$\alpha_n = \det \mathcal{H}_n, \quad F(2^{-n}) = 2^{-\binom{n}{2}} \det \mathcal{H}_n. \quad (28.63)$$

*Proof.* Set  $D_0 := 1$  and  $D_n := \det \mathcal{H}_n$ . Note that  $\mathcal{H}_k$  is the leading  $k \times k$  principal submatrix of  $\mathcal{H}_n$  for  $k \leq n$ . Expand  $D_n$  along the last row, whose entries are  $\beta_{n,k}$  for  $0 \leq k \leq n - 1$ , and let  $M_{n,k}$  be the minor obtained by deleting row  $n$  and column  $k$ . In  $M_{n,k}$  the rows  $1, \dots, k$  have zero entries in every column of index greater than  $k$ , because the superdiagonal entry  $-1$  of row  $i \leq k$  sits in column  $i \leq k$ . Hence  $M_{n,k}$  is block lower triangular with diagonal blocks

$$(\text{rows } 1..k, \text{ columns } 0..k-1) = \mathcal{H}_k, \quad (\text{rows } k+1..n-1, \text{ columns } k+1..n-1),$$

and the second block is lower triangular with every diagonal entry equal to  $-1$ . Therefore  $\det M_{n,k} = (-1)^{n-1-k} D_k$ . The cofactor sign of the entry in row  $n$  and column  $k$ , that is, in matrix position  $(n, k+1)$ , is  $(-1)^{n+k+1}$ , so

$$D_n = \sum_{k=0}^{n-1} \beta_{n,k} (-1)^{n+k+1} (-1)^{n-1-k} D_k = \sum_{k=0}^{n-1} \beta_{n,k} D_k.$$

This is (28.11) written as  $\alpha_n = \sum_{k < n} \beta_{n,k} \alpha_k$ , with the same initial value. Induction gives  $D_n = \alpha_n$ , and Theorem 28.9 gives the second formula.  $\square$

*Remark 28.40.* Expanding  $\det \mathcal{H}_n$  over its nonzero Hessenberg paths reproduces (28.61): a path that descends from row  $n$  to row 0 using the steps  $r_1, \dots, r_\ell$  contributes exactly the corresponding composition term. The determinant and the composition sum are two descriptions of the same triangular resolvent, and the recursion (28.11) is its unrolled form.

## 28.4 A Fourier formula for the Fabius function

**Corollary 28.41** (Sinc-product representation of  $F$ ). *For  $0 \leq x \leq 1$ ,*

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\sin((2x-1)t)}{t} \prod_{k=1}^\infty \operatorname{sinc}_{\text{rad}}\left(\frac{t}{2^k}\right) dt, \quad (28.64)$$

*the integral being absolutely convergent.*

*Proof.* Let  $Y = 2X - 1$ , whose density is  $\text{up}$  and whose Fourier transform is  $\widehat{\text{up}}_{\text{ang}}(-t) = \prod_{k \geq 1} \operatorname{sinc}_{\text{rad}}(t/2^k)$  by Theorem 27.16. Since  $\widehat{\text{up}}_{\text{ang}}(-t) = e^{-it} \varphi_{1/2}(2t)$ , Theorem 27.7 shows that  $\widehat{\text{up}}_{\text{ang}}(-\cdot)$  is integrable, and indeed decays faster than every negative power of  $t$ . Fourier inversion and evenness of  $\text{up}$  give

$$\text{up}(y) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-ity} \widehat{\text{up}}_{\text{ang}}(-t) dt = \frac{1}{\pi} \int_0^\infty \widehat{\text{up}}_{\text{ang}}(-t) \cos(ty) dt.$$

Because  $\text{up}$  is a probability density and is symmetric,  $F(x) = \mathbb{P}(Y \leq 2x - 1) = \frac{1}{2} + \int_0^{2x-1} \text{up}(y) dy$  for  $0 \leq x \leq 1$ . Integrating the last display in  $y$  over  $[0, 2x - 1]$  and interchanging the order of integration, which is permitted by absolute convergence, replaces  $\cos(ty)$  by  $\sin((2x - 1)t)/t$  and proves (28.64).  $\square$

*Remark 28.42.* The two exact descriptions of  $F$  are complementary. Formula (28.64) is global, analytic, and well suited to quadrature and to smoothness statements; formula (28.50) is finite, rational, and adapted to dyadic arguments. Their agreement encodes a highly nonlocal cancellation in the sinc integral. At  $x = 1/4$ , for instance, numerical evaluation of (28.64) reproduces  $5/72$  to more than twenty digits.

## 28.5 Digital cancellation in an arbitrary radix

The signs  $\tau_r = (-1)^{\text{wt}_2(r)}$  are the case  $b = 2, \omega = -1$  of a roots-of-unity construction available in every radix. Separating the radix from the special real signs that exist only at  $b = 2$  shows which half of Theorems 26.7 and 26.8 comes from binary digits and which half comes from having a digit expansion at all.

For  $b \geq 2$  let  $s_b(r)$  denote the sum of the base- $b$  digits of  $r$ , and for  $\omega \in \mathbb{C}$  put

$$\tau_r^{(b)} := \omega^{s_b(r)} \quad (r \geq 0), \quad (28.65)$$

so that  $\tau_r^{(2)} = \tau_r$  when  $\omega = -1$ .

**Theorem 28.43** (Radix- $b$  Prouhet filter). *Let  $b \geq 2$  and let  $\omega \in \mathbb{C}$  satisfy*

$$\omega^b = 1, \quad \omega \neq 1. \quad (28.66)$$

*Then for every  $k \geq 0$  the polynomial identity*

$$\sum_{r=0}^{b^k-1} \tau_r^{(b)} z^r = \prod_{j=0}^{k-1} \left( \sum_{a=0}^{b-1} \omega^a z^{ab^j} \right) \quad (28.67)$$

holds, and for every  $\theta \in \mathbb{C}$ ,

$$\sum_{r=0}^{b^k-1} \tau_r^{(b)}(r+\theta)^m = 0 \quad (0 \leq m < k), \quad (28.68)$$

$$\sum_{r=0}^{b^k-1} \tau_r^{(b)}(r+\theta)^k = \frac{k! b^{k(k+1)/2}}{(\omega-1)^k}. \quad (28.69)$$

Consequently, for every polynomial  $p$  of degree at most  $k$  and all  $\alpha, \beta \in \mathbb{C}$ ,

$$\sum_{r=0}^{b^k-1} \tau_r^{(b)} p(\alpha + \beta r) = \frac{k! b^{k(k+1)/2}}{(\omega-1)^k} \beta^k [X^k] p(X). \quad (28.70)$$

*Proof.* Every  $r$  with  $0 \leq r < b^k$  has a unique expansion  $r = a_0 + a_1 b + \cdots + a_{k-1} b^{k-1}$  with  $0 \leq a_j < b$ , and then  $s_b(r) = a_0 + \cdots + a_{k-1}$ . Hence

$$\tau_r^{(b)} z^r = \prod_{j=0}^{k-1} \omega^{a_j} z^{a_j b^j},$$

and summing independently over the  $k$  digits gives (28.67).

Substitute  $z \mapsto e^z$  in (28.67) and multiply by  $e^{\theta z}$ . The left side becomes the exponential generating function

$$\sum_{r=0}^{b^k-1} \tau_r^{(b)} e^{(r+\theta)z} = \sum_{m \geq 0} \left( \sum_{r=0}^{b^k-1} \tau_r^{(b)} (r+\theta)^m \right) \frac{z^m}{m!},$$

and the right side becomes  $e^{\theta z} \prod_{j=0}^{k-1} G_j(z)$ , where  $G_j(z) = \sum_{a=0}^{b-1} \omega^a e^{ab^j z}$ .

Because  $\omega^b = 1$  and  $\omega \neq 1$ , the finite geometric sum gives

$$G_j(0) = \sum_{a=0}^{b-1} \omega^a = \frac{\omega^b - 1}{\omega - 1} = 0.$$

Differentiating the identity  $\sum_{a=0}^{b-1} Z^a = (Z^b - 1)/(Z - 1)$  and multiplying by  $Z$  gives

$$\sum_{a=0}^{b-1} a Z^a = \frac{b Z^b (Z - 1) - Z(Z^b - 1)}{(Z - 1)^2},$$

so at  $Z = \omega$ , where  $\omega^b = 1$ ,

$$\sum_{a=0}^{b-1} a \omega^a = \frac{b}{\omega - 1} \neq 0. \quad (28.71)$$

Hence  $G_j'(0) = b^j \sum_{a < b} a \omega^a = b^{j+1}/(\omega - 1)$ , so each  $G_j$  has a zero of order exactly one at the origin and  $\prod_{j < k} G_j$  has a zero of order exactly  $k$ , with

$$[z^k] \prod_{j=0}^{k-1} G_j(z) = \prod_{j=0}^{k-1} \frac{b^{j+1}}{\omega - 1} = \frac{b^{k(k+1)/2}}{(\omega - 1)^k}.$$

Multiplying by  $e^{\theta z} = 1 + \mathcal{O}(z)$  changes neither the order of the zero nor the coefficient of  $z^k$ . Comparing the coefficients of  $z^m/m!$  for  $0 \leq m \leq k$  gives (28.68) and (28.69).

For (28.70), expand  $q(X) := p(\alpha + \beta X) = \sum_{m=0}^k \gamma_m X^m$ , a polynomial of degree at most  $k$  with  $\gamma_k = \beta^k [X^k] p$ . Applying (28.68) with  $\theta = 0$  to each  $m < k$  and (28.69) to  $m = k$  leaves only the term  $\gamma_k$ .  $\square$

**Corollary 28.44** (The binary case). *For  $b = 2$  and  $\omega = -1$ , identity (28.67) is (26.8), identity (28.70) is (26.10), and the top value (28.69) is*

$$\frac{k! 2^{k(k+1)/2}}{(-2)^k} = (-1)^k k! 2^{\binom{k}{2}},$$

*in agreement with (26.9).*

*Proof.* Substitute  $b = 2$  and  $\omega = -1$ , and use  $k(k+1)/2 - k = \binom{k}{2}$ .  $\square$

**Remark 28.45** (Rigidity of the binary radix). It is worth separating what in (28.46) needs the radix 2 from what does not. Less needs it than one might expect.

*The  $q$ -calculus is radix-free.* Chapter 25 is written at general  $q$ : for real  $b > 1$  the annihilator  $(z; 1/b)_n = \prod_{j=0}^{n-1} (1 - zb^{-j})$  has the roots  $1, b, \dots, b^{n-1}$  and supplies divided-difference weights on the nodes  $-1, -b, \dots, -b^n$ . This is Corollary 28.15 at  $q = 1/b$ , whose top moment (28.23) is

$$\mathfrak{m}_n^{(b)} := \prod_{j=1}^n (b^j - 1),$$

the Mersenne product  $\mathfrak{m}_n$  at  $b = 2$ . When  $b$  is a prime power the same  $b$  carries in addition a finite-field meaning through  $\left[ \begin{smallmatrix} n \\ k \end{smallmatrix} \right]_b$  (Theorem 6.1), so that at such a radix three constructions share one parameter: digital cancellation (28.70), geometric interpolation on the nodes  $-b^k$ , and subspace enumeration. For  $b = 2$  this is the coincidence exploited in Principle 26.10.

*The digital signs are radix-free.* Theorem 28.43 produces, for every  $b \geq 2$  and every  $\omega$  with  $\omega^b = 1$ ,  $\omega \neq 1$ , a sign system  $\tau_r^{(b)} = \omega^{s_b(r)}$  whose block exponential sum has a zero of order exactly  $k$  at the origin – the property of (26.9) that removes the apparent pole in (28.40). Since (28.66) does not force  $\omega$  to be primitive, a composite  $b$  carries several inequivalent such systems;  $b = 4$ ,  $\omega = -1$  gives  $\tau_r^{(4)} = (-1)^{s_4(r)}$  with constant  $k! 4^{k(k+1)/2} / (-2)^k$ .

*The coefficient formula is radix-free as well, in a shape that must be stated carefully.* Put

$$Y_b := \sum_{j \geq 1} b^{-j} V_j = \frac{X_{1/b}}{b-1}, \quad a_n^{(b)} := \frac{\mathbb{E}(Y_b^n)}{n!},$$

with  $V_1, V_2, \dots$  independent and uniform on  $[0, 1]$ , so that  $Y_2 = X$  and  $a_n^{(2)} = a_n$ . Everything in Corollary 28.5 and lemma 28.25 runs verbatim with 2 replaced by  $b$ : the moment generating function satisfies  $A_b(bz) = \Phi(z)A_b(z)$ , and  $\Psi_k^{(b)}(z) = e^{-z} \prod_{j < k} \Phi(-b^j z)$  satisfies the bridge relation (28.30) at  $q = 1/b$ , since then  $q^{-k} = b^k$ . Expanding  $\prod_{j < k} (1 - e^{-b^j z})$  over subsets and applying Theorem 28.21 together with Corollary 28.23 gives, for every  $\theta \in \mathbb{C}$ ,

$$a_n^{(b)} = \frac{(-1)^n}{\mathfrak{m}_n^{(b)}} \sum_{k=0}^n \frac{\left[ \begin{smallmatrix} n \\ k \end{smallmatrix} \right]_{1/b}}{b^{2\binom{k}{2}} (n+k)!} \sum_{r \in \mathcal{B}_k^{(b)}} (-1)^{\sigma(r)} (r + \theta)^{n+k}, \quad (28.72)$$

where  $\mathcal{B}_k^{(b)}$  is the set of  $r = \sum_{j < k} \varepsilon_j b^j$  with every  $\varepsilon_j \in \{0, 1\}$ , and  $\sigma(r) := \varepsilon_0 + \dots + \varepsilon_{k-1}$ . The inner sums of (28.72) are *not* the radix- $b$  Prouhet sums of Theorem 28.43: the refinement equation delivers the factor  $1 - y^{b^j}$ , whose digit alphabet is  $\{0, 1\}$ , whereas  $\tau^{(b)}$  arises from  $\sum_{a < b} \omega^a y^{ab^j}$ , whose digit alphabet is  $\{0, \dots, b-1\}$ . The two coincide exactly when  $\{0, 1\} = \{0, \dots, b-1\}$ , that is when  $b = 2$ ; there  $\mathcal{B}_k^{(2)} = \{0, 1, \dots, 2^k - 1\}$ ,  $(-1)^{\sigma(r)} = \tau_r$ ,  $b^{2\binom{k}{2}} = 4^{\binom{k}{2}}$ , and (28.72) is (28.44).

*What is genuinely rigid is the passage from coefficients to values.* The simplex cut in the proof of Theorem 28.30 uses only  $b^{n-j} \geq 1$  and a threshold at most 1, so it survives at every radix:

$$\mathbb{P}(Y_b \leq b^{-n}) = \frac{\mathbb{E}[(1 - Y_b)^n]}{n! b^{\binom{n}{2}}}.$$

Turning the right-hand side into the single moment  $\mathbb{E}(Y_b^n)$  uses the reflection  $1 - Y_b \stackrel{d}{=} Y_b$ . But  $Y_b$  has support  $[0, 1/(b-1)]$  while  $1 - Y_b$  has support  $[1 - 1/(b-1), 1]$ , and these agree only when  $1/(b-1) = 1$ . Only at  $b = 2$ , where the geometric mass  $\sum_{j \geq 1} b^{-j}$  is exactly 1 and  $Y_2 = X_{1/2}$  is symmetric about  $1/2$  by (27.2), does (28.72) evaluate a distribution function. For  $b \geq 3$  one still gets every value, since

$$F_{1/b}((b-1)b^{-n}) = \mathbb{P}(Y_b \leq b^{-n}) = \frac{1}{b^{\binom{n}{2}}} \sum_{m=0}^n (-1)^m \binom{n}{m} \frac{m!}{n!} a_m^{(b)} \cdot n! / n!,$$

a binomial combination of  $a_0^{(b)}, \dots, a_n^{(b)}$  rather than the single coefficient  $a_n^{(b)}$ . The  $q$ -binomial and the digital ingredients are therefore automatic at every radix; what singles out  $b = 2$  is the one normalization in which the dilation factor of (27.3), the digit alphabet of the refinement factor, and the right endpoint of the support all agree.

## 28.6 The geometric row as a summability kernel

The Lagrange row  $\lambda_{n,k}^{(q)}$  of (25.4) was used in Chapter 25 for exactness. It also has an exact and uniformly bounded  $\ell^1$  norm, which turns it into a regular summability kernel.

**Theorem 28.46** (Row mass, total variation, and the geometric Lebesgue constant). *Let  $0 < q < 1$ . The weights (25.4) alternate in sign,*

$$\operatorname{sgn} \lambda_{n,k}^{(q)} = (-1)^{n-k} \quad (0 \leq k \leq n), \quad (28.73)$$

and for every  $n \geq 0$

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} = 1, \quad (28.74)$$

$$\sum_{k=0}^n |\lambda_{n,k}^{(q)}| = \frac{(-q; q)_n}{(q; q)_n} = \frac{(q^2; q^2)_n}{(q; q)_n^2}. \quad (28.75)$$

The right-hand side of (28.75) increases strictly with  $n$  to the finite limit

$$\mathcal{L}(q) := \frac{(-q; q)_\infty}{(q; q)_\infty} = \frac{(q^2; q^2)_\infty}{(q; q)_\infty^2}, \quad (28.76)$$

so that  $\sum_{k=0}^n |\lambda_{n,k}^{(q)}| < \mathcal{L}(q) < \infty$  for every  $n$ . At the half base,

$$\sum_{k=0}^n |\lambda_{n,k}^{(1/2)}| = 3 \prod_{r=2}^n \frac{1+2^{-r}}{1-2^{-r}} < \mathcal{L}(1/2) = 8.2559879357782500655 \dots < 9 \quad (n \geq 1). \quad (28.77)$$

*Proof.* By Theorem 25.1 and (25.4),

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} z^k = \frac{1}{(q; q)_n} \prod_{j=1}^n (z - q^j). \quad (28.78)$$

Evaluating at  $z = 1$  gives  $(q; q)_n / (q; q)_n = 1$ , which is (28.74). For  $0 < q < 1$  the three factors  $(q; q)_n$ ,  $q^{\binom{n-k+1}{2}}$  and  $\begin{bmatrix} n \\ k \end{bmatrix}_q$  occurring in (25.4) are positive, which is (28.73); hence  $|\lambda_{n,k}^{(q)}| = (-1)^n (-1)^k \lambda_{n,k}^{(q)}$  and, by (28.78) at  $z = -1$ ,

$$\sum_{k=0}^n |\lambda_{n,k}^{(q)}| = \frac{(-1)^n}{(q; q)_n} \prod_{j=1}^n (-1 - q^j) = \frac{(-q; q)_n}{(q; q)_n}.$$

The second form in (28.75) follows from

$$(-q; q)_n (q; q)_n = \prod_{j=1}^n (1 + q^j) (1 - q^j) = \prod_{j=1}^n (1 - q^{2j}) = (q^2; q^2)_n.$$

Passing from  $n$  to  $n + 1$  multiplies the quotient by  $(1 + q^{n+1})/(1 - q^{n+1}) > 1$ , so it increases strictly; it converges because

$$\sum_{j \geq 1} \log \frac{1 + q^j}{1 - q^j} = \sum_{j \geq 1} \log \left( 1 + \frac{2q^j}{1 - q^j} \right) \leq \frac{2}{1 - q} \sum_{j \geq 1} q^j < \infty,$$

both products in (28.76) being convergent for  $0 < q < 1$ .

At  $q = 1/2$  the factor  $r = 1$  contributes  $(1 + 1/2)/(1 - 1/2) = 3$ , which is the displayed prefactor. For the bound, split the infinite product at  $r = 4$ : the factors  $r = 1, 2, 3$  give  $3 \cdot \frac{5}{3} \cdot \frac{9}{7} = \frac{45}{7}$ , while for  $r \geq 4$  one has  $x = 2^{-r} \leq 1/16$  and

$$\frac{1 + x}{1 - x} = 1 + \frac{2x}{1 - x} \leq \exp\left(\frac{2x}{1 - x}\right) \leq \exp\left(\frac{32x}{15}\right),$$

so, since  $\sum_{r \geq 4} 2^{-r} = 1/8$ , the tail is at most  $e^{4/15} < 1.3063$  and  $\mathcal{L}(1/2) \leq \frac{45}{7} e^{4/15} < 8.4 < 9$ . The decimal expansion in (28.77) is obtained by evaluating the product numerically.  $\square$

**Corollary 28.47** (Conditioning of geometric extrapolation). *Let  $0 < q < 1$ , let  $p$  be a polynomial of degree at most  $n$ , and let  $y_0, \dots, y_n$  be data with  $|y_k - p(q^k)| \leq \varepsilon$  for every  $k$ . Then*

$$\left| \sum_{k=0}^n \lambda_{n,k}^{(q)} y_k - p(0) \right| \leq \mathcal{L}(q) \varepsilon. \quad (28.79)$$

*The amplification of data error is therefore bounded independently of the number of nodes.*

*Proof.* By Theorem 25.3,  $\sum_k \lambda_{n,k}^{(q)} p(q^k) = p(0)$ ; subtract, take absolute values, and apply (28.75) together with (28.76).  $\square$

**Remark 28.48** (Geometric versus harmonic node sequences). The uniform bound is special to geometric nodes. For the harmonic sequence  $v_k = 1/(k + 1)$ ,  $0 \leq k \leq n$ , the corresponding evaluation weights  $\ell_k(0) = \prod_{j \neq k} v_j / (v_j - v_k)$  have absolute sums

$$9, \quad 91.67, \quad 1.007 \cdot 10^3, \quad 1.151 \cdot 10^4, \quad 1.600 \cdot 10^6, \quad 3.495 \cdot 10^{10}$$

at  $n = 2, 4, 6, 8, 12, 20$  (computed in exact rational arithmetic), growing geometrically in  $n$ , whereas at  $q = 1/2$  the same quantity never reaches 9. Combined with Theorem 25.9, this says that the filters  $\mathcal{G}_{n,q}$  are exact and simultaneously uniformly bounded on the sampled values: raising the order of a geometric filter costs nothing in stability. This is the analytic half of Principle 28.52.

**Remark 28.49** (The half-base row in reversed coordinates). Put

$$W_{n,j} := \frac{(-1)^j 2^{-\binom{j+1}{2}}}{(1/2; 1/2)_n} \left[ \begin{matrix} n \\ j \end{matrix} \right]_{1/2}, \quad 0 \leq j \leq n.$$

The symmetry  $\left[ \begin{matrix} n \\ j \end{matrix} \right]_{1/2} = \left[ \begin{matrix} n \\ n-j \end{matrix} \right]_{1/2}$  gives  $W_{n,j} = \lambda_{n,n-j}^{(1/2)}$ , and (28.78) becomes

$$\sum_{j=0}^n W_{n,j} z^j = \frac{(z/2; 1/2)_n}{(1/2; 1/2)_n}.$$

The two rows are reversals of one another, so Theorem 28.46 applies verbatim to either.



**Theorem 28.50** (Bounded-variation tail rows). *Let  $(x_m)_{m \geq 0}$  converge to  $L$  in a normed vector space. For each  $n$  let  $u_{n,0}, \dots, u_{n,n}$  be scalars and  $r(n,0), \dots, r(n,n)$  nonnegative integers with*

$$\sum_{j=0}^n u_{n,j} = 1, \quad \sup_n \sum_{j=0}^n |u_{n,j}| \leq M < \infty, \quad \min_{0 \leq j \leq n} r(n,j) \rightarrow \infty.$$

*Then  $\sum_{j=0}^n u_{n,j} x_{r(n,j)} \rightarrow L$  as  $n \rightarrow \infty$ .*

*Proof.* Unit row mass gives

$$\left\| \sum_{j=0}^n u_{n,j} x_{r(n,j)} - L \right\| = \left\| \sum_{j=0}^n u_{n,j} (x_{r(n,j)} - L) \right\| \leq \sum_{j=0}^n |u_{n,j}| \|x_{r(n,j)} - L\|.$$

Given  $\varepsilon > 0$ , choose  $N$  with  $\|x_m - L\| < \varepsilon/M$  for all  $m \geq N$ . For all  $n$  large enough,  $r(n,j) \geq N$  for every  $j$ , so the right side is at most  $M \cdot \varepsilon/M = \varepsilon$ .  $\square$

**Corollary 28.51** (Gaussian averages of the spline approximants). *Let  $0 < q < 1$  and let  $F_{q,N}$  be the distribution function of  $X_{q,N}$ . For any integers  $r(n,j) \geq n$ ,*

$$\sup_{x \in \mathbb{R}} \left| \sum_{j=0}^n \lambda_{n,j}^{(q)} F_{q,r(n,j)}(x) - F_q(x) \right| \rightarrow 0 \quad (n \rightarrow \infty).$$

*Proof.* Apply [Theorem 28.50](#) in the Banach space of bounded functions on  $\mathbb{R}$  with the supremum norm, to the sequence  $x_m := F_{q,m}$ , which converges to  $F_q$  by (27.29). The hypotheses hold by [Theorem 28.46](#), with  $M = (-q; q)_\infty / (q; q)_\infty$ , and  $\min_j r(n,j) \geq n \rightarrow \infty$ .  $\square$

**Principle 28.52** (One polynomial, two roles). *The single finite product  $\prod_{j=1}^n (z - q^j)$ , normalized by  $(q; q)_n$ , controls both exactness and stability of the geometric row:*

1. *its roots  $q, q^2, \dots, q^n$  make the row an order- $n$  geometric finite-difference stencil, annihilating  $q^{kd}$  for  $1 \leq d \leq n$  ([Theorem 25.3](#)) and extracting one leading coefficient ([Theorem 26.4](#));*
2. *its endpoint values at  $z = 1$  and  $z = -1$  give unit row mass and the exact  $\ell^1$  norm  $(-q; q)_n / (q; q)_n$  ([Theorem 28.46](#)), which are precisely the two hypotheses of [Theorem 28.50](#).*

*Roots control vanishing moments; endpoint values control norms.*

*Proof.* Combine [Theorems 25.3](#), [26.4](#), [28.46](#) and [28.50](#).  $\square$

**Remark 28.53** (Where the analysis actually enters). The dyadic results of this chapter split along the boundary recommended in [Principle 24.19](#), and the split is sharper than it may look. Each of [Theorems 26.4](#), [28.21](#) and [28.30](#), [corollaries 28.23](#) and [28.31](#), and [proposition 28.35](#) is a finite identity between rational numbers, proved in polynomial rings and formal power series over  $\mathbb{Q}$ ; none of them uses a convergence hypothesis, and each survives mapping into an arbitrary commutative  $\mathbb{Q}$ -algebra, exactly as in [Remark 25.2](#). Analysis enters at precisely two points: the identification of  $\mathcal{A}$  with  $\mathbb{E}^{zX}$  in [Theorem 28.3](#), and [Theorem 28.9](#), which converts the formal coefficient  $\alpha_n$  into the value  $F(2^{-n})$ . Every closed formula above is the algebraic core composed with that single analytic bridge. The formalization status of the individual statements is recorded in [Chapter E](#).

### 28.6.1 Negative parameters

The series (27.1) and the convergence estimate in the proof of [Theorem 27.1](#) use only  $|q| < 1$ , so  $X_q$  is a well-defined bounded random variable for every real  $q \in (-1, 1)$ . For negative  $q$  it is no longer supported in  $[0, 1]$ , but it is an affine image of a positive-parameter member of the same family.



**Proposition 28.54** (Negative parameters as affine positive laws). *Let  $0 < a < 1$  and set*

$$\lambda_a := \frac{1+a}{1-a}, \quad \beta_a := \frac{a}{1-a}, \quad \text{so that } \lambda_a = 1 + 2\beta_a. \quad (28.80)$$

*Then*

$$X_{-a} \stackrel{d}{=} \lambda_a X_a - \beta_a, \quad M_{-a}(t) = e^{-\beta_a t} M_a(\lambda_a t). \quad (28.81)$$

*Consequently  $\mathbb{E}X_{-a} = \frac{1}{2}$  and  $X_{-a} \stackrel{d}{=} 1 - X_{-a}$ , and the topological support of  $X_{-a}$  is the interval*

$$\left[ -\frac{a}{1-a}, \frac{1}{1-a} \right]. \quad (28.82)$$

*Proof.* Put  $V_j := U_j - \frac{1}{2}$ . The  $V_j$  are independent and symmetric, so the sequences  $(V_j)_{j \geq 0}$  and  $((-1)^j V_j)_{j \geq 0}$  have the same joint law. Since  $(1-q) \sum_{j \geq 0} q^j = 1$  for every  $|q| < 1$ ,

$$X_q - \frac{1}{2} = (1-q) \sum_{j \geq 0} q^j V_j.$$

Taking  $q = -a$  and using  $(1+a)(-a)^j = (1+a)a^j \cdot (-1)^j$ ,

$$\begin{aligned} X_{-a} - \frac{1}{2} &= (1+a) \sum_{j \geq 0} a^j (-1)^j V_j \stackrel{d}{=} (1+a) \sum_{j \geq 0} a^j V_j \\ &= \frac{1+a}{1-a} \left[ (1-a) \sum_{j \geq 0} a^j V_j \right] = \lambda_a \left( X_a - \frac{1}{2} \right), \end{aligned}$$

all three series converging absolutely. Hence  $X_{-a} \stackrel{d}{=} \lambda_a X_a - \frac{\lambda_a - 1}{2}$ , and  $\frac{\lambda_a - 1}{2} = \beta_a$  by (28.80). Taking moment generating functions gives the second identity in (28.81).

For the remaining assertions, recall from Theorem 27.1 that  $X_a$  has support  $[0, 1]$  and satisfies  $X_a \stackrel{d}{=} 1 - X_a$ . An increasing affine map carries the support onto its image, so the support of  $X_{-a}$  is  $[-\beta_a, \lambda_a - \beta_a]$ , and  $\lambda_a - \beta_a = \frac{(1+a)-a}{1-a} = \frac{1}{1-a}$ , which is (28.82). Finally  $\mathbb{E}X_{-a} = \frac{\lambda_a}{2} - \beta_a = \frac{(1+a)-2a}{2(1-a)} = \frac{1}{2}$ , and, using  $\lambda_a - 1 - \beta_a = \beta_a$ ,

$$1 - X_{-a} \stackrel{d}{=} 1 - \lambda_a X_a + \beta_a = \lambda_a(1 - X_a) - \beta_a \stackrel{d}{=} \lambda_a X_a - \beta_a \stackrel{d}{=} X_{-a}. \quad \square$$

For instance  $X_{-1/2} \stackrel{d}{=} 3X_{1/2} - 1$ , so the law of  $X_{-1/2}$  is the Fabius law stretched by 3 and supported on  $[-1, 2]$ ; likewise  $X_{-1/4} \stackrel{d}{=} \frac{5}{3}X_{1/4} - \frac{1}{3}$ .

### 28.6.2 Reciprocal germs and reciprocity

For  $|q| > 1$  the series (27.1) diverges and there is no random variable to expand. The functional equation (28.6) survives, however, and pins down a unique normalized germ.

**Theorem 28.55** (The Mahler equation with a normalization is rigid). *Let  $q \in \mathbb{C}$  with  $q \neq 0$  and  $|q| \neq 1$ , and put  $\rho := 2\pi \min(1, |q|)/|1 - q|$ . Then there is exactly one function  $M$ , analytic on the disc  $|t| < \rho$ , with*

$$M(0) = 1, \quad M(t) = \mathcal{E}((1-q)t)M(qt) \text{ for } |t| < \rho, |qt| < \rho, \quad (28.83)$$

*namely*

$$M(t) = \begin{cases} \prod_{j=0}^{\infty} \mathcal{E}((1-q)q^j t), & |q| < 1, \\ \left[ \prod_{j=1}^{\infty} \mathcal{E}((1-q)q^{-j} t) \right]^{-1}, & |q| > 1. \end{cases} \quad (28.84)$$

*Both products converge locally uniformly, and in the second case no factor vanishes on  $|t| < \rho$ .*

*Proof.* Convergence of the first product is [Theorem 28.3](#); the second is identical after replacing  $q$  by  $q^{-1}$  in the estimate, since  $\sum_{j \geq 1} |q|^{-j} < \infty$ . When  $|q| > 1$  one has  $\rho = 2\pi/|1 - q|$ , so for  $|t| < \rho$  and  $j \geq 1$

$$|(1 - q)q^{-j}t| \leq |1 - q| |q|^{-1} |t| < |1 - q| |q|^{-1} \rho = \frac{2\pi}{|q|} < 2\pi.$$

Since every zero of  $\mathcal{E}$  has modulus at least  $2\pi$ , no factor vanishes on that disc and the reciprocal is analytic there.

That [\(28.84\)](#) solves [\(28.83\)](#) is the computation of [Theorem 28.3](#) in the first case. In the second case write  $P(t) = \prod_{j \geq 1} \mathcal{E}((1 - q)q^{-j}t)$  and split off  $j = 1$  after replacing  $t$  by  $qt$ :

$$P(qt) = \prod_{j \geq 1} \mathcal{E}((1 - q)q^{1-j}t) = \mathcal{E}((1 - q)t)P(t).$$

Taking reciprocals gives  $M(t) = \mathcal{E}((1 - q)t)M(qt)$  with  $M = P^{-1}$ .

For uniqueness, let  $M$  be any solution. If  $|q| < 1$ , then  $|q^k t| < \rho$  for all  $k \geq 0$ , so [\(28.83\)](#) may be iterated  $N$  times:

$$M(t) = \left[ \prod_{j=0}^{N-1} \mathcal{E}((1 - q)q^j t) \right] M(q^N t).$$

Let  $N \rightarrow \infty$ :  $q^N t \rightarrow 0$ , so  $M(q^N t) \rightarrow M(0) = 1$  by continuity, and the bracket converges to the first product in [\(28.84\)](#).

If  $|q| > 1$ , apply [\(28.83\)](#) at the point  $q^{-1}t$ , which again lies in the disc, to get  $M(q^{-1}t) = \mathcal{E}((1 - q)q^{-1}t)M(t)$ , that is,

$$M(t) = \frac{M(q^{-1}t)}{\mathcal{E}((1 - q)q^{-1}t)}.$$

Iterating  $N$  times,

$$M(t) = M(q^{-N}t) \left[ \prod_{j=1}^N \mathcal{E}((1 - q)q^{-j}t) \right]^{-1},$$

and letting  $N \rightarrow \infty$  gives the second line of [\(28.84\)](#).  $\square$

**Definition 28.56** (The reciprocal germ). For  $|q| > 1$ ,  $M_q$  denotes the unique germ of [Theorem 28.55](#), that is, the reciprocal product in [\(28.84\)](#); its Taylor coefficients are again written  $a_n(q) = d_n(q)/n!$ .

*Remark 28.57.* The symbol  $M_q$  is used in two senses. For real  $q$  with  $|q| < 1$  it is a genuine moment generating function and  $d_n(q) = \mathbb{E}X_q^n$ . For  $|q| > 1$  there is no underlying random variable, and  $d_n(q)$  are merely normalized Taylor coefficients. What is common to both regimes is [\(28.6\)](#), and therefore also [\(28.7\)](#): the derivation of the recurrence used the Mahler equation and nothing else, so a single rational function  $a_n$  of  $q$  governs both regimes.

**Theorem 28.58** (Reciprocity). For every  $q \in \mathbb{C}$  with  $q \neq 0$  and  $|q| \neq 1$ ,

$$M_q(t) M_{1/q}(-t) = 1 \tag{28.85}$$

as analytic germs at  $t = 0$ . Equivalently, at the level of coefficients,

$$\sum_{k=0}^n \binom{n}{k} (-1)^{n-k} d_k(q) d_{n-k}(1/q) = \delta_{n0} \quad (n \geq 0). \tag{28.86}$$

*Proof.* By symmetry it suffices to treat  $|q| < 1$ , so that  $|1/q| > 1$  and  $M_{1/q}$  is the germ of [Definition 28.56](#). Writing  $Q = 1/q$  in [\(28.84\)](#) and evaluating at  $-t$ ,

$$M_{1/q}(-t)^{-1} = \prod_{j=1}^{\infty} \mathcal{E}((1 - Q)Q^{-j}(-t)) = \prod_{j=1}^{\infty} \mathcal{E}(-(1 - q^{-1})q^j t).$$

Now  $-(1 - q^{-1})q^j = (q^{-1} - 1)q^j = q^{j-1} - q^j = (1 - q)q^{j-1}$ , so after the substitution  $i = j - 1$  the right-hand side is  $\prod_{i \geq 0} \mathcal{E}((1 - q)q^i t) = M_q(t)$ , which is (28.85).

For (28.86), multiply the two Taylor series

$$M_q(t) = \sum_{k \geq 0} d_k(q) \frac{t^k}{k!}, \quad M_{1/q}(-t) = \sum_{m \geq 0} (-1)^m d_m(1/q) \frac{t^m}{m!},$$

and read off the coefficient of  $t^n/n!$  in the product; by (28.85) it must equal  $\delta_{n0}$ .  $\square$

### 28.6.3 The spectral $q^2$ -Pochhammer factorization

Splitting  $\mathcal{E}$  into its centered part is the step that converts the geometric ratio  $q$  of the random series into the nome  $q^2$  of a  $q$ -Pochhammer product. From (27.5),

$$\mathcal{E}(z) = e^{z/2} \frac{e^{z/2} - e^{-z/2}}{z} = e^{z/2} \frac{\sinh(z/2)}{z/2}, \quad (28.87)$$

and Euler's product for the sine, in its hyperbolic form, is

$$\frac{\sinh(z/2)}{z/2} = \prod_{m=1}^{\infty} \left( 1 + \frac{z^2}{4\pi^2 m^2} \right). \quad (28.88)$$

**Theorem 28.59** (Spectral Pochhammer factorization). *For  $|q| < 1$  and every  $t \in \mathbb{C}$ ,*

$$M_q(t) = e^{t/2} \prod_{m=1}^{\infty} \left( -\frac{(1-q)^2 t^2}{4\pi^2 m^2}; q^2 \right)_{\infty}, \quad (28.89)$$

*and the associated double product converges normally on compact subsets of the  $t$ -plane.*

*For  $|q| > 1$ , put*

$$\mathcal{D}_q(t) := \prod_{m=1}^{\infty} \left( -\frac{(1-q^{-1})^2 t^2}{4\pi^2 m^2}; q^{-2} \right)_{\infty}.$$

*Then  $\mathcal{D}_q$  is entire, its defining double product converges normally on compact subsets of  $\mathbb{C}$ , and, on the germ disk of Definition 28.56,*

$$M_q(t) = \frac{e^{t/2}}{\mathcal{D}_q(t)}. \quad (28.90)$$

*Consequently the quotient in (28.90) is the meromorphic continuation of  $M_q$  to  $\mathbb{C}$ . Its pole divisor is the multiset*

$$t = \frac{2\pi i k q^j}{1 - q}, \quad k \in \mathbb{Z} \setminus \{0\}, \quad j \geq 1,$$

*where multiplicities add when listed points coincide. The reciprocal product converges locally uniformly on compact subsets avoiding this divisor.*

*Proof.* Let  $|q| < 1$  and substitute (28.87) and (28.88) with  $z = (1 - q)q^j t$  into (28.5). The exponential factors multiply to

$$\exp\left(\frac{t}{2}(1 - q) \sum_{j \geq 0} q^j\right) = e^{t/2},$$

leaving the double product  $\prod_{j \geq 0} \prod_{m \geq 1} (1 + (1 - q)^2 q^{2j} t^2 / (4\pi^2 m^2))$ . On  $|t| \leq T$  the sum of the absolute values of the perturbations is at most

$$\sum_{j \geq 0} \sum_{m \geq 1} \frac{|1 - q|^2 |q|^{2j} T^2}{4\pi^2 m^2} = \frac{|1 - q|^2 T^2}{24(1 - |q|^2)} < \infty,$$

using  $\sum_m m^{-2} = \pi^2/6$ . The double product is therefore normally convergent and may be rearranged. Fixing  $m$  and multiplying over  $j$ ,

$$\prod_{j \geq 0} \left(1 + \frac{(1-q)^2 t^2}{4\pi^2 m^2} q^{2j}\right) = \left(-\frac{(1-q)^2 t^2}{4\pi^2 m^2}; q^2\right)_\infty$$

by the definition  $(x; p)_\infty = \prod_{j \geq 0} (1 - xp^j)$  with  $p = q^2$ . This proves (28.89).

Now let  $|q| > 1$  and write

$$P(t) = \prod_{j \geq 1} \mathcal{E}((1-q)q^{-j}t), \quad M_q = P^{-1}$$

on the germ disk. The same estimate, with  $q^{-1}$  in place of  $q$ , proves normal convergence on all compacta of the direct product obtained from the quadratic factors; hence  $\mathcal{D}_q$  is entire. The linear arguments sum to

$$\frac{t}{2}(1-q) \sum_{j \geq 1} q^{-j} = \frac{t}{2}(1-q) \frac{q^{-1}}{1-q^{-1}} = \frac{t}{2} \frac{1-q}{q-1} = -\frac{t}{2},$$

and, for the quadratic part, reindexing by  $i = j - 1 \geq 0$  gives

$$(1-q)^2 q^{-2j} = (1-q)^2 q^{-2} q^{-2i} = (1-q^{-1})^2 (q^{-2})^i,$$

because  $(1-q)^2 q^{-2} = (q^{-1} - 1)^2 = (1 - q^{-1})^2$ . Thus

$$P(t) = e^{-t/2} \mathcal{D}_q(t),$$

which proves (28.90) wherever the reciprocal germ is defined and supplies its meromorphic continuation.

The zeros of  $P$  arise from the simple nonzero zeros  $2\pi ik$  of  $\mathcal{E}$ : they are precisely the solutions of

$$(1-q)q^{-j}t = 2\pi ik, \quad k \in \mathbb{Z} \setminus \{0\}, \quad j \geq 1.$$

This is the stated pole divisor, with multiplicities added at collisions; no cancellation is possible because  $e^{t/2}$  never vanishes. Finally, on a compact set disjoint from the divisor,  $\mathcal{D}_q$  has a positive minimum. Uniform convergence of the direct partial products then implies uniform convergence of their reciprocals.  $\square$

**Lean boundary for Theorem 28.59.** The current formal counterpart proves the general inside Pochhammer product, but not the whole compound theorem. The module `RvachevPochhammerFactorization.lean` defines `complexQPochhammerInf` and proves its defining identity, multipliability, named product, and finite-prefix convergence through `complexQPochhammerInf_eq_tprod`, `multipliable_one_sub_mul_pow_complex`, `hasProd_complexQPochhammerInf`, and `tendsto_finiteQPochhammerInf_complex`; the last three statements allow arbitrary complex  $a$  and assume exactly  $\|q\| < 1$ . For arbitrary complex  $q, z$  with  $\|q\| < 1$ , the next three theorems, `summable_norm_sineTerm_qpows_pair`, `geometricSincProduct_eq_tprod_pair`, and `geometricSincProduct_eq_tprod_complexQPochhammerInf`, prove absolute summability of the paired Euler perturbations, identify the scale-zero product, and justify the rearrangement

$$S_q(z) = \prod_{k \geq 0} \left( \frac{z^2}{(k+1)^2}; q^2 \right)_\infty.$$

This is global in  $z$ , includes  $q = 0$ , negative and nonreal contractions, and requires no nonvanishing premise. The last two theorems, `rvachevFourierProduct_eq_tprod_complexQPochhammerI`

nf and rvachevFourier\_eq\_tprod\_complexQPochhammerInf, specialize to dyadic scale and nome  $1/4$  and transfer the product to a bounded Fabius witness satisfying IsFabius.

In centered characteristic-function coordinates,  $z = (1 - q)t/(2\pi)$  identifies the general theorem with the inside product. At the dyadic scale this is  $t = 4\pi z$ . In the uncentered MGF normalization of (28.89), the corresponding dyadic substitution is  $t = 4\pi iz$ : its Pochhammer product becomes  $\Phi(z)$  and the retained phase gives  $M_{1/2}(4\pi iz) = e^{2\pi iz}\Phi(z)$ .

The complementary sinc-product API supplies the analytic convergence and characteristic-function bridge around this general factorization. For complex  $q, z$  with  $\|q\| < 1$ , the theorems hasProdLocallyUniformly\_geometricSincProduct, geometricSincProductFactors\_multipliable, hasProd\_geometricSincProduct, and geometricSincProduct\_differe ntiable prove locally uniform convergence, pointwise Multipliable/HasProd, and entire-ness of  $S_q(z) = \prod_{n \geq 0} \text{sinc}_{\text{rad}}(\pi q^n z)$ . For real  $|q| < 1$ ,  $t \in \mathbb{R}$ , and  $z_q(t) = (1 - q)t/(2\pi)$ , the two theorems charFun\_geometricUniformDistribution\_eq\_phase\_mul\_geometricSincProduct and charFun\_geometricUniformDistribution\_eq\_phase\_mul\_geometricReciprocalGamma give

$$\varphi_q(t) = e^{it/2} S_q(z_q(t)) = e^{it/2} G_q(z_q(t)) G_q(-z_q(t)),$$

where  $G_q(z) = \text{geometricReciprocalGamma}(q, z)$ . This includes  $q = 0$  and negative contractions. The additional declarations tendstoLocallyUniformly\_prefix\_sinc\_charFun and tendstoUniformlyOn\_prefix\_sinc\_charFun give local uniform convergence of the full prefixes on the real frequency line and uniform convergence on every compact real frequency set. There is still no named centered or MGF bridge and no outside formula or zero–pole exchange; named local-uniform or normal convergence of the Pochhammer-product right-hand side and a packaged complex-frequency full-prefix statement are also absent. Thus Theorem 28.59 remains only partially formalized.

*Remark 28.60* (The nome, and where the  $q$ -objects finally appear). The geometric ratio  $q$  of the random series (27.1) becomes the nome  $q^2$  of the spectral product: at  $q = \pm \frac{1}{2}$  the nome is  $\frac{1}{4}$ , and at  $q = \pm \frac{1}{4}$  it is  $\frac{1}{16}$ . Reciprocal parameters exchange the zeros of (28.89) with the poles of (28.90), in accordance with Theorem 28.58. Theorem 28.59 is the point at which the two halves of this Part meet: the transform of the geometric uniform sum — and hence, at  $q = 1/2$ , of the Fabius law — is literally an infinite product of  $q$ -Pochhammer symbols in the nome  $q^2$ , with the  $m$ th factor supplied by the  $m$ th spectral line of Euler’s sine product.

Taking logarithms in (28.89) and expanding  $\log(1 + x)$  gives, for small  $|t|$ ,

$$\log M_q(t) - \frac{t}{2} = \sum_{r \geq 1} \frac{(-1)^{r-1}}{r} \frac{\zeta(2r)}{1 - q^{2r}} \left( \frac{(1 - q)^2 t^2}{4\pi^2} \right)^r,$$

which, by  $\zeta(2r) = (-1)^{r+1} B_{2r} (2\pi)^{2r} / (2(2r)!)$ , is exactly the cumulant expansion (27.8) of Theorem 27.3. The spectral factorization and the Bernoulli cumulants are two readings of the same product.

## 28.7 The moment polynomial and the endpoint values

The recurrence (28.7) carries apparent poles at all roots of unity. A single finite  $q$ -Pochhammer factor removes them.

**Definition 28.61** (The moment polynomial). For  $n \geq 0$  set

$$\mathcal{P}_n(q) := \frac{(q; q)_n}{(1 - q)^n} a_n(q) = \frac{(q; q)_n}{(1 - q)^n} \frac{d_n(q)}{n!}. \quad (28.91)$$

**Theorem 28.62** (Polynomiality and recurrence). *Each  $\mathcal{P}_n$  lies in  $\mathbb{Q}[q]$ . One has  $\mathcal{P}_0 = 1$  and, for  $n \geq 1$ ,*

$$\mathcal{P}_n(q) = \sum_{k=0}^{n-1} \frac{q^k}{(n-k+1)!} \frac{(q; q)_{n-1}}{(q; q)_k} \mathcal{P}_k(q), \quad (28.92)$$

and moreover

$$\deg \mathcal{P}_n \leq \binom{n}{2}, \quad \mathcal{P}_n(0) = \frac{1}{(n+1)!}. \quad (28.93)$$

The first cases are

$$\begin{aligned} \mathcal{P}_0 &= 1, & \mathcal{P}_1 &= \frac{1}{2}, & \mathcal{P}_2 &= \frac{q+2}{12}, & \mathcal{P}_3 &= \frac{q^2+q+1}{24}, \\ \mathcal{P}_4 &= \frac{(q^2+q+1)(6+3q+5q^2+2q^3-q^4)}{720}. \end{aligned}$$

*Proof.* Inverting (28.91) gives  $a_k(q) = (1-q)^k \mathcal{P}_k(q) / (q; q)_k$ . Substituting this into (28.7) and multiplying through by  $(q; q)_{n-1} / (1-q)^n$  turns the left-hand side into

$$\frac{(1-q^n)(q; q)_{n-1}}{(1-q)^n} a_n(q) = \frac{(q; q)_n}{(1-q)^n} a_n(q) = \mathcal{P}_n(q),$$

and the  $k$ th term on the right into  $\frac{q^k}{(n-k+1)!} \frac{(q; q)_{n-1}}{(q; q)_k} \mathcal{P}_k(q)$ . This is (28.92).

Each quotient  $(q; q)_{n-1} / (q; q)_k = \prod_{j=k+1}^{n-1} (1-q^j)$  occurring in (28.92) is a polynomial of degree  $\sum_{j=k+1}^{n-1} j = \binom{n}{2} - \binom{k+1}{2}$ . Assuming inductively that  $\mathcal{P}_k \in \mathbb{Q}[q]$  with  $\deg \mathcal{P}_k \leq \binom{k}{2}$  for all  $k < n$ , the  $k$ th summand is a polynomial of degree at most

$$k + \left( \binom{n}{2} - \binom{k+1}{2} \right) + \binom{k}{2} = \binom{n}{2},$$

because  $\binom{k+1}{2} - \binom{k}{2} = k$ . Hence  $\mathcal{P}_n \in \mathbb{Q}[q]$  and  $\deg \mathcal{P}_n \leq \binom{n}{2}$ . Setting  $q = 0$  in (28.92) kills every term with  $k \geq 1$  and leaves  $\mathcal{P}_n(0) = \mathcal{P}_0(0) / (n+1)! = 1 / (n+1)!$ . The displayed values follow by evaluating (28.92) for  $n \leq 4$ .  $\square$

**Proposition 28.63** (Exact degree and the two leading coefficients). *Let  $B_m$  be the Bernoulli numbers in the convention (27.6), so that  $B_1 = -\frac{1}{2}$ . For every  $n \geq 0$ ,*

$$[q^{\binom{n}{2}}] \mathcal{P}_n(q) = \frac{(-1)^n B_n}{n!}, \quad (28.94)$$

and for every  $n \geq 2$ ,

$$[q^{\binom{n}{2}-1}] \mathcal{P}_n(q) = -\frac{(-1)^n B_n}{n!} + \frac{(-1)^{n-1} B_{n-1}}{2(n-1)!}. \quad (28.95)$$

Consequently

$$\deg \mathcal{P}_n = \begin{cases} \binom{n}{2}, & n \text{ even, or } n = 1, \\ \binom{n}{2} - 1, & n \geq 3 \text{ odd.} \end{cases} \quad (28.96)$$

*Proof.* Write  $R_n(q) := (q; q)_n / (1-q)^n = \prod_{j=2}^n (1+q+\cdots+q^{j-1})$ , a polynomial of degree  $\sum_{j=2}^n (j-1) = \binom{n}{2}$ . Factoring the top power out of each of its  $n-1$  factors,

$$R_n(q) = q^{\binom{n}{2}} \prod_{j=2}^n (1+q^{-1}+\cdots+q^{-(j-1)}) = q^{\binom{n}{2}} (1+(n-1)q^{-1}+O(q^{-2})) \quad (28.97)$$

as  $q \rightarrow \infty$ . Since  $\mathcal{P}_n = R_n a_n$  is a polynomial of degree at most  $\binom{n}{2}$ , the rational function  $a_n = \mathcal{P}_n / R_n$  is regular at  $q = \infty$ ; let

$$a_n(q) = L_n + M_n q^{-1} + O(q^{-2}) \quad (q \rightarrow \infty) \quad (28.98)$$

be its expansion there. Multiplying (28.97) by (28.98) and comparing coefficients gives

$$[q^{\binom{n}{2}}] \mathcal{P}_n = L_n, \quad [q^{\binom{n}{2}-1}] \mathcal{P}_n = M_n + (n-1)L_n \quad (n \geq 2). \quad (28.99)$$

It remains to compute  $L_n$  and  $M_n$ . Multiply (28.7) by  $q^{-n}$ :

$$(q^{-n} - 1) a_n(q) = \sum_{k=0}^{n-1} \frac{(-1)^{n-k} (1 - q^{-1})^{n-k}}{(n-k+1)!} a_k(q), \quad (28.100)$$

using  $q^{-n} q^k (1 - q)^{n-k} = (-1)^{n-k} (1 - q^{-1})^{n-k}$ . Letting  $q \rightarrow \infty$  in (28.100) gives

$$-L_n = \sum_{k < n} (-1)^{n-k} \frac{L_k}{(n-k+1)!},$$

that is,

$$\sum_{k=0}^n \frac{(-1)^{n-k}}{(n-k+1)!} L_k = 0 \quad (n \geq 1), \quad L_0 = 1. \quad (28.101)$$

Set  $L(s) = \sum_{n \geq 0} L_n s^n$  and  $\Psi(s) = \sum_{r \geq 0} (-s)^r / (r+1)! = (1 - e^{-s})/s$ . Relation (28.101) says precisely that  $L(s)\Psi(s) = 1$ , whence

$$L(s) = \frac{s}{1 - e^{-s}} = \frac{-s}{e^{-s} - 1} = \sum_{n \geq 0} B_n \frac{(-s)^n}{n!}$$

by (27.6). Thus  $L_n = (-1)^n B_n / n!$ , which with (28.99) is (28.94).

For  $M_n$ , note first that  $a_1 \equiv \frac{1}{2}$ , so  $M_0 = M_1 = 0$ . For  $n \geq 2$  the term  $q^{-n} a_n(q)$  on the left of (28.100) is  $O(q^{-2})$ , so comparing coefficients of  $q^{-1}$  there gives

$$-M_n = \sum_{k=0}^{n-1} \frac{(-1)^{n-k} (M_k - (n-k)L_k)}{(n-k+1)!},$$

that is,  $[s^n](M\Psi) = [s^n](L\Theta)$  for  $n \geq 2$ , where  $M(s) = \sum M_n s^n$  and  $\Theta(s) = \sum_{r \geq 0} (-1)^r r s^r / (r+1)!$ . Since  $\sum_{r \geq 0} (r+1)(-s)^r / (r+1)! = e^{-s}$ , we have  $\Theta = e^{-s} - \Psi$ . The two remaining coefficients are checked directly:  $[s^0](M\Psi) = M_0 = 0 = [s^0](L\Theta)$ , while  $[s^1](M\Psi) = M_1 = 0$  and  $[s^1](L\Theta) = L_0 \Theta_1 = -\frac{1}{2}$ . Hence  $M\Psi = L\Theta + \frac{s}{2}$ , and since  $1/\Psi = L$ ,

$$M = L^2 \Theta + \frac{sL}{2} = L^2 e^{-s} - L + \frac{sL}{2}.$$

Differentiating  $L = s/(1 - e^{-s})$  gives  $sL' = L - L^2 e^{-s}$ , so  $L^2 e^{-s} = L - sL'$  and therefore  $M = -sL' + \frac{sL}{2}$ . Reading off coefficients,  $M_n = -nL_n + \frac{1}{2}L_{n-1}$ , and substituting into (28.99),

$$[q^{\binom{n}{2}-1}] \mathcal{P}_n = -nL_n + \frac{1}{2}L_{n-1} + (n-1)L_n = -L_n + \frac{1}{2}L_{n-1},$$

which is (28.95).

Finally,  $B_n \neq 0$  for  $n$  even and  $B_n = 0$  for odd  $n \geq 3$ , while  $L_1 = \frac{1}{2} \neq 0$ . So (28.94) is nonzero exactly for  $n$  even and for  $n = 1$ ; and when  $n \geq 3$  is odd, (28.95) reduces to  $\frac{1}{2}L_{n-1} = B_{n-1}/(2(n-1)!) \neq 0$ , since  $n-1$  is even and at least 2. This is (28.96).  $\square$



**Example 28.64.** For  $n = 3, \dots, 8$  the degrees of  $\mathcal{P}_n$  are

$$2, 6, 9, 15, 20, 28, \quad \text{against} \quad \binom{n}{2} = 3, 6, 10, 15, 21, 28,$$

so the bound (28.93) is missed by exactly one at each odd  $n$ . The leading coefficients of  $\mathcal{P}_4, \mathcal{P}_6, \mathcal{P}_8$  are  $-\frac{1}{720}, \frac{1}{30240}, -\frac{1}{1209600}$ , in agreement with  $B_4/4!, B_6/6!, B_8/8!$ .

**Corollary 28.65** (Pochhammer-cleared endpoint form and the half-base dictionary). For  $0 < q \leq \frac{1}{2}$  and  $n \geq 0$ ,

$$F_q(q^n) = \frac{q^{\binom{n+1}{2}}}{(q; q)_n} \mathcal{P}_n(q). \quad (28.102)$$

At  $q = 1/2$  this is a dictionary between the moment polynomial and the values of the Fabius function:

$$\mathcal{P}_n(1/2) = \mathfrak{m}_n F(2^{-n}), \quad (28.103)$$

where  $\mathfrak{m}_n = \prod_{j=1}^n (2^j - 1)$  is the Mersenne product (26.1).

*Proof.* By (28.91),  $\alpha_n^{(q)} / (1 - q)^n = \mathcal{P}_n(q) / (q; q)_n$ , so (28.102) is (28.15) rewritten. At  $q = \frac{1}{2}$ , (26.2) gives  $(1/2; 1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{m}_n$ , and the two powers of 2 in (28.102) cancel, leaving  $F(2^{-n}) = \mathcal{P}_n(1/2) / \mathfrak{m}_n$ .  $\square$

The first values, with  $F(2^{-n}) = 2^{-\binom{n}{2}} \alpha_n$  from Theorem 28.9 and  $\alpha_n$  as in Example 28.6, are

| $n$                  | 1             | 2              | 3               | 4                     | 5                     |
|----------------------|---------------|----------------|-----------------|-----------------------|-----------------------|
| $\mathcal{P}_n(1/2)$ | $\frac{1}{2}$ | $\frac{5}{24}$ | $\frac{7}{96}$  | $\frac{1001}{46080}$  | $\frac{4123}{737280}$ |
| $\mathfrak{m}_n$     | 1             | 3              | 21              | 315                   | 9765                  |
| $F(2^{-n})$          | $\frac{1}{2}$ | $\frac{5}{72}$ | $\frac{1}{288}$ | $\frac{143}{2073600}$ | $\frac{19}{33177600}$ |

*Remark 28.66* (The Pochhammer denominator is not imported notation). Corollary 28.65 closes the circle of this part. The finite  $q$ -Pochhammer symbol  $(q; q)_n$  in (28.102) was not chosen for cosmetic reasons: by Definition 28.61 and Theorem 28.62 it is exactly the pole-clearing factor of the recurrence (28.7), whose denominators  $1 - q^n$  are forced by the self-similarity (27.3). At  $q = 1/2$  it specializes, through Theorem 26.1, to the Mersenne product  $\mathfrak{m}_n$  of Chapter 26 – the same constant that Theorem 28.21 produces on the right of (28.32), and that normalizes the dyadic numerator  $\mathcal{N}_n(\theta) = \mathfrak{m}_n \alpha_n$  of Corollary 28.31. The three occurrences are one occurrence: the geometric Vandermonde denominator of the order- $n$  stencil on the nodes  $-2^k$ , the pole-clearing factor of the moment recurrence, and the Mersenne product are the same object read three ways.

## 28.8 The universal division-free definition

The factorial quotient is a formula, not a definition: it presupposes that every factor  $1 - q^j$  with  $j \leq n$  is invertible. Either recurrence of Theorem 2.3 is free of that hypothesis, and free of subtraction as well. It therefore defines the Gaussian coefficient over an arbitrary commutative semiring, and it is this universal object, rather than the quotient, that makes the root-of-unity specializations of Chapter 19 and the layered architecture of Principle 24.19 legitimate.

**Definition 28.67** (Universal Gaussian polynomials). Let  $u$  be an indeterminate. Define  $G_{n,k}(u)$  for  $n \in \mathbb{N}_0$  and  $0 \leq k \leq n$  by  $G_{0,0}(u) = 1$  and

$$G_{n+1,k}(u) = G_{n,k}(u) + u^{n+1-k} G_{n,k-1}(u) \quad (0 \leq k \leq n+1), \quad (28.104)$$

with the convention that  $G_{n,k}(u) = 0$  whenever  $k < 0$  or  $k > n$ . In the displayed range the exponent  $n+1-k$  is nonnegative, so no negative power occurs. For a commutative semiring  $R$  and  $a \in R$ , write  $G_{n,k}(a) \in R$  for the element produced by the same recursion with  $u$  replaced by  $a$ .



**Theorem 28.68** (Existence, uniqueness, and functoriality). (i) *There is exactly one family satisfying (28.104) with  $G_{0,0} = 1$ , and every  $G_{n,k}(u)$  lies in  $\mathbb{N}_0[u]$ .*

(ii)  $G_{n,k}(u) = \begin{bmatrix} n \\ k \end{bmatrix}_q$  under  $q = u$ ; that is, the factorial quotient is the value of the universal polynomial.

(iii) *If  $\phi: R \rightarrow S$  is a homomorphism of commutative semirings, then*

$$\phi(G_{n,k}(a)) = G_{n,k}(\phi(a)) \quad (a \in R, 0 \leq k \leq n).$$

*Hence the Gaussian coefficient is definable, and specializable, with no division, no subtraction, and no invertibility hypothesis on the factors  $1 - a^j$ .*

*Proof.* (i) Row  $n + 1$  is determined by row  $n$  through (28.104), and row 0 is prescribed; induction on  $n$  gives existence and uniqueness. The recursion uses only 0, 1, addition, multiplication, and nonnegative powers of  $u$ , so every  $G_{n,k}$  lies in  $\mathbb{N}_0[u]$ .

(ii) Replacing  $n$  by  $n + 1$  in the second recurrence (2.2) of Theorem 2.3 gives

$$\begin{bmatrix} n+1 \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n+1-k} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q,$$

which is (28.104) with  $u = q$ , and the boundary value  $\begin{bmatrix} 0 \\ 0 \end{bmatrix}_q = 1$  agrees. The uniqueness argument of (i) is valid over any commutative ring, in particular over the field  $\mathbb{Q}(u)$ ; hence the two families coincide there, and therefore in  $\mathbb{N}_0[u]$  by Corollary 2.4.

(iii) Apply  $\phi$  to (28.104). A semiring homomorphism preserves 0, 1, sums, products, and hence powers, so the family  $(\phi(G_{n,k}(a)))$  satisfies the same boundary condition and the same recursion in  $S$  with parameter  $\phi(a)$ . Uniqueness in  $S$ , again by (i), gives the assertion.  $\square$

**Corollary 28.69** (Denominator-free factorial identities). *Let  $R$  be a commutative ring and let  $a \in R$ . For  $0 \leq k \leq n$ ,*

$$(a; a)_k G_{n,k}(a) = (a^{n-k+1}; a)_k, \quad (28.105)$$

$$(a; a)_n = (a; a)_k (a; a)_{n-k} G_{n,k}(a). \quad (28.106)$$

*No factor  $1 - a^j$  is assumed invertible; both identities hold even when every product displayed in them vanishes. Thus Proposition 2.2 is a consequence of a polynomial identity rather than a prerequisite for the definition.*

*Proof.* By Theorem 28.68(ii),  $G_{n,k}(u) = (u; u)_n / ((u; u)_k (u; u)_{n-k})$  in  $\mathbb{Q}(u)$ . Both sides of (28.106) with  $a = u$  are polynomials, so the equality holds in  $\mathbb{Z}[u]$ . By Proposition 1.2 with  $m = n - k$ ,

$$(u; u)_n = (u; u)_{n-k} (u^{n-k+1}; u)_k;$$

since  $\mathbb{Z}[u]$  is an integral domain and  $(u; u)_{n-k} \neq 0$ , cancelling it in (28.106) gives (28.105) in  $\mathbb{Z}[u]$ . Finally apply the unique ring homomorphism  $\mathbb{Z}[u] \rightarrow R$  with  $u \mapsto a$ .  $\square$

**Example 28.70** (A primitive root of unity). Let  $a = e^{2\pi i/3}$ , so  $(a; a)_3 = (1 - a)(1 - a^2)(1 - a^3) = 0$  while  $(a; a)_2 = 3$ . The quotient  $\begin{bmatrix} 6 \\ 3 \end{bmatrix}_q$  at  $q = a$  displays 0/0, but the polynomial value is unambiguous:  $G_{6,3}(a) = 2$ , in agreement with the  $q$ -Lucas congruence of Chapter 19. Identity (28.106) reads  $0 = 0 \cdot 0 \cdot 2$ , and it is true. For a nondegenerate instance take  $n = 5$ ,  $k = 2$ : here  $G_{5,2}(a) = 1$  and (28.105) reads  $(a; a)_2 \cdot 1 = (a^4; a)_2$ , both sides equal to 3.

**Remark 28.71.** The transport statement Theorem 28.68(iii) is what licenses steps 1–2 and 6 of Principle 24.19: an identity established without division descends to finite characteristic, to quotient rings, to truncated power-series rings, and to roots of unity with no cancellation theorem interposed. The same observation applies verbatim to Theorem 3.1, whose proof is an induction from Theorem 2.3 and therefore valid over every commutative ring, and to the reversed row of Chapter 25; compare Remark 25.2.

## 28.9 The Möbius function of the subspace lattice

Counting is only the first half of the finite-field picture. The subspaces of  $\mathbb{F}_Q^n$  form a lattice, and its Möbius function is again a Gaussian datum: a sign and a quadratic power of  $Q$ . This is the structural reason for the shape of the inverse kernel in [Theorem 3.12](#).

**Theorem 28.72** (Möbius function of the subspace lattice). *Let  $Q$  be a prime power and let  $V$  be a  $d$ -dimensional vector space over  $\mathbb{F}_Q$ . In the lattice of subspaces of  $V$ , ordered by inclusion,*

$$\mu(0, V) = (-1)^d Q^{\binom{d}{2}}. \quad (28.107)$$

More generally, if  $U \subseteq W$  are subspaces and  $r = \dim(W/U)$ , then

$$\mu(U, W) = (-1)^r Q^{\binom{r}{2}}.$$

*Proof.* The correspondence  $X \mapsto X/U$  is an isomorphism from the interval  $[U, W]$  onto the lattice of subspaces of  $W/U$ , and the Möbius function of a poset depends only on the isomorphism type of the interval. It therefore suffices to prove (28.107).

Induct on  $d$ . For  $d = 0$  the value is  $\mu(0, 0) = 1$ , as asserted. Let  $d \geq 1$  and assume the formula for all smaller dimensions. The defining relation for the Möbius function of the interval  $[0, V]$  is

$$\sum_{X \subseteq V} \mu(0, X) = 0,$$

the sum running over all subspaces of  $V$ . Grouping the subspaces by dimension and using [Theorem 6.1](#) together with the induction hypothesis,

$$\mu(0, V) = - \sum_{j=0}^{d-1} \left[ \begin{matrix} d \\ j \end{matrix} \right]_Q (-1)^j Q^{\binom{j}{2}}.$$

By [Corollary 3.2](#) at base  $Q$ , which is (3.1) at  $z = 1$ ,

$$\sum_{j=0}^d \left[ \begin{matrix} d \\ j \end{matrix} \right]_Q (-1)^j Q^{\binom{j}{2}} = (1; Q)_d = 0 \quad (d \geq 1),$$

since the factor with index 0 in  $(1; Q)_d$  vanishes. Hence the truncated sum through  $j = d - 1$  equals the negative of the omitted term  $j = d$ , and therefore  $\mu(0, V) = (-1)^d Q^{\binom{d}{2}}$ .  $\square$

*Remark 28.73* (Where this kernel is met again). The inverse kernel  $(-1)^{n-k} q^{\binom{n-k}{2}} \left[ \begin{matrix} n \\ k \end{matrix} \right]_q$  of [Theorem 3.12](#) is exactly  $\mu(U, W)$  of [Theorem 28.72](#) with  $r = n - k$ , multiplied by the number  $\left[ \begin{matrix} n \\ k \end{matrix} \right]_Q$  of intermediate subspaces: for  $q = Q$  a prime power,  $q$ -binomial inversion is Möbius inversion over the subspace lattice for functions depending only on dimension, and the polynomial identity then extends it to an indeterminate  $q$ . The reversed Gaussian row  $N_{n,k}(q)$  of (25.1) carries the additional monomial  $q^{n-k}$ , because  $\binom{n-k}{2} + (n-k) = \binom{n-k+1}{2}$  and its nodes are  $q^1, \dots, q^n$  rather than  $q^0, \dots, q^{n-1}$ . This accounts for the sign and the quadratic exponent that recur throughout [Chapter 25](#).

**Proposition 28.74** (Recurrence for the translated Thue–Morse power sums). *For all  $k, d \geq 0$ ,*

$$M_{0,d}(c) = c^d, \quad (28.108)$$

and

$$M_{k+1,d}(c) = - \sum_{j=1}^d \binom{d}{j} 2^{kj} M_{k,d-j}(c). \quad (28.109)$$

Consequently all the quantities  $M_{k,d}(c)$  with  $k \leq n$  and  $d \leq 2n$  needed in (28.46) can be tabulated with  $O(n^3)$  arithmetic operations, rather than the  $2^k$  terms of the defining sum (28.37).

*Proof.* The base case is the single term  $r = 0$ . For the recurrence, split  $\{0, \dots, 2^{k+1} - 1\}$  into  $\{0, \dots, 2^k - 1\}$  and its translate by  $2^k$ . Adding  $2^k$  to an integer  $r < 2^k$  turns off no binary digit and turns on exactly one, so  $\tau_{r+2^k} = -\tau_r$ . Hence

$$M_{k+1,d}(c) = \sum_{r=0}^{2^k-1} \tau_r \left( (r+c)^d - (r+c+2^k)^d \right) = - \sum_{j=1}^d \binom{d}{j} 2^{kj} \sum_{r=0}^{2^k-1} \tau_r (r+c)^{d-j},$$

by the binomial theorem, the term  $j = 0$  cancelling. This is (28.109). Filling the table in increasing  $k$  and  $d$  costs  $O(d)$  operations per entry, hence  $O(n \cdot n \cdot n)$  in total for the stated ranges.  $\square$

## 28.10 A ring-level residual principle

**Theorem 25.6** is stated for the Lagrange weights of  $N$  distinct nodes in a field, and the free target  $x$  carried by both sides of (25.7) is what makes it a statement about interpolation. The mechanism it uses is narrower than its hypotheses. The field, the distinctness of the nodes, the Lagrange normalization of the weights, and the invertibility of the differences  $v_i - v_j$  can all be discarded if one asks only for the moments of a finite weight system: the single hypothesis that the first  $p$  moments vanish already determines every later moment. In that form the statement holds over an arbitrary commutative ring, and both exact tails of this part become instances of it.

**Theorem 28.75** (All-degree residual from the first surviving moment). *Let  $R$  be a commutative ring and let  $x_0, \dots, x_p \in R$  and  $w_0, \dots, w_p \in R$  be arbitrary elements. Put*

$$M_m := \sum_{i=0}^p w_i x_i^m \quad (m \geq 0), \quad (28.110)$$

with the convention  $x^0 = 1$  for every  $x \in R$ . If

$$M_0 = M_1 = \dots = M_{p-1} = 0, \quad M_p = K, \quad (28.111)$$

then

$$M_{p+r} = K h_r(x_0, \dots, x_p) \quad (r \geq 0), \quad (28.112)$$

where  $h_r$  is the complete homogeneous symmetric polynomial in the  $p+1$  displayed variables, evaluated in  $R$ . No node is assumed distinct from another, no weight and no difference  $x_i - x_j$  is assumed invertible, and  $R$  may have zero divisors. For  $p = 0$  the hypothesis (28.111) is empty,  $K = w_0$ , and (28.112) is the definition of  $M_r$ .

*Proof.* Work in  $R[[t]]$  and put

$$M(t) = \sum_{m \geq 0} M_m t^m, \quad E(t) = \prod_{i=0}^p (1 - x_i t).$$

Each factor  $1 - x_i t$  is invertible in  $R[[t]]$ , with inverse  $\sum_{m \geq 0} x_i^m t^m$ . Interchanging the finite sum over  $i$  with the sum over  $m$  in (28.110) gives  $M(t) = \sum_{i=0}^p w_i (1 - x_i t)^{-1}$ , and therefore

$$E(t)M(t) = \sum_{i=0}^p w_i \prod_{j \neq i} (1 - x_j t). \quad (28.113)$$

The right-hand side of (28.113) is a polynomial in  $t$  of degree at most  $p$ , each of its  $p+1$  summands being a product of  $p$  linear factors.

Compute the same coefficients from the left. By (4.3) with  $z = -t$  one has  $E(t) = \sum_{l=0}^{p+1} (-1)^l e_l t^l$ , where  $e_l = e_l(x_0, \dots, x_p)$ , so

$$[t^m](E(t)M(t)) = \sum_{l=0}^m (-1)^l e_l M_{m-l}$$

involves only  $M_0, \dots, M_m$ . For  $m < p$  every moment occurring here has index at most  $m < p$  and vanishes by (28.111); for  $m = p$  the only surviving term is  $e_0 M_p = K$ . Comparing with the degree bound just obtained,

$$E(t)M(t) = Kt^p.$$

Finally  $E(t)$  has constant term 1, and by (4.4) its inverse in  $R[[t]]$  is  $\sum_{r \geq 0} h_r(x_0, \dots, x_p) t^r$ . Multiplying the last display by that inverse gives

$$M(t) = Kt^p \sum_{r \geq 0} h_r(x_0, \dots, x_p) t^r,$$

and comparison of the coefficients of  $t^{p+r}$  is (28.112).  $\square$

**Remark 28.76** (The exact scope of the hypotheses). Three comments delimit the statement.

- (i) *The hypotheses are satisfiable with non-invertible data.* In  $R = \mathbb{Z}/12\mathbb{Z}$  take nodes  $(x_0, x_1, x_2) = (0, 1, 2)$  and weights  $(w_0, w_1, w_2) = (1, -2, 1)$ . Then  $M_0 = M_1 = 0$  and  $K = M_2 = 2$ , a zero divisor; no weight and no node difference is a unit. Here  $h_r(0, 1, 2) = h_r(1, 2) = 2^{r+1} - 1$  and  $M_{2+r} = 2^{r+2} - 2 = 2h_r(0, 1, 2)$ , an identity of integers and hence one in every quotient of  $\mathbb{Z}$ .
- (ii) *Coincident nodes make the conclusion empty rather than false.* Group the nodes by their distinct values  $v_1, \dots, v_s$  and let  $W_j$  be the sum of the weights carried by the value  $v_j$ , so that  $M_m = \sum_j W_j v_j^m$ . If two nodes coincide then  $s \leq p$ , and the first  $s$  hypotheses say that the  $s \times s$  Vandermonde matrix of  $v_1, \dots, v_s$  annihilates  $(W_1, \dots, W_s)$ . Whenever its determinant  $\prod_{i < j} (v_j - v_i)$  is not a zero divisor – in particular over a field – the adjugate identity forces  $W_1 = \dots = W_s = 0$ , hence  $K = 0$ , and both sides of (28.112) vanish. Distinctness is thus not needed as a hypothesis, and adding it would exclude nothing of substance.
- (iii) *Neither residual theorem contains the other.* [Theorem 25.6](#) is not a corollary of [Theorem 28.75](#): its two sides carry a free target  $x$ , and its right-hand side involves the  $(N+1)$ -variable symmetric function  $h_r(x, v_1, \dots, v_N)$ , of which (28.112) sees only the specialization  $x = 0$ . Conversely (28.112) is not a corollary of (25.7), which presupposes a field, distinct nodes, and one particular weight vector. The present development uses (25.7) only at  $x = 0$ , in the proof of [Theorem 25.7](#), and [Remark 28.80](#) (i) recovers exactly that use from (28.112).

**Corollary 28.77** (Geometric Gaussian residual for arbitrary weights). *Let  $R$  be a commutative ring, let  $c, q \in R$ , and let  $w_0, \dots, w_p \in R$  satisfy (28.111) on the nodes  $x_i = cq^i$ . Then*

$$\sum_{i=0}^p w_i (cq^i)^{p+r} = K c^r \begin{bmatrix} p+r \\ p \end{bmatrix}_q \quad (r \geq 0), \quad (28.114)$$

*the Gaussian coefficient being read in  $R$  as the value at  $q$  of the universal polynomial of [Definition 28.67](#), so that no factor  $1 - q^j$  is assumed invertible.*

*Proof.* By [Theorem 28.75](#) the left-hand side of (28.114) equals  $K h_r(c, cq, \dots, cq^p)$ . Since  $h_r$  is homogeneous of degree  $r$ , scaling every variable by  $c$  multiplies it by  $c^r$ , so  $h_r(c, cq, \dots, cq^p) = c^r h_r(1, q, \dots, q^p)$ .

It remains to identify  $h_r(1, q, \dots, q^p)$ . In the notation of [Chapter 4](#) this is  $S_r^{p+1}(\mathbf{w})$  for  $w_i = q^{i-1}$ , which by [Theorem 4.6](#) (ii) with  $n = p + 1$  and  $k = r$  equals  $\begin{bmatrix} p+r \\ r \end{bmatrix}_q$ , hence  $\begin{bmatrix} p+r \\ p \end{bmatrix}_q$  by the symmetry [Theorem 2.5](#) (i). Both sides of

$$h_r(1, q, \dots, q^p) = \begin{bmatrix} p+r \\ p \end{bmatrix}_q \quad (28.115)$$

lie in  $\mathbb{Z}[q]$  – the left one by inspection of (4.4), the right one by [Theorem 28.68](#) (i),(ii) – and they agree in  $\mathbb{Q}(q)$ , so they agree in  $\mathbb{Z}[q]$ . Applying the evaluation homomorphism  $\mathbb{Z}[q] \rightarrow R$  with  $q \mapsto q$  and using the functoriality [Theorem 28.68](#) (iii) transports (28.115) to  $R$ .  $\square$

*Remark 28.78.* [Corollary 25.8](#) is the case of (28.114) in which the weights are the Lagrange row  $\lambda_{n,k}^{(q)}$ , after the deflation described in [Remark 28.80](#) (i); what is new here is that no property of the weight row is used beyond (28.111), and that  $c, q$  and the  $w_i$  need not lie in a field.

**Corollary 28.79** (Formal-series filter for arbitrary weights). *Let  $R$  be a commutative ring and let  $x_0, \dots, x_p$  and  $w_0, \dots, w_p$  in  $R$  satisfy (28.111). For every  $f(z) = \sum_{m \geq 0} a_m z^m \in R[[z]]$ ,*

$$\sum_{i=0}^p w_i f(x_i z) = K \sum_{r \geq 0} a_{p+r} h_r(x_0, \dots, x_p) z^{p+r}. \quad (28.116)$$

*On a geometric node set  $x_i = cq^i$  the multiplier of  $a_{p+r}$  is  $Kc^r \begin{bmatrix} p+r \\ p \end{bmatrix}_q$ .*

*Proof.* The left-hand side is a sum of *finitely many* elements of  $R[[z]]$ , so coefficient extraction commutes with it:

$$[z^m] \sum_{i=0}^p w_i f(x_i z) = a_m \sum_{i=0}^p w_i x_i^m = a_m M_m.$$

By (28.111) this vanishes for  $m < p$ , and for  $m = p + r$  it is  $K a_{p+r} h_r(x_0, \dots, x_p)$  by [Theorem 28.75](#). The last assertion is [Corollary 28.77](#).  $\square$

*Remark 28.80* (One principle, two exact tails). The two exact tails of this part are instances of (28.116). This does not merge them – what keeps them apart is recorded in [Remark 28.18](#) – but it does replace two separate residual computations by one.

- (i) *Evaluation at zero, after deflation.* The Lagrange row  $\lambda_{n,k}^{(q)}$  has  $M_0 = 1$ , so (28.111) fails for it. Adjoin one further node  $x_{n+1} = 0$  with weight  $w_{n+1} = -1$ . The augmented system has  $n + 2$  nodes, so  $p = n + 1$ ; its moments are  $M_0 = 1 - 1 = 0$  and  $M_m = 0$  for  $1 \leq m \leq n$  by (25.5), since  $0^m = 0$  there. Its value of  $K$  can be read off (28.113) without any prior knowledge of the row: among the  $n + 2$  summands only the one indexed by the adjoined node has degree  $n + 1$ , namely  $-\prod_{k=0}^n (1 - q^k t)$ , whose coefficient of  $t^{n+1}$  is  $-(-1)^{n+1} q^{\binom{n+1}{2}}$ , so

$$K = (-1)^n q^{\binom{n+1}{2}}.$$

A variable equal to 0 contributes the factor 1 to (4.4), so

$$h_r(1, q, \dots, q^n, 0) = h_r(1, q, \dots, q^n),$$

and (28.114) returns (25.8) while (28.116) returns (25.12). Running the same argument on the nodes  $cq^k$  together with 0 returns [Corollary 25.8](#). This is a proof of [Theorem 25.7](#) that does not pass through [Theorem 25.6](#) and uses no symmetric-function orthogonality.

- (ii) *Extraction of the top coefficient.* The data (28.20) satisfy (28.111) with  $p = n$  and  $K = q^{-\binom{n+1}{2}}(q; q)_n$ , by (28.22) and (28.23); no deflation is needed, because that row already annihilates the constants. Its nodes are  $-q^{-k} = c\tilde{q}^k$  with  $c = -1$  and  $\tilde{q} = q^{-1}$ , so (28.115) gives

$$h_r(-1, -q^{-1}, \dots, -q^{-n}) = (-1)^r \begin{bmatrix} n+r \\ n \end{bmatrix}_{q^{-1}} = (-1)^r q^{-nr} \begin{bmatrix} n+r \\ n \end{bmatrix}_q,$$

the last step being the reciprocity Theorem 2.5 (ii) with  $k(n-k) = nr$ . Multiplying (28.116) by  $q^{\binom{n+1}{2}}/(q; q)_n$  therefore reproduces (28.27) exactly, giving a second proof of Theorem 28.17 in which no shifted factorial  $(q^{-n-r}; q)_n$  is ever rewritten.

The two instances differ in precisely one structural respect: the first has a nonzero moment of order 0 and must be deflated by one extra node, the second does not. Everything else — the vanishing block, the single surviving constant  $K$ , and the Gaussian row that multiplies the tail — is supplied by Theorem 28.75.

## 28.11 Cells, a local reduction, and the digit expansion

Formula (28.49) evaluates  $F$  exactly, but only at dyadic rationals; formula (28.64) evaluates it at every point, but only by quadrature. This section supplies the missing third description: an exact, digit-driven expansion valid at every point of  $[0, 1)$ , whose partial sums are finite rational combinations of the moments of  $X$  and whose remainder is itself a value of  $F$ , at an argument smaller by a factor  $2^{-N}$ . The mechanism is the moment functional of Corollary 28.31 carried through with a fractional part attached.

**Definition 28.81** (Shifted moment polynomials). For  $n \geq 0$  and  $\theta \in \mathbb{C}$  put

$$\mathfrak{a}_n(\theta) := \mathbb{E}(X + \theta)^n = \sum_{j=0}^n \binom{n}{j} \mathbb{E}X^{n-j} \theta^j, \quad (28.117)$$

a monic polynomial of degree  $n$  with

$$\mathfrak{a}_n(0) = \mathbb{E}X^n = n! \alpha_n$$

and exponential generating function

$$\sum_{n \geq 0} \mathfrak{a}_n(\theta) z^n / n! = e^{\theta z} \mathcal{A}(z).$$

Thus  $(\mathfrak{a}_n)$  is the Appell sequence attached to  $\mathcal{A}$ : differentiating the generating function gives  $\mathfrak{a}'_n = n \mathfrak{a}_{n-1}$ , multiplying it by  $e^{cz}$  gives the binomial translation law, of which (27.13) is the instance  $c = 1/2$  for the centred variable, and  $X \stackrel{d}{=} 1 - X$  from (27.2) gives the reflection  $\mathfrak{a}_n(-1 - \theta) = (-1)^n \mathfrak{a}_n(\theta)$ , the moment counterpart of the  $k$ -independent digit-complementation sign  $(-1)^n$  computed at the end of the proof of Corollary 28.31. Only (28.117) itself is used below; in this notation the numerator coefficient (28.52) reads  $p_{m,n} = \frac{1}{n!} \sum_{h < m} \tau_h \mathfrak{a}_n(m - h - 1)$ .

**Lemma 28.82** (Spline expectation at an arbitrary argument). Write  $y_+ := \max(y, 0)$ . For every  $n \geq 1$  and every real  $t$ ,

$$F\left(\frac{t}{2^n}\right) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{r=0}^{2^n-1} \tau_r \mathbb{E}(t - r - X)_+^n. \quad (28.118)$$



*Proof.* Use the scaled splitting (28.18):  $2^n X = \sum_{j=1}^n c_j V_j + X'$  with  $c_j = 2^{n-j}$ , with  $V_1, \dots, V_n$  independent and uniform on  $[0, 1]$ , and with  $X'$  independent of them and distributed as  $X$ . Since  $(V_1, \dots, V_n)$  is uniform on the unit cube, conditioning on  $X'$  gives

$$\mathbb{P}(2^n X \leq t \mid X') = \mathbb{P}\left(\sum_{j=1}^n c_j V_j \leq t - X' \mid X'\right) = V_c(t - X'),$$

where  $V_c$  is the weighted cube cut (27.19). For the integer weights  $c_j = 2^{n-j}$  that cut is (27.21), an identity valid at every real argument. Substituting it, averaging over  $X'$ , and exchanging the expectation with the finite sum – legitimate because the sum has  $2^n$  terms, each bounded on the bounded range of  $X'$  – gives (28.118), with  $X$  written for  $X'$ .  $\square$

**Remark 28.83** (What the unrestricted argument buys). At an integer  $t = m$  with  $0 \leq m \leq 2^n$  every term with  $r \geq m$  of (28.118) vanishes and every term with  $r < m$  loses its positive part; that is exactly the computation performed in Step 1 of the proof of Corollary 28.31, and it produces (28.53). The content of the lemma is that no restriction on  $t$  is needed. At a non-integer  $t$  one term keeps its positive part, and Theorem 28.84 identifies it.

**Theorem 28.84** (Exact formula on every dyadic cell). *Let  $n \geq 1$ , let  $m$  be an integer with  $0 \leq m \leq 2^n$ , and let  $\theta$  be real with  $0 \leq \theta < 1$  and  $m + \theta \leq 2^n$ . Then*

$$F\left(\frac{m + \theta}{2^n}\right) = \tau_m F\left(\frac{\theta}{2^n}\right) + \frac{2^{-\binom{n}{2}}}{n!} \sum_{h=0}^{m-1} \tau_h \mathfrak{a}_n(m - h - 1 + \theta), \quad (28.119)$$

the sum being empty when  $m = 0$ . At  $\theta = 0$  the first term vanishes and (28.119) is (28.53).

*Proof.* Apply Lemma 28.82 at  $t = m + \theta$  and classify the  $2^n$  terms, using  $0 \leq X \leq 1$  and  $0 \leq \theta < 1$ .

If  $r \geq m + 1$ , then  $t - r - X \leq m + \theta - (m + 1) = \theta - 1 < 0$  and the term vanishes.

If  $r \leq m - 1$ , then  $t - r - X \geq m + \theta - (m - 1) - 1 = \theta \geq 0$ , so the positive part is inactive and

$$\mathbb{E}(m + \theta - r - X)_+^n = \mathbb{E}((1 - X) + m - r - 1 + \theta)^n = \mathfrak{a}_n(m - r - 1 + \theta),$$

because  $1 - X \stackrel{d}{=} X$  by (27.2). These terms contribute the sum in (28.119).

There remains the term  $r = m$ , which occurs in the range  $0 \leq r \leq 2^n - 1$  precisely when  $m \leq 2^n - 1$ . In that case its contribution is  $\tau_m 2^{-\binom{n}{2}} \mathbb{E}(\theta - X)_+^n / n!$ . Apply Lemma 28.82 once more, now at  $t = \theta$ : since  $0 \leq \theta < 1$ , every index  $r \geq 1$  gives  $\theta - r - X \leq \theta - 1 < 0$  and drops out, so only  $r = 0$  survives and

$$F\left(\frac{\theta}{2^n}\right) = \frac{2^{-\binom{n}{2}}}{n!} \mathbb{E}(\theta - X)_+^n. \quad (28.120)$$

The  $r = m$  term is therefore  $\tau_m F(\theta/2^n)$ , which is the first summand of (28.119).

If instead  $m = 2^n$ , then  $\theta = 0$  by the hypothesis  $m + \theta \leq 2^n$ ; the index  $r = m$  is absent from the range of summation, and the first summand of (28.119) is  $\tau_{2^n} F(0) = 0$ , so the formula again holds. This proves (28.119) in all cases. Finally, at  $\theta = 0$  one has  $F(0) = 0$  and  $\mathfrak{a}_n(m - h - 1)$  in place of  $\mathfrak{a}_n(m - h - 1 + \theta)$ , so (28.119) becomes  $F(m/2^n) = 2^{-\binom{n}{2}} p_{m,n}$  by (28.52), which is (28.53).  $\square$

**Remark 28.85** (Both hypotheses on  $\theta$  are needed, and how sharp they are). The restriction  $m + \theta \leq 2^n$  cannot be dropped: at  $n = 1$ ,  $m = 2$ ,  $\theta = 1/4$  the right-hand side of (28.119) is  $-F(1/8) + \mathfrak{a}_1(5/4) - \mathfrak{a}_1(1/4) = 1 - \frac{1}{288} = \frac{287}{288}$ , whereas  $F(9/8) = 1$ . The restriction  $\theta < 1$  can be relaxed to  $\theta \leq 1$  but no further. At  $\theta = 1$  with  $m + 1 \leq 2^n$  the identity persists, because  $\mathfrak{a}_n(0) = n! \alpha_n$  and (28.13) turn the extra term  $\tau_m 2^{-\binom{n}{2}} \mathfrak{a}_n(0)/n!$  of (28.119) at the numerator  $m + 1$  into  $\tau_m F(2^{-n})$ . At  $\theta = 3/2$  it fails: for  $n = 2$ ,  $m = 1$  the right-hand side is  $-F(3/8) + \frac{1}{4} \mathfrak{a}_2(3/2) = -\frac{73}{288} + \frac{145}{144} = \frac{217}{288}$ , whereas  $F(5/8) = \frac{215}{288}$ .

Let  $x \in [0, 1)$  and let  $x = 0.b_1b_2b_3 \dots$  in base two be its canonical binary expansion, the one that is not eventually constant equal to 1. Put

$$m_j := \lfloor 2^j x \rfloor, \quad \theta_j := 2^j x - m_j \in [0, 1) \quad (j \geq 0), \quad (28.121)$$

so that  $m_0 = 0$  and  $\theta_0 = x$ . The binary digits of  $m_j$  are  $b_1 \dots b_j$ ; hence

$$m_j = 2m_{j-1} + b_j, \quad \tau_{m_j} = (-1)^{b_j} \tau_{m_{j-1}}, \quad \theta_{j-1} = \frac{b_j + \theta_j}{2} \quad (j \geq 1), \quad (28.122)$$

the middle relation because appending a digit 0 leaves  $\text{wt}_2$  unchanged and appending a digit 1 raises it by one.

**Example 28.86** (A non-dyadic argument). Let  $x = 1/3$ , whose binary expansion is  $0.\overline{01}$ . Then  $b_j = 1$  exactly for even  $j$ , and  $\theta_j = 1/3$  for even  $j$ ,  $\theta_j = 2/3$  for odd  $j$ , while  $m_1 = 0$ ,  $m_3 = 2$ ,  $m_5 = 10$ , with  $\tau_{m_1} = 1$ ,  $\tau_{m_3} = -1$ ,  $\tau_{m_5} = 1$ . Hence (28.152) reads

$$F\left(\frac{1}{3}\right) = \frac{\mathfrak{a}_2(1/3)}{4} - \frac{\mathfrak{a}_4(1/3)}{2^6 \cdot 4!} + \frac{\mathfrak{a}_6(1/3)}{2^{15} \cdot 6!} - \dots = \frac{13}{72} - \frac{2429}{6220800} + \frac{198007}{8427641241600} - \dots,$$

whose partial sums are

$$S_2 = \frac{13}{72} = 0.180555\dots, \quad S_4 = 0.1801650913\dots, \quad S_6 = 0.18016511480153\dots,$$

against  $F(1/3) = 0.1801651148014819\dots$  obtained from (28.64). Three terms already agree with  $F(1/3)$  to within  $5.2 \cdot 10^{-14}$ . The guaranteed bound of (28.153) at  $N = 6$  is  $2^{-15}/6! = 4.2 \cdot 10^{-8}$ ; it is far from sharp here because the residual is  $F(1/192)$  and  $1/192$  lies well inside the interval  $[0, 2^{-6}]$  on which  $F$  is extremely flat.

*Remark 28.87* (Three exact descriptions of  $F$ ). The remark following [Corollary 28.41](#) contrasts two exact descriptions of  $F$ : the global analytic one (28.64) and the finite rational one (28.49), which applies only at dyadic arguments. Formula (28.152) is a third, and it is the only one that is simultaneously exact, elementary, and valid at every argument. Its cost structure is also different from both. The  $j$ th term needs the moments  $\mathbb{E}X^0, \dots, \mathbb{E}X^j$ , which (28.11) produces through  $\mathbb{E}X^n = n! \alpha_n$  at one rational division each; the accuracy is governed by the superfactorial weight  $2^{-\binom{N}{2}}/N!$  of (28.153), which is  $7.8 \cdot 10^{-21}$  at  $N = 10$  and  $2.6 \cdot 10^{-76}$  at  $N = 20$ , uniformly in  $x$ ; and only the indices  $j$  with  $b_j = 1$  contribute a term, so the number of nonzero terms is the binary weight of the truncated argument and not its length. By contrast (28.49) needs a Thue–Morse block of length  $m2^k$ , and (28.64) needs quadrature of an oscillatory integral.

The truncation  $X_{q,N}$  of (27.17) discards a nonnegative tail, so [Corollary 27.13](#) is a one-sided coupling and its rate  $q^N$  is the length of the discarded interval. Adding back the *mean* of that tail centres the coupling and removes the first-order term; the resulting rate is second order in  $q^N$ , and it can be computed exactly.

**Proposition 28.88** (Midpoint-centred truncations, and an exact second-order rate). *Let  $0 < q < 1$  and  $N \geq 1$ , and put*

$$X_{q,N}^{\text{mid}} := X_{q,N} + \frac{1}{2}q^N, \quad F_{q,N}^{\text{mid}}(x) := \mathbb{P}(X_{q,N}^{\text{mid}} \leq x) = F_{q,N}\left(x - \frac{1}{2}q^N\right). \quad (28.123)$$

*At the half base write  $S_p := F_{1/2,p}^{\text{mid}}$ . Then:*

1. *for every  $p \geq 1$  and every real  $x$ ,*

$$S_p(x) = \frac{2^{-\binom{p}{2}}}{p!} \sum_{r=0}^{2^p-1} \tau_r \left(2^p x - \frac{1}{2} - r\right)_+^p, \quad \tau_r = (-1)^{\text{wt}_2(r)}; \quad (28.124)$$



2. for every  $N \geq 2$ ,

$$\sup_{x \in \mathbb{R}} |F_{q,N}^{\text{mid}}(x) - F_q(x)| \leq \frac{q^{2N-1}}{24(1-q^2)}; \quad (28.125)$$

3. if moreover  $0 < q \leq 1/2$ , then (28.125) is an equality, the extreme value being attained at  $x^* := q(1-q)$ :

$$F_{q,N}^{\text{mid}}(x^*) - F_q(x^*) = -\frac{q^{2N-1}}{24(1-q^2)}. \quad (28.126)$$

At  $q = 1/2$  this reads  $\sup_{x \in \mathbb{R}} |S_p(x) - F(x)| = \frac{1}{9} 4^{-p}$  for  $p \geq 2$ , attained at  $x = 1/4$ ;

4. the hypothesis  $N \geq 2$  cannot be dropped: at  $q = 1/2$  and  $N = 1$  the variable  $X_{1/2,1}^{\text{mid}} = \frac{1}{2}U_0 + \frac{1}{4}$  is uniform on  $[1/4, 3/4]$ , so  $S_1(1/4) = 0$  while  $F(1/4) = 5/72$  by Remark 28.32, whereas the right-hand side of (28.125) would be  $1/36 < 5/72$ .

*Proof.* (i). By Remark 27.10 with  $n = p$ , the weights  $c_j = 2^{p-j}$  are  $2^p$  times the half-base weights  $a_{j-1} = 2^{-j}$ , so  $V_c(t) = \mathbb{P}(2^p X_{1/2,p} \leq t) = F_{1/2,p}(2^{-p}t)$  and (27.21) reads

$$F_{1/2,p}(2^{-p}t) = \frac{2^{-\binom{p}{2}}}{p!} \sum_{r=0}^{2^p-1} \tau_r(t-r)_+^p.$$

By (28.123),  $S_p(x) = F_{1/2,p}(x - 2^{-p-1})$ , which is the above at  $t = 2^p x - \frac{1}{2}$ . This is (28.124).

An exact error integral. Iterating (27.3)  $N$  times gives  $X_q \stackrel{d}{=} X_{q,N} + q^N X'_q$  with  $X'_q$  an independent copy of  $X_q$ ; at  $q = 1/2$  this is (28.18) after scaling. Write  $Y := X_{q,N}$ , let  $g_{q,N}$  be its density (Theorem 27.9), and put

$$H_q(u) := \mathbf{1}_{[1/2, \infty)}(u) - F_q(u),$$

which vanishes for  $u < 0$  and for  $u > 1$ . Conditioning on  $Y$ ,

$$F_q(x) = \mathbb{E}\left[F_q(q^{-N}(x - Y))\right], \quad F_{q,N}^{\text{mid}}(x) = \mathbb{P}(Y \leq x - \tfrac{1}{2}q^N) = \mathbb{E}\left[\mathbf{1}_{[1/2, \infty)}(q^{-N}(x - Y))\right],$$

so, substituting  $u = q^{-N}(x - y)$ ,

$$F_{q,N}^{\text{mid}}(x) - F_q(x) = \mathbb{E}\left[H_q(q^{-N}(x - Y))\right] = q^N \int_0^1 g_{q,N}(x - q^N u) H_q(u) du. \quad (28.127)$$

The two moments of  $H_q$ . Since  $\mathbb{E}X_q = \int_0^1 (1 - F_q)$  and  $\mathbb{E}X_q = 1/2$  by (27.2),

$$\int_0^1 H_q(u) du = \frac{1}{2} - \int_0^1 F_q(u) du = \frac{1}{2} - (1 - \mathbb{E}X_q) = 0. \quad (28.128)$$

Integrating by parts,  $\int_0^1 u F_q(u) du = \frac{1}{2} - \frac{1}{2} \mathbb{E}X_q^2$ , whence

$$\int_0^1 (u - \tfrac{1}{2}) H_q(u) du = \int_{1/2}^1 (u - \tfrac{1}{2}) du - \int_0^1 (u - \tfrac{1}{2}) F_q(u) du = \tfrac{1}{2} \mathbb{E}X_q^2 - \tfrac{1}{8} = \tfrac{1}{2} \text{Var}(X_q).$$

By (27.7) at  $m = 1$ , with  $B_2 = 1/6$ ,  $\text{Var}(X_q) = \kappa_2(Y_q) = (1 - q)/(12(1 + q))$ , so

$$\int_0^1 (u - \tfrac{1}{2}) H_q(u) du = \frac{1 - q}{24(1 + q)}. \quad (28.129)$$

Finally  $H_q \leq 0$  on  $[0, 1/2)$  and  $H_q \geq 0$  on  $[1/2, 1]$ , so  $(u - \frac{1}{2})H_q(u) = |u - \frac{1}{2}| |H_q(u)|$  pointwise and (28.129) also computes  $\int_0^1 |u - \frac{1}{2}| |H_q(u)| du$ .

*The density and its Lipschitz constant.* For  $M \geq 1$  write  $X_{q,M} = (1-q)U_0 + qX_{q,M-1}$  with the two summands independent and  $X_{q,0} := 0$ . Conditioning on the second summand, which contributes the uniform density  $1/(1-q)$  on a window of length  $1-q$ , gives

$$g_{q,M}(x) = \frac{1}{1-q} \left[ F_{q,M-1}\left(\frac{x}{q}\right) - F_{q,M-1}\left(\frac{x-1+q}{q}\right) \right], \quad (28.130)$$

the difference of two values of a distribution function; hence  $0 \leq g_{q,M} \leq 1/(1-q)$ , the finite- $M$  form of Proposition 27.12(ii). If  $M \geq 2$  then  $F_{q,M-1}$  has the bounded density  $g_{q,M-1}$  and is therefore Lipschitz with constant  $1/(1-q)$ , so (28.130) shows that  $g_{q,M}$  is Lipschitz with constant

$$\text{Lip}(g_{q,M}) \leq \frac{1}{q(1-q)^2} \quad (M \geq 2). \quad (28.131)$$

(ii). Let  $N \geq 2$ . By (28.128) we may subtract the constant  $g_{q,N}(x - \frac{1}{2}q^N)$  inside (28.127), so that

$$\begin{aligned} |F_{q,N}^{\text{mid}}(x) - F_q(x)| &\leq q^N \int_0^1 \left| g_{q,N}(x - q^N u) - g_{q,N}(x - \tfrac{1}{2}q^N) \right| \\ &\quad \cdot |H_q(u)| \, du \\ &\leq \frac{q^{2N}}{q(1-q)^2} \int_0^1 \left| u - \tfrac{1}{2} \right| |H_q(u)| \, du. \end{aligned}$$

by (28.131). Now insert (28.129):

$$\frac{q^{2N-1}}{(1-q)^2} \cdot \frac{1-q}{24(1+q)} = \frac{q^{2N-1}}{24(1-q^2)},$$

which is (28.125).

(iii). Let  $0 < q \leq 1/2$  and  $N \geq 2$ . We claim that  $g_{q,N}$  is affine with slope exactly  $1/(q(1-q)^2)$  on  $I := [x^* - q^N, x^*]$ , where  $x^* = q(1-q)$ .

First, for  $M \geq 1$ ,

$$g_{q,M}(y) = \frac{1}{1-q} \quad \text{for (almost) every } y \in [1-q-q^M, 1-q]. \quad (28.132)$$

For  $M = 1$  the variable  $X_{q,1} = (1-q)U_0$  is uniform on  $[0, 1-q]$  and  $1-q-q \geq 0$  because  $q \leq 1/2$ , so the interval lies in the support and the density is  $1/(1-q)$  almost everywhere on it. For  $M \geq 2$  use (28.130): such a  $y$  has  $y-1+q \leq 0$ , and  $X_{q,M-1}$  has a density, so  $F_{q,M-1}((y-1+q)/q) = 0$ ; moreover  $y/q \geq (1-q)/q - q^{M-1} \geq 1 - q^{M-1}$  because  $(1-q)/q \geq 1$ , while  $X_{q,M-1} \leq 1 - q^{M-1}$ , so  $F_{q,M-1}(y/q) = 1$ . This proves (28.132), and hence  $F_{q,M}$  is affine with slope  $1/(1-q)$  on  $[1-q-q^M, 1-q]$ .

Now let  $x \in I$ . Then  $x \leq q(1-q) < 1-q$ , so  $x-1+q < 0$  and the second term of (28.130) vanishes, giving  $g_{q,N}(x) = F_{q,N-1}(x/q)/(1-q)$ . As  $x$  runs through  $I$ , the point  $x/q$  runs through  $[1-q-q^{N-1}, 1-q]$ , on which  $F_{q,N-1}$  is affine with slope  $1/(1-q)$  by (28.132) with  $M = N-1 \geq 1$ . Composing the two affine maps proves the claim.

Write therefore  $g_{q,N}(x^* - q^N u) = g_{q,N}(x^*) - q^N u/(q(1-q)^2)$  for  $0 \leq u \leq 1$  and substitute into (28.127). The constant term is annihilated by (28.128), which also gives  $\int_0^1 u H_q = \int_0^1 (u - \frac{1}{2}) H_q$ , so by (28.129)

$$F_{q,N}^{\text{mid}}(x^*) - F_q(x^*) = -\frac{q^{2N}}{q(1-q)^2} \int_0^1 (u - \tfrac{1}{2}) H_q(u) \, du = -\frac{q^{2N-1}}{24(1-q^2)},$$

which is (28.126); with (ii) it makes (28.125) an equality. At  $q = 1/2$  one has  $x^* = 1/4$  and  $q^{2N-1}/(24(1-q^2)) = 2^{1-2N}/18 = \frac{1}{9}4^{-N}$ .

(iv). The computation is the one displayed in the statement.  $\square$

*Remark 28.89* (What the midpoint buys). Against (27.30) the gain is a full extra factor  $q^{N-1}(1 - q)/(24(1 + q))$ , and at the half base the gain is  $9 \cdot 2^N$ : the plain truncation obeys  $\sup_x |F_{1/2,N} - F| = 2^{-N}$ , while  $\sup_x |S_N - F| = \frac{1}{9}4^{-N}$ . The mechanism is visible in (28.127): the shift  $\frac{1}{2}q^N$  is exactly the one that makes the kernel  $H_q$  have vanishing mean, after which only its first moment  $\frac{1}{2}\text{Var}(X_q)$  survives. The same identity shows why no further shift helps: the next obstruction is the second moment of  $H_q$ , which cannot be removed by a translation.

## 28.12 An off-grid continuation of the dyadic formula

**Corollary 28.31** evaluates  $F$  at the dyadic points  $m/2^n$  with  $0 \leq m \leq 2^n$ , and **Remark 28.32** shows that the hypothesis  $m \leq 2^n$  cannot simply be dropped: at  $n = 2, m = 5$  the right-hand side of (28.49) returns  $67/72$  instead of  $F(5/4) = 1$ . Both restrictions disappear at once if the block length  $m2^k$  is replaced by a cutoff that is *clamped* at the top. The resulting function is defined for every real argument, agrees with  $F$  on the grid, and is an explicit Gaussian average of the midpoint splines of **Proposition 28.88**.

**Definition 28.90** (The clamped midpoint cutoff). For an integer  $p \geq 1$  and  $x \in \mathbb{R}$  put

$$N_p(x) := \min\left(2^p, \max\left(0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor\right)\right) \in \{0, 1, \dots, 2^p\}. \quad (28.133)$$

**Lemma 28.91** (Cutoff form of the midpoint spline). Let  $p \geq 1$ . For every real  $x$ ,

$$\frac{1}{p!} \sum_{r=0}^{N_p(x)-1} \tau_r\left(r - 2^p x + \frac{1}{2}\right)^p = (-1)^p 2^{\binom{p}{2}} S_p(x), \quad (28.134)$$

an empty sum being 0. Moreover, if  $0 \leq n \leq p$  and  $m$  is an integer with  $0 \leq m \leq 2^n$ , then

$$N_p\left(\frac{m}{2^n}\right) = m 2^{p-n}. \quad (28.135)$$

*Proof.* Put  $t := 2^p x - \frac{1}{2}$ . In (28.124) the term of index  $r$  is nonzero exactly when  $t - r > 0$ ; since  $r$  is an integer this says  $r < \lceil t \rceil$ , so the terms that survive are those with  $0 \leq r < \min(2^p, \max(0, \lceil t \rceil))$ .

We check that  $\lceil t \rceil$  may be replaced by  $\lfloor t + 1 \rfloor = \lfloor 2^p x + \frac{1}{2} \rfloor$ . The two integers agree unless  $t \in \mathbb{Z}$ , in which case  $\lfloor t + 1 \rfloor = \lceil t \rceil + 1$ ; the single extra index is then  $r = t$ , whose contribution to (28.134) is  $\tau_t(t - t)^p = 0$  because  $p \geq 1$ . Neither side of (28.134) changes, so the summation range in (28.134) is exactly the active range of (28.124).

Consequently, using  $(t - r)^p = (-1)^p (r - t)^p$  and  $r - t = r - 2^p x + \frac{1}{2}$ ,

$$S_p(x) = \frac{2^{-\binom{p}{2}}}{p!} \sum_{r=0}^{2^p-1} \tau_r(t - r)_+^p = \frac{(-1)^p 2^{-\binom{p}{2}}}{p!} \sum_{r=0}^{N_p(x)-1} \tau_r\left(r - 2^p x + \frac{1}{2}\right)^p,$$

which is (28.134) after multiplying by  $(-1)^p 2^{\binom{p}{2}}$ .

For (28.135),  $2^p \cdot m/2^n = m 2^{p-n}$  is a nonnegative integer, so  $\lfloor m 2^{p-n} + \frac{1}{2} \rfloor = m 2^{p-n}$ ; and  $0 \leq m 2^{p-n} \leq 2^n 2^{p-n} = 2^p$ , so neither clamp is active.  $\square$

*Remark 28.92* (The clamp repairs the hypothesis  $m \leq 2^n$ ). The upper clamp in (28.133) is not cosmetic. Without it the right-hand side of (28.134) would continue to accumulate terms for  $x > 1$  and would describe a signed global extension rather than a bounded distribution function; the failure recorded in **Remark 28.32** at  $n = 2, m = 5$  is precisely the case  $N_{n+k}(x)$  replaced by an unclamped  $m 2^k > 2^{n+k}$ . The lower clamp plays the symmetric role for  $x < 0$ .

**Theorem 28.93** (Off-grid continuation of the dyadic formula). *For  $n \geq 1$  define, for every  $x \in \mathbb{R}$ ,*

$$D_n(x) := \frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{N_{n+k}(x)-1} \tau_r \left( r - 2^{n+k} x + \frac{1}{2} \right)^{n+k}, \quad (28.136)$$

with  $N_p$  as in (28.133), and put

$$\omega_{n,j} := \frac{(-1)^j 2^{-\binom{j+1}{2}}}{(1/2; 1/2)_n} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} \quad (0 \leq j \leq n). \quad (28.137)$$

Then:

1. for every  $x \in \mathbb{R}$ ,

$$D_n(x) = \sum_{j=0}^n \omega_{n,j} S_{2n-j}(x); \quad (28.138)$$

2.  $\omega_{n,j} = W_{n,j} = \lambda_{n,n-j}^{(1/2)}$  in the notation of Remark 28.49; in particular

$$\sum_{j=0}^n \omega_{n,j} = 1, \quad \text{sgn } \omega_{n,j} = (-1)^j,$$

and

$$\sum_{j=0}^n \omega_{n,j} z^j = \frac{(z/2; 1/2)_n}{(1/2; 1/2)_n}, \quad \sum_{j=0}^n |\omega_{n,j}| = \frac{(-1/2; 1/2)_n}{(1/2; 1/2)_n},$$

3.  $D_n$  interpolates  $F$  on the grid of resolution  $2^{-n}$ :

$$D_n\left(\frac{m}{2^n}\right) = F\left(\frac{m}{2^n}\right) \quad (m \in \mathbb{Z}, 0 \leq m \leq 2^n); \quad (28.139)$$

4.  $D_n$  is a continuous piecewise polynomial of degree at most  $2n$ , its knots are the odd multiples of  $2^{-2n-1}$ , it satisfies  $D_n(1-x) = 1 - D_n(x)$ , and

$$D_n(x) = 0 \quad (x \leq 2^{-2n-1}), \quad D_n(x) = 1 \quad (x \geq 1 - 2^{-2n-1}). \quad (28.140)$$

*Proof.* (i). Apply Lemma 28.91 with  $p = n + k$ : the inner sum of (28.136), divided by  $(n+k)!$ , equals  $(-1)^{n+k} 2^{\binom{n+k}{2}} S_{n+k}(x)$ . Hence

$$D_n(x) = \frac{1}{(1/2; 1/2)_n} \sum_{k=0}^n (-1)^{n+k} 2^{\binom{n+k}{2} - n^2 - 2\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} S_{n+k}(x),$$

since  $4^{\binom{k}{2}} = 2^{2\binom{k}{2}}$ . Substitute  $j = n - k$ , so that  $n + k = 2n - j$ . The sign is  $(-1)^{2n-j} = (-1)^j$ ; the Gaussian coefficient is unchanged because  $\begin{bmatrix} n \\ n-j \end{bmatrix}_{1/2} = \begin{bmatrix} n \\ j \end{bmatrix}_{1/2}$ , by the symmetry visible in (26.3); and the exponent of 2 becomes

$$\begin{aligned} \binom{2n-j}{2} - 2\binom{n-j}{2} - n^2 &= \frac{1}{2}(2n-j)(2n-j-1) \\ &\quad - (n-j)(n-j-1) - n^2 \\ &= -\frac{1}{2}j(j+1) = -\binom{j+1}{2}, \end{aligned}$$

an identity in  $n$  and  $j$  verified by expanding both sides. This is exactly (28.137), and (28.138) follows.

(ii). The displayed  $\omega_{n,j}$  is the row  $W_{n,j}$  of Remark 28.49, where it is identified with  $\lambda_{n,n-j}^{(1/2)}$  and its generating polynomial is computed. Unit mass is (28.74), the sign pattern is (28.73) read in reversed coordinates, and the  $\ell^1$  norm is (28.75), all at  $q = 1/2$ .

(iii). Let  $0 \leq m \leq 2^n$ . By (28.135) with  $p = n + k \geq n$  we have  $N_{n+k}(m/2^n) = m2^k$ , while  $2^{n+k} \cdot m/2^n = m2^k$ ; hence (28.136) becomes

$$\frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{m2^k-1} \tau_r \left( r - m2^k + \frac{1}{2} \right)^{n+k},$$

which is the right-hand side of (28.49) at  $\theta = \frac{1}{2}$ . By Corollary 28.31 its value is  $F(m/2^n)$ .

(iv). Each  $S_p$  is, by (28.124), a continuous piecewise polynomial of degree  $p$  whose knots are the numbers  $(2r+1)/2^{p+1}$ ; the degrees occurring in (28.138) are  $n, \dots, 2n$  and the finest knot set is that of  $S_{2n}$ . The variable  $X_{1/2,p}^{\text{mid}}$  of (28.123) is symmetric about  $\frac{1}{2}$ , because  $X_{1/2,p}$  is symmetric about its mean  $(1-2^{-p})/2$  and  $(1-2^{-p})/2 + 2^{-p-1} = \frac{1}{2}$ ; as it has no atoms,  $S_p(1-x) = 1 - S_p(x)$ , and the unit row mass of (ii) transfers this to  $D_n$ . Finally  $X_{1/2,p}^{\text{mid}}$  takes values in  $[2^{-p-1}, 1-2^{-p-1}]$ , so  $S_p = 0$  on  $(-\infty, 2^{-p-1}]$  and  $S_p = 1$  on  $[1-2^{-p-1}, \infty)$ ; both hold simultaneously for all  $p \in \{n, \dots, 2n\}$  exactly on the two rays of (28.140), the binding index being  $p = 2n$ .  $\square$

*Remark 28.94* (Which part of the dyadic formula is being continued). The separation of Remark 28.32 survives the continuation: in (28.136) the half-base Gaussian row  $[n]_{1/2}/4^{\binom{k}{2}}$  still depends only on the resolution  $2^{-n}$ , and the argument  $x$  enters only through the block length  $N_{n+k}(x)$  and the common shift. What changes is that the block length is now a step function of a real variable instead of an integer parameter, and (28.138) identifies the resulting object as a *fixed* Gaussian average of box splines — one Lagrange row of Chapter 25 applied to the family  $S_{2n}, S_{2n-1}, \dots, S_n$ . The choice  $\theta = \frac{1}{2}$  of the free shift in (28.49) is what forces the midpoint centring: no other value of  $\theta$  produces a combination of distribution functions.

**Theorem 28.95** (Uniform half-base Toeplitz limit, with the sharp rate). *For every  $n \geq 2$ ,*

$$\sup_{x \in \mathbb{R}} |D_n(x) - F(x)| \leq \frac{1}{9 \cdot 16^n} \frac{(-2; 1/2)_n}{(1/2; 1/2)_n} < \frac{2\mathcal{L}(1/2)}{3} 16^{-n} < 6 \cdot 16^{-n}, \quad (28.141)$$

with  $\mathcal{L}(1/2) = 8.2559879357782500655 \dots$  as in (28.77). In particular  $D_n \rightarrow F$  uniformly on the whole real line, and the convergence is geometric of ratio  $16^{-1}$ .

*Proof.* By Theorem 28.93(i) and the unit row mass of Theorem 28.93(ii),

$$D_n(x) - F(x) = \sum_{j=0}^n \omega_{n,j} (S_{2n-j}(x) - F(x)) \quad (x \in \mathbb{R}).$$

For  $0 \leq j \leq n$  one has  $2n-j \geq n \geq 2$ , so Proposition 28.88(iii) at  $q = 1/2$  applies to every summand and gives  $\sup_x |S_{2n-j} - F| = \frac{1}{9} 4^{-(2n-j)}$ . Hence

$$\sup_{x \in \mathbb{R}} |D_n(x) - F(x)| \leq \frac{1}{9} \sum_{j=0}^n |\omega_{n,j}| 4^{j-2n} = \frac{16^{-n}}{9} \sum_{j=0}^n |\omega_{n,j}| 4^j.$$

By Theorem 28.93(ii) the signs alternate,  $|\omega_{n,j}| = (-1)^j \omega_{n,j}$ , so the last sum is the value at  $z = -4$  of the row polynomial of Remark 28.49:

$$\sum_{j=0}^n |\omega_{n,j}| 4^j = \sum_{j=0}^n \omega_{n,j} (-4)^j = \frac{(-2; 1/2)_n}{(1/2; 1/2)_n}. \quad (28.142)$$

This proves the first inequality in (28.141).

For the second, split off the first two factors of the numerator: for  $n \geq 2$ ,

$$(-2; 1/2)_n = \prod_{i=0}^{n-1} (1 + 2^{1-i}) = 3 \cdot 2 \cdot \prod_{h=1}^{n-2} (1 + 2^{-h}) = 6 (-1/2; 1/2)_{n-2}.$$

Both infinite products in (28.76) converge at  $q = 1/2$ , by Theorem 28.46; since  $(-1/2; 1/2)_m$  increases in  $m$  to  $(-1/2; 1/2)_\infty$  and  $(1/2; 1/2)_m$  decreases in  $m$  to  $(1/2; 1/2)_\infty > 0$ , we get

$$\frac{(-2; 1/2)_n}{(1/2; 1/2)_n} = 6 \frac{(-1/2; 1/2)_{n-2}}{(1/2; 1/2)_n} < 6 \frac{(-1/2; 1/2)_\infty}{(1/2; 1/2)_\infty} = 6 \mathcal{L}(1/2).$$

Dividing by 9 gives  $\frac{2}{3} \mathcal{L}(1/2) = 5.5039919571 \dots < 6$ .  $\square$

*Remark 28.96* (The first row, and what the abstract theorem gives). The hypothesis  $n \geq 2$  in Theorem 28.95 is necessary. For  $n = 1$  the decomposition (28.138) reads  $D_1 = 2S_2 - S_1$ , and  $S_1$  is outside the range of Proposition 28.88(iii); indeed  $S_1(1/4) = 0$  and  $S_2(1/4) = 1/16$ , so  $D_1(1/4) = 1/8$ , whereas  $F(1/4) = 5/72$  by Remark 28.32. Thus

$$D_1(1/4) - F(1/4) = \frac{1}{8} - \frac{5}{72} = \frac{1}{18},$$

against the value  $6/144 = 1/24$  that the right-hand side of (28.141) would take at  $n = 1$ . Replacing the term  $j = 1$  by the exact bound  $\sup_x |S_1 - F| = 5/72$  restores a valid estimate  $|D_1 - F| \leq 1/12$ .

It is worth separating what is and is not an instance of the abstract machinery. For any exponents  $r(n, j) \geq n$ , the uniform convergence

$$\sum_j \omega_{n,j} F_{1/2, r(n,j)} \longrightarrow F$$

is Corollary 28.51 at  $q = 1/2$ , itself an application of Theorem 28.50 with the two hypotheses supplied by Theorem 28.46; that argument gives no rate beyond the rate of its input, and applied to (27.31) it yields only  $O(2^{-n})$ . What Theorem 28.95 adds is, first, that the average being estimated is the concrete off-grid formula (28.136) – an object that exists only because of the clamped cutoff – and second, that feeding the sharp midpoint rate (28.126) into the same sum improves the exponent from  $2^{-n}$  to  $16^{-n}$ . The improvement is entirely due to the input: the row is unchanged, and (28.142) is the same endpoint evaluation as (28.75), taken at  $z = -4$  instead of  $z = -1$ .

Both shift patterns evaluated so far – the common shift of (28.50) and the shift  $2^k + \theta$  of (28.44) – belong to a single two-parameter family, on which the weighted sum is not constant at all, but is a polynomial: the Appell polynomial attached to the Fabius law. The mechanism is the decoupling  $U \neq V$  permitted by Theorem 28.21.

**Definition 28.97** (Fabius Appell polynomials). For  $n \geq 0$  and  $\theta \in \mathbb{C}$  put

$$A_n(\theta) := \mathbb{E} (X + \theta)^n = \sum_{j=0}^n \binom{n}{j} \mathbb{E} X^{n-j} \theta^j, \quad (28.143)$$

a monic polynomial of degree  $n$  in  $\theta$  with  $A_n(0) = \mathbb{E} X^n = n! \alpha_n$ . Since  $0 \leq X \leq 1$ , the exchange of expectation and summation in

$$\sum_{n \geq 0} A_n(\theta) \frac{z^n}{n!} = \mathbb{E} e^{z(X+\theta)} = e^{\theta z} \mathcal{A}(z) \quad (28.144)$$

is legitimate for every  $z \in \mathbb{C}$ , by (28.8) at  $q = 1/2$ . The roman  $A_n$  is not the calligraphic  $\mathcal{A}$  of Corollary 28.5: the latter is one entire function of  $z$ , the former the coefficient sequence of its exponential shift, and  $A_n(0)/n! = \alpha_n$  recovers it.

**Proposition 28.98** (Appell and reflection laws). *For  $n \geq 1$  and  $\theta, c \in \mathbb{C}$ ,*

$$A'_n(\theta) = n A_{n-1}(\theta), \quad A_n(\theta + c) = \sum_{j=0}^n \binom{n}{j} A_{n-j}(\theta) c^j, \quad (28.145)$$

$$A_n(-1 - \theta) = (-1)^n A_n(\theta). \quad (28.146)$$

*Proof.* Differentiating (28.144) in  $\theta$  multiplies it by  $z$ , which shifts the coefficient index by one; multiplying it by  $e^{cz}$  and expanding the product of the two exponential series gives the translation law. For the reflection, (27.2) at  $q = 1/2$  gives  $1 - X \stackrel{d}{=} X$ , whence

$$A_n(-1 - \theta) = \mathbb{E}(X - 1 - \theta)^n = (-1)^n \mathbb{E}(1 - X + \theta)^n = (-1)^n \mathbb{E}(X + \theta)^n. \quad \square$$

**Theorem 28.99** (Geometric block shifts and the Appell polynomials). *For every  $n \geq 0$  and all  $\theta, \eta \in \mathbb{C}$ ,*

$$\sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \tau_r (r - 2^k + \eta - 2^k \theta)^{n+k} = \mathfrak{m}_n \frac{A_n(\theta)}{n!}; \quad (28.147)$$

*the left-hand side depends on  $\theta$ , but not on  $\eta$ . Equivalently, in the centred normalization  $\eta = 0$ ,*

$$\boxed{\frac{A_n(\theta)}{n!} = \frac{2^{-(n+1)}}{\left(\frac{1}{2}; \frac{1}{2}\right)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \tau_r (r - 2^k(1 + \theta))^{n+k}}, \quad (28.148)}$$

*and, in the raw normalization, for all  $\theta, \eta \in \mathbb{C}$ ,*

$$\boxed{\frac{A_n(\theta)}{n!} = \frac{(-1)^n 2^{-(n+1)}}{\left(\frac{1}{2}; \frac{1}{2}\right)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \tau_r (r + \eta + 2^k \theta)^{n+k}}. \quad (28.149)}$$

*The specialization  $\theta = 0$  of (28.147) is  $\mathcal{N}_n(\eta) = \mathfrak{m}_n \alpha_n$ , that is (28.50); the specialization  $\theta = -1$  is (28.44), with  $\eta$  in the role of the dummy parameter there.*

*Proof.* Apply Theorem 28.21 at  $q = 1/2$ , where  $q^{-k} = 2^k$ , with

$$U(z) = \mathcal{A}(-z), \quad V(u) = e^{\theta u} \mathcal{A}(u).$$

The normalization  $u_0 = 1$  holds because  $\mathcal{A}(0) = 1$  (Corollary 28.5), and  $v_n = A_n(\theta)/n!$  by (28.144). The unique family satisfying the bridge relation (28.30) is

$$\Psi_k^\theta(z) := \frac{e^{-z} V(-2^k z)}{\mathcal{A}(-z)} = e^{-2^k \theta z} \frac{e^{-z} \mathcal{A}(-2^k z)}{\mathcal{A}(-z)} = e^{-2^k \theta z} \Psi_k(z),$$

the last step by the dilation bridge (28.39).

Multiply the centred Thue–Morse identity (28.42) by  $e^{(\eta - 2^k \theta)z}$ :

$$\sum_{r=0}^{2^k-1} \tau_r e^{(r - 2^k + \eta - 2^k \theta)z} = (-1)^k 2^{\binom{k}{2}} z^k e^{\eta z} \Psi_k^\theta(z).$$

The left-hand side is the exponential generating function  $\sum_{d \geq 0} T_{k,d}(\eta - 2^k \theta) z^d / d!$  of the centred power sums (28.37), so comparing the coefficient of  $z^{n+k}$  gives

$$[z^n](e^{\eta z} \Psi_k^\theta(z)) = \frac{(-1)^k}{2^{\binom{k}{2}} (n+k)!} T_{k,n+k}(\eta - 2^k \theta).$$



By [Corollary 28.23](#) with  $\alpha = \eta$ , together with (28.32) and the Mersenne evaluation (26.2),

$$\sum_{k=0}^n W_{n,k}(1/2) [z^n] (e^{\eta z} \Psi_k^\theta) = 2^{\binom{n+1}{2}} \left(\frac{1}{2}; \frac{1}{2}\right)_n \frac{A_n(\theta)}{n!} = \mathfrak{m}_n \frac{A_n(\theta)}{n!},$$

and the same corollary states that this value does not depend on  $\eta$ . Inserting

$$W_{n,k}(1/2) = (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$$

makes the two signs cancel and combines the two triangular powers of two into  $4^{-\binom{k}{2}}$ ; that is (28.147).

Putting  $\eta = 0$ , writing  $r - 2^k - 2^k \theta = r - 2^k(1 + \theta)$  and dividing by  $\mathfrak{m}_n = 2^{\binom{n+1}{2}} \left(\frac{1}{2}; \frac{1}{2}\right)_n$  gives (28.148). For (28.149), replace  $\theta$  by  $-1 - \theta$  in (28.147): the inner argument becomes  $r - 2^k + \eta + 2^k(1 + \theta) = r + \eta + 2^k \theta$ , while the right-hand side becomes  $\mathfrak{m}_n A_n(-1 - \theta)/n! = (-1)^n \mathfrak{m}_n A_n(\theta)/n!$  by (28.146).

Finally,  $\theta = 0$  in (28.147) is (28.50), because  $A_n(0) = n! \alpha_n$  and the inner argument is then  $r - 2^k + \eta$ ; and  $\theta = -1$  turns the inner argument into  $r + \eta$  and the right-hand side into  $(-1)^n \mathfrak{m}_n \alpha_n$ , which is (28.44).  $\square$

*Remark 28.100* (Why  $\theta$  is a variable and  $\eta$  a dummy). [Corollary 28.23](#) says that a shift common to all  $n + 1$  blocks is invisible; [Theorem 28.99](#) says that a shift scaled by  $2^k$  in the  $k$ th block is visible, and is seen exactly once, in the top coefficient of  $e^{\theta u} \mathcal{A}(u)$ . The distinction is precisely the decoupling  $U \neq V$  of [Theorem 28.21](#): the self-referential instance  $V = \mathcal{A}$  attached to  $U(z) = \mathcal{A}(-z)$  can only ever return  $\alpha_n$ , whereas the factor  $e^{\theta u}$  inserted into  $V$  alone survives the double cancellation of [Remark 28.24](#). The two displays (28.148) and (28.149) are one identity in the two coordinate systems of [Remark 28.27](#), exchanged by the digit complementation carried out in the last paragraph of the proof of [Corollary 28.31](#).

**Example 28.101** (The case  $n = 2$ ). Here  $\mathfrak{m}_2 = 3$  and the half-base Gaussian row is  $1, \frac{3}{2}, 1$ . At  $\eta = 0$  the three inner sums of (28.148) are

$$(1 + \theta)^2, \quad -8(1 + \theta)^3 + (1 + 2\theta)^3, \quad c^4 - (1 - c)^4 - (2 - c)^4 + (3 - c)^4 \quad \text{with } c = 4(1 + \theta),$$

that is  $\theta^2 + 2\theta + 1$ ,  $-12\theta^2 - 18\theta - 7$  and  $384\theta^2 + 480\theta + 160$ . Weighted by  $1/2!$ ,  $\frac{3}{2}/3!$  and  $1/(4 \cdot 4!)$  they sum to

$$\frac{3}{2}\theta^2 + \frac{3}{2}\theta + \frac{5}{12} = 3 \left( \frac{\theta^2}{2} + \frac{\theta}{2} + \frac{5}{36} \right) = \mathfrak{m}_2 \frac{A_2(\theta)}{2!},$$

as (28.147) asserts. At  $\theta = 1$  the inner sums are the integers 4,  $-37$ , 1024, the terms are  $2, -\frac{37}{4}, \frac{32}{3}$ , the total is  $41/12$ , and  $A_2(1)/2! = 41/36 = \frac{1}{2} + \frac{1}{2} + \frac{5}{36}$ . At  $\theta = 0$  they are 1,  $-7$ , 160 with total  $5/12 = 3\alpha_2$ , which is [Example 28.29](#) once more.

### 28.13 A digit-sparse expansion at an arbitrary argument

The dyadic formula (28.49) is exact but confined to the grid; the sinc integral (28.64) reaches every  $x$  but is an integral of an infinite product. Between them lies a third representation: an exact series whose  $j$ th term is a finite rational expression attached to the  $j$ th binary digit of  $x$ , which terminates on the grid and converges superexponentially off it.

The unrestricted spline identity (28.118) already gives this representation for every real  $t$ , and therefore in particular on  $0 \leq t \leq 2^n$ . The binary expansion below uses that stronger canonical lemma directly.



**Corollary 28.102** (Local reduction identity). *For every  $n \geq 1$  and every  $\theta \in [0, 1)$ ,*

$$F\left(\frac{1+\theta}{2^n}\right) + F\left(\frac{\theta}{2^n}\right) = \frac{2^{-\binom{n}{2}}}{n!} A_n(\theta). \quad (28.150)$$

*Proof.* Both  $t = \theta$  and  $t = 1 + \theta$  lie in  $[0, 2] \subseteq [0, 2^n]$ , so [Lemma 28.82](#) applies to each. Since  $0 \leq X \leq 1$ , a term with  $r \geq 1$  in the sum for  $t = \theta$  has  $\theta - r - X \leq \theta - 1 < 0$  and drops out, leaving

$$F\left(\frac{\theta}{2^n}\right) = \frac{2^{-\binom{n}{2}}}{n!} \mathbb{E}(\theta - X)_+^n.$$

In the sum for  $t = 1 + \theta$  a term with  $r \geq 2$  has  $1 + \theta - r - X \leq \theta - 1 < 0$  and drops out; the term  $r = 0$  has  $1 + \theta - X \geq \theta \geq 0$ , so its positive part is the plain power, and  $1 - X \stackrel{d}{=} X$  by (27.2) turns  $\mathbb{E}(1 + \theta - X)^n$  into  $\mathbb{E}(X + \theta)^n = A_n(\theta)$ ; the term  $r = 1$  carries  $\tau_1 = -1$  and contributes  $-\mathbb{E}(\theta - X)_+^n$ . Hence

$$F\left(\frac{1+\theta}{2^n}\right) = \frac{2^{-\binom{n}{2}}}{n!} \left( A_n(\theta) - \mathbb{E}(\theta - X)_+^n \right),$$

and adding the two displays proves (28.150).  $\square$

Let  $x \in [0, 1)$  and let  $x = \sum_{j \geq 1} b_j 2^{-j}$ ,  $b_j \in \{0, 1\}$ , be its canonical binary expansion, the one that is not eventually constant equal to 1. Put

$$m_j := \lfloor 2^j x \rfloor, \quad \theta_j := 2^j x - m_j \in [0, 1), \quad m_0 = 0, \quad \theta_0 = x, \quad (28.151)$$

the binary *prefix* and *tail* of  $x$  at depth  $j$ ; then  $m_j = 2m_{j-1} + b_j$  and  $\theta_{j-1} = (b_j + \theta_j)/2$ , and the floor in (28.151) selects the canonical expansion automatically.

**Theorem 28.103** (Digit-sparse Appell expansion). *For every  $x \in [0, 1)$ , with the notation (28.151),*

$$F(x) = \sum_{j=1}^{\infty} b_j \tau_{m_{j-1}} \frac{2^{-\binom{j}{2}}}{j!} A_j(\theta_j). \quad (28.152)$$

Writing  $S_N(x)$  for the sum of the first  $N$  terms,

$$F(x) - S_N(x) = \tau_{m_N} F\left(\frac{\theta_N}{2^N}\right), \quad |F(x) - S_N(x)| \leq \frac{2^{-\binom{N}{2}}}{N!}. \quad (28.153)$$

The series therefore converges absolutely and uniformly on  $[0, 1)$ , and at a dyadic rational  $m/2^n$ , taken in its canonical terminating expansion, it terminates after  $n$  terms.

*Proof.* Put  $R_N := \tau_{m_N} F(\theta_N/2^N)$ , so that  $R_0 = \tau_0 F(x) = F(x)$ . Fix  $j \geq 1$ . If  $b_j = 0$  then  $m_j = 2m_{j-1}$ , and doubling shifts every binary digit without creating or destroying one, so  $\tau_{m_j} = \tau_{m_{j-1}}$ ; moreover  $\theta_{j-1} = \theta_j/2$ , hence  $\theta_{j-1}/2^{j-1} = \theta_j/2^j$  and  $R_{j-1} = R_j$ , which is also the  $j$ th term of (28.152), namely 0.

If  $b_j = 1$  then  $m_j = 2m_{j-1} + 1$ , so  $\text{wt}_2(m_j) = \text{wt}_2(m_{j-1}) + 1$  and  $\tau_{m_j} = -\tau_{m_{j-1}}$ , while  $\theta_{j-1}/2^{j-1} = (1 + \theta_j)/2^j$ . Therefore

$$R_{j-1} - R_j = \tau_{m_{j-1}} \left[ F\left(\frac{1+\theta_j}{2^j}\right) + F\left(\frac{\theta_j}{2^j}\right) \right] = \tau_{m_{j-1}} \frac{2^{-\binom{j}{2}}}{j!} A_j(\theta_j)$$

by [Corollary 28.102](#), which is again the  $j$ th term. Summing over  $j = 1, \dots, N$  telescopes to  $R_0 - R_N = S_N(x)$ , which is the first half of (28.153).

For the bound,  $0 \leq \theta_N < 1$  and  $F$  is nondecreasing, so  $F(\theta_N/2^N) \leq F(2^{-N}) = 2^{-\binom{N}{2}} \alpha_N$  by [Theorem 28.9](#), and  $\alpha_N = \mathbb{E}X^N/N! \leq 1/N!$  because  $0 \leq X \leq 1$ . This bound is independent of  $x$  and tends to 0, so  $S_N \rightarrow F$  uniformly on  $[0, 1)$  and [\(28.152\)](#) holds. The convergence is absolute because  $0 \leq X + \theta_j < 2$  gives  $|A_j(\theta_j)| \leq 2^j$ , so the terms are dominated by the summable sequence  $2^{j-\binom{j}{2}}/j!$ .

Finally, if  $x = m/2^n$  then  $2^j x \in \mathbb{Z}$  for every  $j \geq n$ , so  $b_j = 0$  and  $\theta_j = 0$  there; hence  $R_N = \tau_{m_N} F(0) = 0$  for  $N \geq n$  and  $F(x) = S_N(x) = S_n(x)$ .  $\square$

**Corollary 28.104** (Digit-sparse half-base  $q$ -series). *For every  $x \in [0, 1)$ , with the notation [\(28.151\)](#),*

$$F(x) = \sum_{j=1}^{\infty} \frac{b_j \tau_{m_{j-1}}}{2^{j^2} \left(\frac{1}{2}; \frac{1}{2}\right)_j} \sum_{k=0}^j \frac{\begin{bmatrix} j \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}} (j+k)!} \sum_{r=0}^{2^k-1} \tau_r (r - 2^k(1 + \theta_j))^{j+k}, \quad (28.154)$$

with the convergence, residual and termination properties of [Theorem 28.103](#).

*Proof.* Insert [\(28.148\)](#) with  $n = j$  and  $\theta = \theta_j$  into [\(28.152\)](#). The two triangular powers of two combine as  $2^{-\binom{j}{2}} 2^{-\binom{j+1}{2}} = 2^{-j^2}$ . This is a termwise replacement by an exact finite identity, so every analytic assertion is unchanged.  $\square$

**Remark 28.105** (Three ways to evaluate  $F$ ). The monograph now evaluates  $F$  by three mechanisms. [Corollary 28.31](#) is finite, rational and exact, but only on the dyadic grid; [Corollary 28.41](#) is global and analytic, an integral of an infinite sinc product; [Corollary 28.104](#) is global and *arithmetic*, each term being a finite half-base Gaussian sum indexed by one binary digit of  $x$ . The three agree where they overlap: at  $x = 2^{-n}$  only the digit  $b_n$  is nonzero, with  $m_{n-1} = 0$  and  $\theta_n = 0$ , so [\(28.152\)](#) collapses to  $F(2^{-n}) = 2^{-\binom{n}{2}} A_n(0)/n! = 2^{-\binom{n}{2}} \alpha_n$ , which is [Theorem 28.9](#).

At  $x = \frac{1}{3} = 0.\overline{01}_2$  the odd digits vanish and  $\theta_j = \frac{2}{3}$  or  $\frac{1}{3}$  according as  $j$  is odd or even, so the first six terms contain only three nonzero summands:

$$S_2 = \frac{13}{72} = 0.18055\dots, \quad S_4 = \frac{1120771}{6220800} = 0.18016509\dots, \quad S_6 = 0.1801651148015\dots,$$

against  $F(\frac{1}{3}) = 0.180165114801481906955\dots$ : thirteen correct digits from three terms, where [\(28.153\)](#) guarantees only  $2^{-15}/6! < 4.3 \cdot 10^{-8}$ . Unlike [\(28.46\)](#), whose conditioning is analysed in [Remark 28.34](#), the outer series here has no cancellation at all for  $x < \frac{1}{2}$  beyond the Thue–Morse signs  $\tau_{m_{j-1}}$ ; the ill-conditioning of the inner blocks is inherited unchanged, so for numerical work one evaluates  $A_j(\theta_j)$  from [\(28.143\)](#) and the recurrence [\(28.11\)](#) rather than from [\(28.154\)](#).

## **Part VII**

# **Branch-aware inverse theory**

## Chapter 29

# Scope, status, and inverse conventions

### 29.1 What this volume consolidates

The source corpus consisted of six independently delivered packages: *Inverse  $q$ -Analog Functions*, *Inverse  $q$ -Analog Jet Atlas*, *Inverse  $q$ -Analog: Extended Report*, two reports entitled *Inverse  $q$ -Analog*, and  *$q$ -Pochhammer and  $q$ -Binomial Expansions*. They treated largely the same inverse maps in different orders, repeatedly reproved the same reversion formulas, and occasionally assigned incompatible hypotheses or coefficients to the same claim. The present volume is an editorial and mathematical synthesis, not a mechanical concatenation.

The forward-expansion report supplies the strongest systematic engine for finite products, Gaussian coefficients, roots of unity, and the  $q \rightarrow 1$  regimes. The jet atlas supplies the most useful all-order inverse, Bell-polynomial, mixed-parameter, and Lagrange–Good formulations. The function report supplies the clearest branchwise narrative. The extended and two earlier inverse reports contribute certification arguments, special regimes, and the Fabius–Rvachev applications. Repeated material appears once, with the strongest correct hypotheses and the most transparent proof.

### 29.2 Status vocabulary

Three statuses are used and are not interchangeable.

**Lean-proved.** The statement has a named, compiler-validated declaration in `Analysis/FabiusFunction`. Its exact module and declaration name are recorded in the reference appendix.

**Human-proved frontier result.** The statement is proved completely in this volume but does not yet have an exact Lean counterpart. It belongs in this research-frontier document rather than in the primary formalization-backed exposition.

**Conjecture.** The statement is plausible and mathematically meaningful, but the argument currently has a genuine missing step. Conjectures are isolated in explicitly named *Conjecture* environments. A numerical experiment, symbolic check, or manuscript label is never treated as a proof.

Definitions, notation, and descriptions of an algorithm are not themselves assertions of correctness. Whenever correctness, convergence, error, or complexity is claimed, it is stated as a theorem or proposition and proved.

### 29.3 Basic normalizations

For  $n \in \mathbb{N}_0$ , complex parameters  $a, q$ , and an empty product equal to one, the finite shifted factorial is

$$(a; q)_n := \prod_{j=0}^{n-1} (1 - aq^j).$$

When  $|q| < 1$ , the infinite shifted factorial is

$$(a; q)_\infty := \prod_{j=0}^{\infty} (1 - aq^j).$$

The convergence and holomorphy assertions implicit in the latter notation are proved in [Section 32.1](#); outside  $|q| < 1$ , the symbol is used only after an explicit continuation or reciprocal normalization has been specified.

For integers  $0 \leq k \leq n$ , the Gaussian coefficient is normalized by

$$\begin{bmatrix} n \\ k \end{bmatrix}_q := \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}},$$

with its polynomial continuation at points where this quotient is formally indeterminate. This convention has degree  $k(n - k)$ , constant coefficient one, and satisfies the reciprocity law

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}$$

for  $q \neq 0$ . A proof independent of quotient cancellation appears in [Section 33.6](#).

We use

$$q = e^{-t} \quad (t \rightarrow 0)$$

for logarithmic expansions near  $q = 1$ , and  $h = q - 1$  only when an ordinary Taylor coordinate is genuinely preferable. Conversion between the two is performed by formal composition, never by silently replacing  $t$  with  $-h$ . At a root of unity  $\zeta$ , radial limits are written  $q = \zeta e^{-t}$  with  $t \downarrow 0$ ; this specifies a sectorial or radial asymptotic branch and does not assert the existence of an analytic germ across the unit circle.

### 29.4 What an inverse symbol means

An expression such as  $a = a(y; q)$  or  $q = q(y; a)$  is shorthand, not a globally defined function. Every inverse used below records the following data:

1. the forward map and which parameters are held fixed;
2. the parameter being recovered;
3. the base point, limiting regime, or real interval;
4. the local sheet or global monotone branch;
5. the logarithm or fractional-power determination when one is needed.

Thus a scalar equation with several free parameters defines a level set, not a simultaneous inverse. Several parameters can be recovered only after enough independent observables have been supplied and the corresponding Jacobian has been shown invertible. This elementary distinction removes a substantial fraction of the apparent disagreements among the source reports.

## 29.5 Corrections made during consolidation

The synthesis corrects, rather than propagates, the following recurrent defects from the source corpus:

- for  $0 < q < 1$ , the real zero lattice of  $a \mapsto (a; q)_\infty$  is ordered  $1 = q^0 < q^{-1} < q^{-2} < \dots$ ;
- a centered-palindromic inverse theorem requires nonzero equal endpoint coefficients, not palindromy alone;
- continuous lower-parameter Gaussian inversion must stay on the branch where the target lies strictly between the endpoint value and the central maximum;
- zeros introduced by a cyclotomic factor are simple unless a separate factor collision is proved; multiplicity cannot be inferred from the factor's mere presence;
- analytic-germ language at a natural boundary is replaced by radial or sectorial asymptotics with an explicit approach region;
- Newton and residual certificates include the differentiability, simple-root, derivative-separation, and domain-control hypotheses on which their conclusions depend;
- multivariate Lagrange–Good product formulas are used only for systems in the required product form; arbitrary locally invertible systems instead use the general total-degree recursion;
- conjectures contradicted by elementary counterexamples are removed rather than relabelled.

Later chapters give the corrected formulas and proofs at the point of use.

## Chapter 30

# Universal inversion calculus

This part collects the inversion machinery used throughout the volume. Its purpose is not merely to quote familiar formulas: coefficient conventions, domains, branches, and certification hypotheses are made explicit so that later applications can invoke the results without changing their meaning. Unless otherwise stated, a power series is based at the origin and  $[z^n]A(z)$  denotes its  $z^n$ -coefficient.

### 30.1 Regular inverse germs

**Definition 30.1** (Partial inverse). Let  $F(p, \lambda)$  be a scalar observable, where  $p$  is the variable to be recovered and  $\lambda = (\lambda_1, \dots, \lambda_s)$  consists of passive parameters. A *partial inverse germ in  $p$*  at  $(p_0, \lambda_0)$ , with  $y_0 = F(p_0, \lambda_0)$ , is a germ  $G(y, \lambda)$  at  $(y_0, \lambda_0)$  satisfying

$$F(G(y, \lambda), \lambda) = y, \quad G(y_0, \lambda_0) = p_0.$$

Thus the word *inverse* always includes the recovered variable, the base point, and the selected local branch.

**Theorem 30.2** (Regular analytic partial inverse). *Let  $F$  be holomorphic in a neighbourhood of  $(p_0, \lambda_0) \in \mathbb{C} \times \mathbb{C}^s$ , put  $y_0 = F(p_0, \lambda_0)$ , and suppose*

$$F_p(p_0, \lambda_0) \neq 0.$$

*Then there are neighbourhoods  $U$  of  $(y_0, \lambda_0)$  and  $V$  of  $p_0$ , and a unique holomorphic map  $G: U \rightarrow V$ , after shrinking them if necessary, which is the partial inverse germ of [definition 30.1](#). For each fixed nearby  $\lambda$ , the two maps  $p \mapsto F(p, \lambda)$  and  $y \mapsto G(y, \lambda)$  are inverse biholomorphic germs.*

*Proof.* Apply the holomorphic implicit-function theorem to

$$\Psi(p, y, \lambda) := F(p, \lambda) - y.$$

At  $(p_0, y_0, \lambda_0)$  one has  $\Psi = 0$  and  $\Psi_p = F_p \neq 0$ , so the theorem supplies a unique holomorphic solution  $p = G(y, \lambda)$ . Its defining identity is  $F(G(y, \lambda), \lambda) = y$ . Holding  $\lambda$  fixed and applying the same theorem with  $p$  as input shows that this identity holds between inverse germs in both directions. Uniqueness is the uniqueness clause of the implicit-function theorem.  $\square$

**Remark 30.3.** The identical statement holds in the real  $C^r$  category,  $1 \leq r \leq \infty$ , with *holomorphic* replaced by  $C^r$ . This is the one-dimensional parameterized inverse-function theorem.

## 30.2 Formal triangular inversion

**Theorem 30.4** (Triangular inverse recurrence). *Let  $R$  be a commutative ring and*

$$f(z) = \sum_{n \geq 1} c_n z^n \in R[[z]]$$

*with  $c_1$  a unit. There is a unique series  $g(w) = \sum_{n \geq 1} b_n w^n$  satisfying both*

$$f(g(w)) = w, \quad g(f(z)) = z.$$

*Its coefficients are determined without division except by  $c_1$ :*

$$b_1 = c_1^{-1},$$

*and, for  $n \geq 2$ ,*

$$b_n = -c_1^{-1} \sum_{k=2}^n c_k \sum_{\substack{j_1+\dots+j_k=n \\ j_1, \dots, j_k \geq 1}} b_{j_1} \cdots b_{j_k}. \quad (30.1)$$

*Proof.* The coefficient of  $w$  in  $f(g(w))$  is  $c_1 b_1$ , which forces  $b_1 = c_1^{-1}$ . For  $n \geq 2$ , the coefficient of  $w^n$  is

$$c_1 b_n + \sum_{k=2}^n c_k \sum_{\substack{j_1+\dots+j_k=n \\ j_i \geq 1}} b_{j_1} \cdots b_{j_k}.$$

Every  $j_i$  in the second term is at most  $n-1$ , so setting this coefficient equal to zero gives (30.1) and constructs a unique right compositional inverse  $g$ .

It remains to check the other composition. Put  $h(z) = g(f(z))$ . Then

$$f(h(z)) = (f \circ g)(f(z)) = f(z),$$

and the linear coefficient of  $h$  is  $b_1 c_1 = 1$ . If  $h(z) \neq z$ , let  $r \geq 2$  be the first degree at which they differ and write  $h(z) = z + dz^r + \mathcal{O}(z^{r+1})$  with  $d \neq 0$ . The first degree at which  $f(h(z))$  and  $f(z)$  differ is then  $r$ , with difference  $c_1 d$ . Since  $c_1$  is a unit this is nonzero, contradicting  $f \circ h = f$ . Hence  $g \circ f$  is also the identity.  $\square$

**Corollary 30.5** (The first five ordinary coefficients). *Over a field, the first coefficients in Theorem 30.4 are*

$$\begin{aligned} b_1 &= \frac{1}{c_1}, & b_2 &= -\frac{c_2}{c_1^3}, \\ b_3 &= \frac{2c_2^2 - c_1 c_3}{c_1^5}, & b_4 &= \frac{-5c_2^3 + 5c_1 c_2 c_3 - c_1^2 c_4}{c_1^7}, \\ b_5 &= \frac{14c_2^4 - 21c_1 c_2^2 c_3 + 6c_1^2 c_2 c_4 + 3c_1^2 c_3^2 - c_1^3 c_5}{c_1^9}. \end{aligned}$$

*Proof.* The recurrence specializes successively to

$$\begin{aligned} b_2 &= -c_1^{-1} c_2 b_1^2, \\ b_3 &= -c_1^{-1} (2c_2 b_1 b_2 + c_3 b_1^3), \\ b_4 &= -c_1^{-1} \{c_2(2b_1 b_3 + b_2^2) + 3c_3 b_1^2 b_2 + c_4 b_1^4\}, \\ b_5 &= -c_1^{-1} \{c_2(2b_1 b_4 + 2b_2 b_3) + c_3(3b_1^2 b_3 + 3b_1 b_2^2) + 4c_4 b_1^3 b_2 + c_5 b_1^5\}. \end{aligned}$$

Substituting  $b_1 = c_1^{-1}$  from top to bottom and collecting terms gives the five displayed formulas.  $\square$

**Remark 30.6.** These are ordinary-power-series coefficients. Factorial-normalized derivatives obey the Bell recurrence in Theorem 30.12.



### 30.3 Formal and analytic Lagrange–Bürmann inversion

**Lemma 30.7** (Formal residue under a change of coordinate). *Let  $R$  be a commutative  $\mathbb{Q}$ -algebra, let  $f(z) = zU(z)$  with  $U(0)$  a unit, and let  $\text{res}_z A(z) := [z^{-1}]A(z)$ . For every formal Laurent series  $A(w) \in R((w))$ ,*

$$\text{res}_w A(w) = \text{res}_z (A(f(z))f'(z)).$$

*Proof.* Both sides are  $R$ -linear, and in any requested coefficient only finitely many terms of  $A$  contribute, so it suffices to take  $A(w) = w^k$ . If  $k \neq -1$ , then

$$f(z)^k f'(z) = \frac{1}{k+1} \frac{d}{dz} f(z)^{k+1},$$

whose residue is zero, as is the residue of  $w^k$ . Division by  $k+1$  is legitimate because  $R$  is a  $\mathbb{Q}$ -algebra. If  $k = -1$ , then

$$\frac{f'(z)}{f(z)} = \frac{1}{z} + \frac{U'(z)}{U(z)}.$$

The second summand is an ordinary power series and has zero residue, while the first has residue one. This is also  $\text{res}_w w^{-1}$ .  $\square$

**Theorem 30.8** (Formal Lagrange–Bürmann formula). *Let  $R$  be a commutative  $\mathbb{Q}$ -algebra, let  $\phi(z) \in R[[z]]$  have invertible constant term, and put*

$$f(z) = \frac{z}{\phi(z)}.$$

*Let  $g$  be the compositional inverse of  $f$ . Then for every  $H(z) \in R[[z]]$  and every  $n \geq 1$ ,*

$$[w^n]H(g(w)) = \frac{1}{n} [z^{n-1}]H'(z)\phi(z)^n. \quad (30.2)$$

*In particular,*

$$[w^n]g(w) = \frac{1}{n} [z^{n-1}]\phi(z)^n.$$

*Proof.* By lemma 30.7, with  $w = f(z)$ ,

$$\begin{aligned} [w^n]H(g(w)) &= \text{res}_w H(g(w))w^{-n-1} \\ &= \text{res}_z H(z)f'(z)f(z)^{-n-1}. \end{aligned}$$

The residue of a formal derivative is zero. Applying this to  $H(z)f(z)^{-n}$  gives

$$0 = \text{res}_z (H'(z)f(z)^{-n} - nH(z)f'(z)f(z)^{-n-1}).$$

Consequently the preceding residue equals

$$\frac{1}{n} \text{res}_z H'(z)f(z)^{-n} = \frac{1}{n} \text{res}_z H'(z)z^{-n}\phi(z)^n,$$

which is (30.2). Taking  $H(z) = z$  proves the special case.  $\square$

**Theorem 30.9** (Analytic Lagrange–Bürmann formula). *Under the hypotheses of Theorem 30.2 with the passive parameter fixed, write  $y_0 = F(p_0)$ . The selected inverse germ is*

$$G(y) = p_0 + \sum_{n \geq 1} \frac{(y - y_0)^n}{n!} \left[ \frac{d^{n-1}}{dp^{n-1}} \left( \frac{p - p_0}{F(p) - F(p_0)} \right)^n \right]_{p=p_0}.$$

*The quotient in parentheses is understood by its removable value  $1/F'(p_0)$ .*

*Proof.* Set  $z = p - p_0$ ,  $w = y - y_0$ , and

$$f(z) = F(p_0 + z) - F(p_0), \quad \phi(z) = \frac{z}{f(z)}.$$

Since  $f'(0) = F'(p_0) \neq 0$ ,  $\phi$  is holomorphic at zero with nonzero constant term. The analytic inverse-function theorem gives a convergent inverse germ  $z = g(w)$ , while [Theorem 30.8](#) gives

$$[w^n]g(w) = \frac{1}{n}[z^{n-1}]\phi(z)^n = \frac{1}{n!} \left[ \frac{d^{n-1}}{dz^{n-1}} \phi(z)^n \right]_{z=0}.$$

Translating back from  $z$  and  $w$  proves the formula.  $\square$

**Warning 30.10** (Radius is branchwise). Lagrange–Bürmann determines a germ, not a global sheet. If  $g$  is the selected inverse germ at the expansion value  $w_0$ , its Taylor radius is precisely

$$R = \sup\{r > 0 : g \text{ extends as a single-valued holomorphic function to } |w - w_0| < r\}.$$

Thus  $R$  is the distance from  $w_0$  to the boundary of the maximal centered disk of single-valued holomorphy of that germ. A critical value belonging only to another sheet, or one reachable only after leaving this disk and crossing a chosen cut, need not control  $R$ ; conversely, any obstruction to single-valued holomorphic continuation on the centered disk can control it. The phrase *the nearest critical value* is therefore not a valid unqualified radius formula.

## 30.4 Bell-polynomial packaging

The complete and partial exponential Bell polynomials are fixed by

$$\exp\left(\sum_{j \geq 1} x_j \frac{t^j}{j!}\right) = \sum_{n \geq 0} B_n(x_1, \dots, x_n) \frac{t^n}{n!}, \quad (30.3)$$

$$\frac{1}{r!} \left(\sum_{j \geq 1} x_j \frac{t^j}{j!}\right)^r = \sum_{n \geq r} B_{n,r}(x_1, \dots, x_{n-r+1}) \frac{t^n}{n!}. \quad (30.4)$$

These identities are definitions and fix all factorial conventions.

**Lemma 30.11** (Exponentiating logarithmic jets). *Let  $R$  be a commutative  $\mathbb{Q}$ -algebra and let  $A(z) \in R[[z]]$  have invertible constant term. If*

$$\log \frac{A(z)}{A(0)} = \sum_{j \geq 1} \ell_j \frac{z^j}{j!},$$

*then*

$$\frac{A(z)}{A(0)} = \sum_{n \geq 0} B_n(\ell_1, \dots, \ell_n) \frac{z^n}{n!}.$$

*Proof.* Exponentiate the displayed logarithmic identity. The right-hand side is exactly the defining generating function (30.3).  $\square$

**Theorem 30.12** (Bell recurrence for inverse derivatives). *Let  $R$  be a commutative  $\mathbb{Q}$ -algebra and write*

$$f(z) = \sum_{r \geq 1} f_r \frac{z^r}{r!}, \quad g(w) = \sum_{n \geq 1} g_n \frac{w^n}{n!},$$

where  $f_1$  is a unit and  $f(g(w)) = w$ . Then

$$g_1 = f_1^{-1}$$

and, for  $n \geq 2$ ,

$$g_n = -f_1^{-1} \sum_{r=2}^n f_r B_{n,r}(g_1, \dots, g_{n-r+1}). \quad (30.5)$$

*Proof.* Substituting  $g(w)$  into the  $r$ -th term of  $f$ , and using (30.4), shows that the coefficient multiplying  $w^n/n!$  in  $f(g(w))$  is

$$\sum_{r=1}^n f_r B_{n,r}(g_1, \dots, g_{n-r+1}).$$

Here  $B_{n,1}(g_1, \dots, g_n) = g_n$ . For  $n = 1$ , comparison with  $w$  gives  $f_1 g_1 = 1$ . For every  $n \geq 2$ , the displayed coefficient must vanish; isolating  $f_1 g_n$  gives (30.5). Each right-hand side involves only  $g_j$  with  $j < n$ , so the recurrence is triangular.  $\square$

### 30.5 Passive-parameter sensitivities

**Theorem 30.13** (First and second inverse sensitivities). *Let  $F$  and its partial inverse  $G$  be  $C^2$  (or holomorphic) and suppose  $F_p \neq 0$  on the germ under consideration. All derivatives of  $F$  below are evaluated at  $(G(y, \lambda), \lambda)$ . Then*

$$G_y = \frac{1}{F_p}, \quad G_{\lambda_i} = -\frac{F_{\lambda_i}}{F_p},$$

$$G_{yy} = -\frac{F_{pp}}{F_p^3}, \quad G_{y\lambda_i} = \frac{F_{pp}F_{\lambda_i} - F_p F_{p\lambda_i}}{F_p^3},$$

and

$$G_{\lambda_i \lambda_j} = -\frac{F_{pp}G_{\lambda_i}G_{\lambda_j} + F_{p\lambda_i}G_{\lambda_j} + F_{p\lambda_j}G_{\lambda_i} + F_{\lambda_i \lambda_j}}{F_p}.$$

If  $F$  and  $G$  are  $C^r$ , the same differentiation gives all mixed derivatives through total order  $r$  recursively: at every new order the highest derivative of  $G$  occurs once, multiplied by  $F_p$ , and all remaining terms involve lower derivatives of  $G$ . In the holomorphic case this continues to every order.

*Proof.* Differentiate  $F(G(y, \lambda), \lambda) = y$  with respect to  $y$  and to  $\lambda_i$ :

$$F_p G_y = 1, \quad F_p G_{\lambda_i} + F_{\lambda_i} = 0.$$

These give the first two formulas. Differentiating the first identity once more with respect to  $y$  gives

$$F_{pp}G_y^2 + F_p G_{yy} = 0,$$

and substituting  $G_y = 1/F_p$  gives the formula for  $G_{yy}$ . Differentiating  $F_p G_y = 1$  with respect to  $\lambda_i$  at fixed  $y$  gives

$$(F_{pp}G_{\lambda_i} + F_{p\lambda_i})G_y + F_p G_{y\lambda_i} = 0.$$

Substitution of the first derivatives yields the stated mixed formula. Finally, differentiate  $F_p G_{\lambda_i} + F_{\lambda_i} = 0$  with respect to  $\lambda_j$ . The chain rule gives

$$F_p G_{\lambda_i \lambda_j} + F_{pp}G_{\lambda_i}G_{\lambda_j} + F_{p\lambda_j}G_{\lambda_i} + F_{p\lambda_i}G_{\lambda_j} + F_{\lambda_i \lambda_j} = 0,$$

which proves the last formula. Repeating the chain rule proves the triangular assertion at every order.  $\square$

**Remark 30.14** (Conditioning). For perturbations of the observed value with passive parameters fixed, the absolute local condition number is  $|F_p|^{-1}$ . When  $p \neq 0$  and  $y \neq 0$ , the corresponding relative condition number is

$$\left| \frac{y}{pF_p} \right|.$$

The blow-up as  $F_p \rightarrow 0$  is a conditioning statement; by itself it is not a proof of nonidentifiability.

## 30.6 Finite ramification and Puiseux inversion

**Theorem 30.15** (Finite-ramification normal form). *Let  $F$  be holomorphic near  $p_0$ , let  $y_0 = F(p_0)$ , and suppose, for some  $m \geq 2$ ,*

$$F(p_0 + z) = y_0 + c_m z^m + c_{m+1} z^{m+1} + \mathcal{O}(z^{m+2}), \quad c_m \neq 0.$$

*Choose a local value*

$$\xi = \left( \frac{y - y_0}{c_m} \right)^{1/m}$$

*on a sufficiently small simply connected punctured  $y$ -domain. Then the  $m$  inverse branches near  $p_0$ , indexed by the  $m$ -th roots of unity, are*

$$p - p_0 = \omega \xi - \frac{c_{m+1}}{m c_m} \omega^2 \xi^2 + \mathcal{O}(\xi^3), \quad \omega^m = 1. \quad (30.6)$$

*For every sufficiently small  $y \neq y_0$ , these are precisely the  $m$  solutions near  $p_0$ , counted without multiplicity; at  $y = y_0$  they coalesce.*

*Proof.* Factor

$$F(p_0 + z) - y_0 = c_m z^m H(z), \quad H(z) = 1 + \frac{c_{m+1}}{c_m} z + \mathcal{O}(z^2).$$

On a small disk,  $H$  is nonzero and has a holomorphic  $m$ -th root with value one at zero. Define  $v = zH(z)^{1/m}$ . Then  $v(0) = 0$  and  $v'(0) = 1$ , so  $v$  has a regular holomorphic inverse  $z = \psi(v)$ . Moreover,

$$\frac{F(p_0 + z) - y_0}{c_m} = v^m.$$

Thus the solutions are  $z = \psi(\omega \xi)$ ,  $\omega^m = 1$ . Since

$$v = z + \frac{c_{m+1}}{m c_m} z^2 + \mathcal{O}(z^3), \quad \psi(v) = v - \frac{c_{m+1}}{m c_m} v^2 + \mathcal{O}(v^3),$$

substitution gives (30.6). For nonzero  $\xi$ , the  $m$  values  $\omega \xi$  are distinct, and the injectivity of  $\psi$  on a smaller disk makes their images distinct. Conversely every nearby solution yields one of these  $m$  roots of  $v^m = \xi^m$ , proving exhaustion.  $\square$

**Corollary 30.16** (Universal fold jet through order two). *Let a holomorphic germ, in the notation of Theorem 30.15 with  $m = 2$ , have the expansion*

$$y - y_* = Az^2 + Bz^3 + Cz^4 + Dz^5 + \mathcal{O}(z^6), \quad A \neq 0,$$

*and choose a square root  $\sqrt{A}$ . The two inverse branches, indexed by  $\sigma \in \{1, -1\}$ , are*

$$\begin{aligned} z = \sigma \frac{(y - y_*)^{1/2}}{\sqrt{A}} - \frac{B}{2A^2} (y - y_*) \\ + \sigma \frac{5B^2 - 4AC}{8A^{7/2}} (y - y_*)^{3/2} - \frac{A^2 D - 3ABC + 2B^3}{2A^5} (y - y_*)^2 + \mathcal{O}((y - y_*)^{5/2}). \end{aligned} \quad (30.7)$$

*The powers  $A^{7/2}$  use the same chosen square root as the leading term.*

*Proof.* Put  $t = (y - y_*)^{1/2}$  on the chosen ramified coordinate and seek

$$z = at + bt^2 + ct^3 + dt^4 + \mathcal{O}(t^5).$$

Matching the coefficient of  $t^2$  gives  $Aa^2 = 1$ , hence  $a = \sigma/\sqrt{A}$ . Successively setting the coefficients of  $t^3, t^4, t^5$  equal to zero gives

$$\begin{aligned} 2Aab + Ba^3 &= 0, \\ A(b^2 + 2ac) + 3Ba^2b + Ca^4 &= 0, \\ A(2ad + 2bc) + B(3a^2c + 3ab^2) + 4Ca^3b + Da^5 &= 0. \end{aligned}$$

Solving these triangular equations yields

$$b = -\frac{B}{2A^2}, \quad c = \sigma \frac{5B^2 - 4AC}{8A^{7/2}}, \quad d = -\frac{A^2D - 3ABC + 2B^3}{2A^5}.$$

Substitution into the ansatz proves (30.7).  $\square$

*Warning 30.17* (What a singular Jacobian does not say). The condition  $F_p(p_0, \lambda_0) = 0$  says only that the regular inverse-function theorem is unavailable. A finite first nonzero derivative can produce finitely ramified branches as above; an injective real map such as  $p \mapsto p^3$  remains identifiable despite its zero derivative; and more degenerate behaviour is also possible. Neither nonidentifiability nor Puiseux behaviour follows from a singular Jacobian alone.

### 30.7 Asymptotic inverse transfer and Lambert $W$

**Proposition 30.18** (Residual transfer along a monotone inverse branch). *Let  $F$  be real  $C^1$  and injective on an interval  $I$ , with inverse  $G$  on  $F(I)$ . Let  $\tilde{G}(y) \in I$ . If  $|F'(p)| \geq m(y) > 0$  on the segment joining  $\tilde{G}(y)$  to  $G(y)$ , then*

$$|\tilde{G}(y) - G(y)| \leq \frac{|F(\tilde{G}(y)) - y|}{m(y)}.$$

*In particular, a forward residual estimate transfers to an inverse asymptotic estimate once the derivative scale is controlled on the selected branch.*

*Proof.* The mean-value theorem supplies a point  $\theta$  between  $\tilde{G}(y)$  and  $G(y)$  such that

$$F(\tilde{G}(y)) - F(G(y)) = F'(\theta)(\tilde{G}(y) - G(y)).$$

Use  $F(G(y)) = y$  and the assumed lower bound for  $|F'(\theta)|$ .  $\square$

**Theorem 30.19** (Reciprocal-scale reversion). *Let  $A \neq 0$  and suppose, on a real or complex sector as  $t \rightarrow 0$ ,*

$$x = X(t) = \frac{A}{t} + B + Ct + Dt^2 + \mathcal{O}(t^3).$$

*For any inverse branch  $t = T(x) \rightarrow 0$ , put  $S = x - B$ . Then  $t \sim A/S$  and, in the corresponding sector as  $S \rightarrow \infty$ ,*

$$T(x) = \frac{A}{S} + \frac{CA^2}{S^3} + \frac{DA^3}{S^4} + \mathcal{O}(S^{-5}). \quad (30.8)$$

*In particular, the coefficient of  $S^{-2}$  is zero.*

*Proof.* After subtracting  $B$  and multiplying by  $t$ , the inverse equation is

$$St = A + Ct^2 + Dt^3 + \mathcal{O}(t^4).$$

Since  $t \rightarrow 0$ , this first gives  $St \rightarrow A$ , hence  $t \sim A/S$ . Dividing the same identity by  $S$  and using  $t = \mathcal{O}(S^{-1})$  gives

$$t = \frac{A}{S} + \mathcal{O}(S^{-3}).$$

It follows that

$$t^2 = \frac{A^2}{S^2} + \mathcal{O}(S^{-4}), \quad t^3 = \frac{A^3}{S^3} + \mathcal{O}(S^{-5}).$$

Substituting these two estimates once more into

$$t = \frac{A}{S} + \frac{C}{S}t^2 + \frac{D}{S}t^3 + \mathcal{O}(t^4/S)$$

gives (30.8). This purely algebraic bootstrap does not differentiate the asymptotic remainder.  $\square$

*Remark 30.20.* The theorem deliberately excludes logarithmic corrections. If the forward expansion contains  $t \log t$ , substitution of  $t \sim A/S$  generally produces a  $(\log S)/S^3$  term in the inverse. Such a term must not be hidden inside an ordinary power-series ansatz.

**Theorem 30.21** (Lambert- $W$  preconditioner with branch compatibility). *Fix a simply connected domain  $D \subset \mathbb{C} \setminus \{0\}$  and a holomorphic branch of  $\log t$  on  $D$ . Consider the truncated equation*

$$\Lambda = \frac{A}{t} + B \log t + C, \quad A \neq 0, \quad t \in D. \quad (30.9)$$

*If  $B = 0$ , its solution is  $t = A/(\Lambda - C)$ , provided the denominator is nonzero and this value lies in  $D$ . If  $B \neq 0$ , put*

$$Z = -\frac{A}{B} \exp\left(-\frac{\Lambda - C}{B}\right).$$

*For every branch  $W_k$  defined at  $Z$  with  $W_k(Z) \neq 0$ , set*

$$t_k = -\frac{A}{B W_k(Z)}.$$

*Then  $t_k$  solves (30.9) if and only if  $t_k \in D$  and the branch-compatibility identity*

$$\log t_k = \frac{\Lambda - C}{B} + W_k(Z) \quad (30.10)$$

*holds for the chosen logarithm. Conversely, every solution of (30.9) has this form for some value of the multivalued relation  $ue^u = Z$ , and hence for a local branch of  $W$  whenever that relation is locally regular.*

*Proof.* The case  $B = 0$  is immediate. Suppose  $B \neq 0$  and set

$$u = -\frac{A}{Bt}.$$

If  $t$  solves (30.9), then

$$\log t = \frac{\Lambda - C}{B} + u.$$

Exponentiating and using  $t = -A/(Bu)$  gives

$$ue^u = -\frac{A}{B} \exp\left(-\frac{\Lambda - C}{B}\right) = Z.$$

Hence  $u$  is a value of  $W(Z)$ ; wherever  $1 + u \neq 0$ , it belongs to a locally holomorphic branch  $W_k$ . The displayed formula for  $t_k$  follows, and (30.10) is the original, unexponentiated equation. Conversely, if  $u = W_k(Z)$ ,  $t_k = -A/(Bu)$ , and (30.10) holds, then

$$\frac{A}{t_k} + B \log t_k + C = -Bu + B \left( \frac{\Lambda - C}{B} + u \right) + C = \Lambda.$$

□

**Warning 30.22.** The compatibility test in (30.10) is essential over  $\mathbb{C}$ . Exponentiation forgets integer multiples of  $2\pi i$ ; choosing a  $W$ -branch without also fixing and checking the logarithm branch can therefore create spurious solutions or omit genuine ones.

**Corollary 30.23** (Accuracy of the Lambert preconditioner). *On a fixed real or complex branch, suppose the full equation is*

$$\Lambda = \frac{A}{t} + B \log t + C + R(t), \quad A \neq 0, \quad R(t) = \mathcal{O}(t)$$

*uniformly in the chosen branch domain. Let  $t_0 \rightarrow 0$  be a branch-compatible solution of the truncated equation (30.9), and let  $t$  solve the full equation with  $t/t_0 \rightarrow 1$ . In the complex case assume the segment joining  $t_0$  and  $t$  stays in the chosen logarithm domain. Then*

$$t = t_0 + \mathcal{O}(t_0^3).$$

*Proof.* Write  $L(t) = A/t + B \log t + C$ . The two equations give

$$L(t) - L(t_0) = -R(t) = \mathcal{O}(t_0).$$

On the joining segment, the assumption  $t/t_0 \rightarrow 1$  gives uniformly

$$L'(t) = -\frac{A}{t^2} + \frac{B}{t} = -\frac{A}{t_0^2}(1 + o(1)).$$

The real mean-value theorem, or integration of  $L'$  along the complex segment, therefore gives

$$L(t) - L(t_0) = (t - t_0) \left( -\frac{A}{t_0^2} \right) (1 + o(1)).$$

Since its left-hand side is  $\mathcal{O}(t_0)$ , division proves  $t - t_0 = \mathcal{O}(t_0^3)$ .

□

## 30.8 Multivariate triangular inversion

**Theorem 30.24** (Total-degree inverse recurrence). *Let  $R$  be a commutative ring and let*

$$\mathbf{F}(\mathbf{p}) = A\mathbf{p} + \sum_{r \geq 2} \mathbf{F}_r(\mathbf{p})$$

*be a  $d$ -component formal map, where each  $\mathbf{F}_r$  is homogeneous of total degree  $r$  and  $A \in R^{d \times d}$  is invertible. There is a unique two-sided formal inverse  $\mathbf{G}(\mathbf{y}) = \sum_{n \geq 1} \mathbf{G}_n(\mathbf{y})$ , with each  $\mathbf{G}_n$  homogeneous of degree  $n$ . It is determined by*

$$\mathbf{G}_1 = A^{-1}\mathbf{y}$$

*and, for  $n \geq 2$ ,*

$$\mathbf{G}_n = -A^{-1}[\text{total degree } n] \sum_{r=2}^n \mathbf{F}_r \left( \sum_{j=1}^{n-1} \mathbf{G}_j \right). \quad (30.11)$$

*If  $\mathbf{F}$  is analytic over  $\mathbb{R}$  or  $\mathbb{C}$ , then after translating a base point  $\mathbf{p}_0$  and its value to the origin, the hypothesis  $\det D\mathbf{F}(\mathbf{p}_0) \neq 0$  makes the formal series the Taylor series of the local analytic inverse, and*

$$D\mathbf{G}(\mathbf{y}) = D\mathbf{F}(\mathbf{G}(\mathbf{y}))^{-1}.$$

*Proof.* In total degree one, the equation  $F(G(y)) = y$  is  $AG_1 = y$ . At degree  $n \geq 2$ , the only occurrence of the unknown  $G_n$  is the linear term  $AG_n$ : every nonlinear  $F_r$  uses at least two positive-degree inputs, so its degree- $n$  part involves only  $G_1, \dots, G_{n-1}$ . Isolating  $G_n$  gives (30.11) and constructs a unique right inverse.

For two-sidedness, let  $H = G \circ F$ . Then  $F \circ H = F$  and  $DH(0) = I$ . If  $H - \text{id}$  first has a nonzero homogeneous part  $K_n$ , then  $F \circ H - F$  first has homogeneous part  $AK_n$ , which cannot vanish because  $A$  is invertible. Thus  $H = \text{id}$ .

The analytic inverse-function theorem proves convergence in the analytic case. Differentiating  $F(G(y)) = y$  gives  $DF(G(y))DG(y) = I$ , and hence the derivative formula.  $\square$

*Remark 30.25.* When  $\det DF = 0$ , the regular multivariate theorem is simply inapplicable. Rank loss alone does not identify a ramification type; a separate normal-form or elimination argument is required.

### 30.9 The product-form Lagrange–Good formula

**Lemma 30.26** (Multivariate formal residue substitution). *Let  $R$  be a commutative  $\mathbb{Q}$ -algebra and*

$$T_i(\mathbf{x}) = x_i U_i(\mathbf{x}), \quad U_i(\mathbf{0}) \in R^\times, \quad 1 \leq i \leq d.$$

*For a formal Laurent series  $A(\mathbf{T})$ , whose support is bounded below in each coordinate, define*

$$\text{Res}_{\mathbf{T}} A(\mathbf{T}) \, dT_1 \cdots dT_d := [T_1^{-1} \cdots T_d^{-1}] A(\mathbf{T}).$$

*Then*

$$\text{Res}_{\mathbf{T}} A(\mathbf{T}) \, dT_1 \cdots dT_d = \text{Res}_{\mathbf{x}} A(\mathbf{T}(\mathbf{x})) \det \left( \frac{\partial T_i}{\partial x_j} \right) dx_1 \cdots dx_d.$$

*Proof.* By linearity it is enough to consider a Laurent monomial  $\prod_i T_i^{k_i}$ . If some  $k_i \neq -1$ , its associated top form is exact:

$$\left( \prod_j T_j^{k_j} \right) dT_1 \wedge \cdots \wedge dT_d = d \left( \frac{(-1)^{i-1} T_i^{k_i+1}}{k_i + 1} \prod_{j \neq i} T_j^{k_j} \wedge_{j \neq i} dT_j \right).$$

Formal differentiation has zero residue, and pullback commutes with  $d$ , so both residues vanish. If every  $k_i = -1$ , the form is

$$\bigwedge_{i=1}^d \frac{dT_i}{T_i} = \bigwedge_{i=1}^d \left( \frac{dx_i}{x_i} + \frac{dU_i}{U_i} \right).$$

The product of all  $dx_i/x_i$  has residue one. Every other wedge term replaces at least one logarithmic pole by a regular power-series differential, so its full  $x_1^{-1} \cdots x_d^{-1}$  residue is zero. Thus the pullback also has residue one, proving the assertion.  $\square$

**Theorem 30.27** (Lagrange–Good inversion). *Let  $R$  be a commutative  $\mathbb{Q}$ -algebra and let  $\phi_i(\mathbf{x}) \in R[[x_1, \dots, x_d]]$  have invertible constant terms. The product-form system*

$$w_i = t_i \phi_i(\mathbf{w}), \quad 1 \leq i \leq d,$$

*has a unique formal solution  $\mathbf{w}(t)$  with zero constant term. For every  $H(\mathbf{x}) \in R[[x_1, \dots, x_d]]$ ,*

$$H(\mathbf{w}(t)) = \sum_{\mathbf{n} \in \mathbb{Z}_{\geq 0}^d} t^n [\mathbf{x}^{\mathbf{n}}] H(\mathbf{x}) \prod_{i=1}^d \phi_i(\mathbf{x})^{n_i} \det \left( \delta_{ij} - \frac{x_j}{\phi_i(\mathbf{x})} \frac{\partial \phi_i}{\partial x_j}(\mathbf{x}) \right)_{i,j=1}^d. \quad (30.12)$$



*Proof.* Existence and uniqueness follow from [Theorem 30.24](#) applied to  $t_i = w_i/\phi_i(\mathbf{w})$ . Fix  $\mathbf{n} \in \mathbb{Z}_{\geq 0}^d$ . Coefficient extraction as a formal residue gives

$$[t^{\mathbf{n}}]H(\mathbf{w}(t)) = \text{Res}_{\mathbf{t}} H(\mathbf{w}(t)) \prod_i t_i^{-n_i} \frac{dt_i}{t_i}.$$

Use the formal coordinate change

$$t_i = \frac{x_i}{\phi_i(\mathbf{x})}.$$

Since  $\mathbf{w}(t(\mathbf{x})) = \mathbf{x}$ , one has

$$t_i^{-n_i} = x_i^{-n_i} \phi_i(\mathbf{x})^{n_i}$$

and

$$\frac{dt_i}{t_i} = \sum_{j=1}^d \left( \delta_{ij} - \frac{x_j}{\phi_i(\mathbf{x})} \frac{\partial \phi_i}{\partial x_j}(\mathbf{x}) \right) \frac{dx_j}{x_j}.$$

Taking the wedge product produces the determinant in (30.12); taking the remaining residue is exactly  $[\mathbf{x}^{\mathbf{n}}]$ . Summing over  $\mathbf{n}$  proves the formula.  $\square$

*Remark 30.28* (Equivalent determinant conventions). If  $D = \text{diag}(x_1, \dots, x_d)$  and  $Q_{ij} = \phi_i^{-1} \partial \phi_i / \partial x_j$ , the determinant in (30.12) is  $\det(I - QD)$ . Some sources instead print  $\det(I - DQ)$ , whose entries contain  $x_i$  rather than  $x_j$ . The two are equal because  $\det(I - AB) = \det(I - BA)$  for square matrices over a commutative ring. For completeness, this identity follows by taking the determinant after reducing

$$\begin{pmatrix} I & A \\ B & I \end{pmatrix}$$

in two different ways: the operation  $\text{row}_2 \leftarrow \text{row}_2 - B\text{row}_1$  gives diagonal blocks  $(I, I - BA)$ , while  $\text{row}_1 \leftarrow \text{row}_1 - A\text{row}_2$  gives diagonal blocks  $(I - AB, I)$ . Both block row operations have determinant one.

*Remark 30.29* (Scope of Good's determinant). [Theorem 30.27](#) applies to systems already written in the componentwise product form  $w_i = t_i \phi_i(\mathbf{w})$ . It is not a coefficient formula for an arbitrary locally invertible square system. For such a system one should first use the invariant total-degree recurrence (30.11); a reduction to product form requires additional structure and must be justified.

## 30.10 Residual, Rouché, and interval-Newton certificates

The real residual estimate needed here is already the general branchwise transfer principle [Proposition 30.18](#). If  $|F'| \geq m > 0$  on all of an interval, Darboux's theorem forces the derivative to have constant sign, so  $F$  is injective there; taking  $m(y) = m$  gives the familiar bound  $|\tilde{x} - x_*| \leq |F(\tilde{x}) - y|/m$  without a second theorem or proof.

**Theorem 30.30** (Rouché disk certificate). *Let  $r > 0$ , let  $F$  be holomorphic on a neighbourhood of the closed disk  $\overline{D}(z_0, r)$ , let  $y \in \mathbb{C}$ , and assume*

$$F'(z_0) \neq 0, \quad \sup_{\overline{D}(z_0, r)} |F''(z)| \leq M.$$

*If*

$$\frac{Mr^2}{2} < |F'(z_0)|r - |F(z_0) - y|, \tag{30.13}$$

*then  $F(z) = y$  has exactly one solution in  $D(z_0, r)$ , and that solution is simple.*

*Proof.* Put  $H(z) = F(z) - y$  and

$$L(z) = H(z_0) + F'(z_0)(z - z_0).$$

Taylor's formula along the segment from  $z_0$  to  $z$ , valid because a disk is convex, gives

$$|H(z) - L(z)| \leq \frac{M}{2}|z - z_0|^2.$$

On the boundary  $|z - z_0| = r$ ,

$$|L(z)| \geq |F'(z_0)|r - |H(z_0)| > \frac{Mr^2}{2} \geq |H(z) - L(z)|.$$

Rouché's theorem says that  $H$  and  $L$  have the same number of zeros in the disk, counted with multiplicity. Condition (30.13) also implies  $|H(z_0)/F'(z_0)| < r$ , so the affine function  $L$  has exactly one zero there. Hence  $H$  has total zero multiplicity one, which means its unique zero is simple.  $\square$

**Theorem 30.31** (Scalar interval-Newton certificate at an arbitrary target). *Let  $I = [a, b] \subset \mathbb{R}$ , let  $F \in C^1(I)$ , choose  $x_0 \in I$  and  $y \in \mathbb{R}$ , and let  $D = [d_-, d_+]$  be a closed real interval containing  $F'(I)$ , with  $0 \notin D$ . Interpret  $(F(x_0) - y)/D$  as its interval image and define*

$$N = x_0 - \frac{F(x_0) - y}{D}.$$

*If  $N \subseteq I$ , then  $F(x) = y$  has exactly one solution  $x_*$  in  $I$ , and  $x_* \in N$ . If  $N \subseteq \text{int } I$ , the certified solution lies in  $\text{int } I$ .*

*Proof.* Replace  $F$  by  $F - y$ . Its derivative enclosure is still  $D$ , its zero set is the target level set of the original function, and its interval Newton image is exactly the displayed  $N$ . We may therefore assume  $y = 0$ .

Replacing  $F$  by  $-F$  and  $D$  by  $-D$ , if necessary, leaves  $N$  unchanged, so assume  $0 < d_- \leq F'(x) \leq d_+$  on  $I$ . Thus  $F$  is strictly increasing. If  $F(x_0) = 0$ , the conclusion is immediate. If  $F(x_0) > 0$ , then

$$N = \left[ x_0 - \frac{F(x_0)}{d_-}, x_0 - \frac{F(x_0)}{d_+} \right] \subseteq [a, b].$$

The left-end inclusion implies  $F(x_0) \leq d_-(x_0 - a)$ . Since

$$F(x_0) - F(a) = \int_a^{x_0} F'(s) \, ds \geq d_-(x_0 - a),$$

we obtain  $F(a) \leq 0 < F(x_0)$ , so the intermediate-value theorem gives a zero between  $a$  and  $x_0$ . If  $F(x_0) < 0$ , the right-end inclusion similarly gives  $-F(x_0) \leq d_-(b - x_0)$ , hence  $F(b) \geq 0$ , and again a zero exists. Strict monotonicity makes it unique.

For that zero  $x_*$ , the mean-value theorem gives some  $\theta$  between  $x_0$  and  $x_*$  such that

$$F(x_0) = F'(\theta)(x_0 - x_*).$$

Because  $F'(\theta) \in D$ ,

$$x_* = x_0 - \frac{F(x_0)}{F'(\theta)} \in N.$$

The final interior assertion follows from  $x_* \in N$ .  $\square$

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These certificates answer different logical questions. A residual estimate controls the error once existence and the intended branch are already known. The Rouché test proves complex existence, uniqueness, and simplicity in a disk. The scalar interval-Newton test proves real existence, uniqueness, and localization in an interval. A small numerical residual alone must never be reported as an existence or branch certificate.

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## Chapter 31

# Finite $q$ -Pochhammer inversion

### 31.1 The finite product as an inverse covering

Fix  $n \geq 1$  and write

$$P_n(a, q) := (a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j).$$

Throughout,  $[n]_q$  and  $\begin{bmatrix} n \\ r \end{bmatrix}_q$  denote their polynomial continuations at  $q = 0, 1$  and at roots of unity; no quotient with a vanishing denominator is being used there. This section consolidates the finite-product results that were repeated in the six source reports. The results are human-proved frontier results: no Lean formalization is claimed here. We first invert in  $a$ , then in  $q$ , and keep the two branch geometries separate.

Whenever no factor vanishes, logarithmic differentiation gives

$$\partial_a \log P_n(a, q) = - \sum_{j=0}^{n-1} \frac{q^j}{1 - aq^j}, \quad (31.1)$$

$$\partial_q \log P_n(a, q) = -a \sum_{j=1}^{n-1} \frac{j q^{j-1}}{1 - aq^j}. \quad (31.2)$$

Both identities also hold after clearing denominators at a zero. They will be used only on zero-free sets, where the logarithms are unambiguous locally.

#### 31.1.1 Argument inversion at the unit level

Euler's finite  $q$ -binomial identity is

$$P_n(a, q) = \sum_{r=0}^n (-1)^r q^{\binom{r}{2}} \begin{bmatrix} n \\ r \end{bmatrix}_q a^r. \quad (31.3)$$

Set  $s = 1 - y$  and

$$b_r = (-1)^{r+1} q^{\binom{r}{2}} \begin{bmatrix} n \\ r \end{bmatrix}_q, \quad f(a) = 1 - P_n(a, q) = \sum_{r \geq 1} b_r a^r.$$

**Theorem 31.1** (All-order argument inverse at  $y = 1$ ). *If  $[n]_q = b_1 \neq 0$ , there is a unique analytic branch  $a = a(s)$  with  $a(0) = 0$  and  $P_n(a(s), q) = 1 - s$ . Its coefficients are*

$$[s^m]a(s) = \frac{1}{m} [z^{m-1}] \left( \frac{z}{1 - P_n(z, q)} \right)^m, \quad m \geq 1. \quad (31.4)$$

In particular,

$$a = \frac{s}{[n]_q} + \frac{q \begin{bmatrix} n \\ 2 \end{bmatrix}_q}{[n]_q^3} s^2 + \left( \frac{2q^2 \begin{bmatrix} n \\ 2 \end{bmatrix}_q^2}{[n]_q^5} - \frac{q^3 \begin{bmatrix} n \\ 3 \end{bmatrix}_q}{[n]_q^4} \right) s^3 + \mathcal{O}(s^4). \quad (31.5)$$

*Proof.* The linear coefficient of  $f$  is  $f'(0) = [n]_q \neq 0$ , so the analytic inverse-function theorem supplies one and only one local branch. Rewrite  $s = f(a)$  as  $a = s\phi(a)$ , where  $\phi(a) = a/f(a)$ . Lagrange inversion gives  $[s^m]a = m^{-1}[z^{m-1}]\phi(z)^m$ , which is Equation (31.4). Finally

$$s = [n]_q a - q \begin{bmatrix} n \\ 2 \end{bmatrix}_q a^2 + q^3 \begin{bmatrix} n \\ 3 \end{bmatrix}_q a^3 + \mathcal{O}(a^4).$$

Substitute  $a = A_1 s + A_2 s^2 + A_3 s^3 + \cdots$  and compare coefficients. This gives

$$A_1 = [n]_q^{-1}, \quad A_2 = \frac{q \begin{bmatrix} n \\ 2 \end{bmatrix}_q}{[n]_q^3}, \quad A_3 = \frac{2q^2 \begin{bmatrix} n \\ 2 \end{bmatrix}_q^2}{[n]_q^5} - \frac{q^3 \begin{bmatrix} n \\ 3 \end{bmatrix}_q}{[n]_q^4},$$

proving both displayed claims.  $\square$

### 31.1.2 All zero sheets and the complete real atlas

For  $q \neq 0$ , the argument zeros are  $a_j = q^{-j}$ ,  $0 \leq j < n$ . They are distinct precisely when  $q^r \neq 1$  for  $1 \leq r < n$ .

**Proposition 31.2** (Exact normalization at every simple zero). *Assume  $q \neq 0$  and  $q^r \neq 1$  for  $1 \leq r < n$ . Then*

$$\partial_a P_n(q^{-j}, q) = (-1)^{j+1} q^{-j(j-1)/2} (q; q)_j (q; q)_{n-1-j}. \quad (31.6)$$

Consequently the inverse sheet through  $(y, a) = (0, q^{-j})$  begins

$$a(y) = q^{-j} + \frac{(-1)^{j+1} q^{j(j-1)/2}}{(q; q)_j (q; q)_{n-1-j}} y + \mathcal{O}(y^2). \quad (31.7)$$

*Proof.* Differentiate the unique vanishing factor:

$$\partial_a P_n(q^{-j}, q) = -q^j \prod_{\ell \neq j} (1 - q^{\ell-j}).$$

For  $\ell < j$ , write  $1 - q^{\ell-j} = -q^{-(j-\ell)}(1 - q^{j-\ell})$ . Their product is  $(-1)^j q^{-j(j+1)/2} (q; q)_j$ , while the factors with  $\ell > j$  give  $(q; q)_{n-1-j}$ . Multiplication by  $-q^j$  proves Equation (31.6). Its nonvanishing permits ordinary analytic inversion, and division by the derivative gives Equation (31.7).  $\square$

**Theorem 31.3** (Complete real argument atlas). *Let  $q > 0$ ,  $q \neq 1$ , and order the zeros increasingly as  $r_0 < \cdots < r_{n-1}$ . Then:*

1.  $P_n(\cdot, q)$  is positive and strictly decreasing on  $(-\infty, r_0)$ , and maps that interval bijectively onto  $(0, \infty)$ .
2. Every gap  $(r_j, r_{j+1})$  contains exactly one critical point  $c_j$ . It is simple, so the two inverse sheets meeting at  $P_n(c_j, q)$  have square-root ramification.
3. There is no critical point on  $(r_{n-1}, \infty)$ .

For  $n \geq 2$ , the maximal real intervals on which  $P_n(\cdot, q)$  is one-to-one are

$$(-\infty, c_0], \quad [c_{j-1}, c_j] \quad (1 \leq j \leq n-2), \quad [c_{n-2}, \infty).$$

Each contains exactly one simple zero  $r_j$ , and together they exhaust the maximal real injective branches; their interiors are the maximal open domains of regular real inversion. Equivalently, each zero-free gap  $(r_j, r_{j+1})$  consists of two half-branches meeting at the fold  $c_j$ . When  $n = 1$ , the sole maximal branch is all of  $\mathbb{R}$ .

*Proof.* Since  $1 - aq^j = -q^j(a - q^{-j})$ , away from the roots

$$\frac{\partial_a P_n(a, q)}{P_n(a, q)} = \sum_{j=0}^{n-1} \frac{1}{a - r_j}, \quad \frac{d}{da} \frac{\partial_a P_n}{P_n} = - \sum_{j=0}^{n-1} \frac{1}{(a - r_j)^2} < 0. \quad (31.8)$$

On the leftmost interval every summand in the first expression is negative. All product factors are positive there, the product tends to  $+\infty$  at  $-\infty$ , and it tends to zero at  $r_0$ ; this proves the first item. On  $(r_j, r_{j+1})$ , the logarithmic derivative decreases continuously from  $+\infty$  to  $-\infty$ , so it has exactly one zero. At that zero,  $P_n''/P_n$  equals the strictly negative second expression in Equation (31.8); hence  $P_n'' \neq 0$ . Taylor's formula and reversion of its first nonzero quadratic term give two square-root sheets. To the right of the largest root every summand in Equation (31.8) is positive. The preceding argument, together with the nonvanishing derivatives at the simple zeros, proves that  $c_0, \dots, c_{n-2}$  are all the real zeros of  $\partial_a P_n$ . Its sign is therefore fixed and nonzero on each component of their complement. The function is strictly monotone on the closure of each such component. Because every  $c_j$  is a strict local extremum, no closure can be extended past one of its finite endpoints while preserving injectivity. These closures are therefore precisely the maximal injective intervals, while their interiors are the maximal regular inverse domains. The interlacing  $c_{j-1} < r_j < c_j$ , with the evident one-sided versions at the extremes, shows that each contains exactly one simple zero; analytic inversion continues through that zero. For  $n = 1$ , the polynomial is linear and the same conclusion is immediate.  $\square$

Put  $N = \binom{n}{2}$ ,  $A_n = (-1)^n q^N$ , and  $e_1 = \sum_{j=0}^{n-1} q^{-j}$ . Factoring the leading power of  $a$  gives

$$P_n(a, q) = A_n a^n \left( 1 - \frac{e_1}{a} + \mathcal{O}(a^{-2}) \right).$$

Thus, on each choice  $T = (y/A_n)^{1/n}$ ,

$$a = T + \frac{e_1}{n} + \mathcal{O}(T^{-1}). \quad (31.9)$$

Indeed, substitution of  $a = T + c + \mathcal{O}(T^{-1})$  makes the coefficient of  $T^{-1}$  equal to  $nc - e_1$ , which forces  $c = e_1/n$ .

### 31.1.3 Cyclotomic collisions in the argument

**Theorem 31.4** (Exact root-of-unity factorization and ramification). *Let  $\zeta$  be primitive of order  $d$ , and write  $n = Md + r$  with  $0 \leq r < d$ . Then*

$$(a; \zeta)_n = (1 - a^d)^M (a; \zeta)_r. \quad (31.10)$$

At  $a_s = \zeta^{-s}$ ,  $0 \leq s < d$ , the zero multiplicity is

$$\nu_s = M + \mathbf{1}_{s < r}. \quad (31.11)$$

For every  $s$  with  $\nu_s \geq 1$ , the point  $a_s$  is a zero of multiplicity  $\nu_s$ . On the ramified cover of a sufficiently small punctured target neighborhood, the inverse at target zero then has exactly  $\nu_s$  local Puiseux sheets

coalescing at  $a_s$ . Equivalently, after choosing a  $\nu_s$ -th root on a simply connected punctured target domain, these are the  $\nu_s$  determinations obtained by multiplying that root by the  $\nu_s$ -th roots of unity. When  $\nu_s = 0$ ,  $a_s$  is not a zero and contributes no inverse germ at target zero. At the unit target  $y = 1$ , the branch through  $a = 0$  is regular when  $r > 0$ . When  $r = 0$ , it has the  $d$  local sheets

$$a = \omega \left( \frac{1-y}{M} \right)^{1/d} + \mathcal{O}((1-y)^{2/d}), \quad \omega^d = 1. \quad (31.12)$$

*Proof.* Partition the indices into  $M$  complete residue blocks modulo  $d$  and one residual block. Each complete block contributes  $\prod_{j=0}^{d-1} (1 - a\zeta^j) = 1 - a^d$ , proving Equation (31.10). A factor vanishes at  $a_s$  exactly when its index is congruent to  $s \pmod{d}$ . That residue occurs  $M$  times in the complete blocks and once more precisely when  $s < r$ , proving Equation (31.11). When  $\nu_s \geq 1$ , removing the nonvanishing factors leaves a nonzero constant times  $(a - a_s)^{\nu_s}$ , whose equation with  $y$  has  $\nu_s$  Puiseux determinations. When  $\nu_s = 0$ , no factor vanishes at  $a_s$ , so there is no inverse germ over target zero there.

At  $a = 0$ , the derivative is  $-\sum_{j=0}^{n-1} \zeta^j = -\sum_{j=0}^{r-1} \zeta^j$ , which is nonzero for  $r > 0$ . If  $r = 0$ , Equation (31.10) gives  $(a; \zeta)_n = (1 - a^d)^M = 1 - Ma^d + \mathcal{O}(a^{2d})$ ; solving its leading equation gives Equation (31.12).  $\square$

At  $q = -1$ , this includes

$$(a; -1)_{2M} = (1 - a^2)^M, \quad (a; -1)_{2M+1} = (1 - a)(1 - a^2)^M.$$

For even order every argument sheet is obtained from  $a = \pm \sqrt{1 - \eta y^{1/M}}$ , where  $\eta^M = 1$ .

## 31.2 Finite-product inversion in the base

### 31.2.1 The canonical real base branch

**Theorem 31.5** (Global finite base inverse on  $0 \leq q \leq 1$ ). *Let  $n \geq 2$ .*

1. *If  $0 < a < 1$ , then  $q \mapsto P_n(a, q)$  is strictly decreasing from  $1 - a$  to  $(1 - a)^n$ . Every  $(1 - a)^n < y < 1 - a$  has exactly one inverse  $q \in (0, 1)$ .*
2. *If  $a < 0$ , the map is strictly increasing from  $1 - a$  to  $(1 - a)^n$ . Every  $1 - a < y < (1 - a)^n$  has exactly one inverse  $q \in (0, 1)$ .*

*For  $a = 0$ ,  $a = 1$ , or  $n = 1$ , the map is constant in  $q$ .*

*Proof.* For  $0 < q < 1$ , every denominator in Equation (31.2) is positive. Its sign is therefore  $-\text{sgn}(a)$ , and the nonzero derivative proves strict monotonicity. Direct substitution gives  $P_n(a, 0) = 1 - a$  and  $P_n(a, 1) = (1 - a)^n$ . Continuity and strict monotonicity now give the stated ranges and uniqueness. The three constant cases follow immediately.  $\square$

### 31.2.2 All-order inversion at $q = 0$

Assume  $n \geq 2$  and  $a \notin \{0, 1\}$ , and normalize the target by

$$\eta = \frac{(1 - a) - y}{a(1 - a)}. \quad (31.13)$$

Define  $F_{n,a}(q) = (1 - P_n(a, q))/(1 - a)/a$ . Then  $F_{n,a}(q) = q + \mathcal{O}(q^2)$ , and  $P_n(a, q) = y$  is precisely  $F_{n,a}(q) = \eta$ .

**Theorem 31.6** (Exact coefficient engine at  $q = 0$ ). *The inverse branch  $q = Q_{n,a}(\eta)$  is analytic at zero and satisfies*

$$[\eta^m]Q_{n,a}(\eta) = \frac{1}{m}[z^{m-1}] \left( \frac{z}{F_{n,a}(z)} \right)^m. \quad (31.14)$$

Moreover,

$$\log \frac{P_n(a, q)}{1-a} = \sum_{r \geq 1} \ell_r^{(n)}(a) q^r, \quad \ell_r^{(n)}(a) = - \sum_{\substack{d|r \\ 1 \leq d \leq n-1}} \frac{a^{r/d}}{r/d}. \quad (31.15)$$

For  $n \geq 6$ ,

$$q = \eta - \eta^2 + (1+a)\eta^3 - (1+4a)\eta^4 + (1+11a+3a^2)\eta^5 + \mathcal{O}(\eta^6). \quad (31.16)$$

For  $n = 2$ ,  $q = \eta$  exactly; for  $n = 3$ , the cubic coefficient is  $2 + a$ ; and for every  $n \geq 4$  it is  $1 + a$ .

*Proof.* The coefficient of  $q$  in  $F_{n,a}$  is one, so analytic inversion and Lagrange's formula give Equation (31.14). For the divisor formula, expand

$$\log \frac{P_n(a, q)}{1-a} = - \sum_{j=1}^{n-1} \sum_{m \geq 1} \frac{a^m q^{jm}}{m}.$$

The coefficient of  $q^r$  comes from  $jm = r$ , proving Equation (31.15).

When  $n \geq 6$ , subset expansion gives

$$F_{n,a}(q) = q + q^2 + (1-a)q^3 + (1-a)q^4 + (1-2a)q^5 + \mathcal{O}(q^6).$$

Substitute  $q = \eta + A_2\eta^2 + \cdots + A_5\eta^5$  and compare powers. Successively,  $A_2 = -1$ ,  $A_3 = 1 + a$ ,  $A_4 = -(1 + 4a)$ , and  $A_5 = 1 + 11a + 3a^2$ . The same calculation with the available positive indices gives the stated low-order exceptions.  $\square$

### 31.2.3 The complete logarithmic jet at $q = 1$

Put  $q = e^{-t}$ ,  $S_m(n) = \sum_{j=0}^{n-1} j^m$ , and choose the local logarithm of  $y/(1-a)^n$ .

**Theorem 31.7** (All-order regular base inverse at  $q = 1$ ). *Let  $n \geq 2$  and  $a \in \mathbb{C} \setminus \{0, 1\}$ . With*

$$u = \log \frac{y}{(1-a)^n}, \quad c_m = \frac{(-1)^{m+1}}{m!} \text{Li}_{1-m}(a) S_m(n),$$

*the forward germ and its inverse are*

$$u = \sum_{m \geq 1} c_m t^m, \quad (31.17)$$

$$t = \frac{u}{c_1} - \frac{c_2}{c_1^3} u^2 + \left( \frac{2c_2^2}{c_1^5} - \frac{c_3}{c_1^4} \right) u^3 + \mathcal{O}(u^4), \quad q = e^{-t}. \quad (31.18)$$

Here  $c_1 = a \binom{n}{2} / (1-a) \neq 0$ , and all subsequent inverse coefficients are obtained uniquely by triangular coefficient comparison.

*Proof.* For  $|a| < 1$ ,

$$\log(1 - ae^{-x}) = - \sum_{r \geq 1} \frac{a^r e^{-rx}}{r} = \log(1-a) + \sum_{m \geq 1} \frac{(-1)^{m+1}}{m!} \text{Li}_{1-m}(a) x^m.$$

Substitute  $x = jt$  and sum over  $0 \leq j < n$ . Both sides are analytic in  $a$  away from  $a = 1$ , and the nonpositive-index polylogarithms are rational, so analytic continuation proves the formula for every stated  $a$ . Since  $S_1(n) = \binom{n}{2}$ , the linear coefficient is nonzero. Substitution of  $t = A_1 u + A_2 u^2 + A_3 u^3 + \cdots$  gives  $A_1 = c_1^{-1}$ ,  $A_2 = -c_2 c_1^{-3}$ , and  $A_3 = 2c_2^2 c_1^{-5} - c_3 c_1^{-4}$ . Every later equation is linear in the new coefficient.  $\square$

For  $n \geq 2$ , holding  $y \neq 0$  fixed while  $q \rightarrow 1$  gives a useful mixed inverse. Choose  $r = y^{1/n}$ , put  $a_0 = 1 - r$ , and keep that root determination fixed. Substitution in the logarithm of the level equation gives

$$a(t) = a_0 + \frac{n-1}{2}a_0t + \frac{a_0(n-1)(2n-4-3a_0(n-1))}{24(1-a_0)}t^2 + \mathcal{O}(t^3). \quad (31.19)$$

Indeed, writing  $a = a_0 + At + Bt^2$  makes the constant term  $n \log(1 - a_0) = \log y$ ; the linear and quadratic equations give  $A = (n-1)a_0/2$  and the displayed  $B$ .

### 31.2.4 The exceptional slice $q = -1$

**Theorem 31.8** (Complete zero-free criticality at  $q = -1$ ). *Let  $n \geq 3$  and  $a \notin \{0, \pm 1\}$ . Then  $\partial_q P_n(a, -1) = 0$  if and only if*

$$a = a_*(n) = \begin{cases} 1/(n-1), & n \text{ even}, \\ -1/n, & n \text{ odd}. \end{cases} \quad (31.20)$$

For  $n = 2s$ ,  $s \geq 2$ ,

$$y_* = \left( \frac{4s(s-1)}{(2s-1)^2} \right)^s, \quad (31.21)$$

$$L_2 := \partial_q^2 \log P_{2s}(a_*, -1) = -\frac{(s+1)(2s-1)^2}{12s(s-1)}. \quad (31.22)$$

For  $n = 2s+1$ ,  $s \geq 2$ ,

$$y_* = \left( \frac{2s+2}{2s+1} \right)^{s+1} \left( \frac{2s}{2s+1} \right)^s, \quad (31.23)$$

$$L_2 = -\frac{(s-1)(2s+1)^2}{12s(s+1)}. \quad (31.24)$$

All these slices are quadratic folds, with local inverse

$$q = -1 \pm \sqrt{\frac{2(y-y_*)}{y_*L_2}} + \mathcal{O}(y-y_*). \quad (31.25)$$

Within this zero-free family, the sole higher degeneracy is

$$P_3(-1/3, q) = \frac{32}{27} + \frac{4}{27}(q+1)^3. \quad (31.26)$$

*Proof.* Let  $L = \log P_n$ . At  $q = -1$ , separation of even and odd indices in Equation (31.2) gives

$$L_q(a, -1) = a \left( \frac{E}{1-a} - \frac{O}{1+a} \right),$$

where  $E$  and  $O$  are the sums of the positive even and odd indices below  $n$ . For  $n = 2s$ ,  $E = s(s-1)$ ,  $O = s^2$ , giving  $a = 1/(2s-1)$ . For  $n = 2s+1$ ,  $E = s(s+1)$ ,  $O = s^2$ , giving  $a = -1/(2s+1)$ . Substitution in  $P_n(a, -1) = (1-a)^{\lceil n/2 \rceil} (1+a)^{\lfloor n/2 \rfloor}$  gives the two critical values.

Differentiating Equation (31.2) once more gives

$$L_{qq} = -a \sum_{j=1}^{n-1} \left( \frac{j(j-1)q^{j-2}}{1-aq^j} + \frac{aj^2q^{2j-2}}{(1-aq^j)^2} \right).$$



For completeness, the four parity sums needed here are

|                       |                          |                           |
|-----------------------|--------------------------|---------------------------|
|                       | $\sum j(j-1)$            | $\sum j^2$                |
| $2, 4, \dots, 2(s-1)$ | $\frac{s(s-1)(4s-5)}{3}$ | $\frac{2s(s-1)(2s-1)}{3}$ |
| $1, 3, \dots, 2s-1$   | $\frac{s(s-1)(4s+1)}{3}$ | $\frac{s(4s^2-1)}{3}$     |
| $2, 4, \dots, 2s$     | $\frac{s(s+1)(4s-1)}{3}$ | $\frac{2s(s+1)(2s+1)}{3}$ |

At  $q = -1$ , the first summand in  $L_{qq}$  uses the even row with a plus sign and the odd row with a minus sign, whereas its  $j^2$  summand uses both rows with a plus sign. For  $n = 2s$ , insert the first two rows and  $a = 1/(2s-1)$ ; for  $n = 2s+1$ , insert the last two rows and  $a = -1/(2s+1)$ . Cancelling the common factors gives exactly [Equations \(31.22\) and \(31.24\)](#). The even expression never vanishes for  $s \geq 2$ ; the odd expression vanishes only at  $s = 1$ , where direct multiplication proves [Equation \(31.26\)](#). At a critical point  $P_{qq} = P_n L_{qq} = y_* L_2 \neq 0$ . Taylor expansion and reversion of the quadratic term give [Equation \(31.25\)](#).  $\square$

The exclusions in [Theorem 31.8](#) are structural, not hidden exceptions. At  $a = 0$  the product is identically one, and at  $a = 1$  it is identically zero. At  $a = -1$ ,

$$P_n(-1, q) = \prod_{j=0}^{n-1} (1 + q^j)$$

has at  $q = -1$  a zero of exact order  $r = \lfloor n/2 \rfloor$ , because exactly the odd-indexed factors vanish and each does so simply. More precisely, with  $e = \lceil n/2 \rceil$  and  $J_n = \prod_{\substack{1 \leq j < n \\ j \text{ odd}}} j$ ,

$$P_n(-1, q) = 2^e J_n (q+1)^r + \mathcal{O}((q+1)^{r+1}). \quad (31.27)$$

Indeed, every even-indexed factor tends to two and  $1 + q^j = j(q+1) + \mathcal{O}((q+1)^2)$  for odd  $j$ . Thus, on the ramified target cover,

$$q = -1 + \omega \left( \frac{y}{2^e J_n} \right)^{1/r} + \mathcal{O}(y^{2/r}), \quad \omega^r = 1.$$

This zero-target cyclotomic ramification is the repeated-factor phenomenon seen in [Theorem 31.4](#); it is different from the nonzero-level critical points classified above.

**Corollary 31.9** (The cubic slice lies on a Morse saddle). *For every odd  $n = 2s + 1$ ,  $s \geq 1$ , the point  $(a, q) = (-1/(2s+1), -1)$  is a nondegenerate saddle of  $(a, q) \mapsto P_n(a, q)$ . Thus [Equation \(31.26\)](#) is a tangent one-dimensional slice, not a cusp of the full parameter surface.*

*Proof.* At this point both  $L_a$  and  $L_q$  vanish. Direct differentiation of  $L = \sum_{j=0}^{2s} \log(1 - aq^j)$ , followed by the parity sums, gives

$$L_{aa} = -\frac{(2s+1)^3}{4s(s+1)}, \quad L_{aq} = -\frac{(2s+1)^2}{4(s+1)}, \quad L_{qq} = -\frac{(s-1)(2s+1)^2}{12s(s+1)}.$$

For instance,

$$L_{aa} = -\frac{s+1}{(1-a)^2} - \frac{s}{(1+a)^2}, \quad L_{aq} = \frac{s(s+1)}{(1-a)^2} - \frac{s^2}{(1+a)^2},$$

which yield the first two entries after substituting  $a = -1/(2s+1)$ ; the third is [Equation \(31.24\)](#). Hence

$$\det \text{Hess } L = -\frac{(2s+1)^4(s^2+s+1)}{48s^2(s+1)^2} < 0. \quad (31.28)$$

Since  $P_n \neq 0$  and its gradient vanishes, its Hessian is  $P_n$  times that of  $L$ . It is therefore also nonsingular and indefinite.  $\square$

### 31.2.5 Reciprocal-base inversion at infinity

Let  $n \geq 2$ ,  $a \notin \{0, 1\}$ ,  $N = \binom{n}{2}$ ,  $C = (1 - a)(-a)^{n-1}$ , and

$$H(z) = \prod_{j=1}^{n-1} (1 - a^{-1}z^j).$$

The exact factorization  $P_n(a, q) = Cq^N H(q^{-1})$  converts infinity into an ordinary point.

**Theorem 31.10** (All-order finite base inverse at infinity). *Choose an  $N$ -th root  $x = (C/y)^{1/N}$ . The corresponding branch  $s = q^{-1}$  is the convergent solution of*

$$s = xH(s)^{1/N}, \quad [x^m]s(x) = \frac{1}{m}[z^{m-1}]H(z)^{m/N}. \quad (31.29)$$

In particular,

$$q = x^{-1} + \frac{1}{aN} + \mathcal{O}(x). \quad (31.30)$$

All  $N$  complex sheets arise from the  $N$  choices of  $x$ .

*Proof.* The level equation is  $y = Cs^{-N}H(s)$ , hence  $x = sH(s)^{-1/N}$ , equivalently the first equation in Equation (31.29). Since  $H(0) = 1$ , a consistent analytic  $N$ -th root exists near zero and the analytic inverse-function theorem gives a unique convergent branch. Lagrange inversion proves the coefficient formula. Finally  $H(s) = 1 - a^{-1}s + \mathcal{O}(s^2)$ , so  $s = x - a^{-1}x^2/N + \mathcal{O}(x^3)$ . Taking the reciprocal gives Equation (31.30).  $\square$

**Warning 31.11** (Claims deliberately not promoted). The source reports contain exact-looking secondary-discriminant factorizations for a proposed first Maxwell collision at  $n = 5$ . No independent exact certificate for the large factorization is currently part of the canonical package, so that special assertion is not stated here as a theorem. The proved branch atlas remains valid, and a certified resultant/Sturm computation would suffice to add the collision theorem later.

## Chapter 32

# Infinite $q$ -Pochhammer inverse geometry

### 32.1 The infinite product: analytic domain and certified tails

For  $|q| < 1$ , put

$$P(a, q) := (a; q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j).$$

The domain condition is part of the definition. Values on  $|q| = 1$ , radial limits toward that boundary, and exterior reciprocal normalizations are different objects and will be treated separately.

**Theorem 32.1** (Normal convergence, zeros, and derivative kernels). *Let  $0 < \rho < 1$  and  $R > 0$ . Uniformly for  $|q| \leq \rho$ ,  $|a| \leq R$ , and  $N \geq 0$ ,*

$$\left| \prod_{j=N}^{\infty} (1 - aq^j) - 1 \right| \leq \exp \left( \frac{R\rho^N}{1 - \rho} \right) - 1. \quad (32.1)$$

If  $R\rho^N < 1$ , the logarithm obtained by continuation from  $a = 0$  satisfies

$$\left| \sum_{j=N}^{\infty} \log(1 - aq^j) \right| \leq \frac{R\rho^N}{(1 - \rho)(1 - R\rho^N)}. \quad (32.2)$$

Consequently  $P$  is holomorphic on  $\{(a, q) \in \mathbb{C}^2 : |q| < 1\}$ , and is entire in  $a$  for fixed  $q$ . For  $0 < |q| < 1$ , its zeros as a function of  $a$  are exactly  $a = q^{-m}$ ,  $m \geq 0$ , and they are simple. Away from those zeros,

$$\partial_a \log P(a, q) = - \sum_{j \geq 0} \frac{q^j}{1 - aq^j}, \quad (32.3)$$

$$\partial_a^2 \log P(a, q) = - \sum_{j \geq 0} \frac{q^{2j}}{(1 - aq^j)^2}, \quad (32.4)$$

$$\partial_q \log P(a, q) = -a \sum_{j \geq 1} \frac jq^{j-1} \frac{1}{1 - aq^j}. \quad (32.5)$$

All three series converge locally uniformly on the indicated zero-free set.

*Proof.* Put  $u_j = -aq^j$ . Expanding finite products and applying the triangle inequality gives

$$\left| \prod_{j=N}^M (1 + u_j) - 1 \right| \leq \prod_{j=N}^M (1 + |u_j|) - 1 \leq \exp \left( \sum_{j=N}^M |u_j| \right) - 1.$$

Since  $\sum_{j \geq N} |u_j| \leq R\rho^N/(1-\rho)$ , the finite tails form a uniformly Cauchy family and their limit obeys Equation (32.1). If  $R\rho^N < 1$ , then

$$|\log(1-z)| \leq \sum_{k \geq 1} \frac{|z|^k}{k} \leq \frac{|z|}{1-|z|}$$

for  $|z| < 1$ . With  $|aq^j| \leq R\rho^N \rho^{j-N}$ , summation proves Equation (32.2).

Locally uniform limits of the finite holomorphic products are holomorphic. If no factor vanishes, choose  $N$  so that  $|aq^j| < 1/2$  for  $j \geq N$ . The tail is the exponential of an absolutely convergent logarithmic sum and is therefore nonzero; hence the only zeros are the factor zeros. At  $a = q^{-m}$ , exactly one factor vanishes, so the zero is simple. On a compact zero-free set, the denominators are separated from zero for the finite head, while the tails in Equations (32.3) to (32.5) are bounded by convergent geometric series with at most a polynomial factor in  $j$ . Normal convergence therefore permits termwise differentiation and proves all three identities.  $\square$

**Lemma 32.2** (Lambert logarithm). *For  $|a| < 1$  and  $|q| < 1$ ,*

$$\log(a; q)_\infty = - \sum_{m \geq 1} \frac{a^m}{m(1-q^m)}, \quad (32.6)$$

where the logarithm is the branch equal to zero at  $a = 0$ . The double series used in its derivation is absolutely and locally uniformly convergent.

*Proof.* For each factor,  $\log(1 - aq^j) = -\sum_{m \geq 1} a^m q^{jm}/m$ . On compact subdiscs  $|a| \leq \alpha < 1$ ,  $|q| \leq \rho < 1$ ,

$$\sum_{j \geq 0} \sum_{m \geq 1} \frac{|a|^m |q|^{jm}}{m} \leq \sum_{m \geq 1} \frac{\alpha^m}{m(1-\rho^m)} \leq \frac{-\log(1-\alpha)}{1-\rho} < \infty.$$

Thus Fubini's theorem applies, and summing  $\sum_{j \geq 0} q^{jm} = (1 - q^m)^{-1}$  gives Equation (32.6).  $\square$

## 32.2 Argument inversion and the complete real sheet atlas

**Theorem 32.3** (Principal inverse, all gaps, and exact zero normalization). *Fix  $0 < q < 1$ , write  $P_q(a) = (a; q)_\infty$ , and put  $r_m = q^{-m}$ .*

1.  $P_q$  is positive and strictly decreasing from  $+\infty$  to 0 on  $(-\infty, 1)$ . It therefore has a unique real-analytic inverse  $\mathcal{A}_q : (0, \infty) \rightarrow (-\infty, 1)$ .
2. Every  $r_m$  is a simple zero, with

$$d_m := P'_q(r_m) = (-1)^{m+1} q^{-m(m-1)/2} (q; q)_m (q; q)_\infty. \quad (32.7)$$

Hence its inverse germ is  $a_m(y) = r_m + y/d_m + \mathcal{O}(y^2)$ .

3. On the gap  $I_m = (r_m, r_{m+1})$ , the sign of  $P_q$  is  $\sigma_m = (-1)^{m+1}$ . There is exactly one critical point  $c_m \in I_m$ , and  $P''_q(c_m) \neq 0$ . If  $M_m = |P_q(c_m)|$ , then  $I_m$  contains two roots of  $P_q(a) = y$  when  $0 < \sigma_m y < M_m$ , one double root when  $\sigma_m y = M_m$ , and no root when  $\sigma_m y < 0$  or  $\sigma_m y > M_m$ . At  $y = 0$ , its two endpoint roots are  $r_m, r_{m+1}$ .

The principal inverse is strictly decreasing and convex, and, with

$$S_k(a) = \sum_{j \geq 0} \frac{q^{kj}}{(1 - aq^j)^k},$$

it satisfies

$$\mathcal{A}'_q(y) = -\frac{1}{yS_1(\mathcal{A}_q(y))}, \quad (32.8)$$

$$\mathcal{A}''_q(y) = \frac{S_1(\mathcal{A}_q(y))^2 - S_2(\mathcal{A}_q(y))}{y^2 S_1(\mathcal{A}_q(y))^3} > 0. \quad (32.9)$$

*Proof.* For  $a < 1$ , all factors are positive and  $\partial_a \log P_q < 0$  by Equation (32.3). As  $a \uparrow 1$ ,

$$P_q(a) = (1-a)(aq; q)_\infty \sim (q; q)_\infty (1-a) \rightarrow 0,$$

where normal convergence gives the limit of the second factor. If  $a = -x$ , then  $P_q(-x) = \prod_{j \geq 0} (1 + xq^j) \geq 1 + x \rightarrow \infty$ . This proves the first bijection and, since its derivative never vanishes, real analyticity of its inverse.

To compute the derivative at  $r_m$ , remove the vanishing factor:

$$P'_q(r_m) = -q^m \prod_{j \neq m} (1 - q^{j-m}).$$

The indices  $j < m$  contribute  $(-1)^m q^{-m(m+1)/2} (q; q)_m$ ; the indices  $j > m$  contribute  $(q; q)_\infty$ . This proves Equation (32.7) and the inverse germ.

On every compact subset avoiding the zeros, Equation (32.3) may be written

$$\frac{P'_q(a)}{P_q(a)} = \sum_{j \geq 0} \frac{1}{a - r_j}, \quad \left( \frac{P'_q}{P_q} \right)'(a) = - \sum_{j \geq 0} \frac{1}{(a - r_j)^2} < 0.$$

In  $I_m$ , the first sum tends to  $+\infty$  at the left endpoint and to  $-\infty$  at the right endpoint. It therefore has one and only one zero. At that point  $P'_q/P_q$  equals the strictly negative second sum, so the critical point is nondegenerate. There are  $m+1$  negative factors in the gap, giving the sign  $\sigma_m$ . The logarithmic derivative has opposite signs on the two sides of  $c_m$ ; hence the absolute value rises strictly from zero to  $M_m$  and falls strictly to zero. The root count follows.

Implicit differentiation gives Equation (32.8). Since

$$S_1(a)^2 - S_2(a) = 2 \sum_{0 \leq i < j} \frac{q^{i+j}}{(1 - aq^i)(1 - aq^j)} > 0$$

for  $a < 1$ , a second differentiation gives Equation (32.9) and proves convexity.  $\square$

### 32.2.1 Endpoint and central inverse jets

Euler's expansion

$$(a; q)_\infty = \sum_{m \geq 0} \frac{(-1)^m q^{\binom{m}{2}}}{(q; q)_m} a^m$$

and Lagrange inversion give, with  $s = 1 - y$ ,

$$\begin{aligned} \mathcal{A}_q(y) &= (1-q)s + \frac{q(1-q)}{1+q} s^2 \\ &\quad + \frac{q^2(1-q)(q^2+q+2)}{(1+q)^2(q^2+q+1)} s^3 + \mathcal{O}(s^4), \end{aligned} \quad (32.10)$$

$$[s^m] \mathcal{A}_q(1-s) = \frac{1}{m} [z^{m-1}] \left( \frac{z}{1-(z; q)_\infty} \right)^m. \quad (32.11)$$

This follows exactly as in [Theorem 31.1](#): the forward linear coefficient is  $1/(1 - q)$ , and direct coefficient comparison gives the three displayed terms.

At the other endpoint, let

$$C_q = (q; q)_\infty, \quad L_1(q) = \sum_{j \geq 1} \frac{q^j}{1 - q^j}, \quad v = \frac{y}{C_q}.$$

Writing  $a = 1 - \delta$  gives

$$(a; q)_\infty = C_q \delta (1 + L_1(q) \delta + \mathcal{O}(\delta^2)).$$

Reversion therefore yields

$$\mathcal{A}_q(y) = 1 - v + L_1(q) v^2 + \mathcal{O}(v^3). \quad (32.12)$$

The radius of either inverse series is bounded by the nearest singular value encountered by its chosen sheet; such a singular value may be critical or asymptotic, and need not be the numerically nearest critical value belonging to some other sheet.

### 32.2.2 Remote real critical points

**Theorem 32.4** (Remote-gap asymptotics). *Let  $m = k + 1$ . As  $k \rightarrow \infty$ ,*

$$c_k(q) = q^{-m} \left( 1 - \frac{1}{m+1} + \mathcal{O}(m^{-2}) \right), \quad (32.13)$$

$$\begin{aligned} |P_q(c_k)| &= (q; q)_\infty^2 q^{-m(m+1)/2} \frac{m^m}{(m+1)^{m+1}} (1 + \mathcal{O}(m^{-1})) \\ &= \frac{(q; q)_\infty^2}{e(m+1)} q^{-m(m+1)/2} (1 + \mathcal{O}(m^{-1})). \end{aligned} \quad (32.14)$$

*The error constants are locally uniform for  $q$  in compact subsets of  $(0, 1)$ .*

*Proof.* Parameterize the gap by  $a = q^{-m}(1 - \varepsilon)$ ,  $0 < \varepsilon < 1 - q$ . Separating the  $m$  factors before the right endpoint gives the exact identity

$$\begin{aligned} P_q(a) &= (-1)^m q^{-m(m+1)/2} \varepsilon (1 - \varepsilon)^m G_m(\varepsilon), \\ G_m(\varepsilon) &= \left( \frac{q}{1 - \varepsilon}; q \right)_m ((1 - \varepsilon)q; q)_\infty. \end{aligned} \quad (32.15)$$

If  $\varepsilon = \mathcal{O}(m^{-1})$ , logarithmic differentiation of the two products shows uniformly that

$$\partial_\varepsilon \log G_m(\varepsilon) = \mathcal{O}(1).$$

Indeed, the resulting summands are bounded by constant multiples of  $q^j/(1 - q^j)$ , whose sum converges. The same logarithmic estimate, together with the omitted tail  $\sum_{j > m} q^j = \mathcal{O}(q^m)$ , gives

$$G_m(\varepsilon) = (q; q)_\infty^2 (1 + \mathcal{O}(m^{-1})).$$

It remains to verify that the unknown extremum lies in this range. The logarithmic derivative of [Equation \(32.15\)](#) is positive as  $\varepsilon \downarrow 0$ . At  $\varepsilon = 2/(m+1)$ , which lies inside the gap for all sufficiently large  $m$ , the first two terms below have size  $-m/2 + \mathcal{O}(1)$ , while the already established  $G'_m/G_m = \mathcal{O}(1)$  estimate applies there. The derivative is therefore negative. Uniqueness of the gap extremum gives  $\varepsilon_* < 2/(m+1)$ , so  $\varepsilon_* = \mathcal{O}(m^{-1})$  without circular reasoning.

At that unique extremum, logarithmic differentiation of [Equation \(32.15\)](#) now gives

$$0 = \frac{1}{\varepsilon} - \frac{m}{1 - \varepsilon} + \partial_\varepsilon \log G_m(\varepsilon).$$

The bounded last term perturbs the zero of the first two terms by  $\mathcal{O}(m^{-2})$ , so  $\varepsilon = (m+1)^{-1} + \mathcal{O}(m^{-2})$ . Substitution proves [Equations \(32.13\)](#) and [\(32.14\)](#); finally  $(1 - 1/(m+1))^m = e^{-1}(1 + \mathcal{O}(m^{-1}))$ .  $\square$

### 32.3 Large targets and the log-periodic principal inverse

**Theorem 32.5** (Theta-modulated growth and its inverse). *Let  $q = e^{-\tau}$ ,  $\tau > 0$ . For  $x = e^L \geq 1$ , write uniquely  $x = q^{-m}r$ ,  $m \in \mathbb{N}_0$ ,  $1 \leq r < q^{-1}$ , and  $\rho = \log r \in [0, \tau)$ . Define on that interval*

$$\Psi_q(\rho) = \log \left( (-e^\rho; q)_\infty (-qe^{-\rho}; q)_\infty \right) - \frac{\rho}{2} - \frac{\rho^2}{2\tau}, \quad (32.16)$$

*and extend it  $\tau$ -periodically. Then  $\Psi_q$  is smooth and*

$$\log(-e^L; q)_\infty = \frac{L^2}{2\tau} + \frac{L}{2} + \Psi_q(L \bmod \tau) + \mathcal{O}(e^{-L}). \quad (32.17)$$

*If  $Y = (-x; q)_\infty \rightarrow \infty$ , set*

$$R = \sqrt{2\tau \log Y}, \quad \rho_0 = (R - \tau/2) \bmod \tau.$$

*Then*

$$\log x = R - \frac{\tau}{2} + \frac{\tau(\tau/8 - \Psi_q(\rho_0))}{R} + \mathcal{O}((\log Y)^{-1}). \quad (32.18)$$

*Proof.* The finite shift is exact:

$$(-q^{-m}r; q)_\infty = r^m q^{-m(m+1)/2} (-q/r; q)_m (-r; q)_\infty.$$

Uniformly for  $1 \leq r \leq q^{-1}$ , replacing the finite product by  $(-q/r; q)_\infty$  has relative error  $1 + \mathcal{O}(q^m)$ , by Equation (32.1). Since  $L = m\tau + \rho$ ,

$$m\rho + \frac{\tau m(m+1)}{2} = \frac{L^2}{2\tau} + \frac{L}{2} - \frac{\rho^2}{2\tau} - \frac{\rho}{2}.$$

Taking logarithms proves Equation (32.17). Normal convergence permits all  $\rho$ -derivatives on the compact fundamental interval. Under  $\rho \mapsto \rho + \tau$ , the product inside the logarithm is multiplied by  $e^{\rho+\tau}$ , exactly the change of the two subtracted terms, so the extension is smooth and periodic.

Completing the square rewrites the forward equation as

$$2\tau \log Y = (L + \tau/2)^2 + 2\tau(\Psi_q(\rho) - \tau/8) + \mathcal{O}(e^{-L}).$$

Taking the positive square root first gives  $L = R - \tau/2 + \mathcal{O}(R^{-1})$ . Insert this once into the bounded correction. The exact phase differs from  $\rho_0$ , on the periodic circle, by  $\mathcal{O}(R^{-1})$ ; smoothness of  $\Psi_q$  changes the  $R^{-1}$  correction by only  $\mathcal{O}(R^{-2})$ . Since  $R^{-2} = \mathcal{O}((\log Y)^{-1})$ , this is Equation (32.18).  $\square$

### 32.4 Inversion in the base at $q = 0$ and on the real interval

**Theorem 32.6** (Real base atlas and the stable small-base jet). *Fix real  $a$ .*

1. *If  $0 < a < 1$ ,  $q \mapsto (a; q)_\infty$  decreases strictly on  $[0, 1)$  from  $1 - a$  to 0.*
2. *If  $a < 0$ , it increases strictly from  $1 - a$  to  $+\infty$ .*
3. *If  $a = 0$  it is identically 1, and if  $a = 1$  it is identically 0.*
4. *If  $a > 1$ , it has the distinct simple base zeros  $q_j = a^{-1/j}$ ,  $j \geq 1$ , accumulating at 1; hence only interval-by-interval base inverses are possible.*

*For  $a \notin \{0, 1\}$ , set  $\eta = ((1 - a) - y)/(a(1 - a))$ . The branch through  $q = 0$  is*

$$\begin{aligned} q = & \eta - \eta^2 + (1 + a)\eta^3 - (1 + 4a)\eta^4 \\ & + (1 + 11a + 3a^2)\eta^5 - (1 + 26a + 22a^2)\eta^6 + \mathcal{O}(\eta^7). \end{aligned} \quad (32.19)$$

*Every coefficient is obtained by finite exact arithmetic: through order  $\eta^M$ , it agrees with every finite product of order  $n > M$ .*

*Proof.* On  $0 < q < 1$ , Equation (32.5) has sign  $-\operatorname{sgn}(a)$  whenever  $a < 1$  and  $a \neq 0$ . If  $0 < a < 1$ , then for any  $M$ , and  $q$  sufficiently close to one, the first  $M$  nonconstant factors are at most  $1 - a/2 < 1$ ; the remaining factors lie in  $(0, 1)$ . Thus the product is at most  $(1 - a/2)^M$ , and its limit is zero. If  $a < 0$ , the first  $M$  factors are at least  $1 + |a|/2$ , while all remaining factors exceed one, so the limit is  $+\infty$ . The values at  $q = 0$  and the two constant cases are immediate. For  $a > 1$ , exactly the  $j$ -th factor vanishes at  $q_j$ , proving simplicity and the asserted accumulation.

Near  $q = 0$ , normal convergence makes the product analytic and  $\partial_q P(a, 0) = -a(1 - a) \neq 0$ , so the analytic inverse exists. The coefficient of  $q^M$  uses only factors with index at most  $M$ ; hence it equals the stable finite-product coefficient. Direct subset expansion gives

$$\frac{1 - (aq; q)_\infty}{a} = q + q^2 + (1 - a)q^3 + (1 - a)q^4 + (1 - 2a)q^5 + (1 - 2a + a^2)q^6 + \mathcal{O}(q^7).$$

Substitution of  $q = \eta + \sum_{j=2}^6 c_j \eta^j$ , followed by comparison of successive powers, gives

$$(c_2, c_3, c_4, c_5, c_6) = (-1, 1 + a, -1 - 4a, 1 + 11a + 3a^2, -1 - 26a - 22a^2),$$

which is Equation (32.19). Because the subset calculation through degree  $M$  uses no factor beyond index  $M$ , it also proves the stated finite exactness mechanism.  $\square$

## 32.5 Fixed-argument inversion as $q \rightarrow 1$

**Theorem 32.7** (Uniform all-order fixed-argument expansion). *Let  $q = e^{-t}$ , where  $t \rightarrow 0$  in a closed subsector of  $\Re t > 0$ . Uniformly for  $|a| \leq \rho < 1$ ,*

$$\log(a; e^{-t})_\infty \sim -\frac{\operatorname{Li}_2(a)}{t} + \frac{1}{2} \log(1 - a) - \sum_{r \geq 1} \frac{B_{2r}}{(2r)!} \operatorname{Li}_{2-2r}(a) t^{2r-1}. \quad (32.20)$$

*More precisely, for every  $M \geq 1$ , retain the leading and constant terms and the Bernoulli terms with  $1 \leq r \leq M - 1$  (an empty sum when  $M = 1$ ). The remainder is then  $\mathcal{O}(|t|^{2M-1})$ , uniformly on the stated sets.*

*Proof.* Insert, in Equation (32.6),

$$\frac{1}{1 - e^{-x}} = \frac{1}{x} + \frac{1}{2} + \sum_{r=1}^{M-1} \frac{B_{2r}}{(2r)!} x^{2r-1} + R_M(x).$$

The displayed terms sum respectively to the dilogarithm, logarithm, and polylogarithms in Equation (32.20). To control the remainder, choose a small  $\delta > 0$  and split the Lambert index at  $m \leq \delta/|t|$ . Taylor's theorem is uniform on this first range and gives a contribution bounded by

$$C|t|^{2M-1} \sum_{m \geq 1} \rho^m m^{2M-2} = \mathcal{O}(|t|^{2M-1}).$$

On the complementary range,  $\rho^m \leq \exp(-c/|t|)$ . Because  $t$  stays in a closed subsector of the right half-plane,  $1 - e^{-mt}$  is uniformly separated from zero there; the tail is exponentially small and hence smaller than every power of  $|t|$ .  $\square$

**Theorem 32.8** (Rigorous large-log base inverse). *Fix  $a$  with  $|a| < 1$  and  $A = \operatorname{Li}_2(a) \neq 0$ , and choose compatible branches of all logarithms. Put*

$$D = \frac{\operatorname{Li}_0(a)}{12}, \quad E = \frac{\operatorname{Li}_{-2}(a)}{720}, \quad F = \frac{\operatorname{Li}_{-4}(a)}{30240}, \quad X = -\log y + \frac{1}{2} \log(1 - a).$$



Let  $X \rightarrow \infty$  so that  $A/X$  remains in a closed subsector of  $\Re t > 0$ . On the inverse sheet with  $t \sim A/X$ ,

$$t = \frac{A}{X} + \frac{DA^2}{X^3} + \frac{A^3(2D^2 - AE)}{X^5} + \frac{A^4(A^2F - 6ADE + 5D^3)}{X^7} + \mathcal{O}(X^{-9}), \quad q = e^{-t}. \quad (32.21)$$

For real  $0 < a < 1$ , this is the unique real base inverse as  $y \downarrow 0$ .

*Proof.* The preceding theorem gives

$$X = \frac{A}{t} + Dt - Et^3 + Ft^5 + \mathcal{O}(t^7).$$

Insert an ansatz  $t = A/X + \alpha X^{-3} + \beta X^{-5} + \gamma X^{-7}$ . Comparison of  $X^{-1}, X^{-3}, X^{-5}$  gives

$$\alpha = DA^2, \quad \beta = A^3(2D^2 - AE), \quad \gamma = A^4(A^2F - 6ADE + 5D^3).$$

This proves the formal coefficients, but the error also needs transfer. For the displayed approximation  $t_0$ , substitution in the forward equation leaves a residual  $\mathcal{O}(X^{-7})$ , while

$$\frac{d}{dt} \left( \frac{A}{t} + Dt - Et^3 + Ft^5 \right) = -\frac{A}{t^2} + \mathcal{O}(1)$$

has modulus comparable with  $|X|^2$ . The remainder is holomorphic and uniformly  $\mathcal{O}(t^7)$  on a slightly wider closed subsector, so Cauchy's estimate makes its derivative  $\mathcal{O}(t^6)$ . Taylor expansion of the exact target equation on  $|t - t_0| = K|X|^{-9}$  therefore consists of a constant  $\mathcal{O}(X^{-7})$ , a linear term of modulus comparable with  $K|X|^{-7}$ , and a quadratic error  $\mathcal{O}(|X|^3|X|^{-18})$ . For fixed sufficiently large  $K$ , the linear term dominates the other two. Rouché's theorem gives one zero inside that circle, proving the  $\mathcal{O}(X^{-9})$  remainder. The real uniqueness follows from [Theorem 32.6](#).  $\square$

## 32.6 The coalescing parameter $a = q^x$

**Theorem 32.9** (All-order coalescing expansion). *Let  $\Re x > 0$ ,  $q = e^{-t}$ , and let  $t \rightarrow 0$  in a closed subsector of  $|\arg t| < \pi/2$ . Use the logarithm obtained by analytic continuation from positive  $t$ . Then, locally uniformly for  $x$  in compact subsets of  $\Re x > 0$ ,*

$$\begin{aligned} \log(q^x; q)_\infty \sim & -\frac{\pi^2}{6t} + \left(\frac{1}{2} - x\right) \log t + \frac{1}{2} \log(2\pi) - \log \Gamma(x) \\ & + \frac{B_2(x)}{4}t - \sum_{r \geq 1} \frac{B_{2r}B_{2r+1}(x)}{2r(2r+1)!}t^{2r}. \end{aligned} \quad (32.22)$$

*Proof.* For  $c > 1$ , initially with  $t > 0$ , Mellin inversion of the exponential in the Lambert sum gives

$$\log(e^{-xt}; e^{-t})_\infty = -\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(s) \zeta(s+1) \zeta(s, x) t^{-s} ds. \quad (32.23)$$

Absolute convergence for  $c > 1$  justifies the interchange with the Lambert sum. Both sides then continue analytically to the stated sector, on the logarithm sheet fixed in the theorem. Indeed, uniformly for  $x$  in a fixed compact subset of  $\Re x > 0$ , sufficiently small  $t$  in a smaller closed subsector gives no factor zero: for bounded  $j$  the exponent  $(x+j)t$  has modulus below  $2\pi$ , while for large  $j$  its argument stays strictly between  $-\pi/2$  and  $\pi/2$ , so it cannot be a nonzero multiple of  $2\pi i$ . We

may therefore shift the continued contour left. The pole at  $s = 1$  contributes  $-\pi^2/(6t)$ . At  $s = 0$  the double pole, together with

$$\zeta(0, x) = \frac{1}{2} - x, \quad \zeta'(0, x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi),$$

gives the logarithmic and constant terms. At  $s = -1$ , the residue of  $\Gamma$ ,  $\zeta(0) = -1/2$ , and  $\zeta(-1, x) = -B_2(x)/2$  give  $B_2(x)t/4$ . At  $s = -2r$ ,

$$\operatorname{Res}_{s=-2r} \Gamma(s) = \frac{1}{(2r)!}, \quad \zeta(1-2r) = -\frac{B_{2r}}{2r}, \quad \zeta(-2r, x) = -\frac{B_{2r+1}(x)}{2r+1},$$

which yields the remaining coefficients, including the outer minus sign in Equation (32.23). At negative odd integers below  $-1$ , a trivial zero of  $\zeta(s+1)$  cancels the gamma pole.

Here is the required uniform remainder estimate. Fix a compact set  $K \subseteq \{\Re x > 0\}$ , a closed subsector  $|\arg t| \leq \pi/2 - \delta$ , and a vertical line  $\Re s = \sigma$  avoiding the poles just listed. Uniformly for  $x \in K$ , Stirling's estimate and the standard polynomial vertical bounds for the two zeta functions give constants  $C, N$  such that

$$|\Gamma(\sigma + iu)\zeta(\sigma + 1 + iu)\zeta(\sigma + iu, x)| \leq C(1 + |u|)^N e^{-\pi|u|/2}.$$

For the chosen branch of  $t^{-s}$ ,

$$|t^{-(\sigma+iu)}| = |t|^{-\sigma} e^{u \arg t} \leq |t|^{-\sigma} e^{(\pi/2-\delta)|u|}.$$

Thus the integrand on the shifted vertical line is bounded by

$$C|t|^{-\sigma}(1 + |u|)^N e^{-\delta|u|},$$

which is integrable uniformly for  $x \in K$ . The same exponential factor makes the horizontal sides of the truncating rectangles tend to zero. The shifted contour is therefore  $\mathcal{O}_K(|t|^{-\sigma})$ . Shifting successively past the displayed poles, and choosing  $\sigma$  strictly between the last retained pole and the next uncanceled pole, proves the asserted sectorial Poincaré remainder locally uniformly in  $x$ .  $\square$

The specialization  $x = 1$  has more structure than its power asymptotic: Dedekind modularity restores the exponentially small remainder exactly, both at the positive cusp and on the radial path toward  $-1$ .

**Theorem 32.10** (Exact eta completions at  $q = 1$  and  $q = -1$ ). *For  $t > 0$ ,*

$$(e^{-t}; e^{-t})_{\infty} = \sqrt{\frac{2\pi}{t}} \exp\left(-\frac{\pi^2}{6t} + \frac{t}{24}\right) (e^{-4\pi^2/t}; e^{-4\pi^2/t})_{\infty}. \quad (32.24)$$

If  $Q_c = \exp(-c\pi^2/t)$ , then

$$(-e^{-t}; -e^{-t})_{\infty} = \sqrt{\frac{\pi}{t}} \exp\left(-\frac{\pi^2}{24t} + \frac{t}{24}\right) \frac{(Q_2; Q_2)_{\infty}^3}{(Q_4; Q_4)_{\infty}(Q_1; Q_1)_{\infty}}. \quad (32.25)$$

Consequently the relative errors after deleting the final product factor in Equation (32.24) and the final product quotient in Equation (32.25) are, respectively,  $\mathcal{O}(e^{-4\pi^2/t})$  and  $\mathcal{O}(e^{-\pi^2/t})$ .

*Proof.* Recall

$$\eta(\tau) = e^{\pi i \tau / 12} \prod_{n \geq 1} (1 - e^{2\pi i n \tau})$$

and the modular identity  $\eta(-1/\tau) = \sqrt{-i\tau} \eta(\tau)$ , with the positive square root when  $\tau$  is positive imaginary. Set  $\tau = it/(2\pi)$ . Then

$$\begin{aligned}\eta(\tau) &= e^{-t/24} (e^{-t}; e^{-t})_{\infty}, \\ \eta(-1/\tau) &= e^{-\pi^2/(6t)} (e^{-4\pi^2/t}; e^{-4\pi^2/t})_{\infty}, \\ \sqrt{-i\tau} &= \sqrt{t/(2\pi)}.\end{aligned}$$

Solving the modular identity for the first product proves Equation (32.24).

For the negative radial path put  $x = e^{-t}$ . Separating the even and odd powers in the defining product, and then separating the odd powers in the ordinary positive product, gives

$$\begin{aligned}(-x; -x)_{\infty} &= (x^2; x^2)_{\infty} (-x; x^2)_{\infty}, \\ (-x; x^2)_{\infty} &= \frac{(x^2; x^4)_{\infty}}{(x; x^2)_{\infty}}, \\ (x; x^2)_{\infty} &= \frac{(x; x)_{\infty}}{(x^2; x^2)_{\infty}}, \quad (x^2; x^4)_{\infty} = \frac{(x^2; x^2)_{\infty}}{(x^4; x^4)_{\infty}}.\end{aligned}$$

Thus

$$(-x; -x)_{\infty} = \frac{(x^2; x^2)_{\infty}^3}{(x; x)_{\infty} (x^4; x^4)_{\infty}}. \quad (32.26)$$

Apply Equation (32.24) with  $t, 2t, 4t$ . The square-root factors simplify to  $\sqrt{\pi/t}$ , the elementary exponent to  $-\pi^2/(24t) + t/24$ , and the three complementary nomes to  $Q_4, Q_2, Q_1$ , respectively. Substitution in Equation (32.26) proves Equation (32.25).

Finally, as  $Q \downarrow 0$ , expansion of the first factor and the product tail estimate Equation (32.1) give  $(Q; Q)_{\infty} = 1 + \mathcal{O}(Q)$ ; the same estimate keeps its reciprocal bounded. Applying this observation to the exact final factors proves both relative error assertions.  $\square$

For real  $x > 0$ , the product decreases from 1 to 0 as  $q$  runs from 0 to 1, since every factor  $1 - q^{x+j}$  does. It therefore has a unique real base inverse. Put

$$\Lambda = -\log y, \quad A = \frac{\pi^2}{6}, \quad B = x - \frac{1}{2}, \quad C = \log \Gamma(x) - \frac{1}{2} \log(2\pi).$$

The leading equation is  $\Lambda = A/t + B \log t + C$ . If  $B \neq 0$ , its exact real Lambert- $W$  solution for sufficiently large  $\Lambda$  is

$$t_0 = -\frac{A}{B W_k\left(-\frac{A}{B} \exp\left(-\frac{\Lambda-C}{B}\right)\right)}, \quad (32.27)$$

where  $k = -1$  when  $B > 0$ , and  $k = 0$  when  $B < 0$ . When  $B = 0$ ,  $t_0 = A/(\Lambda - C)$ . Exponentiating the leading equation and putting  $w = -A/(Bt)$  proves Equation (32.27); all quantities are real, so this exponentiation loses no logarithm-sheet information. The omitted part of Equation (32.22) is  $\mathcal{O}(t)$ , while the derivative of  $A/t + B \log t + C$  has magnitude comparable with  $t^{-2}$ . The real mean-value theorem, applied between the exact inverse and  $t_0$ , therefore gives  $t = t_0 + \mathcal{O}(t_0^3)$ .

## 32.7 Fixed-target double scaling at $q = 1$

**Theorem 32.11** (All-order fixed-target inverse and recurrence). *Fix  $L > 0$ . The unique real solution*

$$(a; e^{-\varepsilon})_{\infty} = e^{-L}, \quad 0 < a < 1,$$

*has, for every  $M$ , an asymptotic expansion  $a(\varepsilon) = \sum_{k=1}^M A_k(L) \varepsilon^k + \mathcal{O}(\varepsilon^{M+1})$ . Define*

$$C_{r,n} = [\varepsilon^n] \left( \sum_{k \geq 1} A_k \varepsilon^k \right)^r.$$

Then  $A_1 = L$ , and for  $n \geq 1$ ,

$$\begin{aligned} A_{n+1} = & - \sum_{r=2}^{n+1} \frac{C_{r,n+1}}{r^2} - \frac{1}{2} \sum_{r=1}^n \frac{C_{r,n}}{r} \\ & - \sum_{j=1}^{\lfloor n/2 \rfloor} \frac{B_{2j}}{(2j)!} \sum_{r=1}^{n-2j+1} r^{2j-2} C_{r,n-2j+1}. \end{aligned} \quad (32.28)$$

In particular,

$$\begin{aligned} a = L\varepsilon - \frac{L(L+2)}{4}\varepsilon^2 + \frac{L(L^2+9L+12)}{72}\varepsilon^3 \\ - \frac{L(L^3+4L^2+12L+24)}{576}\varepsilon^4 \\ - \frac{L(31L^4-75L^3-800L^2-720)}{86400}\varepsilon^5 + \mathcal{O}(\varepsilon^6). \end{aligned} \quad (32.29)$$

Every  $A_k$  is a rational polynomial in  $L$  divisible by  $L$ .

*Proof.* Taking the negative logarithm gives the exact positive Lambert equation

$$\Phi_\varepsilon(a) := \sum_{r \geq 1} \frac{a^r}{r(1 - e^{-r\varepsilon})} = L. \quad (32.30)$$

Since the first summand is at least  $a/\varepsilon$ , every solution obeys  $0 < a \leq L\varepsilon$ . Moreover

$$\partial_a \Phi_\varepsilon(a) = \sum_{r \geq 1} \frac{a^{r-1}}{1 - e^{-r\varepsilon}} \geq \frac{1}{1 - e^{-\varepsilon}} \geq \frac{1}{\varepsilon}.$$

Thus  $\Phi_\varepsilon$  is strictly increasing, and existence and uniqueness also follow from the principal real argument atlas.

Expand

$$\frac{1}{1 - e^{-r\varepsilon}} = \frac{1}{r\varepsilon} + \frac{1}{2} + \sum_{j \geq 1} \frac{B_{2j}}{(2j)!} (r\varepsilon)^{2j-1}.$$

The coefficient of  $\varepsilon^n$  in Equation (32.30) contains the new unknown  $A_{n+1}$  only in  $a/\varepsilon$ , with coefficient one. The remaining contributions are, respectively, the three sums on the right of Equation (32.28); setting the coefficient to zero proves the recurrence. It gives the five displayed polynomials by successive substitution. Inductively its operations preserve rational polynomiality, and every term vanishes at  $L = 0$ , proving the factor  $L$ .

It remains to justify the asymptotic remainder. For a truncation  $a_M = \sum_{k=1}^M A_k \varepsilon^k = \mathcal{O}(\varepsilon)$ , use

$$1 - e^{-r\varepsilon} \geq c \min(r\varepsilon, 1)$$

for real  $\varepsilon > 0$ . Splitting at  $r = \varepsilon^{-1}$  then bounds the Lambert tail  $r \geq M + 1$  by

$$\frac{C}{\varepsilon} \sum_{M+1 \leq r \leq \varepsilon^{-1}} \frac{|a_M|^r}{r^2} + C \sum_{r > \varepsilon^{-1}} \frac{|a_M|^r}{r} = \mathcal{O}(\varepsilon^M).$$

For the finitely many retained indices, Taylor's theorem for the elementary kernel, together with the recurrence, gives  $\Phi_\varepsilon(a_M) - L = \mathcal{O}(\varepsilon^M)$ . The derivative lower bound and the real mean-value theorem then imply

$$|a - a_M| \leq \varepsilon |\Phi_\varepsilon(a) - \Phi_\varepsilon(a_M)| = \mathcal{O}(\varepsilon^{M+1}).$$

This completes both the formal and analytic parts of the proof.  $\square$

**Warning 32.12** (Complex double scaling). For complex  $L = -\log y$ , the recurrence still defines a formal branch, but the poles  $2\pi i k/r$  of the Lambert kernels accumulate at  $\varepsilon = 0$ . The real theorem above does not imply a convergent complex Taylor series. A complex version requires a sector, a logarithm branch, and uniform remainder estimates; absent those data it remains a formal asymptotic construction.

## 32.8 Raw root-of-unity endpoints and radial inverses

**Proposition 32.13** (Raw boundary partial products). *Let  $\zeta$  be primitive of order  $d$ , and write  $N = kd + s$ ,  $0 \leq s < d$ . Then*

$$(a; \zeta)_N = (1 - a^d)^k \prod_{j=0}^{s-1} (1 - a\zeta^j). \quad (32.31)$$

*Hence the partial products tend to zero when  $|1 - a^d| < 1$ , are unbounded when  $|1 - a^d| > 1$ , and do not converge when  $|1 - a^d| = 1$  and  $a \neq 0$ . For  $a = 0$ , they are identically one.*

*Proof.* Every complete residue block contributes  $\prod_{j=0}^{d-1} (1 - a\zeta^j) = 1 - a^d$ , proving Equation (32.31). The first two conclusions follow from the geometric factor. In the unit-modulus case, the subsequence  $s = 0$  is  $(1 - a^d)^k$ ; it could converge only if  $1 - a^d = 1$ , which forces  $a = 0$ .  $\square$

This proposition concerns raw partial products on the boundary. The next theorem instead approaches the boundary from  $|q| < 1$ , where the analytic product is defined.

**Theorem 32.14** (Uniform radial cyclotomic expansion and inverse). *Let  $\zeta$  be primitive of order  $d$ , let  $|a| \leq \rho < 1$ , and put  $q = \zeta e^{-t}$ , with  $t \rightarrow 0$  in a closed subsector of  $\Re t > 0$ . All logarithms below are the holomorphic determinations obtained by continuation from  $a = 0$ ; in particular,*

$$\log(1 - a^d) = - \sum_{\ell \geq 1} \frac{a^{d\ell}}{\ell} \quad (|a| < 1).$$

*Define*

$$A_\zeta(a) = \frac{\text{Li}_2(a^d)}{d^2}, \quad (32.32)$$

$$C_\zeta(a) = \frac{1}{2d} \log(1 - a^d) - \sum_{\substack{m \geq 1 \\ d \nmid m}} \frac{a^m}{m(1 - \zeta^m)}, \quad (32.33)$$

$$D_\zeta(a) = \frac{a^d}{12(1 - a^d)} - \sum_{\substack{m \geq 1 \\ d \nmid m}} \frac{a^m \zeta^m}{(1 - \zeta^m)^2}. \quad (32.34)$$

*The coefficient series converge absolutely, and*

$$\log(a; \zeta e^{-t})_\infty = -\frac{A_\zeta(a)}{t} + C_\zeta(a) - D_\zeta(a)t + \mathcal{O}(t^2) \quad (32.35)$$

*uniformly on the stated compact sets. If  $A_\zeta(a) \neq 0$ , choose a branch of  $\log y$ , put  $X = -\log y + C_\zeta(a)$ , and suppose that  $X \rightarrow \infty$  with  $A_\zeta(a)/X$  in the chosen closed subsector. The corresponding inverse satisfies*

$$t = \frac{A_\zeta(a)}{X} + \frac{D_\zeta(a)A_\zeta(a)^2}{X^3} + \mathcal{O}(X^{-4}), \quad q = \zeta e^{-t}. \quad (32.36)$$

*Proof.* Split Equation (32.6) into indices  $m = d\ell$  and  $d \nmid m$ . For the multiples,

$$\frac{1}{1 - e^{-d\ell t}} = \frac{1}{d\ell t} + \frac{1}{2} + \frac{d\ell t}{12} + \mathcal{O}((\ell t)^3).$$

Their singular, constant, and linear sums are respectively

$$-\frac{\text{Li}_2(a^d)}{d^2 t}, \quad \frac{1}{2d} \log(1 - a^d), \quad -\frac{a^d}{12(1 - a^d)} t.$$

For nonmultiples,

$$\frac{1}{1 - \zeta^m e^{-mt}} = \frac{1}{1 - \zeta^m} - \frac{m\zeta^m}{(1 - \zeta^m)^2} t + \mathcal{O}(m^2 t^2).$$

These are exactly the remaining terms in Equations (32.33) and (32.34).

Absolute convergence of the coefficient sums is not by itself a remainder proof, so split again at  $m \leq \delta/|t|$ . Choose  $\delta$  below the nearest singularity of each of the finitely many residue-class kernels. On the first range Taylor's theorem gives a uniform remainder, and summation of  $\rho^m m|t|^2$  gives  $\mathcal{O}(t^2)$ . On the tail,  $\rho^m = \mathcal{O}(\exp(-c/|t|))$ ; the sector condition also separates every denominator from zero. This proves Equation (32.35).

The target equation is

$$X = \frac{A_\zeta}{t} + D_\zeta t + \mathcal{O}(t^2).$$

Substitution of  $t_0 = A_\zeta/X + D_\zeta A_\zeta^2/X^3$  leaves residual  $\mathcal{O}(X^{-2})$ , while the derivative has size comparable with  $X^2$ . Uniform holomorphy on a slightly wider subsector and Cauchy's estimate control the derivatives of the  $\mathcal{O}(t^2)$  remainder. Thus, on a circle of radius  $K|X|^{-4}$  around  $t_0$ , the linear term has size comparable with  $K|X|^{-2}$ , the constant residual is  $\mathcal{O}(X^{-2})$ , and all higher variations are smaller. For fixed sufficiently large  $K$ , Rouché's theorem supplies a unique nearby solution and proves Equation (32.36).  $\square$

**Corollary 32.15** (The radial points 1 and  $-1$ ). *For real  $0 < a < 1$ , Equation (32.21) is the unique inverse toward  $q = 1^-$ . For real  $0 < |a| < 1$ , put*

$$A_- = \frac{1}{4} \text{Li}_2(a^2), \quad C_- = \frac{1}{2} \log(1 - a), \quad D_- = \frac{a(a+3)}{12(1-a^2)},$$

$$E_- = \frac{a(15a^4 + 4a^3 + 90a^2 + 4a + 15)}{720(1-a^2)^3}, \quad X = -\log y + C_-.$$

On the real radial branch  $q = -e^{-t}$ ,  $t \downarrow 0$ ,

$$\log(a; -e^{-t})_\infty = -\frac{A_-}{t} + C_- - D_- t + E_- t^3 + \mathcal{O}(t^5), \quad (32.37)$$

$$t = \frac{A_-}{X} + \frac{D_- A_-^2}{X^3} + \frac{2D_-^2 A_-^3 - E_- A_-^4}{X^5} + \mathcal{O}(X^{-7}). \quad (32.38)$$

Here the inverse formula is taken as  $y = (a; -e^{-t})_\infty \downarrow 0$ , equivalently  $X \rightarrow +\infty$ .

*Proof.* Split the canonical Lambert logarithm into even and odd indices:

$$\log(a; -e^{-t})_\infty = -\sum_{m \geq 1} \frac{a^m}{m(1 - (-1)^m e^{-mt})}.$$

For even indices use

$$\frac{1}{1 - e^{-x}} = \frac{1}{x} + \frac{1}{2} + \frac{x}{12} - \frac{x^3}{720} + \mathcal{O}(x^5),$$

and for odd indices use

$$\frac{1}{1+e^{-x}} = \frac{1}{2} + \frac{x}{4} - \frac{x^3}{48} + \mathcal{O}(x^5).$$

Only the even indices  $m = 2r$  contribute a singular term, namely

$$-\sum_{r \geq 1} \frac{a^{2r}}{2r} \frac{1}{2rt} = -\frac{\text{Li}_2(a^2)}{4t}.$$

The constant term is

$$-\frac{1}{2} \sum_{m \geq 1} \frac{a^m}{m} = \frac{1}{2} \log(1-a).$$

The coefficient of  $t$  is

$$-\frac{1}{12} \sum_{r \geq 1} a^{2r} - \frac{1}{4} \sum_{\substack{m \geq 1 \\ m \text{ odd}}} a^m = -\frac{a(a+3)}{12(1-a^2)}.$$

Writing  $z = a^2$ , the coefficient of  $t^3$  is

$$\frac{1}{180} \sum_{r \geq 1} r^2 z^r + \frac{1}{48} \sum_{\substack{m \geq 1 \\ m \text{ odd}}} m^2 a^m.$$

The elementary identities

$$\sum_{r \geq 1} r^2 z^r = \frac{z(1+z)}{(1-z)^3}, \quad \sum_{\substack{m \geq 1 \\ m \text{ odd}}} m^2 a^m = \frac{a(1+a)}{(1-a)^3} - \frac{4a^2(1+a^2)}{(1-a^2)^3}$$

simplify this coefficient to

$$\frac{a(15a^4 + 4a^3 + 90a^2 + 4a + 15)}{720(1-a^2)^3} = E_-.$$

For fixed  $|a| < 1$ , the same finite-index/tail split used in the proof of [Theorem 32.14](#) justifies the expansions under the sum; the summed fifth-order remainders are  $\mathcal{O}(t^5)$ . This proves [Equation \(32.37\)](#) with every displayed coefficient.

Now

$$X = \frac{A_-}{t} + D_- t - E_- t^3 + \mathcal{O}(t^5).$$

Substituting  $t = A_-/X + \alpha X^{-3} + \beta X^{-5}$  and comparing the coefficients of  $X^{-1}$  and  $X^{-3}$  gives

$$\alpha = D_- A_-^2, \quad \beta = 2D_-^2 A_-^3 - E_- A_-^4.$$

The resulting residual is  $\mathcal{O}(X^{-5})$ , whereas the derivative of the target equation is  $-A_-/t^2 + \mathcal{O}(1) \asymp -X^2$ . The real mean-value theorem therefore improves the inverse error to  $\mathcal{O}(X^{-7})$ , proving [Equation \(32.38\)](#). The exclusion  $a = 0$  is necessary: there the product is identically one and  $A_- = 0$ .  $\square$

**Proposition 32.16** (Fixed-level cyclotomic coalescence with winding). *Let  $\zeta$  be primitive of order  $d$ , let  $q = \zeta e^{-t}$ , and let  $t \rightarrow 0$  in a closed subsector of  $\Re t > 0$ . Fix  $y \in \mathbb{C}^\times$ , choose a logarithm  $\ell$  of  $y$ , and, for  $\nu \in \mathbb{Z}$ , put*

$$L_\nu := -\ell - 2\pi i \nu.$$

*Suppose  $L_\nu \neq 0$ . For the Lambert logarithm that is zero at  $a = 0$ , the lifted equation*

$$\log(a; q)_\infty = \ell + 2\pi i \nu$$

has exactly  $d$  solution germs tending to  $a = 0$ . They satisfy

$$a^d \sim d^2 L_\nu t, \quad a \sim \omega(d^2 L_\nu t)^{1/d}, \quad \omega^d = 1. \quad (32.39)$$

Consequently the raw equation  $(a; q)_\infty = y$  has one such  $d$ -sheeted cluster for every fixed winding  $\nu$  with  $L_\nu \neq 0$ . Every continuous raw solution branch tending to zero belongs to exactly one of these clusters. The exceptional lift  $L_\nu = 0$ , when it occurs, is a different degenerate problem and is not covered by the displayed scale.

*Proof.* Write  $\Lambda(a, t)$  for the Lambert logarithm in Equation (32.35). The lifted equation is  $-\Lambda(a, t) = L_\nu$ . Suppose first that a solution germ has  $a \rightarrow 0$ . In the uniform forward formula,  $C_\zeta(a) = o(1)$  and  $D_\zeta(a)t = o(1)$ ; its remainder is uniform for  $|a| \leq \rho$ , so

$$\frac{\text{Li}_2(a^d)}{d^2 t} = L_\nu + o(1).$$

Since  $\text{Li}_2(z)/z \rightarrow 1$ , the last identity first gives  $a^d = \mathcal{O}(t)$  and then  $a^d/(d^2 t) \rightarrow L_\nu$ , proving the first relation and the scale  $a = \mathcal{O}(t^{1/d})$ .

Choose a determination  $\tau = t^{1/d}$  on the subsector and write  $a = \tau b$ . Uniformly for  $b$  in a fixed compact set,

$$-\Lambda(\tau b, t) - L_\nu = \frac{b^d}{d^2} - L_\nu + o(1),$$

because  $\text{Li}_2(tb^d)/(d^2 t) = b^d/d^2 + \mathcal{O}(t)$ , while  $C_\zeta(\tau b) = o(1)$  and  $tD_\zeta(\tau b) = o(1)$ . The limiting polynomial has the  $d$  distinct roots of  $b^d = d^2 L_\nu$ . Rouché's theorem on disjoint small circles about those roots supplies one zero on each circle for all sufficiently small  $t$ . The preceding converse estimate shows that every solution tending to zero lies on one of these circles. Each circle contains one zero counted with multiplicity, so that zero is simple. For every nonzero  $t$  in the subsector, the holomorphic implicit-function theorem therefore makes it a locally holomorphic function of the chosen coordinate  $\tau = t^{1/d}$ . The circles are fixed and pairwise disjoint, and each contains exactly one zero; uniqueness makes these local functions agree on overlaps and patch to  $d$  sectorial solution germs. Conversely, any solution with  $a \rightarrow 0$  has

$$b^d = d^2 L_\nu + o(1), \quad b = a/\tau,$$

and hence eventually lies in exactly one of these circles. This proves the exact germ count, its exhaustivity, and both asymptotic relations.

Finally, exponentiating a lifted solution gives the raw equation. Conversely, on a continuous raw solution branch the quantity  $(\Lambda(a, t) - \ell)/(2\pi i)$  is integer-valued and continuous, hence is one fixed integer  $\nu$ . Thus the winding clusters are disjoint and exhaust all continuous raw branches tending to zero.  $\square$

**Warning 32.17** (Exterior normalization). The defining product does not converge for  $|q| > 1$ . There is therefore no canonical unnormalized value of  $(a; q)_\infty$ , and no canonical inverse at  $q = \infty$ . One may instead specify, for example,  $(a; q^{-1})_\infty$  for  $|q| > 1$ , or a  $q$ -gamma normalization with its explicit powers removed. Different choices can have different inverse asymptotics and must not be conflated.



## Chapter 33

# Gaussian and multinomial inversion

### 33.1 Polynomial structure and the positive real inverse

Fix integers  $0 < k < n$ , put  $b = n - k$ ,  $d = kb$ , and write

$$G_{n,k}(q) := \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

Although the quotient in [Section 29.3](#) is useful away from roots of unity, the polynomial itself is the primary object.

**Proposition 33.1** (Positive palindromic Gaussian polynomial). *The function  $G_{n,k}$  is a monic polynomial of degree  $d$ , with nonnegative integer coefficients, constant coefficient one, and*

$$G_{n,k}(q) = q^d G_{n,k}(q^{-1}) \quad (q \neq 0).$$

*Its coefficient of  $q$  is one.*

*Proof.* The finite  $q$ -binomial recurrence is

$$G_{n,k}(q) = G_{n-1,k}(q) + q^{n-k} G_{n-1,k-1}(q).$$

It follows directly from the product quotient: after putting the two terms over the denominator  $(q; q)_k (q; q)_{n-k}$ , the numerator becomes

$$(q; q)_{n-1} ((1 - q^{n-k}) + q^{n-k}(1 - q^k)) = (q; q)_{n-1} (1 - q^n).$$

Starting from  $G_{m,0} = G_{m,m} = 1$ , the recurrence proves inductively that all coefficients are non-negative integers. The same induction gives degree  $k(n - k)$ , monicity, and constant coefficient one.

For reciprocity, replace  $q$  by  $q^{-1}$  in

$$G_{n,k}(q) = \prod_{j=1}^k \frac{1 - q^{b+j}}{1 - q^j}.$$

Since  $1 - q^{-m} = -q^{-m}(1 - q^m)$ , the signs cancel and the accumulated power is

$$-\sum_{j=1}^k (b + j) + \sum_{j=1}^k j = -kb = -d.$$

Thus  $G_{n,k}(q^{-1}) = q^{-d} G_{n,k}(q)$ . Finally, the recurrence or the partition interpretation shows that the coefficient of  $q$  is one: there is exactly one partition of 1 in a nonempty  $k \times b$  rectangle.  $\square$

**Theorem 33.2** (Global positive-base inverse). *The restriction of  $G_{n,k}$  to  $[0, \infty)$  is a continuous strictly increasing bijection onto  $[1, \infty)$ . Consequently, for every real  $y > 1$  there is exactly one  $q > 0$  satisfying  $G_{n,k}(q) = y$ .*

*Proof.* Write  $G_{n,k}(q) = 1 + \sum_{r=1}^d p_r q^r$ . By [proposition 33.1](#),  $p_r \geq 0$  and  $p_1 = 1$ . Hence

$$G'_{n,k}(q) = \sum_{r=1}^d r p_r q^{r-1} \geq 1 \quad (q \geq 0).$$

The function is therefore strictly increasing, starts at  $G_{n,k}(0) = 1$ , and tends to infinity because it is monic of positive degree. Continuity and the intermediate value theorem give existence; strict increase gives uniqueness.  $\square$

### 33.2 The inverse at $q = 0$

Let  $p_{k,b}(r)$  denote the number of partitions of  $r$  whose Ferrers diagram fits in a  $k \times b$  rectangle. The recurrence in the preceding proof gives the standard generating identity

$$G_{n,k}(q) = \sum_{r=0}^d p_{k,b}(r) q^r.$$

In particular,  $p_{k,b}(0) = p_{k,b}(1) = 1$ . Put

$$c_2 = \mathbf{1}_{k \geq 2} + \mathbf{1}_{b \geq 2}, \quad c_3 = \mathbf{1}_{b \geq 3} + \mathbf{1}_{k \geq 2, b \geq 2} + \mathbf{1}_{k \geq 3}.$$

These are exactly  $p_{k,b}(2)$  and  $p_{k,b}(3)$ .

**Theorem 33.3** (Gaussian reversion at zero). *Let  $u = y - 1$ . The positive inverse from [Theorem 33.2](#) has the convergent local expansion*

$$q = u - c_2 u^2 + (2c_2^2 - c_3) u^3 + \mathcal{O}(u^4).$$

More generally,

$$[u^m]q(u) = \frac{1}{m} [z^{m-1}] \left( \frac{z}{G_{n,k}(z) - 1} \right)^m \quad (m \geq 1).$$

*Proof.* Set  $F(z) = G_{n,k}(z) - 1$ . Then  $F(0) = 0$  and  $F'(0) = 1$ , so the analytic inverse function theorem gives a convergent inverse near zero. The Lagrange–Bürmann formula from [Theorem 30.9](#) gives the coefficient identity. For the first terms, substitute

$$u = q + c_2 q^2 + c_3 q^3 + \mathcal{O}(q^4), \quad q = u + A u^2 + B u^3 + \mathcal{O}(u^4).$$

Comparison of  $u^2$  and  $u^3$  gives  $A = -c_2$  and  $B = 2c_2^2 - c_3$ .  $\square$

### 33.3 Complete expansion and inversion at $q = 1$

Let

$$S_m(N) := \sum_{j=1}^N j^m, \quad \Delta_m(n, k) := S_m(n) - S_m(k) - S_m(n - k).$$

**Lemma 33.4** (The elementary Bernoulli factor). *On  $|x| < 2\pi$ , let  $\ell(x)$  be the unique holomorphic logarithm of  $(1 - e^{-x})/x$  normalized by  $\ell(0) = 0$ . Then*

$$\ell(x) = -\frac{x}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} x^{2r}.$$

*Proof.* The generating function

$$\frac{x}{e^x - 1} = \sum_{m \geq 0} B_m \frac{x^m}{m!}$$

has nearest nonzero singularities at  $\pm 2\pi i$ . Therefore, for  $|x| < 2\pi$ ,

$$\frac{1}{e^x - 1} - \frac{1}{x} = -\frac{1}{2} + \sum_{r \geq 1} B_{2r} \frac{x^{2r-1}}{(2r)!}.$$

The quotient  $(1 - e^{-x})/x$ , extended by the value one at zero, is zero-free in the open disk because its first nonzero zeros are  $\pm 2\pi i$ . Hence its normalized holomorphic logarithm exists there. The left side above is its derivative. Termwise integration from 0 to  $x$  gives the formula, with zero integration constant by the normalization.  $\square$

**Theorem 33.5** (Complete Gaussian logarithmic expansion). *Let  $q = e^{-t}$ . For  $|t| < 2\pi/n$ , take the holomorphic logarithm of the normalized quotient whose value at  $t = 0$  is zero. Then*

$$\log \frac{G_{n,k}(e^{-t})}{\binom{n}{k}} = -\frac{d}{2}t + \sum_{r \geq 1} \frac{B_{2r} \Delta_{2r}(n, k)}{2r(2r)!} t^{2r}.$$

If

$$\lambda_1 = -\frac{d}{2}, \quad \lambda_{2r} = \frac{B_{2r}}{2r} \Delta_{2r}(n, k), \quad \lambda_{2r+1} = 0 \quad (r \geq 1),$$

and the complete exponential Bell polynomials  $\mathcal{B}_m$  are defined by

$$\exp \left( \sum_{j \geq 1} x_j \frac{z^j}{j!} \right) = \sum_{m \geq 0} \mathcal{B}_m(x_1, \dots, x_m) \frac{z^m}{m!},$$

then

$$G_{n,k}(e^{-t}) = \binom{n}{k} \sum_{m \geq 0} \mathcal{B}_m(\lambda_1, \dots, \lambda_m) \frac{t^m}{m!}.$$

*Proof.* Use

$$G_{n,k}(e^{-t}) = \prod_{j=1}^k \frac{1 - e^{-(b+j)t}}{1 - e^{-jt}}.$$

Apply lemma 33.4 to every numerator and denominator. The logarithms of the scale factors  $(b+j)/j$  sum to  $\log \binom{n}{k}$ . The linear difference is

$$\sum_{j=1}^k (b+j-j) = kb = d,$$

and the even differences are precisely  $\Delta_{2r}(n, k)$ . Every factor expansion converges when  $|(b+j)t| < 2\pi$ ; the sufficient common disk is  $|t| < 2\pi/n$ . Exponentiating the resulting series and invoking the defining generating identity for  $\mathcal{B}_m$  proves the second formula.  $\square$

**Corollary 33.6** (Literal Taylor coordinate  $h = q - 1$ ). *As  $h \rightarrow 0$ ,*

$$\frac{G_{n,k}(1+h)}{\binom{n}{k}} = 1 + \frac{d}{2}h + \frac{d(3d+n-5)}{24}h^2 + \frac{d(d-2)(d+n-3)}{48}h^3 + \mathcal{O}(h^4).$$

*Proof.* In [Theorem 33.5](#), substitute

$$t = -\log(1+h) = -h + \frac{h^2}{2} - \frac{h^3}{3} + \mathcal{O}(h^4),$$

use  $\Delta_2 = d(n+1)$ , and exponentiate through degree three. All discarded products have degree at least four. Straight collection gives the displayed coefficients.  $\square$

Put  $v = \log(y/\binom{n}{k})$ . The linear term in [Theorem 33.5](#) is nonzero.

**Corollary 33.7** (Ordinary Gaussian inverse at  $q = 1$ ). *On the local branch through  $q = 1$ ,*

$$t = -\frac{2v}{d} + \frac{\Delta_2}{3d^3}v^2 - \frac{\Delta_2^2}{9d^5}v^3 + \mathcal{O}(v^4), \quad q = e^{-t}.$$

*The coefficient of  $v^4$  is the first one involving  $\Delta_4$ .*

*Proof.* Write

$$v = c_1 t + c_2 t^2 + c_4 t^4 + \cdots, \quad c_1 = -d/2, \quad c_2 = \Delta_2/24.$$

There is no cubic term. Substituting  $t = Av + Bv^2 + Cv^3 + \cdots$  and comparing coefficients gives

$$A = c_1^{-1}, \quad B = -c_2 c_1^{-3}, \quad C = 2c_2^2 c_1^{-5},$$

which are the displayed values. Since  $c_4$  first contributes in degree four,  $\Delta_4$  cannot occur earlier.  $\square$

### 33.4 Centered palindromic inversion

The ordinary inverse above retains the deterministic reciprocity drift. Once that drift is removed,  $q = 1$  becomes a fold.

**Theorem 33.8** (Centered palindromic inverse). *Let*

$$P(q) = \sum_{j=0}^D c_j q^j$$

*be nonconstant, with  $c_j \geq 0$ ,  $c_j = c_{D-j}$ , and  $c_0 = c_D > 0$ . Define*

$$H_P(t) := e^{Dt/2} \frac{P(e^{-t})}{P(1)}.$$

*Then  $H_P$  is even,  $H_P(0) = 1$ , strictly increases on  $(0, \infty)$ , and tends to infinity there. For every  $h > 1$ , the equation  $H_P(t) = h$  therefore has exactly two real solutions  $t = \pm E_P(h)$ . At  $h = 1$  the two branches meet in a square-root fold.*

*Proof.* Let  $X$  take value  $j$  with probability  $c_j/P(1)$ , and set  $Z = X - D/2$ . Palindromy says  $Z$  and  $-Z$  have the same law. Hence

$$H_P(t) = \mathbb{E}(e^{-tZ}) = \mathbb{E}(\cosh(tZ)),$$

which is even and equals one at zero. For  $t > 0$ ,

$$H'_P(t) = \mathbb{E}(Z \sinh(tZ)) > 0.$$

Indeed, the integrand is nonnegative and is positive whenever  $Z \neq 0$ ; the endpoint masses  $c_0 = c_D > 0$  ensure that this happens with positive probability. Those same endpoints yield

$$H_P(t) \geq \frac{c_0}{P(1)} e^{Dt/2} \rightarrow \infty.$$

This proves the global branch count.

Finally,

$$H_P(t) = 1 + \frac{\mathbb{E}(Z^2)}{2}t^2 + \mathcal{O}(t^4),$$

and  $\mathbb{E}(Z^2) > 0$  by nondegeneracy. Thus

$$t = \pm \sqrt{\frac{2(h-1)}{\mathbb{E}(Z^2)}} (1 + \mathcal{O}(h-1)),$$

which is exactly a square-root fold. □

Let  $\kappa_{2r}$  be the even cumulants of  $Z$ , so that

$$u := \log H_P(t) = \frac{\kappa_2}{2!}t^2 + \frac{\kappa_4}{4!}t^4 + \frac{\kappa_6}{6!}t^6 + \dots$$

**Proposition 33.9** (Centered inverse through  $u^{5/2}$ ). *As  $u \downarrow 0$ ,*

$$t = \pm \sqrt{\frac{2u}{\kappa_2}} \left[ 1 - \frac{\kappa_4}{12\kappa_2^2}u + \left( \frac{7\kappa_4^2}{288\kappa_2^4} - \frac{\kappa_6}{180\kappa_2^3} \right) u^2 + \mathcal{O}(u^3) \right].$$

*Proof.* Put  $s = t^2$ . Then

$$u = As + Bs^2 + Cs^3 + \mathcal{O}(s^4), \quad A = \kappa_2/2, \quad B = \kappa_4/24, \quad C = \kappa_6/720.$$

Substitute

$$t = (u/A)^{1/2}(1 + \alpha u + \beta u^2 + \mathcal{O}(u^3)).$$

Comparison in  $u = At^2 + Bt^4 + Ct^6 + \dots$  gives

$$\alpha = -\frac{B}{2A^2}, \quad \beta = \frac{7B^2 - 4AC}{8A^4}.$$

Inserting  $A, B, C$  yields the stated coefficients. The two signs are the two branches supplied by [Theorem 33.8](#). □

For  $P = G_{n,k}$ , [Theorem 33.5](#) gives

$$\kappa_{2r} = \frac{B_{2r}}{2r} \Delta_{2r}(n, k), \quad \kappa_2 = \frac{d(n+1)}{12}.$$

Thus the centered Gaussian inverse is an explicit Bernoulli–Faulhaber Puiseux series.

### 33.5 Cyclotomic roots and the alternating base

**Theorem 33.10** (Square-free Gaussian covering). *The cyclotomic factorization is*

$$G_{n,k}(q) = \prod_{m=2}^n \Phi_m(q)^{e_m}, \quad e_m = \left\lfloor \frac{n}{m} \right\rfloor - \left\lfloor \frac{k}{m} \right\rfloor - \left\lfloor \frac{n-k}{m} \right\rfloor.$$

Every  $e_m$  is either zero or one. Consequently every complex zero of  $G_{n,k}$  is a simple root of unity, and the local inverse at a zero is an ordinary Taylor branch rather than a Puiseux branch of higher order.

*Proof.* The factorization  $1 - q^j = \prod_{m|j} \Phi_m(q)$  shows that the multiplicity of  $\Phi_m$  in  $(q; q)_N$  is  $\lfloor N/m \rfloor$ . Subtracting the denominator multiplicities proves the formula for  $e_m$ . Write  $n = k + b$ . The elementary carry identity

$$\left\lfloor \frac{k+b}{m} \right\rfloor - \left\lfloor \frac{k}{m} \right\rfloor - \left\lfloor \frac{b}{m} \right\rfloor \in \{0, 1\}$$

proves square-freeness. Distinct cyclotomic polynomials are coprime and each is separable over characteristic zero. Thus every zero is simple. The analytic inverse function theorem then supplies an ordinary local inverse.  $\square$

If  $\zeta$  is primitive of order  $m$  and  $e_m = 1$ , put

$$R_m(q) := \prod_{\substack{2 \leq r \leq n \\ r \neq m}} \Phi_r(q)^{e_r}.$$

Then

$$G'_{n,k}(\zeta) = \Phi'_m(\zeta) R_m(\zeta) \neq 0, \quad q(y) = \zeta + \frac{y}{\Phi'_m(\zeta) R_m(\zeta)} + \mathcal{O}(y^2).$$

**Proposition 33.11** (Value and zero derivative at  $q = -1$ ). *One has*

$$G_{n,k}(-1) = \begin{cases} 0, & n \text{ even and } k \text{ odd,} \\ \binom{\lfloor n/2 \rfloor}{\lfloor k/2 \rfloor}, & \text{otherwise.} \end{cases}$$

In the zero case, writing  $n = 2m$  and  $k = 2r + 1$ ,

$$G'_{2m,2r+1}(-1) = (m - r) \binom{m}{r}.$$

*Proof.* Separate even and odd exponents in the product quotient and pair every vanishing factor  $1 - q^{2s}$  with a factor  $1 - q^2$ . Since

$$\lim_{q \rightarrow -1} \frac{1 - q^{2s}}{1 - q^2} = s, \quad \lim_{q \rightarrow -1} (1 - q^{2s+1}) = 2,$$

the surviving factorial ratios reduce to the displayed binomial coefficient. When  $n$  is even and  $k$  odd, there is exactly one unpaired even numerator factor, so the value is zero.

In that case the quotient after removing the unique factor  $1 + q$  has limit

$$\frac{m!}{r!(m - r - 1)!} = (m - r) \binom{m}{r}.$$

Because  $1 + q$  has derivative one at  $-1$ , this limit is exactly  $G'_{2m,2r+1}(-1)$ .  $\square$

### 33.6 The reciprocal and large-target inverse

Write the small- $Q$  expansion

$$G_{n,k}(Q) = 1 + Q + c_2 Q^2 + c_3 Q^3 + \mathcal{O}(Q^4)$$

with the  $c_2, c_3$  of [Section 33.2](#), and choose a branch  $Y = y^{1/d}$ .

**Theorem 33.12** (Correct large-target Gaussian inverse). *On every branch for which  $Y \rightarrow \infty$ ,*

$$q = Y - \frac{1}{d} + \left( \frac{d-1}{2d^2} - \frac{c_2}{d} \right) Y^{-1} \\ + \left( -\frac{(d-1)(d-2)}{3d^3} + \frac{(d-2)c_2}{d^2} - \frac{c_3}{d} \right) Y^{-2} + \mathcal{O}(Y^{-3}).$$

*Proof.* Reciprocity gives

$$y = q^d G_{n,k}(q^{-1}).$$

Set  $x = Y^{-1}$  and

$$q = Y(1 + Ax + Bx^2 + Cx^3 + \mathcal{O}(x^4)).$$

Then

$$q^{-1} = x - Ax^2 + (A^2 - B)x^3 + \mathcal{O}(x^4),$$

and therefore

$$G_{n,k}(q^{-1}) = 1 + x + (c_2 - A)x^2 + (A^2 - B - 2Ac_2 + c_3)x^3 + \mathcal{O}(x^4).$$

Divide the equation by  $Y^d$  and take logarithms. Comparing the coefficients of  $x, x^2, x^3$  in

$$0 = d \log(1 + Ax + Bx^2 + Cx^3) + \log G_{n,k}(q^{-1})$$

gives successively

$$A = -\frac{1}{d}, \quad B = \frac{d-1}{2d^2} - \frac{c_2}{d},$$

and

$$C = -\frac{(d-1)(d-2)}{3d^3} + \frac{(d-2)c_2}{d^2} - \frac{c_3}{d}.$$

Multiplication by  $Y$  produces the displayed expansion and leaves a remainder of order  $Y^{-3}$ .  $\square$

*Remark 33.13.* Several source reports printed  $-(d-1)(2d-1)/(3d^3) + 2c_2(d-1)/d^2 - c_3/d$  at order  $Y^{-2}$ . The coefficient comparison above shows that expression is incorrect. The correct coefficient vanishes in the expected low-degree special cases and is the one retained here.

### 33.7 Multinomial extension

Throughout this section let  $s \geq 1$ , let  $n_1, \dots, n_s \in \mathbb{Z}_{\geq 0}$ , and put  $n = n_1 + \dots + n_s > 0$ . Set

$$M_{\mathbf{n}}(q) := \frac{(q; q)_n}{(q; q)_{n_1} \cdots (q; q)_{n_s}}, \quad D(\mathbf{n}) := \sum_{i < j} n_i n_j,$$

$$\Delta_m(\mathbf{n}) := S_m(n) - \sum_{i=1}^s S_m(n_i).$$

**Theorem 33.14** (Complete multinomial expansion at one). *For  $q = e^{-t}$  and  $|t| < 2\pi/n$ , take the holomorphic logarithm of the normalized quotient whose value at  $t = 0$  is zero. Then*

$$\log \frac{M_{\mathbf{n}}(e^{-t})}{\binom{n}{n_1, \dots, n_s}} = -\frac{D(\mathbf{n})}{2}t + \sum_{r \geq 1} \frac{B_{2r} \Delta_{2r}(\mathbf{n})}{2r(2r)!} t^{2r}.$$

*If  $D(\mathbf{n}) > 0$ , equivalently at least two of the  $n_i$  are positive, then*

$$e^{D(\mathbf{n})t/2} \frac{M_{\mathbf{n}}(e^{-t})}{\binom{n}{n_1, \dots, n_s}}$$

*is a nonconstant even function with a nondegenerate minimum at  $t = 0$ , and its inverse has the two Puiseux branches of [proposition 33.9](#). If  $D(\mathbf{n}) = 0$ , then  $M_{\mathbf{n}} \equiv 1$  and no local inverse exists.*

*Proof.* Apply lemma 33.4 to the numerator  $(q; q)_n$  and every denominator  $(q; q)_{n_i}$ . Since  $|jt| < 2\pi$  for every factor, the normalized factor logarithms sum to the holomorphic branch specified in the statement. Constants give the ordinary multinomial coefficient and the power sums give  $\Delta_{2r}(\mathbf{n})$ . The linear difference is

$$\frac{1}{2} \left( n^2 - \sum_i n_i^2 \right) = \sum_{i < j} n_i n_j = D(\mathbf{n}).$$

It remains to justify the hypotheses for centered inversion. Put  $N_i = n_1 + \cdots + n_i$ . The telescoping factorization

$$M_{\mathbf{n}}(q) = \prod_{i=2}^s \left[ \begin{matrix} N_i \\ n_i \end{matrix} \right]_q$$

proves polynomiality, with the empty product for  $s = 1$  understood as one. Boundary factors with  $n_i = 0$  or  $N_{i-1} = 0$  are one. By proposition 33.1, every nontrivial Gaussian factor has nonnegative coefficients, equal positive endpoints, and is palindromic; hence so does their product. Its degree is

$$\sum_{i=2}^s n_i N_{i-1} = \sum_{j < i} n_i n_j = D(\mathbf{n}).$$

Thus it is nonconstant exactly when  $D(\mathbf{n}) > 0$ , in which case reciprocity makes the centered normalization even and Theorem 33.8 and proposition 33.9 applies. If  $D(\mathbf{n}) = 0$ , at most one part is positive and the same factorization gives  $M_{\mathbf{n}} = 1$ .  $\square$

**Theorem 33.15** (Cyclotomic carry and exact ramification). *Under the standing hypotheses above, let  $m \geq 1$ , let  $\zeta$  be a primitive  $m$ -th root of unity, and write  $n_i = m\ell_i + r_i$  with  $0 \leq r_i < m$ . The multiplicity of  $\zeta$  as a zero of  $M_{\mathbf{n}}$  is*

$$e_m = \left\lfloor \frac{n}{m} \right\rfloor - \sum_{i=1}^s \left\lfloor \frac{n_i}{m} \right\rfloor = \left\lfloor \frac{r_1 + \cdots + r_s}{m} \right\rfloor.$$

Thus  $0 \leq e_m \leq s - 1$ . If  $e_m > 0$ , the inverse at target zero has exactly  $e_m$  local Puiseux determinations, counted with their choices of the  $e_m$ -th root.

*Proof.* As in Theorem 33.10, cyclotomic factorization of each factorial gives the first expression. Since

$$n = m \sum_i \ell_i + \sum_i r_i,$$

substitution gives the residue-carry formula. The residue sum lies between zero and  $s(m - 1)$ , proving  $e_m \leq s - 1$ .

Factor locally

$$M_{\mathbf{n}}(q) = (q - \zeta)^{e_m} U(q), \quad U(\zeta) \neq 0.$$

After choosing an analytic  $e_m$ -th root of  $U$  in a sufficiently small disk, the equation  $M_{\mathbf{n}}(q) = y$  is equivalent to

$$(q - \zeta) U(q)^{1/e_m} = \omega y^{1/e_m},$$

where  $\omega$  runs through the  $e_m$ -th roots of unity. The analytic inverse function theorem applies to the left side, whose derivative at  $\zeta$  is  $U(\zeta)^{1/e_m} \neq 0$ . This produces precisely the stated Puiseux determinations.  $\square$

For  $s = 2$ , the carry count is at most one, recovering the square-free Gaussian case. For many parts, the exact same cyclotomic mechanism produces higher ramification. Centered inversion at  $q = 1$  and cyclotomic inversion therefore represent genuinely different singular mechanisms even when they occur in the same polynomial family.



## Chapter 34

# Inverse $q$ -special functions

### 34.1 $q$ -numbers

Fix a logarithm value  $\ell_q$  of  $q \in \mathbb{C} \setminus \{0, 1\}$ , put  $q^x = e^{x\ell_q}$ , and define

$$[x]_q := \frac{1 - q^x}{1 - q}.$$

**Proposition 34.1** (Exact inverse in the exponent). *Let  $w = 1 - (1 - q)y \neq 0$ , and fix one logarithm value  $\ell_w$  of  $w$ . Then, for every  $k \in \mathbb{Z}$ ,*

$$x_k = \frac{\ell_w + 2\pi i k}{\ell_q}$$

*solves  $[x_k]_q = y$ , and every solution is obtained in this way.*

*Proof.* The equation  $[x]_q = y$  is equivalent to

$$e^{x\ell_q} = w.$$

The complete preimage of  $w$  under the exponential is  $\ell_w + 2\pi i\mathbb{Z}$ . Since  $q \neq 1$ , one has  $\ell_q \neq 0$ , so division by  $\ell_q$  gives precisely the displayed solutions. Conversely, exponentiating any one of them recovers the equation.  $\square$

For an integer  $m \geq 2$ ,

$$[m]_q = 1 + q + \cdots + q^{m-1},$$

so inversion in the base is an algebraic covering.

**Proposition 34.2** (Distinguished base jets of the  $q$ -number). *Fix an integer  $m \geq 2$ , and let  $u = y - 1$ . The inverse branch through  $q = 0$  satisfies*

$$\begin{aligned} m = 2 : \quad q &= u, & \text{exactly,} \\ m = 3 : \quad q &= u - u^2 + 2u^3 + \mathcal{O}(u^4), \\ m \geq 4 : \quad q &= u - u^2 + u^3 + \mathcal{O}(u^4). \end{aligned}$$

For a fixed complex  $x$  and  $q = e^{-t} \rightarrow 1$ ,

$$[x]_{e^{-t}} = x + \frac{x(1-x)}{2}t + \frac{x(2x^2-3x+1)}{12}t^2 - \frac{x^2(x-1)^2}{24}t^3 + \mathcal{O}(t^4).$$

If  $x \notin \{0, 1\}$ , its base inverse is

$$t = \frac{2(y-x)}{x(1-x)} + \frac{2(2x-1)}{3x^2(x-1)^2}(y-x)^2 + \mathcal{O}((y-x)^3).$$

*Proof.* At zero, the forward series begins  $u = q$  for  $m = 2$ ,  $u = q + q^2$  for  $m = 3$ , and  $u = q + q^2 + q^3 + \mathcal{O}(q^4)$  for  $m \geq 4$ . Direct substitution of  $q = u + Au^2 + Bu^3 + \dots$  gives the three formulas.

For  $q = e^{-t}$ , divide the Taylor series of  $1 - e^{-xt}$  by that of  $1 - e^{-t}$ , or equivalently use

$$[x]_{e^{-t}} = x \exp \left( -\frac{x-1}{2}t + \sum_{r \geq 1} \frac{B_{2r}(x^{2r}-1)}{2r(2r)!} t^{2r} \right).$$

Expansion through degree three gives the stated forward jet. Reversion of  $\delta := y - x = at + bt^2 + \dots$ , where

$$a = \frac{x(1-x)}{2}, \quad b = \frac{x(x-1)(2x-1)}{12},$$

gives  $t = \delta/a - b\delta^2/a^3 + \dots$ , which is the displayed inverse. For  $x = 0$  or  $1$ , the forward function is identically  $0$  or  $1$ , so base recovery is impossible rather than singular of finite order.  $\square$

**Proposition 34.3** (Alternating and reciprocal  $q$ -number branches). *For  $r \geq 1$ ,*

$$[2r]_{-1} = 0, \quad \frac{\partial}{\partial q} [2r]_q \Big|_{q=-1} = r,$$

and

$$[2r+1]_{-1} = 1, \quad \frac{\partial}{\partial q} [2r+1]_q \Big|_{q=-1} = -r.$$

For every integer  $m \geq 2$ , at infinity,

$$[m]_q = q^{m-1}(1 + q^{-1} + \dots + q^{-(m-1)}),$$

so the inverse at a large target has  $m - 1$  Puiseux determinations, indexed by the choices of  $y^{1/(m-1)}$ .

*Proof.* Evaluate  $\sum_{j=0}^{m-1} q^j$  and its derivative  $\sum_{j=1}^{m-1} jq^{j-1}$  at  $-1$ , pairing consecutive terms. The reciprocal identity is obtained by factoring out  $q^{m-1}$ . Its parenthesized factor tends to one, so the finite-ramification normal form of [Theorem 30.15](#) gives exactly  $m - 1$  inverse determinations.  $\square$

## 34.2 $q$ -factorials and their exact ramification

For  $n \in \mathbb{N}_0$ , put

$$[n]_q! := \prod_{j=1}^n [j]_q.$$

**Proposition 34.4** (Zero-base factorial inverse). *For  $n \geq 2$ ,*

$$[n]_q! = 1 + (n-1)q + \frac{(n-2)(n+1)}{2}q^2 + \mathcal{O}(q^3).$$

Consequently, with  $u = y - 1$ ,

$$q = \frac{u}{n-1} - \frac{(n-2)(n+1)}{2(n-1)^3}u^2 + \mathcal{O}(u^3).$$

*Proof.* In the product

$$[n]_q! = \prod_{j=2}^n (1 + q + \dots + q^{j-1}),$$

the coefficient of  $q$  is  $n - 1$ . A term of degree two is obtained either by choosing  $q$  from two distinct factors, in  $\binom{n-1}{2}$  ways, or by choosing  $q^2$  from one of the  $n - 2$  factors with  $j \geq 3$ . Their sum is  $(n-2)(n+1)/2$ . Reversion of  $u = (n-1)q + Aq^2 + \dots$  proves the inverse formula.  $\square$

**Theorem 34.5** (Exact root-of-unity ramification of  $q$ -factorials). *For every  $n \geq 0$ , let  $\zeta \neq 1$  be a primitive  $d$ -th root of unity. Then*

$$\text{ord}_{q=\zeta}[n]_q! = \left\lfloor \frac{n}{d} \right\rfloor.$$

*If  $n \geq 2$ , put  $M = \lfloor n/2 \rfloor \geq 1$ . Then*

$$[n]_q! = M!(q+1)^M(1 + \mathcal{O}(q+1)) \quad (q \rightarrow -1),$$

*and the inverse at target zero has the  $M$  branches*

$$q = -1 + \omega_M^j \left( \frac{y}{M!} \right)^{1/M} + \mathcal{O}(y^{2/M}), \quad 0 \leq j < M.$$

*For  $n = 0, 1$ , one has  $[n]_q! \equiv 1$ , so there is no inverse at target zero.*

*Proof.* Because  $\zeta \neq 1$ , the denominator in  $[j]_q = (1 - q^j)/(1 - q)$  is nonzero at  $\zeta$ . The numerator has a simple zero there exactly when  $d \mid j$ . Exactly  $\lfloor n/d \rfloor$  factors therefore vanish, each simply.

For  $d = 2$  and  $n \geq 2$ , the even factors satisfy

$$[2r]_q = r(q+1) + \mathcal{O}((q+1)^2),$$

while the odd factors tend to one. Multiplying  $r = 1, \dots, M$  gives the coefficient  $M!$ . Applying [Theorem 30.15](#) to the resulting exact zero order gives the Puiseux branches and their remainder.  $\square$

For  $n \geq 2$ , write  $N = \binom{n}{2} > 0$ . One also has

$$[n]_q! = q^N \left( 1 + \frac{n-1}{q} + \mathcal{O}(q^{-2}) \right),$$

and the same logarithmic coefficient comparison as in [Theorem 33.12](#) yields, on each choice  $Y = y^{1/N} \rightarrow \infty$ ,

$$q = Y - \frac{n-1}{N} + \mathcal{O}(Y^{-1}) = Y - \frac{2}{n} + \mathcal{O}(Y^{-1}).$$

For  $n = 0, 1$ , the forward function is constant and has no large-target inverse.

### 34.3 The $q$ -gamma function

For  $0 < q < 1$  and  $x > 0$ , define

$$\Gamma_q(x) = (1-q)^{1-x} \frac{(q; q)_\infty}{(q^x; q)_\infty}.$$

It satisfies

$$\Gamma_q(x+1) = [x]_q \Gamma_q(x), \quad \Gamma_q(1) = \Gamma_q(2) = 1.$$

**Proposition 34.6** (Exact generalized expansion at  $q = 0$ ). *For  $0 < q < 1$  and  $x > 0$ ,*

$$\log \Gamma_q(x) = \sum_{m \geq 1} \frac{q^{mx}}{m(1-q^m)} - \sum_{m \geq 1} \frac{q^m}{m(1-q^m)} - (1-x) \sum_{m \geq 1} \frac{q^m}{m}.$$

*All three series converge absolutely. Expanding  $(1-q^m)^{-1} = \sum_{r \geq 0} q^{mr}$  gives a convergent generalized power series with exponents in*

$$\{m(x+r) : m \geq 1, r \geq 0\} \cup \mathbb{N}_0.$$

For rational  $x$  it is a Puiseux series; for irrational  $x$ , its natural ordering is a generalized-power ordering rather than an ordinary Taylor ordering. Before coincident exponents are collected, the  $(m, r) = (1, 0)$  term of the first double series contributes  $q^x$ , while the two integer-power series contribute  $-(2 - x)q$  at exponent one. These are bookkeeping contributions, not necessarily the first two terms in the ordered generalized-power expansion. Whenever exponents coincide, all contributions at that exponent must be collected.

*Proof.* Take logarithms in the product definition and use the absolutely convergent Lambert identity

$$\log(a; q)_\infty = - \sum_{m \geq 1} \frac{a^m}{m(1 - q^m)}$$

from lemma 32.2, together with  $\log(1 - q) = - \sum_{m \geq 1} q^m/m$ . Absolute convergence permits expansion of each geometric denominator and rearrangement. The displayed exponent set is then immediate. The  $(m, r) = (1, 0)$  term and the integer exponent-one terms give the two stated uncollected contributions; collecting all terms with a common exponent is essential when collisions occur.  $\square$

The logarithmic derivative and its derivative are

$$\psi_q(x) = -\log(1 - q) + (\log q) \sum_{j \geq 0} \frac{q^{x+j}}{1 - q^{x+j}},$$

$$\psi'_q(x) = (\log q)^2 \sum_{j \geq 0} \frac{q^{x+j}}{(1 - q^{x+j})^2} > 0.$$

Termwise differentiation is justified locally uniformly for  $x > 0$ .

**Theorem 34.7** (Unique  $q$ -gamma minimum and two real inverse branches). *For every  $0 < q < 1$ , there is a unique  $x_q > 0$  with  $\psi_q(x_q) = 0$ . The function  $\Gamma_q$  strictly decreases on  $(0, x_q)$ , strictly increases on  $(x_q, \infty)$ , and tends to infinity at both endpoints. Thus each target strictly above  $\Gamma_q(x_q)$  has exactly two positive argument inverses, the minimum has one double inverse point, and smaller targets have none. The two branches meet in a square-root fold.*

*Proof.* The formula above makes  $\psi_q$  strictly increasing. As  $x \downarrow 0$ , the  $j = 0$  summand is asymptotic to

$$(\log q) \frac{1}{-x \log q} = -\frac{1}{x},$$

so  $\psi_q(x) \rightarrow -\infty$ . As  $x \rightarrow \infty$ , the series tends to zero and  $\psi_q(x) \rightarrow -\log(1 - q) > 0$ . Hence it has one zero.

Since  $\Gamma'_q/\Gamma_q = \psi_q$ , the derivative has the asserted signs. The factor  $(q^x; q)_\infty$  has a simple zero at  $x = 0$ , so  $\Gamma_q(x) \rightarrow \infty$  there. As  $x \rightarrow \infty$ , the denominator tends to one while  $(1 - q)^{1-x} \rightarrow \infty$ . The branch count follows by continuity and strict monotonicity. At the minimum,

$$(\log \Gamma_q)''(x_q) = \psi'_q(x_q) > 0,$$

so Taylor's theorem and Theorem 30.15 give the square-root fold.  $\square$

**Theorem 34.8** (All-order  $q \rightarrow 1$  expansion of  $\Gamma_q$ ). *Let  $q = e^{-t}$ , with  $t \rightarrow 0$  in a closed subsector of  $|\arg t| < \pi/2$ . Uniformly for  $x$  in compact subsets of  $\Re x > 0$ ,*

$$\log \Gamma_{e^{-t}}(x) \sim \log \Gamma(x) + c_1(x)t + \sum_{r \geq 1} c_{2r}(x)t^{2r},$$

where

$$c_1(x) = \frac{(x-1)(2-x)}{4}$$

and

$$c_{2r}(x) = \frac{B_{2r}}{2r(2r)!} \left( 1 - x + \frac{B_{2r+1}(x)}{2r+1} \right).$$

No odd algebraic power beyond  $t^1$  occurs.

*Proof.* Insert the all-order coalescing expansion of  $(q^x; q)_\infty$  from [Theorem 32.9](#) into

$$\log \Gamma_q(x) = (1-x) \log(1-q) + \log(q; q)_\infty - \log(q^x; q)_\infty.$$

Use the  $x = 1$  specialization for  $(q; q)_\infty$  and

$$\log \frac{1 - e^{-t}}{t} = -\frac{t}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} t^{2r}.$$

The  $t^{-1}$ , logarithmic, and constant normalization terms cancel, leaving  $\log \Gamma(x)$ . The linear coefficient simplifies to

$$\frac{1}{24} - \frac{B_2(x)}{4} - \frac{1-x}{2} = \frac{(x-1)(2-x)}{4}.$$

At every even order, the  $x = 1$  Bernoulli-polynomial term vanishes and the remaining two terms combine to  $c_{2r}(x)$ . The expansions invoked above are locally uniform with uniform differentiated remainders, so subtraction preserves the asserted uniform asymptotic meaning.  $\square$

In particular,

$$c_2(x) = \frac{(x-2)(x-1)(2x+3)}{144},$$

and, for fixed  $x \notin \{1, 2\}$ , if

$$v = \log \frac{y}{\Gamma(x)},$$

ordinary reversion gives

$$t = \frac{v}{c_1(x)} - \frac{c_2(x)}{c_1(x)^3} v^2 + \frac{2c_2(x)^2}{c_1(x)^5} v^3 + \mathcal{O}(v^4), \quad q = e^{-t}.$$

At  $x = 1, 2$ ,  $\Gamma_q(x) \equiv 1$ , so there is no base inverse.

### 34.4 $q$ -digamma inversion

**Proposition 34.9** (Global argument inverse and the exceptional reciprocal fold). *For fixed  $0 < q < 1$ , the function  $\psi_q$  is a strictly increasing bijection*

$$(0, \infty) \longrightarrow (-\infty, -\log(1-q)).$$

For  $q = e^{-t}$ ,

$$\psi_{e^{-t}}(x) \sim \psi(x) + \frac{3-2x}{4}t + \sum_{r \geq 1} c'_{2r}(x)t^{2r}.$$

At fixed  $x \neq 3/2$ , base inversion at  $q = 1$  is ordinary. At  $x = 3/2$ ,

$$\psi_{e^{-t}}(3/2) - \psi(3/2) = -\frac{t^2}{288} + \mathcal{O}(t^4),$$

so, within the original domain  $0 < q < 1$ ,  $q = 1$  is a one-sided quadratic endpoint. More precisely, if

$$\Delta = \psi_{e^{-t}}(3/2) - \psi(3/2),$$

then the unique local inverse with  $t \downarrow 0$  is

$$t = \sqrt{-288\Delta} (1 + \mathcal{O}(\Delta)), \quad \Delta \uparrow 0, \quad \Delta < 0.$$

There is also a natural genuine two-sided statement. Extend the standard normalization to  $q > 1$  by

$$\Gamma_q(x) := q^{(x-1)(x-2)/2} \Gamma_{1/q}(x), \quad \psi_q(x) := \partial_x \log \Gamma_q(x),$$

and use the classical functions at  $q = 1$ . Then

$$\psi_q(x) = \psi_{1/q}(x) + \left(x - \frac{3}{2}\right) \log q.$$

In particular,  $t \mapsto \psi_{e^{-t}}(3/2)$  is reciprocal-even, is  $C^2$  at  $t = 0$ , and has a genuine two-sided quadratic fold there. Its branches are  $t = \pm \sqrt{-288\Delta}(1 + \mathcal{O}(\Delta))$ , equivalently  $q = e^{\mp \sqrt{-288\Delta}(1 + \mathcal{O}(\Delta))}$ .

*Proof.* Strict increase and the two endpoint limits were proved in [Theorem 34.7](#); these give the bijection. Differentiate [Theorem 34.8](#) termwise in  $x$ , which is permitted by its compact local uniformity. The linear coefficient  $(3 - 2x)/4$  vanishes only at  $3/2$ . Direct differentiation of

$$c_2(x) = \frac{(x-2)(x-1)(2x+3)}{144}$$

gives  $c'_2(3/2) = -1/288$ . The inverse function theorem handles the nonexceptional points. The differentiated remainder in [Theorem 34.8](#) gives, at the exceptional point,

$$\partial_t \psi_{e^{-t}}(3/2) = -\frac{t}{144} + \mathcal{O}(t^3) < 0$$

for all sufficiently small  $t > 0$ . Hence the physical one-sided inverse exists and is locally unique. Factoring the forward relation as  $\Delta = -t^2(1 + \mathcal{O}(t^2))/288$  and taking the positive square root gives its displayed expansion.

The twice-differentiated remainder likewise gives

$$\partial_t^2 \psi_{e^{-t}}(3/2) = -\frac{1}{144} + \mathcal{O}(t^2).$$

For the reciprocal extension, logarithmic differentiation in  $x$  gives the stated identity for  $\psi_q$ . At  $x = 3/2$  its correction term vanishes, so replacing  $q$  by  $1/q$ , or  $t$  by  $-t$ , changes nothing. The one-sided expansion and its differentiated remainder patch with this exact reflection to a  $C^2$  function at zero, with second derivative  $-1/144$ . The reflected positive and negative solutions are therefore the asserted two branches. This real argument requires no holomorphic continuation across the boundary  $|q| = 1$ .  $\square$

### 34.5 $q$ -beta coordinate and base inverses

For  $x, y > 0$  and initially  $0 < q < 1$ , define

$$B_q(x, y) := \frac{\Gamma_q(x)\Gamma_q(y)}{\Gamma_q(x+y)}.$$

**Theorem 34.10** (Coordinate monotonicity and base type change). *For every  $0 < q < 1$ ,  $B_q(x, y)$  is strictly decreasing in either positive coordinate while the other is fixed. Hence each coordinate inverse is unique on its range.*

As  $q = e^{-t} \rightarrow 1$ ,

$$\log \frac{B_{e^{-t}}(x, y)}{B(x, y)} = b_1(x, y)t + b_2(x, y)t^2 + \mathcal{O}(t^4),$$

where

$$b_1(x, y) = \frac{xy - 1}{2}, \quad b_2(x, y) = -\frac{x^2y + xy^2 - xy - 1}{24}.$$

The base inverse is ordinary when  $xy \neq 1$ . On  $xy = 1$ ,

$$b_2(x, 1/x) = -\frac{(x-1)^2}{24x};$$

For  $x \neq 1$ , this is a one-sided quadratic endpoint in the original domain. If

$$\Delta = \log \frac{B_{e^{-t}}(x, 1/x)}{B(x, 1/x)},$$

then its unique local inverse with  $t \downarrow 0$  is

$$t = \sqrt{-\frac{24x}{(x-1)^2} \Delta} (1 + \mathcal{O}(\Delta)), \quad \Delta \uparrow 0, \quad \Delta < 0.$$

Under the reciprocal extension of  $\Gamma_q$  from [proposition 34.9](#),

$$B_q(x, y) = q^{1-xy} B_{1/q}(x, y).$$

Consequently, on  $xy = 1$  the extended beta function is exactly reciprocal-even and the one-sided endpoint becomes a genuine two-sided quadratic fold. At  $(x, y) = (1, 1)$ , the identity  $B_q(1, 1) \equiv 1$  makes the base completely unidentifiable.

*Proof.* The logarithmic coordinate derivatives are

$$\partial_x \log B_q(x, y) = \psi_q(x) - \psi_q(x+y) < 0,$$

$$\partial_y \log B_q(x, y) = \psi_q(y) - \psi_q(x+y) < 0,$$

because  $\psi_q$  is strictly increasing. This proves strict coordinate monotonicity and uniqueness of each coordinate inverse.

Subtract the  $x + y$  instance of [Theorem 34.8](#) from the sum of its  $x$  and  $y$  instances. Elementary expansion of  $c_1$  and  $c_2$  gives the displayed  $b_1, b_2$ ; no cubic term exists because the  $q$ -gamma expansion has no odd term beyond the linear one. The inverse function theorem applies when  $b_1 \neq 0$ . On  $xy = 1$ , substitution gives the stated negative  $b_2$ , nonzero unless  $x = 1$ . Uniform differentiated remainders give

$$\partial_t \Delta = -\frac{(x-1)^2}{12x} t + \mathcal{O}(t^3) < 0$$

for sufficiently small  $t > 0$ . Thus the one-sided inverse exists and is unique, and taking the positive square root in  $\Delta = -(x-1)^2 t^2 / (24x) + \mathcal{O}(t^4)$  gives its displayed expansion. The twice-differentiated remainder is

$$\partial_t^2 \Delta = -\frac{(x-1)^2}{12x} + \mathcal{O}(t^2),$$

which will control the regularity at the reflected endpoint.

For  $q > 1$ , substitution of the reciprocal  $q$ -gamma normalization into the definition of  $B_q$  gives an exponent

$$\frac{(x-1)(x-2) + (y-1)(y-2) - (x+y-1)(x+y-2)}{2} = 1 - xy,$$

which proves the reciprocal identity. On  $xy = 1$  it gives exact evenness under  $t \mapsto -t$ ; the differentiated one-sided remainder patches to a  $C^2$  function with nonzero second derivative, and reflection supplies the second branch. At  $(1, 1)$ ,  $\Gamma_q(1) = \Gamma_q(2) = 1$  proves the exact constant identity.  $\square$

**Warning 34.11.** The claim that  $q$ -beta is always strictly decreasing in its base is false:  $B_q(1, 1) = 1$  for every base. The theorem above is the corrected statement. It proves strictness in the two shape coordinates and gives the exact base-inverse type change.

## 34.6 Euler $q$ -exponentials

It is useful to separate the product variable from the classical-limit scaling:

$$\mathbf{e}_q(z) := \frac{1}{(z; q)_\infty}, \quad \mathbf{E}_q(z) := (-z; q)_\infty,$$

$$e_q(x) := \mathbf{e}_q((1-q)x), \quad E_q(x) := \mathbf{E}_q((1-q)x).$$

**Proposition 34.12** (Exact inverse reduction and  $q \rightarrow 1$  generator). *Let  $|q| < 1$ . If  $A_q$  is a selected argument-inverse branch of  $a \mapsto (a; q)_\infty$ , then*

$$\mathbf{e}_q^{-1}(y) = A_q(y^{-1}), \quad \mathbf{E}_q^{-1}(y) = -A_q(y).$$

Writing  $h = 1 - q$ , suppose additionally that  $|hx| < 1$ , and choose the logarithms determined by their power series at  $x = 0$ . Then the scaled functions satisfy the absolutely convergent identities

$$\log e_q(x) = \sum_{m \geq 1} \frac{h^m x^m}{m(1 - (1-h)^m)},$$

$$\log E_q(x) = \sum_{m \geq 1} \frac{(-1)^{m+1} h^m x^m}{m(1 - (1-h)^m)}.$$

For every fixed  $x$ , the condition  $|hx| < 1$  holds for all sufficiently small positive  $h$ . In the resulting  $h \downarrow 0$  asymptotic expansion, only finitely many indices  $m$  contribute at each fixed order.

*Proof.* The inverse formulas are immediate after rewriting the two target equations as  $(z; q)_\infty = y^{-1}$  and  $(-z; q)_\infty = y$ , with the selected branch data retained. Apply [lemma 32.2](#) with  $a = \pm hx$  and  $q = 1 - h$ , remembering the minus sign introduced by the reciprocal in  $e_q$ . Since the  $m$ -th summand begins at order  $h^{m-1}$  after the denominator is expanded, only finitely many  $m$  can affect any prescribed coefficient. For completeness, fix  $K \geq 0$ , take real  $0 < h < 1$ , and shrink  $h$  until  $h|x| \leq 1/2$ . Since  $1 - (1-h)^m \geq h$ , the tail satisfies

$$\sum_{m \geq K+2} \left| \frac{h^m x^m}{m(1 - (1-h)^m)} \right| \leq \frac{1}{h} \sum_{m \geq K+2} \frac{(h|x|)^m}{m} = \mathcal{O}(h^{K+1}).$$

The alternating numerator obeys the same bound. The finitely many terms with  $m \leq K+1$  may be expanded individually, proving the asserted coefficientwise asymptotic statement with a genuine remainder estimate.  $\square$



For example,

$$\begin{aligned}\log e_q(x) &= x + \frac{x^2}{4}h + \left(\frac{x^2}{8} + \frac{x^3}{9}\right)h^2 + \mathcal{O}(h^3), \\ \log E_q(x) &= x - \frac{x^2}{4}h + \left(-\frac{x^2}{8} + \frac{x^3}{9}\right)h^2 + \mathcal{O}(h^3).\end{aligned}$$

Bell exponentiation gives the functions themselves, while  $e_q(z)E_q(-z) = 1$  is exact.

### 34.7 Basic hypergeometric argument inversion

**Theorem 34.13** (Universal local series-variable inverse). *Let a basic hypergeometric function, after all removable parameter cancellations, have a convergent local series*

$$F(z; \mathbf{a}, \mathbf{b}, q) = 1 + c_1 z + c_2 z^2 + c_3 z^3 + \cdots, \quad c_1 \neq 0.$$

Then its argument inverse at  $y = 1$  is

$$z = \frac{y-1}{c_1} - \frac{c_2}{c_1^3}(y-1)^2 + \left(\frac{2c_2^2}{c_1^5} - \frac{c_3}{c_1^4}\right)(y-1)^3 + \mathcal{O}((y-1)^4).$$

All higher coefficients are given by the analytic Lagrange–Bürmann formula.

*Proof.* The condition  $c_1 \neq 0$  gives a local analytic inverse. Put  $u = y - 1$  and

$$z = Au + Bu^2 + Cu^3 + \mathcal{O}(u^4).$$

Substitution in  $u = c_1 z + c_2 z^2 + c_3 z^3 + \cdots$  yields

$$c_1 A = 1, \quad c_1 B + c_2 A^2 = 0, \quad c_1 C + 2c_2 AB + c_3 A^3 = 0.$$

Solving gives the displayed coefficients. The all-order statement is [Theorem 30.9](#).  $\square$

If a numerator parameter reaches  $q^{-N}$ , the series truncates and the argument inverse becomes an algebraic covering. If a denominator parameter approaches the same truncation locus, common finite  $q$ -Pochhammer factors must be cancelled before applying the theorem; otherwise a removable singularity is misclassified as a branch point.

### 34.8 Multiplicative theta products

For  $0 < |q| < 1$ , let

$$\vartheta(z; q) := (z; q)_\infty (q/z; q)_\infty (q; q)_\infty, \quad z \in \mathbb{C}^\times.$$

**Proposition 34.14** (Simple theta zero atlas). *The zeros of  $z \mapsto \vartheta(z; q)$  are exactly  $z = q^m$ ,  $m \in \mathbb{Z}$ , and every zero is simple. Hence the inverse at target zero is ordinary on every sheet:*

$$z = q^m + \frac{y}{\partial_z \vartheta(q^m; q)} + \mathcal{O}(y^2).$$

Moreover,

$$\vartheta(qz; q) = -z^{-1} \vartheta(z; q),$$

so one zero derivative determines all the others.

*Proof.* The factor  $(z; q)_\infty$  has one simple vanishing factor precisely at  $z = q^{-j}$ ,  $j \geq 0$ . The factor  $(q/z; q)_\infty$  has one simple vanishing factor precisely at  $z = q^{j+1}$ ,  $j \geq 0$ . Because  $|q| < 1$ , the two lists do not overlap, and every other factor is nonzero. Their union is  $\{q^m : m \in \mathbb{Z}\}$ , proving the zero and multiplicity claims. The local inverse formula is the analytic inverse function theorem.

Finally,

$$(qz; q)_\infty = \frac{(z; q)_\infty}{1 - z}, \quad (1/z; q)_\infty = (1 - z^{-1})(q/z; q)_\infty.$$

Multiplying and using  $(1 - z^{-1})/(1 - z) = -z^{-1}$  proves quasi-periodicity. Differentiating it at a zero transports the derivative normalization.  $\square$

## Chapter 35

# Cyclotomic and Fabius–Rvachev recovery

### 35.1 The boundary and probability regimes are different

Two uses of a geometric parameter occur in this volume. In [Section 32.8](#), the base  $q$  approaches the unit circle and the Lambert series develops resonant residue classes. In the present part,  $0 < q < 1$  is the contraction ratio of a random series. The Rvachev member is the perfectly regular interior value  $q = 1/2$ . Thus the radial inverse [Theorem 32.14](#) and the finite cyclotomic analysis of [Section 33.5](#) remain available when a product formula reaches the boundary, but they should not be conflated with statistical recovery of the geometric contraction parameter.

The law, support, CDF refinement, smooth density, and half-base identification are represented in Lean by the modules `FabiusFunction.GeometricUniformLaw`, `FabiusFunction.GeometricUniformCDF`, and `FabiusFunction.ProbabilityRepresentation`. In particular, `geometricUniformCDF_one_half_eq_fabiusReal` is the exact formal bridge at  $q = 1/2$ . The phase-bearing sinc-product bridge is formalized separately in `FabiusFunction.GeometricSincCharacteristicFunction`, as `charFun_geometricUniformDistribution_eq_phase_mul_geometricSincProduct`. The arbitrary-power Fourier decay in [Theorem 27.7](#), its parameter-derivative strengthening in [Equation \(35.23\)](#), and the cumulant inversions later in this chapter are complete human proofs at the frontier level; they are not claimed here as existing Lean declarations.

### 35.2 The geometric uniform law and its sinc product

Continue with the independent uniform variables  $U_j$ , the random variable  $X_q$ , its CDF  $F_q$ , density  $f_q$ , and characteristic function  $\varphi_q$  from [Chapter 27](#). Put

$$\varepsilon_j := 2U_j - 1, \quad Z_q := 2X_q - 1 = (1 - q) \sum_{j=0}^{\infty} q^j \varepsilon_j, \quad u_q(z) := \frac{1}{2} f_q\left(\frac{z+1}{2}\right). \quad (35.1)$$

Thus  $Z_q$  has density  $u_q$  and support  $[-1, 1]$ .

**Corollary 35.1** (The symmetric normalization and its refinement). *For real  $t$ ,*

$$\psi_q(t) := \mathbb{E} e^{itZ_q} = e^{-it} \varphi_q(2t) = \prod_{j=0}^{\infty} \text{sinc}_{\text{rad}}((1 - q)q^j t), \quad (35.2)$$

*The function  $\psi_q$  has the log-Gaussian, hence arbitrary-power, decay of [Theorem 27.7](#); the density  $u_q$  is  $C^\infty$  and compactly supported. If  $Z'_q$  is an independent copy, then  $Z_q \stackrel{\text{law}}{=} (1 - q)\varepsilon_0 + qZ'_q$ , and*

consequently

$$u_q(x) = \frac{1}{2(1-q)} \int_{(x-(1-q))/q}^{(x+(1-q))/q} u_q(s) \, ds, \quad (35.3)$$

$$u'_q(x) = \frac{u_q((x+(1-q))/q) - u_q((x-(1-q))/q)}{2q(1-q)}. \quad (35.4)$$

*Proof.* The characteristic-function identity is the affine change  $Z_q = 2X_q - 1$  in [Theorem 27.2](#); the phase cancels exactly. Decay and smoothness follow from [Theorems 27.7](#) and [27.8](#). Splitting off the first centered coordinate gives the displayed fixed point. Conditional on  $Z'_q = s$ , the variable  $(1-q)\varepsilon_0 + qs$  is uniform on  $[qs - (1-q), qs + (1-q)]$ ; averaging its density gives [Equation \(35.3\)](#). Differentiating that identity by the fundamental theorem of calculus gives [Equation \(35.4\)](#).  $\square$

### 35.3 The exact Fabius and Rvachev member

**Definition 35.2** (Normalized bounded Fabius function). A normalized bounded Fabius function is a continuously differentiable map  $G : \mathbb{R} \rightarrow [0, 1]$  such that

$$G(x) = 0 \quad (x \leq 0), \quad G(x) = 1 \quad (x \geq 1), \quad G(1-x) = 1 - G(x),$$

and

$$G'(x) = 2G(2x) \quad (0 \leq x \leq \tfrac{1}{2}).$$

This exterior-tail normalization removes the additive and scaling ambiguities from the differential equation.

**Theorem 35.3** (Uniqueness of the normalized bounded Fabius function). *The function  $F$  of [Theorem 27.15](#) is the unique normalized bounded Fabius function in the sense of [definition 35.2](#).*

*Proof.* Existence is [Theorem 27.15](#): its CDF  $F$  has the constant exterior tails, reflection law, and local differential equation required by [definition 35.2](#). For uniqueness, let  $G$  be any other candidate. It is nondecreasing: its derivative is nonnegative on  $[0, 1/2]$ , and differentiating reflection gives  $G'(1-x) = G'(x)$ . Thus  $G$  is the CDF of a probability measure  $\mu$  on  $[0, 1]$ . For such a CDF define

$$(\mathcal{T}G)(x) = \int_0^1 G(2x-v) \, dv = \int_{2x-1}^{2x} G(s) \, ds.$$

This is the CDF of  $(V+Z)/2$ , where  $V$  is uniform on  $[0, 1]$ ,  $Z$  has CDF  $G$ , and they are independent. On  $(0, 1/2)$ ,

$$(\mathcal{T}G)'(x) = 2G(2x) = G'(x),$$

while on  $(1/2, 1)$ , reflection and its differentiated form give

$$(\mathcal{T}G)'(x) = 2(1 - G(2x-1)) = 2G(2-2x) = G'(1-x) = G'(x).$$

Both functions vanish at zero and have the same constant exterior tails, so  $\mathcal{T}G = G$ : every normalized bounded Fabius law is a fixed point.

On probability measures on  $[0, 1]$ , let  $\mathcal{T}\mu$  denote the law just constructed. Given any coupling  $(Z_1, Z_2)$  of  $\mu_1, \mu_2$ , use the same independent  $V$  in both images. Then

$$\mathbb{E} \left| \frac{V+Z_1}{2} - \frac{V+Z_2}{2} \right| = \frac{1}{2} \mathbb{E} |Z_1 - Z_2|,$$

and taking the infimum over couplings yields  $W_1(\mathcal{T}\mu_1, \mathcal{T}\mu_2) \leq \frac{1}{2} W_1(\mu_1, \mu_2)$ . Two fixed points therefore have zero Wasserstein distance. Finally, separating the first summand shows  $X_{1/2} \stackrel{\text{law}}{=} (U_0 + X'_{1/2})/2$ , so its law is a fixed point. It is the unique one. This is the human-readable uniqueness argument complementing the formal identification `geometricUniformCDF_one_half_eq_fabiusReal`.  $\square$

The centered normalization introduces no second Fabius or Rvachev result:  $u_{1/2} = \text{up}$  is exactly Equation (27.36), its two Fabius branches are proposition 28.1, and its angular Fourier product is Theorem 27.16. With  $\widehat{u}_{2\pi}(z) = \int_{\mathbb{R}} u(s)e^{-2\pi isz} ds$ , substituting  $t = -2\pi z$  there and using evenness gives

$$\widehat{\text{up}}_{2\pi}(z) = \prod_{j=0}^{\infty} \text{sinc}_{\pi}\left(\frac{z}{2^j}\right).$$

### 35.4 Thue–Morse spline approximants

Let  $\text{wt}_2(k)$  be the number of ones in the binary expansion of  $k$ , and put  $\varepsilon_k = (-1)^{\text{wt}_2(k)}$ . The occurrence of these signs is ordinary inclusion–exclusion on the dyadic geometric comb.

**Corollary 35.4** (Exact Thue–Morse partial CDF). *For  $N \geq 1$ , put*

$$F_N(x) = \mathbb{P}(X_{1/2,N} \leq x), \quad r_+ = \max(r, 0).$$

*Then*

$$F_N(x) = \frac{2^{N(N+1)/2}}{N!} \sum_{k=0}^{2^N-1} \varepsilon_k \left(x - \frac{k}{2^N}\right)_+^N. \quad (35.5)$$

*For the Fabius function  $F$  and every real  $x$ ,*

$$F_N(x - 2^{-N}) \leq F(x) \leq F_N(x). \quad (35.6)$$

*In particular the explicit splines converge pointwise to  $F$ , and uniformly with the quantitative bound*

$$0 \leq F_N(x) - F(x) \leq F(x + 2^{-N}) - F(x) \leq \omega_F(2^{-N}), \quad (35.7)$$

*where  $\omega_F(\delta) = \sup_{|a-b| \leq \delta} |F(a) - F(b)|$ . Consequently  $\|F_N - F\|_{\infty} \leq \omega_F(2^{-N}) \rightarrow 0$ .*

*Proof.* Specialize the weighted cube-cut formula Equation (27.19) to  $c_j = 2^{-j-1}$ . The product of the weights is  $2^{-N(N+1)/2}$ , and binary bit reversal identifies each subset sum with a unique  $k/2^N$  without changing the parity of the subset cardinality. This is Equation (35.5).

The tail estimate Equation (27.27) at  $q = 1/2$  says  $0 \leq X - X_{1/2,N} \leq 2^{-N}$ . Hence

$$\{X_{1/2,N} \leq x - 2^{-N}\} \subseteq \{X \leq x\} \subseteq \{X_{1/2,N} \leq x\}.$$

Taking probabilities proves the sandwich. Its right inequality gives  $F(x) \leq F_N(x)$ . Applying its left inequality at  $x + 2^{-N}$  gives  $F_N(x) \leq F(x + 2^{-N})$ , and hence Equation (35.7). The continuous function  $F$  is constant off  $[0, 1]$ , so it is uniformly continuous on  $\mathbb{R}$  and  $\omega_F(2^{-N}) \rightarrow 0$ . This proves the claimed uniform convergence.  $\square$

The finite sign identity behind the theorem is

$$\prod_{j=0}^{N-1} (1 - z^{2^j}) = \sum_{k=0}^{2^N-1} \varepsilon_k z^k. \quad (35.8)$$

It follows immediately by choosing either 1 or  $-z^{2^j}$  from each factor; uniqueness of binary expansion identifies the chosen subset with  $k$ , and its cardinality with  $\text{wt}_2(k)$ .

### 35.5 Cumulants as exact inverse observables

We use the Bernoulli-number convention  $B_2 = 1/6$ ,  $B_4 = -1/30$ .

**Theorem 35.5** (All cumulants and monotone base recovery). *For every  $m \geq 1$ ,*

$$\kappa_{2m}(Z_q) = \frac{2^{2m} B_{2m}}{2m} \frac{(1-q)^{2m}}{1-q^{2m}}, \quad (35.9)$$

*and every odd cumulant vanishes:*

$$\kappa_{2m+1}(Z_q) = 0 \quad (m \geq 0). \quad (35.10)$$

*After division by the nonzero uniform cumulant, set*

$$R_m(q) = \frac{\kappa_{2m}(Z_q)}{\kappa_{2m}(\varepsilon_0)} = \frac{(1-q)^{2m}}{1-q^{2m}}.$$

*Then  $R_m$  is a strictly decreasing bijection from  $(0, 1)$  to  $(0, 1)$ . Thus, given an observed normalized cumulant  $\rho_m \in (0, 1)$ , the unique recovered base is equivalently the unique  $q \in (0, 1)$  satisfying*

$$\rho_m(1 - q^{2m}) = (1 - q)^{2m}. \quad (35.11)$$

*Its endpoint expansions are*

$$q = \frac{1 - \rho_m}{2m} + \mathcal{O}((1 - \rho_m)^2), \quad \rho_m \uparrow 1, \quad (35.12)$$

$$1 - q = (2m\rho_m)^{1/(2m-1)} \left(1 + \mathcal{O}(\rho_m^{1/(2m-1)})\right), \quad \rho_m \downarrow 0. \quad (35.13)$$

*Proof.* The forward chapter writes  $Y_q = X_q - 1/2$ , so Equation (35.1) gives  $Z_q = 2Y_q$ . Homogeneity of cumulants and Theorem 27.3 therefore give

$$\kappa_r(Z_q) = 2^r \kappa_r(Y_q).$$

This is Equation (35.9), and the odd cumulants vanish for the same reason. Equation (27.9) gives  $\kappa_{2m}(U_0 - 1/2) = B_{2m}/(2m)$ ; since  $\varepsilon_0 = 2(U_0 - 1/2)$ , homogeneity gives  $\kappa_{2m}(\varepsilon_0) = 2^{2m} B_{2m}/(2m)$ . Thus the displayed normalization of  $R_m$  follows without a second cumulant derivation.

Logarithmic differentiation gives

$$\frac{R'_m(q)}{R_m(q)} = -\frac{2m}{1-q} + \frac{2mq^{2m-1}}{1-q^{2m}} < 0,$$

because  $q^{2m-1}(1-q) < 1 - q^{2m} = (1-q)(1+q+\dots+q^{2m-1})$ . The endpoint limits are one and zero, so strict monotonicity proves the bijection. Substituting the observed value  $\rho_m = R_m(q)$  gives Equation (35.11) and its unique root in  $(0, 1)$ . At  $q = 0$ ,  $R_m(q) = 1 - 2mq + \mathcal{O}(q^2)$ . At  $q = 1 - \varepsilon$ ,

$$R_m(q) = \frac{\varepsilon^{2m-1}}{2m} (1 + \mathcal{O}(\varepsilon)).$$

Ordinary and Puiseux reversion give Equations (35.12) and (35.13). □

The case  $m = 1$  is especially simple.

**Corollary 35.6** (Exact variance inverse). *The centered variance and its inverse are*

$$V(q) = \text{Var}(Z_q) = \frac{1-q}{3(1+q)}, \quad q = \frac{1-3V}{1+3V}. \quad (35.14)$$

*The Rvachev value  $q = 1/2$  is characterized by  $V = 1/9$ .*

*Proof.* Either put  $m = 1$  in Equation (35.9), or sum  $\frac{1}{3}(1-q)^2 \sum_{j \geq 0} q^{2j}$ . Solving the resulting linear-fractional equation gives the inverse and the specialization. □

### 35.6 Scale-free recovery and an upgraded conjecture

**Theorem 35.7** (Kurtosis inverse and total location–scale identifiability). *The excess kurtosis of  $Z_q$  is*

$$\gamma_2(q) = \frac{\kappa_4(Z_q)}{\kappa_2(Z_q)^2} = -\frac{6}{5} \frac{1 - q^2}{1 + q^2}, \quad (35.15)$$

*and is a strictly increasing bijection from  $(0, 1)$  to  $(-6/5, 0)$ , with inverse*

$$q = \sqrt{\frac{1 + 5\gamma_2/6}{1 - 5\gamma_2/6}}. \quad (35.16)$$

*Consequently the mean, positive variance, and excess kurtosis uniquely identify every member of the normalized family*

$$W_{\mu,s,q} = \mu + sZ_q, \quad \mu \in \mathbb{R}, \quad s > 0, \quad 0 < q < 1.$$

*Explicitly,  $\mu = \mathbb{E}W_{\mu,s,q}$ ,  $q$  is given by Equation (35.16), and*

$$s = \sqrt{\text{Var}(W_{\mu,s,q}) \frac{3(1+q)}{1-q}}. \quad (35.17)$$

*Equivalently, in the raw-series convention*

$$\widetilde{W}_{\mu,a,q} := \mu + a \sum_{j \geq 0} q^j \varepsilon_j, \quad a = s(1 - q), \quad (35.18)$$

*the recovered scale is*

$$a = \sqrt{3 \text{Var}(\widetilde{W}_{\mu,a,q})(1 - q^2)}. \quad (35.19)$$

*Thus the total cumulant identifiability statement left conjectural in one source report is a theorem for the stated three-parameter family.*

*Proof.* Insert  $B_2 = 1/6$  and  $B_4 = -1/30$  into Equation (35.9). Cancellation gives

$$\frac{-2}{15} \frac{(1 - q)^4}{1 - q^4} \left( \frac{3(1 - q^2)}{(1 - q)^2} \right)^2 = -\frac{6}{5} \frac{1 - q^2}{1 + q^2}.$$

The last rational function is strictly increasing, with the stated endpoint limits. Solving it for  $q^2$  gives Equation (35.16). Translation changes only the mean, positive scaling multiplies variance by  $s^2$ , and excess kurtosis is unchanged by both operations. Hence the three displayed reconstruction formulas exist and are unique. Formula Equation (35.17) is just  $\text{Var}(W_{\mu,s,q}) = s^2 V(q)$  with Equation (35.14) substituted. Finally, Equations (35.1) and (35.18) give  $a = s(1 - q)$ ; multiplying the square of Equation (35.17) by  $(1 - q)^2$  gives Equation (35.19).  $\square$

For redundancy checks, the next standardized cumulant is

$$\frac{\kappa_6(Z_q)}{\kappa_2(Z_q)^3} = \frac{48}{7} \frac{(1 - q^2)^2}{1 + q^2 + q^4}. \quad (35.20)$$

At  $q = 1/2$ , the exact diagnostic triple begins

$$\text{Var}(Z_{1/2}) = \frac{1}{9}, \quad \gamma_2(Z_{1/2}) = -\frac{18}{25}, \quad \frac{\kappa_6(Z_{1/2})}{\kappa_2(Z_{1/2})^3} = \frac{144}{49}.$$

Equation Equation (35.20) follows from  $\kappa_6(\varepsilon_0) = 16/63$  and the same geometric-series cancellation as above.

### 35.7 Endpoint regimes and conditioning

**Proposition 35.8** (Uniform and Gaussian endpoint laws). *As  $q \downarrow 0$ , for the chosen pointwise bounded versions,  $Z_q \rightarrow \varepsilon_0$  uniformly in the sample point (and hence almost surely). As  $q \uparrow 1$ ,  $Z_q \rightarrow 0$  in  $L^2$ . After standardization,*

$$\frac{Z_q}{\sqrt{V(q)}} \xrightarrow{\text{law}} \mathcal{N}(0, 1). \quad (35.21)$$

*Proof.* The first limit has the deterministic bound

$$|Z_q - \varepsilon_0| \leq q|\varepsilon_0| + (1 - q) \sum_{j \geq 1} q^j |\varepsilon_j| \leq 2q.$$

The  $L^2$  assertion follows from  $\mathbb{E}Z_q^2 = V(q) \rightarrow 0$ . For the Gaussian limit, put

$$a_{q,j} = \frac{(1 - q)q^j}{\sqrt{V(q)}}.$$

Then

$$\sum_{j \geq 0} a_{q,j}^2 = 3, \quad \max_{j \geq 0} a_{q,j} = a_{q,0} = \sqrt{3(1 - q^2)} \rightarrow 0.$$

The sinc product gives the characteristic function of the standardized law as  $\prod_{j \geq 0} \text{sinc}_{\text{rad}}(ta_{q,j})$ . For fixed real  $t$  and  $q$  sufficiently close to one, every argument lies in a fixed neighborhood of zero on which

$$\log \text{sinc}_{\text{rad}} z = -\frac{z^2}{6} + \mathcal{O}(z^4),$$

with a uniform remainder constant. The resulting logarithmic series is absolutely convergent because  $\sum_j a_{q,j}^2 = 3$ . Consequently

$$\log \prod_{j \geq 0} \text{sinc}_{\text{rad}}(ta_{q,j}) = -\frac{t^2}{6} \sum_{j \geq 0} a_{q,j}^2 + \mathcal{O}\left(t^4 \sum_{j \geq 0} a_{q,j}^4\right) = -\frac{t^2}{2} + o(1),$$

because  $\sum_j a_{q,j}^4 \leq (\max_j a_{q,j}^2) \sum_j a_{q,j}^2 \rightarrow 0$ . Thus the characteristic functions converge pointwise to  $e^{-t^2/2}$ , and Lévy's continuity theorem proves Equation (35.21). This direct Fourier argument also avoids applying a finite-row triangular-array theorem to the countably infinite series.  $\square$

Variance recovery is absolutely well-conditioned on every closed subinterval of  $0 < V < 1/3$ , since

$$\frac{dq}{dV} = -\frac{6}{(1 + 3V)^2}.$$

Its relative condition number diverges as  $q \downarrow 0$ , but that is the inevitable normalization by a recovered parameter tending to zero, not an absolute singularity. Near  $q = 1$ , the Puiseux behavior in Equation (35.13) shows why high-order cumulant recovery is progressively flatter than variance recovery.

### 35.8 Parameter scores, local CDF inversion, and the symmetry obstruction

Put  $w_j(q) = (1 - q)q^j$ . Then

$$w'_j(q) = q^{j-1}(j - (j + 1)q),$$



where the formula at  $j = 0$  is interpreted by cancellation as  $-1$ . Write

$$H_q(x) = \mathbb{P}(Z_q \leq x) = F_q\left(\frac{x+1}{2}\right)$$

for the centered CDF.

**Proposition 35.9** (Fourier score and quantile sensitivity). *Away from the zeros of the sinc factors,*

$$\partial_q \log \psi_q(t) = \sum_{j \geq 0} w'_j(q) \left( t \cot(w_j(q)t) - \frac{1}{w_j(q)} \right). \quad (35.22)$$

*The bracket is interpreted by analytic continuation at  $t = 0$ , where its value is zero. The series converges locally uniformly in  $q \in (0, 1)$  and real  $t$  on the open set where none of those factors vanishes. More quantitatively, for every compact  $K \subseteq (0, 1)$ , every  $A > 0$ , and all integers  $k \geq 0$ , there is a constant  $C_{A,k,K}$  such that*

$$\sup_{q \in K} \left| \partial_q^k \psi_q(t) \right| \leq C_{A,k,K} (1 + |t|)^{-A}. \quad (35.23)$$

*Consequently the characteristic function, density, centered CDF  $H_q$ , and uncentered CDF  $F_q(x) = H_q(2x - 1)$  are  $C^\infty$  in  $q$  on compact subintervals of  $(0, 1)$ , and parameter differentiation may be passed through Fourier inversion. If a centered quantile branch satisfies  $H_q(x_p(q)) = p$  and  $u_q(x_p(q)) > 0$ , then it is locally smooth and*

$$x'_p(q) = -\frac{\partial_q \mathbb{P}(Z_q \leq x)|_{x=x_p(q)}}{u_q(x_p(q))}. \quad (35.24)$$

*Proof.* For a nonzero sinc factor,

$$\frac{d}{dw} \log \text{sinc}_{\text{rad}}(wt) = t \cot(wt) - \frac{1}{w}.$$

At  $t = 0$  the apparent singularity is removable because

$$t \cot(wt) - \frac{1}{w} = -\frac{wt^2}{3} - \frac{w^3 t^4}{45} + \mathcal{O}(w^5 t^6).$$

The chain rule gives each summand in Equation (35.22). For large  $j$ , the bracket equals  $-w_j t^2/3 + \mathcal{O}(w_j^3 t^4)$ , while  $|w'_j| \leq C_K(j+1)q_+^{j-1}$  on any compact  $K \subset (0, 1)$ , where  $q_+ = \max K < 1$ . The resulting majorant is a convergent multiple of  $\sum (j+1)q_+^{2j}$ , proving local uniform convergence away from the zeros.

For completeness, we quantify the product differentiation, including at its zeros. For every  $r \geq 0$ , direct differentiation of  $w_j = (1-q)q^j$  gives constants depending only on  $K, r$  such that

$$\sup_{q \in K} |\partial_q^r w_j(q)| \leq C_{K,r}(j+1)^r q_+^j, \quad \sum_{j \geq 0} \sup_{q \in K} |\partial_q^r w_j(q)| < \infty.$$

The integral representation  $\text{sinc}_{\text{rad}} y = \frac{1}{2} \int_{-1}^1 e^{iyu} du$  shows that every real derivative of  $\text{sinc}_{\text{rad}}$  is bounded. These two estimates, together with the chain rule, give normally convergent differentiated products on compact  $(q, t)$ -sets.

Now fix  $k$  and choose an integer  $R > A + 2k$ . On  $K$ , each of the first  $R$  weights is bounded away from zero, so an undifferentiated leading factor is  $\mathcal{O}_K(|t|^{-1})$ . In a term of the  $k$ -fold Leibniz rule, suppose that  $s \leq k$  of those leading factors are differentiated. Their total derivative order is at most  $k$ , so boundedness of the sinc derivatives makes the term  $\mathcal{O}_{K,k,R}(|t|^{k+s-R})$ . Since  $s \leq k$  and  $R > A + 2k$ , this is  $\mathcal{O}_{K,k,R}(|t|^{-A})$ . Derivatives falling on later factors are summable by the preceding

weight estimates and do not change this worst power. Summing the normally convergent Leibniz expansion proves Equation (35.23); enlarging the constant handles bounded  $t$ .

Given any numbers of spatial and parameter derivatives, choose  $A$  large enough that the corresponding power of  $t$  is integrable. Dominated Fourier inversion then proves smooth dependence of  $u_q$ . Since all centered laws are supported in the common interval  $[-1, 1]$ ,  $H_q(x) = \int_{-1}^x u_q(s) ds$  on that interval, with constant exterior tails; hence the same conclusion holds for  $H_q$  and  $F_q$ . Finally, the implicit-function theorem applies at a quantile where  $u_q(x_p) > 0$ . Differentiating its defining identity and solving  $\partial_q H_q(x_p) + u_q(x_p)x'_p = 0$  gives Equation (35.24).  $\square$

For fixed  $x$ , the  $C^2$  inverse sensitivity theorem Theorem 30.13 applies to  $q \mapsto F_q(x)$  at any point where its  $q$ -derivative is nonzero. If  $A = \partial_q F_q(x)|_{q=q_0}$ ,  $B = \partial_q^2 F_q(x)|_{q=q_0}$ , and  $\Delta = p - F_{q_0}(x)$ , the first terms are

$$q - q_0 = \frac{\Delta}{A} - \frac{B}{2A^3}\Delta^2 + \mathcal{O}(\Delta^3).$$

At  $x = 1/2$ , however, reflection forces  $F_q(1/2) = 1/2$  for every  $q$ . The entire parameter score vanishes there, so the midpoint contains no information about  $q$ . This is an exact identifiability obstruction, not a numerical conditioning accident.

### 35.9 The geometric comb as a $q$ -Pochhammer interpolation basis

**Proposition 35.10** (Newton remainder on geometric nodes). *Let  $n \geq 1$ , let  $0 < q < 1$ , let  $I \subset \mathbb{R}$  be an interval containing  $x, 1, q, \dots, q^{n-1}$ , and let  $g : I \rightarrow \mathbb{R}$  be  $n$ -times continuously differentiable. Let  $N_{n-1}g$  be its interpolation polynomial at those  $n$  distinct geometric nodes. Then some point  $\xi$  in the convex hull of  $x, 1, q, \dots, q^{n-1}$  satisfies*

$$g(x) - N_{n-1}g(x) = \frac{g^{(n)}(\xi)}{n!} \prod_{j=0}^{n-1} (x - q^j) = \frac{g^{(n)}(\xi)}{n!} x^n (x^{-1}; q)_n, \quad (35.25)$$

where the final expression is understood by polynomial continuation at  $x = 0$ .

*Proof.* If  $x = q^j$  for one of the interpolation nodes, both sides of the first equality vanish, and any  $\xi$  in the convex hull may be chosen. Otherwise the  $n + 1$  points are distinct. Put

$$c = \frac{g(x) - N_{n-1}g(x)}{\prod_{j=0}^{n-1} (x - q^j)}, \quad h(t) = g(t) - N_{n-1}g(t) - c \prod_{j=0}^{n-1} (t - q^j).$$

The real-valued function  $h$  has those  $n + 1$  distinct zeros. Applying Rolle's theorem  $n$  times gives a point  $\xi$  in their convex hull with  $h^{(n)}(\xi) = 0$ . Since  $N_{n-1}g$  has degree at most  $n - 1$  and the node polynomial is monic of degree  $n$ , this identity reads  $g^{(n)}(\xi) - n!c = 0$ , proving the first equality. For  $x \neq 0$ , the second equality is the factor-by-factor identity

$$x^n (x^{-1}; q)_n = x^n \prod_{j=0}^{n-1} (1 - x^{-1}q^j) = \prod_{j=0}^{n-1} (x - q^j).$$

The final product is a polynomial in  $x$ , so this identity defines the polynomial continuation of the middle expression at  $x = 0$ .  $\square$

This is the precise interpolation link behind the informal phrase *geometric comb*. The same nodes  $q^j$  govern the random-series weights, the cumulant sums, Newton divided differences, and the finite  $q$ -Pochhammer basis. Parameter recovery from Fabius–Rvachev observables is therefore connected to inverse  $q$ -products structurally, rather than by a mere resemblance of notation.

## Chapter 36

# Certification and research frontiers

### 36.1 Branch-aware numerical certification

An inverse approximation is not a certificate. A certificate must identify the forward map, the active parameter, the target, the domain containing the selected branch, and the arithmetic enclosure used to control every discarded tail and rounding error. In the complex setting it must additionally record the continuation path and every logarithm or fractional-power determination. Changing any one of these data can change the inverse being computed.

The basic workflow is therefore:

1. choose a coordinate in which the selected sheet is regular;
2. evaluate the forward map and its derivatives with outward-rounded enclosures, including explicit infinite tails;
3. obtain a candidate from an asymptotic expansion, continuation, or a floating-point solve;
4. prove existence and uniqueness in a real interval or complex disk;
5. if all sheets are required, certify the remaining zero count by the argument principle.

The candidate-generation step may be heuristic. The last two steps are not.

#### 36.1.1 Certified tails for infinite products

The next estimate controls the value and the first two argument derivatives with one truncation index.

**Theorem 36.1** (Complex  $q$ -Pochhammer tail). *Let  $a, q \in \mathbb{C}$ , put  $\rho = |q| < 1$ , and choose  $N \in \mathbb{N}_0$  so that*

$$\eta := |a|\rho^N < 1.$$

*The tail*

$$T_N(a, q) := \prod_{j=N}^{\infty} (1 - aq^j)$$

*is nonzero and has the logarithm*

$$L_N(a, q) := - \sum_{j=N}^{\infty} \sum_{k=1}^{\infty} \frac{(aq^j)^k}{k}.$$

*With*

$$B_N := \frac{|a|\rho^N}{(1 - \eta)(1 - \rho)},$$

one has

$$|L_N(a, q)| \leq B_N, \quad |T_N(a, q) - 1| \leq e^{B_N} - 1, \quad (36.1)$$

$$|\partial_a L_N(a, q)| \leq \frac{\rho^N}{(1 - \eta)(1 - \rho)}, \quad |\partial_a^2 L_N(a, q)| \leq \frac{\rho^{2N}}{(1 - \eta)^2(1 - \rho^2)}. \quad (36.2)$$

More explicitly, suppose interval arithmetic certifies scalar bounds

$$\rho \leq \bar{\rho} < 1, \quad \eta \leq \bar{\eta} < 1.$$

Then the value bounds hold with

$$\bar{B}_N := \frac{\bar{\eta}}{(1 - \bar{\eta})(1 - \bar{\rho})}$$

in place of  $B_N$ , and the two derivative bounds hold after replacing their right-hand sides respectively by

$$\frac{\bar{\rho}^N}{(1 - \bar{\eta})(1 - \bar{\rho})}, \quad \frac{\bar{\rho}^{2N}}{(1 - \bar{\eta})^2(1 - \bar{\rho}^2)}.$$

For interval balls  $\mathcal{A}$  and  $\mathcal{Q}$  of possible  $a$  and  $q$ , it suffices to certify

$$\bar{\rho} \geq \sup_{q \in \mathcal{Q}} |q|, \quad \bar{\eta} \geq \left( \sup_{a \in \mathcal{A}} |a| \right) \bar{\rho}^N,$$

with both upper endpoints still strictly below one.

*Proof.* For  $|z| < 1$ , the power-series logarithm gives

$$|-\log(1 - z)| \leq \sum_{k \geq 1} |z|^k = \frac{|z|}{1 - |z|}.$$

For  $j \geq N$ ,  $|aq^j| \leq \eta$ . Hence

$$|L_N(a, q)| \leq \sum_{j \geq N} \frac{|a|\rho^j}{1 - \eta} = B_N.$$

The majorant is summable, so the logarithmic series converges absolutely and the product converges to  $T_N = e^{L_N}$ . No tail factor vanishes because  $|aq^j| < 1$ . The elementary estimate  $|e^z - 1| \leq e^{|z|} - 1$  proves the second bound in (36.1).

On a slightly smaller neighbourhood of the given  $a$ , the differentiated series have the summable majorants used below; termwise differentiation is therefore legitimate. Explicitly,

$$\partial_a L_N = - \sum_{j \geq N} \frac{q^j}{1 - aq^j}, \quad \partial_a^2 L_N = - \sum_{j \geq N} \frac{q^{2j}}{(1 - aq^j)^2}.$$

Using  $|1 - aq^j| \geq 1 - \eta$  and summing the two geometric series gives (36.2). Each factor in the three displayed majorants is monotone in  $\rho$  and  $\eta$  on  $[0, 1)$ , so substituting  $\bar{\rho}, \bar{\eta}$  proves the interval versions.  $\square$

Near a zero of a finite prefix, that factor should be removed symbolically. The theorem then controls the nonvanishing tail without hiding the local zero inside a logarithmic sum.

The Rvachev Fourier product has an equally simple real-axis certificate. We use  $\text{sinc}_{\text{rad}} x = \sin(x)/x$  for  $x \neq 0$  and  $\text{sinc}_{\text{rad}}(0) = 1$ .

**Theorem 36.2** (Geometric sinc-product tail). *Let  $0 < q < 1$ ,  $t \in \mathbb{R}$ , and  $w_j = (1 - q)q^j$ . Then*

$$S_N(t) := \prod_{j=N}^{\infty} \text{sinc}_{\text{rad}}(w_j t)$$

*converges, and*

$$|S_N(t) - 1| \leq \exp\left(\frac{t^2(1-q)^2 q^{2N}}{6(1-q^2)}\right) - 1. \quad (36.3)$$

*Proof.* Taylor's formula with integral remainder gives, for real  $x$ ,

$$|\sin x - x| \leq \int_0^{|x|} (|x| - u)u \, du = \frac{|x|^3}{6}.$$

Consequently  $|\text{sinc}_{\text{rad}} x - 1| \leq x^2/6$ , with the value at zero understood by continuity. Put  $u_j = \text{sinc}_{\text{rad}}(w_j t) - 1$ . For a finite product,

$$\left| \prod_{j=N}^M (1 + u_j) - 1 \right| \leq \prod_{j=N}^M (1 + |u_j|) - 1 \leq \exp\left(\sum_{j=N}^M |u_j|\right) - 1.$$

Moreover,

$$\sum_{j \geq N} |u_j| \leq \frac{t^2}{6} \sum_{j \geq N} w_j^2 = \frac{t^2(1-q)^2 q^{2N}}{6(1-q^2)}.$$

The summability proves convergence of the product, and passage to the limit in the finite-product inequality proves (36.3).  $\square$

### 36.1.2 Real branches: brackets, residuals, and interval Newton

On a real branch, monotonicity is the cleanest existence and uniqueness certificate. Newton acceleration may be used inside the bracket, but it is the bracket that carries the proof.

**Theorem 36.3** (Monotone branch certificate). *Let  $f : [\alpha, \beta] \rightarrow \mathbb{R}$  be continuous and strictly monotone, and let  $y$  lie between  $f(\alpha)$  and  $f(\beta)$ . Then there is a unique  $x_* \in [\alpha, \beta]$  with  $f(x_*) = y$ . Sign-preserving bisection produces nested brackets  $X_r$  containing  $x_*$  with*

$$\text{length}(X_r) = 2^{-r}(\beta - \alpha).$$

*If  $f$  is differentiable and  $|f'(x)| \geq m > 0$  throughout the original bracket, then every  $x_0 \in [\alpha, \beta]$  satisfies*

$$|x_0 - x_*| \leq \frac{|f(x_0) - y|}{m}. \quad (36.4)$$

*Proof.* The intermediate value theorem gives existence, and strict monotonicity gives uniqueness. At each bisection step, one of the two half-intervals still has endpoint values bracketing  $y$ ; choosing it preserves  $x_*$  and halves the length. Finally, the mean-value theorem gives

$$f(x_0) - f(x_*) = f'(\xi)(x_0 - x_*)$$

for some  $\xi$  between  $x_0$  and  $x_*$ . Taking absolute values and using the derivative lower bound proves (36.4).  $\square$

The residual estimate assumes that the root already exists. The general arbitrary-target interval-Newton theorem [Theorem 30.31](#) certifies existence, uniqueness, and enclosure as soon as the derivative interval is separated from zero. This is exactly the one-dimensional test formerly repeated here.

Directed rounding is essential here. A derivative sampled at many points is not an interval extension of  $f'(X)$ , and an ordinary floating-point inclusion is not a mathematical inclusion.

### 36.1.3 Complex disks: contraction, Rouché, and total zero counts

The complex analogue of interval Newton is particularly transparent in one variable. It certifies a selected sheet and gives a constructive refinement map.

**Theorem 36.4** (Complex Krawczyk disk). *Let  $f$  be holomorphic on a neighbourhood of the closed disk  $D = \{z : |z - z_0| \leq r\}$ , where  $r > 0$ , and let  $A \in \mathbb{C} \setminus \{0\}$ . If*

$$\kappa := \sup_{z \in D} |1 - Af'(z)| < 1, \quad |Af(z_0)| \leq (1 - \kappa)r,$$

*then  $f$  has exactly one zero in  $D$ . If the second inequality is strict, the zero lies in the open disk.*

*Proof.* Set  $T(z) = z - Af(z)$ . The disk is convex, and integration along the line segment from  $w$  to  $z$  gives

$$|T(z) - T(w)| \leq \kappa|z - w|.$$

Also, for  $z \in D$ ,

$$|T(z) - z_0| \leq |T(z) - T(z_0)| + |Af(z_0)| \leq \kappa r + (1 - \kappa)r = r.$$

Thus  $T$  is a contraction of the complete metric space  $D$  into itself. The contraction theorem supplies a unique fixed point  $z_*$ . Since  $A \neq 0$ ,  $T(z_*) = z_*$  is equivalent to  $f(z_*) = 0$ . Every zero is a fixed point, so there are no others. Under the strict second inequality the displayed estimate is strict on the boundary, hence  $z_*$  is interior.  $\square$

When a second-derivative bound is sharper than an enclosure of the Newton map, apply the arbitrary-target Rouché certificate [Theorem 30.30](#). Its specialization to target zero is exactly the one-zero disk test formerly repeated here.

Disjoint one-zero disks do not by themselves prove that every inverse sheet has been found. A validated argument-principle count closes that gap.

**Theorem 36.5** (Validated argument-principle count). *Let  $\Omega$  be a bounded Jordan domain, let  $f$  be holomorphic on a neighbourhood of  $\Omega$ , and suppose interval evaluation proves  $0 \notin f(\partial\Omega)$ . Write the positively oriented boundary as a finite concatenation of  $C^3$  panels  $\gamma_k : [0, 1] \rightarrow \partial\Omega$ , and put*

$$g_k(s) := \frac{f'(\gamma_k(s))\gamma_k'(s)}{f(\gamma_k(s))}.$$

*Suppose validated bounds  $M_k \geq \sup_{0 \leq s \leq 1} |g_k''(s)|$  are available. Let*

$$Q := \sum_k \frac{g_k(0) + g_k(1)}{2}, \quad E := \frac{1}{12} \sum_k M_k.$$

*Suppose outward-rounded arithmetic provides  $\hat{Q} \in \mathbb{C}$  and  $\varepsilon_Q \geq 0$  with  $|Q - \hat{Q}| \leq \varepsilon_Q$ . If the closed disk*

$$\left\{ z : \left| z - \frac{\hat{Q}}{2\pi i} \right| \leq \frac{E + \varepsilon_Q}{2\pi} \right\}$$

*contains exactly one integer  $n$ , then  $f$  has exactly  $n$  zeros in  $\Omega$ , counted with multiplicity.*

*Proof.* For a twice continuously differentiable complex-valued function  $g$  on  $[0, 1]$ , two integrations by parts give the trapezoidal remainder formula

$$\int_0^1 g(s) \, ds - \frac{g(0) + g(1)}{2} = -\frac{1}{2} \int_0^1 s(1-s)g''(s) \, ds.$$

Since  $\int_0^1 s(1-s) ds = 1/6$ , its modulus is at most  $\sup|g''|/12$ . Summing over the panels gives

$$\left| \int_{\partial\Omega} \frac{f'(z)}{f(z)} dz - Q \right| \leq E.$$

The boundary nonvanishing hypothesis makes the argument principle applicable, so

$$\frac{1}{2\pi i} \int_{\partial\Omega} \frac{f'(z)}{f(z)} dz$$

is the integer equal to the number of zeros of  $f$  in  $\Omega$ . The quadrature and rounding bounds, followed by the triangle inequality, place this integer in the displayed disk. If that disk contains only  $n$ , the zero count is  $n$ .  $\square$

In practice, interval balls for  $f(\gamma_k([0, 1]))$  certify boundary nonvanishing, automatic differentiation bounds  $g_k''$ , and panel subdivision reduces  $E$ . Counting one zero in each selected disk and then the same total number in a containing contour certifies both uniqueness and completeness. For a meromorphic forward map, the same integral counts zeros minus poles; the poles and their multiplicities must be enclosed and added back explicitly.

### 36.1.4 Asymptotic truncation is part of the certificate

An  $O(t^{N+1})$  printed in a manuscript has no numerical content until its constant, domain, and branch are known. For a convergent Taylor expansion, Cauchy's estimate supplies all three.

**Theorem 36.6** (Cauchy tail bound). *Let  $H$  be holomorphic on a neighbourhood of  $\{z : |z| \leq R\}$ , let  $M \geq \max_{|z|=R} |H(z)|$ , and write  $H(z) = \sum_{j \geq 0} c_j z^j$ . For  $N \in \mathbb{N}_0$ ,  $0 \leq r < R$ , and  $|z| \leq r$ ,*

$$\left| H(z) - \sum_{j=0}^N c_j z^j \right| \leq M \frac{(r/R)^{N+1}}{1 - r/R}.$$

*Proof.* Cauchy's coefficient formula gives  $|c_j| \leq M/R^j$ . Hence

$$\left| \sum_{j>N} c_j z^j \right| \leq M \sum_{j>N} (r/R)^j = M \frac{(r/R)^{N+1}}{1 - r/R}.$$

$\square$

For an inverse expansion, it is usually sharper to certify its forward residual than to bound inverse coefficients separately. Apply [Proposition 30.18](#) to each fixed parameter  $s$ , with  $G(y_s) = p(s)$ ,  $\tilde{G}(y_s) = p_N(s)$ , and  $m(y_s) = m_s$ . It gives the branch-specific estimate

$$|p_N(s) - p(s)| \leq \frac{|F_s(p_N(s)) - y_s|}{m_s} \leq \frac{E_N(s)}{m_s}.$$

At a fold,  $m_s$  tends to zero in the original coordinate. One must first pass to the square-root coordinate in which the sheet is regular, or use a Rouché or Krawczyk disk. Reporting a large residual quotient in the singular coordinate does not certify the branch.

Divergent asymptotic series require a different discipline. The elementary calculation below explains optimal truncation for a Gevrey-one envelope.

**Proposition 36.7** (Optimal index for a Gevrey-one remainder envelope). *Let  $A, C > 0$  and  $t \neq 0$ . For every  $N \in \mathbb{N}_0$ , define the candidate remainder envelope*

$$E_N(t) = C(A|t|)^{N+1}(N+1)!.$$

*Suppose the actual remainder after truncation at  $N$  has been proved to have modulus at most  $E_N(t)$  for every  $N$ . Then*

$$\frac{E_{N+1}(t)}{E_N(t)} = A|t|(N+2).$$

*Put  $x = (A|t|)^{-1} - 2$ . The certified envelope decreases exactly when  $N < x$ , is unchanged when  $N = x$ , and increases when  $N > x$ . If  $A|t| > 1/2$ , its unique minimizing index is  $N = 0$ . If  $A|t| = 1/2$ , the minimizing indices are 0 and 1. If  $A|t| < 1/2$  and  $x \notin \mathbb{N}_0$ , the unique minimizing index is  $\lceil x \rceil$ ; if  $x \in \mathbb{N}_0$ , the minimizing indices are  $x$  and  $x + 1$ .*

*Proof.* Cancel the common factors in  $E_{N+1}/E_N$ . The monotonicity and location of the discrete minimum follow immediately from whether this ratio is below or above one.  $\square$

This proposition is not permission to assume a Gevrey bound. Such a bound must first be proved uniformly in the stated sector. A defensible asymptotic certificate therefore records the coordinate and sheet, the allowed domain, the remainder constants, the chosen truncation order, the outward-rounded forward residual, and the final real or complex root enclosure. Merely observing that successive terms initially decrease is evidence, not proof.

## 36.2 Two archived conjectures that are theorems

Two claims repeatedly presented as conjectural in the source reports need no numerical evidence. In their natural precise forms they are elementary theorems.

### 36.2.1 Total cumulants identify the geometric uniform law

Let  $\varepsilon_0, \varepsilon_1, \dots$  be independent random variables, each uniform on  $[-1, 1]$ , and put

$$X_{\mu,c,q} = \mu + c \sum_{j=0}^{\infty} q^j \varepsilon_j, \quad \mu \in \mathbb{R}, \quad c > 0, \quad 0 < q < 1.$$

The series converges absolutely almost surely: with probability one all the inequalities  $|\varepsilon_j| \leq 1$  hold, and then its absolute value is bounded by  $\sum_j q^j$ .

This is exactly the raw-series convention  $\widetilde{W}_{\mu,a,q}$  of Equation (35.18), with  $a = c$ . The complete proof therefore appears once, in Theorem 35.7. Its raw-scale clause proves the full bijection from  $(\mu, c, q)$  to mean, positive variance, and standardized fourth cumulant, together with all three inverse formulas. This subsection records the source normalization and its disposition; it does not state or prove the same identifiability result a second time. Thus total cumulants identify the exact three-parameter model. Stability under noisy cumulants, missing observations, or a perturbed digit law is a different problem; it becomes a theorem or conjecture only after the perturbation model and metric are specified.

### 36.2.2 A transcendental value determines the Gaussian rectangle

For integers  $a, b \geq 0$ , write

$$G_{a,b}(Q) = \left[ \begin{matrix} a+b \\ a \end{matrix} \right]_Q.$$

This is the generating polynomial for partitions whose Ferrers diagrams fit inside an  $a$ -by- $b$  rectangle.



**Theorem 36.8** (Uniqueness at a transcendental base). *Let  $q \in \mathbb{C}$  be transcendental over  $\mathbb{Q}$ . If*

$$0 < k < n, \quad 0 < k' < n', \quad \begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n' \\ k' \end{bmatrix}_q,$$

*then  $n' = n$  and either  $k' = k$  or  $k' = n - k$ .*

*Proof.* Transcendence says that evaluation at  $q$  is injective on  $\mathbb{Z}[Q]$ . The asserted equality of values is therefore an equality of Gaussian polynomials. Put

$$a = k, \quad b = n - k, \quad m = \min(a, b), \quad M = \max(a, b),$$

and define  $a', b', m', M'$  similarly.

The coefficient of  $Q^d$  in  $G_{a,b}$  counts partitions of  $d$  fitting inside the rectangle. Every partition of  $d \leq m$  has at most  $m$  parts and largest part at most  $m$ , so these coefficients equal the unrestricted partition numbers  $p(d)$ . At  $d = m + 1$ , at least one partition is excluded: if  $a = m$ , the partition  $1 + \dots + 1$  with  $m + 1$  parts is too tall; if  $b = m$ , the one-part partition  $(m + 1)$  is too wide. Hence  $m + 1$  is the first degree at which the coefficient is smaller than  $p(d)$ . Equality of the two polynomials therefore gives  $m = m'$ .

The degree of  $G_{a,b}$  is  $ab = mM$ , so equality of degrees and positivity of  $m$  give  $M = M'$ . Thus the unordered pairs  $\{a, b\}$  and  $\{a', b'\}$  agree. Since  $n = a + b$  and  $k = a$ , this says exactly that  $n' = n$  and  $k' = k$  or  $k' = n - k$ .  $\square$

### 36.3 Precisely stated open frontiers

The following list keeps only assertions with a definite truth value. Each conjecture states its domain, normalization, and branch, followed by the present evidence and the missing mathematical obligation.

#### 36.3.1 Generic monodromy of finite $q$ -products

Set

$$P_{n,q}(a) = \prod_{j=0}^{n-1} (1 - aq^j).$$

The group-theoretic implication needed for this frontier is already a theorem.

**Theorem 36.9** (Simple, distinct critical values force full monodromy). *Let  $P \in \mathbb{C}[a]$  have degree  $n \geq 2$ . Suppose  $P'$  has  $n - 1$  simple zeros and their images under  $P$  are pairwise distinct. Then the monodromy group of  $P(a) - y = 0$  over  $\mathbb{C} \setminus \{\text{critical values of } P\}$  is  $S_n$ .*

*Proof.* Near a simple critical point, the holomorphic Morse lemma identifies the map with  $z \mapsto z^2$  in local coordinates. A small loop around its critical value therefore exchanges precisely the two colliding sheets and fixes the others: its monodromy is a transposition. Distinct critical values give one such branch cycle for each critical point. Small positively oriented loops around the finite critical values generate the fundamental group of their complement; the loop at infinity is the inverse of their product. Hence these transpositions generate the monodromy group.

The cover is connected. Algebraically, this follows because  $P(a) - y$  is irreducible in  $\mathbb{C}(y)[a]$ : viewed in  $\mathbb{C}[a, y]$ , it has degree one in  $y$ , and any factor independent of  $y$  would divide the coefficient  $-1$ ; Gauss's lemma then gives irreducibility over  $\mathbb{C}(y)$ . Hence the monodromy action is transitive.

Associate a graph to the sheets, with an edge for every generating transposition. Its connected components are exactly the orbits of the group they generate, so transitivity makes the graph connected. The edge transpositions of a connected graph generate all of  $S_n$ . Thus the monodromy group is  $S_n$ .  $\square$

**Conjecture 36.10** (Generic finite  $q$ -Pochhammer monodromy). *For every  $n \geq 3$  there is a nonzero polynomial  $E_n \in \mathbb{Z}[q]$  such that, whenever  $q \in \mathbb{C}$  and  $qE_n(q) \neq 0$ , the critical points of  $P_{n,q}$  are simple and their critical values are pairwise distinct. In particular, the monodromy group is then  $S_n$ .*

Simplicity is controlled by the discriminant of  $P'_{n,q}$ , and coincidence of two critical values by a second resultant or discriminant, all with integer coefficients in  $q$ . The real obligation is to show that their product is not the zero polynomial for every  $n$ . Once that is known, every excluded nonzero parameter is algebraic. A single exact specialization with simple critical points and distinct critical values for each  $n$ , or a uniform symbolic argument, would suffice. Floating-point plots cannot establish this nonvanishing.

### 36.3.2 Continuous Gaussian monotonicity

For  $0 < q < 1$  and  $u > 0$ , use the standard normalization

$$\Gamma_q(u) = (1 - q)^{1-u} \prod_{j=0}^{\infty} \frac{1 - q^{j+1}}{1 - q^{u+j}}.$$

For  $x > y > 0$ , define the continuous Gaussian coefficient by

$$\mathcal{G}_q(x, y) = \begin{cases} \frac{\Gamma_q(x+1)}{\Gamma_q(y+1)\Gamma_q(x-y+1)}, & 0 < q < 1, \\ \frac{\Gamma(x+1)}{\Gamma(y+1)\Gamma(x-y+1)}, & q = 1, \\ q^{y(x-y)}\mathcal{G}_{1/q}(x, y), & q > 1. \end{cases}$$

The last line fixes the reciprocal-base normalization rather than leaving it implicit.

**Conjecture 36.11** (Continuous Gaussian monotonicity). *For every pair  $x > y > 0$ , the function*

$$q \longmapsto \mathcal{G}_q(x, y)$$

*is strictly increasing on  $(0, 1)$ .*

For integral  $x = n$  and  $y = k$ , this is immediate from the nonnegative coefficients of the Gaussian polynomial, with strictness because  $0 < k < n$ . There is also rigorous local evidence at the upper endpoint. Apply [Theorem 34.8](#) at  $x + 1$ ,  $y + 1$ , and  $x - y + 1$ . Its linear coefficient satisfies

$$c_1(x+1) - c_1(y+1) - c_1(x-y+1) = -\frac{y(x-y)}{2}.$$

Consequently, if  $q = e^{-t}$ , then

$$\log \frac{\mathcal{G}_{e^{-t}}(x, y)}{\mathcal{G}_1(x, y)} = -\frac{y(x-y)}{2}t + O(t^2),$$

so  $\mathcal{G}_{e^{-t}}(x, y) < \mathcal{G}_1(x, y)$  for every sufficiently small  $t > 0$ . A global proof must establish the sign of the  $q$ -derivative of the displayed  $q$ -gamma quotient, most likely by reorganizing the resulting Lambert series into a manifestly positive expression. Conversely, one outward-rounded interval evaluation showing a negative derivative would be a rigorous counterexample.

### 36.3.3 A complete-Bernstein principal inverse

Fix  $0 < q < 1$  and let

$$P_q(a) = (a; q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j), \quad a < 1.$$

Every factor is positive. Locally uniform differentiation of the logarithm gives

$$\frac{P'_q(a)}{P_q(a)} = - \sum_{j=0}^{\infty} \frac{q^j}{1 - aq^j} < 0.$$

Moreover,  $P_q(a) \rightarrow 0$  as  $a \uparrow 1$ , while  $P_q(a) \rightarrow \infty$  as  $a \rightarrow -\infty$ . For the latter limit, retain any fixed number of factors and then let that finite product tend to infinity. Thus  $P_q$  has a decreasing inverse

$$A_q : (0, \infty) \longrightarrow (-\infty, 1), \quad X_q(y) = 1 - A_q(y) > 0.$$

The first two Bernstein signs are theorems. Indeed, if  $r_j = q^j / (1 - aq^j) > 0$ , then

$$\frac{P''_q(a)}{P_q(a)} = \left( \sum_{j \geq 0} r_j \right)^2 - \sum_{j \geq 0} r_j^2 = 2 \sum_{0 \leq i < j} r_i r_j > 0.$$

Implicit differentiation now gives

$$A'_q(y) = \frac{1}{P'_q(A_q(y))} < 0, \quad A''_q(y) = -\frac{P''_q(A_q(y))}{P'_q(A_q(y))^3} > 0,$$

and therefore  $X'_q(y) > 0$  and  $X''_q(y) < 0$ .

**Conjecture 36.12** (Complete-Bernstein principal inverse). *For every  $0 < q < 1$ ,  $X_q$  is a complete Bernstein function. Equivalently, there exist  $\alpha_q, \beta_q \geq 0$  and a positive measure  $\nu_q$  on  $(0, \infty)$ , with*

$$\int_{(0, \infty)} \frac{\nu_q(ds)}{1+s} < \infty,$$

such that

$$X_q(y) = \alpha_q + \beta_q y + \int_{(0, \infty)} \frac{y}{y+s} \nu_q(ds) \quad (y > 0).$$

The conjecture implies all alternating derivative inequalities  $(-1)^{r-1} X_q^{(r)}(y) \geq 0$  for  $r \geq 1$ . The two proved signs above and the higher signs observed in formal series reversion are evidence, not a proof of the infinite family. The decisive obligation is a Pick continuation of  $X_q$  to the slit plane, or an explicit positive representing measure. Notice also that the endpoint above forces  $\alpha_q = X_q(0+) = 0$  if the conjecture holds.

### 36.3.4 The minimum of the $q$ -gamma function

The existence and uniqueness of the minimizer are not conjectural: they are the first assertion of [Theorem 34.7](#), whose proof also gives the complete two-branch description of every real level above the minimum. Write  $x_q^* = x_q$  and  $m_q = \Gamma_q(x_q^*)$ .

**Conjecture 36.13** (Monotonic motion of the  $q$ -gamma minimum). *On  $0 < q < 1$ , the minimizer  $x_q^*$  is strictly increasing and the minimum value  $m_q$  is strictly decreasing.*

The archived safeguarded-Newton computations, which are numerical evidence rather than interval certificates, give

| $q$     | 0.05          | 0.20          | 0.60          | 0.95          |
|---------|---------------|---------------|---------------|---------------|
| $x_q^*$ | 1.37998346149 | 1.42196580342 | 1.45066979499 | 1.46060675866 |
| $m_q$   | 0.98270463141 | 0.95376342969 | 0.91169489999 | 0.88840573828 |

and exhibit both proposed signs. They are compatible with the formal  $q \downarrow 0$  asymptotics and the classical-gamma limit as  $q \uparrow 1$ . A proof requires sign control of the implicit derivative

$$\frac{dx_q^*}{dq} = -\frac{\partial_q \psi_q(x_q^*)}{\partial_x \psi_q(x_q^*)}$$

and, by the envelope identity,

$$\frac{d}{dq} \log m_q = \partial_q \log \Gamma_q(x_q^*).$$

The denominators are already positive by the theorem; the two numerators remain the substantive Lambert-series inequalities.

### 36.3.5 Cyclotomic resurgence with fixed normalization

Let  $\zeta$  be a primitive  $m$ -th root of unity and, for  $\Re t > 0$  and  $|a| < 1$ , take the logarithm determined by the absolutely convergent series

$$L_a(t) = \text{Log}(a; \zeta e^{-t})_\infty = -\sum_{r=1}^{\infty} \frac{a^r}{r(1 - \zeta^r e^{-rt})}.$$

The singular action and constant term forced by this normalization are

$$S_{-1}(a) = -\frac{\text{Li}_2(a^m)}{m^2},$$

$$S_0(a) = \frac{1}{2m} \log(1 - a^m) - \sum_{\substack{r \geq 1 \\ m \nmid r}} \frac{a^r}{r(1 - \zeta^r)}.$$

A formal series  $\widehat{R}(t) = \sum_{n \geq 0} c_n t^n$  is *Gevrey one* here if  $|c_n| \leq C A^n n!$  for some  $C, A > 0$ . Its formal Borel transform is

$$\mathcal{B}\widehat{R}(\xi) = \sum_{n \geq 0} \frac{c_{n+1}}{n!} \xi^n;$$

the constant  $c_0$  is carried separately by Borel–Laplace summation.

**Conjecture 36.14** (Cyclotomic resurgent inverse). *Let  $K \Subset \{a : |a| < 1\}$  and  $0 < \delta < \pi/2$ . Uniformly for  $a \in K$ , as  $t \rightarrow 0$  in  $|\arg t| \leq \pi/2 - \delta$ , the normalized function*

$$R_a(t) = L_a(t) - \frac{S_{-1}(a)}{t} - S_0(a)$$

*has a Gevrey-one asymptotic expansion. Its Borel transform admits endless analytic continuation: for each  $a \in K$ , there is a locally finite set  $\Sigma_{a,\zeta} \subset \mathbb{C}$ , with  $0 \notin \Sigma_{a,\zeta}$ , such that the germ at zero continues along every path in  $\mathbb{C} \setminus \Sigma_{a,\zeta}$  issuing from zero. In every direction whose open Borel ray avoids  $\Sigma_{a,\zeta}$  and for which the Laplace integral converges, its Borel sum equals the exact  $R_a(t)$ .*

*Moreover, fix  $a_0 \in \text{int } K$ , and let  $v(t) \rightarrow 0$  be a resurgent germ on a non-Stokes sector. Let  $a(t) \in K$ , with  $a(t) \rightarrow a_0$ , be the chosen local solution of*

$$L_{a(t)}(t) = L_{a_0}(t) + v(t).$$

If  $\partial_a L_{a(t)}(t)$  is bounded away from zero on every closed subsector under consideration, then  $a(t)$  is resurgent, and its Stokes automorphisms are obtained from those of  $L$  and  $v$  by the same local implicit inversion. Exponentiating the displayed equation gives the corresponding statement for the chosen branch of the product inverse.

The exact Lambert-series identity for  $L_a$ , its residue-class splitting, and the Bernoulli expansion of the  $m$ -divisible terms explain  $S_{-1}$ ,  $S_0$ , and the observed Gevrey growth. They do not prove endless continuation or locate every Borel singularity. The remaining obligations are uniform factorial remainder bounds on  $K$ , analytic continuation and growth estimates in the Borel plane, identification of the Stokes set, equality of Borel sums with the convergent product, and a quantitative implicit-function argument for the inverse clause.

### 36.3.6 Editorial disposition of the remaining proposals

Several former conjectures were deliberately not carried forward.

- The proposed finite-CDF inversion statement combined an elementary simple-fold theorem with an unproved analyticity assertion about a specific density. The fold theorem is covered by the branch-aware certificates above; any density claim must name the density and regularity hypothesis.
- The phrase *generic Gaussian monodromy has the permitted group* did not define either the parameter space or the permitted group. The finite-product conjecture above is its falsifiable replacement.
- Broad inverse-universality and positivity-after-normalization claims did not specify a topology, scaling, normalization, or limiting class.
- Total cumulant identifiability and discrete Gaussian uniqueness at a transcendental base were upgraded to Theorems 35.7 and 36.8.

The omitted themes may still motivate research programs, but they are not mathematical assertions until their quantifiers and normalizations are fixed.

## 36.4 Lean source-status map

This section is a source-presence crosswalk, not compilation evidence. The declarations below occur in the indicated Lean source modules; no fresh build status is asserted by this table. A source declaration supports exactly the topic stated in the last column; it does not silently formalize the stronger complex certification theorems or conjectures of this chapter. Module names suppress the common `FabiusFunction.` prefix, and declaration names suppress the common `Fabius.` namespace; the two `Bell.` declarations retain their different namespace explicitly. Every displayed suffix otherwise matches the source exactly.

| Lean module                | Exact source declarations                                                                                                                                                                   | Legitimate human crosswalk                                                                                     |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <code>FabiusInverse</code> | <code>fabiusReal_fabiusInv;</code><br><code>fabiusInv_fabiusReal;</code><br><code>strictMonoOn_fabiusInv;</code><br><code>continuous_fabiusInv;</code><br><code>fabiusInv_hasDerivAt</code> | Two-sided inversion on the proved domain, strict monotonicity, continuity, and the regular inverse derivative. |

*Continued on the next page*

| Lean module                                                    | Exact source declarations                                                                                                                                                                                                                                                                                             | Legitimate human crosswalk                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FabiusInverseEffectiveContinuity                               | <code>abs_fabiusInv_sub_lt_inverse_two_pow_of_lt_factorialDenominator; fabiusInv_effectivelyUniformContinuous</code>                                                                                                                                                                                                  | An explicit factorial-denominator inverse modulus and effective uniform continuity.                                                                                                                                                                                                                                                                |
| FabiusInverseLogarithmicModulus                                | <code>abs_fabiusInv_sub_lt_inv_nat_of_lt_logarithmicFactorialDenominator; fabiusInv_effectivelyUniformContinuous_logarithmic; fabiusInv_effectivelyUniformContinuous_logarithmicDelta</code>                                                                                                                          | The later logarithmic inverse moduli, including the delta formulation.                                                                                                                                                                                                                                                                             |
| LambertWBranchPointGeometry;<br>LambertWBranchPointAsymptotics | <code>tendsto_deriv_principalLambertW_branchPoint_atTop; tendsto_deriv_lowerLambertW_branchPoint_atBot; principalLambertW_not_differentiableAt_branchPoint; lowerLambertW_not_differentiableAt_branchPoint; principalLambertW_add_one_isEquivalent_branchPoint; lowerLambertW_add_one_isEquivalent_branchPoint</code> | The opposite vertical tangents, failure of finite differentiability, and the signed leading square-root laws at the real Lambert- $W$ branch point. They identify the correct signed leading square-root scale for a possible regularization; they establish neither a regularized analytic coordinate chart nor a higher Puiseux remainder bound. |
| GeometricUniformLaw                                            | <code>hasSum_geometricUniformWeight; geometricUniformSeries_mem_Icc; geometricUniformDistribution_selfSimilar; geometricUniformDistribution_support_eq_Icc</code>                                                                                                                                                     | Convergence, support, and self-similarity of Lean's normalized $[0, 1]$ geometric uniform law. If $Z = (1 - q) \sum_{j \geq 0} q^j V_j$ and $U_j = 2V_j - 1$ , then $\sum_{j \geq 0} q^j U_j = (2Z - 1)/(1 - q)$ . This affine rescaling underlies Theorem 35.7.                                                                                   |
| GeometricSincCharacteristicFunction                            | <code>charFun_geometricUniformDistribution_eq_phase_mul_geometricSincProduct; charFun_geometricUniformDistribution_eq_phase_mul_geometricReciprocalGamma</code>                                                                                                                                                       | The sinc-product characteristic function and its reciprocal-gamma form.                                                                                                                                                                                                                                                                            |
| SincZetaRemainder; FabiusComputability                         | <code>norm_sincZeta_tail_le; rvachevFourierProduct_eq_prefix_mul_correction_mul_tail; abs_uniformPartialCDF_sub_fabiusReal_le; fabiusEffectiveUniformModulus</code>                                                                                                                                                   | Certified sinc-product and CDF tails and an effective forward Fabius modulus. These are nearby formal certificates, not the general-purpose complex Theorems 36.1–36.5.                                                                                                                                                                            |
| QPochhammerElementaryIdentities;<br>QBinomialReciprocity       | <code>finiteQPochhammerIn_base_reversal; finiteQPochhammerIn_inv_base_reversal; gaussianBinomial_adjacent_mul; gaussianBinomial_reciprocity; gaussianBinomial_neg_one_eq_zero_of_odd_degree; gaussianBinomial_neg_one_even_odd_eq_zero</code>                                                                         | Finite base reversal, adjacent ratios, Gaussian reciprocity, and the odd-degree vanishing mechanism at $q = -1$ .                                                                                                                                                                                                                                  |

*Continued on the next page*

| Lean module                                                  | Exact source declarations                                                                                                                                                              | Legitimate human crosswalk                                                                                                                                                             |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GaussianBinomial<br>AtNegOne                                 | gaussianBinomial_neg_one_even_even;<br>gaussianBinomial_neg_one_odd_even;<br>gaussianBinomial_neg_one_odd_odd;<br>finiteQPochhammerIn_neg_one_even;<br>finiteQPochhammerIn_neg_one_odd | The remaining parity evaluations for Gaussian coefficients and the finite $q$ -Pochhammer formulas at $q = -1$ ; together with the preceding row these form the complete parity atlas. |
| BellPolynomialIn<br>version;<br>QuadraticAsymptoticInversion | Bell.complete_cumulant;<br>Bell.cumulant_complete;<br>quadratic_asymptotic_inversion                                                                                                   | Mutual moment-cumulant inversion and the proved quadratic asymptotic inversion schema.                                                                                                 |

No declaration in this audited source crosswalk is cited as proving the exact statements of the general  $q$ -Pochhammer derivative-tail theorem, interval Newton theorem, complex Krawczyk theorem, Rouché disk theorem, validated argument principle, total-cumulant parameter-identification theorem, or transcendental Gaussian uniqueness theorem as stated here. They are complete human-readable proofs in this chapter, not claimed Lean milestones. Likewise, no Lean counterpart is claimed for the five conjectures in Section 36.3.

## Appendix A

# Inverse-source provenance and concordance

### A.1 Source corpus and editorial provenance

The canonical volume is a semantic consolidation of six independently delivered source packages. They are not numbered editions of one manuscript: overlap records independent rediscovery or reuse, and disagreement must be settled mathematically rather than by choosing the newest-looking file. The historical paths in [Table A.1](#) were relative to the former sibling layout under exponents *s*- and -*q*-series; the pinned pre-retirement revision and repository history preserve them. The row counts are the numbers of theorem-like source environments recorded in the machine-readable concordance.

The editorial disposition is governed by five rules.

1. A semantic duplicate is stated once, with the strongest correct hypotheses and one complete proof. Several source rows may therefore point to one canonical label.
2. A useful source theorem may be strengthened, split into logically independent results, or absorbed into a more general lemma. The concordance records the primary destination and explains any one-to-many refinement in its disposition note.
3. A false, ill-posed, redundant, or unfalsifiable assertion is not preserved merely because it carried a theorem or conjecture heading. Interesting statements with genuine missing obligations survive only in explicitly labelled *Conjecture* environments.
4. Numerical and symbolic calculations are evidence only to the extent certified by reproducible exact arithmetic, interval enclosures, or a proved residual bound. Unsupported large factorizations are certification targets, not hidden premises of unconditional theorems.
5. A script, table, captured output, or figure is migrated only when it uniquely supports a retained result or reproducible calculation. Byte-identical duplicates and generated publication byproducts are not separate mathematical sources. Superseded layouts remain recoverable from repository history.

### A.2 The exhaustive result concordance

The file `theorem_concordance.csv` is the machine-readable ledger for the source corpus. It has one header and exactly 260 data rows, one for every environment recognized by the extraction audit. The package totals are

$$49 + 38 + 48 + 59 + 29 + 37 = 260.$$



| Source package and principal file                                            | Rows | Distinctive contribution and canonical disposition                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| inverse_q_analog_functions_report/<br>inverse_q_analog_functions.tex         | 49   | The clearest branchwise narrative, passive-parameter viewpoint, real and complex sheet atlases, and continuation algorithms. Its reusable inverse calculus is absorbed into the universal chapter; its product-specific geometry is distributed among the finite and infinite product chapters.                                     |
| inverse_q_analog_jet_atlas/<br>inverse_q_analog_jet_atlas.tex                | 38   | Formal triangular reversion, Lagrange–Bürmann and Lagrange–Good formulas, Bell-polynomial organization, mixed jets, and coefficient atlases. These are consolidated, with corrected algebraic hypotheses, in the universal inversion calculus.                                                                                      |
| inverse_q_analogs_extended_report/<br>inverse_q_analogs_extended_report.tex  | 48   | The broadest collection of exceptional finite regimes, radial inverses, $q$ -special functions, certification ideas, and Fabius–Rvachev recovery. Correct special cases are retained at their natural topical destinations; overgeneralized claims are corrected or retired.                                                        |
| inverse_q_analogs_report/<br>inverse_q_analogs_report.tex                    | 59   | Discriminants, remote inverse sheets, real branch geometry, numerical certification, and the earliest large conjecture register. Exact results are deduplicated; symbolic collision claims without independently checkable certificates remain conditional targets rather than theorems.                                            |
| inverse_q_analogs_report-2/<br>inverse_q_analogs_report.tex                  | 29   | An independent synthesis emphasizing reciprocal regimes, roots of unity, compact coefficient tables, and algorithmic cross-checks. It supplies corroborating provenance and occasionally sharper normalizations, not a presumption of chronological priority.                                                                       |
| q_pochhammer_q_binomial_expansions_report/<br>q_analog_expansions_report.tex | 37   | The primary forward-expansion engine: finite and infinite $q$ -Pochhammer products, Gaussian and multinomial coefficients, Bernoulli/polylogarithmic expansions, roots of unity, and $q$ -special functions. Its uniform head–tail remainder arguments are retained wherever termwise formal expansion alone would be insufficient. |

Table A.1: The six source packages and their canonical roles.

By kind, the inventory contains 131 theorems, 28 propositions, 16 corollaries, 2 lemmas, 32 conjectures, 28 problems or research problems, 9 definitions, 5 algorithms, 4 examples, 4 principles, and 1 computational result. The extractor found an intervening proof or proofidea environment after 163 rows and no such environment after 97. This last field is a syntactic audit signal, not a verdict on correctness: conjectures and definitions normally have no proof, while a source proof environment may still contain a gap.

| CSV field group                                                                                                                                               | Meaning                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>source_key</code>                                                                                                                                       | Stable key formed from the package name and the source label, or from a deterministic kind-and-ordinal key when the source supplied no label.                                                             |
| <code>source_package</code> ,<br><code>source_file</code> ,<br><code>source_line</code>                                                                       | Exact source location. Line numbers belong to the archived source snapshot and are provenance coordinates, not canonical page references.                                                                 |
| <code>source_label</code> ,<br><code>source_kind</code> ,<br><code>source_title</code> ,<br><code>source_chapter</code> ,<br><code>source_section_path</code> | The source manuscript’s own classification and surrounding hierarchy. These fields preserve what was claimed without endorsing it.                                                                        |
| <code>source_proof_present</code>                                                                                                                             | Whether a proof or proofidea environment occurs before the next recognized result. It does not assert that the proof is sound, complete, or relevant to the exact statement.                              |
| <code>canonical_label</code> ,<br><code>canonical_status</code>                                                                                               | The primary canonical destination and its strict status. Repeated labels across source rows express semantic deduplication. A split or partial retention is described in <code>disposition_notes</code> . |
| <code>lean_module</code> ,<br><code>lean_declaration</code>                                                                                                   | An exact formal counterpart, when one has actually been identified and validated. Related supporting lemmas do not justify filling these fields for a stronger human theorem.                             |
| <code>disposition_notes</code>                                                                                                                                | The reason for strengthening, merging, splitting, correcting, making conditional, retaining as a conjecture, or retiring the source result.                                                               |

Table A.2: Interpretation of the concordance columns.

For an assertion, `canonical_status` uses exactly the three-way vocabulary of [Section 29.2](#): *Lean-proved*, *human-proved frontier result*, or *conjecture*. A definition, notation declaration, or purely descriptive algorithm is not forced into an assertion status; its retention or retirement is explained in the disposition field. Likewise, *merged*, *strengthened*, *corrected*, and *retired* are dispositions, not proof statuses.

The extractor initially generated the five editorial columns empty, because text matching cannot decide mathematical equivalence. In the reviewed ledger, all 260 rows now have a canonical status and disposition note; 180 map to canonical labels, while the remaining 80 are explicitly absorbed, retired, or not separately asserted. Exact Lean module and declaration cells remain empty unless an exact formal counterpart is claimed. The retained reviewed CSV is authoritative. The revision-backed extractor `audit/extract_source_results.py` reproduces and verifies its ten immutable source columns from the commit pinned in `audit/SOURCE_REVISION`; it refuses to overwrite the five editorial columns.

### A.3 Notation and mandatory branch data

An inverse symbol denotes a branch only after its data have been fixed. At minimum one must record the forward map  $F$ , the active variable, all passive parameters, the target or limiting coordinate, the

base point or real domain, the continuation path or sheet, and every logarithm or fractional-power determination. The following table is a compact decoding guide for the notation used throughout the volume.

Table A.3: Notation and the branch information suppressed by a bare inverse symbol.

| Notation                                     | Meaning and domain                                                                                                                    | Branch data that must accompany inversion                                                                                                                                        |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $P_n(a, q) = (a; q)_n$                       | Finite product $\prod_{j=0}^{n-1} (1 - aq^j)$ , with an empty product equal to one.                                                   | State whether $a$ or $q$ is active; fix $n$ and the other parameter; record a base point, real monotonicity interval, or Puiseux sheet.                                          |
| $P(a, q) = (a; q)_\infty$                    | Normally convergent product only for $ q  < 1$ . Boundary radial limits and exterior reciprocal normalizations are different objects. | For argument inversion, give the principal interval or the zero-gap index and side. For base inversion, give the interior path or the radial normalization and target logarithm. |
| $\mathcal{A}_q(y)$                           | The principal real inverse of $a \mapsto (a; q)_\infty$ for $0 < q < 1$ , $y > 0$ , selected by $a < 1$ .                             | No sheet index is needed on this monotone branch; a nonprincipal real or complex inverse must not reuse this symbol without an explicit sheet label.                             |
| $\begin{bmatrix} n \\ k \end{bmatrix}_q$     | Gaussian polynomial, including its polynomial continuation where the factorial quotient reads $0/0$ .                                 | Fix $n, k$ ; distinguish the positive real branch, a reciprocal $q \leftrightarrow q^{-1}$ branch, and a cyclotomic zero or local sheet.                                         |
| $q = e^{-t}$                                 | Logarithmic coordinate near $q = 1$ , with $t = -\text{Log } q$ .                                                                     | Record the logarithm branch and, in a complex asymptotic statement, a closed sector contained in $\Re t > 0$ .                                                                   |
| $q = \zeta e^{-t}$                           | Radial or sectorial approach to a primitive root $\zeta$ of order $d$ .                                                               | Record $\zeta, d$ , the sector or real radial path, and the continuation of every logarithm. This notation never asserts a germ across $ q  = 1$ .                               |
| $s = 1 - y, \eta = ((1 - a) - y)/(a(1 - a))$ | Regular target coordinates at the unit level and at the small-base endpoint.                                                          | Record the source base point and the nonvanishing denominators. A different normalization changes every printed inverse coefficient.                                             |
| $X = -\text{Log } y + C$                     | Reciprocal-log scale used when a forward logarithm begins $-A/t + C + \dots$ .                                                        | Fix the target logarithm, the constant $C$ , the sheet with $t \sim A/X$ , and a sector on which $A/X$ is admissible.                                                            |
| $W_k$                                        | The $k$ -th Lambert- $W$ determination.                                                                                               | Record $k$ and verify the original chosen-log equation. Solving an exponentiated equation alone can admit candidates on the wrong logarithm sheet.                               |
| $(y - y_0)^{1/m}$                            | Ramified coordinate at a zero of order $m$ .                                                                                          | Choose the root on a simply connected punctured target domain, or work on the ramified cover; all $m$ determinations coalesce at $y_0$ .                                         |
| $\Gamma_q, \psi_q, B_q$                      | The $q$ -gamma, $q$ -digamma, and $q$ -beta normalizations specified in the $q$ -special-function chapter.                            | State the real parameter interval or complex continuation, the active parameter, the base regime, and any reciprocal continuation for $q > 1$ .                                  |

| Notation                    | Meaning and domain                                                                              | Branch data that must accompany inversion                                                                                                                                        |
|-----------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\phi_q(t)$ , sinc products | Characteristic functions of the normalized geometric-uniform law and the Fabius–Rvachev member. | Fix the probability normalization, phase convention, real base $0 < q < 1$ , and whether inversion concerns a distribution parameter, a Fourier value, or a cumulant observable. |

## A.4 Source-to-canonical topical crosswalk

The row-level concordance is the authoritative provenance mechanism. For human navigation, [Table A.4](#) records where the major source themes landed. A source can appear in several canonical chapters, and a canonical theorem can synthesize several sources.

| Source package                                         | Principal source themes                                                                                                                                                 | Canonical destinations                                                     |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <code>inverse_q_analog_functions_report</code>         | Branch orientation, regular inverse germs, sensitivities, real and complex sheet atlases, continuation.                                                                 | <a href="#">Sections 30.1, 30.5, 31.1, 32.2 and 36.1.</a>                  |
| <code>inverse_q_analog_jet_atlas</code>                | Triangular coefficient recurrences, Lagrange inversion, Bell polynomials, mixed jets, multivariate and Good inversion.                                                  | <a href="#">Sections 30.2 to 30.4, 30.8 and 30.9.</a>                      |
| <code>inverse_q_analogs_extended_report</code>         | Exceptional finite bases, $q = 0, \pm 1$ , radial cyclotomic inversion, $q$ -special functions, certification, Fabius–Rvachev recovery.                                 | <a href="#">Sections 31.2, 31.2.4, 32.4, 32.8, 34.3, 35.2 and 36.1.</a>    |
| <code>inverse_q_analogs_report</code>                  | Discriminants, real gaps and remote sheets, $q$ -special inverses, certification algorithms, conjecture register.                                                       | <a href="#">Sections 31.1, 32.2, 34.4, 36.1 and 36.3 and Theorem 32.4.</a> |
| <code>inverse_q_analogs_report-2</code>                | Reciprocal regimes, roots of unity, compact inverse tables, independent algorithmic derivations.                                                                        | <a href="#">Sections 31.2, 32.5, 32.8, 33.6, 34.6 and 36.1.</a>            |
| <code>q_pochhammer_q_binomial_expansions_report</code> | Exact forward coefficient engines, uniform remainders, Bernoulli and polylogarithmic expansions, Gaussian/multinomial limits, roots of unity, $q$ -gamma and $q$ -beta. | <a href="#">Sections 31.1, 32.5, 32.6, 33.3, 33.7, 34.3 and 34.5.</a>      |

Table A.4: Topical source-to-canonical crosswalk.

## A.5 Formalization boundary

The exact declaration-level source map is already given in [Section 36.4](#) and is not duplicated here. Its current formal surface has five principal islands:

1. classical Fabius inverse identities, monotonicity, continuity, differentiability, and explicit effective-continuity moduli;
2. the geometric-uniform probability law, its support and self-similarity, the geometric sinc characteristic-function identity, certified sinc and CDF tails, and an effective forward Fabius modulus;
3. selected finite  $q$ -Pochhammer and Gaussian-polynomial identities, including reciprocity and exact values at  $q = -1$ ;
4. Bell/cumulant conversion and a quadratic asymptotic-inversion lemma;
5. real Lambert- $W$  branch-point geometry: the opposite vertical tangents and failure of finite differentiability, together with the signed leading square-root equivalences for the principal and lower branches.

The last island comes from the companion Lambert- $W$  corpus, specifically `LambertWBranchPointGeometry.lean` and `LambertWBranchPointAsymptotics.lean`. It is supporting formalization for the universal fold discussion, not a seventh inverse- $q$  precursor; accordingly it adds no row to the frozen 260-result source ledger. The signed leading equivalences do not include a higher Puiseux coefficient or remainder theorem, still less a formal inversion theorem for a  $q$ -product.

These declarations support parts of the canonical narrative; they do not silently formalize stronger neighbors. In particular, no Lean status is claimed here for the general analytic and formal inverse theorems beyond the named Bell and quadratic support, or for the human proofs of infinite-product normal convergence, remote-gap and log-periodic asymptotics, all-order  $q \rightarrow 1$  and cyclotomic expansions, Dedekind-eta completion, Gaussian and  $q$ -special global inverse atlases, complex certification algorithms, or the conjectures in [Section 36.3](#). A theorem in those families remains a *human-proved frontier result* unless an exact named declaration is listed for that exact statement.

The Lean table in [Section 36.4](#) is explicitly a source inspection, not a build report. No Lean process was run while preparing this appendix, so it contributes no fresh compiler evidence. Likewise, a nonempty `lean_module` or `lean_declaration` field in a future concordance row will be a cross-reference, not by itself proof of a current aggregate build. Compiler validation, source inspection, a focused module build, and a full repository build are distinct evidence levels and must be reported separately.

## A.6 Repository and source note

No citation key occurs in the canonical package at the time of this consolidation, so this volume does not manufacture a bibliography from the longer reading lists of its precursors. Its immediate documentary sources were the six packages in [Table A.1](#). The retained package-level ledger `PROVENANCE.md`, row-level ledger `theorem_concordance.csv`, editorial contract `README.md`, and revision-backed extractor `audit/extract_source_results.py` are the live canonical records. Repository history preserves superseded layouts and retired duplicates.

Classical named tools used in proofs—the inverse-function theorem, Lagrange–Bürmann and Lagrange–Good inversion, Rouché’s theorem, Dedekind-eta modularity, and standard special-function identities—are invoked with their exact hypotheses at the point of use. If external citations are introduced in a later edition, their actual citation keys and bibliographic entries should be added together; an uncited generic reading list would not improve the present provenance record.

# Appendix B

## Formula atlas

This appendix is a navigation table, not a substitute for hypotheses and proofs. Each formula points back to the theorem where its exact domain and derivation are given.

| Name                      | Formula                                                                                                                                                                                                                  | Location                        |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Concatenation             | $(a; q)_{m+n} = (a; q)_m (aq^m; q)_n$                                                                                                                                                                                    | <a href="#">Proposition 1.2</a> |
| Dissection                | $(a; q)_{rn} = \prod_{j=0}^{r-1} (aq^j; q^r)_n$                                                                                                                                                                          | <a href="#">Proposition 1.3</a> |
| Dissection with remainder | $(a; q)_{rn+u} = \prod_{s<u} (aq^s; q^r)_{n+1} \prod_{s\geq u} (aq^s; q^r)_n$                                                                                                                                            | <a href="#">Corollary 1.4</a>   |
| Base reversal             | $(a; q)_n = (-a)^n q^{\binom{n}{2}} (a^{-1} q^{1-n}; q)_n, aq \neq 0$                                                                                                                                                    | <a href="#">Proposition 1.6</a> |
| Negative index            | $(a; q)_{-n} = 1/(aq^{-n}; q)_n$                                                                                                                                                                                         | <a href="#">Proposition 1.8</a> |
| Infinite logarithm        | $\log(a; q)_\infty = -\sum_{r\geq 1} a^r/(r(1-q^r))$                                                                                                                                                                     | <a href="#">Theorem 1.16</a>    |
| Gaussian product          | $\begin{bmatrix} n \\ k \end{bmatrix}_q = (q; q)_n / ((q; q)_k (q; q)_{n-k})$                                                                                                                                            | <a href="#">Proposition 2.2</a> |
| Pascal I                  | $\begin{bmatrix} n \\ k \end{bmatrix}_q = q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q$                                                                                     | <a href="#">Theorem 2.3</a>     |
| Pascal II                 | $\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q$                                                                                 | <a href="#">Theorem 2.3</a>     |
| Reciprocity               | $\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}$                                                                                                                      | <a href="#">Theorem 2.5</a>     |
| Finite theorem            | $(z; q)_n = \sum_k (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k$                                                                                                                                   | <a href="#">Theorem 3.1</a>     |
| Reciprocal theorem        | $(z; q)_{n+1}^{-1} = \sum_{k\geq 0} \begin{bmatrix} n+k \\ k \end{bmatrix}_q z^k$                                                                                                                                        | <a href="#">Theorem 3.6</a>     |
| $q$ -Vandermonde          | $\begin{bmatrix} m+n \\ r \end{bmatrix}_q = \sum_k q^{\binom{m-k}{2} + \binom{r-k}{2}} \begin{bmatrix} m \\ k \end{bmatrix}_q \begin{bmatrix} n \\ r-k \end{bmatrix}_q$                                                  | <a href="#">Theorem 3.8</a>     |
| $q$ -binomial inversion   | $b_n = \sum_k \begin{bmatrix} n \\ k \end{bmatrix}_q a_k \iff a_n = \sum_k (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q b_k$                                                                     | <a href="#">Theorem 3.12</a>    |
| Infinite theorem          | $\sum_{n\geq 0} (a; q)_n z^n / (q; q)_n = (az; q)_\infty / (z; q)_\infty$                                                                                                                                                | <a href="#">Theorem 8.1</a>     |
| Euler I                   | $(z; q)_\infty = \sum_{n\geq 0} (-1)^n q^{\binom{n}{2}} z^n / (q; q)_n$                                                                                                                                                  | <a href="#">Corollary 8.2</a>   |
| Euler II                  | $1/(z; q)_\infty = \sum_{n\geq 0} z^n / (q; q)_n,  z  < 1$                                                                                                                                                               | <a href="#">Corollary 8.2</a>   |
| $q$ -Gauss                | ${}_2\phi_1(a, b; c; q, c/(ab)) = (c/a, c/b; q)_\infty / (c, c/(ab); q)_\infty$                                                                                                                                          | <a href="#">Theorem 9.6</a>     |
| Heine                     | ${}_2\phi_1(a, b; c; q, z) = \frac{(b, az; q)_\infty}{(c, z; q)_\infty} {}_2\phi_1(c/b, z; az; q, b)$                                                                                                                    | <a href="#">Theorem 9.5</a>     |
| $q$ -Pfaff–Saalschütz     | Terminating balanced ${}_3\phi_2$ product evaluation                                                                                                                                                                     | <a href="#">Theorem 9.9</a>     |
| $q$ -Chu–Vandermonde      | Two terminating ${}_2\phi_1$ evaluations                                                                                                                                                                                 | <a href="#">Corollary 9.10</a>  |
| Ramanujan ${}_1\psi_1$    | ${}_1\psi_1(a; b; q, z) = (q, b/a, az, q/(az); q)_\infty / (b, q/a, z, b/(az); q)_\infty$                                                                                                                                | <a href="#">Theorem 11.3</a>    |
| Finite triple product     | $(z; q)_N (q/z; q)_N = \sum_{k=-N}^N (-1)^k q^{\binom{k}{2}} \begin{bmatrix} 2N \\ N+k \end{bmatrix}_q z^k$                                                                                                              | <a href="#">Theorem 14.3</a>    |
| Jacobi triple product     | $\sum_{n\in\mathbb{Z}} (-1)^n q^{\binom{n}{2}} z^n = (q, z, q/z; q)_\infty$                                                                                                                                              | <a href="#">Theorem 14.1</a>    |
| Rogers–Ramanujan pair     | $\sum_{n\geq 0} \frac{q^{n^2}}{(q; q)_n} = \frac{1}{(q, q^4; q^5)_\infty}, \sum_{n\geq 0} \frac{q^{n^2+n}}{(q; q)_n} = \frac{1}{(q^2, q^3; q^5)_\infty}$                                                                 | <a href="#">Theorem 17.9</a>    |
| Andrews–Gordon family     | $\sum_{\substack{n_1, \dots, n_{k-1} \geq 0 \\ N_j = n_j + \dots + n_{k-1}}} \frac{q^{\sum_j N_j^2 + \sum_{j=1}^{k-1} N_j}}{\prod_j (q; q)_{n_j}} = \frac{(q^i, q^{2k+1-i}, q^{2k+1}; q^{2k+1})_\infty}{(q; q)_\infty},$ | <a href="#">Theorem 18.10</a>   |
| Rogers–Ramanujan fraction | $\frac{F_1(q)}{F_0(q)} = \frac{1}{1 + \frac{q}{1 + \frac{q^2}{1 + \ddots}}}$                                                                                                                                             | <a href="#">Theorem 18.8</a>    |

| Name                     | Formula                                                                                                                                               | Location                         |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Pentagonal theorem       | $(q; q)_\infty = \sum_{n \in \mathbb{Z}} (-1)^n q^{n(3n-1)/2}$                                                                                        | <a href="#">Theorem 14.7</a>     |
| $q$ -derivative          | $D_q f(x) = (f(x) - f(qx)) / ((1 - q)x)$                                                                                                              | <a href="#">Proposition 12.1</a> |
| Jackson integral         | $\int_0^a f(x) d_q x = (1 - q)a \sum_{n \geq 0} q^n f(aq^n)$                                                                                          | <a href="#">Theorem 12.11</a>    |
| $q$ -gamma               | $\Gamma_q(x) = (1 - q)^{1-x} (q; q)_\infty / (q^x; q)_\infty$                                                                                         | <a href="#">Theorem 13.2</a>     |
| Cyclotomic factorization | $\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_d \Phi_d(q)^{\lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n-k)/d \rfloor}$                    | <a href="#">Theorem 19.1</a>     |
| $q$ -Lucas               | $\begin{bmatrix} a+d+b \\ rd+s \end{bmatrix}_q \equiv \begin{bmatrix} a \\ r \end{bmatrix}_q \begin{bmatrix} b \\ s \end{bmatrix}_q \pmod{\Phi_d(q)}$ | <a href="#">Theorem 19.7</a>     |
| $q$ -Babbage             | $\begin{bmatrix} an \\ bn \end{bmatrix}_q \equiv \begin{bmatrix} a \\ b \end{bmatrix}_{q^{n^2}} \pmod{\Phi_n(q)^2}$                                   | <a href="#">Theorem 19.11</a>    |
| Negative upper index     | $\begin{bmatrix} -m \\ k \end{bmatrix}_q = (-1)^k q^{-mk - \binom{k}{2}} \begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q$                                  | <a href="#">Proposition 20.2</a> |
| Geometric Newton basis   | $P(x) = \sum_k c_k (-1)^k q^{\binom{k}{2}} (x; q^{-1})_k$                                                                                             | <a href="#">Theorem 20.10</a>    |
| Central reduction        | $\begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} / (-q; q)_{2k} = (q; q^2)_k / (q^2; q^2)_k$                                                              | <a href="#">Lemma 21.1</a>       |
| Bernoulli expansion      | All-orders expansion of $\log(\begin{bmatrix} n \\ k \end{bmatrix}_e^{-t} / \binom{n}{k})$                                                            | <a href="#">Theorem 22.2</a>     |

# Appendix C

## Limit and specialization dictionary

### C.1 Elementary limits

For fixed nonnegative integers, the following transitions are direct consequences of the product formulas:

$$\begin{aligned} \lim_{q \rightarrow 1} [n]_q &= n, & \lim_{q \rightarrow 1} (1 - q)^{-n} (q^a; q)_n &= (a)_n, \\ \lim_{q \rightarrow 1} \begin{bmatrix} n \\ k \end{bmatrix}_q &= \binom{n}{k}, & \lim_{n \rightarrow \infty} \begin{bmatrix} n \\ k \end{bmatrix}_q &= \frac{1}{(q; q)_k} \quad (|q| < 1), \\ \lim_{q \rightarrow 1^-} \Gamma_q(x) &= \Gamma(x), & \lim_{q \rightarrow 0} \begin{bmatrix} n \\ k \end{bmatrix}_q &= 1 \quad (0 \leq k \leq n). \end{aligned}$$

Each has been proved in the main text; the relevant references are [Propositions 1.19](#) and [22.11](#), [corollary 2.9](#), and [Theorems 13.5](#) and [22.6](#).

The dictionary also contains transitions in the size parameter. For example,  $\lim_{n \rightarrow \infty} \begin{bmatrix} n \\ k \end{bmatrix}_q = 1/(q; q)_k$  appears above. Its central counterpart is

$$\lim_{N \rightarrow \infty} \begin{bmatrix} 2N \\ N + k \end{bmatrix}_q = \frac{1}{(q; q)_\infty} \quad (k \in \mathbb{Z} \text{ fixed, } 0 < |q| < 1),$$

which is [Theorem 22.7](#) when  $k = 0$  and follows for every fixed  $k$  by the same argument; it is what carries the finite triple product ([14.3](#)) to Jacobi's ([14.2](#)). See [Theorem 14.3](#) and [corollary 14.5](#), and compare [Proposition 16.6](#), which records the same stabilization coefficientwise.

### C.2 Classical shadows of the main summations

Under the standard scaling  $a = q^A$ ,  $b = q^B$ , and  $q \rightarrow 1^-$ , terminating basic hypergeometric sums tend to ordinary hypergeometric sums. The mechanism is

$$\frac{(q^A; q)_n}{(1 - q)^n} \longrightarrow (A)_n, \quad \frac{(q; q)_n}{(1 - q)^n} \longrightarrow n!.$$

Thus:

1. the finite  $q$ -binomial theorem tends to Newton's binomial theorem;
2.  $q$ -Chu–Vandermonde tends to Chu–Vandermonde;
3.  $q$ -Pfaff–Saalschütz tends to Pfaff–Saalschütz;
4. the Jackson  $q$ -beta integral tends to Euler's beta integral;



5. [Theorem 21.2](#) tends to [Corollary 21.3](#);
6. [Theorem 21.4](#) tends, at the level of its Lambert tail and prefactor, to [Theorem 21.7](#).

### C.3 Roots of unity

At a root of unity, factorial quotients should be replaced by polynomial formulas before evaluation. The basic dictionary is:

$$\begin{aligned} \Phi_d(q) \mid \begin{bmatrix} n \\ k \end{bmatrix}_q &\iff k \bmod d > n \bmod d, \\ \begin{bmatrix} ad + b \\ rd + s \end{bmatrix}_q &\equiv \begin{pmatrix} a \\ r \end{pmatrix} \begin{bmatrix} b \\ s \end{bmatrix}_q \pmod{\Phi_d(q)}, \\ \begin{bmatrix} n \\ k \end{bmatrix}_q \Big|_{q=-1} &\text{ is given by [Corollary 19.8](#).} \end{aligned}$$

These statements turn apparently singular substitutions into finite enumerations and are the algebraic basis of the subset cyclic-sieving theorem.

## Appendix D

# Proof-dependency guide

The monograph can be read linearly, but several shorter routes are useful.

### Route to Gaussian combinatorics

finite products  $\longrightarrow$   $q$ -Pascal  $\longrightarrow$  finite  $q$ -binomial theorem  $\longrightarrow$  partitions, paths, subspaces.

Read [Chapters 1 to 3, 5 and 6](#).

### Route to product–sum identities

infinite  $q$ -binomial theorem  $\longrightarrow$  Heine transformation  $\longrightarrow$   $q$ -Gauss,  
negative indices and bilateral convergence  $\longrightarrow$   ${}_1\psi_1$ ,  
Euler expansions  $\longrightarrow$  Jacobi triple product  $\longrightarrow$  Bailey pairs.

Read [Chapters 8, 9, 11, 14 and 17](#).

### Route to arithmetic

$1 - q^n = \prod_{d|n} \Phi_d(q) \longrightarrow$  carry criterion  $\longrightarrow$   $q$ -Lucas  $\longrightarrow$  cyclic sieving and  $q$ -Babbage.

Read [Chapter 19](#).

### Route to analytic deformation

$\log(a; q)_\infty \longrightarrow$  Lambert series and dilogarithms  $\longrightarrow$   $q$ -gamma  
 $\longrightarrow$  central sums and Bernoulli asymptotics.

Read [Chapters 1, 13, 21 and 22](#).

### Route to geometric filters and the Fabius bridge

reversed Gaussian rows  $\longrightarrow$  geometric Lagrange moments  $\longrightarrow$  exact coefficient filters  
 $\longrightarrow$  dyadic Prouhet cancellation  $\longrightarrow$  Fabius–Rvachev sinc products.

Read [Chapters 25 to 27](#).

## Appendix E

# Formalization status

Every mathematical result in this book is intended, eventually, to have a machine-checked proof in the Lean development that accompanies the Fabius project [43]. The table below is the exhaustive Lean crosswalk for the *forward* part: it covers the labelled results inherited from the forward backbone together with all labelled definitions and results retained from the three general guides. The branch-aware inverse part has a separate exhaustive source-to-canonical register in [Section A.2](#); the machine-readable `theorem_concordance.csv` records its 260 historical source results. Finally, `source_concordance.csv` maps all 547 result environments from the five publications merged into this volume, including duplicates and retired source assertions, to their surviving canonical destinations. These three views are complementary: the table below answers the Lean-status question for each labelled forward result, while the concordances preserve complete merger provenance without counting a repeated theorem twice.

The classification is deliberately severe, and it is the same rule the project uses to decide whether a claim may leave its quarantine boundary. A result counts as formalized only when a current Lean declaration proves the *exact* statement: same hypotheses, same domain, same normalization, same conclusion. A related definition, a theorem with stronger hypotheses or a weaker conclusion, an executable computation, a numerical match, or an apparently immediate derivation is *not* an exact counterpart. In particular a Lean lemma that appears as a step inside the proof below does not formalize the result the proof concludes.

**exact** a named Lean declaration proves the same statement; the declaration and its module are given.

**partial** Lean has the result only in a special case, at a fixed parameter, on a restricted domain, or with a weaker conclusion. The named declaration is the nearest one, and the shortfall is real.

**none** nothing in the development proves it, even partially.

**n/a** the environment states a convention or an interface rather than a mathematical claim.

Of the 284 labelled forward results, 42 are **exact**, 74 are **partial**, 160 are **none**, and 8 are n/a. The original 191-result pre-Fabius core had 36 **exact**, 29 **partial**, 123 **none**, and 3 n/a entries. The four integrated guide chapters add 31 human-proved but not Lean-formalized assertions and five labelled definitions. The later Fabius bridge contributes the remaining 57 rows. The formalized part is concentrated exactly where the Fabius development needed it: the finite Gaussian-coefficient core, the geometric Lagrange and filter calculus, half-base and Thue–Morse cancellation, and the geometric uniform law. The complementary modules `CompleteHomogeneousGenerating` and `SymmetricFunctionGenerating` now formalize both finite algebraic halves of the weighted generating-product theorem as formal-power-series identities; its analytic evaluation and convergence clause is not formalized. The finite-family theorem in `CompleteHomogeneousAsymptotics` separately transfers coordinatewise Big-O bounds through a fixed complete homogeneous degree, without asserting formal-series convergence. Most of the classical analytic layer — nonterminating

basic hypergeometric summation, Jacobi triple product, Bailey inversion and the Bailey lemma, and the Rogers–Ramanujan identities — remains untouched; the algebraic  $q$ -difference annihilation input to that chapter is now exact. The elementary base-reversal, terminating-factor, and adjacent-ratio claims are now exact as well: their field forms record  $a \neq 0$ ,  $q \neq 0$ , and every displayed nonzero denominator, while their cleared and cross-multiplied polynomial forms remain valid at roots of unity; both adjacent cross identities are total in all natural indices by zero extension. The former claim that every infinite  $q$ -Pochhammer symbol in the development was a finite `Finset.range` product is now false: `Fabius.qPochhammerInf` from `LimitConditionNumber.lean` is the genuine real  $\prod'_{j \geq 0} (1 - aq^j)$ ; for  $0 \leq q < 1$  the factors are multipliable for every real  $a$ , the finite symbols converge to it, and the positive-factor specializations used for  $(q; q)_\infty$  and  $(-q; q)_\infty$  are strictly positive. This real contractive-base product layer does not formalize the book’s general complex analytic or basic-hypergeometric infinite-product theory. The separate `RvachevPochhammerFactorization` module now defines the complex symbol and proves its pointwise convergence for every contracting complex nome. It also identifies every geometric sinc product at a strict complex contraction with the corresponding nome- $q^2$  Pochhammer product, pointwise and globally in the complex argument; only its final two `Rvachev` wrappers fix the nome to  $1/4$ . This does not supply the book’s named centered/MGF wrapper, outside-disk reciprocal formula, or local-uniform/normal-convergence theorem for the Pochhammer-product right-hand side.

The promoted, strengthened, and new finite- $q$  and weighted symmetric-function rows in this edition were statically checked against the complete public declaration lists:

```

QBinomialCauchy.lean: one definition and five theorems, including the compatibility spelling of the
    primary identity;
SymmetricFunctionOrthogonality.lean: one definition and six theorems;
CompleteHomogeneousGenerating.lean: one definition and six theorems;
    CompleteHomogeneousAsymptotics.lean: one theorem;
    FiniteTriangularTransform.lean: one definition and one theorem;
SymmetricFunctionTransform.lean: four definitions and five theorems;
SymmetricFunctionGenerating.lean: two definitions and six theorems;
    QDifferenceAnnihilation.lean: four theorems;
QBinomialInversionSpecializations.lean: two definitions and four theorems;
    QPochhammerElementaryIdentities.lean: 13 theorems;
    QBinomialReciprocity.lean: four theorems;
    GaussianBinomialAtNegOne.lean: five theorems;
RvachevPochhammerFactorization.lean: one definition and nine theorems;
    GeometricCompleteHomogeneous.lean: six theorems;
    GeometricLagrangeCompleteHomogeneous.lean: five theorems;
    GeometricLagrangeQMoments.lean: one definition and 37 theorems;
    FinitePolynomialFilterExactness.lean: five theorems;
    QuarterCatalanGerm.lean: two definitions and thirteen theorems;
    FabiusInverseQuarterJet.lean: two theorems;
    QuarterCatalanRichardson.lean: three definitions and 15 theorems.

```

The added finite  $q$ -algebra and geometric layers have the following exhaustive public surfaces.

- The four theorems of `QBinomialReciprocity.lean` are `Fabius.gaussianBinomial_reciprocity_units`, `Fabius.gaussianBinomial_reciprocity`, `Fabius.gaussianBinomial_neg_one_eq_zero_of_odd_degree`, and `Fabius.gaussianBinomial_neg_one_even_odd_eq_zero`. Unit reciprocity is total over every commutative semiring; its nonzero-base wrapper needs only a semifield. The two  $q = -1$  consequences are universal commutative-ring identities, transported from the integer Gaussian polynomial, and the even-row odd-column statement includes columns above the row by zero extension.
- The five theorems of `GaussianBinomialAtNegOne.lean` are `Fabius.gaussianBinomial_neg_one_even_even`, `Fabius.gaussianBinomial_neg_one_odd_even`,

`Fabius.gaussianBinomial_neg_one_odd_odd`, `Fabius.finiteQPochhammerIn_neg_one_even`, and `Fabius.finiteQPochhammerIn_neg_one_odd`. The first three, together with `Fabius.gaussianBinomial_neg_one_even_odd_eq_zero` from `QBinomialReciprocity.lean`, give all four parity cases of [Corollary 19.8](#) exactly. Every identity uses only a commutative ring; the Gaussian formulas are total in all natural parameters, including columns above the row by zero extension, and the last two theorems give the paired even- and odd-length finite  $q$ -Pochhammer products without any domain, characteristic, or nonvanishing assumption.

- The 13 theorems of `QPochhammerElementaryIdentities.lean` are `Fabius.finiteQPochhammerIn_base_reversal_units`, `Fabius.finiteQPochhammerIn_inv_base_reversal_units`, `Fabius.finiteQPochhammerIn_base_reversal`, `Fabius.finiteQPochhammerIn_inv_base_reversal`, `Fabius.prod_pow_sub_pow_eq_finiteQPochhammerIn`, `Fabius.pow_mul_finiteQPochhammerIn_inv_pow_eq`, `Fabius.finiteQPochhammerIn_inv_pow_eq_self_div`, `Fabius.finiteQPochhammerIn_inv_pow_eq_zero_of_lt`, `Fabius.one_sub_mul_gaussianBinomial_one`, `Fabius.gaussianBinomial_adjacent_mul`, `Fabius.gaussianBinomial_row_adjacent_mul`, `Fabius.gaussianBinomial_adjacent_div`, and `Fabius.gaussianBinomial_row_adjacent_div`. The two unit reversals, the terminating numerator, the first-column clearer, and the adjacent cross identities are commutative-ring statements. The field reversals require exactly  $a \neq 0$  and  $q \neq 0$ ; the cleared terminating formula and above-range zero clause require  $q \neq 0$ , and the terminating quotient also requires  $(q; q)_{N-k} \neq 0$ . The adjacent quotients require exactly their displayed Gaussian and linear-factor denominators, retain  $k < n$ , and do not require  $q \neq 0$ ; the two adjacent cross identities are total in all  $n, k$ , including on and above the row boundary.
- `RvachevPochhammerFactorization.lean` has the one definition `Fabius.complexQPochhammerInf` and exactly nine theorems: `Fabius.complexQPochhammerInf_eq_tprod`, `Fabius.multipliable_one_sub_mul_pow_complex`, `Fabius.hasProd_complexQPochhammerInf`, `Fabius.tendsto_finiteQPochhammerIn_complex`, `Fabius.summable_norm_sineTerm_qpow_pair`, `Fabius.geometricSincProduct_eq_tprod_pair`, `Fabius.geometricSincProduct_eq_tprod_complexQPochhammerInf`, `Fabius.rvachevFourierProduct_eq_tprod_complexQPochhammerInf`, and `Fabius.rvachevFourier_eq_tprod_complexQPochhammerInf`. The complex symbol is total; its named convergence results require exactly  $\|q\| < 1$ . For every complex  $q, z$  under that same contraction hypothesis, the three new theorems prove absolute summability of the paired Euler perturbations, the paired scale-spectral-zero tprod, and

$$S_q(z) = \prod_{k \geq 0} \left( \frac{z^2}{(k+1)^2}; q^2 \right)_{\infty}.$$

This identity includes  $q = 0$ , negative and nonreal contractions, and zero factors, and it has no nonvanishing hypothesis. The final two factorizations specialize to the fixed nome  $1/4$ ; the Fourier form assumes exactly a bounded Fabius witness and `IsFabius`. What remains outside the API from [Theorem 28.59](#) is its named centered/MGF packaging, outside-disk reciprocal and zero-pole clauses, and named local-uniform or normal convergence of the Pochhammer-product right-hand side.

- The final geometric-sinc tranche contributes four public theorems in `GeometricReciprocalGamma.lean`, namely `Fabius.hasProdLocallyUniformly_geometricSincProduct`, `Fabius.geometricSincProductFactors_multipliable`, `Fabius.hasProd_geometricSincProduct`, and `Fabius.geometricSincProduct_differentiable`, and the zero-definition, four-theorem module `GeometricSincCharacteristicFunction.lean`

consists of `Fabius.charFun_geometricUniformDistribution_eq_phase_mul_geometricSincProduct` and `Fabius.charFun_geometricUniformDistribution_eq_phase_mul_geometricReciprocalGamma`, `Fabius.tendstoLocallyUniformly_prefix_sinc_charFun`, and `Fabius.tendstoUniformlyOn_prefix_sinc_charFun`. For complex  $q, z$  with  $\|q\| < 1$ , the first four prove locally uniform convergence, pointwise Multipliable/HasProd, and entire-ness of  $S_q(z) = \prod_{n \geq 0} \text{sinc}_{\text{rad}}(\pi q^n z)$ . For real  $|q| < 1$ ,  $t \in \mathbb{R}$ , and  $z_q(t) = (1 - q)t/(2\pi)$ , including  $q = 0$  and negative contractions, the first two characteristic-function declarations state

$$\varphi_q(t) = e^{it/2} S_q(z_q(t)) = e^{it/2} G_q(z_q(t)) G_q(-z_q(t)).$$

Here  $G_q(z) = \text{geometricReciprocalGamma}(q, z)$ . The last two declarations assert local uniform convergence of the full phase-bearing prefixes on  $\mathbb{R}$  and its uniform-on-every-compact-set corollary. These are exact subclaims of [Theorems 27.2](#) and [28.59](#); neither compound label is promoted, because no declaration packages the former's complex-frequency full phase-prefix convergence or the latter's centered/MGF wrapper, reciprocal outside-disk formula, zero-pole exchange, and named local-uniform or normal convergence of the Pochhammer-product right-hand side.

- The six theorems of `GeometricCompleteHomogeneous.lean` are `Fabius.completeHomogeneousEval_geometric`, `Fabius.completeHomogeneousEval_scaled_geometric`, `Fabius.completeHomogeneousEvalOn_range_pow_eq_gaussianBinomial`, `Fabius.completeHomogeneousEvalOn_range_pow_eq_gaussianBinomial_degree`, `Fabius.gaussianBinomial_add_symm`, and `Fabius.gaussianBinomial_symm_via_completeHomogeneous`. They prove both principal-specialization orientations, common scaling, and both generic Gaussian symmetry laws over every commutative semiring, without division, distinctness, or cancellation.
- The five theorems of `GeometricLagrangeCompleteHomogeneous.lean` are `Fabius.completeHomogeneousEvalOn_geometric_range`, `Fabius.sum_geometricLagrangeWeight_mul_pow_succ_add_eq_gaussianBinomial`, `Fabius.geometricLagrangeQMoment_eq_residual_gaussianBinomial`, `Fabius.completeHomogeneousEvalOn_geometric_range_eq_qBinomial`, and `Fabius.geometricLagrangeQMoment_eq_residual_qBinomial_via_completeHomogeneous`. The first is semiring-level; the field statement uses finite-node injectivity; the rational quotient bridges retain their explicit nonzero-Pochhammer or  $0 < q < 1$  assumptions.
- `GeometricLagrangeQMoments.lean` has the one definition `Fabius.geometricLagrangeQMoment` and exactly 37 theorems: `Fabius.geometricLagrangeQMoment_eq_weightPolynomial_eval`, `Fabius.geometricLagrangeQMoment_eq_forwardRichardsOn_eval`, `Fabius.geometricRootPolynomial_inv_eval_pow_mul_signedPowers`, `Fabius.geometricRootPolynomial_inv_eval_pow_mul_triangular`, `Fabius.geometricRootPolynomial_inv_eval_one_mul_triangular`, `Fabius.geometricLagrangeQMoment_eq_qPochhammer`, `Fabius.geometricLagrangeQMoment_zero`, `Fabius.geometricLagrangeQMoment_eq_zero`, `Fabius.geometricRootPolynomial_inv_eval_pow_eq_qPochhammer_of_le`, `Fabius.geometricLagrangeQMoment_eq_residual_qPochhammer`, `Fabius.qPochhammer_self_add`, `Fabius.qPochhammer_self_pos_of_pos_of_lt_one`, `Fabius.qBinomial_pos_of_pos_of_lt_one`, `Fabius.gaussianBinomial_eq_qBinomial_of_pos_of_lt_one`, `Fabius.qPochhammer_pow_pos_of_pos_of_lt_one`, `Fabius.qPochhammer_tail_div_self_eq_q_qBinomial`, `Fabius.geometricLagrangeQMoment_eq_residual_qBinomial`, `Fabius.geometricLagrangeQMoment_firstUncancelled`, `Fabius.negOnePow_mul_geometricLagrangeQMoment_eq_positiveResidual`, `Fabius.negOneP`

ow\_mul\_geometricLagrangeQMoment\_pos, Fabius.qPochhammer\_self\_succ, Fabius.qBinomial\_succ\_succ\_of\_pos\_of\_lt\_one', Fabius.qBinomial\_succ\_succ\_of\_pos\_of\_lt\_one, Fabius.qBinomial\_theorem\_of\_pos\_of\_lt\_one, Fabius.sum\_qBinomial\_triangular\_succ\_eq\_neg\_qPochhammer, Fabius.abs\_geometricLagrangeWeight\_eq\_qBinomial, Fabius.abs\_geometricLagrangeWeight\_eq\_q\_sign\_mul, Fabius.abs\_geometricLagrangeWeight\_complement\_eq\_qBinomial, Fabius.sum\_abs\_geometricLagrangeWeight\_eq\_qPochhammer\_ratio, Fabius.neg\_qPochhammer\_div\_self\_eq\_prod, Fabius.sum\_abs\_geometricLagrangeWeight\_eq\_prod, Fabius.quarterGeometricLagrangeQMoment\_eq\_qPochhammer, Fabius.quarterGeometricLagrangeQMoment\_eq\_zero, Fabius.quarterGeometricLagrangeQMoment\_eq\_residual\_qPochhammer, Fabius.quarterGeometricLagrangeQMoment\_eq\_residual\_qBinomial, Fabius.quarterGeometricLagrangeQMoment\_firstUncancelled, and Fabius.sum\_abs\_quarterGeometricLagrangeWeight\_eq\_qPochhammer\_ratio. These are finite rational identities. Quotient statements retain their nonzero denominators, and positivity, sign, and absolute-value formulas retain  $0 < q < 1$ ; the module makes no analytic convergence or remainder claim.

- The five theorems of `FinitePolynomialFilterExactness.lean` are `Fabius.polynomialFilter_response_eq`, `Fabius.polynomialFilter_exact`, `Fabius.normalizedGeometricRootPolynomial_filter_exact`, `Fabius.forwardGeometricRichardsonPolynomial_filter_exact`, and `Fabius.forwardGeometricRichardsonPolynomial_filter_firstUncancelled`. The first two are arbitrary commutative-semiring response and exactness laws; the field-valued geometric forms retain their nonzero-base and normalization- denominator hypotheses and expose the first surviving mode.
- `QuarterCatalanGerm.lean` has the two definitions `Fabius.quarterCatalanCoefficient` and `Fabius.quarterCatalanGermSeries`, and the thirteen theorems `Fabius.quarterCatalanCoefficient_zero`, `Fabius.quarterCatalanCoefficient_succ_eq_report`, `Fabius.quarterCatalanGermSeries_coeff`, `Fabius.quarterCatalanGermSeries_coeff_succ`, `Fabius.quarterCatalanGermSeries_constantCoeff`, `Fabius.quarterCatalanGermSeries_equation`, `Fabius.powerSeries_quadratic_injectiveOn_zeroConstant`, `Fabius.eq_quarterCatalanGermSeries_of_equation`, `Fabius.existsUnique_quarterCatalanGermSeries`, `Fabius.dyadicGermTwo_functionalEquation`, `Fabius.rescale_dyadicGermTwo_eq_quadraticInverse`, `Fabius.dyadicGermTwo_eq_rescale_quadraticInverse`, and `Fabius.coeff_dyadicGermTwo_succ`. This is the unique zero-constant solution of  $D + 4D^2 = (4/9)X$  in  $\mathbb{Q}[[X]]$ , with its complete coefficient formula and exact rescaling bridge to the distinguished dyadic germ and the quadratic inverse. The downstream module `FabiusInverseQuarterJet.lean` connects this algebraic shadow to the actual smooth inverse through exactly two public theorems, `Fabius.iteratedDeriv_centeredFabiusInv_quarter_eq_quadraticInverse` and `Fabius.iteratedDeriv_fabiusInv_five_seventy_two_succ`. For every bounded Fabius solution they prove the full centered jet identity at  $5/72 = F(1/4)$  and the Catalan formula

$$G^{(m+1)}(5/72) = (m+1)!(-4)^m C_m.$$

These are smooth derivative identities, not convergence of the formal series, equality on a neighborhood, or analyticity of the inverse at the anchor.

- `QuarterCatalanRichardson.lean` has the three definitions `Fabius.finiteRescaleFilter`, `Fabius.geometricRichardsonPowerSeriesFilter`, and `Fabius.quarterCatalanRichardsonFilter`, and the 15 theorems `Fabius.finiteRescaleFilter_coeff`, `Fabius.geometricRichardsonPowerSeriesFilter_coeff`, `Fabius.geometricRichardsonPowerSeriesFilter_coeff_zero`, `Fabius.geometricRichardsonPowerSeriesFilter`

iesFilter\_coeff\_eq\_zero, Fabius.geometricRichardsonPowerSeriesFilter\_coeff\_eq\_qPochhammer, Fabius.geometricRichardsonPowerSeriesFilter\_coeff\_eq\_qBinomial, Fabius.geometricRichardsonPowerSeriesFilter\_firstUncancelled\_coeff\_of\_nonzero, Fabius.geometricRichardsonPowerSeriesFilter\_firstUncancelled\_coeff, Fabius.quarterCatalanRichardsonFilter\_coeff, Fabius.quarterCatalanRichardsonFilter\_coeff\_zero, Fabius.quarterCatalanRichardsonFilter\_coeff\_eq\_zero, Fabius.quarterCatalanRichardsonFilter\_coeff\_eq\_zero\_of\_le, Fabius.quarterCatalanRichardsonFilter\_coeff\_eq\_qBinomial, Fabius.quarterCatalanRichardsonFilter\_coeff\_succ\_eq\_qBinomial, and Fabius.quarterCatalanRichardsonFilter\_firstUncancelled\_coeff. These are coefficientwise formal-power-series identities only: no convergence, real error sign, remainder estimate, or analytic acceleration is asserted.

Finally, Fabius.sum\_lagrangeEvalWeight\_mul\_pow\_card\_add\_zero in LagrangeResidualMoments.lean gives every higher evaluation-at-zero moment of a nonempty distinct field-valued node family as the negative signed nodal product times its complete homogeneous function. Its nonempty premise is exactly what removes the exceptional empty-row  $0^0$  term.

The theorem statements were checked at arbitrary parameters and degrees, including the empty and degree-zero boundaries. Weighted inversion is now **exact**. Weighted generating products and reciprocal finite products are each **partial**: their formal power-series identities are proved, but their analytic evaluation and convergence clauses are not. The nearby  $q$ -Pfaff–Saalschütz summation remains **none** and is not inferred from a related finite identity.



## E.1 By chapter

| Chapter                                                                  | exact     | partial   | none       | n/a      |
|--------------------------------------------------------------------------|-----------|-----------|------------|----------|
| The algebra of $q$ -shifted factorials                                   | 3         | 1         | 11         | 0        |
| $q$ -integers and Gaussian coefficients                                  | 3         | 2         | 4          | 0        |
| Finite $q$ -binomial theorems and inversion                              | 9         | 1         | 0          | 0        |
| Weighted binomial coefficients and symmetric functions                   | 3         | 2         | 3          | 0        |
| Partitions, words, paths, and inversion statistics                       | 1         | 1         | 7          | 0        |
| Finite fields, Grassmannians, and rank enumeration                       | 0         | 1         | 5          | 0        |
| Symmetry and unimodality                                                 | 0         | 0         | 3          | 0        |
| The infinite $q$ -binomial theorem                                       | 0         | 0         | 5          | 0        |
| Basic hypergeometric series                                              | 1         | 0         | 8          | 0        |
| Euler telescoping and very-well-poised summations                        | 0         | 0         | 4          | 0        |
| Bilateral series and Ramanujan's ${}_1\psi_1$ sum                        | 0         | 0         | 2          | 0        |
| $q$ -difference calculus and the quantum binomial                        | 0         | 0         | 10         | 0        |
| The $q$ -gamma and $q$ -beta functions                                   | 0         | 0         | 8          | 0        |
| Jacobi's triple product                                                  | 0         | 0         | 8          | 0        |
| Theta modularity, lattice dissection, and sums of squares                | 0         | 0         | 14         | 4        |
| Partitions and product-sum identities                                    | 0         | 0         | 4          | 0        |
| Bailey pairs and the Rogers–Ramanujan identities                         | 1         | 0         | 6          | 0        |
| Bailey transforms and the Andrews–Gordon identities                      | 0         | 0         | 8          | 1        |
| Cyclotomic factorization and congruences                                 | 1         | 0         | 8          | 0        |
| Negative upper indices and geometric Newton interpolation                | 0         | 2         | 7          | 0        |
| Central $q$ -binomial sums and Lambert series                            | 0         | 0         | 7          | 0        |
| Asymptotic regimes and Bernoulli expansions                              | 0         | 0         | 8          | 0        |
| Exact limits, reciprocity, and research frontiers                        | 0         | 0         | 5          | 0        |
| Computation, certification, and formal proof architecture                | 0         | 3         | 8          | 3        |
| Geometric interpolation and exact $q$ -filters                           | 6         | 2         | 1          | 0        |
| Half-base arithmetic and Prouhet–Thue–Morse cancellation                 | 6         | 3         | 0          | 0        |
| Geometric uniform sums, $q$ -cumulants, and the Fabius–Rvachev functions | 2         | 11        | 0          | 0        |
| Dyadic Gaussian–Thue–Morse structure and the values of $F$               | 6         | 45        | 6          | 0        |
| <b>Forward total</b>                                                     | <b>42</b> | <b>74</b> | <b>160</b> | <b>8</b> |

## E.2 By result

| Result                                                  | Status | Lean counterpart                               |
|---------------------------------------------------------|--------|------------------------------------------------|
| <i>The algebra of <math>q</math>-shifted factorials</i> |        |                                                |
| Proposition 1.2                                         | exact  | finiteQPochhammerIn_add<br>FiniteQBinomialCore |
| Proposition 1.3                                         | none   | —                                              |
| Corollary 1.4                                           | none   | —                                              |

| Result                                          | Status  | Lean counterpart                                                                                                                                                                                                                                                         |
|-------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proposition 1.6                                 | exact   | finiteQPochhammerIn_base_reversal,<br>finiteQPochhammerIn_inv_base_reversal<br>QPochhammerElementaryIdentities; field wrappers with<br>$a \neq 0, q \neq 0$ , strengthened by the two unit-valued ring forms                                                             |
| Proposition 1.8                                 | none    | —                                                                                                                                                                                                                                                                        |
| Corollary 1.9                                   | exact   | finiteQPochhammerIn_inv_pow_eq_self_div,<br>finiteQPochhammerIn_inv_pow_eq_zero_of_<br>lt<br>QPochhammerElementaryIdentities; $q \neq 0$ , with the<br>quotient additionally assuming $(q; q)_{N-k} \neq 0$ ; root-safe<br>numerator and cleared forms are also exported |
| Theorem 1.10                                    | none    | —                                                                                                                                                                                                                                                                        |
| Proposition 1.11                                | none    | —                                                                                                                                                                                                                                                                        |
| Theorem 1.12                                    | none    | —                                                                                                                                                                                                                                                                        |
| Proposition 1.14                                | none    | —                                                                                                                                                                                                                                                                        |
| Proposition 1.15                                | none    | —                                                                                                                                                                                                                                                                        |
| Theorem 1.16                                    | partial | tsum_log_one_sub_geom<br>EulerLogTransform                                                                                                                                                                                                                               |
| Corollary 1.17                                  | none    | —                                                                                                                                                                                                                                                                        |
| Theorem 1.18                                    | none    | —                                                                                                                                                                                                                                                                        |
| Proposition 1.19                                | none    | —                                                                                                                                                                                                                                                                        |
| <i>q-integers and Gaussian coefficients</i>     |         |                                                                                                                                                                                                                                                                          |
| Proposition 2.2                                 | partial | qPochhammer_tail_div_self_eq_qBinomial<br>GeometricLagrangeQMoments                                                                                                                                                                                                      |
| Theorem 2.3                                     | exact   | gaussianBinomial_succ_succ_alt<br>FiniteQBinomialCore                                                                                                                                                                                                                    |
| Corollary 2.4                                   | none    | —                                                                                                                                                                                                                                                                        |
| Theorem 2.5                                     | partial | gaussianBinomial_symm, gaussianBinomial_<br>reciprocity_units<br>FiniteQBinomialCore; QBinomialReciprocity; items<br>(i)–(ii) exact, polynomial degree and coefficient reversal still open                                                                               |
| Corollary 2.7                                   | exact   | gaussianBinomial_neg_one_eq_zero_of_<br>odd_degree, gaussianBinomial_neg_one_<br>even_odd_eq_zero<br>QBinomialReciprocity; arbitrary commutative ring,<br>including above-row zero extension                                                                             |
| Corollary 2.9                                   | none    | —                                                                                                                                                                                                                                                                        |
| Proposition 2.11                                | none    | —                                                                                                                                                                                                                                                                        |
| Proposition 2.12                                | exact   | gaussianBinomial_adjacent_div,<br>gaussianBinomial_row_adjacent_div<br>QPochhammerElementaryIdentities; exact displayed<br>denominators and $k < n$ , with both root-safe cross-multiplied<br>forms exported totally for all $n, k$ by zero extension                    |
| Theorem 2.13                                    | none    | —                                                                                                                                                                                                                                                                        |
| <i>Finite q-binomial theorems and inversion</i> |         |                                                                                                                                                                                                                                                                          |

| Result                                                        | Status  | Lean counterpart                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 3.1                                                   | exact   | <code>finite_qBinomial_theorem</code><br><code>FiniteQBinomialCore</code>                                                                                                                                                                                                                              |
| Corollary 3.2                                                 | exact   | <code>sum_gaussianBinomial_alternating_mul_pow</code> , <code>sum_gaussianBinomial_alternating</code><br><code>QBinomialInversion</code> ; arbitrary commutative ring, with the row-zero boundary included                                                                                             |
| Corollary 3.4                                                 | exact   | <code>sum_pow_sum_powersetCard_eq_gaussianBinomial</code> , <code>sum_pow_sum_powersetCard_Icc_eq_gaussianBinomial</code><br><code>BitPositionQBinomial</code> ; total zero-based and literal one-based forms                                                                                          |
| Theorem 3.6                                                   | partial | <code>prod_one_sub_qPow_X_mul_gaussianBinomialGeneratingSeries</code><br><code>SymmetricFunctionGenerating</code> ; exact formal identity in $R[[X]]$ over every commutative ring, but no analytic evaluation or convergence proof for $ z  < 1$                                                       |
| Theorem 3.8                                                   | exact   | <code>gaussianBinomial_add_vandermonde</code> , <code>gaussianBinomial_add_vandermonde'</code><br><code>QBinomialVandermonde</code>                                                                                                                                                                    |
| Corollary 3.9                                                 | exact   | <code>gaussianBinomial_add_central</code> , <code>gaussianBinomial_add_central_min</code><br><code>QBinomialVandermonde</code>                                                                                                                                                                         |
| Corollary 3.10                                                | exact   | <code>gaussianBinomial_two_mul_int_shifted_central_finsum</code><br><code>QBinomialVandermonde</code> , using <code>gaussianBinomialInt</code> from <code>FiniteQBinomialCore</code> ; total in $k \in \mathbb{Z}$ , with support contained in $0 \leq \ell \leq N$                                    |
| Theorem 3.12                                                  | exact   | <code>gaussianBinomial_inversion</code><br><code>QBinomialTransform</code> ; module-valued, hence in particular ring-valued sequences                                                                                                                                                                  |
| Corollary 3.13                                                | exact   | <code>gaussianBinomialKernel_left_orthogonality</code> , <code>gaussianBinomialKernel_right_orthogonality</code><br><code>QBinomialInversion</code> ; both total <code>Icc</code> convolution orders                                                                                                   |
| Theorem 3.15                                                  | exact   | <code>finite_qCauchy_identity</code> , <code>finiteQPochhammerIn_mul_eq_sum_gaussianBinomial</code><br><code>QBinomialCauchy</code> ; canonical name plus compatibility spelling over an arbitrary commutative ring; reflected identity and q-Bernstein partition-of-unity strengthenings also exposed |
| <i>Weighted binomial coefficients and symmetric functions</i> |         |                                                                                                                                                                                                                                                                                                        |
| Theorem 4.2                                                   | exact   | <code>elementarySymmetricEval_option_succ</code> , <code>completeHomogeneousEval_option_succ</code><br><code>SymmetricFunctionOrthogonality</code> , <code>CompleteHomogeneous</code> ; both adjoining-variable recurrences, all degrees                                                               |

| Result                                                    | Status  | Lean counterpart                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 4.3                                               | partial | <code>elementarySymmetricGeneratingSeries_eq_prod</code> ,<br><code>elementarySymmetricGeneratingSeries_neg_mul</code> ,<br><code>completeHomogeneousGeneratingSeries</code> ,<br><code>completeHomogeneousGeneratingSeries_eq_invOfUnit</code> ,<br><code>elementarySymmetricGeneratingSeries_neg_prod_one_sub_mul</code> ,<br><code>completeHomogeneousGeneratingSeriesOn</code> ,<br><code>completeHomogeneousGeneratingSeriesOn_eq_invOfUnit_prod</code><br><code>SymmetricFunctionGenerating</code> and<br><code>CompleteHomogeneousGenerating</code> ; complementary<br>finite-family formal surfaces prove the elementary product,<br>elementary–complete reciprocity, and the canonical<br>complete-homogeneous inverse, but not analytic evaluation or<br>convergence under $ w_i z  < 1$ |
| Corollary 4.4                                             | exact   | <code>sum_elementarySymmetricEval_mul</code> ,<br><code>completeHomogeneousEval</code><br><code>SymmetricFunctionOrthogonality</code> ; arbitrary finite family,<br>including empty family and degree zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Theorem 4.6                                               | partial | <code>completeHomogeneousEval_geometric</code><br><code>GeometricCompleteHomogeneous</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Lemma 4.8                                                 | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Theorem 4.9                                               | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Corollary 4.10                                            | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Theorem 4.12                                              | exact   | <code>weightedSymmetricFunction_inversion</code><br><code>SymmetricFunctionTransform</code> ; whole-function iff over a<br>commutative ring, strengthened to sequences in any additive<br>commutative monoid carrying the module structure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <i>Partitions, words, paths, and inversion statistics</i> |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Theorem 5.1                                               | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Corollary 5.2                                             | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Theorem 5.3                                               | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Corollary 5.4                                             | partial | <code>qBinomial_symm</code><br><code>HalfQBinomial</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Corollary 5.5                                             | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Theorem 5.7                                               | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Proposition 5.9                                           | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Proposition 5.10                                          | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Corollary 5.11                                            | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <i>Finite fields, Grassmannians, and rank enumeration</i> |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Theorem 6.1                                               | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Theorem 6.2                                               | partial | <code>gaussianBinomial_succ_succ</code><br><code>FiniteQBinomialCore</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Result                                                              | Status | Lean counterpart                                                                                            |
|---------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------|
| Theorem 6.3                                                         | none   | —                                                                                                           |
| Theorem 6.4                                                         | none   | —                                                                                                           |
| Theorem 6.5                                                         | none   | —                                                                                                           |
| Proposition 6.7                                                     | none   | —                                                                                                           |
| <i>Symmetry and unimodality</i>                                     |        |                                                                                                             |
| Lemma 7.1                                                           | none   | —                                                                                                           |
| Proposition 7.2                                                     | none   | —                                                                                                           |
| Theorem 7.3                                                         | none   | —                                                                                                           |
| <i>The infinite <math>q</math>-binomial theorem</i>                 |        |                                                                                                             |
| Theorem 8.1                                                         | none   | —                                                                                                           |
| Corollary 8.2                                                       | none   | —                                                                                                           |
| Corollary 8.3                                                       | none   | —                                                                                                           |
| Proposition 8.4                                                     | none   | —                                                                                                           |
| <i>Basic hypergeometric series</i>                                  |        |                                                                                                             |
| Proposition 9.2                                                     | none   | —                                                                                                           |
| Theorem 9.3                                                         | none   | —                                                                                                           |
| Theorem 9.5                                                         | none   | —                                                                                                           |
| Theorem 9.6                                                         | none   | —                                                                                                           |
| Lemma 9.7                                                           | exact  | finite_qCauchy_second_identity<br>QBinomialCauchy; arbitrary commutative ring and all<br>parameters/degrees |
| Theorem 9.9                                                         | none   | —                                                                                                           |
| Corollary 9.10                                                      | none   | —                                                                                                           |
| Lemma 9.11                                                          | none   | —                                                                                                           |
| Proposition 9.12                                                    | none   | —                                                                                                           |
| <i>Bilateral series and Ramanujan's <math>{}_1\psi_1</math> sum</i> |        |                                                                                                             |
| Proposition 11.2                                                    | none   | —                                                                                                           |
| Theorem 11.3                                                        | none   | —                                                                                                           |
| <i><math>q</math>-difference calculus and the quantum binomial</i>  |        |                                                                                                             |
| Proposition 12.1                                                    | none   | —                                                                                                           |
| Theorem 12.2                                                        | none   | —                                                                                                           |
| Lemma 12.3                                                          | none   | —                                                                                                           |
| Theorem 12.4                                                        | none   | —                                                                                                           |
| Theorem 12.6                                                        | none   | —                                                                                                           |
| Corollary 12.7                                                      | none   | —                                                                                                           |
| Corollary 12.9                                                      | none   | —                                                                                                           |
| Proposition 12.10                                                   | none   | —                                                                                                           |
| Theorem 12.11                                                       | none   | —                                                                                                           |
| Corollary 12.12                                                     | none   | —                                                                                                           |

| Result                                                            | Status | Lean counterpart                                                                                                                                                                                        |
|-------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>The <math>q</math>-gamma and <math>q</math>-beta functions</i> |        |                                                                                                                                                                                                         |
| Theorem 13.2                                                      | none   | —                                                                                                                                                                                                       |
| Proposition 13.3                                                  | none   | —                                                                                                                                                                                                       |
| Theorem 13.4                                                      | none   | —                                                                                                                                                                                                       |
| Theorem 13.5                                                      | none   | —                                                                                                                                                                                                       |
| Theorem 13.7                                                      | none   | —                                                                                                                                                                                                       |
| Corollary 13.8                                                    | none   | —                                                                                                                                                                                                       |
| Theorem 13.9                                                      | none   | —                                                                                                                                                                                                       |
| Proposition 13.10                                                 | none   | —                                                                                                                                                                                                       |
| <i>Jacobi's triple product</i>                                    |        |                                                                                                                                                                                                         |
| Theorem 14.1                                                      | none   | —                                                                                                                                                                                                       |
| Corollary 14.2                                                    | none   | —                                                                                                                                                                                                       |
| Theorem 14.3                                                      | none   | —                                                                                                                                                                                                       |
| Corollary 14.5                                                    | none   | —                                                                                                                                                                                                       |
| Theorem 14.7                                                      | none   | —                                                                                                                                                                                                       |
| Corollary 14.8                                                    | none   | —                                                                                                                                                                                                       |
| Theorem 14.9                                                      | none   | —                                                                                                                                                                                                       |
| Corollary 14.10                                                   | none   | —                                                                                                                                                                                                       |
| <i>Partitions and product–sum identities</i>                      |        |                                                                                                                                                                                                         |
| Theorem 16.2                                                      | none   | —                                                                                                                                                                                                       |
| Proposition 16.4                                                  | none   | —                                                                                                                                                                                                       |
| Corollary 16.5                                                    | none   | —                                                                                                                                                                                                       |
| Proposition 16.6                                                  | none   | —                                                                                                                                                                                                       |
| <i>Bailey pairs and the Rogers–Ramanujan identities</i>           |        |                                                                                                                                                                                                         |
| Lemma 17.2                                                        | exact  | qDifference_sum_eval <sub>2</sub> _eq_zero_of_degree_lt<br>QDifferenceAnnihilation; semiring coefficients, arbitrary scalar extension, commutative-ring target; no division or distinct-node hypothesis |
| Theorem 17.4                                                      | none   | —                                                                                                                                                                                                       |
| Corollary 17.5                                                    | none   | —                                                                                                                                                                                                       |
| Lemma 17.6                                                        | none   | —                                                                                                                                                                                                       |
| Theorem 17.7                                                      | none   | —                                                                                                                                                                                                       |
| Corollary 17.8                                                    | none   | —                                                                                                                                                                                                       |
| Theorem 17.9                                                      | none   | —                                                                                                                                                                                                       |
| <i>Cyclotomic factorization and congruences</i>                   |        |                                                                                                                                                                                                         |
| Theorem 19.1                                                      | none   | —                                                                                                                                                                                                       |
| Corollary 19.2                                                    | none   | —                                                                                                                                                                                                       |
| Corollary 19.4                                                    | none   | —                                                                                                                                                                                                       |
| Lemma 19.6                                                        | none   | —                                                                                                                                                                                                       |
| Theorem 19.7                                                      | none   | —                                                                                                                                                                                                       |

| Result                                                           | Status  | Lean counterpart                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Corollary 19.8                                                   | exact   | gaussianBinomial_neg_one_even_even,<br>gaussianBinomial_neg_one_odd_even,<br>gaussianBinomial_neg_one_odd_odd,<br>gaussianBinomial_neg_one_even_odd_eq_zero<br>GaussianBinomialAtNegOne; QBinomialReciprocity;<br>arbitrary commutative ring, all four parity classes, including total<br>above-row zero extension |
| Theorem 19.10                                                    | none    | —                                                                                                                                                                                                                                                                                                                  |
| Theorem 19.11                                                    | none    | —                                                                                                                                                                                                                                                                                                                  |
| Corollary 19.13                                                  | none    | —                                                                                                                                                                                                                                                                                                                  |
| <i>Negative upper indices and geometric Newton interpolation</i> |         |                                                                                                                                                                                                                                                                                                                    |
| Proposition 20.2                                                 | none    | —                                                                                                                                                                                                                                                                                                                  |
| Theorem 20.3                                                     | partial | gaussianBinomial_succ_succ<br>FiniteQBinomialCore                                                                                                                                                                                                                                                                  |
| Corollary 20.4                                                   | none    | —                                                                                                                                                                                                                                                                                                                  |
| Theorem 20.5                                                     | none    | —                                                                                                                                                                                                                                                                                                                  |
| Theorem 20.7                                                     | none    | —                                                                                                                                                                                                                                                                                                                  |
| Corollary 20.8                                                   | none    | —                                                                                                                                                                                                                                                                                                                  |
| Theorem 20.9                                                     | partial | finite_qBinomial_theorem<br>FiniteQBinomialCore                                                                                                                                                                                                                                                                    |
| Theorem 20.10                                                    | none    | —                                                                                                                                                                                                                                                                                                                  |
| Corollary 20.11                                                  | none    | —                                                                                                                                                                                                                                                                                                                  |
| <i>Central <math>q</math>-binomial sums and Lambert series</i>   |         |                                                                                                                                                                                                                                                                                                                    |
| Lemma 21.1                                                       | none    | —                                                                                                                                                                                                                                                                                                                  |
| Theorem 21.2                                                     | none    | —                                                                                                                                                                                                                                                                                                                  |
| Corollary 21.3                                                   | none    | —                                                                                                                                                                                                                                                                                                                  |
| Theorem 21.4                                                     | none    | —                                                                                                                                                                                                                                                                                                                  |
| Theorem 21.6                                                     | none    | —                                                                                                                                                                                                                                                                                                                  |
| Theorem 21.7                                                     | none    | —                                                                                                                                                                                                                                                                                                                  |
| Proposition 21.8                                                 | none    | —                                                                                                                                                                                                                                                                                                                  |
| <i>Asymptotic regimes and Bernoulli expansions</i>               |         |                                                                                                                                                                                                                                                                                                                    |
| Lemma 22.1                                                       | none    | —                                                                                                                                                                                                                                                                                                                  |
| Theorem 22.2                                                     | none    | —                                                                                                                                                                                                                                                                                                                  |
| Corollary 22.4                                                   | none    | —                                                                                                                                                                                                                                                                                                                  |
| Theorem 22.6                                                     | none    | —                                                                                                                                                                                                                                                                                                                  |
| Theorem 22.7                                                     | none    | —                                                                                                                                                                                                                                                                                                                  |
| Proposition 22.9                                                 | none    | —                                                                                                                                                                                                                                                                                                                  |
| Corollary 22.10                                                  | none    | —                                                                                                                                                                                                                                                                                                                  |
| Proposition 22.11                                                | none    | —                                                                                                                                                                                                                                                                                                                  |
| <i>Computation, certification, and formal proof architecture</i> |         |                                                                                                                                                                                                                                                                                                                    |
| Algorithm 24.1                                                   | n/a     |                                                                                                                                                                                                                                                                                                                    |

| Result          | Status  | Lean counterpart                                                                                                                                                                                  |
|-----------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 24.2    | partial | gaussianBinomial<br>FiniteQBinomialCore                                                                                                                                                           |
| Algorithm 24.4  | n/a     |                                                                                                                                                                                                   |
| Theorem 24.5    | none    | —                                                                                                                                                                                                 |
| Theorem 24.7    | none    | —                                                                                                                                                                                                 |
| Corollary 24.8  | none    | —                                                                                                                                                                                                 |
| Lemma 24.10     | none    | —                                                                                                                                                                                                 |
| Corollary 24.11 | none    | —                                                                                                                                                                                                 |
| Lemma 24.12     | partial | finiteQPochhammerIn_self_mul_<br>gaussianBinomial<br>FiniteQBinomialCore                                                                                                                          |
| Theorem 24.13   | none    | —                                                                                                                                                                                                 |
| Corollary 24.15 | none    | —                                                                                                                                                                                                 |
| Theorem 24.18   | none    | —                                                                                                                                                                                                 |
| Principle 24.19 | n/a     |                                                                                                                                                                                                   |
| Principle 24.20 | partial | gaussianBinomial_succ_succ_alt,<br>qBinomial_half_sum_eq_zero_iff<br>FiniteQBinomialCore; HalfQBinomial; several tests are<br>formalized separately, but not the six-part falsification principle |

#### *Geometric interpolation and exact $q$ -filters*

|                 |         |                                                                                                 |
|-----------------|---------|-------------------------------------------------------------------------------------------------|
| Theorem 25.1    | exact   | reversed_finite_qBinomial_theorem<br>GeometricQBinomialLagrange                                 |
| Theorem 25.3    | exact   | geometricLagrangeWeight_eq_<br>gaussianBinomial_div<br>GeometricQBinomialLagrange               |
| Corollary 25.4  | exact   | geometricLagrangeWeight_eq_product<br>GeometricLagrangeWeights                                  |
| Theorem 25.6    | exact   | sum_lagrangeEvalWeight_mul_pow_card_add<br>LagrangeResidualMoments                              |
| Theorem 25.7    | exact   | sum_geometricLagrangeWeight_mul_pow_eq_<br>gaussianBinomial<br>GeometricQBinomialLagrange       |
| Corollary 25.8  | partial | sum_geometricLagrangeWeight_mul_scaled_<br>geometric_pow_of_pos<br>GeometricResidualMoments     |
| Theorem 25.9    | exact   | coeff_geometricSeriesFilter<br>GeometricPowerSeriesFilter                                       |
| Corollary 25.10 | partial | geometricLagrangeAnalyticSeriesFilter_<br>eq_constant_add_gaussian_tsum<br>AnalyticSeriesFilter |
| Theorem 25.11   | none    | —                                                                                               |

#### *Half-base arithmetic and Prouhet–Thue–Morse cancellation*

|              |       |                                                  |
|--------------|-------|--------------------------------------------------|
| Theorem 26.1 | exact | halfQPochhammer_eq_mersenne_div<br>HalfQBinomial |
| Theorem 26.2 | exact | qPochhammer_two_pow_eq_ite<br>HalfQBinomial      |



| Result                                                                                    | Status  | Lean counterpart                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Corollary 26.3                                                                            | partial | halfQBinomial_sum_eq_zero_iff<br>HalfQBinomial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Theorem 26.4                                                                              | exact   | halfQBinomial_negativeDyadic_<br>polynomial_sum_eq_mersenne<br>HalfQBinomial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Corollary 26.5                                                                            | partial | halfQBinomial_negativeDyadic_<br>polynomial_sum_eq_mersenne<br>HalfQBinomial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Theorem 26.7                                                                              | exact   | prod_one_sub_pow_two_pow<br>ThueMorseProductIdentity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Theorem 26.8                                                                              | exact   | sum_thueMorseSign_mul_affine_pow_eq_<br>zero<br>ThueMorseSparseProuhet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Corollary 26.9                                                                            | partial | sum_thueMorseSign_mul_affine_pow_eq_<br>zero<br>ThueMorseSparseProuhet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Principle 26.10                                                                           | exact   | halfQBinomial_negativeDyadic_<br>polynomial_sum_eq_mersenne<br>HalfQBinomial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <i>Geometric uniform sums, <math>q</math>-cumulants, and the Fabius–Rvachev functions</i> |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Theorem 27.1                                                                              | partial | geometricUniformSeries_mem_Icc<br>GeometricUniformLaw                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Theorem 27.2                                                                              | partial | charFun_geometricUniformDistribution_<br>eq_phase_mul_geometricSincProduct,<br>charFun_geometricUniformDistribution_<br>eq_phase_mul_geometricReciprocalGamma,<br>tendstoLocallyUniformly_prefix_sinc_<br>charFun, tendstoUniformlyOn_prefix_sinc_<br>charFun<br>GeometricSincCharacteristicFunction; together with<br>hasProdLocallyUniformly_geometricSincProduct,<br>geometricSincProductFactors_multipliable,<br>hasProd_geometricSincProduct, and<br>geometricSincProduct_differentiable in<br>GeometricReciprocalGamma; exact real- $t$ formula for every<br>$ q  < 1$ , real-frequency local and compact uniform convergence<br>of the full prefixes, and standalone locally uniform entire<br>complex product, but no single complex-frequency full<br>phase-prefix package |
| Theorem 27.3                                                                              | partial | centeredRvachevEvenCumulant_eq_<br>bernoulliMersenne_formula,<br>centeredRvachevFullCumulant_odd<br>SinhDivBernoulliLog; CenteredMomentParity; proved at<br>$q = 1/2$ in the scaled $Z = 2Y$ normalization, not for arbitrary $q$<br>or as the displayed analytic log-MGF identity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Corollary 27.5                                                                            | partial | completeBellPolynomial<br>MomentCumulantAlgebra                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Result                                                                       | Status  | Lean counterpart                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Corollary 27.6                                                               | partial | centeredRvachevEvenCumulant_eq_bernoulliMersenne_formula, centeredRvachevFullCumulant_odd SinhDivBernoulliLog; CenteredMomentParity; the exact scaled $Z = 2Y_{1/2}$ cumulants are proved, but the displayed $Y_{1/2}$ normalization and the raw-moment rationality sentence are not packaged verbatim |
| Theorem 27.7                                                                 | partial | norm_rvachevFourierProduct_le_decayGauge_abs GlobalDecayEnvelope                                                                                                                                                                                                                                       |
| Theorem 27.8                                                                 | partial | geometricUniformDistribution_eq_withDensity GeometricUniformCDF                                                                                                                                                                                                                                        |
| Theorem 27.9                                                                 | partial | fabiusUniformSpline_eq_centeredPartialCDF_all, fabiusUniformSpline_eq_tsum_positivePart FabiusUniformSpline; proves the midpoint-corrected $q = 1/2$ spline/CDF and its global Thue–Morse positive-part formula, not the arbitrary-weight density, cube-volume, and piecewise-support package          |
| Theorem 27.11                                                                | partial | fabiusUniformSpline_eq_tsum_positivePart FabiusUniformSpline                                                                                                                                                                                                                                           |
| Proposition 27.12                                                            | partial | geometricUniformDistribution_eq_withDensity, isGreatest_deriv_fabiusReal GeometricUniformCDF; Convexity; the general- $q$ central-band formula and sharp supremum statement are absent                                                                                                                 |
| Corollary 27.13                                                              | partial | abs_uniformPartialSum_sub_weightedCoordinateSum_le FabiusUniformSpline                                                                                                                                                                                                                                 |
| Theorem 27.15                                                                | exact   | weightedSumCDF_eq_fabiusReal ProbabilityRepresentation                                                                                                                                                                                                                                                 |
| Theorem 27.16                                                                | exact   | rvachevFourier_eq_product FourierProduct                                                                                                                                                                                                                                                               |
| <i>Dyadic Gaussian–Thue–Morse structure and the values of <math>F</math></i> |         |                                                                                                                                                                                                                                                                                                        |
| Proposition 28.1                                                             | partial | weightedSumDistribution_real_Ioi_eq_rvachevUp ProbabilityLaplaceMoments; Lean has the strict survival event, not the stated closed-tail equality                                                                                                                                                       |
| Corollary 28.2                                                               | partial | intervalIntegral_rvachevUp_smul_eq_integral ProbabilityLaplaceMoments; the integral bridge is formalized, but not the displayed tail-moment formula                                                                                                                                                    |
| Theorem 28.3                                                                 | partial | complexMGF_weightedSumDistribution_eq_complexGeneratingFunction FabiusComplexMGF; only the half-base transform is present, not the general complex- $q$ product and recurrence                                                                                                                         |

| Result            | Status  | Lean counterpart                                                                                                                                                                                  |
|-------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Corollary 28.5    | partial | fabiusRecurrenceSequence_recurrence<br>FabiusRecurrenceSequence; coefficient-level half-base consequences are present, not the stated entire-function dilation package                            |
| Lemma 28.8        | partial | iteratedDeriv_fabiusReal_of_lt_one<br>BoundedDerivatives; the Lean domain and endpoint packaging differ from the displayed statement                                                              |
| Theorem 28.9      | exact   | fabiusAtInverseTwoPow_eq_halfMoment,<br>unitEndpointMoment_<br>weightedSumDistribution_eq_halfMoment<br>MomentPowerSeries; ProbabilityLaplaceMoments                                              |
| Proposition 28.10 | partial | fabiusAtInverseTwoPow_eq_halfMoment<br>MomentPowerSeries; only the $q = 1/2$ specialization is formalized                                                                                         |
| Corollary 28.12   | partial | completeBellPolynomial,<br>unitEndpointMoment_<br>weightedSumDistribution_eq_halfMoment<br>MomentCumulantAlgebra; ProbabilityLaplaceMoments;<br>the displayed Bell specialization is not packaged |
| Theorem 28.14     | partial | halfQBinomial_negativeDyadic_<br>polynomial_sum_eq_coeff<br>HalfQBinomial; only the half-base polynomial transform is present                                                                     |
| Corollary 28.15   | partial | halfQBinomial_negativeDyadic_<br>polynomial_sum_eq_coeff<br>HalfQBinomial; the general reciprocal-base statement is absent                                                                        |
| Theorem 28.17     | partial | coeff_finitePowerSeriesFilter<br>FinitePowerSeriesFilter; Lean has the diagonal filter, not this reciprocal Gaussian tail formula                                                                 |
| Corollary 28.19   | partial | qBinomial_half_eq, gaussianBinomial_symm<br>HalfQBinomial; the odd-integer valuation conclusion is not formalized                                                                                 |
| Theorem 28.21     | partial | qBinomialThueMorseTranslatedNumeratorIn_eq_centered<br>FabiusQBinomialTaylor; only the specialized half-base transfer used by the Fabius formula is present                                       |
| Corollary 28.23   | partial | qBinomialThueMorseTranslatedNumeratorIn_eq_centered<br>FabiusQBinomialTaylor; common-shift invariance is formalized only for the half-base numerator                                              |
| Lemma 28.25       | partial | thueMorseBlock_exp_neg_eq_laplacePrefix,<br>complexLaplacePrefix_eq_exp_mul_<br>shiftedComplexSincPrefix<br>ThueMorseComplexProductBridge; the full coefficient-series bridge is not stated       |
| Corollary 28.26   | partial | qBinomialThueMorseRawTranslatedNumeratorIn_eq_centered<br>FabiusRawQBinomialScalar; the component identities exist, not the displayed $\Psi_k$ normalization dictionary                           |

| Result            | Status  | Lean counterpart                                                                                                                                                        |
|-------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 28.28     | exact   | qBinomialThueMorseRawTranslatedNumeratorIn_eq_wolfram_sum,<br>qBinomialThueMorseTranslatedNumeratorIn_div_eq_halfMoment                                                 |
| Theorem 28.30     | exact   | FabiusRawQBinomialScalar; FabiusQBinomialTaylor<br>fabius_inverse_two_pow_eq_<br>qBinomialThueMorse_rawTranslated_sum_complex                                           |
| Corollary 28.31   | exact   | FabiusRawQBinomialScalar<br>globalFabius_dyadic_eq_qBinomialThueMorseDyadic_translated_sum_complex,<br>qBinomialThueMorseDyadicTranslatedFormulaIn_independent          |
| Proposition 28.35 | partial | FabiusDyadicQBinomialScalar<br>qBinomialFabiusReductionPolynomial_eq_reductionSumIn, iteratedDeriv_<br>fabiusReal_inv_two_pow_all                                       |
| Lemma 28.37       | none    | FabiusQBinomialTaylor; BoundedDerivatives; the two pieces are not the proposition's single Taylor-polynomial identity<br>—                                              |
| Theorem 28.38     | partial | fabiusAtInverseTwoPow_eq_composition_formula<br>FabiusInverseDyadicClosedForm; only the half-base specialization is formalized                                          |
| Theorem 28.39     | none    | —                                                                                                                                                                       |
| Corollary 28.41   | partial | rvachev_fourier_inversion_analytic,<br>rvachevFourier_eq_product<br>FourierAnalytic; FourierProduct; inversion and the product are separate, not the stated CDF formula |
| Theorem 28.43     | partial | sum_thueMorseSign_mul_affine_pow_eq_zero<br>ThueMorseSparseProuhet; only radix two is formalized                                                                        |
| Corollary 28.44   | exact   | prod_one_sub_pow_two_pow, sum_<br>thueMorseSign_mul_affine_pow_card<br>ThueMorseProductIdentity; ThueMorseSparseProuhet                                                 |
| Theorem 28.46     | partial | sum_abs_qToeplitzWeight, tendsto_sum_abs_qToeplitzWeight<br>GeneralQConditionNumber; LimitConditionNumber; the strict increase and full stated bound package are absent |
| Corollary 28.47   | partial | sum_abs_geometricLagrangeWeight_eq_prod<br>GeometricLagrangeQMoments; the error inequality is not stated and this counterpart is rational-valued                        |
| Theorem 28.50     | partial | tendsto_weighted_rows_of_tendsto<br>FabiusDiscreteLimitToeplitz; Lean is restricted to a normed ring rather than an arbitrary normed vector space                       |
| Corollary 28.51   | partial | fabiusDiscreteLimitApproximationComplex_eq_weighted_shiftSpline<br>FabiusDiscreteLimitIntegration; only the half-base discrete-limit spline layer is present            |

| Result            | Status  | Lean counterpart                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Principle 28.52   | partial | reversed_finite_qBinomial_theorem, sum_abs_qToeplitzWeight<br>GeometricQBinomialLagrange;<br>GeneralQConditionNumber; no declaration packages the same row and all conclusions of the principle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Proposition 28.54 | partial | geometricUniformDistribution_reflection<br>GeometricUniformLaw; only positive-base reflection is formalized, not the negative-base affine law                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Theorem 28.55     | partial | mgf_id_self_similar<br>GeometricCgfTails; a refinement identity exists, but not the general normalized analytic-germ uniqueness theorem                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Theorem 28.58     | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Theorem 28.59     | partial | charFun_geometricUniformDistribution_eq_phase_mul_geometricSincProduct,<br>charFun_geometricUniformDistribution_eq_phase_mul_geometricReciprocalGamma,<br>tendstoLocallyUniformly_prefix_sinc_charFun, tendstoUniformlyOn_prefix_sinc_charFun<br>GeometricSincCharacteristicFunction; with<br>hasProdLocallyUniformly_geometricSincProduct,<br>geometricSincProductFactors_multipliable,<br>hasProd_geometricSincProduct, and<br>geometricSincProduct_differentiable from<br>GeometricReciprocalGamma, plus summable_norm_sineTerm_qpow_pair, geometricSincProduct_eq_tprod_pair, and geometricSincProduct_eq_tprod_complexQPochhammerInf from<br>RvachevPochhammerFactorization; general uncentered real-frequency sinc bridge, locally uniform entire $S_q$ , the pointwise general- $q$ nome- $q^2$ Pochhammer factorization, and real-frequency local and compact uniform convergence of the full prefixes are exact, but the named centered/MGF wrapper, reciprocal outside-disk formula, zero-pole exchange, and named local-uniform or normal convergence of the Pochhammer-product right-hand side are not packaged |
| Theorem 28.62     | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Proposition 28.63 | none    | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Corollary 28.65   | partial | fabiAtInverseTwoPow_eq_halfMoment,<br>halfQPochhammer_eq_mersenne_div<br>MomentPowerSeries; HalfQBinomial; the polynomial $\mathcal{P}_n(q)$ and general- $q$ endpoint formula are absent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Theorem 28.68     | partial | map_gaussianBinomial<br>FiniteQBinomialCore; functoriality is formalized, but not the theorem's full universal characterization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Corollary 28.69   | exact   | finiteQPochhammerIn_self_mul_gaussianBinomial, finiteQPochhammerIn_self_eq_mul_mul_gaussianBinomial<br>FiniteQBinomialCore                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Result            | Status  | Lean counterpart                                                                                                                                                                                                                                                                                                                          |
|-------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 28.72     | none    | —                                                                                                                                                                                                                                                                                                                                         |
| Proposition 28.74 | partial | <code>thueMorsePowerSum_succ</code><br><code>ThueMorsePrefix</code> ; Lean has the untranslated recurrence, not the arbitrary-shift and complexity statement                                                                                                                                                                              |
| Theorem 28.75     | partial | <code>sum_weight_mul_pow_card_add</code><br><code>LagrangeResidualMoments</code> ; the hypotheses and ordering of the general residual theorem differ                                                                                                                                                                                     |
| Corollary 28.77   | partial | <code>sum_geometricLagrangeWeight_mul_pow_eq_gaussianBinomial</code><br><code>GeometricQBinomialLagrange</code> ; this is the special Lagrange-weight case, not the arbitrary-weight corollary                                                                                                                                            |
| Corollary 28.79   | partial | <code>coeff_finitePowerSeriesFilter</code><br><code>FinitePowerSeriesFilter</code> ; the diagonal identity is formalized, not the displayed complete tail formula                                                                                                                                                                         |
| Lemma 28.82       | partial | <code>fabiusUniformSpline_eq_tsum_positivePart</code><br><code>FabiusUniformSpline</code> ; the spline formula exists without the same unrestricted expectation identity                                                                                                                                                                  |
| Theorem 28.84     | partial | <code>globalFabius_dyadic_eq_qBinomialThueMorseDyadic_translated_sum_complex</code><br><code>FabiusDyadicQBinomialScalar</code> ; exact dyadic-grid values are formalized, not the off-grid cell formula                                                                                                                                  |
| Proposition 28.88 | partial | <code>fabiusUniformSpline_eq_centeredPartialCDF_all</code><br><code>FabiusUniformSpline</code> ; the midpoint identity is present, but the general- $q$ sharp second-order rate is absent                                                                                                                                                 |
| Lemma 28.91       | partial | <code>fabiusDiscreteLimitRangeLength_eq_floor_add_one</code> , <code>fabiusUniformSpline_eq_tsum_positivePart</code><br><code>FabiusDiscreteLimitToeplitz</code> ; <code>FabiusUniformSpline</code> ; no declaration states the same cutoff-to-spline equality                                                                            |
| Theorem 28.93     | partial | <code>fabiusDiscreteLimitApproximationComplex_eq_weighted_shiftSpline</code><br><code>FabiusDiscreteLimitIntegration</code> ; the convolutional approximation is present without the theorem's full interpolation geometry                                                                                                                |
| Theorem 28.95     | partial | <code>norm_fabiusDiscreteLimitApproximationComplex_center_sub_globalFabius_le</code><br><code>norm_fabiusDiscreteLimitApproximationComplex_center_sub_globalFabius_le_sixtyFour</code><br><code>FabiusDiscreteLimitRates</code> ; an explicit global $4^{-n}$ rate is formalized, but not the displayed sharp $16^{-n}$ rate and constant |
| Proposition 28.98 | partial | <code>Appell.derivative_poly</code> , <code>Appell.poly_translate</code><br><code>AppellSequence</code> ; the generic derivative and addition laws exist, but the Fabius-moment reflection law is absent                                                                                                                                  |

| Result           | Status  | Lean counterpart                                                                                                                                                                                                                                                                                                         |
|------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 28.99    | partial | qBinomialThueMorseTranslatedNumeratorIn_eq_centered,<br>qBinomialThueMorseRawTranslatedNumeratorIn_eq_centered<br>FabiusQBinomialTaylor; FabiusRawQBinomialScalar;<br>only the $\theta = 0$ and $\theta = -1$ specializations are present                                                                                |
| Corollary 28.102 | partial | extendedFabius_reduction<br>TaylorReduction; Lean has the binary reduction identity, not this Appell two-cell formulation                                                                                                                                                                                                |
| Theorem 28.103   | partial | extendedFabius_eq_<br>globalBinaryReductionSum_add_remainder,<br>hasSum_globalBinaryReductionSummand_succ_all<br>FabiusBinaryReductionSeries; the binary telescope is formalized, but not the stated Appell summand, factorial residual bound, and termination package                                                   |
| Corollary 28.104 | partial | globalFabius_eq_qBinomialThueMorse_<br>global_sum_complex, globalFabius_eq_<br>qBinomialThueMorse_teslescope_add_<br>remainder_complex<br>FabiusGlobalQBinomialSeries;<br>FabiusDiscreteLimitIntegration; the formalized global series has a different convolutional summand and lacks the stated sharp residual package |

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*Euler telescoping and very-well-poised summations*


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|                |      |   |
|----------------|------|---|
| Lemma 10.1     | none | — |
| Lemma 10.2     | none | — |
| Theorem 10.3   | none | — |
| Corollary 10.5 | none | — |

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*Theta modularity, lattice dissection, and sums of squares*


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|                   |      |                      |
|-------------------|------|----------------------|
| Theorem 15.1      | none | —                    |
| Definition 15.3   | n/a  | notation only        |
| Proposition 15.4  | none | —                    |
| Proposition 15.5  | none | —                    |
| Definition 15.6   | n/a  | transform convention |
| Theorem 15.7      | none | —                    |
| Lemma 15.8        | none | —                    |
| Theorem 15.9      | none | —                    |
| Definition 15.10  | n/a  | notation only        |
| Theorem 15.11     | none | —                    |
| Definition 15.12  | n/a  | notation only        |
| Proposition 15.13 | none | —                    |
| Theorem 15.14     | none | —                    |
| Proposition 15.16 | none | —                    |
| Lemma 15.18       | none | —                    |

| Result                                                     | Status | Lean counterpart |
|------------------------------------------------------------|--------|------------------|
| Theorem 15.19                                              | none   | —                |
| Corollary 15.20                                            | none   | —                |
| Theorem 15.21                                              | none   | —                |
| <i>Bailey transforms and the Andrews–Gordon identities</i> |        |                  |
| Definition 17.1                                            | n/a    | notation only    |
| Theorem 18.1                                               | none   | —                |
| Lemma 18.3                                                 | none   | —                |
| Lemma 18.5                                                 | none   | —                |
| Corollary 18.6                                             | none   | —                |
| Lemma 18.7                                                 | none   | —                |
| Theorem 18.8                                               | none   | —                |
| Theorem 18.10                                              | none   | —                |
| Corollary 18.11                                            | none   | —                |
| <i>Exact limits, reciprocity, and research frontiers</i>   |        |                  |
| Theorem 23.1                                               | none   | —                |
| Proposition 23.2                                           | none   | —                |
| Proposition 23.3                                           | none   | —                |
| Proposition 23.4                                           | none   | —                |
| Proposition 23.5                                           | none   | —                |



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